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LINEAR ALGEBRA



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TO THE READER

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Preface

This textbook is a comprehensive united course in linear algebra and analytic geometry based on lectures read by the author for many years at various institutes to future specialists in computational mathematics.

It is intended mainly for those in whose education computational mathematics is to occupy a substantial place. Much of the instruction in this speciality is connected with the traditional mathematical courses. Nevertheless the interests of computational mathematics make it necessary to introduce large enough changes in both the methods of presentation of these courses and their content.

There are a lot of good textbooks of linear algebra and analytic geometry, including English textbooks. Their direct use for training specialists in computational mathematics proves difficult, however. To our mind, this is mainly due to computers to be requiring much more facts from linear algebra than is usually given in the available books. And the necessary additional facts fail as a rule to appear in the conventional texts of linear algebra and are instead contained in either papers scattered in journals or books relating to other branches of mathematics.

Computer students begin to get familiar with linear algebra and analytic geometry sufficiently early. At this same time their scientific world outlook begins to shape. Therefore what is read in this course and how it is read determine to a great extent the future perception of entire computational mathematics by the students. Of course computer students must get a systematic and rigorous presentation of all the fundamentals of algebra and geometry. But they must be made familiar as early as possible at least briefly with those problems and methods which computational algebra has accumulated.

Introduction to problems of computations allows the lecture course to be effectively accentuated in the interests of computational mathematics and a close relation to be established between theory and numerical methods in linear algebra. The basic material for this is provided by the simplest facts pertaining to such topics as round-off errors, perturbation instability of many basic notions of linear algebra, stability of orthonormal systems, metric and

normed spaces, singular decomposition, bilinear forms and their relation to computational processes, and so on.

Of course incorporation of novel and sufficiently large material is impossible without substantial restructuring of the traditional course. This book is an attempt at such a restructuring.

V. V. Voyevodin

May, 6 1982

PART I

Vector Spaces

CHAPTER 1

Sets, Elements, Operations

1. Sets and elements

In all areas of activity we have continually to deal with various collections of objects united by some common feature.

Thus, studying the design of some mechanism we may consider the totality of its parts. An individual object of the collection may be any of its parts, the feature uniting all the objects being the fact that they all belong to a quite definite mechanism.

Speaking of the collection of points of a circle in the plane we actually speak of objects, points of a plane, that are united by the property that they are all the same distance from some fixed point.

It is customary in mathematics to call a collection of objects united by some common feature a *set* and to term the objects *elements* of the set. It is impossible to give the notion of set a rigorous definition. We may of course say (as we did!) that the set is a "collection", a "system", a "class" and so on. It looks, however, very much like a formal utilization of the rich vocabulary of our language.

To define a concept it is first of all necessary to point out how it is related to more general notions. For the concept of set this cannot be done, since there is no more general concept for it in mathematics. Instead of defining it we are compelled to resort to illustrations.

One of the simplest ways of describing a set is to give a complete list of elements constituting the set. For example, the set of all books available to a reader of a library is completely defined by their lists in the library catalogues, the set of all prices of goods is completely defined by a price-list and so on. However, this method applies only to *finite sets*, i.e. such that contain a finite number of elements. But *infinite sets*, i.e. sets containing infinitely many elements, cannot be defined with the aid of a list. How, for example, can we compile a list of all real numbers?

When it is impossible or inconvenient to give a set by using a list, it can be given by pointing out a *characteristic property*,

i.e. a property possessed only by the elements of the set. In problems in defining loci, for example, a characteristic property of the set of points which is the solution of the problem is nothing than the collection of conditions these points must satisfy according to the requirements of the problem.

The description of a set may be very simple and cause no difficulties. For example, if we take a set consisting of two numbers, 1 and 2, then clearly neither the number 3 nor a notebook or a car will be contained in that set. But in the general case, giving sets by their characteristic properties sometimes results in complications. The reasons for these are rather numerous.

One reason seems to be the insufficient definiteness of the concepts used to describe sets. Suppose we are considering the set of all planets of the solar system. What is the question? There are nine major planets known. But there are over a thousand minor planets, or asteroids, turning round the sun. The diameters of some measure hundreds of kilometres, but there are also such whose diameter is under one kilometre. As methods of observation improve, smaller and smaller planets will be discovered, and finally the question will arise as to where the minor planets end and the meteorites and solar dust begin.

These are not the only difficulties with the definition of the structure of a set. Sometimes sets, quite well defined at first sight, turn out to be very poorly defined, if defined at all. Suppose, for example, some set consists of one number. Let us define that number as *the smallest integer not definable in under a hundred words*. Assume that only words taken from some dictionary and their grammatical forms are used and that the dictionary contains such words as "one", "two" and so on.

Notice that on the one hand such an integer must not exist for it is defined in under a hundred words, italicized above, and according to the meaning of the words it cannot be defined in such a way. But on the other hand since the number of the words used in the language is finite this means that there are integers that cannot be defined in under a hundred words and hence there is a smallest one among these integers.

The area of mathematics called *set theory* has accumulated many examples where the definition of the set is intrinsically contradictory. The study of the question under what conditions this may happen has led to deep investigations in logic, we shall lay them aside, however. Throughout the following it will be assumed that we are considering only the sets that are defined precisely and without contradictions and have a structure that raises no doubts.

As a rule, we shall denote sets by capital Latin letters $A, B, \dots$ and elements of sets by small letters $a, b, \dots$. We shall write $x \in A$ if an element x is in a set A and $x \notin A$ if it is not.

We shall sometimes introduce into consideration the so-called *empty set*, or *null set*, i.e. the set that contains no elements. It is convenient to use this set where it is not known in advance whether there is at least one element in the collection under consideration.

Exercises

1. Construct finite and infinite sets. What properties are characteristic of them?
2. Construct sets whose descriptions contain a contradiction.
3. Is the set of real roots of the polynomial $x^4 + 4x^3 + 7x^2 + 4x + 1$ empty?
4. Construct sets whose elements are sets.
5. Construct sets that contain themselves as an element.

2. Algebraic operation

Among all possible sets there are such on whose elements some operations are allowed to be performed. Suppose we are considering the set of all real numbers. Then for each of its elements such operations as calculation of the absolute value of that element and calculation of the sine of that element can be defined, and for every pair of elements addition and multiplication can be defined.

In the above example, note especially the following features of the operations. One is the definiteness of all operations for *any* element of the given set, another is the *uniqueness* of all operations and the final feature is that the result of any operation belongs to the elements of the *same* set. Such a situation takes place by far not always.

An operation may be defined not for all elements of a set; for example, calculation of logarithms is not defined for negative numbers. Taking the square root of positive numbers is defined, but not uniquely. However, even if an operation is uniquely defined for every element, its result may not be an element of the given set. Consider division on the set of positive integers. It is clear that for any two numbers of the set division is realizable but its result is not necessarily an integer.

Let A be a set containing at least one element. We shall say that an *algebraic operation* is defined in A if a law is indicated by which any pair of elements, a and b , taken from A in a definite order, is uniquely assigned a third element, c , also from that set.

This operation may be called addition, and c will then be called the sum of a and b and designated $c = a + b$; it may be called multiplication, and c will then be called the product of a and b and designated $c = ab$.

In general terminology and notation for an operation defined in A will not play any significant part in what follows. As a rule,

we shall employ the notation of the sum and product regardless of the way the operation is in fact defined. But if it is necessary to emphasize some general properties of the algebraic operation, then it will be designated $*$.

Consider some simple examples to see what features an algebraic operation may have. Let A be the set of all positive rational numbers. We introduce for the elements of A the usual multiplication and division of numbers and use the conventional notation. It is not hard to check that both operations on A are algebraic. But while for multiplication $ab = ba$ for every element of A , i.e. the order of the elements is immaterial, for division, on the contrary, it is very essential, for the equation $a : b = b : a$ is possible only if $a = b$. Thus, although the algebraic operation is defined for an ordered pair of elements the ordering of the elements is inessential.

An algebraic operation is said to be *commutative* if its result is independent of the order of choosing the elements, i.e. for any two elements a and b of a given set $a * b = b * a$. It is obvious that of the conventional arithmetical operations on numbers addition and multiplication are commutative and subtraction and division are noncommutative.

Suppose now that three arbitrary elements, a , b and c , are taken. Then the question naturally arises as to what meaning should be given to the expression $a * b * c$. How can the algebraic operation defined for two elements be applied to three?

Since we can apply an algebraic operation only to a pair of elements, the expression $a * b * c$ may be given a definite meaning by bracketing either the first two or the last two elements. In the first case the expression becomes $(a * b) * c$ and in the second we get $a * (b * c)$. Consider the elements $d = a * b$ and $e = b * c$. Since they are members of the original set, $(a * b) * c$ and $a * (b * c)$ may be considered as the result of applying the algebraic operation to the elements d , c and a , e , respectively.

In general, the elements $d * c$ and $a * e$ may turn out to be different. Consider again the set of positive rational numbers with the algebraic operation of division. It is easy to see that as a rule $(a : b) : c \neq a : (b : c)$. For example, $((3/2) : 3) : (3/4) = 2/3$, but $(3/2) : (3 : (3/4)) = 3/8$.

An algebraic operation is said to be *associative* if for any three elements, a , b and c , of the original set $a * (b * c) = (a * b) * c$.

The associativity of an operation allows one to speak of a uniquely defined result of applying an algebraic operation to any three elements, a , b and c , meaning by it any of the equivalent expressions $a * (b * c)$ and $(a * b) * c$, and write $a * b * c$ without brackets.

In the case of the associative operation one can also speak of the uniqueness of the expression $a_1 * a_2 * \dots * a_n$ containing any finite number of elements $a_1, a_2, \dots, a_n$. By $a_1 * a_2 * \dots * a_n$

we shall mean the following. Arrange brackets in this expression in an arbitrary way, if only it may be defined by successive application of an algebraic operation to pairs of elements. For example, for the five elements a_1, a_2, a_3, a_4 and a_5 the brackets may be arranged either like this: $a_1 * ((a_2 * a_3) * (a_4 * a_5))$ or like this: $((a_1 * a_2) * a_3) * (a_4 * a_5)$ or in many other ways.

We prove that for an associative operation the result of a calculation is independent of bracket arrangement. Indeed, for $n = 3$ this assertion follows from the definition of an associative operation. Therefore we set $n > 3$ and assume that for all numbers less than n our assertion is already proved.

Let elements $a_1, a_2, \dots, a_n$ be given and suppose that brackets indicating the order in which to perform the operation are arranged in some way. Notice that the final step is always to perform the operation on the two elements $a_1 * a_2 * \dots * a_k$ and $a_{k+1} * a_{k+2} * \dots * a_n$ for some k satisfying the condition $1 \leq k \leq n - 1$. Since both expressions contain fewer than n elements, by assumption they are uniquely defined and it remains for us to prove that for any positive integers $k, l, l \geq 1$,

$$\begin{aligned} (a_1 * a_2 * \dots * a_k) * (a_{k+1} * a_{k+2} * \dots * a_n) \\ = (a_1 * a_2 * \dots * a_{k+l}) * (a_{k+l+1} * a_{k+l+2} * \dots * a_n). \end{aligned}$$

Letting

$$\begin{aligned} a_1 * a_2 * \dots * a_k &= b, \\ a_{k+1} * a_{k+2} * \dots * a_{k+l} &= c, \\ a_{k+l+1} * a_{k+l+2} * \dots * a_n &= d, \end{aligned}$$

we get, on the basis of the associativity of the operation,

$$b * (c * d) = (b * c) * d,$$

and our assertion is proved.

If an operation is commutative as well as associative, then the expression $a_1 * a_2 * \dots * a_n$ is independent of the order of its elements. It is left as an exercise for the reader to prove that this assertion is true.

It should not be supposed that the commutativity and associativity of an operation are in some way related to each other. It is possible to construct operations with very different combinations of these properties. We have already seen from the examples of multiplication and division of numbers that an operation may be commutative and associative or noncommutative and nonassociative. Consider two more examples. Let a set consist of three elements, a, b , and c .

Give algebraic operations by these tables:

$$\begin{array}{c|c|c|c} * & a & b & c \\ \hline a & a & c & b \\ \hline b & c & b & a \\ \hline c & b & a & c \end{array} \qquad \begin{array}{c|c|c|c} * & a & b & c \\ \hline a & a & a & a \\ \hline b & b & b & b \\ \hline c & c & c & c \end{array} \qquad (2.1)$$

and let the first element be always chosen from the column and the second from the row, and let the result of the operation be taken at the intersection of the corresponding row and column. In the first case the operation is obviously commutative but not associative, since, for example,

$$\begin{aligned} (a * b) * c &= c * c = c, \\ a * (b * c) &= a * a = a. \end{aligned}$$

In the second case the operation is not commutative, but associative, which is easy to show by a straightforward check.

Exercises

1. Is the operation of calculating $\tan x$ on the set of all real numbers x algebraic?
2. Consider the set of real numbers x satisfying the inequality $|x| \leq 1$. Are the operations of multiplication, addition, division and subtraction algebraic on this set?
3. Is the algebraic operation $x * y = x^3 + y$ commutative and associative on the set of all real numbers x and y ?
4. Let a set consist of a single element. How can the algebraic operation be defined on that set?
5. Construct algebraic operations on a set whose elements are sets. Are these operations commutative, associative?

3. Inverse operation

Let A be a set with some algebraic operation. As we know, it assigns to any two elements a and b of A a third element $c = a * b$. Consider the collection C of the elements of A that can be represented as the result of the given algebraic operation. It is clear that regardless of the algebraic operation all the elements of C are at the same time the elements of A . It is quite optional, however, for all the elements of A to be in C .

Indeed, fix in A some element f and assign it to any pair of elements a and b of A . It is obvious that the resulting correspondence is an algebraic operation, commutative and associative. The set C will contain only one element f regardless of the number of elements in A .

Exactly what elements of A are in C depends on the algebraic operation. Let it be such that C coincides with A , i.e. let both sets contain the same elements. Then each element of A can be represented as the result of the given algebraic operation on some two elements of the same set A . Of course, such a representation may be nonunique. Nevertheless we conclude that each element of A may be assigned definite pairs of elements of A .

Thus, the original algebraic operation generates on A some other operation. This may not be unique, since one element may be assigned to more than one pair. But even if it is unique, it will not be algebraic since it is defined not for any *pair* of elements, but for only *one* element, although this may be arbitrary. In respect to the given algebraic operation it would be natural to call this new operation the "inverse" one. In fact, however, by the inverse operation we shall mean something different, something closer to the concept of algebraic operation.

Notice that investigating the "inverse" operation is equivalent to investigating those elements u and v which satisfy the equation

$$u * v = b \quad (3.1)$$

for different elements b . Investigation of this equation for the two elements u and v is easy to reduce to investigation of two equations for one element. To do this it suffices to fix one of them and to determine the other from equation (3.1). So investigating the "inverse" operation is mathematically equivalent to solving the equations

$$a * x = b, \quad y * a = b \quad (3.2)$$

for the elements x and y of A , with different elements a and b of A fixed.

Suppose (3.2) have unique solutions for any a and b . Then each ordered pair of elements a and b of A can be assigned uniquely defined elements x and y of A , i.e. *two* algebraic operations can be introduced. These are called respectively the *right* and the *left inverse* of the basic operation. If they exist, we shall say that the basic operation has an *inverse*. Note that the above example shows that an algebraic operation, even commutative and associative one, may lack both the right and the left inverse.

The existence of an inverse implies in fact the existence of two, in general different, algebraic operations, the right and the left inverse. We are compelled therefore to speak of different elements x and y . If, however, the algebraic operation is commutative and has an inverse, then obviously $x = y$ and the right inverse coincides with the left inverse.

Consider some examples. Let A be the real axis with the usual multiplication of numbers as algebraic operation. This has no inverse on the *given* set, since when, for example, $a = 0$ and $b = 1$,

equations (3.2) cannot hold for any numbers x and y . But if we consider the operation of multiplication given only on the set of positive numbers, then the operation will *now* have an inverse.

Indeed, for any positive numbers a and b there are unique positive numbers x and y satisfying equations (3.2). The inverse in this case is nothing but division of numbers. The fact that in reality $x = y$ is of no interest to us now.

The operation of addition has no inverse if it is given on the set of *positive* numbers, since equations (3.2), for example, can hold for no positive x and y if $a = 2$ and $b = 1$. But if the operation of addition is given on the *entire* real axis, then its inverse exists and is nothing but subtraction of numbers.

The example of addition and multiplication shows that a direct operation and its inverse may have quite different *properties*. From the associativity and commutativity of an algebraic operation need not necessarily follow the associativity or commutativity of its inverse, even if the inverse exists. Moreover, as already noted above, a commutative and associative algebraic operation may simply have neither the right nor the left inverse.

These simple examples show yet another important fact. Consider again multiplication on the set of positive numbers. Its right and left inverse coincide for this operation and are division of numbers. At first it may seem that now for division of numbers the inverse is multiplication of numbers. This is not quite so, however.

Indeed, write the corresponding equations (3.2)

$$a : x = b, \quad y : a = b.$$

It is then obvious that

$$x = a : b, \quad y = a \cdot b.$$

Consequently, the right inverse of division is again division, and the left inverse is multiplication. Thus, the inverse of an inverse does not necessarily coincide with the original algebraic operation.

Exercises

1. Are there right and left inverses of the algebraic operations given by tables (2.1)?
2. What are the right and left inverse of the algebraic operation $x \bullet y = x^y$ defined on the set of positive numbers x and y ?
3. Prove that if the right and the left inverse coincide, then the original algebraic operation is commutative.
4. Prove that if an algebraic operation has an inverse, then the right and the left inverse have inverses too. What are these?
5. Construct an algebraic operation for which all four inverses of the inverse operations coincide with the original operation.

4. Equivalence relation

Notice that in discussing above the properties of the algebraic operation we implicitly assumed the possibility of checking any two elements of a set for coincidence or noncoincidence. Moreover, we treated coinciding elements rather freely never making any distinction between them. We did not assume anywhere that the coinciding elements were indeed one element rather than different objects. But actually we only used the fact that some group of elements, which we called equal, are the same in certain contexts.

This situation occurs fairly often. Investigating general properties of similar triangles we in fact make no distinction between any triangles having the same angles. In terms of the properties preserved under a similarity transformation, these triangles are indistinguishable and could be called "equal". Investigating the criteria of the equality of triangles we make no difference between the triangles that are situated in different places of the plane but can be made to coincide if displaced.

In many different problems we shall be faced with the necessity of partitioning one set or another into groups of elements united according to some criterion. If none of the elements is in two different groups, then we shall say that the set is partitioned into *disjoint*, or *nonoverlapping*, groups or *classes*.

Although the criteria according to which the elements of a set are partitioned into classes may be very different, they are not entirely arbitrary. Suppose, for example, that we want to divide into classes all real numbers, including numbers a and b into the same class if and only if $b > a$. Then no number a can be in the same class with itself, since a is not greater than a itself. Consequently, no partitioning into classes according to this criterion is possible.

Let some criterion be given. We assume that with regard to any pair of elements a and b of a set A it can be said that either a is related to b by the given criterion or not. If a is related to b , then we shall write $a \sim b$ and say that a is *equivalent* to b .

Even the analysis of the simplest examples suggests the conditions a criterion must satisfy for partitioning a set A into classes according to it to be possible. Namely:

1. Reflexivity: $a \sim a$ for all $a \in A$.
2. Symmetry: if $a \sim b$, then $b \sim a$.
3. Transitivity: if $a \sim b$ and $b \sim c$, then $a \sim c$.

A criterion satisfying these conditions is called an *equivalence relation*.

We prove that any equivalence relation partitions a set into classes. Indeed, let K_a be a group of elements of A equivalent to

a fixed element a . By reflexivity $a \in K_a$. We show that two groups K_a and K_b either coincide or have no elements in common.

Let some element c be in K_a and K_b , i.e. let $c \sim a$ and $c \sim b$. By symmetry $a \sim c$ and by transitivity $a \sim b$ and, of course, $b \sim a$. If now $x \in K_a$, then $x \sim a$ and hence $x \sim b$, i.e. $x \in K_b$. Similarly, if $x \in K_b$, then it follows that $x \in K_a$. Thus, two groups having at least one element in common completely coincide and we have indeed obtained a partition of the set A into classes.

Any two elements may be equivalent or nonequivalent in terms of the criterion in question. Nothing will happen if we call equivalent elements *equal* (with respect to a given criterion!) and nonequivalent elements *unequal* (with respect to the same criterion!).

It may seem that in doing so we ignore the meaning of the word "equal", for now elements equal according to one criterion may prove unequal according to another. There is nothing unnatural to it, however. In every particular problem we distinguish the elements or do not only in relation to their properties that are of interest to us in this particular problem, and in different problems we may be concerned with different properties of the same elements.

It will be assumed in what follows that, whenever necessary, for the elements of the set an equality criterion must be defined saying that an element a is or is not equal to an element b . If a is equal to b , then we shall write $a = b$, and $a \neq b$ otherwise. It will also be assumed that the equality criterion is an equivalence relation. The reflexivity, symmetry and transitivity conditions may be regarded as reflecting the most general properties of the usual equality relation of numbers.

The equality relation allows us to partition an entire set into classes of elements which we have decided for some reasons to consider equal. This means that the difference between the elements of the same class is of no importance to us. Consequently, in all situations to be considered in what follows the elements called equal must exhibit sameness.

If the equality relation is introduced *axiomatically*, i.e. without reference to the particular nature of the elements, it will be agreed to assume that the equality sign merely implies that the elements on its sides simply coincide, that is that it is the same element. When the equality sign is used in this way, the properties of reflexivity, symmetry and transitivity require no particular convention. Partitioning a set into classes of equal elements will make each class consist of only one element.

Where the equality relation is introduced relying on a *particular nature of elements* it may happen that some or all classes of equal elements will consist of more than one element. This makes us impose additional requirements on the operations on elements to be introduced.

Indeed, as we have agreed, equal elements must exhibit sameness. Therefore every operation to be introduced must now necessarily give equal results when applied to equal elements. In fact we shall never verify this requirement, and it will be left for the reader to see for himself that the given property holds for the operations to be introduced.

Exercises

1. Is it possible to divide all the countries of the world into classes, placing two countries in the same class if and only if they have a common border? If not, why?

2. Consider a set of cities with motorway communication. Say that two cities A and B are connected if one can go from A to B by motorway. Can the cities be divided into classes according to this criterion? If they can, what are the classes?

3. Say that two complex numbers a and b are equal in absolute value if $|a| = |b|$. Is this criterion an equivalence relation? What is this partition into classes?

4. Consider the algebraic operations of addition and multiplication of complex numbers. How do they act on classes of elements equal in absolute value?

5. Construct algebraic operations on the set defined in Exercise 2. How do they act on the classes?

5. Directed line segments

The foregoing examples may give an impression that all talk about operations on elements of sets concerns only operations on various number sets. It is not so, however. In what follows we construct many examples of other kinds of sets with operations, but for the present consider just one example which will be constantly referred to throughout the course.

The most fundamental concepts of physics are such notions as force, displacement, velocity, acceleration. They are all characterized not only by a number giving their magnitude but also by some direction. We now construct a geometrical analogue of such notions.

Let A and B be two distinct points in space. On the straight line through them they define in a natural way some line segment. It is assumed that the points are always given in a definite order, for example, first A is given and then B is. Now we can state a direction on the constructed line segment, namely, the direction from the first point A to the second point B .

A line segment together with the direction stated on it is called a *directed line segment* with initial point A and terminal point B . It will be otherwise termed a *vector*, and A will be called the *point of application* of the vector. A vector with point of application A will be said to be *fixed* at A .

For directed line segments or vectors double notation will be used. If it must be stressed that a directed line segment with initial point A and terminal point B is meant, we shall write $\overrightarrow{AB}$. But if we do not care exactly what points of the directed line segment are limiting points, then we shall use some simpler notation, small Latin letters, for example. In drawings directed line segments will be denoted by arrows, with arrowheads always at the terminal point of the line segment.

In a directed line segment it is essential which of the limiting points is initial and which is terminal. Directed line segments $\overrightarrow{AB}$

and $\overrightarrow{BA}$ will therefore be considered different.

So we can construct different sets whose elements are directed line segments. Before introducing operations on elements we define what directed line segments will be considered equal.

Consider first the (*parallel*) *translation* of a directed line segment

$\overrightarrow{AB}$ to a point C . Let C be off the straight line through A and B

(Fig. 5.1). Draw the straight line through A and C , then the straight line through C parallel to AB , and finally the straight line through B parallel to AC . Denote the point of the intersection of the last two lines by D . The directed line segment $\overrightarrow{CD}$ will be precisely the result of the translation of $\overrightarrow{AB}$ to C . But if C is on the straight line through A and B , then the directed line segment $\overrightarrow{CD}$ is obtained

by shifting the directed line segment $\overrightarrow{AB}$ along the straight line containing it until the point A coincides with C .

Now we can give a definition of the equality of vectors. Two vectors are said to be *equal* if they can be made to coincide under a translation. It is not hard to see that this definition of equality is an equivalence relation, i.e. possesses the properties of reflexivity, symmetry and transitivity.

Thus the collection of all vectors can be broken down in a natural way into classes of equal vectors. It is simply sufficient to describe each of the classes. It is obtained by translating any of the vectors of a class to each point of space.

Notice that there is one and only one vector, in every class of equal vectors, fixed at any point of space. In comparing vectors

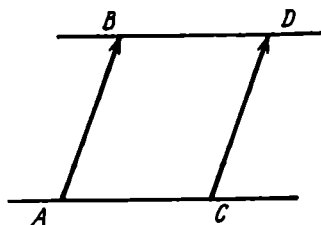


Fig. 5.1

a and b therefore we can use the following device. Fix some point and translate to it the vectors a and b . If they completely coincide, then $a = b$, and if not, $a \neq b$.

Besides the set consisting of all vectors of a space we shall often deal with other sets. These will mainly be sets of vectors either parallel to some straight line or lying on it or parallel to some plane or lying in it. Such vectors will be called respectively *collinear* and *coplanar*. Of course, on the sets of collinear and coplanar vectors the above definition of the equality of vectors is preserved.

We shall also consider the so-called *zero directed line segments* whose initial and terminal points coincide. The direction of zero vectors is not defined and they are all considered equal by definition. If it is not necessary to specify the limiting points of a zero vector, then we shall denote that vector by 0 .

Also it will be assumed by definition that any zero vector is parallel to any straight line and any plane. Throughout the following therefore, unless otherwise specified, the set of vectors of a space, as well as any set of collinear or coplanar vectors, will be assumed to include the set of all zero vectors. This should not be forgotten.

Exercises

1. Prove that the nonzero vectors of a space can be partitioned into classes of nonzero collinear vectors.
2. Prove that any class of nonzero collinear vectors can be partitioned into classes of nonzero equal vectors.
3. Prove that any class of nonzero equal vectors is entirely in one and only one class of nonzero collinear vectors.
4. Can the nonzero vectors of a space be partitioned into classes of coplanar vectors? If not, why?
5. Prove that any set of nonzero coplanar vectors can be partitioned into classes of nonzero collinear vectors.
6. Prove that any pair of different classes of nonzero collinear vectors is entirely in one and only one set of nonzero coplanar vectors.

6. Addition of directed line segments

As already noted, force, displacement, velocity and acceleration are the originals of the directed line segments we have constructed. If these line segments are to be useful in solving various physical problems, we must take into account the corresponding physical analogies when introducing operations.

Well known is the operation of addition of forces performed by the so-called parallelogram law. The same law is used to add displacements, velocities and accelerations. According to the introduced terminology this operation is algebraic, commutative and associative. Our immediate task is to construct a similar operation on directed line segments.

The operation of *vector addition* is defined as follows. Suppose it is necessary to add vectors a and b . Translate the vector b to the terminal point of a (Fig. 6.1). Then the sum $a + b$ is the vector whose initial point coincides with the initial point of a and whose terminal point coincides with the terminal point of b . This rule is usually called the *triangle law*.



Fig. 6.1

It is obvious that vector addition is an algebraic operation. We shall prove that it is commutative and associative.

To establish the commutativity of addition suppose first that a and b are not collinear. Apply them to a common origin O (Fig. 6.2). Denote by A and B the terminal points of a and b respectively and consider the parallelogram $OBCA$. It follows from the definition of the equality of vectors that

$$\overrightarrow{BC} = \overrightarrow{OA} = a, \quad \overrightarrow{AC} = \overrightarrow{OB} = b.$$

But then the same diagonal $\overrightarrow{OC}$ of the parallelogram $OBCA$ is simultaneously $a + b$ and $b + a$. The collinearity of a and b is obvious.

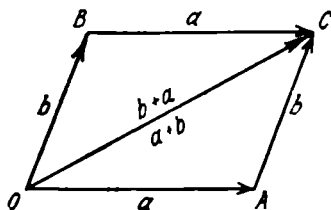


Fig. 6.2

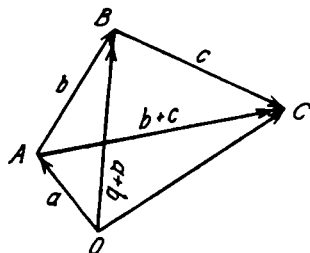


Fig. 6.3

Notice that incidentally we have obtained another way of constructing a vector sum. Namely, if on vectors a and b fixed at one point we construct a parallelogram, then its diagonal fixed at the same point will be the sum $a + b$.

To prove the associativity of addition, apply a to an arbitrary point O , b to the terminal point of a , and c to the terminal point of b (Fig. 6.3). Denote by A , B and C the terminal points of a , b and c . Then

$$(a + b) + c = (\overrightarrow{OA} + \overrightarrow{AB}) + \overrightarrow{BC} = \overrightarrow{OB} + \overrightarrow{BC} = \overrightarrow{OC},$$

$$a + (b + c) = \overrightarrow{OA} + (\overrightarrow{AB} + \overrightarrow{BC}) = \overrightarrow{OA} + \overrightarrow{AC} = \overrightarrow{OC}.$$

From the transitivity of the equality relation of vectors we conclude that the operation is also associative.

These properties of vector addition allow us to calculate the sum of any number of vectors. If we apply a vector a_2 to the terminal point of a_1 , a_3 to the terminal point of a_2 and so on, and finally a vector a_n to the terminal point of a_{n-1} , then the sum $a_1 + a_2 + \dots + a_n$ will be a vector whose initial point coincides with the initial point of a_1 and whose terminal point coincides with the terminal point of a_n . This rule of constructing a vector sum is called the *polygon law*.

We now discuss the existence of an inverse for vector addition. As is known, to answer this question it is necessary to investigate the existence and uniqueness of the solution of the equations

$$a + x = b, \quad y + a = b$$

for arbitrary vectors a and b . By virtue of the commutativity of the basic operation it is obvious that it suffices to examine only one of the equations.

Take an arbitrary directed line segment $\overrightarrow{AB}$. Using an elementary geometric construction we establish that always

$$\overrightarrow{AB} + \overrightarrow{BA} = 0, \quad \overrightarrow{AB} + 0 = \overrightarrow{AB}. \quad (6.1)$$

Therefore the equation

$$\overrightarrow{AB} + x = \overrightarrow{CD} \quad (6.2)$$

for any vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ will clearly have at least one solution, for example,

$$x = \overrightarrow{BA} + \overrightarrow{CD}. \quad (6.3)$$

Suppose (6.2) holds for some other vector z as well, i.e.

$$\overrightarrow{AB} + x = \overrightarrow{CD}, \quad \overrightarrow{AB} + z = \overrightarrow{CD}.$$

Then adding $\overrightarrow{BA}$ to both sides of these equations we get, in view of (6.1), $x = \overrightarrow{BA} + \overrightarrow{CD}$, $z = \overrightarrow{BA} + \overrightarrow{CD}$ and hence $x = z$.

Thus, the operation of vector addition has an inverse. It is *vector subtraction*. If for vectors a , b and c we have $a + c = b$, then we write in symbols $c = b - a$. The vector $b - a$ uniquely determined by the vectors a and b is called the *vector difference*. The justification of this notation will be given somewhat later.

It is easy to show a rule for constructing the difference of two given vectors a and b . Apply these to a common point and construct

a parallelogram on them (Fig. 6.4). We have already shown above that one of the parallelogram diagonals is the sum of the given vectors. The other diagonal is easily seen to be the difference of the same vectors. This rule of constructing the sum and the difference of vectors is usually called the *parallelogram law*.

Notice that we could define addition not for the set of all vectors of a space but only for one of the sets of collinear or coplanar vectors.

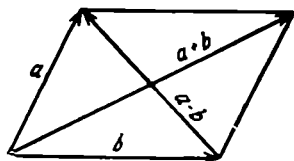


Fig. 6.4

The sum of two vectors of any such set will again be in the *same* set. The operation of vector addition therefore remains algebraic in this case too. Moreover, it preserves all its properties and, what is especially important, it has as before an inverse. The validity of the last assertion follows from formula (6.3). If

vectors $\vec{AB}$ and $\vec{CD}$ are parallel to some straight line or plane, then it is obvious that so is the vector $\vec{BA} + \vec{CD}$ or equivalently the difference vector $\vec{CD} - \vec{AB}$.

Thus, the operation of vector addition is algebraic, commutative and associative, and has an inverse on the sets of three types: on the set of vectors of a space, on the set of collinear vectors and on the set of coplanar vectors.

Exercises

1. Three forces equal in magnitude and directed along the edges of a cube are applied to one of its vertices. What is the direction of the sum of these forces?
2. Let three different classes of collinear vectors be given. When can any vector of a space be represented as the sum of three vectors of these classes?
3. Applied to the vertices of a regular polygon are forces equal in magnitude and directed to its centre. What is the sum of these forces?
4. What is the set of the sum of vectors taken from two different classes of collinear vectors?

7. Groups

Sets with one algebraic operation are in a sense the simplest and it is therefore natural to begin our studies just with such sets. We shall assume the properties of an operation to be axioms and then deduce their consequences. This will allow us *later on* to immediately apply the results of our studies to all sets where the operations have similar properties, regardless of specific features.

A *group* is a set G with one algebraic operation, associative (although not necessarily commutative), for which there must exist an inverse.

Notice that the inverse of an operation cannot be considered to be a second independent operation in a group, since it is defined in terms of the basic operation. As is customary in group theory, we call the operation given in G *multiplication* and use the corresponding notation. Before considering the various examples of groups we deduce the simplest consequences following from the definition.

Take an element a of a group G . The existence of an inverse in the group implies the existence of a unique element e_a such that $ae_a = a$. Consequently, this element plays the same part in multiplying by it the element a on the right as unity does in multiplying numbers. Suppose further that b is any other element of the group. It is obvious that there is an element y satisfying $ya = b$. We now get

$$b = ya = y(ae_a) = (ya)e_a = be_a.$$

So e_a plays the part of the right unity with respect to all elements of G , not only with respect to a . An element with such a property must be unique. Indeed, all such elements satisfy $ax = a$, but by the definition of the inverse of an operation this equation has a unique solution. Denote the resulting element by e' .

Similarly we can prove the existence and uniqueness in G of e'' satisfying $e''b = b$ for every b in G . In fact e' and e'' coincide, which follows from $e''e' = e''$ and $e''e' = e'$.

Thus we have obtained a first important consequence: in any group G there is a *unique element* e satisfying

$$ae = ea = a$$

for every a in G . It is called the *identity* (or *identity element*) of a group G .

The definition of the inverse also implies the existence and uniqueness for any a of elements a' and a'' such that

$$aa' = e, \quad a''a = e.$$

They are called the *right* and the *left inverse element* respectively. It is easy to show that in this case they coincide. Indeed, consider an element $a''aa'$ and calculate it in two different ways. We have

$$\begin{aligned} a''aa' &= a''(aa') = a''e = a'', \\ a''aa' &= (a''a)a' = ea' = a'. \end{aligned}$$

Consequently, $a'' = a'$. This element is the *inverse* of a and denoted by a^{-1} .

Now we have obtained another important consequence: in any group G every element a has a *unique inverse element* a^{-1} for which

$$aa^{-1} = a^{-1}a = e. \quad (7.1)$$

Because of the associativity of the group operation we can speak of the uniqueness of the product of any finite number of elements of a group given (in view of the possible non-commutativity of the group operation) in a definite order. Taking into account (7.1) it is not hard to indicate the general formula for the inverse element of the product. Namely,

$$(a_1 a_2 \dots a_n)^{-1} = a_n^{-1} a_{n-1}^{-1} \dots a_1^{-1}. \quad (7.2)$$

From (7.1) it follows that the inverse element of a^{-1} is the element a and the inverse of the identity element is the identity element, i.e.

$$(a^{-1})^{-1} = a, \quad e^{-1} = e. \quad (7.3)$$

Verifying that a set with one associative operation is a group is greatly facilitated by the fact that in the definition of a group the requirement that the inverse operation should hold can be replaced by the assumption about the existence of an identity and inverse elements, on only one side (say, right) and without the assumption that they are unique. More precisely, we have the following

Theorem 7.1. *A set G with one associative operation is a group if G has at least one element e with the property $ae = a$ for every a in G and with respect to that element any element a in G has at least one right inverse element a^{-1} , i.e. $aa^{-1} = e$.*

Proof. Let a^{-1} be one of the right inverses of a . We have

$$aa^{-1} = e = ee = eaa^{-1}.$$

Multiply both sides of this equation on the right by one of the right inverses of a^{-1} . Then $ae = eae$, from which $a = ea$ since e is the right identity for G . Thus the element e is found to be also the left identity for G .

If now e' is an arbitrary right identity and e'' is an arbitrary left identity, then it follows from $e''e' = e'$ and $e''e' = e''$ that $e' = e''$, i.e. any right identity is equal to any left identity. This proves the existence and uniqueness in G of an identity element which we again denote by e .

Further, for any right inverse element a^{-1}

$$a^{-1} = a^{-1}e = a^{-1}aa^{-1}.$$

Multiply both sides of this equation on the right by a right inverse of a^{-1} . Then $e = a^{-1}a$, i.e. a^{-1} is simultaneously a left inverse of a . If now $a^{-1'}$ is an arbitrary right inverse of a and $a^{-1''}$ is an arbitrary left inverse, then from

$$a^{-1''}aa^{-1'} = (a^{-1''}a)a^{-1'} = ea^{-1'} = a^{-1'},$$

$$a^{-1''}aa^{-1'} = a^{-1''}(aa^{-1'}) = a^{-1''}e = a^{-1''}$$

it follows that $a^{-1'} = a^{-1''}$. This implies the existence and uniqueness for any element a in G of an inverse element a^{-1} .

Now it is easy to show that the set G is a group. Indeed, $ax = b$ and $ya = b$ clearly hold for the elements

$$x = a^{-1}b, \quad y = ba^{-1}.$$

Suppose that there are other solutions, for example, an element z for the first equation. Then $ax = b$ and $az = b$ yield $ax = az$. Multiplying both sides on the left by a^{-1} we get $x = z$. So the set G is a group.

A group is said to be *commutative* or *Abelian* if the group operation is commutative. In that case the operation is as a rule called *addition* and the summation symbol $a + b$ is written instead of the product notation ab . The identity of an Abelian group is called the *zero element* and designated 0. The inverse of the operation is called *subtraction*, and the inverse element is called the *negative element*. It is denoted by $-a$. It will be assumed that by definition the difference symbol $a - b$ denotes the sum $a + (-b)$.

But if for some reason we shall call the operation in a commutative group *multiplication*, then its inverse will be assumed to be *division*. The now equal products $a^{-1}b$ and ba^{-1} will be denoted by b/a and called a quotient of b by a .

Exercises

Prove that the following sets are Abelian groups. Everywhere the name of the operation reflects its content rather than notation.

1. The set consists of integers; the operation is addition of numbers.
2. The set consists of complex numbers, except zero; the operation is multiplication of numbers.
3. The set consists of integer multiples of 3; the operation is addition of numbers.
4. The set consists of positive rationals; the operation is multiplication of numbers.
5. The set consists of numbers of the form $a + b\sqrt{2}$, where a and b are nonzero rationals; the operation is multiplication of numbers.
6. The set consists of a single element a ; the operation is called addition and defined by $a + a = a$.
7. The set consists of integers $0, 1, 2, \dots, n-1$; the operation is called *mod n addition* and consists in calculating the nonnegative remainder less than n of the division of the sum of two numbers by the number n .
8. The set consists of integers $1, 2, 3, \dots, n-1$, where n is a prime; the operation is called *mod n multiplication* and consists in calculating the nonnegative remainder less than n of the division of the product of two numbers by the number n .
9. The set consists of collinear directed line segments; the operation is addition of directed line segments.
10. The set consists of coplanar directed line segments; the operation is addition of directed line segments.
11. The set consists of directed line segments of a space; the operation is addition of directed line segments.

As regards the last three examples, notice that the zero element of an Abelian group of directed line segments is a zero directed line segment and that the inverse line segment to $\overrightarrow{AB}$ is $\overrightarrow{BA}$. It follows from what was proved above that they are unique. Examples of noncommutative groups will be given later.

8. Rings and fields

Consider a set K in which two operations are introduced. Call one of them addition and the other multiplication, and use the corresponding notation. Assume that both are related by the *distributive law*, i.e. for any three elements a , b and c of K

$$(a + b)c = ac + bc, \quad a(b + c) = ab + ac.$$

The set K is said to be a *ring* if two operations are defined in it, addition and multiplication, both associative as well as related by the distributive law, addition being commutative and possessing an inverse. A ring is said to be *commutative* if multiplication is commutative and *noncommutative* otherwise.

Notice that any ring is an additive Abelian group. Consequently, there is a unique zero element 0 in it. The element possesses the property that for any element a of the ring

$$a + 0 = a.$$

We gave the definition of the zero element only with respect to the operation of addition. But it plays a particular role with respect to multiplication as well. Namely, in any ring the product of any element by the zero element is a zero element. Indeed, let a be any element of K ; then

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding to each side an element $-a \cdot 0$ we get $a \cdot 0 = 0$. It can be proved similarly that $0 \cdot a = 0$.

Using this property of the zero element it can be established that in any ring for any elements a and b

$$(-a)b = -(ab).$$

Indeed,

$$ab + (-a)b = (a + (-a))b = 0 \cdot b = 0,$$

i.e. the element $(-a)b$ is the negative of ab . According to our notation we may write it as $-(ab)$.

Now it is easy to show that the distributive law is true for the difference of elements. We have

$$\begin{aligned} (a - b)c &= (a + (-b))c = ac + (-b)c = ac + (-(bc)) = ac - bc, \\ a(b - c) &= a(b + (-c)) = ab + a(-c) = ab + (-(ac)) = ab - ac. \end{aligned}$$

The distributive law, i.e. the usual rule of removing brackets, is the only requirement in the definition of a ring relating addition and multiplication. Only due to this law a simultaneous study of the two operations gives more than could be obtained if they were studied separately.

We have just proved that algebraic operations in a ring possess the many customary properties of operations on numbers. It should not be supposed, however, that any property of addition and multiplication of numbers is preserved in any ring, be it even a commutative one. Thus multiplication of numbers has a property converse to that of multiplication by a zero element. Namely, if a product of two numbers is equal to zero, then at least one of the multipliers equals zero. In an arbitrary commutative ring this property does not necessarily hold, i.e. a product of elements not equal to the zero element may be zero.

Nonzero elements whose product is a zero element are called *zero divisors*. Their existence in a ring makes investigating them substantially more difficult and prevents one from drawing a deep analogy between numbers and elements of a commutative ring. This analogy can be drawn, however, for rings having no zero divisors.

Suppose in a commutative ring with respect to the operation of multiplication there is an identity element e and each nonzero element a has an inverse element a^{-1} . It is not hard to prove that both the identity and the inverse element are unique, but what is most important is the fact that now the ring has no zero divisors. Indeed, let $ab = 0$, but $a \neq 0$. Multiplying both sides of this equation on the left by a^{-1} we get

$$a^{-1}ab = (a^{-1}a)b = eb = b$$

and certainly $a^{-1}0 = 0$. Consequently, $b = 0$.

From the absence of zero divisors it follows that from any equation we can cancel the nonzero common multiplier. If $ca = cb$ and $c \neq 0$, then $c(a - b) = 0$, from which we conclude that $a - b = 0$, i.e. $a = b$.

A commutative ring P in which there is an identity element and each nonzero element has an inverse is called a *field*.

Writing the quotient a/b as the product ab^{-1} , it is easy to show that *any field preserves all the usual rules of handling fractions, in terms of addition, subtraction, division and multiplication*. Namely,

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \frac{-a}{b} = -\frac{a}{b}.$$

Besides, $a/b = c/d$ if and only if $ad = bc$, provided, of course, $b \neq 0$ and $d \neq 0$. It is left as an exercise for the reader to check that these assertions are true.

So in terms of the usual rules of handling fractions all fields are indistinguishable from the set of numbers. For this reason the elements of *any* field will be called numbers if of course this name does not lead to any ambiguity. As a rule, the zero element of any field will be designated as 0 and the identity element as 1.

We shall now list all the general facts we need about the elements of any field in what follows.

A. To every pair of elements a and b there corresponds an element $a + b$, called the sum of a and b , and

- (1) addition is commutative, $a + b = b + a$,
- (2) addition is associative, $a + (b + c) = (a + b) + c$,
- (3) there is a unique zero element 0 such that $a + 0 = a$ for any element a ,
- (4) for every element a there is a unique negative element $-a$ such that $a + (-a) = 0$.

B. To every pair of elements a and b there corresponds an element ab , called the product of a and b , and

- (1) multiplication is commutative, $ab = ba$,
- (2) multiplication is associative, $a(bc) = (ab)c$,
- (3) there is a unique identity element 1 such that $a \cdot 1 = 1 \cdot a = a$ for any element a ,
- (4) for every nonzero element a there is a unique inverse element a^{-1} such that $aa^{-1} = a^{-1}a = 1$.

C. The operations of addition and multiplication are connected by the following relation: multiplication is distributive over addition, $(a + b)c = ac + bc$.

These facts lay no claim to logical independence and are but a convenient way of characterizing elements. Properties A describe the field in terms of the operation of addition and say that with respect to this operation the field is an Abelian group. Properties B describe the field in terms of the operation of multiplication and say that with respect to this operation the field becomes an Abelian group if we eliminate from it the zero element. Property C describes the relation of the two operations to each other.

Exercises

Prove that sets 1-7 are rings and not fields and that sets 8-13 are fields. Everywhere the name of the operation reflects its content rather than notation.

1. The set consists of integers; the operations are addition and multiplication of numbers.
2. The set consists of integer multiples of some number n ; the operations are addition and multiplication of numbers.
3. The set consists of real numbers of the form $a + b\sqrt{2}$, where a and b are integers; the operations are addition and multiplication of numbers.

4. The set consists of polynomials with real coefficients in a single variable t , including constants; the operations are addition and multiplication of polynomials.

5. The set consists of a single element a ; the operations are defined by $a + a = a$ and $a \cdot a = a$.

6. The set consists of integers $0, 1, 2, \dots, n-1$, where n is a composite number; the operations are mod n addition and mod n multiplication.

7. The set consists of pairs (a, b) of integers; the operations are defined by the formulas

$$(a, b) + (c, d) = (a + c, b + d); \quad (a, b) \cdot (c, d) = (ac, bd).$$

8. The set consists of rational numbers; the operations are addition and multiplication of numbers.

9. The set consists of real numbers; the operations are addition and multiplication of numbers.

10. The set consists of complex numbers; the operations are addition and multiplication of numbers.

11. The set consists of real numbers of the form $a + b\sqrt{2}$, where a and b are rationals; the operations are addition and multiplication of numbers.

12. The set consists of two elements a and b ; the operations are defined by the equations

$$a + a = b + b = a, \quad a + b = b + a = b, \\ a \cdot a = a \cdot b = b \cdot a = a, \quad b \cdot b = b.$$

13. The set consists of integers $0, 1, 2, \dots, n-1$, where n is a prime; the operations are mod n addition and mod n multiplication.

The reader should note that one of the examples gives a ring with zero divisors. Which example is it? What is the general form of zero divisors?

9. Multiplication of directed line segments by a number

We stress once again that an algebraic operation was defined by us as an operation on two elements of the same set. Numerous examples from physics suggest, however, that it is sometimes reasonable to consider operations on elements of different sets. One of such operations is suggested by the concepts of force, displacement, velocity and acceleration, and we shall again use the example of directed line segments to consider it.

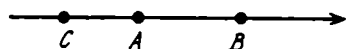
It has been customary for a long time now in physics to make use of line segments. If, for example, a force is said to have increased by a factor of five, then the line segment representing it is "extended" by a factor of five without changing the general direction. If, however, the force direction is said to have changed, then the initial and terminal points of the corresponding line segment are interchanged. Proceeding from these considerations we introduce multiplication of a directed line segment by a real number.

We first discuss some general questions. Suppose an arbitrary straight line is given in the plane or in space. We agree to consider one of the directions on the line to be positive and the other to be

negative. A straight line on which a direction is specified will be called an *axis*.

Suppose now that some axis is given and, in addition, a unit line segment is indicated, which can be used to measure any other line segment and thus determine its length. With every directed line segment on the axis we associate its numerical characteristic, the so-called magnitude of the directed line segment.

The magnitude $\{\overrightarrow{AB}\}$ of a directed line segment $\overrightarrow{AB}$ is a number equal to the length of the line segment $\overrightarrow{AB}$ taken with a plus sign if the direction of $\overrightarrow{AB}$ coincides with the positive direction of the axis, and with a minus sign if the direction of $\overrightarrow{AB}$ coincides with the negative direction of the axis. The magnitudes of all zero directed line segments are considered equal to zero, i.e.



$$\{\overrightarrow{AA}\} = 0.$$

Fig. 9.1

Regardless of which direction on the axis is taken to be positive, $\overrightarrow{AB}$ is opposite in direction to $\overrightarrow{BA}$ and the lengths of $\overrightarrow{AB}$ and $\overrightarrow{BA}$ are equal; consequently,

$$\{\overrightarrow{AB}\} = -\{\overrightarrow{BA}\}. \quad (9.1)$$

The magnitude of a directed line segment, unlike its length, may have any sign. Since the length of $\overrightarrow{AB}$ is the absolute value of its magnitude, we shall use the symbol $|\overrightarrow{AB}|$ to designate it. It is clear that in contrast to (9.1)

$$|\overrightarrow{AB}| = |\overrightarrow{BA}|.$$

Let A , B and C be any three points on the axis determining three directed line segments $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AC}$. Whatever the location of the points, the magnitudes of these directed line segments satisfy the relation

$$\{\overrightarrow{AB}\} + \{\overrightarrow{BC}\} = \{\overrightarrow{AC}\}. \quad (9.2)$$

Indeed, let the direction of the axis and the location of the points be such as in Fig. 9.1, for example. Then obviously

$$|\overrightarrow{CA}| + |\overrightarrow{AB}| = |\overrightarrow{CB}|. \quad (9.3)$$

According to the definition of the magnitude of a directed line segment and equation (9.1)

$$\begin{aligned} |\overrightarrow{CA}| &= \{\overrightarrow{CA}\} = -\{\overrightarrow{AC}\}, & |\overrightarrow{AB}| &= \{\overrightarrow{AB}\}, \\ |\overrightarrow{CB}| &= \{\overrightarrow{CB}\} = -\{\overrightarrow{BC}\}. \end{aligned} \quad (9.4)$$

Therefore (9.3) yields

$$-\{\overrightarrow{AC}\} + \{\overrightarrow{AB}\} = -\{\overrightarrow{BC}\}$$

which coincides essentially with (9.2).

In our proof we used only relations (9.3) and (9.4) which depend only on relative positions of the points A , B and C on the axis and are independent of their coincidence or noncoincidence with one another. It is clear that for any other location of the points the proof is similar.

Identity (9.2) is the *basic identity*. In terms of the operation of vector addition, for vectors on the same axis, it means that

$$\{\overrightarrow{AB} + \overrightarrow{BC}\} = \{\overrightarrow{AB}\} + \{\overrightarrow{BC}\}. \quad (9.5)$$

The magnitude of a directed line segment determines the 'line segment on the axis to within translation. But if we consider that equal directed line segments are also determined to within translation, then this means that the magnitude of a directed line segment uniquely determines on a given axis the entire collection of equal directed line segments.

Now let $\overrightarrow{AB}$ be a directed line segment and let α be a number. The product $\alpha \cdot \overrightarrow{AB}$ of the directed line segment $\overrightarrow{AB}$ by the real number α is a directed line segment lying on the axis through the points A and B and having a magnitude equal to $\alpha \cdot \{\overrightarrow{AB}\}$. Thus by definition

$$\{\alpha \cdot \overrightarrow{AB}\} = \alpha \cdot \{\overrightarrow{AB}\}. \quad (9.6)$$

For any numbers α and β and any directed line segments a and b the multiplication of a directed line segment by a number possesses the following properties:

$$\begin{aligned} 1a &= a, & \alpha(\beta a) &= (\alpha\beta)a, \\ (\alpha + \beta)a &= \alpha a + \beta a, & \alpha(a + b) &= \alpha a + \alpha b. \end{aligned}$$

The first three properties are very simple. To prove them it suffices to note that on the left- and right-hand sides of the equations we

have vectors lying on the same axis and use relations (9.5) and (9.6). We prove the fourth property. Suppose for simplicity that

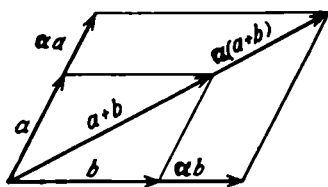


Fig. 9.2

$\alpha > 0$. Apply vectors a and b to a common point and construct on them a parallelogram whose diagonal is equal to $a + b$ (Fig. 9.2). When a and b are multiplied by α , the parallelogram diagonal, by the similitude of figures, is also multiplied by α . But this means that $\alpha a + \alpha b = \alpha(a + b)$.

Note in conclusion that the magnitude of a directed line segment may be treated as some "function"

$$\xi = \{x\} \quad (9.7)$$

whose "independent variable" are vectors x of the same axis and whose "value" are real numbers ξ , with

$$\begin{aligned} \{x + y\} &= \{x\} + \{y\}, \\ \{\lambda x\} &= \lambda \{x\} \end{aligned} \quad (9.8)$$

for any vectors x and y on the axis and any number λ .

Exercises

1. Prove that the result of multiplication by a number does not depend on the choice of the positive direction on the axis.
2. Prove that the result of multiplication by a number does not depend on the way the unit line segment is specified on the axis.
3. Prove that if we perform multiplication by a number defined on any set of collinear line segments, the result will remain in the same set.
4. Prove that if we perform multiplication by a number defined on any set of coplanar line segments, the result will remain in the same set.
5. What are the zero and the negative directed line segment in terms of multiplication by a number?

10. Vector spaces

Solving any problems reduces in the final analysis to the study of some sets and, in the first place, to the study of the structure of those sets. The structure of sets can be studied by various methods, for example, starting from the characteristic property the elements possess, as it is done in problems in constructing loci, or starting from the properties of operations if they are defined for the elements.

The last method seems particularly tempting by virtue of its generality. Indeed, we have already seen at various times that various sets allow introduction of various operations possessing

nevertheless the same properties. It is obvious therefore that if in investigating sets we obtain some result relying only on the properties of the operation, then that result will hold in all sets where operations possess the same properties. The specific nature of both the elements and the operations on them may be quite different.

Somewhat earlier we introduced new mathematical objects, called directed line segments or vectors, and defined operations on them. It is well known that in fact there are quite real physical objects behind vectors. Therefore a detailed study of the structure of vector sets is of interest at least to physics.

Even now we have three types of sets where operations have the same properties. These are the set of collinear vectors, the set of coplanar vectors and the set of vectors in the whole of space. In spite of the fact that the same operations are introduced in these sets we are justified in expecting that the structure of the sets must be different.

There is some temptation, due to their simplicity, to study the sets relying only on the specific features of their elements. One cannot help noticing, however, that they have very much in common. It is therefore appropriate to attack them from some general positions, in the hope of at least *avoiding the tedious and monotonous repetitions* in going from one set to another. But in addition we hope of course that if we obtain some set with similar properties *we shall be able to carry over all the results of the studies already made.*

We now list the familiar general facts about vectors forming any of the three sets in question.

A. To every pair of vectors x and y there corresponds a vector $x + y$, called the sum of x and y , and

- (1) addition is commutative, $x + y = y + x$,
- (2) addition is associative, $x + (y + z) = (x + y) + z$,
- (3) there is a unique zero vector 0 such that $x + 0 = x$ for any vector x ,
- (4) for every vector x there is a unique negative vector $-x$ such that $x + (-x) = 0$.

B. To every pair α and x , where α is a number and x is a vector, there corresponds a vector αx , called the product of α and x , and

- (1) multiplication by a number is associative, $\alpha(\beta x) = (\alpha\beta)x$,
- (2) $1 \cdot x = x$ for any vector x .

C. Addition and multiplication are connected by the following relations:

- (1) multiplication by a number is distributive over vector addition, $\alpha(x + y) = \alpha x + \alpha y$,
- (2) multiplication by a vector is distributive over addition of numbers, $(\alpha + \beta)x = \alpha x + \beta x$.

These facts, as in the case of the field, lay no claim to logical independence. Properties A describe a set of vectors in terms of

addition and say that it is an Abelian group under addition. Properties B describe a set of vectors in terms of multiplication of a vector by a number. Properties C describe the relation of the two operations to each other.

Now consider a set K and a field P of an arbitrary nature. We shall say that K is a *linear* or *vector space* over P if for all elements of K addition and multiplication by a number from P are defined, axioms A, B and C holding. Using this terminology we can say that the set of collinear vectors, the set of coplanar vectors and the set of vectors in the whole of space are vector spaces over the field of real numbers.

The elements of *any* vector space will be called *vectors* although in their specific nature they may not at all resemble directed line segments. Geometrical ideas associated with the name "vectors" will help us understand and often anticipate the required results as well as help us find the not always obvious geometrical meaning in the various facts.

Vectors of a vector space will as before be denoted by small Latin letters and numbers by small Greek letters. We shall call a vector space *rational*, *real* or *complex* according as the field P is the field of rational, real or complex numbers, and denote it by $\mathbf{D}$, $\mathbf{R}$ or $\mathbf{C}$ respectively. The fact that the name and notation lack any reference to the elements of the set has a deep meaning, but we shall discuss it much later.

Before proceeding to a detailed study of vector spaces consider the simplest consequences following from the existence of addition and multiplication by a number. They will mainly concern the zero and negative vectors.

In *any* vector space, for every element x

$$0 \cdot x = 0,$$

where at the right 0 stands for the zero vector and at the left 0 is the number zero. To prove this relation consider the element $0 \cdot x + x$. We have

$$0 \cdot x + x = 0 \cdot x + 1 \cdot x = (0 + 1) x = 1 \cdot x = x.$$

Consequently,

$$x = 0 \cdot x + x.$$

Adding to each side $-x$ we find

$$\begin{aligned} 0 &= x + (-x) = (0 \cdot x + x) + (-x) = 0 \cdot x + (x + (-x)) \\ &= 0 \cdot x + 0 = 0 \cdot x. \end{aligned}$$

Now it is easy to show an *explicit* expression for the negative element $-x$ in terms of the element x . Namely,

$$-x = (-1) x.$$

This formula follows from the simple relations

$$x + (-1)x = 1 \cdot x + (-1)x = (1 - 1)x = 0 \cdot x = 0.$$

This in turn establishes the relations

$$-(\alpha x) = (-\alpha)x = \alpha(-x),$$

since

$$-(\alpha x) = (-1)(\alpha x) = (-\alpha)x = \alpha((-1)x) = \alpha(-x).$$

Recall that by the definition of the operation of subtraction $x - y = x + (-y)$ for any vectors x and y . The explicit expression for the negative vector shows the validity of distributive laws for a *difference* too. Indeed, regardless of the numbers α and β and the vectors x and y we have

$$\begin{aligned} (\alpha - \beta)x &= \alpha x + (-\beta)x = \alpha x + (-\beta x) = \alpha x - \beta x, \\ \alpha(x - y) &= \alpha(x + (-1)y) = \alpha x + (-\alpha)y = \alpha x + (-\alpha y) \\ &= \alpha x - \alpha y. \end{aligned}$$

It follows in particular that for *any* number α

$$\alpha \cdot 0 = 0,$$

since

$$\alpha \cdot 0 = \alpha(x - x) = \alpha x - \alpha x = \alpha x + (-\alpha x) = 0.$$

And, finally, the last consequence. If for any number α and any vector x

$$\alpha x = 0, \tag{10.1}$$

then *either* $\alpha = 0$ or $x = 0$. Indeed, if (10.1) holds, then there are two possibilities: either $\alpha = 0$ or $\alpha \neq 0$. The case $\alpha = 0$ supports our assertion. Now let $\alpha \neq 0$. Then

$$x = 1 \cdot x = \left(\frac{1}{\alpha} \cdot \alpha\right)x = \frac{1}{\alpha}(\alpha x) = \frac{1}{\alpha} \cdot 0 = 0.$$

It follows in particular that in any vector space the common nonzero multiplier can formally be *cancelled* from any equation, whether it is a number or a vector. Indeed, if $\alpha x = \beta x$ and $x \neq 0$, then $(\alpha - \beta)x = 0$ and hence $\alpha - \beta = 0$, i.e. $\alpha = \beta$. If $\alpha x = \alpha y$ and $\alpha \neq 0$, then $\alpha(x - y) = 0$ and hence $x - y = 0$, i.e. $x = y$.

So *in terms of multiplication, addition and subtraction all the rules of equivalent transformations of algebraic expressions formally hold*. We shall no longer state them explicitly in what follows.

We have broached the subject of vector spaces. To conclude, it is no mere chance that we have used a single notation for the properties of operations in the field and in the vector space. There are features of striking resemblance (as well as difference) between the axioms of the field and those of the vector space over a field. The reader should ponder on them.

Exercises

Prove that the following sets are vector spaces. Everywhere the name of the operation reflects its content rather than notation.

1. The field consists of real numbers; the set consists of real numbers; addition is the addition of real numbers; multiplication by a number is the multiplication of a real number by a real number.

2. The field consists of real numbers; the set consists of complex numbers; addition is the addition of complex numbers; multiplication by a number is the multiplication of a complex number by a real number.

3. The field consists of rational numbers; the set consists of real numbers; addition is the addition of real numbers; multiplication by a number is the multiplication of a real number by a rational number.

4. The field consists of any number; the set consists of a single vector a ; addition is defined by the rule $a + a = a$; multiplication of the vector a by any number α is defined by the rule $\alpha a = a$.

5. The field consists of real numbers; the set consists of polynomials with real coefficients in a single variable t , including constants; addition is the addition of polynomials; multiplication by a number is the multiplication of a polynomial by a real number.

6. The field consists of rational numbers; the set consists of numbers of the form $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{5}$, where a, b, c and d are rationals; addition is the addition of numbers of the indicated form; multiplication by a number is the multiplication of a number of the indicated form by a rational number.

7. The field is any field; the set is the same field; addition is the addition of elements (vectors!) of the field; multiplication by a number is the multiplication of an element (a vector!) of the field by an element (a number!) of the field.

11. Finite sums and products

Fields and vector spaces are the main sets with which we shall have to deal in what follows. Two operations, addition and multiplication, are introduced in these sets. If a large number of operations on elements is performed, then there appear expressions containing a considerable number of summands and factors. For notational convenience we introduce the appropriate symbols. It will be assumed that addition and multiplication are *commutative* and *associative* operations.

Given a finite number of not necessarily different elements, assume that all the elements are numbered in some way and have indices taking on all consecutive values from some integer k to an integer p . Denote elements by a single letter with index. The *index* may be placed anywhere in notation. It may be put in parentheses near the letter, at the bottom of the letter, at its top and so on. This is of no consequence. Most often we shall write it at the lower right of the letter.

We shall denote the sum of elements $a_k, a_{k+1}, \dots, a_p$ by the symbol of the following form:

$$a_k + a_{k+1} + \dots + a_p = \sum_{i=k}^p a_i. \quad (11.1)$$

The index i in the formula is the *summation index*. Of course nothing will happen if we denote it by any other letter. Sometimes under the summation sign the collection of indices is explicitly stated over which summation is carried out. For example, the sum in question could be written as:

$$a_k + a_{k+1} + \dots + a_p = \sum_{k \leq i \leq p} a_i.$$

It is obvious that if every element a_i equals the product of an element b_i and an element α , where α is independent of the summation index i , then

$$\sum_{i=k}^p \alpha b_i = \alpha \sum_{i=k}^p b_i,$$

i.e. the multiplier independent of the summation index may be taken outside the summation sign.

Suppose now that the elements have two indices, each changing independently. We agree to use for these elements common notation a_{ij} and let, for example, $k \leq i \leq p$ and $m \leq j \leq n$. Arrange the elements as a rectangular array:

$$\begin{array}{cccc} a_{km} & a_{k, m+1} & \dots & a_{kn}, \\ a_{k+1, m} & a_{k+1, m+1} & \dots & a_{k+1, n}, \\ \cdot & \cdot & \cdot & \cdot \\ a_{pm} & a_{p, m+1} & \dots & a_{pn}. \end{array}$$

It is clear that whatever the order of summation the result will be the same. Therefore, taking into account the above notation for the sum, we have

$$\begin{aligned} & (a_{km} + a_{k, m+1} + \dots + a_{kn}) + (a_{k+1, m} + a_{k+1, m+1} + \dots + a_{k+1, n}) + \\ & \dots + (a_{pm} + a_{p, m+1} + \dots + a_{pn}) \\ &= \sum_{j=m}^n a_{kj} + \sum_{j=m}^n a_{k+1, j} + \dots + \sum_{j=m}^n a_{pj} = \sum_{i=k}^p \left(\sum_{j=m}^n a_{ij} \right). \end{aligned}$$

On the other hand, the same sum equals

$$\begin{aligned} & (a_{km} + a_{k+1, m} + \dots + a_{pm}) \\ & + (a_{k, m+1} + a_{k+1, m+1} + \dots + a_{p, m+1}) + \dots + (a_{kn} + a_{k+1, n} + \dots + a_{pn}) \\ &= \sum_{i=k}^p a_{im} + \sum_{i=k}^p a_{i, m+1} + \dots + \sum_{i=k}^p a_{in} = \sum_{j=m}^n \left(\sum_{i=k}^p a_{ij} \right). \end{aligned}$$

Consequently,

$$\sum_{i=h}^p \left(\sum_{j=m}^n a_{ij} \right) = \sum_{j=m}^n \left(\sum_{i=h}^p a_{ij} \right).$$

If we agree that we shall always take summation consecutively over the summation indices arranged from right to left, then the brackets may be dropped and we finally get

$$\sum_{i=h}^p \sum_{j=m}^n a_{ij} = \sum_{j=m}^n \sum_{i=h}^p a_{ij}.$$

This means that we may change the order of summation in summing over two indices. If, for example, $a_{ij} = \alpha_i b_{ij}$, where α_i is independent of the index j , then

$$\sum_{i=h}^p \sum_{j=m}^n \alpha_i b_{ij} = \sum_{i=h}^p \alpha_i \sum_{j=m}^n b_{ij}.$$

Similar results hold for sums over any finite number of indices.

A product of elements $a_h, a_{h+1}, \dots, a_p$ will be designated by a symbol of the following form:

$$a_h a_{h+1} \dots a_p = \prod_{i=h}^p a_i.$$

Now if $a_i = \alpha b_i$, then

$$\prod_{i=h}^p \alpha b_i = \alpha^{p-h+1} \prod_{i=h}^p b_i.$$

As in the case of summation, we may change the order of calculating the products over two indices, i.e.

$$\prod_{i=h}^p \prod_{j=m}^n a_{ij} = \prod_{j=m}^n \prod_{i=h}^p a_{ij}.$$

All these facts can be proved according to the same scheme as in the case of summation of numbers.

Exercises

Calculate the following expressions:

$$\sum_{i=1}^n 1, \quad \sum_{i=1}^n t, \quad \sum_{i=1}^n t^3 \sum_{l=1}^n 8(l-1),$$

$$\sum_{r=1}^n \sum_{j=1}^m rj, \quad \sum_{i=1}^n \sum_{s=1}^m (i+5s), \quad \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^p (2j-i)^2 k,$$

$$\prod_{i=1}^n 2, \quad \prod_{p=1}^n 10^p, \quad \prod_{i=1}^n \prod_{j=1}^m 2^{1-j}, \quad \prod_{i=1}^n \prod_{j=1}^m \prod_{k=1}^p 2^{1+j+k}.$$

12. Approximate calculations

The sets discussed above are very widely used in various theoretical studies. To obtain a result it is almost always necessary to perform some operations on the elements of the sets. Especially frequently a need arises to carry out manipulations with elements of number fields. We want to note a very important *feature of practical realizations* of such computations.

Suppose for definiteness that we are dealing with a field of real numbers. Let every number be represented as an infinite decimal fraction. Neither man nor the most modern computer can handle infinite fractions. In practice therefore every such fraction is replaced by a finite decimal fraction close to it or by a suitable rational number.

So an exact real number is replaced by an approximate one. In theoretical studies implying precise assignment of numbers one expression or another is fairly often replaced by an equal expression, possibly written in another form. Of course in this case such a substitution can raise neither any objections nor even questions. But if we want to calculate some expression using approximate numbers, then the form of the expression is no longer irrelevant.

Consider a simple example. It is easy to check that in the case of the exact assignment of the number $\sqrt{2}$

$$(\sqrt{2}-1)^6 = 99 - 70\sqrt{2}. \quad (12.1)$$

Since $\sqrt{2} = 1.4142 \dots$, the numbers $7/5 = 1.4$ and $17/12 = 1.4166 \dots$ may be considered to be approximate values for $\sqrt{2}$. But substituting $7/5$ on the left and right of (12.1) we get $0.00509 \dots$ and 1.0 respectively. For $17/12$ we have $0.00523 \dots$ and $-0.1666 \dots$. The results of the substitutions considerably differ, and it is not immediately apparent which is closer to truth. This shows *how careful one must be in handling approximate numbers*.

We have discussed only one possible source of approximate numbers, the rounding of exact numbers. In fact there are many other sources. For example, initial data for calculations often result from experiments and every experiment may produce a result only to a limited precision. Even in such simple operations as multiplication and division the number of digits in fractions may greatly increase. We are compelled therefore to discard part of the digits in the results of intermediate calculations, i.e. we are again compelled to replace some numbers by approximate ones and so on.

A detailed study of operations with approximate numbers is beyond the scope of this course. However, we shall fairly frequently return to the discussion of the difference between theoretical and

practical calculations. The need for such a discussion arises from the fact that *theoretical calculations cannot as a rule be realized in exact form.*

Exercises

1. What finite decimal fraction must be used to approximate $\sqrt{2}$ for the first six digits to coincide in the results of computing the left- and right-hand sides of (12.1)?
2. Let the result of each operation on two real numbers be rounded according to any rule you know to t decimal places. Are the commutative and associative properties of the operations preserved?
3. Will the distributive laws hold under the hypotheses of the preceding exercise?
4. To what conclusion do you come if the answer in Exercises 2 and 3 is no?

The Structure of a Vector Space

13. Linear combinations and spans

Let $e_1, e_2, \dots, e_n$ be a finite number of arbitrary, not necessarily distinct vectors from a vector space K over a field P . We shall call them a *system of vectors*. One system of vectors is said to be a *subsystem* of a second system if the first system contains only some vectors of the second and no other vectors.

The vectors of a given system and those obtained from them will be subjected to the operations of addition and multiplication by a number. It is clear that any vector x of the form

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n, \quad (13.1)$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are some numbers from P , is obtained from the vectors of a given system $e_1, e_2, \dots, e_n$ with the aid of the two operations. Moreover, in whatever order the operations are performed, we shall obtain vectors of the form (13.1).

A vector x in (13.1) is said to be *linearly expressible* in terms of the vectors $e_1, e_2, \dots, e_n$. The right-hand side of (13.1) is called a *linear combination* of these vectors and the numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ are the *coefficients of the linear combination*.

Fix a system of vectors $e_1, e_2, \dots, e_n$ and allow the coefficients of linear combinations take on any values from the field P . This will determine some set of vectors in K . This set is the *span* of vectors $e_1, e_2, \dots, e_n$ and is designated $L(e_1, e_2, \dots, e_n)$.

Our interest in spans is accounted for by two circumstances. First, any span has a simple structure, being a collection of all linear combinations of vectors of a given system. Second, the span of any system of vectors from any vector space is itself a vector space.

Indeed, all the axioms of a vector space are almost obvious. Some explanation may only be required by the axioms relating to the zero and the negative vector. The zero vector is clearly in any span and corresponds to the zero values of the coefficients of a linear combination, i.e.

$$0 = 0 \cdot e_1 + 0 \cdot e_2 + \dots + 0 \cdot e_n.$$

The negative of (13.1) is

$$-x = (-\alpha_1) e_1 + (-\alpha_2) e_2 + \dots + (-\alpha_n) e_n.$$

The uniqueness of the zero and negative vectors follows from their uniqueness as vectors of the vector space K .

Notice that the span of vectors $e_1, e_2, \dots, e_n$ is the "smallest" vector space containing those vectors. Indeed, the span consists of only linear combinations of vectors $e_1, e_2, \dots, e_n$ and any vector space containing $e_1, e_2, \dots, e_n$ must contain all their linear combinations.

So any vector space contains in the general case an infinite number of other vector spaces, the spans. Now the following questions arise:

What are the conditions under which the spans of two *distinct* systems of vectors consist of the *same* vectors of the original space?

What *minimum* number of vectors determines the same span?

Is the original vector space the span of some of *its* vectors?

We shall soon get answers to these and other questions. To do this a very wide use will be made of the concept of linear combination, and in particular of its *transitive property*. Namely, if some vector z is a linear combination of vectors $x_1, x_2, \dots, x_r$ and each of them in turn is a linear combination of vectors $y_1, y_2, \dots, y_s$, then z too may be represented as a linear combination of $y_1, y_2, \dots, y_s$. We prove this property. Let

$$z = \sum_{i=1}^r \beta_i x_i \quad (13.2)$$

and in addition for every index i , $1 \leq i \leq r$, let

$$x_i = \sum_{j=1}^s \gamma_{ij} y_j,$$

where β_i and γ_{ij} are some numbers from P .

Substituting the expression for x_i on the right of (13.2) and using the corresponding properties of finite sums we get

$$\begin{aligned} z &= \sum_{i=1}^r \beta_i x_i = \sum_{i=1}^r \beta_i \sum_{j=1}^s \gamma_{ij} y_j = \sum_{i=1}^r \sum_{j=1}^s \beta_i \gamma_{ij} y_j \\ &= \sum_{j=1}^s \sum_{i=1}^r \beta_i \gamma_{ij} y_j = \sum_{j=1}^s \left(\sum_{i=1}^r \beta_i \gamma_{ij} \right) y_j = \sum_{j=1}^s \xi_j y_j, \end{aligned}$$

where the coefficients ξ_j imply the following expressions:

$$\xi_j = \sum_{i=1}^r \beta_i \gamma_{ij}.$$

So the concept of linear combination is indeed transitive.

Exercises

1. What are in a space of directed line segments the spans of systems of one, two, three and a larger number of directed line segments?
2. Consider a vector space of polynomials in t over the field of real numbers. What is the span of the system of vectors $t^2 + 1$, $t^2 + t$ and 1 ?
3. In what space do all spans coincide with the space?
4. Prove that the vector space of all directed line segments cannot be the span of any two directed line segments.

14. Linear dependence

Consider again arbitrary vectors $e_1, e_2, \dots, e_n$ in a vector space. It may happen that one of them is linearly expressible in terms of the others. For example, let it be e_1 . Then each vector of $e_1, e_2, \dots, e_n$ is linearly expressible in terms of $e_2, e_3, \dots, e_n$. Therefore any linear combination of vectors $e_1, e_2, \dots, e_n$ is also a linear combination of vectors $e_2, e_3, \dots, e_n$. Consequently, the spans of the vectors $e_1, e_2, \dots, e_n$ and $e_2, e_3, \dots, e_n$ coincide.

Suppose further that among the vectors $e_2, e_3, \dots, e_n$ there is some vector, say, e_2 which is also linearly expressible in terms of the rest. Repeating our reasoning we conclude that now any linear combination of vectors $e_1, e_2, \dots, e_n$ is also a linear combination of $e_3, e_4, \dots, e_n$. Continuing this process we finally come from the system $e_1, e_2, \dots, e_n$ to a system from which *none* of the vectors can any longer be eliminated. The span of the new system of vectors obviously *coincides* with that of the vectors $e_1, e_2, \dots, e_n$. In addition we can say that if there were at least one nonzero vector among $e_1, e_2, \dots, e_n$, then the new system of vectors would either consist of only one nonzero vector or none of its vectors would be linearly expressible in terms of the others.

Such a system of vectors is called *linearly independent*.

If a system of vectors is not linearly independent, then it is said to be *linearly dependent*. In particular, by definition a system consisting of a zero vector alone is linearly dependent. Linear dependence or independence are properties of a system of vectors. Nevertheless the corresponding adjectives are very often used to refer to the vectors themselves. Instead of a "linearly independent system of vectors" we shall sometimes say a "system of linearly independent vectors" and so on.

In terms of the notions just introduced this means that we proved

Lemma 14.1. *If not all of the vectors $e_1, e_2, \dots, e_n$ are zero and this system is linearly dependent, then we can find in it a linearly independent subsystem of vectors in terms of which any of the vectors $e_1, e_2, \dots, e_n$ is linearly expressible.*

Whether the system of vectors $e_1, e_2, \dots, e_n$ is linearly dependent or linearly independent is determined by one, seemingly unexpected, fact. We have already noted that the zero vector is in the span and is clearly a linear combination (13.1) with zero values of the coefficients. In spite of this it can be linearly expressed in terms of the vectors $e_1, e_2, \dots, e_n$ and in other ways, i.e. defined by another set of the coefficients of a linear combination. The linear independence of $e_1, e_2, \dots, e_n$ is very closely related to the uniqueness of representing the zero element in terms of them. Namely, we have

Theorem 14.1. *A system of vectors $e_1, e_2, \dots, e_n$ is linearly independent if and only if*

$$\alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n = 0 \quad (14.1)$$

implies the equality to zero of all the coefficients of the linear combination.

Proof. Let $n = 1$. If $e_1 \neq 0$, then, as already noted above, $\alpha_1 e_1 = 0$ must yield $\alpha_1 = 0$. But if it follows from $\alpha_1 e_1 = 0$ that α_1 is zero, then e_1 obviously cannot be zero.

Consider now the case $n \geq 2$. Let a system of vectors be linearly independent. Suppose (14.1) is true for some set of coefficients among which there is at least one different from zero. For example, let $\alpha_1 \neq 0$. Then (14.1) yields

$$e_1 = \left(-\frac{\alpha_2}{\alpha_1}\right) e_2 + \left(-\frac{\alpha_3}{\alpha_1}\right) e_3 + \dots + \left(-\frac{\alpha_n}{\alpha_1}\right) e_n,$$

i.e. e_1 is linearly expressible in terms of the other vectors of the system. This contradicts the condition that the system be linearly independent, and therefore it is impossible that there should be nonzero coefficients among those satisfying (14.1).

If (14.1) implies that all coefficients are equal to zero, then the system of vectors cannot be linearly dependent. Indeed, suppose the contrary and let e_1 , for example, be linearly expressible in terms of the other vectors, i.e. let

$$e_1 = \beta_2 e_2 + \beta_3 e_3 + \dots + \beta_n e_n.$$

Then (14.1) will clearly hold for the coefficients $\alpha_1 = -1$, $\alpha_2 = \beta_2$, $\dots$, $\alpha_n = \beta_n$ among which at least one is not equal to zero. Thus the theorem is proved.

This theorem is so widely used in various studies that it is most often regarded just as definition of linear independence.

Note two simple properties of systems of vectors associated with linear independence.

Lemma 14.2. *If some of the vectors $e_1, e_2, \dots, e_n$ are linearly dependent, then so is the entire system $e_1, e_2, \dots, e_n$.*

Proof. We may assume without loss of generality that it is the first vectors $e_1, e_2, \dots, e_h$ that are linearly dependent. Consequently, there are numbers $\alpha_1, \alpha_2, \dots, \alpha_h$, not all zero, such that

$$\alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_h e_h = 0.$$

This yields

$$\alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_h e_h + 0 \cdot e_{h+1} + \dots + 0 \cdot e_n = 0.$$

But this equation implies the linear dependence of $e_1, e_2, \dots, e_n$ since there are nonzero numbers among $\alpha_1, \alpha_2, \dots, \alpha_h, 0, \dots, 0$.

Lemma 14.3. *If there is at least one zero vector among $e_1, e_2, \dots, e_n$, then the entire system $e_1, e_2, \dots, e_n$ is linearly dependent.*

Proof. Indeed, a system of one zero vector is linearly dependent. Therefore it follows from the property just proved that the entire system is linearly dependent.

The following theorem is the most important result relating to linear dependence:

Theorem 14.2. *Vectors $e_1, e_2, \dots, e_n$ are linearly dependent if and only if either $e_1 = 0$ or some vector e_k , $2 \leq k \leq n$, is a linear combination of the preceding vectors.*

Proof. Suppose $e_1, e_2, \dots, e_n$ are linearly dependent. Then in (14.1) not all coefficients are zero. Let the last nonzero coefficient be α_k . If $k = 1$, then this means that $e_1 = 0$. Now let $k > 1$. Then from

$$\alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_k e_k = 0$$

we find that

$$e_k = \left(-\frac{\alpha_1}{\alpha_k}\right) e_1 + \left(-\frac{\alpha_2}{\alpha_k}\right) e_2 + \dots + \left(-\frac{\alpha_{k-1}}{\alpha_k}\right) e_{k-1}.$$

This proves the necessity of the statement formulated in the theorem. Sufficiency is obvious since both the case where $e_1 = 0$ and the case where e_k is linearly expressible in terms of the preceding vectors imply the linear dependence of the first vectors in $e_1, e_2, \dots, e_n$. But this implies the linear dependence of the entire system of vectors.

Exercises

1. Prove that if any vector of a vector space can be uniquely represented as a linear combination of vectors $e_1, e_2, \dots, e_n$, then that system of vectors is linearly independent.

2. Prove that if a system of vectors $e_1, e_2, \dots, e_n$ is linearly independent, then any vector of the span of those vectors can be uniquely represented as a linear combination of them.

3. Prove that a system of vectors $e_1, e_2, \dots, e_n$ is linearly dependent if and only if either $e_n = 0$ or some vector e_k , $1 \leq k \leq n-1$, is a linear combination of the subsequent vectors.

4. Consider a vector space of polynomials in a variable t over the field of real numbers. Prove that the system of vectors $1, t, t^2, \dots, t^n$ is linearly independent for any n .

5. Prove that a system of two noncollinear directed line segments is linearly independent.

15. Equivalent systems of vectors

Consider two systems of vectors of a vector space K . Suppose their spans coincide and constitute some set L . Any vector of each system is clearly in L and in addition each vector of L can be represented in this case as a linear combination of both the vectors of one system and those of the other. Consequently:

Two systems of vectors possess the property that any vector of each system is linearly expressible in terms of the vectors of the other.

Such systems are called *equivalent*.

It follows from the foregoing that if the spans of two systems of vectors coincide, then those systems are equivalent. Now let any two equivalent systems be given. Then by the transitivity of the concept of linear combination any linear combination of vectors of one system can be represented as a linear combination of vectors of the other system, i.e. the spans of both systems coincide. So we have

Lemma 15.1. *For the spans of two systems of vectors to coincide it is necessary and sufficient that those systems should be equivalent.*

Notice that the concept of equivalence of two systems of vectors is an *equivalence relation*. Reflexivity is obvious since any system is equivalent to itself, symmetry follows from the definition of equivalent systems, and the transitivity of the notion follows from that of the concept of linear combination. Therefore *the set of all systems of vectors of any vector space can be divided into classes of equivalent systems*. It is important to stress that all the systems of the same class have the same span.

Nothing can be said in the general case about the number of vectors in equivalent systems. But if at least one of two equivalent systems is linearly independent, then it is possible to make quite definite conclusions concerning the number of vectors. They are based on

Theorem 15.1. *If each of the vectors in a linearly independent system $e_1, e_2, \dots, e_n$ is linearly expressible in terms of vectors $y_1, y_2, \dots, y_m$, then $n \leq m$.*

Proof. Under the hypothesis of the theorem e_n is linearly expressible in terms of $y_1, y_2, \dots, y_m$ and hence the system

$$e_n, y_1, y_2, \dots, y_m \quad (15.1)$$

is linearly dependent. The vector e_n is not equal to zero and therefore by Theorem 14.2 some vector y_i in (15.1) is a linear combination

of the preceding vectors. On eliminating this vector we obtain the following system:

$$e_n, y_1, \dots, y_{l-1}, y_{l+1}, \dots, y_m. \quad (15.2)$$

Using the transitivity of the concept of linear combination it is now easy to show that each of the vectors $e_1, e_2, \dots, e_n$ is linearly expressible in terms of vectors (15.2).

We join to vectors (15.2) on the left a vector e_{n-1} . We again conclude that the system

$$e_{n-1}, e_n, y_1, \dots, y_{l-1}, y_{l+1}, \dots, y_m \quad (15.3)$$

is linearly dependent. The vector e_{n-1} is not equal to zero and therefore by Theorem 14.2 one of the other vectors (15.3) is a linear combination of the preceding vectors. This vector cannot be e_n since this would imply the linear dependence of the system of two vectors e_{n-1}, e_n and hence of the entire system of vectors $e_1, e_2, \dots, e_n$. Thus some vector y_j in (15.3) is linearly expressible in terms of the preceding ones. If we eliminate it, then we again obtain a system in terms of which each of the vectors $e_1, e_2, \dots, e_n$ is linearly expressible.

Continuing this process notice that the vectors $y_1, y_2, \dots, y_m$ cannot be exhausted before we have joined all vectors $e_1, e_2, \dots, e_n$. Otherwise it will turn out that each of the vectors $e_1, e_2, \dots, e_n$ is linearly expressible in terms of some of the vectors of the same system, i.e. that the entire system must be linearly dependent. Since this contradicts the hypothesis of the theorem, it follows that $n \leq m$.

Consider consequences of the theorem. Suppose we are given two equivalent linearly independent systems of vectors. By Theorem 15.1 each of the systems contains at most as many vectors as the other. Consequently:

Equivalent linearly independent systems consist of the same number of vectors.

Take further n arbitrary vectors, construct on them the span and choose on it any $n + 1$ vectors. Since the number of those vectors is greater than that of the given vectors, they cannot be linearly independent. Therefore:

Any $n + 1$ vectors in the span of a system of n vectors are linearly dependent.

In terms of equivalent systems Lemma 14.1 implies that whatever a system of vectors not all equal to zero may be there is an equivalent linearly independent subsystem in it. This subsystem is called a *basis* of the original system.

Of course, any system may have more than one basis. All the bases of equivalent systems are themselves equivalent systems. It follows from the first consequence of Theorem 15.1 that they consist

of the same number of vectors. That number is a characteristic of all equivalent systems and is called their *rank*. By definition the rank of systems of zero vectors is considered to be equal to zero.

Consider now two linearly independent systems consisting of the same number of vectors. Replace some vector of one system by some vector of the other. In the resulting system again replace a vector of the first system by some of the remaining vectors of the second system and so on. The replacement process is carried on until one system is replaced by the other. If replacement is carried out in an arbitrary manner, then the intermediate systems may turn out to be linearly dependent. However, we have

Theorem 15.2. *The process of successive replacement may be carried out so that intermediate systems will all be linearly independent.*

Proof. Let $y_1, y_2, \dots, y_n$ and $z_1, z_2, \dots, z_n$ be two linearly independent systems of vectors. Suppose k steps of the process have been carried out, with $k \geq 0$. We may assume without loss of generality that the vectors $y_1, \dots, y_k$ have been replaced by $z_1, \dots, z_k$ and that all the systems obtained, including the system

$$z_1, \dots, z_k, y_{k+1}, \dots, y_n,$$

are linearly independent. This assumption obviously holds for $k = 0$.

Suppose further that when y_{k+1} is replaced by any of the vectors $z_{k+1}, \dots, z_n$ all systems

$$z_1, \dots, z_k, z_i, y_{k+2}, \dots, y_n$$

are linearly dependent for $i = k + 1, \dots, n$. Since the system

$$z_1, \dots, z_k, y_{k+2}, \dots, y_n \tag{15.4}$$

is linearly independent, it follows that the vectors z_i for $i = k + 1, \dots, n$ are linearly expressible in terms of it. But so are the vectors z_i for $i = 1, 2, \dots, k$. Consequently, all vectors $z_1, \dots, z_n$ must be linearly expressible in terms of (15.4). This is impossible by virtue of Theorem 15.1. Therefore the replacement process indicated in Theorem 15.2 does indeed hold.

Exercises

Prove that the following transformations of a system of vectors, called *elementary*, result in an equivalent system.

1. Addition to a system of vectors of any linear combination of those vectors.
2. Elimination from a system of vectors of any vector which is a linear combination of the remaining vectors.
3. Multiplication of any vector of a system by a number other than zero.
4. Addition to any vector of a system of any linear combination of the remaining vectors.
5. Interchanging of two vectors.

16. The basis

Suppose we are given a vector space consisting of *not only a zero vector*. In such a space there is clearly at least one nonzero vector and consequently there is a linearly independent system of at least one vector. There are two possibilities now: either there is a linearly independent system containing an arbitrarily large number of vectors or there is a linearly independent system containing a maximum number of vectors. In the former case the vector space is called *infinite dimensional* and in the latter it is called *finite dimensional*.

With the exception of some episodic examples, *our attention will be devoted throughout this book to finite dimensional spaces*. In particular, a finite dimensional vector space is any span constructed on a finite number of vectors of an arbitrary (not necessarily finite dimensional) space.

So let vectors $e_1, e_2, \dots, e_n$ constitute in a finite dimensional vector space K a linearly independent system with a maximum number of vectors. This means that for any vector x in K the system $e_1, e_2, \dots, e_n, x$ will be linearly dependent. By Theorem 14.2 the vector x is linearly expressible in terms of $e_1, e_2, \dots, e_n$. Since x is arbitrary and $e_1, e_2, \dots, e_n$ are fixed, we may say that

Any finite dimensional vector space is the span of a finite number of its vectors.

In studying finite dimensional vector spaces now we can use any properties relating to spans and equivalent systems of vectors. We introduce the following definition:

A linearly independent system of vectors in terms of which each vector of a space is expressible is called a *basis of the space*.

Our concept of basis is associated with a linearly independent system containing a maximum number of vectors. It is obvious, however, that all bases of the same finite dimensional vector space are equivalent linearly independent systems. As we know, such systems contain the same number of vectors. Therefore the number of vectors in a basis is a characteristic of a finite dimensional vector space. This number is called the *dimension* of a vector space K and designated $\dim K$. If $\dim K = n$, then the space K is *n-dimensional*. It is clear that:

In an n-dimensional vector space any linearly independent system of n vectors forms a basis and any system of $n + 1$ vectors is linearly dependent.

Notice that throughout the foregoing we have assumed that a vector space consists of not only a zero vector. A space consisting of a zero vector alone has no basis in the above sense, and by definition its dimension is assumed to equal zero.

The concept of basis plays a great role in the study of finite dimensional vector spaces and we shall continually use it for this purpose. It allows a very easy description of the structure of any vector space over an arbitrary field P . In addition it can be used to construct a very efficient method reducing operations on elements of a space to the corresponding operations on numbers from a field P .

As shown above, any vector x of a vector space K may be represented as a linear combination

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n, \quad (16.1)$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are some numbers from P and $e_1, e_2, \dots, e_n$ constitute a basis of K . The linear combination (16.1) is called the *expansion of a vector x with respect to a basis* and the numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ are the *coordinates of x relative to that basis*. The fact that x is given by its coordinates $\alpha_1, \alpha_2, \dots, \alpha_n$ will be written as follows:

$$x = (\alpha_1, \alpha_2, \dots, \alpha_n).$$

As a rule, we shall not indicate to which basis the given coordinates relate, unless any ambiguity arises.

It is easy to show that for any vector x in K its *expansion with respect to a basis is unique*. This can be proved by a device very often used in solving problems concerning linear dependence. Suppose there is another expansion

$$x = \beta_1 e_1 + \beta_2 e_2 + \dots + \beta_n e_n. \quad (16.2)$$

Subtracting term by term (16.2) from (16.1) we get

$$(\alpha_1 - \beta_1) e_1 + (\alpha_2 - \beta_2) e_2 + \dots + (\alpha_n - \beta_n) e_n = 0.$$

Since $e_1, e_2, \dots, e_n$ are linearly independent, it follows that all coefficients of the linear combination are zero and hence expansions (16.1) and (16.2) coincide.

Thus, with a basis of a vector space K fixed, every vector in K is uniquely determined by the collection of its coordinates relative to that basis.

Now let any two vectors x and y in K be given by their coordinates relative to the same basis $e_1, e_2, \dots, e_n$, i.e.

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n,$$

$$y = \gamma_1 e_1 + \gamma_2 e_2 + \dots + \gamma_n e_n,$$

then

$$x + y = (\alpha_1 + \gamma_1) e_1 + (\alpha_2 + \gamma_2) e_2 + \dots + (\alpha_n + \gamma_n) e_n.$$

Also, for any number λ in the field P ,

$$\lambda x = (\lambda \alpha_1) e_1 + (\lambda \alpha_2) e_2 + \dots + (\lambda \alpha_n) e_n.$$

It follows that *in adding two vectors of a vector space their coordinates relative to any basis are added and in multiplying a vector by a number all its coordinates are multiplied by that number.*

Exercises

1. Prove that the rank of a system of vectors coincides with the dimension of its span.
2. Prove that equivalent systems of vectors have the same rank.
3. Prove that if a span L_1 is constructed on the vectors of a span L_2 , then $\dim L_1 \leq \dim L_2$.
4. Prove that if a span L_1 is constructed on the vectors of a span L_2 and $\dim L_1 = \dim L_2$, then the spans coincide.
5. Prove that a vector space of polynomials with real coefficients given over a field of real numbers is infinite dimensional.

17. Simple examples of vector spaces

The fundamental concepts of linear dependence and of basis can be illustrated by very simple but instructive examples if we take as vector spaces numerical sets with the usual operations of addition and multiplication. That the axioms of a vector space hold for such sets is quite obvious and therefore we shall not verify their validity. As before elements of a space will be called *vectors*.

Consider the *complex vector space* which is an additive group of all complex numbers with multiplication over the field of complex numbers. It is clear that any nonzero number z_1 is a linearly independent vector. But even any two nonzero vectors z_1 and z_2 are always linearly dependent. To prove this it suffices to find two complex numbers α_1 and α_2 , not both equal to zero, such that $\alpha_1 z_1 + \alpha_2 z_2 = 0$. But this equation is obvious for $\alpha_1 = -z_2$ and $\alpha_2 = z_1$. Therefore the vector space considered is one-dimensional.

Somewhat different is the *real vector space* which is an additive group of all complex numbers with multiplication over the field of real numbers. As coefficients of a linear combination we can now use only real numbers and therefore this vector space cannot be one-dimensional. Indeed, there are no real numbers α_1 and α_2 , all non-zero, such that for them the linear combination $\alpha_1 z_1 + \alpha_2 z_2$ would vanish, for example, when $z_1 = 1$ and $z_2 = i$. It is left as an exercise for the reader to prove that this vector space is two-dimensional.

It is important to stress that although both of the above spaces consist of the *same* elements they are fundamentally *different* from each other.

It is now clear that the real vector space which is an additive group of all real numbers with multiplication over the field of real numbers is a one-dimensional space. We consider further the *rational vector space* which is an additive group of all real numbers with multiplication over the field of rational numbers.

We shall try, as before, to construct a system containing a maximum number of linearly independent vectors $r_1, r_2, r_3, \dots$. It is clear that we may take, for example, $r_1 = 1$. Since only rational numbers are allowed to be the coefficients of linear combinations $\alpha_1, \alpha_2, \alpha_3, \dots$, it is clear that no number of the form $\alpha_1 \cdot 1$ can represent, for example, $\sqrt[4]{2}$. Therefore this space cannot be one-dimensional. Consequently, it is $\sqrt[4]{2}$ that can be taken as a second vector linearly independent of the identity element. A number of the form $\alpha_1 \cdot 1 + \alpha_2 \cdot \sqrt[4]{2}$ cannot, however, represent, for example, $\sqrt[4]{2}$.

Indeed, let $\sqrt[4]{2} = \alpha_1 + \alpha_2 \sqrt[4]{2}$ hold for some rational numbers α_1 and α_2 . Squaring both sides of the equation we get

$$\sqrt{2} = (\alpha_1^2 + 2\alpha_2^2) + 2\alpha_1\alpha_2 \sqrt{2}$$

or

$$\frac{2(1 - 2\alpha_1\alpha_2)}{\alpha_1^2 + 2\alpha_2^2} = \sqrt{2}.$$

This is impossible since the left-hand side has a rational and the right-hand side has an irrational.

So the space under consideration cannot be two-dimensional either. But then what sort is it? Surprising as it may be, it is *infinite dimensional*. However, the proof of this fact is beyond the scope of this book.

The particular attention we have given to the examples of vector spaces of small dimensions is due to the possibility of using them to construct vector spaces of any dimensions. But we shall discuss this later on.

Exercises

1. What is the dimension of the vector space of rational numbers over the field of rational numbers?
2. Construct linearly independent systems of vectors in the space of complex numbers over the field of rational numbers.
3. Is an additive group of rational numbers over the field of real numbers a vector space? If not, why?

18. Vector spaces of directed line segments

We have already noted earlier that the sets of collinear directed line segments, of coplanar directed line segments and of directed line segments in the whole of space form vector spaces over the field of real numbers. Our immediate task is to show their dimensions and to construct their bases.

Lemma 18.1. *A necessary and sufficient condition of the linear dependence of two vectors is that they should be collinear.*

Proof. Notice that the lemma is obvious if of two vectors at least one is zero. It will therefore be assumed that both vectors are non-zero.

Let a and b be linearly dependent vectors. Then there are numbers α and β such that

$$\alpha a + \beta b = 0.$$

Since under the hypothesis $a \neq 0$ and $b \neq 0$, we have $\alpha \neq 0$ and $\beta \neq 0$ and therefore

$$b = \left(-\frac{\alpha}{\beta}\right) a.$$

Consequently, by the definition of multiplication of a directed line segment by a number, a and b are collinear.

Suppose now that a and b are collinear vectors. Apply them to a common point O . They will be on some straight line which is turned into an axis by specifying on it a direction. The vectors a and b are nonzero and therefore there is a real λ such that the magnitude of the directed line segment a equals the product of the magnitude of b by λ , i.e. $\{a\} = \lambda \{b\}$. But by the definition of multiplication of a directed line segment by a number this means that $a = \lambda b$. So the vectors a and b are linearly dependent.

It follows from the above lemma that a vector space of collinear directed line segments is a *one-dimensional* space and that *any nonzero vector* may serve as its basis.

Lemma 18.1 allows us to deduce one useful consequence. Namely, if vectors a and b are collinear and $a \neq 0$, then there is a number λ such that $b = \lambda a$. Indeed, these vectors are linearly dependent, i.e. for some numbers α and β , not both zero, $\alpha a + \beta b = 0$. If it is assumed that $\beta \neq 0$, then it follows that $\alpha = 0$. Therefore $\beta \neq 0$ and we may take $\lambda = (-\alpha)/\beta$ as a number λ .

Lemma 18.2. *A necessary and sufficient condition for three vectors to be linearly dependent is that they should be coplanar.*

Proof. We may assume without loss of generality that no pair of the three vectors is collinear since otherwise the lemma follows immediately from Lemma 18.1.

So let a , b and c be three linearly dependent vectors. Then we can find real numbers α , β and γ , not all zero, such that

$$\alpha a + \beta b + \gamma c = 0.$$

If, for example, $\gamma \neq 0$, then this equation yields

$$c = \left(-\frac{\alpha}{\gamma}\right) a + \left(-\frac{\beta}{\gamma}\right) b.$$

Apply a , b and c to a common point O . Then it follows from the last equation that the vector c is equal to the diagonal of the parallelogram constructed on the vectors $(-\alpha/\gamma)a$ and $(-\beta/\gamma)b$. This means that after translation to the common point the vectors a , b and c are found to be in the same plane and consequently they are coplanar.

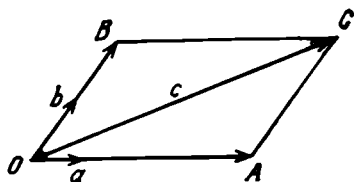


Fig. 18.1

Suppose now that a , b and c are coplanar vectors. Translate them to the same plane and apply them to the common point O (Fig. 18.1). Draw through the terminal point of c the straight lines parallel to

a and b and consider the parallelogram $OACB$. The vectors a , $\vec{OA}$ and b , $\vec{OB}$ are collinear by construction and nonzero and therefore there are numbers λ and μ such that

$$\vec{OA} = \lambda a, \quad \vec{OB} = \mu b.$$

But $\vec{OC} = \vec{OA} + \vec{OB}$, which means that $c = \lambda a + \mu b$ or

$$\lambda a + \mu b + (-1)c = 0.$$

Since λ , μ and -1 are clearly different from zero, the last equation implies that a , b and c are linearly dependent.

We can now solve the question concerning the dimension of the vector space of coplanar directed line segments. By Lemma 18.2 the dimension of this space must be less than three. But any two noncollinear directed line segments of this space are linearly independent. Therefore the vector space of coplanar directed line segments is a *two-dimensional* space and *any two noncollinear vectors* may serve as its basis.

Lemma 18.3. *Any four vectors are linearly dependent.*

Proof. We may assume without loss of generality that no triple of the four vectors are coplanar since otherwise the lemma follows immediately from Lemma 18.2. Apply vectors a , b , c and d to a common origin O and draw through the terminal point D of d the planes parallel to the planes determined respectively by the pairs of vectors b, c ; a, c ; a, b (Fig. 18.2). It follows from the parallelogram law of vector addition that

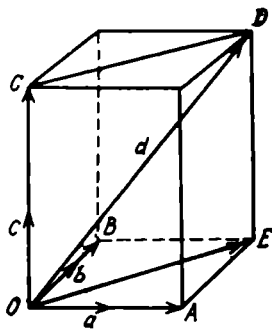


Fig. 18.2

$$\vec{OD} = \vec{OC} + \vec{OE}, \quad \vec{OE} = \vec{OA} + \vec{OB},$$

therefore

$$\vec{OD} = \vec{OA} + \vec{OB} + \vec{OC}. \quad (18.1)$$

The vectors a , $\vec{OA}$, as well as b , $\vec{OB}$ and c , $\vec{OC}$, are collinear by construction, with a , b and c being nonzero. Therefore there are numbers λ , μ and ν such that

$$\vec{OA} = \lambda a, \quad \vec{OB} = \mu b, \quad \vec{OC} = \nu c.$$

Considering (18.1) this yields

$$d = \lambda a + \mu b + \nu c$$

from which it follows that a , b , c and d are linearly dependent.

From Lemma 18.3 we conclude that the dimension of the vector space of all directed line segments must be less than four. But it cannot be less than three since by Lemma 18.2 any three noncoplanar directed line segments are linearly independent. Therefore the vector space of all directed line segments is a *three-dimensional* space and *any three noncoplanar vectors* may serve as its basis.

The vector spaces considered are not very obvious geometrically, since they allow the existence of infinitely many equal vectors. They become much more obvious if we choose one representative from each class of equal vectors and always mean by "vector" a directed line segment from the collection of only these representatives.

One of the most convenient ways of choosing a vector is through considering the set of directed line segments fixed at some point O . Then instead of the vector space of collinear directed line segments we obtain a space of collinear line segments fixed at O and lying on a straight line passing through O ; instead of the vector space of coplanar directed line segments we obtain a space of directed line segments fixed at O and lying in the plane through O ; and finally instead of the vector space of all directed line segments we obtain a space of directed line segments fixed at O .

In what follows we shall deal mostly with only fixed vectors. The corresponding vector spaces will be denoted by V_1 , V_2 and V_3 , where the subscript stands for dimension. A vector space consisting of only a zero directed line segment will be denoted by V_0 .

These spaces allow us to establish a 1-1 correspondence between points and directed line segments. To do this it suffices to assign to every vector its terminal point. Bearing this *geometrical interpretation* in mind we shall sometimes call elements of an abstract vector space *points* rather than vectors.

Exercises

In V_1 , V_2 and V_3 establish the geometrical meaning of such notions as:

1. Span.
2. Linear dependence and independence.
3. Equivalent systems of vectors.
4. Elementary equivalent transformations of a system of vectors.
5. Rank of a system of vectors.

19. The sum and intersection of subspaces

The introduction of spans has shown that every vector space contains an infinite number of other vector spaces. The significance of these spaces is not limited to the questions considered above.

Spans were given by us by directly indicating their *structure*. We could use another way, that of defining "smaller" spaces in terms of the *properties* of vectors. Let L be a set of vectors in a vector space K . If under the same operations as in K the set L is a vector space, then it is said to be a linear *subspace*, thus stressing the fact that a subspace consists of vectors of some space. It is clear that the smallest subspace is that consisting of only a zero vector. This will be called a *zero subspace* and designated 0 . The largest subspace is the space K . These two subspaces are *trivial*, the others are *nontrivial*. It is also obvious that together with every pair of its elements x and y any subspace contains all their linear combinations $\alpha x + \beta y$. The converse is also true. Namely:

If a set L of vectors of the vector space K contains together with every pair of its elements x and y all their linear combinations $\alpha x + \beta y$, then it is a subspace.

Indeed, of all vector space axioms it is necessary to verify only the axioms of the zero and negative vectors. The rest of the axioms are obvious. Take $\alpha = 0$ and $\beta = 0$. From the consequences of the operations for vectors of K we conclude that $0 \cdot x + 0 \cdot y = 0$, i.e. that the zero vector is in L . Now take $\alpha = -1$ and $\beta = 0$. We have $(-1)x + 0 \cdot y = (-1)x$ and therefore together with every vector x the set L contains its negative. So L is a subspace.

The existence of a basis says that in any finite dimensional space any subspace is a span. In finite dimensional vector spaces therefore the span is the most general way of giving linear subspaces. It is not so in an infinite dimensional space. Nevertheless it should not be forgotten that there is very much in common between concepts and facts from finite dimensional spaces and those from infinite dimensional spaces. To emphasize this, even in finite dimensional spaces we shall oftener use the term *linear subspace* than *span*.

Let K be an n -dimensional space. As in the space K , a basis can be constructed in any of its subspaces L . If a basis $e_1, e_2, \dots, e_n$ is chosen in K , then in the general case one cannot choose basis vectors of a subspace L directly among the vectors $e_1, e_2, \dots, e_n$ if only because L can have none of them. We have, however, the converse, in a sense,

Lemma 19.1. *If in some subspace L of dimension s an arbitrary basis $t_1, \dots, t_s$ has been chosen, then we can always choose vectors $t_{s+1}, \dots, t_n$ in a space K of dimension n in such a way that the system of vectors $t_1, \dots, t_s, t_{s+1}, \dots, t_n$ is a basis in the whole of K .*

Proof. Consider only the linearly independent systems of vectors in K that contain the vectors $t_1, \dots, t_s$. It is clear that among these systems there is a system $t_1, \dots, t_s, t_{s+1}, \dots, t_p$ with a maximum number of vectors. But then whatever the vector x in K may be the system $t_1, \dots, t_p, x$ must be linearly dependent. Consequently, x must be linearly expressible in terms of the vectors $t_1, \dots, t_p$. This means that $t_1, \dots, t_s, t_{s+1}, \dots, t_p$ form a basis K and that $p = n$.

Consider again an arbitrary vector space K . It generates the set of all its subspaces, which we denote by U . On U we can define two algebraic operations allowing some subspaces to be constructed from other subspaces.

The *sum* $L_1 + L_2$ of linear subspaces L_1 and L_2 is the set of all vectors of the form $z = x + y$, where $x \in L_1$ and $y \in L_2$.

The *intersection* $L_1 \cap L_2$ of linear subspaces L_1 and L_2 is the set of all vectors simultaneously lying in L_1 and L_2 .

Notice that both the sum of subspaces and their intersection are nonempty sets, since they clearly contain the zero vector of K . We prove that they are subspaces.

Indeed, take two arbitrary vectors z_1 and z_2 from the sum $L_1 + L_2$. This means that $z_1 = x_1 + y_1$ and $z_2 = x_2 + y_2$, where $x_1, x_2 \in L_1$ and $y_1, y_2 \in L_2$. Consider now an arbitrary linear combination $\alpha z_1 + \beta z_2$. We have $\alpha z_1 + \beta z_2 = (\alpha x_1 + \beta x_2) + (\alpha y_1 + \beta y_2)$. Since $\alpha x_1 + \beta x_2 \in L_1$ and $\alpha y_1 + \beta y_2 \in L_2$, we have $\alpha z_1 + \beta z_2 \in L_1 + L_2$. Therefore, $L_1 + L_2$ is a subspace. Now let $z_1, z_2 \in L_1 \cap L_2$, i.e. $z_1, z_2 \in L_1$ and $z_1, z_2 \in L_2$. It is clear that $\alpha z_1 + \beta z_2 \in L_1$ and $\alpha z_1 + \beta z_2 \in L_2$, i.e. $\alpha z_1 + \beta z_2 \in L_1 \cap L_2$. Hence $L_1 \cap L_2$ is also a subspace.

Thus the operations of addition of subspaces and of their intersection are algebraic. They are obviously commutative and associative. Moreover, for any subspace L of K

$$L + 0 = L, \quad L \cap K = L.$$

There are no distributive laws relating the two operations.

As can easily be seen even from the simplest examples, the dimension of the sum of two arbitrary subspaces depends not only on those of the subspaces but also on the size of their common part. We have

Theorem 19.1. *For any two finite dimensional subspaces L_1 and L_2*
 $\dim(L_1 \cap L_2) + \dim(L_1 + L_2) = \dim L_1 + \dim L_2.$ (19.1)

Proof. Denote the dimensions of L_1 , L_2 and $L_1 \cap L_2$ by r_1 , r_2 and m respectively. Choose at the intersection $L_1 \cap L_2$ some basis $c_1, \dots, c_m$. These vectors are linearly independent and are in L_1 . By Lemma 19.1 there are vectors $a_1, \dots, a_k$ in L_1 such that the system $a_1, \dots, a_k, c_1, \dots, c_m$ is a basis in L_1 . Similarly there are vectors $b_1, \dots, b_p$ in L_2 such that $b_1, \dots, b_p, c_1, \dots, c_m$ is a basis in L_2 . We have

$$r_1 = k + m, \quad r_2 = p + m.$$

If we prove that

$$a_1, \dots, a_k, c_1, \dots, c_m, b_1, \dots, b_p \quad (19.2)$$

is a basis of the subspace $L_1 + L_2$, then the theorem holds since

$$m + (k + m + p) = (k + m) + (p + m).$$

Any vector in L_1 and L_2 is linearly expressible in terms of the vectors of its basis and of course any is linearly expressible in terms of vectors (19.2). Therefore any vector in the sum $L_1 + L_2$ is also linearly expressible in terms of these vectors. It remains for us to show that system (19.2) is linearly independent. Let

$$\alpha_1 a_1 + \dots + \alpha_k a_k + \gamma_1 c_1 + \dots + \gamma_m c_m + \beta_1 b_1 + \dots + \beta_p b_p = 0 \quad (19.3)$$

and let

$$b = \beta_1 b_1 + \dots + \beta_p b_p. \quad (19.4)$$

It is clear that $b \in L_2$. But from (19.3) it follows that $b \in L_1$. Consequently, $b \in L_1 \cap L_2$, i.e.

$$b = v_1 c_1 + \dots + v_m c_m \quad (19.5)$$

for some numbers $v_1, \dots, v_m$. Comparing (19.4) and (19.5) we get

$$\beta_1 b_1 + \dots + \beta_p b_p + (-v_1) c_1 + \dots + (-v_m) c_m = 0.$$

The system of vectors $b_1, \dots, b_p, c_1, \dots, c_m$ is linearly independent by construction and therefore

$$\beta_1 = \dots = \beta_p = v_1 = \dots = v_m = 0.$$

By virtue of the linear independence of $a_1, \dots, a_k, c_1, \dots, c_m$ it now follows from (19.3) that

$$\alpha_1 = \dots = \alpha_k = \gamma_1 = \dots = \gamma_m = 0.$$

Thus the theorem is proved.

Exercises

1. Consider a vector space V_3 to establish the geometrical meaning of the sum and intersection of subspaces.
2. What is the sum of subspaces V_1 and V_2 ?
3. What is the intersection of subspaces V_1 and V_2 ?
4. Prove that the dimension of the intersection of any number of subspaces does not exceed the minimum dimension of those subspaces.
5. Prove that the dimension of the sum of any number of subspaces is not less than the maximum dimension of those subspaces.

20. The direct sum of subspaces

Let $L_1, L_2, \dots, L_m$ be subspaces of some vector space. By the definition of the operation of addition any vector x in the sum

$$K = L_1 + L_2 + \dots + L_m \quad (20.1)$$

may be represented as

$$x = x_1 + x_2 + \dots + x_m, \quad (20.2)$$

where $x_i \in L_i$ for every i . In general this representation is not unique. But if every vector in K allows the unique representation (20.2), then sum (20.1) is called a *direct sum* and is designated as follows:

$$K = L_1 \dot{+} L_2 \dot{+} \dots \dot{+} L_m. \quad (20.3)$$

Direct sums possess many special properties. But we shall be concerned not so much with these properties as with the *common features of representation* (20.2) and *expansion with respect to a basis*. Suppose some space K may be represented as a direct sum (20.3) of its subspaces $L_1, L_2, \dots, L_m$. Then by virtue of the uniqueness of representation (20.2) the system of subspaces $L_1, L_2, \dots, L_m$ may be regarded as some "generalized basis" of K and representation (20.2) as an expansion with respect to the "generalized basis". Such an interpretation of a direct sum is especially helpful in the study of vector spaces of higher dimensions, since in those spaces one has as a rule to study not all the components in the expansion with respect to a basis but only a small portion of them. Using direct sums makes it possible to avoid both cumbersome expansions and investigating unnecessary details.

Let K be an n -dimensional vector space. Take its arbitrary basis $e_1, e_2, \dots, e_n$ and construct a collection of spans $L_1 = L_1(e_1), L_2 = L_2(e_2), \dots, L_n = L_n(e_n)$. It is then obvious that K is the direct sum of these n one-dimensional subspaces. But K may be represented in various ways as direct sum of subspaces of other dimension. Such a representation is based on

Exercises

1. Under what conditions is V_3 a direct sum of its subspaces V_1 and V_2 ?
2. Under what conditions is V_3 a direct sum of two of its subspaces of the type V_1 ?
3. Can V_3 be a direct sum of two of its subspaces of the type V_2 ? If not, why?
4. Prove that for sum (20.1) to be direct it is necessary and sufficient that representation (20.2) should be unique for the zero vector.
5. Prove that for sum (20.1) to be direct it is necessary and sufficient that the intersection of each of the subspaces L_i , $i = 1, \dots, m$, with the sum of the others should contain only a zero vector.

21. Isomorphism of vector spaces

Consider the set of all vector spaces over the same field P . It is natural to ask in what they are similar and in what different.

The description of every vector space contains two, essentially different, parts. First, a vector space is a collection of specific objects called vectors. Second, the operations of addition and multiplication by a number that have some properties are defined on those specific objects. We may be concerned therefore either with the nature of vectors and their properties or with the properties of the operations regardless of the nature of the elements.

We were concerned with the nature of vectors only when we studied directed line segments and only to an extent necessary for introducing the operations and establishing their properties. After that our investigation of directed line segments was based solely on the properties of operations. We shall proceed in a similar way in every particular case too. Therefore two spaces with the same structure of addition and multiplication by a number will be assumed to possess the same properties or to be *isomorphic*. More precisely:

Two vector spaces over the same field are said to be isomorphic if between their vectors a 1-1 correspondence can be established such that to the sum of any two vectors of one space there corresponds the sum of the corresponding vectors of the other and to the product of some number by a vector of one space there corresponds the product of the same number by the corresponding vector of the other.

Let K and K' be two isomorphic spaces. The fact that every vector x in K is assigned a definite vector x' in K' may be understood as an introduction of some "function"

$$x' = \omega(x) \quad (21.1)$$

whose "independent variable (or argument)" is a vector x in K and whose "value" is a vector x' in K' . Both properties of that function can now be written as follows. For any x and y in K and any

number λ

$$\omega(x + y) = \omega(x) + \omega(y), \quad (21.2)$$

$$\omega(\lambda x) = \lambda \omega(x).$$

The 1-1 correspondence between K and K' implies that to any different independent variables of the function (21.1) there correspond different values, i.e. if

$$x \neq y, \quad (21.3)$$

then

$$\omega(x) \neq \omega(y). \quad (21.4)$$

Consequently, the equality or nonequality of the values of the function implies respectively the equality or nonequality of its independent variables.

Isomorphic spaces have much in common. In particular, to a zero vector there corresponds a zero vector, for

$$\omega(0) = \omega(0 \cdot x) = 0 \cdot \omega(x) = 0 \cdot x' = 0'.$$

The most important consequence, however, is that a linearly independent system of vectors is sent into a linearly independent system.

Indeed, let $x_1, x_2, \dots, x_n$ be n linearly independent vectors. Consider now a linear combination $\alpha_1 \omega(x_1) + \alpha_2 \omega(x_2) + \dots + \alpha_n \omega(x_n)$ and equate it to zero. By the property of an isomorphism

$$\begin{aligned} 0' &= \alpha_1 \omega(x_1) + \alpha_2 \omega(x_2) + \dots + \alpha_n \omega(x_n) \\ &= \omega(\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n) = \omega(0), \end{aligned}$$

from which we have

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n = 0.$$

Since $x_1, x_2, \dots, x_n$ are linearly independent, all the coefficients must be zero.

The consequence we have proved makes it possible to state that if two finite dimensional vector spaces are isomorphic, then they have the same dimension. The converse is also true. Namely, we have

Theorem 21.1. *Any two finite dimensional vector spaces having the same dimension and given over the same field are isomorphic.*

Proof. Let K and K' be two vector spaces of dimension n . Choose a basis $e_1, e_2, \dots, e_n$ in K and a basis $e'_1, e'_2, \dots, e'_n$ in K' . Using these systems of vectors construct an isomorphism ω as follows. To every vector

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n$$

in K assign a vector

$$\omega(x) = \alpha_1 e'_1 + \alpha_2 e'_2 + \dots + \alpha_n e'_n$$

in K' . The correspondence will be 1-1 since an expansion with respect to a basis is unique.

Take then any two vectors x and y in K and an arbitrary number λ and assume that

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n,$$

$$y = \beta_1 e_1 + \beta_2 e_2 + \dots + \beta_n e_n.$$

We have

$$\begin{aligned}\omega(x+y) &= \omega((\alpha_1 + \beta_1)e_1 + (\alpha_2 + \beta_2)e_2 + \dots + (\alpha_n + \beta_n)e_n) \\ &= (\alpha_1 + \beta_1)e'_1 + (\alpha_2 + \beta_2)e'_2 + \dots + (\alpha_n + \beta_n)e'_n \\ &= (\alpha_1 e'_1 + \alpha_2 e'_2 + \dots + \alpha_n e'_n) \\ &\quad + (\beta_1 e'_1 + \beta_2 e'_2 + \dots + \beta_n e'_n) = \omega(x) + \omega(y), \\ \omega(\lambda x) &= \omega((\lambda\alpha_1)e_1 + (\lambda\alpha_2)e_2 + \dots + (\lambda\alpha_n)e_n) \\ &= (\lambda\alpha_1)e'_1 + (\lambda\alpha_2)e'_2 + \dots + (\lambda\alpha_n)e'_n \\ &= \lambda(\alpha_1 e'_1 + \alpha_2 e'_2 + \dots + \alpha_n e'_n) = \lambda\omega(x).\end{aligned}$$

These equations prove the theorem.

This theorem is very important. It is this theorem that allows us to say with certainty now that in terms of all the consequences of the axioms any two vector spaces having the same dimension and given over the same field are indistinguishable. Consequently, we could construct just one n -dimensional vector space over a given field and show the regularities common to all finite dimensional spaces by investigating just that single space.

Let P be some field. Consider a set whose elements are all possible ordered collections of n numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ from P . If x is an element of that set, then we shall write

$$x = (\alpha_1, \alpha_2, \dots, \alpha_n). \quad (21.5)$$

The operations of addition and multiplication by a number λ from the field P will be defined as follows:

$$\begin{aligned}(\alpha_1, \alpha_2, \dots, \alpha_n) + (\beta_1, \beta_2, \dots, \beta_n) \\ = (\alpha_1 + \beta_1, \alpha_2 + \beta_2, \dots, \alpha_n + \beta_n), \\ \lambda(\alpha_1, \alpha_2, \dots, \alpha_n) = (\lambda\alpha_1, \lambda\alpha_2, \dots, \lambda\alpha_n).\end{aligned} \quad (21.6)$$

It is easy to check that the axioms of a vector space hold. In particular, the zero vector is defined by a set of zeros alone, i.e.

$$0 = (0, 0, \dots, 0),$$

and the negative of vector (21.5) is like this:

$$-x = (-\alpha_1, -\alpha_2, \dots, -\alpha_n).$$

This is an n -dimensional space and one of its bases is easy to show at once. Namely,

$$\begin{aligned} e_1 &= (1, 0, \dots, 0, 0), \\ e_2 &= (0, 1, \dots, 0, 0), \\ &\vdots \\ e_n &= (0, 0, \dots, 0, 1). \end{aligned} \tag{21.7}$$

Since for element (21.5) we have the expansion

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n,$$

the numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ will be called the *coordinates* of the vector x .

We shall call a space of such a type an *arithmetical space* and denote it by P_n , thus emphasizing its relation to the field P . If P is the field of complex, real or rational numbers, then such n -dimensional spaces will be denoted by C_n , R_n and D_n respectively.

It may now seem that there is no need to study arbitrary n -dimensional vector spaces. Indeed, we know that in terms of the consequences of the axioms isomorphic vector spaces are indistinguishable and therefore we can always successfully study, for example, P_n alone. However, general arguments allow us to show the most important properties of vector spaces, i.e. the ones that *are independent of basis systems* or, in other words, *are invariant under isomorphisms*.

Studying spaces P_n alone we should always be tied to a *particular basis* and therefore it would not always be easy to see the invariance of deductions. Besides, it is necessary to see to it that particular properties of a space P_n are not referred to the general properties of vector spaces. This is not always sufficiently easy to do.

In conclusion note one more fact. By analogy with a space P_n consider the space P_∞ whose elements are all possible ordered infinitely large collections of numbers $\alpha_1, \alpha_2, \dots$ of the field P . An element x of that set is by analogy with (21.5) designated

$$x = (\alpha_1, \alpha_2, \dots)$$

and by analogy with (21.6) we introduce operations on elements.

Now P_∞ is an infinite dimensional space. If we assume that infinite dimensional spaces are isomorphic to P_∞ , it is not hard to see that infinite dimensional and finite dimensional spaces must have much in common. This example should not be forgotten.

Exercises

1. Construct an isomorphism from a space V_1 to the space of reals over the field of reals.
2. Construct an isomorphism from a space V_2 to the space of complex numbers over the field of reals.

to this question since it requires a deep understanding of the problems one solves. *It is with this question that the differences distinguishing "exact" mathematics from "approximate" mathematics begin.*

Exercises

1. Prove that if system (22.2) is compatible, then it has a unique solution if and only if the system of vectors $a_1, a_2, \dots, a_m$ is linearly independent.

2. Prove that if a system of vectors $a_1, a_2, \dots, a_m$ has rank r , then system (22.5) consists of r equations.

3. Assume that the solutions of a system are vectors of a space P_m . Let $b = 0$ and let the system of vectors $a_1, a_2, \dots, a_m$ has rank r . Prove that in this case the set of all solutions of (22.2) forms an $(m - r)$ -dimensional subspace of P_m .

4. Find all solutions of the system of linear algebraic equations

$$\sqrt{2}z_1 + 1 \cdot z_2 = \sqrt{3},$$

$$2 \cdot z_1 + \sqrt{2}z_2 = \sqrt{6}.$$

Solve the same system giving $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{6}$ to various accuracy. Compare the results.

5. Establish the relation of the Gauss method to elementary transformations of a system of vectors.

Measurements in Vector Space

23. Affine coordinate systems

An enormous number of science and engineering problems require a precise description in space of various geometrical objects such as points, figures, curves, surfaces and so on. For a complex object it is very important to know not only a general characterization of its location, such as indication of the centre of gravity, but also the position of each of its individual points.

As an example recall that the prediction of lunar and solar eclipses is possible because we know the position of celestial bodies at every moment. Television broadcasts over great distances are possible because the position of each point of the image being transmitted is defined.

It is obviously necessary to give a method of describing the position of only one individual point, since any geometrical object can be given as some collection of points. Probably it would be useful to consider independently the position of a point on a straight line, in the plane or in space because a spatial description of an object is by far not always appropriate. For example, a photograph can obviously be considered only in the plane while the motion of a particle, with no forces acting upon it, can be considered on a straight line.

One of the most common descriptions of the position of a point is based on a very simple idea. We have already noted that it is possible to establish a 1-1 correspondence between all points and fixed directed line segments. The description of the position of a point therefore can be replaced by the description of the position of the corresponding directed line segment. But the position of this line segment is characterized by its coordinates relative to any basis, i.e. by some ordered collections of numbers. Consequently, the position of a point must also be characterized by ordered collections of numbers. We now proceed to explore this idea.

Given some straight line, fix on it an arbitrary point O and consider a space V_1 of vectors lying on the given straight line and fixed at the point O . Choose in that space some basis vector a . Now turn the straight line into an axis by specifying on it a direction so that the magnitude of the segment a is positive (Fig. 23.1).

The axis with the given point O and basis vector a forms an *affine coordinate system on the straight line*. The point O is called the *origin*, and the length of the vector a is the *scale unit*.

The position of any point M on the straight line is uniquely determined by that of the vector $\vec{OM}$. The vectors a and $\vec{OM}$ are collinear, with $a \neq 0$, and so according to the consequence of Lemma 18.1 there is a real α such that

$$\vec{OM} = \alpha a. \quad (23.1)$$

That number is called an *affine coordinate* of the point M on the straight line. The point M with a coordinate α is designated $M(\alpha)$.

Notice that with a fixed affine coordinate system on the straight line relation (23.1)

uniquely defines the affine coordinate α of any point M of that straight line. Obviously the converse is also true. Namely, relation

(23.1) makes every number α uniquely define some point M of the straight line. Thus given a fixed affine coordinate system *there is a 1-1 correspondence between all real numbers and the points of a straight line*.

Giving points by their coordinates allows us to calculate the magnitudes of directed line segments and the distances between points. Let $M_1(\alpha_1)$ and $M_2(\alpha_2)$ be given points. We have

$$\begin{aligned} \{\overrightarrow{M_1 M_2}\} &= \{\vec{OM}_2 - \vec{OM}_1\} = \{\alpha_2 a - \alpha_1 a\} = \{(\alpha_2 - \alpha_1) a\} \\ &= (\alpha_2 - \alpha_1) \{a\} = (\alpha_2 - \alpha_1) |a|. \end{aligned} \quad (23.2)$$

If $\rho(M_1, M_2)$ denotes the distance between points M_1 and M_2 , then

$$\rho(M_1, M_2) = |\{\overrightarrow{M_1 M_2}\}| = |\alpha_2 - \alpha_1| |a|. \quad (23.3)$$

The formulas become particularly simple if the length of the basis vector equals unity. In this case

$$\begin{aligned} \{\overrightarrow{M_1 M_2}\} &= \alpha_2 - \alpha_1, \\ \rho(M_1, M_2) &= |\alpha_2 - \alpha_1|. \end{aligned} \quad (23.4)$$

Now let some plane be given. Fix on it an arbitrary point O and consider a space V_2 of vectors lying in the plane and fixed at O . Choose in that space some pair of basis vectors a and b . Specify directions on the straight lines containing those vectors so that the magnitudes of a and b are positive (Fig. 23.2).

The two axes in the plane intersecting at the same point O and having the basis vectors a and b given on them form an *affine coordi-*



Fig. 23.1

nate system in the plane. The axis containing a is called the x axis or the axis of abscissas; the axis containing b is the y axis or the axis of ordinates.

Again the position of any point M in the plane is uniquely determined by the vector $\vec{OM}$ and in turn there is a unique vector decomposition of the form

$$\vec{OM} = \alpha a + \beta b. \quad (23.5)$$

The real numbers α and β are again called the *affine coordinates* of the point M . The first coordinate is called the *abscissa* and the second the *ordinate* of M . The point M with coordinates α and β is designated $M(\alpha, \beta)$.

On the x and y axes there are unique points M_x and M_y such that

$$\vec{OM} = \vec{OM}_x + \vec{OM}_y. \quad (23.6)$$

They are at the intersection of the coordinate axes with the straight lines parallel to the axes and passing through M . We call them the *affine projections of the point M onto the coordinate axes*. The vectors

$\vec{OM}_x$ and $\vec{OM}_y$ are the *affine projections of $\vec{OM}$* . From the uniqueness of decompositions (23.5) and (23.6) we conclude that

$$\vec{OM}_x = \alpha a, \quad \vec{OM}_y = \beta b. \quad (23.7)$$

Thus if M has the coordinates $M(\alpha, \beta)$, then M_x and M_y , as points of the plane, have the coordinates $M_x(\alpha, 0)$ and $M_y(0, \beta)$. Moreover if

$$\vec{OM} = (\alpha, \beta),$$

then

$$\vec{OM}_x = (\alpha, 0), \quad \vec{OM}_y = (0, \beta).$$

Every basis vector forms a proper coordinate system on its axis. The points M_x and M_y therefore may be regarded as points of the x and y axes given in those proper coordinate systems. It follows from (23.7), however, that the coordinate of M_x on the x axis is equal to the abscissa of M . Similarly, the coordinate of M_y on the y axis

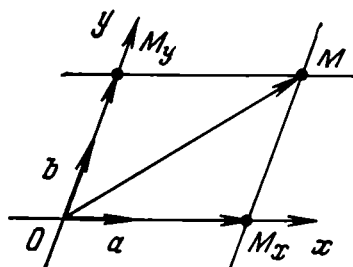


Fig. 23.2

is equal to the ordinate of M . Obvious as they may be, these assertions are very important since they allow us to use formulas (23.2) to (23.4).

Specifying an ordered pair of numbers α and β uniquely defines some point. Indeed, relations (23.7) allow one to construct in a unique way the affine projections of a point that uniquely determine the point in the plane. Consequently, given a fixed affine coordinate system there is a 1-1 correspondence between all ordered pairs of real numbers and points in the plane.

Similarly we can introduce an affine coordinate system in space. Fix a point O and consider a space V_3 of vectors fixed at O . Choose in that space some triple of basis vectors a , b and c . Specify directions on the straight lines containing those vectors so that the magnitudes of the directed line segments a , b and c are positive (Fig. 23.3).

The three axes in space intersecting at the same point O and having the basis vectors a , b and c given on them form an *affine coordinate system in space*. The axis containing a is called the *x axis* or *axis of x coordinates* or *abscissas*, the axis containing b is the *y axis* or *axis of y coordinates* or *ordinates* and the third axis is the *z axis* or *axis of z coordinates*. Pairs of coordinate axes determine the so-called *coordinate planes* designated x, y ; y, z ; x, z planes.

The position of any point M in space is again uniquely determined by the vector $\vec{OM}$ for which there is a unique decomposition

$$\vec{OM} = \alpha a + \beta b + \gamma c.$$

The real numbers α , β and γ are called the *affine coordinates* of a point M in space. The first coordinate is called an *x coordinate* or *abscissa*, the second is called a *y coordinate* or *ordinate* and the third is a *z coordinate* of M . The point M with coordinates α , β and γ is designated $M(\alpha, \beta, \gamma)$.

Draw through the point M the planes parallel to the coordinate planes. The points of intersection of these planes with the x , y and z axes are denoted by M_x , M_y and M_z and called the *affine projections of the point M onto the coordinate axes*. The intersection of the coordinate planes with pairs of planes passing through M determines points M_{yz} , M_{xz} and M_{xy} , called the *affine projections of the point M onto the coordinate planes*. Accordingly the vectors $\vec{OM}_{yz}$ and $\vec{OM}_x$ and

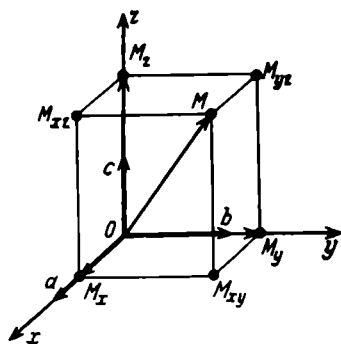


Fig. 23.3

so on are called the *affine projections of the vector* $\vec{OM}$. It is obvious that

$$\begin{aligned}\vec{OM} &= \vec{OM}_x + \vec{OM}_y + \vec{OM}_z, \\ \vec{OM}_{yz} &= \vec{OM}_y + \vec{OM}_z, \\ \vec{OM}_{xz} &= \vec{OM}_x + \vec{OM}_z, \\ \vec{OM}_{xy} &= \vec{OM}_x + \vec{OM}_y.\end{aligned}$$

We conclude, as in the case of the plane, that if a point M has the coordinates

$$M(\alpha, \beta, \gamma),$$

then the affine projections of that point will have the coordinates:

$$\begin{aligned}M_x(\alpha, 0, 0), & \quad M_y(0, \beta, 0), & \quad M_z(0, 0, \gamma), \\ M_{yz}(0, \beta, \gamma), & \quad M_{xz}(\alpha, 0, \gamma), & \quad M_{xy}(\alpha, \beta, 0).\end{aligned}\quad (23.8)$$

Similarly, if

$$\vec{OM} = (\alpha, \beta, \gamma),$$

then

$$\begin{aligned}\vec{OM}_x &= (\alpha, 0, 0), & \vec{OM}_y &= (0, \beta, 0), & \vec{OM}_z &= (0, 0, \gamma), \\ \vec{OM}_{yz} &= (0, \beta, \gamma), & \vec{OM}_{xz} &= (\alpha, 0, \gamma), & \vec{OM}_{xy} &= (\alpha, \beta, 0).\end{aligned}$$

Again every basis vector and every pair of basis vectors form proper affine systems on the coordinate axes and coordinate planes. And again the coordinates of points in those systems coincide with the affine coordinates of the same points regarded as points in space. Now, given a fixed affine coordinate system *there is a 1-1 correspondence between all ordered triples of real numbers and points in space.*

Of the affine coordinate systems on the straight line, in the plane and in space the most widely used are the so-called *rectangular Cartesian coordinate systems*. They are characterized by all basis vectors having unit length and the coordinate axes being mutually perpendicular in the case of the plane and space. In a Cartesian system basis vectors are usually denoted by i, j and k . In what follows we shall use only these systems as a rule.

Exercises

1. Which of the points $A(\alpha)$ and $B(-\alpha)$ is to the right of the other on the coordinate axis of Fig. 23.1?
2. What is the locus of points $M(\alpha, \beta, \gamma)$ for which the affine projections M_{xy} have coordinates $M_{xy}(-3, 2, 0)$?

3. Do the coordinates of a point depend on the choice of direction on the coordinate axes?
4. How will the coordinates of a point change if the length of the basis vectors is changed?
5. What coordinates has the centre of a parallelepiped if the origin coincides with one of its vertices and the basis vectors coincide with its edges?

24. Other coordinate systems

Coordinate systems used in mathematics allow us to give with the aid of numbers the position of any point in space, in the plane or on a straight line. This makes it possible to carry out any calculations with coordinates and, what is very important, to apply modern computers not only to all sorts of numerical computations but also to the solution of geometrical problems and to the investigation of any geometrical objects and relations. Besides the affine coordinate systems considered above some others are not infrequently used.

The polar coordinate system. Choose in the plane some straight line and fix on it a Cartesian system. Call the origin O of the system the *pole* and the coordinate axis the *polar axis*. Assume further that the unit segment of the coordinate system on the straight line is used to measure the lengths of any line segments in the plane. Consider an arbitrary point M in the plane. It is obvious that its position will be completely defined if we specify the distance ρ between the points M and O and the angle φ through which it is necessary to turn the ray Ox *counterclockwise* about the point O until its direction coincides with that of the line segment $\vec{OM}$ (Fig. 24.1).

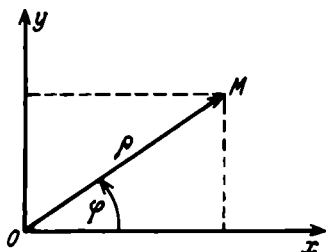


Fig. 24.1

The *polar coordinates* of the point M in the plane are the two numbers ρ and φ . The number ρ is the *polar radius* and the number φ is the *polar angle*. It is usually assumed that

$$0 \leq \rho < +\infty, \quad 0 \leq \varphi < 2\pi. \quad (24.1)$$

If the point M coincides with the pole O , then the polar angle is considered to be *undefined*.

Associated in a natural way with every polar coordinate system is a rectangular Cartesian system. In this the origin coincides with the pole, the axis of abscissas coincides with the polar axis and the axis of ordinates is obtained by rotating the polar axis through an angle of $\pi/2$ about O .

Denote the coordinates of the point M in the Cartesian x, y system by α and β . We have the obvious formulas

$$\alpha = \rho \cos \varphi, \quad \beta = \rho \sin \varphi.$$

From these we obtain the inverse relations

$$\rho^2 = \alpha^2 + \beta^2, \quad \cos \varphi = \frac{\alpha}{+(\alpha^2 + \beta^2)^{1/2}}, \quad \sin \varphi = \frac{\beta}{+(\alpha^2 + \beta^2)^{1/2}}.$$

They allow us to calculate from the Cartesian coordinates of a point its polar coordinates and vice versa.

Cylindrical coordinates. Choose in space some plane π and fix on it a polar coordinate system. Through the pole O draw the z axis

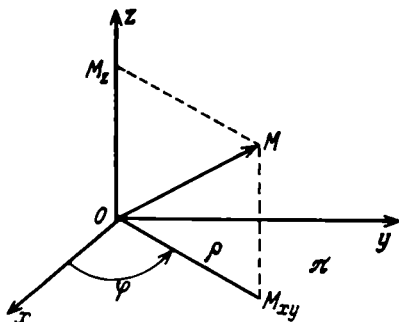


Fig. 24.2

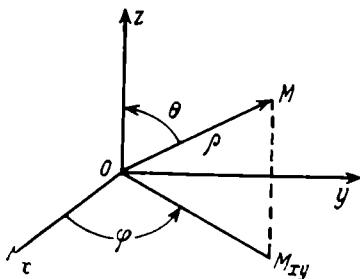


Fig. 24.3

perpendicular to π (Fig. 24.2). Assume again that to measure the lengths of any line segments in space we use the same unit segment. Introduce in π a Cartesian system corresponding to the polar system. Together with the z axis it forms a Cartesian system in space.

Consider the projections M_z and M_{xy} of the point M onto the z axis and x, y plane. The point M_{xy} as a point of π has polar coordinates ρ and φ . The point M_z as a point of the z axis has a z coordinate.

The *cylindrical coordinates* of the point M in space are the three numbers ρ , φ and z . It is again assumed that

$$0 \leq \rho < +\infty, \quad 0 \leq \varphi < 2\pi.$$

For the points of the z axis the angle φ is *not defined*.

Cartesian coordinates in the x, y, z system and cylindrical coordinates are connected by the relations

$$x = \rho \cos \varphi, \quad y = \rho \sin \varphi, \quad z = z.$$

Spherical coordinates. Consider in space a Cartesian x, y, z system and the corresponding polar coordinate system in the x, y plane (Fig. 24.3). Let M be any point in space other than O , and let M_{xy}

be its projection onto the x, y plane. Denote by ρ the distance from M to O , by θ the angle between the vector $\vec{OM}$ and the basis vector of the z axis, and finally by φ the polar angle of the projection M_{xy} .

The *spherical coordinates* of the point M in space are the three numbers ρ , φ and θ . The number ρ is a *radius vector*, the number φ is a *longitude*, and the number θ is a *colatitude*. It is assumed that

$$0 \leq \rho < +\infty, \quad 0 \leq \varphi < 2\pi, \quad 0 \leq \theta \leq \pi.$$

The longitude is *undefined* for all the points of the z axis, and the colatitude is *not defined* for the point O .

Cartesian coordinates in the x, y, z system and spherical coordinates are connected by the relations

$$x = \rho \sin \theta \cos \varphi, \quad y = \rho \sin \theta \sin \varphi, \quad z = \rho \cos \theta.$$

Exercises

1. Construct a curve whose points in polar coordinates satisfy the relation $\rho = \cos 3\varphi$.

2. Construct a curve whose points in cylindrical coordinates satisfy the relations $\rho = \varphi^{-1}$ and $z = \varphi$.

3. Construct a surface whose points in spherical coordinates satisfy the relations $0 \leq \rho \leq 1$, $\varphi = \pi/2$ and $0 \leq \theta \leq \pi/2$.

25. Some problems

Consider several simple problems in applying Cartesian systems. For definiteness we shall consider problems in space. Similar problems in the plane differ from those in space only in minor details. It will be assumed throughout that some coordinate system is fixed whose origin is a point O and whose basis vectors are i, j and k .

The coordinates of a vector. Let $M_1(\alpha_1, \beta_1, \gamma_1)$ and $M_2(\alpha_2, \beta_2, \gamma_2)$ be two points in space. They determine a vector $\vec{M_1M_2}$ which has some coordinates relative to the basis i, j and k . We establish the relation between the coordinates of $\vec{M_1M_2}$ and those of the points M_1 and M_2 . We have

$$\vec{M_1M_2} = \vec{OM_2} - \vec{OM_1}.$$

Further, by the definition of the affine coordinates of M_1 and M_2

$$\vec{OM_1} = \alpha_1 i + \beta_1 j + \gamma_1 k, \quad \vec{OM_2} = \alpha_2 i + \beta_2 j + \gamma_2 k.$$

Therefore it follows that

$$\vec{M_1M_2} = (\alpha_2 - \alpha_1) i + (\beta_2 - \beta_1) j + (\gamma_2 - \gamma_1) k$$

or according to the accepted notation

$$\overrightarrow{M_1 M_2} = (\alpha_2 - \alpha_1, \beta_2 - \beta_1, \gamma_2 - \gamma_1). \quad (25.1)$$

Coordinate projections of a vector. Again consider the directed line segment $\overrightarrow{M_1 M_2}$ in space. On projecting the points M_1 and M_2 onto the same coordinate plane or the same coordinate axis we obtain a new directed line segment. It is called a *coordinate projection of the vector* $\overrightarrow{M_1 M_2}$.

Every vector in space has six coordinate projections—three projections onto the coordinate axes and three projections onto the coordinate planes. It is easy to find the coordinate projections in the basis i, j and k from the coordinate points $M_1 (\alpha_1, \beta_1, \gamma_1)$ and $M_2 (\alpha_2, \beta_2, \gamma_2)$. To do this it suffices to use formulas (23.8) and (25.1).

For example, let us find the coordinates of the projection $\overrightarrow{M_{1x} M_{2x}}$. Considering that the points M_{1x} and M_{2x} have the coordinates

$$M_{1x} (\alpha_1, 0, 0), \quad M_{2x} (\alpha_2, 0, 0),$$

we find that

$$\overrightarrow{M_{1x} M_{2x}} = (\alpha_2 - \alpha_1, 0, 0). \quad (25.2)$$

Similarly

$$\overrightarrow{M_{1xz} M_{2xz}} = (\alpha_2 - \alpha_1, 0, \gamma_2 - \gamma_1)$$

and so on for all the remaining projections.

Comparing the first of the formulas (23.4) with formulas of the type (25.2) we conclude that

$$\{\overrightarrow{M_{1x} M_{2x}}\} = \alpha_2 - \alpha_1, \quad \{\overrightarrow{M_{1y} M_{2y}}\} = \beta_2 - \beta_1, \quad \{\overrightarrow{M_{1z} M_{2z}}\} = \gamma_2 - \gamma_1.$$

Therefore the magnitudes of the projections of a vector onto the coordinate axes coincide with the coordinates of that vector.

The second of the formulas (23.4) allows us to calculate the lengths of the projections of $\overrightarrow{M_1 M_2}$ onto the coordinate axes from the coordinates of M_1 and M_2 . Namely,

$$|\overrightarrow{M_{1x} M_{2x}}| = |\alpha_2 - \alpha_1|, \quad |\overrightarrow{M_{1y} M_{2y}}| = |\beta_2 - \beta_1|, \\ |\overrightarrow{M_{1z} M_{2z}}| = |\gamma_2 - \gamma_1|.$$

The length of a vector. We establish a formula for the length of a vector in space. It is obvious that the length $|\overrightarrow{M_1 M_2}|$ of $\overrightarrow{M_1 M_2}$ equals the distance $\rho (M_1, M_2)$ between M_1 and M_2 and the length

of the diagonal of the rectangular parallelepiped whose faces are parallel to the coordinate planes and pass through M_1 and M_2 (Fig. 25.1). The length of any edge of the parallelepiped equals that of the projection of the vector $\overrightarrow{M_1 M_2}$ onto the coordinate axis parallel

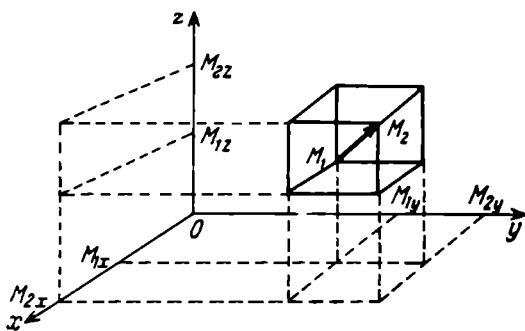


Fig. 25.1

to the edge. Using therefore the Pythagorean theorem we conclude that

$$|\overrightarrow{M_1 M_2}| = (|\overrightarrow{M_{1x} M_{2x}}|^2 + |\overrightarrow{M_{1y} M_{2y}}|^2 + |\overrightarrow{M_{1z} M_{2z}}|^2)^{1/2}.$$

If M_1 and M_2 are now given by their coordinates $M_1(\alpha_1, \beta_1, \gamma_1)$ and $M_2(\alpha_2, \beta_2, \gamma_2)$, then

$$\rho(M_1, M_2) = ((\alpha_2 - \alpha_1)^2 + (\beta_2 - \beta_1)^2 + (\gamma_2 - \gamma_1)^2)^{1/2}. \quad (25.3)$$

If the vector $\overrightarrow{M_1 M_2}$ is given by its x , y and z coordinates relative to the basis i , j and k , then

$$|\overrightarrow{M_1 M_2}| = (x^2 + y^2 + z^2)^{1/2}. \quad (25.4)$$

Of similar form are the corresponding formulas for the plane. If the points $M_1(\alpha_1, \beta_1)$ and $M_2(\alpha_2, \beta_2)$ or the vector $\overrightarrow{M_1 M_2} = (x, y)$ are given by their coordinates, then

$$\rho(M_1, M_2) = ((\alpha_2 - \alpha_1)^2 + (\beta_2 - \beta_1)^2)^{1/2}, \quad |\overrightarrow{M_1 M_2}| = (x^2 + y^2)^{1/2}.$$

The angle between vectors. Consider in space nonzero vectors a and b . Apply them to a point O . Denote by π the plane passing through O and containing both vectors. The *angle between a and b* is the *smallest* angle through which one of the vectors must be turned about O in the plane π for its direction to coincide with that of the other vector. If at least one of the vectors is zero, then the angle is

undefined. Our task is to calculate the cosine of the angle between the vectors from the coordinates of the vectors. We denote a cosine by $\cos \{a, b\}$.

Denote by A and B the terminal points of a and b in π . It is obvious that the angle between a and b is nothing than the angle AOB of the triangle AOB whose sides are the vectors a and b and $b - a$ (Fig. 25.2).

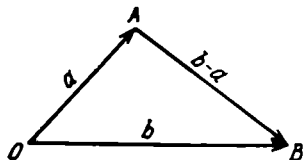


Fig. 25.2

Suppose a and b are given by their coordinates

$$a = (x_1, y_1, z_1), \quad b = (x_2, y_2, z_2).$$

Then

$$b - a = (x_2 - x_1, y_2 - y_1, z_2 - z_1).$$

As is known from elementary geometry, the square of the length of one side of a triangle is equal to the sum of the squares of the lengths of its other two sides minus the double product of the lengths of those sides by the cosine of the angle between them. Therefore

$$|b - a|^2 = |a|^2 + |b|^2 - 2|a| \cdot |b| \cos \{a, b\}$$

or, taking into account formula (25.4),

$$\begin{aligned} (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 &= x_1^2 + y_1^2 + z_1^2 \\ &+ x_2^2 + y_2^2 + z_2^2 - 2(x_1^2 + y_1^2 + z_1^2)^{1/2} (x_2^2 + y_2^2 + z_2^2)^{1/2} \cos \{a, b\}. \end{aligned}$$

Performing elementary transformations we find

$$\cos \{a, b\} = \frac{x_1 x_2 + y_1 y_2 + z_1 z_2}{(x_1^2 + y_1^2 + z_1^2)^{1/2} (x_2^2 + y_2^2 + z_2^2)^{1/2}}. \quad (25.5)$$

Changes in the formula for the plane are obvious.

Dividing a line segment in a given ratio. Suppose a straight line is given in space and let M_1 and M_2 be two distinct points on that straight line. Choose a positive direction on the straight line. On the resulting axis M_1 and M_2 determine a directed line segment $\overrightarrow{M_1 M_2}$. Let M be any point on the axis other than M_2 . The number

$$\lambda = \frac{\overrightarrow{\{M_1 M\}}}{\overrightarrow{\{M M_2\}}} \quad (25.6)$$

is called the *ratio in which the point M divides the directed line segment $\overrightarrow{M_1 M_2}$* .

Changing the direction on the axis makes the numbers $\overrightarrow{\{M_1 M\}}$ and $\overrightarrow{\{M M_2\}}$ change sign simultaneously. Hence ratio (25.6) is in-

dependent of the choice of positive direction on the axis. Also when changing the scale of measuring the length of line segments on the axis the numbers $\overrightarrow{M_1M}$ and $\overrightarrow{MM_2}$ are multiplied by the same number. Hence ratio (25.6) is independent of the choice of unit length. It follows that ratio (25.6) is independent of the choice of coordinate system on the axis.

The problem is to calculate the coordinates of the point M dividing

$\overrightarrow{M_1M_2}$ in a ratio λ given the coordinates of the points M_1 and M_2 and the number λ , with $\lambda \neq -1$. So let $M_1(\alpha_1, \beta_1, \gamma_1)$ and $M_2(\alpha_2, \beta_2, \gamma_2)$ be given and let $M(\alpha, \beta, \gamma)$ be unknown. Project these points onto the coordinate axes, for example the

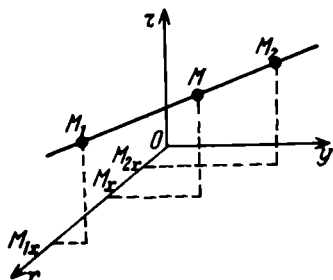


Fig. 25.3

x axis (Fig. 25.3). It is clear from similarity considerations that the point M_x also divides the directed line segment $\overrightarrow{M_{1x}M_{2x}}$ in a ratio λ . Therefore

$$\lambda = \frac{\overrightarrow{M_{1x}M_x}}{\overrightarrow{M_xM_{2x}}}. \quad (25.7)$$

By formula (23.4) $\{\overrightarrow{M_{1x}M_x}\} = \alpha - \alpha_1$ and $\{\overrightarrow{M_xM_{2x}}\} = \alpha_2 - \alpha$. Now taking into account (25.7), we find that $\alpha = (\alpha_1 + \lambda\alpha_2)/(1 + \lambda)$. The calculation of the coordinates β and γ is similar. So

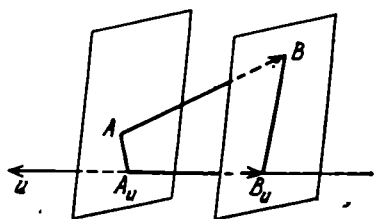
$$\alpha = \frac{\alpha_1 + \lambda\alpha_2}{1 + \lambda}, \quad \beta = \frac{\beta_1 + \lambda\beta_2}{1 + \lambda}, \quad \gamma = \frac{\gamma_1 + \lambda\gamma_2}{1 + \lambda}.$$

Notice that $\lambda \geq 0$ if M is inside the line segment $\overrightarrow{M_1M_2}$, $\lambda < 0$ if M is outside $\overrightarrow{M_1M_2}$, and $\lambda = 0$ if M coincides with M_1 . If M moves from M_1 to M_2 (excluding the coincidence with M_2), the ratio λ first takes on a zero value and then all possible positive values successively in increasing order. If M moves from M_1 in the positive direction of the axis (see Fig. 25.3), then λ first assumes a zero value and then negative values in decreasing order approaching arbitrarily closely $\lambda = -1$ but always remaining greater than $\lambda = -1$. If M moves in the negative direction from M_2 , then λ takes on all possible negative values in increasing order but always remains less than $\lambda = -1$.

Thus a 1-1 correspondence could be established between all real numbers and the points on the straight line if the straight line contained a point M dividing the line segment $\overrightarrow{M_1M_2}$ in the ratio

$\lambda = -1$ and the point M coinciding with M_2 could be assigned some number. This question is usually solved by joining to the straight line an ideal extra "point" and by joining to the number an ideal extra "number". Such a point is called a "point at infinity" and such a number is called an "infinitely large" number.

Orthogonal projections of a vector. Let u be some axis in space and let $\overrightarrow{AB}$ be a directed line segment. Draw through the points A and B the planes perpendicular to u (Fig. 25.4). The intersection of the planes with the axis determines the points A_u and B_u , with A_u lying in the same plane as A and B_u in the same plane as B .



The directed line segment $\overrightarrow{A_u B_u}$ is called the *orthogonal projection* of $\overrightarrow{AB}$ onto the axis u . The following notation is used to denote it:

$$\overrightarrow{A_u B_u} = \text{pr}_u \overrightarrow{AB}.$$

Fig. 25.4

For a fixed axis u every vector x in space uniquely determines its orthogonal projection x' . It may be assumed therefore that some "function"

$$x' = \text{pr}_u x \quad (25.8)$$

is given whose "independent variable" may be any vector in space and whose "value" is the vector on the axis u . We shall now prove that that function possesses the following properties:

$$\begin{aligned} \text{pr}_u (x + y) &= \text{pr}_u x + \text{pr}_u y, \\ \text{pr}_u (\lambda x) &= \lambda \text{pr}_u x, \end{aligned} \quad (25.9)$$

true for any vectors x and y and any number λ .

Indeed, fix some Cartesian system in which the axis u coincides with the axis of abscissas. In that system let

$$\begin{aligned} x &= (\alpha_1, \beta_1, \gamma_1), \\ y &= (\alpha_2, \beta_2, \gamma_2), \end{aligned}$$

then

$$\begin{aligned} x + y &= (\alpha_1 + \alpha_2, \beta_1 + \beta_2, \gamma_1 + \gamma_2), \\ \lambda x &= (\lambda \alpha_1, \lambda \beta_1, \lambda \gamma_1). \end{aligned}$$

In the chosen coordinate system the orthogonal projection of a vector onto the axis u coincides with its coordinate projection onto the axis of abscissas. As already noted earlier, the first coordinate of the projection of any vector onto the axis of abscissas coincides

with the first coordinate of the vector, the other coordinates equalling zero. Therefore

$$\begin{aligned}\text{pr}_u(x + y) &= (\alpha_1 + \alpha_2, 0, 0), \\ \text{pr}_u(\lambda x) &= (\lambda\alpha_1, 0, 0), \\ \text{pr}_u x &= (\alpha_1, 0, 0), \\ \text{pr}_u y &= (\alpha_2, 0, 0).\end{aligned}\tag{25.10}$$

According to the rules of vector addition and of multiplication of vectors by a number, it follows from the last two equations of (25.10) that

$$\begin{aligned}\text{pr}_u x + \text{pr}_u y &= (\alpha_1 + \alpha_2, 0, 0), \\ \lambda \text{pr}_u x &= (\lambda\alpha_1, 0, 0).\end{aligned}$$

Comparing the right-hand sides of these with the right-hand sides of the first two equations of (25.10) we see that both properties of (25.9) are true.

Now let π be some plane in space and let $\overrightarrow{AB}$ be a directed line segment. On dropping from the points A and B perpendiculars to π we obtain in π two points A_π and B_π which determine a directed line segment $\overrightarrow{A_\pi B_\pi}$. This is called the *orthogonal projection of $\overrightarrow{AB}$ onto π* . The same notation is used to denote it, i.e.

$$\overrightarrow{A_\pi B_\pi} = \text{pr}_\pi \overrightarrow{AB}.$$

Of course, for orthogonal projections onto the same plane we have relations similar to (25.9). To prove this we may fix some Cartesian system where π is a coordinate plane and use again the corresponding properties of projections onto a coordinate plane.

We have discussed orthogonal projections of vectors in space. Undoubtedly a close analogy holds for vectors in the plane.

Exercises

1. Two nonzero vectors are given by their Cartesian coordinates. When are they perpendicular?

2. Find the coordinates of the centre of gravity of three particles, given their Cartesian coordinates and masses.

3. Find the area of a triangle, given the Cartesian coordinates of its three vertices.

4. Let x , a , b and c be nonzero vectors in space, with a , b and c mutually perpendicular. Prove that

$$\cos^2\{x, a\} + \cos^2\{x, b\} + \cos^2\{x, c\} = 1.$$

5. Denote by π any coordinate plane, and by u denote any coordinate axis in π . Prove that for any vector x

$$\text{pr}_u(\text{pr}_\pi x) = \text{pr}_u x.$$

26. Scalar product

The use of directed line segments to represent forces and displacements leads to a very important notion of a scalar product of vectors.

It is known from physics that if a vector a represents a force whose point of application moves from the initial point of a vector b to its terminal point, then the work ω of such a force is defined by the equation

$$\omega = |a| |b| \cos \{a, b\}. \quad (26.1)$$

The right-hand side of the equation is called the *scalar product* of a and b and is generally designated (a, b) . So

$$(a, b) = |a| |b| \cos \{a, b\}. \quad (26.2)$$

Strictly speaking, this definition of a scalar product refers only to nonzero vectors a and b , since it is only for such vectors that an angle is defined. Taking into account the above interpretation of the scalar product, however, it is easy to see that it must be defined when at least one of the vectors is zero. If either a force or a displacement is given by a zero vector, then the work done equals zero. It will be assumed therefore that $(a, b) = 0$ if at least one of the vectors a and b is zero.

Formula (26.2) yields some geometrical properties of a scalar product. For example, the angle between two nonzero vectors is acute (obtuse) if and only if the scalar product of the vectors is positive (negative).

If the angle between vectors on a straight line or at least one of the vectors is zero, then the scalar product of the vectors equals zero. Such vectors are called *orthogonal*.

Orthogonal vectors of unit length are called *orthonormal vectors*. In particular, orthonormal vectors are the basis vectors i, j and k of a Cartesian system. It follows from formula (26.2) that

$$\begin{aligned} (i, i) &= 1, & (i, j) &= 0, & (i, k) &= 0, \\ (j, i) &= 0, & (j, j) &= 1, & (j, k) &= 0, \\ (k, i) &= 0, & (k, j) &= 0, & (k, k) &= 1. \end{aligned} \quad (26.3)$$

Consider nonzero vectors a and b . Draw through a an axis u , specifying on it a direction such that the magnitude of a is positive. It is then obvious that

$$\{\text{pr}_u b\} = |b| \cos \{a, b\}.$$

The projection of b onto the axis, which is thus constructed, is called the *projection of the vector b onto the vector a* and designated $\text{pr}_a b$. Of course, the projection of one vector onto the other preserves prop-

erties (25.9). In new symbols

$$(a, b) = |a| \{ \text{pr}_a b \} = |b| \{ \text{pr}_b a \}. \quad (26.4)$$

These formulas establish very important algebraic properties of a scalar product. Namely, for any vectors a , b and c and any real number α

$$\begin{aligned} (1) \quad & (a, b) = (b, a), \\ (2) \quad & (\alpha a, b) = \alpha (a, b), \\ (3) \quad & (a + b, c) = (a, c) + (b, c), \\ (4) \quad & (a, a) > 0 \text{ for } a \neq 0; \quad (0, 0) = 0. \end{aligned} \quad (26.5)$$

Notice that relations (26.5) clearly hold if at least one of the vectors is zero. In the general case Properties 1 and 4 follow immediately from (26.2). To establish Properties 2 and 3 we use formulas (26.4) and the properties of projections. We have

$$\begin{aligned} (\alpha a, b) &= |b| \{ \text{pr}_b (\alpha a) \} = |b| \{ \alpha \cdot \text{pr}_b a \} \\ &= \alpha |b| \{ \text{pr}_b a \} = \alpha (a, b), \\ (a + b, c) &= |c| \{ \text{pr}_c (a + b) \} = |c| \{ \text{pr}_c a + \text{pr}_c b \} \\ &= |c| \{ \text{pr}_c a \} + |c| \{ \text{pr}_c b \} = (a, c) + (b, c). \end{aligned}$$

Properties 2 and 3 are associated with the first factor of a scalar product only. Similar properties hold for the second factor. Indeed,

$$(a, \alpha b) = (\alpha b, a) = \alpha (b, a) = \alpha (a, b),$$

$$(a, b + c) = (b + c, a) = (b, a) + (c, a) = (a, b) + (a, c).$$

In addition, by virtue of the equation $a - b = a + (-1)b$

$$\begin{aligned} (a - b, c) &= (a, c) - (b, c), \\ (a, b - c) &= (a, b) - (a, c), \end{aligned}$$

since

$$\begin{aligned} (a - b, c) &= (a + (-1)b, c) = (a, c) + ((-1)b, c) \\ &= (a, c) + (-1)(b, c) = (a, c) - (b, c). \end{aligned}$$

Theorem 26.1. *If two vectors a and b are given by their Cartesian coordinates, then the scalar product of those vectors is equal to the sum of pairwise products of the corresponding coordinates.*

Proof. Suppose for definiteness that the vectors are given in space, i.e. $a = (x_1, y_1, z_1)$ and $b = (x_2, y_2, z_2)$. Since

$$\begin{aligned} a &= x_1 i + y_1 j + z_1 k, \\ b &= x_2 i + y_2 j + z_2 k, \end{aligned}$$

carrying out algebraic transformations of the scalar product we find

$$\begin{aligned}(a, b) = & x_1x_2(i, i) + x_1y_2(i, j) + x_1z_2(i, k) + y_1x_2(j, i) \\ & + y_1y_2(j, j) + y_1z_2(j, k) + z_1x_2(k, i) \\ & + z_1y_2(k, j) + z_1z_2(k, k).\end{aligned}$$

Now by (26.3) we have

$$(a, b) = x_1x_2 + y_1y_2 + z_1z_2 \quad (26.6)$$

and the theorem is proved.

Formula (26.6) allows expressions (25.4) and (25.5) obtained earlier for the length of a vector and the angle between vectors to be written in terms of a scalar product.

Namely,

$$\begin{aligned}|a| &= (a, a)^{1/2}, \\ \cos \{a, b\} &= \frac{(a, b)}{|a| \cdot |b|}.\end{aligned} \quad (26.7)$$

It may seem that these formulas are trivial, since they follow immediately from (26.2), without any reference to formulas (25.4) and (25.5). We shall not jump at a conclusion but note a very important fact.

Notice that as a matter of fact our investigation proceeded in three steps. Using (26.2) we first proved that properties (26.5) are true. Then relying only on these properties and the orthonormality of the basis vectors of a coordinate system we established formula (26.6). And finally we obtained formulas (26.7) by using formulas (25.4) and (25.5) which were derived without using the concept of a scalar product of vectors.

Starting from this we could now introduce the scalar product not by giving its explicit form but *axiomatically*, as some numerical function defined for every pair of vectors, requiring that properties (26.5) should necessarily hold. Relation (26.6) would then hold for any coordinate systems where the basis vectors are orthonormal in the sense of the axiomatic scalar product. Consequently, bearing in mind the model of a Cartesian system we could *axiomatically* assume that the lengths of vectors and the angles between them are calculated from formulas (26.7). It would be necessary of course to show that the lengths and angles introduced in this way possess the necessary properties.

Exercises

1. Given two vectors a and b , under what conditions on the number α are the vectors a and $b + \alpha a$ orthogonal? What is the geometrical interpretation of the problem?

2. Let a be a vector given in V_3 by its Cartesian coordinates. Find two linearly independent vectors orthogonal to a .
3. Let a and b be linearly independent vectors given in V_3 by their Cartesian coordinates. Find a nonzero vector orthogonal to both vectors.
4. What is the locus of vectors orthogonal to a given vector?

27. Euclidean space

Abstract vector spaces studied earlier are in a sense poorer in concepts and properties than spaces of directed line segments, first of all because they do not reflect the most important facts associated with measurement of lengths, angles, areas, volumes and so on. Metric notions can be extended to abstract vector spaces in different ways. The most efficient method of specifying measurements is through *axiomatic* introduction of a scalar product of vectors. We shall begin our studies with real vector spaces.

A real vector space E is said to be *Euclidean* if every pair of vectors x and y in E is assigned a real number (x, y) called a *scalar product*, with the following axioms holding:

- (1) $(x, y) = (y, x)$,
 - (2) $(\lambda x, y) = \lambda (x, y)$,
 - (3) $(x + y, z) = (x, z) + (y, z)$,
 - (4) $(x, x) > 0$ for $x \neq 0$; $(0, 0) = 0$
- (27.1)

for arbitrary vectors x, y and z in E and an arbitrary real number λ .

As is already known, it follows from these axioms that we can carry out formal algebraic transformations using a scalar product, i.e.

$$\left(\sum_{i=1}^r \alpha_i x_i, \sum_{j=1}^s \beta_j y_j \right) = \sum_{i=1}^r \sum_{j=1}^s \alpha_i \beta_j (x_i, y_j)$$

for any vectors x_i and y_j , any numbers α_i and β_j and any numbers r and s of summands.

Any linear subspace L of a Euclidean space E is a Euclidean space with the scalar product introduced in E .

It is easy to show a general method of introducing a scalar product in an arbitrary real space K . Let $e_1, e_2, \dots, e_n$ be some basis of K . Take two arbitrary vectors x and y in K and suppose that

$$x = \xi_1 e_1 + \xi_2 e_2 + \dots + \xi_n e_n,$$

$$y = \eta_1 e_1 + \eta_2 e_2 + \dots + \eta_n e_n.$$

A scalar product of vectors can now be introduced, for example, as follows:

$$(x, y) = \xi_1 \eta_1 + \xi_2 \eta_2 + \dots + \xi_n \eta_n. \quad (27.2)$$

It is not hard to check that all the axioms hold. Therefore the vector space K with scalar product (27.2) is Euclidean.

Notice that a scalar product can be introduced in K in other ways. For example, a scalar product in K is the following expression:

$$(x, y) = \alpha_1 \xi_1 \eta_1 + \alpha_2 \xi_2 \eta_2 + \dots + \alpha_n \xi_n \eta_n$$

for any fixed positive numbers $\alpha_1, \alpha_2, \dots, \alpha_n$. We should not be confused by this lack of uniqueness. For is there anything strange in the fact that lengths can be measured in metres or inches, angles can be measured in degrees or radians and so on? It is this lack of uniqueness that makes it possible to take the fullest account of the properties of particular spaces when introducing a scalar product in them.

Introducing a scalar product in spaces of directed line segments we had to define it separately when at least one of the segments was zero. The scalar product was assumed to be zero. Now this fact is a property arising from axioms (27.1). If x is an arbitrary vector in E , then

$$(0, x) = (0x, x) = 0 (x, x) = 0.$$

Of course, by the first axiom of (27.1) $(x, 0) = 0$.

A vector x of a Euclidean space is said to be *normed* if $(x, x) = 1$. Any *nonzero* vector y can be normed by multiplying it by some number λ . Indeed, under the hypothesis

$$(\lambda y, \lambda y) = \lambda^2 (y, y) = 1,$$

and therefore as a normalization factor we may take

$$\lambda = (y, y)^{-1/2}.$$

A system of vectors is said to be *normed* if all its vectors are normed. It follows from the foregoing that any system of nonzero vectors can be normed.

One of the most important properties of a scalar product is stated by the following

Theorem 27.1 (the Cauchy-Buniakowski-Schwarz inequality). For any two vectors x and y of a Euclidean space

$$(x, y)^2 \leq (x, x) (y, y).$$

Proof. The theorem is clearly true if $y = 0$, and therefore we assume that $y \neq 0$. Consider a vector $x - \lambda y$, where λ is an arbitrary real number. We have

$$(x - \lambda y, x - \lambda y) = (x, x) - 2\lambda (x, y) + \lambda^2 (y, y).$$

The left-hand side of the equation contains a scalar product of equal vectors. Therefore the second-degree trinomial at the right

is nonnegative for any λ , in particular for

$$\lambda = \frac{(x, y)}{(y, y)}. \quad (27.3)$$

Thus

$$(x, x) - 2 \frac{(x, y)}{(y, y)} (x, y) + \frac{(x, y)^2}{(y, y)^2} (y, y) = (x, x) - \frac{(x, y)^2}{(y, y)} \geq 0,$$

which proves the theorem.

As in the case of spaces of directed line segments two vectors x and y of any vector space are said to be *collinear* if either $x = \lambda y$ or $y = \mu x$ for some numbers λ and μ . From $0 = 0x$ we conclude that two vectors are obviously collinear if there is at least one zero vector among them. A very convenient means of testing vectors for collinearity is the Cauchy-Buniakowski-Schwarz inequality. Namely, we have

Theorem 27.2. *The Cauchy-Buniakowski-Schwarz inequality becomes an equation if and only if vectors x and y are collinear.*

Proof. Let x and y be collinear. Suppose for definiteness that $x = \lambda y$. We find

$$(x, y)^2 = (\lambda y, y)^2 = \lambda^2 (y, y)^2,$$

$$(x, x) (y, y) = (\lambda y, \lambda y) (y, y) = \lambda^2 (y, y)^2.$$

Comparing these equations shows that the sufficiency of the statement of the theorem holds.

Suppose now that for some vectors x and y

$$(x, y)^2 = (x, x) (y, y). \quad (27.4)$$

If $y = 0$, then the vectors are collinear. If $y \neq 0$, then taking λ according to (27.3) and considering (27.4) we get

$$(x - \lambda y, x - \lambda y) = 0.$$

By virtue of the last of the axioms (27.1) this means that $x - \lambda y = 0$ or $x = \lambda y$, i.e. that x and y are collinear. Necessity also holds.

As an example consider a space R_n . It can be made Euclidean if for the vectors

$$x = (\alpha_1, \alpha_2, \dots, \alpha_n),$$

$$y = (\beta_1, \beta_2, \dots, \beta_n)$$

a scalar product is introduced as follows:

$$(x, y) = \sum_{i=1}^n \alpha_i \beta_i. \quad (27.5)$$

It is obvious that axioms (27.1) hold. The Cauchy-Buniakowski-Schwarz inequality then means that

$$\left(\sum_{i=1}^n \alpha_i \beta_i\right)^2 \leq \left(\sum_{i=1}^n \alpha_i^2\right) \left(\sum_{i=1}^n \beta_i^2\right) \quad (27.6)$$

for any real numbers α_i and β_i .

Exercises

1. Introduce a scalar product in a space of polynomials with real coefficients in a single variable.

2. Will a space R_n be Euclidean if the scalar product is introduced in it as follows:

$$(x, y) = \sum_{i=1}^n |\alpha_i| |\beta_i|?$$

3. What is the geometrical meaning of the Cauchy-Buniakowski-Schwarz inequality in spaces of directed line segments?

4. Prove that $x = y$ if and only if $(x, d) = (y, d)$ for every vector d .

28. Orthogonality

The most important relation between the vectors of a Euclidean space is orthogonality.

By definition vectors x and y are said to be *orthogonal* if $(x, y) = 0$. By the first axiom of (27.1) the orthogonality relation of two vectors is symmetrical. In fact in a space of directed line segments the concept of orthogonality coincides with that of perpendicularity. Orthogonality may therefore be regarded as an extension of the notion of perpendicularity to abstract Euclidean spaces.

A system of vectors of a Euclidean space is said to be *orthogonal* if either it consists of a single vector or its vectors are mutually orthogonal. If an orthogonal system consists of nonzero vectors, then it can be normed. A normed orthogonal system is called *orthonormal*.

Interest in orthogonal and orthonormal systems is due to the advantages they offer in investigating Euclidean spaces.

Thus, for example, any orthogonal system of nonzero vectors and of course any orthonormal system is linearly independent. Indeed, let a system $x_1, x_2, \dots, x_k$ be orthogonal and let $x_i \neq 0$ for every i . This means that $(x_i, x_j) = 0$ for $i \neq j$, but that $(x_i, x_j) \neq 0$ for $i = j$. We write

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_k x_k = 0.$$

On performing a scalar multiplication of this equation by any vector x_l we find

$$\alpha_1 (x_l, x_1) + \alpha_2 (x_l, x_2) + \dots + \alpha_k (x_l, x_k) = 0.$$

Consequently,

$$\alpha_i (x_i, x_i) = 0 \quad (28.1)$$

and of course $\alpha_i = 0$. Thus the system of vectors $x_1, x_2, \dots, x_k$ is linearly independent.

From (28.1) we deduce, in particular, that if a sum of mutually orthogonal vectors is zero, then all the vectors are zero.

Especially many useful consequences arise from the assumption that some orthonormal system $e_1, e_2, \dots, e_s$ may form a basis of a Euclidean space E . In this case every vector x in E must be uniquely represented as a linear combination

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_s e_s.$$

But on performing a scalar multiplication of this equation by e_i we obtain an explicit expression for the coefficients of the expansion with respect to a basis. Namely,

$$\alpha_i = (x, e_i). \quad (28.2)$$

If for the other vector, y ,

$$y = \beta_1 e_1 + \beta_2 e_2 + \dots + \beta_s e_s,$$

then on carrying out simple transformations we find that

$$(x, y) = \alpha_1 \beta_1 + \alpha_2 \beta_2 + \dots + \alpha_s \beta_s. \quad (28.3)$$

In particular,

$$(x, x) = \alpha_1^2 + \alpha_2^2 + \dots + \alpha_s^2. \quad (28.4)$$

Before going on with such studies we shall see if there is a basis consisting of orthonormal vectors.

A basis whose vectors form an orthonormal system is called *orthonormal*. The existence of such a basis in a Euclidean space is proved by

Theorem 28.1. *There is an orthonormal basis in any finite dimensional Euclidean space E .*

Proof. Let $\dim E = n$. An orthonormal system is linearly independent and therefore it cannot contain more than n vectors. Suppose a system $e_1, e_2, \dots, e_s$ contains a maximum number of orthonormal vectors. This means that in E there is no nonzero vector orthogonal to all vectors $e_1, e_2, \dots, e_s$. If some vector is orthogonal to these, then it must be zero.

Take an arbitrary vector x in E . If the orthonormal system $e_1, e_2, \dots, e_s$ were a basis, then the vector x would have to coincide with a vector y , where

$$y = (x, e_1) e_1 + (x, e_2) e_2 + \dots + (x, e_s) e_s.$$

Therefore consider the vector $x - y$. We have

$$\begin{aligned} (x - y, e_i) &= (x - \sum_{p=1}^s (x, e_p) e_p, e_i) \\ &= (x, e_i) - \sum_{p=1}^s (x, e_p) (e_p, e_i) = (x, e_i) - (x, e_i) = 0. \end{aligned}$$

The vector $x - y$ turns out to be orthogonal to all vectors $e_1, e_2, \dots, e_s$. Consequently, $x - y = 0$ or $x = y$.

So the linearly independent system $e_1, e_2, \dots, e_s$ possesses the property that any vector of E is linearly expressible in terms of its vectors, i.e. it forms a basis.

Corollary. Any orthonormal system of vectors $e_1, e_2, \dots, e_k$ may be supplemented to an orthonormal basis.

Indeed, choose among the orthonormal systems containing a given system the one that has a maximum number of vectors. Let it be a system $e_1, \dots, e_k, e_{k+1}, \dots, e_s$. Repeating then word for word the proof of Theorem 28.1 we establish that the new system is a basis.

Besides orthogonal vectors in a Euclidean space we shall discuss *orthogonal sets of vectors*. Two sets F and G of vectors of E are said to be orthogonal if every vector in F is orthogonal to every vector in G . This is designated $F \perp G$.

Of course a set may consist of a single vector. If some vector of a set is orthogonal to the entire set, then it is, in particular, orthogonal to itself. Consequently, it may be only zero.

Lemma 28.1. For a vector x to be orthogonal to a subspace L it is necessary and sufficient that it should be orthogonal to each vector of some basis of L .

Proof. Fix a basis $y_1, y_2, \dots, y_k$ of a subspace L . If $x \perp L$, then x is orthogonal to every vector in L and in particular to $y_1, y_2, \dots, y_k$. Now let $(x, y_i) = 0$ for every i . Take an arbitrary vector z in L and expand it with respect to the basis vectors. If

$$z = \alpha_1 y_1 + \alpha_2 y_2 + \dots + \alpha_k y_k$$

for some numbers $\alpha_1, \alpha_2, \dots, \alpha_k$, then

$$\begin{aligned} (x, z) &= (x, \alpha_1 y_1 + \alpha_2 y_2 + \dots + \alpha_k y_k) \\ &= \alpha_1 (x, y_1) + \alpha_2 (x, y_2) + \dots + \alpha_k (x, y_k) = 0. \end{aligned}$$

This means that $x \perp L$.

Corollary. For two subspaces to be orthogonal it is necessary and sufficient that each vector of some basis of one of the subspaces should be orthogonal to each vector of some basis of the other.

The sum K of linear subspaces $L_1, L_2, \dots, L_m$ is said to be *orthogonal* if the subspaces are mutually orthogonal. To denote an orthogonal sum the following notation will be used:

$$K = L_1 \oplus L_2 \oplus \dots \oplus L_m.$$

Lemma 28.2. *An orthogonal sum of nonzero subspaces is always a direct sum.*

Proof. Choose in each subspace an orthonormal basis and consider the system of vectors which is the union of bases of all subspaces. It is clear that each vector of the orthogonal system is linearly expressible in terms of the vectors of the system constructed. But this is linearly independent since it consists of nonzero mutually orthogonal vectors. Now the lemma follows from Theorem 20.1.

Let a Euclidean space K be represented as an orthogonal sum of its subspaces $L_1, L_2, \dots, L_m$. Then the collection of these subspaces may be regarded as a generalized orthogonal basis. In particular, if for any vectors x and y in K we write their expansions with respect to the subspaces $L_1, L_2, \dots, L_m$, i.e. represent them as

$$x = x_1 + x_2 + \dots + x_m,$$

$$y = y_1 + y_2 + \dots + y_m,$$

where $x_i, y_i \in L_i$, then it is easy to establish that

$$(x, y) = (x_1, y_1) + (x_2, y_2) + \dots + (x_m, y_m). \quad (28.5)$$

This formula is similar to formula (28.3).

Consider an arbitrary nonempty set F of vectors of a Euclidean space E . The collection of all vectors orthogonal to F is called the *orthogonal complement* of F and designated $F^\perp$. The orthogonal complement is a subspace. Indeed, if vectors $x, y \in F^\perp$, then $x, y \perp F$. But then $\alpha x + \beta y \perp F$ for any numbers α and β , i.e. $\alpha x + \beta y \in F^\perp$.

Theorem 28.2. *A Euclidean space E is the orthogonal sum of any of its linear subspaces, L , and its orthogonal complement $L^\perp$, i.e.*

$$E = L \oplus L^\perp.$$

Proof. Let $\dim L = s$, $\dim L^\perp = m$. Choose some orthonormal basis $e_1, \dots, e_s$ of a subspace L and some orthonormal basis $r_1, \dots, r_m$ of $L^\perp$. The system of vectors $e_1, \dots, e_s, r_1, \dots, r_m$ is orthonormal and hence linearly independent.

If the system is not a basis of E , then it can be supplemented to an orthonormal basis of E . Let e be one of the complementary vectors. It is orthogonal to the vectors $e_1, \dots, e_s$ and therefore $e \perp L$, i.e. $e \in L^\perp$. But, on the other hand, e is orthogonal to $r_1, \dots, r_m$ and therefore $e \perp L^\perp$. So e is both in $L^\perp$ and orthogonal to $L^\perp$. Consequently, $e = 0$, which proves the theorem.

Decomposing a space as an orthogonal sum of its subspaces allows many studies to be efficiently carried out. The following example will serve as an illustration.

Consider a Euclidean space E with some fixed system of vectors $x_1, x_2, \dots, x_k$. If the rank of that system equals the dimension of E , then it is obvious that the only vector in E orthogonal to all vectors of the given system is the zero vector. We have the converse

Lemma 28.3. *If in a Euclidean space E some system of vectors $x_1, x_2, \dots, x_k$ is given and the only vector in E orthogonal to those vectors is the zero vector, then the rank of the system equals the dimension of E .*

Proof. Denote by L the span of the system of vectors $x_1, x_2, \dots, x_k$. Any vector orthogonal to these vectors is orthogonal to L , i.e. is in the orthogonal complement $L^\perp$. According to the hypothesis of the lemma $L^\perp$ consists of the zero vector only. Since $E = L \oplus L^\perp$, it follows that the dimension of L coincides with that of E . But the dimension of L equals the rank of the system of vectors $x_1, x_2, \dots, x_k$. Thus the lemma is proved.

Exercises

1. Prove that if the scalar product of any two vectors of a Euclidean space is expressed by equation (28.3), then the basis relative to which the coordinates are taken is orthonormal.

2. Prove that if the scalar product of any vector of a Euclidean space by itself is expressed by equation (28.4), then the basis relative to which the coordinates are taken is orthonormal.

3. Prove that if two sets consisting of a finite number of vectors are orthogonal, then so are the spans constructed on those sets.

4. Prove that the intersection of two orthogonal subspaces consists but of a zero vector.

5. Prove that if a Euclidean space is the direct sum of its subspaces and (28.5) holds for any two vectors, then the subspaces are mutually orthogonal.

6. Prove that for any subspaces L and M of a Euclidean space E

$$\dim L + \dim L^\perp = \dim E,$$

$$(L^\perp)^\perp = L,$$

$$(L + M)^\perp = L^\perp \cap M^\perp,$$

$$(L \cap M)^\perp = L^\perp + M^\perp.$$

29. Lengths, angles, distances

We now extend to the elements of a Euclidean space such notions as length, angle and distance. We shall proceed from the analogy with spaces of directed line segments.

The length $|x|$ of a vector x of a Euclidean space E is

$$|x| = + (x, x)^{1/2}.$$

Every vector has length. According to the last axiom of (27.1) it is positive for nonzero vectors and equal to zero for a zero vector.

Also the equation

$$|\lambda x| = (\lambda x, \lambda x)^{1/2} = (\lambda^2 (x, x))^{1/2} = |\lambda| |x|$$

shows that it is possible to take the absolute value of the numerical factor λ outside the vector-length sign. As already noted a nonzero vector can be normed, i.e. multiplied by a number such that the length of the resulting vector is equal to unity.

The angle $\{x, y\}$ between nonzero vectors x and y of a Euclidean space E is the angle defined by the relations

$$\cos \{x, y\} = \frac{(x, y)}{|x| |y|}, \quad 0 \leq \{x, y\} \leq \pi.$$

If there is at least one zero vector among the vectors x and y , then the angle between such vectors is said to be *undefined*.

The Cauchy-Buniakowski-Schwarz inequality allows us to state that the expression which we have called the cosine of the angle between vectors does not exceed unity in absolute value. Therefore the angle between any nonzero vectors is always uniquely defined. It remains unchanged under multiplication of the vectors by any positive numbers and by Theorem 27.2 equals 0 or π if and only if the nonzero vectors are collinear. All this is in full agreement with the concept of the angle between directed line segments.

Take two nonzero vectors x and y . Bearing in mind the analogy with directed line segments we shall assume them to be two sides of some triangle. It is natural to take the vector $x - y$ as its third side. Using the definition of the length of a vector and of the angle between vectors we find

$$\begin{aligned} |x - y|^2 &= (x - y, x - y) = (x, x) - 2(x, y) + (y, y) \\ &= |x|^2 + |y|^2 - 2|x||y|\cos \{x, y\}. \end{aligned} \quad (29.1)$$

So we have shown that in a Euclidean space the square of the length of any side of a triangle is equal to the sum of the squares of the lengths of its two other sides minus the doubled product of the lengths of those sides by the cosine of the angle between them.

If it is a right triangle, i.e. if the angle between the vectors x and y is a right one, then obviously

$$|x - y|^2 = |x|^2 + |y|^2. \quad (29.2)$$

This is nothing than a formal expression of the well-known Pythagorean theorem.

Consider again an arbitrary triangle. Since the cosine of the angle between vectors does not exceed unity in absolute value, it follows from (29.1) that

$$\begin{aligned} |x - y|^2 &\leq (|x| + |y|)^2, \\ |x - y|^2 &\geq (|x| - |y|)^2 \end{aligned}$$

or

$$\begin{aligned} |x - y| &\leq |x| + |y|, \\ |x - y| &\geq ||x| - |y||. \end{aligned} \quad (29.3)$$

Thus in a Euclidean space the length of a triangle side does not exceed the sum of the lengths of the other two sides but is not less than the difference of their lengths.

The *distance* $\rho(x, y)$ between vectors x and y of a Euclidean space is the quantity

$$\rho(x, y) = |x - y|. \quad (29.4)$$

It satisfies the three natural properties of distances between vectors (in point interpretation!) in spaces of directed line segments. Namely, for any vectors x, y and z of a Euclidean space

$$\begin{aligned} (1) \quad &\rho(x, y) = \rho(y, x), \\ (2) \quad &\rho(x, y) > 0 \text{ if } x \neq y, \rho(x, y) = 0 \text{ if } x = y, \\ (3) \quad &\rho(x, y) \leq \rho(x, z) + \rho(z, y). \end{aligned} \quad (29.5)$$

The first two properties are obvious. The last one is nothing than a generalization of the well-known "triangle inequality". It follows from the first of the inequalities (29.3) if we replace x by $x - z$ and y by $y - z$.

The *distance* $\rho(A, B)$ between sets A and B of vectors of the same space is the quantity

$$\rho(A, B) = \inf_{x \in A, y \in B} \rho(x, y).$$

In conclusion note the following fact. Let $e_1, e_2, \dots, e_s$ be an orthonormal basis fixed in a Euclidean space E . For any two vectors x and y given by their coordinates

$$x = (\alpha_1, \alpha_2, \dots, \alpha_s), \quad y = (\beta_1, \beta_2, \dots, \beta_s)$$

relative to that basis we have by (28.3)

$$|x| = (\alpha_1^2 + \alpha_2^2 + \dots + \alpha_s^2)^{1/2}.$$

Consequently,

$$\cos \{x, y\} = \frac{\alpha_1 \beta_1 + \alpha_2 \beta_2 + \dots + \alpha_s \beta_s}{(\alpha_1^2 + \dots + \alpha_s^2)^{1/2} (\beta_1^2 + \dots + \beta_s^2)^{1/2}}.$$

Close analogy with formulas (25.4) and (25.5) is obvious.

Thus the concepts of length, angle and distance we have introduced fully agree with similar notions in spaces of directed line segments.

Exercises

1. Prove that the length of the sum of any number of vectors does not exceed the sum of the lengths of those vectors.
2. Prove that the square of the length of the sum of any number of orthogonal vectors is equal to the sum of the squares of the lengths of those vectors.
3. Given a Euclidean space of polynomials in a single variable t , find the angles of a triangle formed by the vectors 1 , t^2 and $1 - t^2$.
4. What is the distance between the polynomials $3t^2 + 6$ and $2t^3 + t + 1$?
5. Prove that a triangle in a Euclidean space is a right triangle if and only if the length of one of its sides is equal to the product of the length of another side of the cosine of the angle between them.

30. Inclined line, perpendicular, projection

Before extending to abstract Euclidean spaces the concepts of inclined line, perpendicular and projection we consider these notions in the space of directed line segments.

Let L be a plane. Drop to it from some point M a perpendicular and denote its foot by M_L (Fig. 30.1). To give this problem a vector interpretation choose on L a point O and consider a space V_3 of directed line segments fixed at O . The plane L forms a subspace. Therefore the construction of the perpendicular dropped from M to L reduces to decomposing the vector $\overrightarrow{OM}$ of the space as the sum

$$\overrightarrow{OM} = \overrightarrow{OM_L} + \overrightarrow{M_LM}, \quad (30.1)$$

where $\overrightarrow{OM_L} \in L$ and $\overrightarrow{M_LM} \perp L$. From geometrical considerations it is clear that decomposition (30.1) always exists and is unique.

This example suggests how to set the problem of a perpendicular in the general case. Suppose in a Euclidean space E some subspace L is fixed. Take an arbitrary vector f in E and study the possibility of decomposing it as a sum

$$f = g + h, \quad (30.2)$$

where $g \in L$ and $h \perp L$.

We have already encountered this problem. Indeed, the condition $h \perp L$ is equivalent to $h \in L^\perp$. By Theorem 28.2 a Euclidean space E is the direct sum of subspaces L and $L^\perp$. Therefore decomposition (30.2) always exists and is unique.

Bearing in mind the analogy with decomposition (30.1) the vector g in (30.2) will be called the *projection of the vector f onto L* , h will be

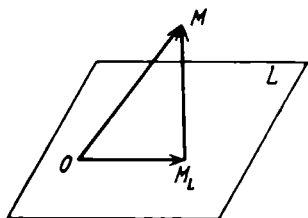


Fig. 30.1

called the *perpendicular from f to L* , and f will be called the *inclined line to L* .

In elementary geometry the length of a perpendicular is known never to exceed that of an inclined line. A similar situation occurs in a Euclidean space. The vectors g and h in decomposition (30.2) are orthogonal. By the Pythagorean theorem therefore

$$|f|^2 = |g|^2 + |h|^2,$$

and so

$$|h| \leq |f|.$$

It is clear that the length of the perpendicular h to a subspace L is equal to the length of an inclined line f to the same subspace if and only if $f \perp L$.

The problem of a perpendicular may be given another interpretation. Consider again an arbitrary vector f in E . It is not necessarily in L . Consequently, we may require to find in L a vector such that is closest to f in the sense of the earlier introduced distance.

Take an arbitrary vector z in L . Subtracting it from both sides of (30.2) we get

$$f - z = (g - z) + h.$$

Since h is orthogonal to $g - z$, by the Pythagorean theorem

$$|f - z|^2 = |g - z|^2 + |h|^2.$$

Therefore

$$|f - z| \geq |h|,$$

equality holding if and only if $z = g$.

So of all the vectors in L the projection of f onto L is closest to f . This means that

$$\rho(f, L) = \rho(f, g).$$

By analogy with directed line segments we can say that the *angle between f and L* is the smallest of the angles between f and the vectors z in L . Taking into account the Cauchy-Buniakowski-Schwarz inequality and decomposition (30.2) we find

$$\cos \{f, z\} = \frac{(f, z)}{|f| |z|} = \frac{(g + h, z)}{|f| |z|} = \frac{(g, z)}{|f| |z|} \leq \frac{|g|}{|f|}.$$

It is obvious that this inequality becomes an equation if and only if z makes a zero angle with g .

Thus the angle between f and L coincides with that between f and its projection onto L .

The above properties of a perpendicular and a projection reflect the geometrical aspect of these concepts. Now we shall consider them from an algebraic point of view. Given a fixed subspace L ,

each vector f of a Euclidean space E uniquely determines relative to L two of its components. Consequently, it may be assumed that decomposition (30.2) gives two functions

$$\begin{aligned} g &= \text{pr}_L f, \\ h &= \text{ort}_L f. \end{aligned}$$

The "independent variable" of a function may be any vector from E , the "value" of the function $\text{pr}_L f$ being a vector from L and the "value" of $\text{ort}_L f$ a vector from $L^\perp$.

By virtue of $(L^\perp)^\perp = L$ the perpendicular and the projection are related by the following equations:

$$\begin{aligned} \text{pr}_L f &= \text{ort}_{L^\perp} f, \\ \text{ort}_L f &= \text{pr}_{L^\perp} f. \end{aligned} \tag{30.3}$$

Therefore in fact the study of these functions always reduces to the study of one of them.

Take two arbitrary vectors x and y in E . According to (30.2)

$$\begin{aligned} x &= \text{pr}_L x + \text{ort}_L x, \\ y &= \text{pr}_L y + \text{ort}_L y. \end{aligned} \tag{30.4}$$

Adding termwise these equations and multiplying the first of them by an arbitrary real number λ yields

$$\begin{aligned} x + y &= (\text{pr}_L x + \text{pr}_L y) + (\text{ort}_L x + \text{ort}_L y), \\ \lambda x &= (\lambda \text{pr}_L x) + (\lambda \text{ort}_L x). \end{aligned}$$

A straightforward check shows that the vectors in the first parentheses are from L and those in the second parentheses are perpendicular to L . According to the uniqueness of decomposition of the type (30.2) this means that the relations

$$\begin{aligned} \text{pr}_L (x + y) &= \text{pr}_L x + \text{pr}_L y, \\ \text{pr}_L (\lambda x) &= \lambda \text{pr}_L x \end{aligned} \tag{30.5}$$

hold for the function pr_L and that of course the similar relations

$$\begin{aligned} \text{ort}_L (x + y) &= \text{ort}_L x + \text{ort}_L y, \\ \text{ort}_L (\lambda x) &= \lambda \text{ort}_L x \end{aligned} \tag{30.6}$$

hold for ort_L . Formulas (25.9) and (30.5) fully coincide.

Notice that $\text{ort}_L z = 0$ for any vector z in L . Therefore it follows from the first of the equations (30.6) that

$$\text{ort}_L (x + z) = \text{ort}_L x.$$

Consequently, the value of the function ort_L remains unchanged if we add to the independent variable any vector from L . In partic-

ular, if we take $z = -\text{pr}_L x$, then, taking into account (30.4),

$$\text{ort}_L (\text{ort}_L x) = \text{ort}_L x. \quad (30.7)$$

A similar relation holds for a projection. Namely,

$$\text{pr}_L (\text{pr}_L x) = \text{pr}_L x. \quad (30.8)$$

Now let a subspace L be the orthogonal sum of subspaces L_1 and L_2 . Take an arbitrary vector x in E and represent it as the sum

$$x = (\text{pr}_{L_1} x + \text{pr}_{L_2} x) + (x - \text{pr}_{L_1} x - \text{pr}_{L_2} x).$$

The vector in the first parentheses is obviously in the subspace $L_1 \oplus L_2$. The vector in the second parentheses is orthogonal to $L_1 \oplus L_2$, which is easy to show by transforming it with the aid of (30.4) as follows:

$$x - \text{pr}_{L_1} x - \text{pr}_{L_2} x = \text{ort}_L (x - \text{pr}_{L_1} x - \text{pr}_{L_2} x) = \text{ort}_L x - \text{pr}_{L_1} x. \quad (30.9)$$

Therefore we conclude that

$$\text{pr}_{L_1 \oplus L_2} x = \text{pr}_{L_1} x + \text{pr}_{L_2} x.$$

The perpendicular from x to $L_1 \oplus L_2$ is equal to one of the expressions (30.9). If in particular $x \perp L_1$, then

$$\text{ort}_{L_1 \oplus L_2} x = \text{ort}_{L_2} x. \quad (30.10)$$

Exercises

1. Does the analogue of the theorem on three perpendiculars hold in a Euclidean space?

2. Prove that the sum of two angles between a vector f and subspaces L and $L^\perp$ is equal to $\pi/2$.

3. Find the perpendicular and projection of a vector f onto trivial subspaces.

4. Prove that if for fixed subspaces L_1 and L_2 and any vector x

$$\text{pr}_{L_1 + L_2} x = \text{pr}_{L_1} x + \text{pr}_{L_2} x,$$

then the sum $L_1 + L_2$ is orthogonal.

5. Prove that if subspaces $L_1, L_2, \dots, L_m$ are mutually orthogonal, then for any vector x in E

$$|x|^2 \geq \sum_{i=1}^m |\text{pr}_{L_i} x|^2.$$

31. Euclidean isomorphism

Carrying out our studies we have repeatedly noted the coincidence of the properties of an abstract Euclidean space and spaces of directed line segments. We could carry over to Euclidean space the other facts and theorems of elementary geometry. But there is no need for this.

We introduce the concept of Euclidean isomorphism. We shall say that Euclidean spaces E and E' are *Euclidean isomorphic* if they are isomorphic as real vector spaces and if in addition for any pair of vectors x and y in E and the corresponding vectors x' and y' in E'

$$(x, y) = (x', y').$$

Theorem 31.1. *For two Euclidean spaces to be Euclidean isomorphic it is necessary and sufficient that they should be of equal dimension.*

Proof. If two Euclidean spaces E and E' are Euclidean isomorphic, then they are isomorphic as real vector spaces as well. But such vector spaces have the same dimension.

Consider now two Euclidean spaces E and E' of the same dimension n . Let $e_1, e_2, \dots, e_n$ be an orthonormal basis in E and let $e'_1, e'_2, \dots, e'_n$ be an orthonormal basis in E' . Assign to each vector

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n$$

of E a vector

$$x' = \alpha_1 e'_1 + \alpha_2 e'_2 + \dots + \alpha_n e'_n$$

of E' . This correspondence was proved earlier to be an isomorphism. Now take another pair of corresponding vectors in E and E'

$$y = \beta_1 e_1 + \beta_2 e_2 + \dots + \beta_n e_n,$$

$$y' = \beta_1 e'_1 + \beta_2 e'_2 + \dots + \beta_n e'_n.$$

By (28.3)

$$(x, y) = \alpha_1 \beta_1 + \alpha_2 \beta_2 + \dots + \alpha_n \beta_n = (x', y').$$

Thus the theorem is proved.

We are concerned throughout only with such properties of vector spaces that are consequences of the basic operations acting in spaces. From this point of view Euclidean isomorphic spaces have the same properties. Therefore any geometrical theorem proved in V_3 will be true in any three-dimensional subspace of a Euclidean space. Consequently, it will be also true in any Euclidean space. Of course, the arithmetical space R_n with a scalar product introduced according to (27.2) may serve as a standard Euclidean space.

Exercises

1. Construct a Euclidean isomorphism from V_2 to R_2 .
2. Prove that in Euclidean isomorphic spaces an orthonormal system of vectors goes over into an orthonormal system.
3. Prove that in Euclidean isomorphic spaces the angles between pairs of corresponding vectors are equal.
4. Prove that in Euclidean isomorphic spaces a perpendicular and a projection go over respectively into a perpendicular and a projection.

32. Unitary spaces

We have extended the basic metric concepts only to real vector spaces. Similar results hold for the complex vector space.

A complex vector space U is said to be *unitary* if every pair of vectors x and y in U is assigned a complex number (x, y) called a *scalar product*, with the following axioms holding:

$$(1) (x, y) = \overline{(y, x)},$$

$$(2) (\lambda x, y) = \lambda (x, y),$$

$$(3) (x + y, z) = (x, z) + (y, z),$$

$$(4) (x, x) > 0 \text{ for } x \neq 0; (0, 0) = 0$$

for arbitrary vectors x, y and z in U and an arbitrary complex number λ .

The bar in the first axiom signifies complex conjugation. This single distinction from the axioms of a Euclidean space involves no profound differences but we should not forget about it. Thus, while in a Euclidean space $(x, \lambda y) = \lambda (x, y)$, in a unitary space $(x, \lambda y) = \overline{\lambda} (x, y)$.

In a unitary space U we can introduce some metric concepts. The length of a vector, as in the real case, will be the quantity

$$|x| = + (x, x)^{1/2}.$$

Every nonzero vector has a positive length and the length of a zero vector is zero. For any complex λ

$$|\lambda x| = |\lambda| \cdot |x|.$$

The Cauchy-Buniakowski-Schwarz inequality

$$|(x, y)|^2 \leq (x, x)(y, y)$$

is also true. The proof is similar to that for the real case.

The concept of the angle between vectors is not introduced in a unitary space as a rule. Only the case where vectors x and y are orthogonal is considered. As in the real case it is understood that

$$(x, y) = 0.$$

It is obvious that $(y, x) = \overline{(x, y)} = 0$.

Essentially the entire theory of Euclidean spaces discussed above can be carried over without changes in the definitions and general schemes of the proofs to unitary spaces.

The arithmetical space C_n may serve as the standard unitary space if for the vectors

$$x = (\alpha_1, \alpha_2, \dots, \alpha_n),$$

$$y = (\beta_1, \beta_2, \dots, \beta_n)$$

their scalar product is introduced as follows:

$$(x, y) = \sum_{i=1}^n \alpha_i \bar{\beta}_i. \quad (32.1)$$

Using this space it is easy to show the part played by complex conjugation in the first axiom. If the scalar product were introduced in C_n according to formula (27.2), then in C_3 , for example, for the vector

$$x = (3, 4, 5i)$$

we would have

$$(x, x) = 9 + 16 + 25i^2 = 0.$$

The fourth axiom would be found to have failed.

Exercises

1. Compare a Euclidean space R_2 and a unitary space C_2 .

2. Write the Cauchy-Buniakowski-Schwarz inequality for a space C_n .

3. If in a complex space the scalar product is introduced according to axioms (27.1), can the Cauchy-Buniakowski-Schwarz inequality hold in such a space?

4. If in a complex space the scalar product is introduced according to axioms (27.1), can there be an orthogonal basis in such a space?

33. Linear dependence and orthonormal systems

We have already noted in Section 22 that the linear independence of a system of basis vectors may be violated by a *small* change in the vectors. This phenomenon leads to great difficulties in the use of the concept of basis in solving practical problems. It is important to stress, however, that not all bases possess so unpleasant a feature. In particular, it is *lacking* in any orthonormal basis.

Let $e_1, e_2, \dots, e_n$ be an arbitrary orthonormal basis chosen in a Euclidean or a unitary space. If for some vector b

$$b = \sum_{i=1}^n \alpha_i e_i,$$

then by (28.4)

$$|b|^2 = \sum_{i=1}^n |\alpha_i|^2. \quad (33.1)$$

Consider now a system of vectors $e_1 + e_1, e_2 + e_2, \dots, e_n + e_n$ and suppose that it is linearly dependent. This means that there

are numbers $\beta_1, \beta_2, \dots, \beta_n$ not all zero such that

$$\sum_{i=1}^n \beta_i (e_i + \varepsilon_i) = 0$$

It follows that

$$\sum_{i=1}^n \beta_i e_i = - \sum_{i=1}^n \beta_i \varepsilon_i.$$

Using (33.1) and (27.6) we get

$$\begin{aligned} \sum_{i=1}^n |\beta_i|^2 &= \left| \sum_{i=1}^n \beta_i e_i \right|^2 = \left| \sum_{i=1}^n \beta_i \varepsilon_i \right|^2 \leq \left(\sum_{i=1}^n |\beta_i| |\varepsilon_i| \right)^2 \\ &\leq \left(\sum_{i=1}^n |\beta_i|^2 \right) \left(\sum_{i=1}^n |\varepsilon_i|^2 \right). \end{aligned}$$

Comparing the left- and right-hand sides of these relations we deduce that

$$\sum_{i=1}^n |\varepsilon_i|^2 \geq 1.$$

Thus the inequality obtained means that if the condition

$$\sum_{i=1}^n |\varepsilon_i|^2 < 1 \tag{33.2}$$

holds, then the system of vectors

$$e_1 + \varepsilon_1, e_2 + \varepsilon_2, \dots, e_n + \varepsilon_n$$

is clearly linearly independent.

The indicated feature of orthonormal systems has determined their wide use in constructing various computational algorithms associated with expansions with respect to bases.

Exercises

1. Let $e_1, e_2, \dots, e_n$ be an orthogonal basis of a Euclidean space. Prove that a system of vectors $x_1, x_2, \dots, x_n$ is linearly independent if

$$\sum_{i=1}^n \cos(e_i, x_i) > n - \frac{1}{2}.$$

2. Let vectors $x_i = (x_{i1}, x_{i2}, \dots, x_{in})$, for $i = 1, 2, \dots, n$, be given by their coordinates in an arbitrary basis. Prove that if

$$|x_{ii}|^2 > n \sum_{k \neq i} |x_{ik}|^2$$

for every i , then the system $x_1, x_2, \dots, x_n$ is linearly independent.

The Volume of a System of Vectors in Vector Space

34. Vector and triple scalar products

Again we begin our studies with a space of directed line segments. As always we assume some fixed Cartesian coordinate system with origin O and basis i, j and k .

Three vectors are said to be a *triple* if it is stated which vector is the first, which is the second and which is the third. In writing we shall arrange the vectors of a triple consecutively from left to right.

A triple of noncoplanar vectors a, b and c is said to be *right-(left)-handed* if they are arranged the way the thumb, the unbent forefinger and the middle finger of the *right (left)* hand can be held respectively.

Of any three noncoplanar vectors a, b and c it is possible to compose the following six triples:

$$abc, bca, cab, bac, acb, cba.$$

The first three are of the same sign as abc , and the others are of the opposite sign. Notice that interchanging any two vectors in any triple makes the triple change sign.

An affine or Cartesian coordinate system is said to be *right-(left)-handed* if its basis vectors form a right- (left-)handed triple. So far our studies have been independent of the sign the basis of the coordinate system had. Now some differences will appear in our investigations. For the sake of definiteness therefore in what follows we shall consider only right-handed coordinate systems.

Suppose two noncollinear vectors a and b are given. We assign to them a third vector c satisfying the following conditions:

- (1) c is orthogonal to each of the vectors a and b ,
- (2) abc is a right-handed triple,
- (3) the length of c equals the area S of the parallelogram constructed on the vectors a and b applied to a common origin. If a and b are collinear, then we assign to such a pair of vectors a zero vector.

The resulting correspondence is an algebraic operation in a space V_3 . It is called *vector multiplication of vectors a and b* and designated

$$a \cdot b = [a, b].$$

Consider the basis vectors i, j and k . By the definition of a vector product

$$\begin{aligned} [i, i] &= 0, & [i, j] &= k, & [i, k] &= -j, \\ [j, i] &= -k, & [j, j] &= 0, & [j, k] &= i, \\ [k, i] &= j, & [k, j] &= -i, & [k, k] &= 0. \end{aligned} \quad (34.1)$$

From these relations it follows in particular that the operation of vector product is noncommutative.

Every triple abc of noncoplanar vectors applied to a common point O determines some parallelepiped. The point O is one of its vertices, and the vectors a, b and c are its edges. We designate the volume of that parallelepiped as $V(a, b, c)$ thus emphasizing its

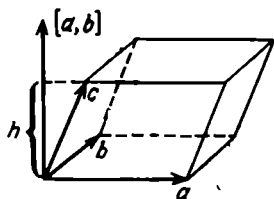


Fig. 34.1

dependence on the vectors a, b and c . If a triple a, b and c is coplanar, then the volume is assumed to be zero. We now assign to the volume a plus sign if the noncoplanar triple abc is right-handed and a minus sign if it is left-handed. The new concept thus defined will be called the *oriented volume* of the parallelepiped and designated $V^\pm(a, b, c)$.

A volume and an oriented volume may be regarded as some numerical functions of three independent vector variables assuming certain real values for every vector triple abc . A volume is always nonnegative, and an oriented volume may have any sign. We shall later see that separating these concepts does make sense.

Let a, b and c be three arbitrary vectors. If we carry out vector multiplication of a on the right by b and then perform scalar multiplication of the vector $[a, b]$ by c , then the resulting number $([a, b], c)$ is said to be a *triple scalar product of the vectors a, b and c* .

Theorem 34.1. *A triple scalar product $([a, b], c)$ is equal to the oriented volume of the parallelepiped constructed on vectors a, b and c applied to a common origin.*

Proof. We may assume without loss of generality that a and b are noncollinear since otherwise $[a, b] = 0$ and the statement of the theorem is obvious. Let S be as before the area of the parallelogram constructed on a and b . By (26.4)

$$([a, b], c) = |[a, b]| \{pr_{[a, b]} c\} = S \{pr_{[a, b]} c\}. \quad (34.2)$$

Suppose a, b and c are noncoplanar. Then $\{pr_{[a, b]} c\}$ is up to a sign equal to the altitude h of the parallelepiped constructed on the vectors a, b and c applied to a common origin provided that the base of the parallelepiped is the parallelogram constructed on a and b (Fig. 34.1). Thus the right-hand side of (34.2) is up to a sign equal to the volume of the parallelepiped constructed on a, b and c .

It is obvious that $\{pr_{[a,b]} c\} = +h$ if the vectors $[a, b]$ and c are on the same side of the plane determined by a and b . But in this case the triple abc is also right-handed. Otherwise $\{pr_{[a,b]} c\} = -h$. If the vectors abc are coplanar, then c is in the plane given by a and b , and therefore $\{pr_{[a,b]} c\} = 0$. Thus the theorem is proved.

Corollary. For any three vectors a, b and c

$$([a, b], c) = (a, [b, c]). \quad (34.3)$$

Indeed, from the symmetry of a scalar product it follows that $(a, [b, c]) = ([b, c], a)$, and therefore it suffices to show that $([a, b], c) = ([b, c], a)$. But the last equation is obvious since the triples abc and bca are of the same sign and their parallelepipeds coincide.

Relation (34.3) allows efficient algebraic studies to be carried out. We first prove that for any vectors a, b and c and any real number α the following properties of vector multiplication hold:

- (1) $[a, b] = -[b, a]$,
- (2) $[\alpha a, b] = \alpha [a, b]$,
- (3) $[a + b, c] = [a, c] + [b, c]$,
- (4) $[a, a] = 0$.

Property 4 follows in an obvious way from the definition. To prove the remaining properties we shall use the fact that vectors x and y are equal if and only if

$$(x, d) = (y, d)$$

for any vector d .

Let d be an arbitrary vector. The triples abd and bad are of opposite signs. Consequently, from Theorem 34.1 and the properties of a scalar product we conclude that

$$([a, b], d) = -([b, a], d) = (-[b, a], d).$$

Since d is an arbitrary vector, this means that $[a, b] = -[b, a]$ and the first property is proved.

To prove the second and the third property we proceed similarly but in addition take into account (34.3). We have

$$\begin{aligned} ([\alpha a, b], d) &= (\alpha a, [b, d]) = \alpha (a, [b, d]) \\ &= \alpha ([ab], d) = (\alpha [a, b], d), \end{aligned}$$

which shows the validity of Property 2. Further

$$\begin{aligned} ([a + b, c], d) &= (a + b, [c, d]) = (a, [c, d]) + (b, [c, d]) \\ &= ([a, c], d) + ([b, c], d) = ([a, c] + [b, c], d), \end{aligned}$$

and Property 3 is also true. Similarly for the second factor

$$[a, \alpha b] = -[\alpha b, a] = -\alpha [b, a] = \alpha [a, b],$$

$$[a, b + c] = -[b + c, a] = -[b, a] - [c, a] = [a, b] + [a, c].$$

Now we can investigate the algebraic properties of an oriented volume as a function given on triples of vectors. For example, let a be a linear combination of some vectors a' and a'' . Then

$$\begin{aligned} ([\alpha a' + \beta a'', b], c) &= (\alpha [a', b] + \beta [a'', b], c) \\ &= \alpha ([a', b], c) + \beta ([a'', b], c). \end{aligned}$$

Consequently

$$V^\pm(\alpha a' + \beta a'', b, c) = \alpha V^\pm(a', b, c) + \beta V^\pm(a'', b, c)$$

for any vectors a' and a'' and any real numbers α and β .

When two independent variables are interchanged, the oriented volume only changes sign and therefore a similar property for a linear combination holds in each independent variable. Bearing this in mind we shall say that an oriented volume is a *linear function* in each of the independent variables.

If vectors abc are linearly dependent, then they are coplanar and therefore the oriented volume is equal to zero. Also, considering (34.1) we find that

$$V^\pm(i, j, k) = ([i, j], k) = (k, k) = 1.$$

So it may be concluded that as a function an oriented volume possesses the following properties:

- (A) an oriented volume is a linear function in each of the independent variables,
 - (B) an oriented volume is equal to zero on all linearly dependent systems,
 - (C) an oriented volume is equal to unity at least on one fixed orthonormal system of vectors.
- (34.4)

Of course we have formulated not all the properties by far of an oriented volume. Both the properties listed in (34.4) and others can easily be established if we know explicit expressions of vector and triple scalar products in terms of the coordinates of vectors a , b and c .

Theorem 34.2. *If vectors a and b are given by their Cartesian coordinates*

$$a = (x_1, y_1, z_1),$$

$$b = (x_2, y_2, z_2),$$

then the vector product will have the following coordinates:

$$[a, b] = (y_1 z_2 - y_2 z_1, z_1 x_2 - z_2 x_1, x_1 y_2 - x_2 y_1). \quad (34.5)$$

Proof. Considering that giving the coordinates of the vectors determines the decompositions

$$a = x_1i + y_1j + z_1k,$$

$$b = x_2i + y_2j + z_2k,$$

and relying on the algebraic properties of a vector product we find

$$\begin{aligned} [a, b] = & x_1x_2 [i, i] + x_1y_2 [i, j] + x_1z_2 [i, k] + y_1x_2 [j, i] \\ & + y_1y_2 [j, j] + y_1z_2 [j, k] + z_1x_2 [k, i] + z_1y_2 [k, j] + z_1z_2 [k, k]. \end{aligned}$$

The statement of the theorem now follows from (34.1).

Corollary. *If a vector c is also given by coordinates x_3, y_3 and z_3 in the same Cartesian system, then*

$$\begin{aligned} ([a, b], c) = & x_1y_2z_3 + x_2y_3z_1 + x_3y_1z_2 - x_1y_3z_2 \\ & - x_2y_1z_3 - x_3y_2z_1. \end{aligned} \quad (34.6)$$

The introduction of an oriented volume and the study of its algebraic properties allow us to make important conclusions concerning length, area and volume.

Notice that right- and left-handed bases determine partition of the set of all bases of a space into two classes. The term "right- and left-handed" has no deep meaning of its own but is merely a convenient way of identifying the class to which one basis or another belongs. The concept of oriented volume is actually also related to these two classes.

We have already met with such facts. All bases on a straight line can also be divided into two classes by combining in the same class vectors pointing in the same direction. It turns out that the magnitude of a directed line segment is closely similar to an oriented volume if both notions are considered as functions on systems of vectors. Property A holds according to relations (9.8). Property B is true since the magnitude of a zero line segment is zero. That Property C holds is obvious.

A similar investigation could be carried out independently in the case of the plane too. It is more simple, however, to use the results already obtained. Fix some x, y Cartesian system. Supplement it to a right-handed x, y, z coordinate system in space. Note that depending on the location of the x and y axes the z axis may have one of the two possible directions. This again determines partition of the set of bases of the plane into two classes. The oriented area $S^\pm(a, b)$ of the parallelogram constructed on vectors a and b in the x, y plane may be defined, for example, by the equation $S^\pm(a, b) = V^\pm(a, b, k)$. Properties A, B and C again hold of course.

Thus, assigning to lengths, areas and volumes some signs and considering them as functions given on systems of vectors we can make all these functions have the same algebraic properties A, B and C of (34.4).

Exercises

1. Prove that a , b and c are coplanar if and only if their triple scalar product is zero.

2. Prove that for any three vectors a , b and c

$$[a, [b, c]] = (a, c) b - (a, b) c.$$

3. Prove that vector multiplication is not an associative operation.

4. Find an expression of the oriented area of a parallelogram in terms of the Cartesian coordinates of vectors in the plane.

5. Will formulas (34.5) and (34.6) change if the coordinate system relative to which the vectors are given is left-handed?

35. Volume and oriented volume of a system of vectors

In vector spaces of directed line segments area and volume are derived concepts of the length of a line segment. We have already extended the concept of length to abstract Euclidean spaces. We now consider a similar problem for area and volume.

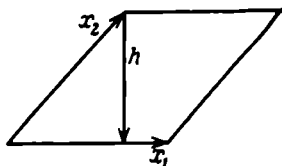


Fig. 35.1

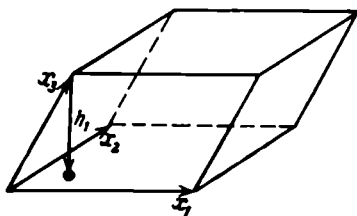


Fig. 35.2

Let x_1 and x_2 be two noncollinear vectors in the plane. Construct on them a parallelogram, taking x_1 as its base (Fig. 35.1). Drop from the terminal point of x_2 to the base a perpendicular h . The area $S(x_1, x_2)$ of the parallelogram will be defined by the formula

$$S(x_1, x_2) = |x_1| |h|. \quad (35.1)$$

Denote by L_0 a zero subspace and by L_1 the span constructed on x_1 . Since

$$|x_1| = |\text{ort}_{L_0} x_1|,$$

formula (35.1) can be written as

$$S(x_1, x_2) = |\text{ort}_{L_0} x_1| |\text{ort}_{L_1} x_2|. \quad (35.2)$$

Take then three noncoplanar vectors x_1 , x_2 and x_3 in space. Construct on them a parallelepiped, taking as its base the parallelogram formed by x_1 and x_2 (Fig. 35.2). Drop from the terminal point of x_3 to the base a perpendicular h_1 . The volume $V(x_1, x_2, x_3)$ of the parallelepiped will be defined by

$$V(x_1, x_2, x_3) = S(x_1, x_2) |h_1|.$$

If by L_2 we denote the span constructed on x_1 and x_2 , then by (35.2)

$$V(x_1, x_2, x_3) = |\text{ort}_{L_2} x_1| |\text{ort}_{L_2} x_2| |\text{ort}_{L_2} x_3|.$$

Thus the length of a vector, the area of a parallelogram and the volume of a parallelepiped are expressed in vector spaces V_1 , V_2 and V_3 by formulas in which one cannot but see a certain regularity:

$$|x_1| = |\text{ort}_{L_0} x_1|,$$

$$S(x_1, x_2) = |\text{ort}_{L_0} x_1| |\text{ort}_{L_1} x_2|, \quad (35.3)$$

$$V(x_1, x_2, x_3) = |\text{ort}_{L_0} x_1| |\text{ort}_{L_1} x_2| |\text{ort}_{L_2} x_3|.$$

In particular, the number of factors coincides everywhere with the dimension of the space.

These formulas suggest how to introduce the concept of volume in a Euclidean space E_n of dimension n . Let $x_1, x_2, \dots, x_n$ be an arbitrary system of vectors in E_n . Denote by L_0 a zero subspace and by L_i the span formed by vectors $x_1, \dots, x_i$. Then by analogy with spaces of directed line segments we say that:

The *volume* $V(x_1, x_2, \dots, x_n)$ of a system of vectors $x_1, x_2, \dots, x_n$ of a Euclidean space E_n is the value on that system of a real-valued function of n independent vector variables in E_n defined by the following equation:

$$V(x_1, x_2, \dots, x_n) = \prod_{i=0}^{n-1} |\text{ort}_{L_i} x_{i+1}|. \quad (35.4)$$

Of course we cannot as yet say that the volume of a system of vectors possesses all the properties inherent in a volume for any n . But for Euclidean spaces of dimensions 1, 2 and 3 respectively, by virtue of Euclidean isomorphism and relations (35.3), it clearly has the same properties as have the length of a line segment, the area of a parallelogram and the volume of a parallelepiped.

We now try to approach the concept of the volume of a system of vectors of a Euclidean space E_n from another point of view. As already noted, assigning definite signs turns length, area and volume into algebraic functions possessing some common properties. Therefore it may be expected that there are similar properties in an arbitrary Euclidean space, too. Bearing this in mind we give the following definition:

The *oriented volume* $V^{\pm}(x_1, x_2, \dots, x_n)$ of a system of vectors $x_1, x_2, \dots, x_n$ of a Euclidean space E_n is the value on that system of a real-valued function of n independent vector variables in E_n possessing properties (34.4).

Much is unclear about this definition too. Thus we do not know if there is an oriented volume for any system of vectors in an arbitrary Euclidean space for $n \geq 4$. But even if there is, is it uniquely defined by properties (34.4)? And finally what relation is there between volume and oriented volume in general? Now we can answer only the last question, and only for the case $n = 1, 2, 3$.

We shall sometimes have to consider volume and oriented volume in a space E_n for systems containing fewer than n vectors. This will mean that in fact we are dealing not with the whole space but with some of its subspaces from which a given system is taken. Accordingly properties (34.4) will be considered only in relation to vectors of the same subspace. A need may arise to consider volume and oriented volume for systems containing more than n vectors. According to formula (35.4) and Property B of (34.4) both functions must be zero on such systems.

In conclusion note that the use of two different notions associated with volume will substantially simplify their investigation, since one concept reflects the geometrical aspect of the problem concerned and the other reflects its algebraic aspect. We shall soon see that there is a very close relation between them. We shall also see that it is important to introduce these notions because they generate a mathematical tool whose significance is not limited to the volume problem.

Exercises

1. Prove that in spaces of directed line segments oriented length, area and volume are defined uniquely by conditions (34.4).

2. Will the same concepts be uniquely defined if one of the conditions (34.4) is excluded?

3. Prove that in any Euclidean space $V(x_1, x_2) = |x_1| \cdot |x_2|$ if and only if vectors x_1 and x_2 are orthogonal.

4. Prove that in any Euclidean space $V(x_1, x_2) = V(x_2, x_1)$.

36. Geometrical and algebraic properties of a volume

We begin our study of the concept of volume in a Euclidean space E_n by exploring its geometrical and algebraic properties following from its definition.

Property 1. *Always $V(x_1, x_2, \dots, x_n) \geq 0$. The equation $V(x_1, x_2, \dots, x_n) = 0$ holds if and only if the system of vectors $x_1, x_2, \dots, x_n$ is linearly dependent.*

The first part of the statement follows in an obvious way from (35.4), and therefore only its second part needs to be proved. Let $x_1, x_2, \dots, x_n$ be a linearly dependent system. If $x_1 = 0$, then by definition so is the volume. If $x_1 \neq 0$, then some vector x_{k+1} is linearly expressible in terms of the preceding vectors $x_1, \dots, x_k$. But then $\text{ort}_{L_k} x_{k+1} = 0$ and again the volume is zero.

Suppose now that the volume is zero. According to the definition this means that so is one of the multipliers on the right of (35.4). For that multiplier let $i = k$. If $k = 0$, then $x_1 = 0$. If $k \neq 0$, then the condition $\text{ort}_{L_k} x_{k+1} = 0$ implies that x_{k+1} is in the span formed by the vectors $x_1, \dots, x_k$, i.e. that the system $x_1, \dots, x_{k+1}$ is linearly dependent. So is the entire system of vectors $x_1, x_2, \dots, x_n$ in both cases.

Property 2. For any system of vectors $x_1, x_2, \dots, x_n$

$$V(x_1, x_2, \dots, x_n) \leq \prod_{i=0}^{n-1} |x_{i+1}|, \quad (36.1)$$

(Hadamard's inequality), with equality holding if and only if the system $x_1, x_2, \dots, x_n$ is orthogonal or contains a zero vector.

According to the properties of a perpendicular and a projection clearly

$$|\text{ort}_{L_i} x_{i+1}| \leq |x_{i+1}|, \quad (36.2)$$

the inequality becoming an equation if and only if $x_{i+1} \perp L_i$ or equivalently if x_{i+1} is orthogonal to the vectors $x_1, x_2, \dots, x_i$. Consider the product of the left- and right-hand sides of inequalities of the form (36.2) for all i . We have

$$\prod_{i=0}^{n-1} |\text{ort}_{L_i} x_{i+1}| \leq \prod_{i=0}^{n-1} |x_{i+1}|.$$

If all vectors of the system $x_1, x_2, \dots, x_n$ are nonzero, then this inequality becomes an equation if and only if the system is orthogonal. The case of a zero vector is trivial.

It is possible to deduce several useful properties from Hadamard's inequality. Let $x_1, x_2, \dots, x_n$ be a normed system. Then it is obvious that

$$V(x_1, x_2, \dots, x_n) \leq 1.$$

The following statement is also true. If the system $x_1, x_2, \dots, x_n$ is normed and its volume equals unity, then it is orthonormal. Since the volume of any normed system is not greater than unity, this means that of all normed systems the orthonormal system has a maximum volume.

Property 3. *For any two orthogonal sets of vectors $x_1, x_2, \dots, x_p$ and $y_1, y_2, \dots, y_r$*

$$V(x_1, x_2, \dots, x_p, y_1, y_2, \dots, y_r) \\ = V(x_1, x_2, \dots, x_p) V(y_1, y_2, \dots, y_r).$$

Denote by L_t the span formed by the first t vectors of a joined system $x_1, \dots, x_p, y_1, \dots, y_r$ and by K_t the span formed by vectors $y_1, \dots, y_t$. Under the hypothesis each of the vectors in the system $y_1, \dots, y_r$ is orthogonal to all vectors of the system $x_1, \dots, x_p$. Therefore

$$L_{p+t} = L_p \oplus K_t$$

for every t from 0 to r . Now, taking into account (30.10), we have

$$V(x_1, \dots, x_p, y_1, \dots, y_r) = \left(\prod_{i=0}^{p-1} |\text{ort}_{L_i} x_{i+1}| \right) \left(\prod_{i=0}^{r-1} |\text{ort}_{L_{p+i}} y_{i+1}| \right) \\ = \left(\prod_{i=0}^{p-1} |\text{ort}_{L_i} x_{i+1}| \right) \left(\prod_{i=0}^{r-1} |\text{ort}_{K_i} y_{i+1}| \right) \\ = V(x_1, \dots, x_p) V(y_1, \dots, y_r).$$

Before proceeding to further studies we make a remark. The volume of a system is only expressible in terms of the perpendiculars dropped to spans formed by the preceding vectors. Taking into account the properties of perpendiculars, it may therefore be concluded that the volume of the system will remain unaffected if to any vector any linear combination of the preceding vectors is added. In particular, the volume will remain unaffected if any vector is replaced by the perpendicular from that vector to any span formed by the preceding vectors.

Property 4. *The volume of a system of vectors remains unaffected by any rearrangement of vectors in the system.*

Consider first the case where in a system of vectors $x_1, \dots, x_n$ two adjacent vectors x_{p+1} and x_{p+2} are interchanged. According to the above remark the volume will remain unaffected if x_{p+1} and x_{p+2} are replaced by the vectors $\text{ort}_{L_p} x_{p+1}$ and $\text{ort}_{L_p} x_{p+2}$, and $x_{p+3}, \dots, x_n$ are replaced by $\text{ort}_{L_{p+2}} x_{p+3}, \dots, \text{ort}_{L_{p+2}} x_n$. But now three sets of vectors

$$x_1, \dots, x_p, \\ \text{ort}_{L_p} x_{p+1}, \text{ort}_{L_p} x_{p+2}, \\ \text{ort}_{L_{p+2}} x_{p+3}, \dots, \text{ort}_{L_{p+2}} x_n$$

are mutually orthogonal and in view of Property 3 we have

$$V(x_1, \dots, x_{p+1}, x_{p+2}, \dots, x_n) = V(x_1, \dots, x_p) \\ \times V(\text{ort}_{L_p} x_{p+1}, \text{ort}_{L_p} x_{p+2}) \cdot V(\text{ort}_{L_{p+2}} x_{p+3}, \dots, \text{ort}_{L_{p+2}} x_n).$$

It is clear that the spans of the vectors $x_1, \dots, x_p, x_{p+1}, x_{p+2}$ and $x_1, \dots, x_p, x_{p+2}, x_{p+1}$ coincide; consequently

$$V(x_1, \dots, x_{p+2}, x_{p+1}, \dots, x_n) = V(x_1, \dots, x_p) \\ \times V(\text{ort}_{L_p} x_{p+2}, \text{ort}_{L_p} x_{p+1}) \cdot V(\text{ort}_{L_{p+2}} x_{p+3}, \dots, \text{ort}_{L_{p+2}} x_n).$$

By virtue of Euclidean isomorphism the volume of a system of two vectors possesses the same properties as the area of a parallelogram does. In particular, it is independent of the order of the vectors of the system. Comparing the right-hand sides of the last two equations we now conclude that

$$V(x_1, \dots, x_{p+1}, x_{p+2}, \dots, x_n) \\ = V(x_1, \dots, x_{p+2}, x_{p+1}, \dots, x_n).$$

A little later we shall prove that any permutation $x_{j_1}, x_{j_2}, \dots, x_{j_n}$ of vectors of a system $x_1, x_2, \dots, x_n$ can be obtained by successive interchanging of adjacent vectors. For an arbitrary permutation therefore Property 4 follows from the above special case.

Property 5. *The volume of a system of vectors is an absolutely homogeneous function, i.e.*

$$V(x_1, \dots, \alpha x_p, \dots, x_n) = |\alpha| V(x_1, \dots, x_p, \dots, x_n)$$

for any p .

By Property 4 we may assume without loss of generality that $p = n$. But then in view of (30.6) we get

$$V(x_1, \dots, x_{n-1}, \alpha x_n) = \left(\prod_{i=0}^{n-2} |\text{ort}_{L_i} x_{i+1}| \right) |\text{ort}_{L_{n-1}}(\alpha x_n)| \\ = |\alpha| \prod_{i=0}^{n-1} |\text{ort}_{L_i} x_{i+1}| = |\alpha| V(x_1, \dots, x_{n-1}, x_n).$$

Property 6. *The volume of a system of vectors remains unaffected if to some one of the vectors of the system a linear combination of the remaining vectors is added.*

Again by Property 4 we may assume that to the last vector a linear combination of the preceding vectors is added. But, as already noted, in this case the volume remains unaffected.

The volume of a system of vectors is a real-valued function. This function possesses some properties part of which we have already established. They have confirmed our supposition that the volume

of a system of vectors we have defined possesses in a Euclidean space all properties inherent in a volume for any n . But the most important thing is perhaps that the established properties uniquely define a volume. More precisely, we have

Theorem 36.1. *If a real-valued function $F(x_1, x_2, \dots, x_n)$ of n independent vector variables in E_n possesses the following properties:*

- (A) *it remains unchanged by addition, to any independent variable, of any linear combination of the remaining independent variables,*
- (B) *it is absolutely homogeneous,*
- (C) *it equals unity for all orthonormal systems,*

(36.3)

then it coincides with the volume of the system of vectors.

Proof. If there is at least one zero independent variable among $x_1, \dots, x_n$, then by Property B

$$F(x_1, x_2, \dots, x_n) = V(x_1, x_2, \dots, x_n) = 0. \quad (36.4)$$

Now let $x_1, x_2, \dots, x_n$ be an arbitrary system. Subtracting from each vector x_i its projection onto the subspace formed by the vectors $x_1, \dots, x_{i-1}$ and taking into account Property A we conclude that

$$F(x_1, x_2, \dots, x_n) = F(\text{ort}_{L_0} x_1, \text{ort}_{L_1} x_2, \dots, \text{ort}_{L_{n-1}} x_n). \quad (36.5)$$

If the system $x_1, x_2, \dots, x_n$ is linearly dependent, then there is at least one zero vector among $\text{ort}_{L_{i-1}} x_i$ and (36.4) again holds. Suppose the system $x_1, x_2, \dots, x_n$ is linearly independent. Then all vectors of the system

$$\text{ort}_{L_0} x_1, \text{ort}_{L_1} x_2, \dots, \text{ort}_{L_{n-1}} x_n$$

are nonzero. Since in addition this system is orthogonal, there is an orthonormal system $e_1, e_2, \dots, e_n$ such that

$$\text{ort}_{L_{i-1}} x_i = |\text{ort}_{L_{i-1}} x_i| e_i.$$

By Property C

$$F(e_1, e_2, \dots, e_n) = 1.$$

It follows from B therefore that

$$\begin{aligned} F(x_1, x_2, \dots, x_n) &= \left(\prod_{i=0}^{n-1} |\text{ort}_{L_i} x_{i+1}| \right) F(e_1, e_2, \dots, e_n) \\ &= V(x_1, x_2, \dots, x_n). \end{aligned}$$

The theorem allows us to state that if we construct in some way a function possessing properties (36.3), then it will be precisely the volume of a system of vectors.

Exercises

1. Give a geometrical interpretation of Properties 2, 3 and 6 in spaces of directed line segments.
2. Give a geometrical interpretation of equation (36.5) in spaces of directed line segments.
3. Can a function satisfying conditions (36.3) be zero on any linearly independent system of vectors?
4. Suppose relative to an orthonormal basis $e_1, e_2, \dots, e_n$ a system of vectors $x_1, x_2, \dots, x_n$ possesses the property

$$(x_i, e_j) = 0$$

for $i = 2, 3, \dots, n$ and $j < i$ ($i = 1, 2, \dots, n-1$ and $j > i$). Find the expression for the volume $V(x_1, x_2, \dots, x_n)$ in terms of the coordinates of vectors $x_1, x_2, \dots, x_n$ in the basis $e_1, e_2, \dots, e_n$.

5. What will change in the concept of volume for a complex space?

37. Algebraic properties of an oriented volume

We now proceed to study the algebraic properties of an oriented volume, laying aside for the time being the question of its existence. Our study will be based on Conditions A, B and C of (34.4).

Property 1. *The oriented volume of a system of vectors is zero if any of its two vectors coincide.*

This property is a direct consequence of Condition B. It is not hard to prove that if A holds, then B and Property 1 are equivalent.

Property 2. *The oriented volume of a system of vectors changes sign if some two vectors are interchanged.*

The proof is similar for any two vectors and therefore for the sake of simplicity we restrict ourselves to the case where the first and second vectors are interchanged. By Property 1

$$V^\pm(x_1 + x_2, x_1 + x_2, x_3, \dots, x_n) = 0.$$

But on the other hand according to A

$$\begin{aligned} V^\pm(x_1 + x_2, x_1 + x_2, x_3, \dots, x_n) \\ = V^\pm(x_1, x_1, x_3, \dots, x_n) + V^\pm(x_2, x_2, x_3, \dots, x_n) \\ + V^\pm(x_1, x_2, x_3, \dots, x_n) + V^\pm(x_2, x_1, x_3, \dots, x_n). \end{aligned}$$

On the right of this equation the first two terms are zero from which it follows that Property 2 is valid. It is again not hard to prove that if A holds, then B and Property 2 are equivalent.

Property 3. *The oriented volume of a system of vectors remains unaffected by addition, to any vector, of any linear combination of the remaining vectors.*

Again for simplicity, consider only the first vector. According to A

$$\begin{aligned} V^\pm(x_1 + \sum_{i=2}^n \alpha_i x_i, x_2, \dots, x_n) \\ = V^\pm(x_1, x_2, \dots, x_n) + \sum_{i=2}^n \alpha_i V^\pm(x_i, x_2, \dots, x_n). \end{aligned}$$

In this equation all terms at the right but the first are zero according to Property 1.

Property 4. *The oriented volume is a homogeneous function, i.e.*

$$V^\pm(x_1, \dots, \alpha x_p, \dots, x_n) = \alpha V^\pm(x_1, \dots, x_p, \dots, x_n)$$

for any p .

This property is a direct consequence of A.

Property 5. *The equation $V^\pm(x_1, x_2, \dots, x_n) = 0$ holds if and only if the system of vectors $x_1, x_2, \dots, x_n$ is linearly dependent.*

Obviously it is only necessary to prove that $V^\pm(x_1, x_2, \dots, x_n) = 0$ implies a linear dependence of the vectors $x_1, x_2, \dots, x_n$. Suppose the contrary. Let the oriented volume be zero for some linearly independent system $y_1, y_2, \dots, y_n$. This system is a basis of E_n and therefore for any vector z in E_n

$$z = \alpha_1 y_1 + \alpha_2 y_2 + \dots + \alpha_n y_n.$$

Now replace in the system $y_1, y_2, \dots, y_n$ any vector, for example y_1 , by a vector z . Using in succession Properties 3 and 4 we find that

$$\begin{aligned} V^\pm(z, y_2, \dots, y_n) &= V^\pm(\alpha_1 y_1 + \sum_{i=2}^n \alpha_i y_i, y_2, \dots, y_n) \\ &= V^\pm(\alpha_1 y_1, y_2, \dots, y_n) = \alpha_1 V^\pm(y_1, y_2, \dots, y_n) = 0. \end{aligned}$$

An oriented volume is by definition not zero at least on one linearly independent system $z_1, z_2, \dots, z_n$. But replacing in turn vectors $y_1, y_2, \dots, y_n$ by $z_1, z_2, \dots, z_n$ we conclude from Theorem 15.2 that on that system the oriented volume is zero. This contradiction proves the property in point.

Property 6. *If two oriented volumes coincide on at least one linearly independent system of vectors, then they coincide identically.*

Suppose it is known that oriented volumes $V_1^\pm(x_1, x_2, \dots, x_n)$ and $V_2^\pm(x_1, x_2, \dots, x_n)$ coincide on a linearly independent system $z_1, z_2, \dots, z_n$. Consider the difference $F(x_1, x_2, \dots, x_n) = V_1^\pm(x_1, x_2, \dots, x_n) - V_2^\pm(x_1, x_2, \dots, x_n)$. This function satisfies Properties 3 and 4 of an oriented volume. Besides it is zero on all linearly dependent systems and at least on one linearly independent system $z_1, z_2, \dots, z_n$. Repeating the arguments used in proving

Property 5 we conclude that $F(x_1, x_2, \dots, x_n)$ is zero on all linearly independent systems, i.e. that it is identically zero.

It follows from Property 6 that an oriented volume is uniquely defined by conditions (34.4) if we fix the orthonormal system on which it must equal unity.

Property 7. *The absolute value of the oriented volume of a system of vectors coincides with the volume of the same system.*

Let an oriented volume equal unity on an orthonormal system $z_1, z_2, \dots, z_n$. Consider functions $|V^\pm(x_1, x_2, \dots, x_n)|$ and $V(x_1, x_2, \dots, x_n)$. They both satisfy A and B of (36.3) and coincide on a linearly independent system $z_1, z_2, \dots, z_n$. The function

$$\Phi(x_1, x_2, \dots, x_n) = |V^\pm(x_1, x_2, \dots, x_n)| - V(x_1, x_2, \dots, x_n)$$

also satisfies A and B of (36.3) and is zero on all linearly dependent systems and at least on one linearly independent system $z_1, z_2, \dots, z_n$. Repeating again the arguments used in proving Property 5 we conclude that $\Phi(x_1, x_2, \dots, x_n)$ is identically zero.

The last property is very important, since it allows us to state that the absolute value of an oriented volume must have all the properties a volume has. In particular, it must equal unity on all orthonormal systems, and not only on one. Hadamard's inequality holds for it, and so on. This property supplies a final answer to all questions posed by us concerning volume and oriented volume. The only thing we lack is the proof of the *existence* of an oriented volume.

Exercises

1. Prove that if A of (34.4) holds, then B is equivalent to both Property 1 and Property 2.

2. Prove that whatever the real number α may be there is a system of vectors such that its oriented volume equals α .

3. Suppose C of (34.4) is replaced by the condition of equality to any fixed number on any fixed linearly independent system. How is an oriented volume affected?

4. Were the presence of a scalar product in a vector space and the reality of an oriented volume used in deriving the properties of an oriented volume? What will change if we consider an oriented volume in a complex space?

38. Permutations

Consider a system $x_1, x_2, \dots, x_n$ and a system $x_{j_1}, x_{j_2}, \dots, x_{j_n}$ obtained from the first using several permutations of vectors. Suppose these systems may be transformed into each other by successive transformations of only pairs of elements. Then the volumes of the systems will be the same and their oriented volumes will be either the same or differ in sign depending on the number of transpositions required.

In the questions of permutations we are going to discuss, individual properties of vectors will play no part, what will be important is their order. Therefore instead of vectors we shall consider their indices $1, 2, \dots, n$. A collection of numbers

$$j_1, j_2, \dots, j_n$$

among which there are no equal numbers and each of which is one of the numbers $1, 2, \dots, n$ is called a *permutation* of those numbers. The permutation $1, 2, \dots, n$ is called *normal*.

It is easy to show that a set of n numbers has $n!$ possible permutations in all. Indeed, for $n = 1$ this is obvious. Let the statement be true for any set of $n - 1$ numbers. All permutations of n numbers can be grouped into n classes by placing in the same class only permutations that have the same number in the first place. The number of permutations in each class coincides with the number of permutations of $n - 1$ numbers, i.e. is equal to $(n - 1)!$. Consequently, the number of all permutations of n numbers is $n!$.

It is said that in a given permutation numbers i and j form an *inversion* if $i > j$ but i precedes j in the permutation. A permutation is said to be *even* if its numbers constitute an even number of inversions and *odd* otherwise. If in some permutation we interchange any two numbers, not necessarily adjacent, leaving all the others in their places, we obtain a new permutation. This transformation of a permutation is called a *transposition*.

We prove that any transposition changes the parity of the permutation. For adjacent numbers this statement is obvious. Their relative positions with respect to other numbers remain the same and permutation of the adjacent numbers changes the total number of inversions by unity.

Suppose now that between numbers i and j to be interchanged there are s other numbers $k_1, k_2, \dots, k_s$, i.e. the permutation is of the form

$$\dots, i, k_1, k_2, \dots, k_s, j, \dots$$

We shall interchange the number i successively with the adjacent numbers $k_1, k_2, \dots, k_s, j$. Then the number j now preceding i is transferred to the left by s transpositions with numbers $k_s, k_{s-1}, \dots, k_1$. We thus carry out $2s + 1$ transpositions of adjacent numbers in all. Consequently, the permutation will change its parity.

Theorem 38.1. *All $n!$ permutations of n numbers can be arranged in such an order that each subsequent permutation is obtained from the preceding one by a single transposition, beginning with any permutation.*

Proof. This is true for $n = 2$. If it is required to begin with the permutation $1, 2$, then the desired arrangement is $1, 2, 2, 1$, if, however, we are to begin with the permutation $2, 1$, then the desired arrangement is $2, 1, 1, 2$.

Suppose the theorem has already been proved for any permutations containing no more than $n - 1$ numbers. Consider permutations of n numbers. Suppose we are to begin with a permutation $i_1, i_2, \dots, i_n$. We shall arrange permutations according to the following principle. We begin with permutations with i_1 in the first place. According to the assumption all these permutations can be ordered in accordance with the requirements of the theorem, since in fact it is necessary to arrange in the required order all permutations of $n - 1$ numbers.

In the last permutation obtained in this way we make one transposition, transferring to the first place the number i_2 . We then put in order, as in the preceding case, all permutations with a given number in the first place and so on. In this way it is possible to look over all permutations of n numbers.

With such a system of arranging permutations of n numbers adjacent permutations will have opposite parities. Considering that $n!$ is even for $n \geq 2$ we can conclude that in this case the number of even permutations of n numbers equals that of odd permutations and is $n!/2$.

Exercises

1. What is the parity of the permutation 5, 2, 3, 1, 4?
2. Prove that no even (odd) permutation can be transformed into a normal one in an odd (even) number of transpositions.
3. Consider a pair of permutations $i_1, i_2, \dots, i_n$ and $1, 2, \dots, n$. We transform into the normal form the first permutation employing transpositions, making for each of them one transposition of any elements in the second permutation. Prove that after the process is over the second permutation will have the same parity as the permutation $i_1, i_2, \dots, i_n$.

39. The existence of an oriented volume

We now discuss the existence of an oriented volume of a system of vectors. Choose in a space E_n an orthonormal system $z_1, z_2, \dots, z_n$ on which an oriented volume must equal unity by Condition C of (34.4). Take an arbitrary system $x_1, x_2, \dots, x_n$ of vectors in E_n . Since the system $z_1, z_2, \dots, z_n$ is a basis in E_n , for every vector x_i there is an expansion

$$x_i = a_{i1}z_1 + a_{i2}z_2 + \dots + a_{in}z_n \quad (39.1)$$

with respect to that basis, where a_{ij} are some numbers.

If an oriented volume exists, then according to A of (34.4) we can transform it successively, taking into account expansions (39.1).

Namely,

$$\begin{aligned}
 V^{\pm}(x_1, x_2, \dots, x_n) &= V^{\pm}\left(\sum_{j_1=1}^n a_{1j_1}z_{j_1}, \sum_{j_2=1}^n a_{2j_2}z_{j_2}, \dots, \sum_{j_n=1}^n a_{nj_n}z_{j_n}\right) \\
 &= \sum_{j_1=1}^n a_{1j_1}V^{\pm}\left(z_{j_1}, \sum_{j_2=1}^n a_{2j_2}z_{j_2}, \dots, \sum_{j_n=1}^n a_{nj_n}z_{j_n}\right) \\
 &= \sum_{j_1=1}^n \sum_{j_2=1}^n a_{1j_1}a_{2j_2}V^{\pm}\left(z_{j_1}, z_{j_2}, \dots, \sum_{j_n=1}^n a_{nj_n}z_{j_n}\right) \\
 &= \dots = \sum_{j_1=1}^n \sum_{j_2=1}^n \dots \sum_{j_n=1}^n a_{1j_1}a_{2j_2} \dots a_{nj_n}V^{\pm}(z_{j_1}, z_{j_2}, \dots, z_{j_n}).
 \end{aligned} \tag{39.2}$$

In the last n -fold sum most of the terms are zero, since by Property 1 the oriented volume of a system of vectors is zero if any two vectors of the system coincide. Therefore only those of the systems $z_{j_1}, z_{j_2}, \dots, z_{j_n}$ should be considered for which the set of indices $j_1, j_2, \dots, j_n$ is a permutation of n numbers $1, 2, \dots, n$. But in this case

$$V^{\pm}(z_{j_1}, z_{j_2}, \dots, z_{j_n}) = \pm 1$$

depending on the evenness or oddness of the permutation of indices.

Thus if an oriented volume exists, then it must be expressible in terms of the coordinates of vectors $x_1, x_2, \dots, x_n$ in the basis $z_1, z_2, \dots, z_n$ by the following formula:

$$V^{\pm}(x_1, x_2, \dots, x_n) = \sum \pm a_{1j_1}a_{2j_2} \dots a_{nj_n}. \tag{39.3}$$

Here summation is taken over all permutations of indices $j_1, j_2, \dots, j_n$ of numbers $1, 2, \dots, n$ and a plus or a minus sign is taken according as the permutation is even or odd.

We prove that the function given by the right-hand side of (39.3) satisfies all conditions defining an oriented volume. Let a vector x_p be a linear combination of vectors x'_p and x''_p , i.e.

$$x_p = \alpha x'_p + \beta x''_p$$

for some numbers α and β . Denote by a'_{pj} and a''_{pj} respectively the coordinates of x'_p and x''_p in a basis $z_1, z_2, \dots, z_n$. Then it is obvious that

$$a_{pj} = \alpha a'_{pj} + \beta a''_{pj}$$

for every j in the range from 1 to n . We further find

$$\begin{aligned}
 \sum \pm a_{1j_1} \dots a_{pj_p} \dots a_{nj_n} \\
 &= \sum \pm a_{1j_1} \dots (\alpha a'_{pj_p} + \beta a''_{pj_p}) \dots a_{nj_n} \\
 &= \alpha \sum \pm a_{1j_1} \dots a'_{pj_p} \dots a_{nj_n} + \beta \sum \pm a_{1j_1} \dots a''_{pj_p} \dots a_{nj_n},
 \end{aligned}$$

and thus A of (34.4) holds.

Suppose we permute some two vectors of a system $x_1, x_2, \dots, x_n$. In this case function (39.3) changes sign since the parity of each permutation is changed. As already noted, if the property of linearity in each independent variable holds, the property just proved is equivalent to Condition B of (34.4).

And finally consider the value of the constructed function on a system of vectors $z_1, z_2, \dots, z_n$. For that system coordinates a_{ij} have the following form:

$$a_{ij} = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases}$$

Consequently, of terms (39.3) only one term $a_{11}a_{22} \dots a_{nn}$ is non-zero. The permutation $1, 2, \dots, n$ is an even one and the elements $a_{11}, a_{22}, \dots, a_{nn}$ equal unity, so the value of the function on the orthonormal system $z_1, z_2, \dots, z_n$ is 1.

Thus all Conditions (34.4) hold and function (39.3) is an expression of the oriented volume of a system of vectors in terms of their coordinates. That expression is *unique* by virtue of the uniqueness of an oriented volume.

Exercises

1. Was the orthonormality of the system $z_1, z_2, \dots, z_n$ actually used in deriving formula (39.3)?

2. What changes will occur in formula (39.3) if in Condition C of (34.4) the oriented volume is not assumed to equal unity?

3. To what extent Condition B of (34.4) was actually used in deriving (39.3)?

4. Will the form of (39.3) be affected if we consider an oriented volume in a complex space?

40. Determinants

Let vectors $x_1, x_2, \dots, x_n$ of a Euclidean space R_n be given by their coordinates

$$x_i = (a_{i1}, a_{i2}, \dots, a_{in})$$

in basis (21.7). Arrange numbers a_{ij} as an array A :

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}.$$

This array is called a *square $n \times n$ matrix* and the numbers a_{ij} are *matrix elements*. If the matrix rows are numbered in succession from top to bottom and the columns are numbered from left to right, then the first index of an element stands for the number of the row

the element is in and the second index is the number of the column. The elements $a_{11}, a_{22}, \dots, a_{nn}$ are said to form the *principal diagonal* of the matrix A .

Any n^2 numbers can be arranged as a square $n \times n$ matrix. If the row elements of a matrix are assumed to be the coordinates of a vector of R_n in basis (21.7), then a 1-1 correspondence is established between all square $n \times n$ matrices and ordered systems of n vectors of the space R_n .

In R_n , as in any other space, there is an oriented volume. It will be unique if we require that Condition C of (34.4) should hold on the system of vectors (21.7). Taking into account the above 1-1 correspondence we conclude that a well-defined function is generated on the set of all square matrices. Considering (39.3) we arrive at the following definition of that function.

An n th-order *determinant* corresponding to a matrix A is an algebraic sum of $n!$ terms which is made up as follows. The terms of the determinant are all possible products of n matrix elements taken an element from each row and each column. The term is taken with a plus sign if the indices of the columns of its elements form an even permutation, provided the elements are arranged in increasing order of the row indices, and a minus sign otherwise.

To designate a determinant we shall use the following symbol:

$$\det \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} \quad (40.1)$$

if it is necessary to give matrix elements in explicit form. If, however, this is not necessary, we shall use a simpler symbol,

$$\det A,$$

restricting ourselves to the notation of a matrix A . The elements of the matrix of a determinant will also be called the *elements of the determinant*.

The determinant coincides with the oriented volume of the system of matrix rows. In investigating it therefore it is possible to use all known facts pertaining to volumes and oriented volumes. In particular, the determinant is zero if and only if the matrix rows are linearly dependent, the determinant changes sign and so on. Now our studies will concern those of its properties which are difficult to prove without using an explicit expression of the determinant in terms of matrix elements.

The *transpose* of a matrix is a transformation such that its rows become columns and its columns become rows with the same indices. The transpose of a matrix A is denoted by A' . Accordingly the deter-

minant

$$\det \begin{pmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \dots & \dots & \dots & \dots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{pmatrix}$$

or $\det A'$ is said to be obtained by transposing determinant (40.1). As to transposition the determinant possesses the following important property:

The determinant of any matrix remains unaffected when transposed.

Indeed, the determinant of a matrix A consists of terms of the following form:

$$a_{1j_1} a_{2j_2} \dots a_{nj_n} \quad (40.2)$$

whose sign depends on the parity of the permutation $j_1, j_2, \dots, j_n$. In the transpose A' all multipliers of product (40.2) remain in different rows and different columns, i.e. their product is a term of the transposed determinant. We denote the elements of A' by a'_{ij} . It is clear that $a'_{ij} = a_{ji}$ and therefore

$$a_{1j_1} a_{2j_2} \dots a_{nj_n} = a'_{j_1 1} a'_{j_2 2} \dots a'_{j_n n}. \quad (40.3)$$

We put the elements of the right-hand side of (40.3) in increasing order of row indices. Then the permutation of column indices will have the same parity as that of the permutation $j_1, j_2, \dots, j_n$. But this means that the sign of term (40.2) in the transposed determinant is the same as that in the original determinant. Consequently, both determinants consist of the same terms with the same signs, i.e. they coincide.

It follows from the above property that the rows and columns of a determinant are equivalent. Therefore all the properties proved earlier for the rows will hold for the columns.

Consider a determinant d of order n . Choose in its matrix k arbitrary rows and k arbitrary columns. The elements at the intersection of the chosen rows and columns form a $k \times k$ matrix. The determinant of the matrix is called the k th-order *minor* of the determinant d . The minor in the first k columns and the first k rows is called the *principal minor*.

Suppose now that in an n th-order determinant d a minor M of order k is taken. If we eliminate the rows and columns at whose intersection the minor M lies, we are left with a minor N of order $n - k$. This is called the *complementary minor* of M . If on the contrary we eliminate the rows and columns occupied by the elements of the minor N , then the minor M is obviously left. It is thus possible to speak of a pair of mutually complementary minors.

If a minor M of order k is in the rows with indices $i_1, i_2, \dots, i_k$ and the columns with indices $j_1, j_2, \dots, j_k$, then the number

$$(-1)^{\sum_{p=1}^k (i_p + j_p)} N \quad (40.4)$$

will be called the *algebraic adjunct* or *cofactor* of M .

Theorem 40.1 (Laplace). *Suppose in a determinant d of order n , k arbitrary rows (columns) are chosen, with $1 \leq k \leq n - 1$. Then the sum of the products of all k th-order minors contained in the chosen rows (columns) by their algebraic adjuncts is equal to the determinant d .*

Proof. Assume the columns of the matrix of d to be vectors $x_1, x_2, \dots, x_n$ of a space R_n . The sum of the products of all k th-order minors contained in the chosen rows by their algebraic adjuncts may be considered as some function $F(x_1, x_2, \dots, x_n)$ of the vectors $x_1, x_2, \dots, x_n$.

This function is obviously linear in each independent variable, since this property is true of both minors and algebraic adjuncts. It equals unity on the orthonormal system (21.7), which is easy to show by direct check. If we prove that $F(x_1, x_2, \dots, x_n)$ changes sign when any two vectors are interchanged, then this will establish that it coincides with the oriented volume of the vector system $x_1, x_2, \dots, x_n$. But the oriented volume coincides with the determinant of a matrix in which the coordinates of vectors are contained in the rows. Since the determinant of a matrix coincides with the determinant of the transpose of the matrix, the proof of the theorem is complete.

Obviously it suffices to consider only the case where two adjacent vectors are interchanged, for permutation of any two vectors always reduces to an odd number of permutations of adjacent vectors. The proof of this fact was given in Section 38.

Suppose vectors x_i and x_{i+1} are interchanged. We establish a 1-1 correspondence between the minors in the chosen rows of the original determinant and of the determinant with interchanged columns. We denote by ω a set of column indices defining a minor. The following cases are possible:

- (1) $i, i + 1 \in \omega$,
- (2) $i, i + 1 \notin \omega$,
- (3) $i \in \omega, i + 1 \notin \omega$,
- (4) $i + 1 \in \omega, i \notin \omega$.

In cases (1) and (2) each minor is assigned a minor on the columns with ω and in cases (3) and (4) a minor on the columns with the set of indices obtained from ω by replacing i by $i + 1$ and $i + 1$ by i respectively.

Note that in all cases the corresponding minors are defined by the same set of elements. Moreover, in cases (2) to (4) they coincide and in case (1) they only differ in sign, since each of them is obtained from the other by interchanging two columns. For similar reasons, the corresponding complementary minors differ in sign in case (2) and coincide in the remaining cases. The algebraic adjuncts and complementary minors coincide up to a sign which depends on the parity of the sum of the indices of the rows and columns containing the minor. They are the same in cases (1) and (2) and differ by unity in cases (3) and (4).

Comparing now the corresponding terms of $F(x_1, x_2, \dots, x_n)$ and those of the function resulting from permuting x_i and x_{i+1} we notice that they coincide up to a sign. Consequently, if two adjacent vectors are interchanged $F(x_1, x_2, \dots, x_n)$ changes sign.

The theorem is often used where only one row or one column is chosen. The determinant of a 1×1 matrix coincides with its only element. Therefore the minor at the intersection of the i th row and j th column is equal to the element a_{ij} . Denote by A_{ij} the algebraic adjunct of a_{ij} . By Laplace's theorem, for every i

$$a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in} = d. \quad (40.5)$$

This formula is called the *expansion of the determinant with respect to the i th row*. Similarly for every j

$$a_{1j}A_{1j} + a_{2j}A_{2j} + \dots + a_{nj}A_{nj} = d, \quad (40.6)$$

which gives the *expansion of the determinant with respect to the j th column*.

We replace in (40.5) the elements of the i th row by a collection of n arbitrary numbers $b_1, b_2, \dots, b_n$. The expression

$$b_1A_{i1} + b_2A_{i2} + \dots + b_nA_{in}$$

is the expansion with respect to the i th row of the determinant

$$\det \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \dots & \dots & \dots & \dots \\ b_1 & b_2 & \dots & b_n \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix} \quad (40.7)$$

obtained from the determinant d by replacing the i th row with the row of the numbers $b_1, b_2, \dots, b_n$. We now take as these numbers the elements of the k th row of d , with $k \neq i$. The corresponding determinant (40.7) is zero since it has two equal rows. Consequently,

$$a_{k1}A_{i1} + a_{k2}A_{i2} + \dots + a_{kn}A_{in} = 0, \quad k \neq i. \quad (40.8)$$

Similarly

$$a_{1k}A_{1j} + a_{2k}A_{2j} + \dots + a_{nk}A_{nj} = 0, \quad k \neq j. \quad (40.9)$$

So the sum of the products of all elements of any row (column) of a determinant by the algebraic adjuncts of the corresponding elements of another row (column) of the same determinant is zero.

In conclusion note that the entire theory of determinants can be extended without change to the case of complex matrices. The only thing lost is the visualization associated with the concept of volume.

Exercises

1. Write expressions for determinants of the second and third orders in terms of matrix elements. Compare them with expression (34.6).

2. Write Hadamard's inequality for the determinant of matrices A and A' .

3. A determinant of the n th order all elements of which are equal to unity in absolute value equals $n^{n/2}$. Prove that its rows (columns) form an orthogonal basis.

4. Find the determinant whose elements satisfy the conditions $a_{ij} = 0$ for $i > j$ ($i < j$, $i \geq j$, $i \leq j$).

5. The elements of a determinant satisfy the conditions $a_{ij} = 0$ for $i > k$ and $j \leq k$. Prove that the determinant is the product of the principal minor of order k and its complementary minor.

6. Let the elements of a complex matrix A satisfy the conditions $a_{ij} = \bar{a}_{ji}$ for all i and j . Prove that the determinant of such a matrix is a real number.

41. Linear dependence and determinants

One of the most common applications of determinants is in problems connected with linear dependence. Given m vectors $x_1, x_2, \dots, x_m$ in a space K_n of dimension n , determine their basis. Choose some basis in K_n and consider the rectangular array

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad (41.1)$$

where the rows represent the coordinates of the given vectors in the chosen basis.

Such an array is called a *rectangular matrix*. As before the first index of an element a_{ij} stands for the number of the matrix row the element is in, and the second index is the number of the column. If we want to stress what number of rows and columns a matrix A has, we shall write $A (m \times n)$ or say that the matrix A is an m

by n matrix. A matrix A ($n \times n$) will as before be called a square n by n matrix. Along with the matrix A we shall consider its transpose A' . If A is m by n , then A' is n by m .

In a rectangular matrix A ($m \times n$) it is again possible to indicate different minors whose order of course does not exceed the smaller of the numbers m and n . If A has not only zero elements, then the highest order r of nonzero minors is said to be the *rank* of A . Any nonzero minor of order r is called a *basis minor*, and its rows and columns are *basis rows* and *columns*. It is clear that there may be more basis minors than one. The rank of a zero matrix is zero by definition.

We shall regard the rows of A as vectors. It is obvious that if we find a basis of these row vectors, then the corresponding vectors of K_n will form a basis of vectors $x_1, x_2, \dots, x_m$.

Theorem 41.1. *Any basis rows of a matrix form a basis of row vectors of that matrix.*

Proof. To see that the theorem is true it is necessary to show that basis rows are linearly independent and that any matrix row is linearly expressible in terms of them.

If basis rows were linearly dependent, then one of them would be linearly expressible in terms of the remaining basis rows. But then the basis minor would be equal to zero, which contradicts the hypothesis.

Now add to the basis rows any other row of A . Then by the definition of a basis minor all minors of order $r + 1$ in those rows will be zero. Suppose the rows are linearly independent. By supplementing them to a basis we construct some square matrix whose determinant must not be zero. But, on the other hand, expanding that determinant with respect to the original $r + 1$ rows we conclude that it is zero. The resulting contradiction means that any row in A is linearly expressible in terms of the basis rows.

The theorem reduces the problem of finding a basis of a system of vectors to that of finding a basis minor of a matrix. Since the determinant of the transpose of a matrix coincides with that of the original matrix, it is clear that Theorem 41.1 is true not only for the rows but also for the columns. This means that for any rectangular matrix the rank of its system of row vectors equals the rank of its system of column vectors, a fact that is not obvious if we have in mind only the concept of the rank of a system of vectors.

In a space with scalar product the linear dependence or independence of a system of vectors $x_1, x_2, \dots, x_m$ can be established without expanding with respect to a basis. Consider the determinant

$$G(x_1, x_2, \dots, x_m) = \det \begin{pmatrix} (x_1, x_1) & (x_1, x_2) & \dots & (x_1, x_m) \\ (x_2, x_1) & (x_2, x_2) & \dots & (x_2, x_m) \\ \dots & \dots & \dots & \dots \\ (x_m, x_1) & (x_m, x_2) & \dots & (x_m, x_m) \end{pmatrix}$$

called the *Gram determinant* or *Gramian* of the system of vectors^s $x_1, x_2, \dots, x_m$.

Theorem 41.2. *A system of vectors is linearly dependent if and only if its Gramian is zero.*

Proof. Let $x_1, x_2, \dots, x_m$ be a linearly dependent system of vectors. Then there are numbers $\alpha_1, \alpha_2, \dots, \alpha_m$, not all zero, such that

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_m x_m = 0.$$

Performing scalar multiplication of this equation by x_i for every i we conclude that the columns of the Gramian are also linearly dependent, i.e. the Gramian is zero.

Suppose now that the Gramian is zero. Then its columns are linearly dependent, i.e. there are numbers $\alpha_1, \alpha_2, \dots, \alpha_m$, not all zero, such that

$$\alpha_1 (x_i, x_1) + \alpha_2 (x_i, x_2) + \dots + \alpha_m (x_i, x_m) = 0$$

for every i . We rewrite these equations as follows:

$$(x_i, \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_m x_m) = 0.$$

Multiplying them termwise by $\alpha_1, \alpha_2, \dots, \alpha_m$ and adding we get

$$|\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_m x_m|^2 = 0.$$

This means that

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_m x_m = 0,$$

i.e. that the vectors $x_1, x_2, \dots, x_m$ are linearly dependent.

Exercises

1. What is a matrix all of whose minors are zero?
2. For a square matrix, are its basis rows and basis columns equivalent systems of vectors?
3. Do the elementary transformations of its rows and columns, discussed in Section 15, affect the rank of a matrix?
4. Prove the inequality

$$0 \leq G(x_1, x_2, \dots, x_n) \leq \prod_{i=1}^n (x_i, x_i).$$

In what cases the inequality becomes an equation?

5. It is obvious that

$$\det \begin{pmatrix} \sqrt{2} & 1 \\ 2 & \sqrt{2} \end{pmatrix} = 0.$$

Prove that for any approximation of the number $\sqrt{2}$ by a finite decimal fraction p

$$\det \begin{pmatrix} p & 1 \\ 2 & p \end{pmatrix} \neq 0.$$

42. Calculation of determinants

A straightforward calculation of a determinant using its explicit expression in terms of matrix elements is rarely employed in practice because of its being laborious. A determinant of the n th order consists of $n!$ terms and for each term to be calculated and added to the other terms it is necessary to carry out $n \cdot n!$ arithmetical operations. Even carrying out all these computations on a modern computer performing 10^6 arithmetical operations per second to compute the determinant of, for example, the hundredth order would require many million years.

One of the most efficient ways of calculating determinants is based on the following idea. Let a_{kp} be a nonzero element in a matrix A . We call it the *leading element*. Adding to any i th row, $i \neq k$, a k th row multiplied by an arbitrary number α_i is known to leave the determinant unaffected. Take

$$\alpha_i = -\frac{a_{ip}}{a_{kp}}$$

and carry out the indicated procedure for every $i \neq k$. All the elements of the p th column of the new matrix, except the leading one, will then be zero. Expanding the new determinant with respect to the p th column we reduce calculating the determinant of the n th order to calculating a single determinant of order $(n - 1)$. We proceed in a similar way with this determinant and so on.

This algorithm is called the *Gauss method*. To calculate the determinant of the n th order by this method would require carrying out a total of $2n^3/3$ arithmetical operations. Now the determinant of the hundredth order could be computed, on a computer performing 10^6 arithmetical operations per second, in less than a second.

In conclusion note that with arithmetical operations approximately performed and information approximately given, the results of computing determinants should be regarded with some caution. If conclusions about linear dependence or independence of a system of vectors are made only on the strength of the determinant being zero or nonzero, then in the presence of instability pointed out in Section 22 they may turn out to be false. This should be borne in mind whenever a determinant is used.

Exercises

1. What accounts for a faster rate of calculation of a determinant by the Gauss method as compared with straightforward calculation?

2. Let all elements of a determinant equal at most unity in absolute value and suppose that in calculating each element we make an error of an order of ϵ . For what n does a straightforward calculation of the determinant make sense in terms of accuracy?

3. Construct an algorithm for calculating the rank of a rectangular matrix, using the Gauss method. What is the result of applying this algorithm to approximate calculations?

The Straight Line and the Plane in Vector Space

43. The equations of a straight line and of a plane

The main object of our immediate studies are the straight line and the plane in spaces of directed line segments. Given some coordinate system, the coordinates of the points on the straight line or in the plane can no longer be arbitrary and must satisfy certain relations. Now we proceed to derive those relations.

Given in the plane a fixed Cartesian x, y system and a straight line L , consider a nonzero vector

$$n = (A, B) \quad (43.1)$$

perpendicular to L . It is obvious that all other vectors perpendicular to L will be collinear with n .

Take a point $M_0(x_0, y_0)$ on L . All points $M(x, y)$ of L and only those points possess the property that the vectors $\overrightarrow{M_0M}$ and n are perpendicular, i.e.

$$(\overrightarrow{M_0M}, n) = 0. \quad (43.2)$$

Since

$$\overrightarrow{M_0M} = (x - x_0, y - y_0),$$

from (43.1) and (43.2) it follows that

$$A(x - x_0) + B(y - y_0) = 0.$$

Letting

$$-Ax_0 - By_0 = C,$$

we conclude that in the given x, y system the coordinates of the points of L and only those coordinates satisfy

$$Ax + By + C = 0. \quad (43.3)$$

Among the numbers A and B there is a nonzero one. Therefore equation (43.3) will be called a *first-degree equation* in variables x and y .

We now prove that any first-degree equation (43.3) defines relative to a fixed x, y coordinate system some straight line. Since (43.3) is a first-degree equation, of the constants A and B at least one is not

zero. Hence (43.3) has at least one solution x_0, y_0 , for example,

$$x_0 = -\frac{AC}{A^2 + B^2}, \quad y_0 = -\frac{BC}{A^2 + B^2},$$

with

$$Ax_0 + By_0 + C = 0.$$

Subtracting from (43.3) this identity yields

$$A(x - x_0) + B(y - y_0) = 0$$

equivalent to (43.3). But it means that any point $M(x, y)$ whose coordinates satisfy the given equation (43.3) is on the straight line passing through $M_0(x_0, y_0)$ and perpendicular to vector (43.1).

So, given a fixed coordinate system in the plane, any first-degree equation defines a straight line and the coordinates of the points of any straight line satisfy a first-degree equation. Equation (43.3) is called the *general equation of a straight line in the plane* and the vector n in (43.1) is the *normal vector* to that straight line.

Without any fundamental changes we can carry out a study of the plane in space. Fix a Cartesian x, y, z system and consider a plane π . Again take a nonzero vector

$$n = (A, B, C) \quad (43.4)$$

perpendicular to π . Repeating the above reasoning we conclude that all points $M(x, y, z)$ of π and only those points satisfy the equation

$$Ax + By + Cz + D = 0 \quad (43.5)$$

which is also called a *first-degree equation* in variables x, y and z .

If we again consider an arbitrary first-degree equation (43.5), then we shall see that it also has at least one solution x_0, y_0, z_0 , for example,

$$x_0 = -\frac{AD}{A^2 + B^2 + C^2}, \quad y_0 = -\frac{BD}{A^2 + B^2 + C^2}, \quad z_0 = -\frac{CD}{A^2 + B^2 + C^2}.$$

We further establish that any point $M(x, y, z)$ whose coordinates satisfy the given equation (43.5) is in the plane through the point $M_0(x_0, y_0, z_0)$ perpendicular to vector (43.4).

Thus, given a fixed coordinate system in space, any first-degree equation defines a plane and the coordinates of the points of any plane satisfy a first-degree equation. Equation (43.5) is called the *general equation of a plane in space* and the vector n in (43.4) is the *normal vector* to that plane.

We shall now show how two general equations defining the same straight line or plane are related. Suppose for definiteness that we are given two equations of a plane π

$$\begin{aligned} A_1x + B_1y + C_1z + D_1 &= 0, \\ A_2x + B_2y + C_2z + D_2 &= 0. \end{aligned} \quad (43.6)$$

The vectors

$$n_1 = (A_1, B_1, C_1), \quad n_2 = (A_2, B_2, C_2)$$

are perpendicular to the same plane π and therefore they are collinear. Since furthermore they are nonzero, there is a number t such that, for example,

$$n_1 = tn_2$$

or

$$A_1 = tA_2, \quad B_1 = tB_2, \quad C_1 = tC_2. \quad (43.7)$$

Multiplying the second of the equations (43.6) by t and subtracting from it the first we get by virtue of (43.7)

$$D_1 = tD_2.$$

Consequently, the coefficients of the general equations defining the same straight line or plane are proportional.

A general equation is said to be *complete* if all of its coefficients are nonzero. An equation which is not complete is called *incomplete*. Consider the complete equation of a straight line (43.3). Since all the coefficients are nonzero, that equation can be written as

$$\frac{x}{-\frac{C}{A}} + \frac{y}{-\frac{C}{B}} = 1.$$

If we let

$$a = -\frac{C}{A}, \quad b = -\frac{C}{B},$$

then we obtain a new equation of a straight line

$$\frac{x}{a} + \frac{y}{b} = 1.$$

This is the *intercept form of the equation of a straight line*. The numbers a and b have a simple geometrical meaning. They are equal to the magnitudes of the intercepts of the straight line on the coordinate semiaxes (Fig. 43.1). Of course the complete equation of a plane can be reduced to a similar form

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$$

Different incomplete equations define special cases of location of a straight line and a plane. It is useful to remember them since they occur fairly often. For example, if $C = 0$, equation (43.3) defines a

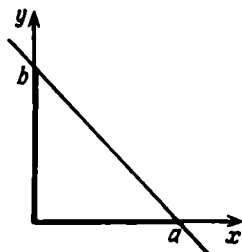


Fig. 43.1

straight line through the origin, if $B = C = 0$, the straight line coincides with the y axis and so on. If $A = 0$, equation (43.5) defines a plane parallel to the x axis, if $A = B = D = 0$, the plane coincides with the x, y plane and so on.

Any nonzero vector parallel to a straight line will be called its *direction vector*. Consider, for example, the case of a space and find the equation of a straight line through a given point $M_0(x_0, y_0, z_0)$ with a given direction vector

$$q = (l, m, n).$$

Obviously, $M(x, y, z)$ is on the straight line if and only if the vectors $\overrightarrow{M_0M}$ and q are collinear, i.e. if and only if the coordinates of these vectors are proportional, i.e.

$$\frac{x-x_0}{l} = \frac{y-y_0}{m} = \frac{z-z_0}{n}. \quad (43.8)$$

These equations are precisely the desired equations of a straight line. They are usually called the *canonical equations of a straight line*. It is clear that in the case of a straight line in the plane the equation will be of the form

$$\frac{x-x_0}{l} = \frac{y-y_0}{m} \quad (43.9)$$

if the straight line passes through the point $M_0(x_0, y_0)$ and has a direction vector $q = (l, m)$.

Using the canonical equations it is easy to obtain the equation of a straight line through two given points M_0 and M_1 . To do this it suffices to take as a direction vector the vector $\overrightarrow{M_0M_1}$, express its coordinates in terms of the coordinates of M_0 and M_1 and substitute them in equations (43.8) and (43.9). For example, in the case of a straight line in the plane we shall have the following equation:

$$\frac{x-x_0}{x_1-x_0} = \frac{y-y_0}{y_1-y_0},$$

and in the case of a space we shall have

$$\frac{x-x_0}{x_1-x_0} = \frac{y-y_0}{y_1-y_0} = \frac{z-z_0}{z_1-z_0}.$$

Notice that in canonical equations of a straight line the denominators may turn out to be zero. Therefore in what follows the proportion $a/b = c/d$ will be understood to be the equation $ad = bc$. Consequently, the vanishing of one of the coordinates of a direction vector implies the vanishing of the corresponding numerator in the canonical equations.

To represent a straight line analytically it is common practice to write the coordinates of its points as functions of some auxiliary

parameter t . Take as the parameter t each of the equal relations of (43.8) and (43.9). Then for the case of a space we shall have the following equations of a straight line:

$$\begin{aligned}x &= x_0 + lt, \\y &= y_0 + mt, \\z &= z_0 + nt\end{aligned}\tag{43.10}$$

and similar equations for the case of a straight line in the plane

$$\begin{aligned}x &= x_0 + lt, \\y &= y_0 + mt.\end{aligned}\tag{43.11}$$

These are called the *parametric equations of a straight line*. Assigning to t different values yields different points of a straight line.

Providing great possibilities and convenience in writing the various equations of a straight line and of a plane is the use of the concept of determinant. We introduce, for example, the equation of a plane through three different points not on the same straight line. So, let $M_1(x_1, y_1, z_1)$, $M_2(x_2, y_2, z_2)$ and $M_3(x_3, y_3, z_3)$ be points. Since they are not on the same straight line, the vectors

$$\overrightarrow{M_1M_2} = (x_2 - x_1, y_2 - y_1, z_2 - z_1), \quad \overrightarrow{M_1M_3} = (x_3 - x_1, y_3 - y_1, z_3 - z_1)$$

are not collinear. Therefore $M(x, y, z)$ is in the same plane with M_1 , M_2 and M_3 if and only if the vectors $\overrightarrow{M_1M_2}$ and $\overrightarrow{M_1M_3}$ and

$$\overrightarrow{M_1M} = (x - x_1, y - y_1, z - z_1)$$

are coplanar, i.e. if and only if the determinant made up of their coordinates is zero. Consequently,

$$\det \begin{pmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{pmatrix} = 0\tag{43.12}$$

is the equation of the desired plane through three given points.

Consider, finally, the equation of a straight line in space through a given point perpendicular to two nonparallel straight lines. Suppose both straight lines are given by their canonical equations

$$\begin{aligned}\frac{x - x_1}{l_1} &= \frac{y - y_1}{m_1} = \frac{z - z_1}{n_1}, \\ \frac{x - x_2}{l_2} &= \frac{y - y_2}{m_2} = \frac{z - z_2}{n_2}.\end{aligned}$$

The direction vector q of the desired straight line must be perpendicular to two vectors

$$q_1 = (l_1, m_1, n_1), \quad q_2 = (l_2, m_2, n_2).$$

These are not collinear, and therefore it is possible to take as q , for example, the vector product $[q_1, q_2]$. Recalling the expression for the coordinates of the vector product in terms of the coordinates of the factors and using for notation the determinants of the second order we get

$$q = \left(\det \begin{pmatrix} m_1 & n_1 \\ m_2 & n_2 \end{pmatrix}, \det \begin{pmatrix} n_1 & l_1 \\ n_2 & l_2 \end{pmatrix}, \det \begin{pmatrix} l_1 & m_1 \\ l_2 & m_2 \end{pmatrix} \right).$$

If the desired straight line passes through the point $M_0(x_0, y_0, z_0)$, then its canonical equations will be as follows:

$$\frac{x-x_0}{\det \begin{pmatrix} m_1 & n_1 \\ m_2 & n_2 \end{pmatrix}} = \frac{y-y_0}{\det \begin{pmatrix} n_1 & l_1 \\ n_2 & l_2 \end{pmatrix}} = \frac{z-z_0}{\det \begin{pmatrix} l_1 & m_1 \\ l_2 & m_2 \end{pmatrix}}.$$

Of course, fundamentally many conclusions concerning the equations of a straight line and of a plane remain valid for any affine coordinate system. Our desire to use Cartesian coordinate systems is mainly due to their ensuring simpler calculations.

Exercises

1. Write the equation of a straight line in the plane through two given points using a second-order determinant. Compare it with (43.12).

2. Is it correct to say that (43.12) is always the equation of a plane?

3. By analogy with equations (43.10) write the parametric equations of a plane in space. How many parameters must they contain?

4. Find the coordinates of the normal vector to the plane through three given points not on the same straight line.

5. What is the locus of points in space whose coordinates are solutions of a system of two linear algebraic equations in three unknowns?

44. Relative positions

Simultaneous consideration of several straight lines and planes gives rise to various problems, first of all to problems of determining their relative positions.

Let two crossing planes be given in space by their general equations

$$A_1x + B_1y + C_1z + D_1 = 0,$$

$$A_2x + B_2y + C_2z + D_2 = 0.$$

They form two adjacent angles adding up to two right angles. We find one of them. The vectors $n_1 = (A_1, B_1, C_1)$ and $n_2 = (A_2, B_2, C_2)$ are normal ones, so determining the angle between the planes reduces to determining the angle φ between n_1 and n_2 . By (25.5)

$$\cos \varphi = \frac{A_1 A_2 + B_1 B_2 + C_1 C_2}{(A_1^2 + B_1^2 + C_1^2)^{1/2} (A_2^2 + B_2^2 + C_2^2)^{1/2}}.$$

Quite similarly derived is the formula for the angle between two straight lines in the plane given by their general equations

$$A_1 x + B_1 y + C_1 = 0,$$

$$A_2 x + B_2 y + C_2 = 0.$$

One of the angles φ made by these straight lines is calculated from the formula

$$\cos \varphi = \frac{A_1 A_2 + B_1 B_2}{(A_1^2 + B_1^2)^{1/2} (A_2^2 + B_2^2)^{1/2}}.$$

The condition of the parallelism of straight lines given by their general equations is that of the collinearity of their normal vectors, i.e. the condition of the proportionality of their coordinates

$$\frac{A_1}{A_2} = \frac{B_1}{B_2}.$$

The condition that straight lines should be perpendicular coincides with the condition that $\cos \varphi = 0$ or equivalently with the condition that

$$A_1 A_2 + B_1 B_2 = 0.$$

Of course, of a similar form is the condition

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

of the parallelism of planes and the condition

$$A_1 A_2 + B_1 B_2 + C_1 C_2 = 0$$

of the perpendicularity of planes also given by their general equations.

Suppose now that two straight lines, for example in space, are given by their canonical equations

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1},$$

$$\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2}.$$

Since the direction vectors of those straight lines are vectors $q_1 = (l_1, m_1, n_1)$ and $q_2 = (l_2, m_2, n_2)$, we conclude again that one of the angles φ between the straight lines will coincide with that be-

tween q_1 and q_2 . Consequently,

$$\cos \varphi = \frac{l_1 l_2 + m_1 m_2 + n_1 n_2}{(l_1^2 + m_1^2 + n_1^2)^{1/2} (l_2^2 + m_2^2 + n_2^2)^{1/2}}.$$

Accordingly, the proportionality of the coordinates

$$\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$$

is the condition of the parallelism of the straight lines, and the equation

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

is the condition of their perpendicularity.

It is clear that if straight lines and planes are given in such a way that their direction vector or normal vector is explicitly indicated, then the finding of the angle between them always reduces to the finding of the angle between these vectors. Suppose, for example, a plane π is given in space by its general equation

$$Ax + By + Cz + D = 0$$

and a straight line L is given by its canonical equation

$$\frac{x-x_0}{l} = \frac{y-y_0}{m} = \frac{z-z_0}{n}.$$

Since the angle φ between the straight line and the plane is complementary to the angle ψ between the direction vector of the straight line and the normal vector to the plane (Fig. 44.1), we have

$$\sin \varphi = \frac{|Al + Bm + Cn|}{(A^2 + B^2 + C^2)^{1/2} (l^2 + m^2 + n^2)^{1/2}}.$$

It is obvious that

$$Al + Bm + Cn = 0,$$

the condition of the parallelism of the straight line and the plane, and that

$$\frac{A}{l} = \frac{B}{m} = \frac{C}{n},$$

the condition of the perpendicularity of the straight line and the plane.

Assigning a straight line and a plane in the form of general equations allows us to solve very efficiently an important problem, that of calculating the distance from a point to the straight line and

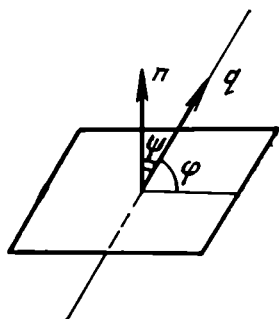


Fig. 44.1

from a point to the plane. The derivation of formulas is quite similar in both cases and we again confine ourselves to a detailed discussion of just one of them.

Let π be a plane in space, given by its general equation (43.5). Take a point $M_0(x_0, y_0, z_0)$. Drop from M_0 a perpendicular to the plane and denote by $M_1(x_1, y_1, z_1)$ its foot. It is clear that the distance $\rho(M_0, \pi)$ from M_0 to the plane is equal to the length of $\overrightarrow{M_0M_1}$.

The vectors $n = (A, B, C)$ and $\overrightarrow{M_0M_1} = (x_1 - x_0, y_1 - y_0, z_1 - z_0)$ are perpendicular to the same plane and hence collinear. Therefore there is a number t such that $\overrightarrow{M_0M_1} = tn$, i.e.

$$x_1 - x_0 = tA,$$

$$y_1 - y_0 = tB,$$

$$z_1 - z_0 = tC.$$

The point $M_1(x_1, y_1, z_1)$ is in π . Expressing its coordinates in terms of the relations obtained and substituting them in the equation of a plane we find

$$t = -\frac{Ax_0 + By_0 + Cz_0 + D}{A^2 + B^2 + C^2}.$$

But the length of n is $(A^2 + B^2 + C^2)^{1/2}$, and therefore $|\overrightarrow{M_0M_1}| = |t| (A^2 + B^2 + C^2)^{1/2}$. Consequently,

$$\rho(M_0, \pi) = \frac{|Ax_0 + By_0 + Cz_0 + D|}{(A^2 + B^2 + C^2)^{1/2}}.$$

In particular

$$\rho(0, \pi) = \frac{|D|}{(A^2 + B^2 + C^2)^{1/2}}.$$

Along with the general equation (43.5) of a plane we consider the following of its equations:

$$\pm (A^2 + B^2 + C^2)^{-1/2} (Ax + By + Cz + D) = 0.$$

Of the two possible signs at the left we choose the one opposite to the sign of D . If $D = 0$, then we choose any sign. Then the free term of that equation is a nonpositive number $-p$, and the coefficients of x , y and z are the cosines of the angles between the normal vector and the coordinate axes. The equation

$$x \cos \alpha + y \cos \beta + z \cos \gamma - p = 0 \quad (44.1)$$

is called the *normed equation of a plane*. It is obvious that

$$\rho(M_0, \pi) = |x_0 \cos \alpha + y_0 \cos \beta + z_0 \cos \gamma - p|,$$

$$\rho(0, \pi) = p.$$

The distance $\rho(M_0, L)$ from a point $M_0(x_0, y_0)$ to a straight line L in the plane, given by its general equation (43.3), is determined from a similar formula

$$\rho(M_0, L) = \frac{|Ax_0 + By_0 + C|}{(A^2 + B^2)^{1/2}}.$$

The *normed equation of a straight line* is like this:

$$x \cos \alpha + y \sin \alpha - p = 0. \quad (44.2)$$

Here α is the angle made by the normal vector with the x axis.

Exercises

1. Under what condition on the coordinates of normal vectors do two straight lines in the plane (three planes in space) intersect in a single point?
2. Under what condition is the straight line (43.8) in plane (43.5)?
3. Under what condition are two straight lines in space, given by their canonical equations, in the same plane?
4. Calculate the angles between a diagonal of a cube and its faces.
5. Derive the formula for the distance between a point and a straight line in space, given by its canonical equations.

45. The plane in vector space

We have repeatedly stressed that a straight line and a plane passing through an origin can be identified in spaces of directed line segments with the geometrical representation of a subspace. But in their properties they differ but little from any other straight lines and planes obtained by translating or shifting these subspaces. Wishing to extend this fact to arbitrary vector spaces we arrive at the concept of plane in vector space.

Let L be some subspace of a vector space K . Fix in K a vector x_0 . In particular it may be in L . A set H of vectors z obtained from the formula

$$z = x_0 + y, \quad (45.1)$$

where y is any vector of L , is called a *plane* in K . The vector x_0 is a *translation vector* and the subspace L is a *direction subspace*. As to H it will be said to be formed by translating L by a vector x_0 .

Formally the concept of plane includes those of straight lines and planes (in vector interpretation!) in spaces of directed line segments. But whether it possesses similar properties is as yet unknown to us.

Each vector of H can be *uniquely* represented as a sum (45.1). If $z = x_0 + y$ and $z = x_0 + y'$, where $y, y' \in L$, then it follows that $y = y'$. In addition, from (45.1) it follows that the difference of any two vectors of H is in L .

Choose in H a vector z_0 . Let $z_0 = x_0 + y_0$. Represent (45.1) as

$$z = z_0 + (y - y_0).$$

The sets of vectors y and $(y - y_0)$ describe the same subspace L . Therefore the last equation means that H may be obtained by translating L by *any* fixed vector of the plane.

A plane H is some set of vectors of K generated by a subspace L and a translation vector x_0 according to (45.1). It is a very important fact that any plane may be generated by only one subspace. Suppose that this is not the case, i.e. that there is another direction subspace L' and another translation vector x'_0 forming the same plane H . Then for any $z \in H$ we have $z = x_0 + y$, where $y \in L$, and at the same time $z = x'_0 + y'$, where $y' \in L'$. It follows that L' is a collection of vectors of K defined by the formula

$$y' = (x_0 - x'_0) + y.$$

Since the zero vector is in L' , it follows from the last formula that the vector $(x_0 - x'_0)$ is in L . But this means that L' consists of the same vectors as L .

We have already noted that the translation vector is *not uniquely* defined by a plane of course. However, here too the question of uniqueness can be dealt with in a quite natural way.

Assume that a scalar product is introduced in K . It is clear that we obtain the same plane if we take $\text{ort}_L x_0$ instead of x_0 . We may therefore assume without loss of generality that $x_0 \perp L$. The vector x_0 is then called the *orthogonal translation vector*. We can now prove that every plane is generated by only *one* translation vector.

Indeed, suppose there are two translation vectors, x'_0 and x''_0 , orthogonal to L but generating nevertheless the same plane H . Then for any vector $y' \in L$ there must be a vector $y'' \in L$ such that $x'_0 + y' = x''_0 + y''$. It follows that $x'_0 - x''_0 \in L$. But under the hypothesis $x'_0 - x''_0 \perp L$. Consequently, $x'_0 - x''_0 = 0$, i.e. $x'_0 = x''_0$.

This means in particular that in a space with scalar product any plane has only one vector orthogonal to the direction subspace.

Two planes are said to be *parallel* if the direction subspace of one is in the direction subspace of the other.

It is easy to justify this definition. Any two parallel planes H_1 and H_2 either have no vector in common or one of them is in the other. Suppose H_1 and H_2 have in common a vector z_0 . Since any plane can be obtained by translating the direction subspace by any of its vectors, both H_1 and H_2 can be obtained by translating the corresponding subspaces by a vector z_0 . But one of the subspaces is in the other and therefore one of the planes is in the other.

A subspace is a special case of a plane. It is obvious that a subspace L is parallel to any plane H obtained by translating L by some

vector x_0 . From the established property of parallel planes it follows that H coincides with L if and only if $x_0 \in L$.

Consider now two nonparallel planes H_1 and H_2 . They either have no vector in common or have a vector in common. In the former case H_1 and H_2 are called *crossing planes* and in the latter they are called *intersecting planes*.

Just as in the case of subspaces a set of vectors which are in both H_1 and H_2 is called the *intersection* of the planes and designated $H_1 \cap H_2$. Let H_1 be formed by translating a subspace L_1 and let H_2 be formed by translating a subspace L_2 . Denote

$$H = H_1 \cap H_2, \quad L = L_1 \cap L_2.$$

Theorem 45.1. *If an intersection H contains a vector z_0 , then it is a plane formed by translating an intersection L by that vector.*

Proof. Under the hypothesis of the theorem there is a vector z_0 in H . Suppose there is another vector $z_1 \in H$. We represent it as

$$z_1 = z_0 + (z_1 - z_0).$$

Now from the sequence of relations

$$z_1, z_0 \in H \rightarrow z_1 - z_0 \in L_1;$$

$$z_1, z_0 \in H_2 \rightarrow z_1 - z_0 \in L_2;$$

$$z_1 - z_0 \in L_2 \rightarrow z_1 - z_0 \in L$$

we conclude that any vector of H can be represented as a sum of z_0 and some vector of L .

Take then a vector f of L . We have

$$f \in L \rightarrow f \in L_1; f \in L_2 \rightarrow z_0 + f \in H_1; \quad z_0 + f \in H_2 \rightarrow z_0 + f \in H,$$

i.e. any vector of L translated by a vector z_0 is in H . Thus the theorem is proved.

A plane is not necessarily a subspace. Nevertheless it can be assigned *dimension* equal to that of the direction subspace. A plane of zero dimension contains only one vector, the translation vector. In determining the dimension of an intersection of planes Theorem 19.1 is useful. From Theorem 19.1 and 45.1 it follows that the dimension of an intersection H does not exceed the minimum one of the dimensions of H_1 and H_2 .

If in spaces of directed line segments two (three) vectors are given, then with some additional conditions it is possible to construct only one plane of dimension 1 (2) containing the given vectors. Those additional conditions can be formulated as follows. If two vectors are given, then they must not coincide, i.e. they must not be in the same plane of zero dimension. If three vectors are given, then they must not be in the same plane of dimension one.

Similar facts hold in an arbitrary vector space.

Let $x_0, x_1, \dots, x_k$ be vectors in a vector space. We shall say that they are in a *general position* if they are not in the same plane of dimension $k - 1$.

Theorem 45.2. *If vectors $x_0, x_1, \dots, x_k$ are in a general position, then there is a unique plane H of dimension k containing those vectors.*

Proof. Consider the vectors $x_1 - x_0, x_2 - x_0, \dots, x_k - x_0$. If they were linearly dependent, then they would be in some subspace of dimension $k - 1$ at most. Consequently, the vectors $x_0, x_1, \dots, x_k$ themselves would be in the plane obtained by translating that subspace by a vector x_0 , which contradicts the hypothesis of the theorem.

So the vectors $x_1 - x_0, x_2 - x_0, \dots, x_k - x_0$ are linearly independent. Denote by L their span. The subspace L has dimension k . By translating it by a vector x_0 we obtain some plane H of the same dimension which contains all the given vectors $x_0, x_1, \dots, x_k$.

The constructed plane H is unique. Indeed, let the vectors $x_0, x_1, \dots, x_k$ be in two planes H_1 and H_2 of dimension k . The plane remains the same if the translation vector is replaced by any other vector of the plane. We may therefore assume without loss of generality that H_1 and H_2 are obtained by translating the corresponding subspaces L_1 and L_2 by the same vector x_0 . But then it follows that both subspaces coincide since they have dimension k and contain the same linearly independent system $x_1 - x_0, x_2 - x_0, \dots, x_k - x_0$.

Exercises

1. Let H_1 and H_2 be any two planes. Define the *sum* $H_1 + H_2$ to be the set of all vectors of the form $z_1 + z_2$, where $z_1 \in H_1$ and $z_2 \in H_2$. Prove that the sum of the planes is a plane.

2. Let H be a plane and let λ be a number. Define the *product* λH of H by λ to be the set of all vectors of the form λz , where $z \in H$. Prove that λH is a plane.

3. Will the set of all planes of the same space, with the operations introduced above on them, be a vector space?

4. Prove that vectors $x_0, x_1, \dots, x_k$ are in a general position if and only if the vectors $x_1 - x_0, x_2 - x_0, \dots, x_k - x_0$ are linearly independent.

5. Prove that a plane of dimension k containing vectors of a general position $x_0, x_1, \dots, x_k$ is a subspace if and only if those vectors are linearly dependent.

46. The straight line and the hyperplane

In a vector space K of dimension m two classes of planes occupy a particular place. These are planes of dimension 1 and planes of dimension $m - 1$. Geometrically any plane of dimension 1 in spaces of directed line segments is a *straight line*. A plane of dimension $m - 1$ is a *hyperplane*.

Consider a straight line H in a vector space K . Denote by x_0 the translation vector and by q the basis vector of a one-dimensional direction subspace. Let these vectors be given by their coordinates

$$\begin{aligned}x_0 &= (x_1, x_2, \dots, x_m), \\q &= (q_1, q_2, \dots, q_m)\end{aligned}$$

relative to some basis of K . It is obvious that any vector z of H can be given by

$$z = x_0 + tq, \quad (46.1)$$

where t is some number. Therefore relation (46.1) may be assumed to be a vector equation of H in K . If z has in the same basis the coordinates

$$z = (z_1, z_2, \dots, z_m),$$

then, writing (46.1) coordinatewise, we get

$$\begin{aligned}z_1 &= x_1 + q_1 t, \\z_2 &= x_2 + q_2 t, \\&\dots \dots \dots \\z_m &= x_m + q_m t.\end{aligned} \quad (46.2)$$

Comparing now these equations with (43.10) and (43.11) it is natural to call them *parametric equations* of a straight line H . We shall say that a straight line H passes through a vector x_0 and has a *direction vector* q .

By Theorem 45.2, it is always possible to draw one and only one straight line through any two distinct vectors x_0 and y_0 . Let x_0 and y_0 be vectors given in some basis of a space K by their coordinates

$$\begin{aligned}x_0 &= (x_1, x_2, \dots, x_m), \\y_0 &= (y_1, y_2, \dots, y_m).\end{aligned}$$

Since it is possible to take, for example, the vector $y_0 - x_0$ as direction vector, equations (46.2) yield the parametric equations

$$\begin{aligned}z_1 &= x_1 + (y_1 - x_1) t, \\z_2 &= x_2 + (y_2 - x_2) t, \\&\dots \dots \dots \\z_m &= x_m + (y_m - x_m) t\end{aligned} \quad (46.3)$$

of the straight line through two given vectors.

When $t = 0$ these equations define the vector x_0 and when $t = 1$ they define the vector y_0 . If K is a real space, then the set of vectors given by (46.3) for $0 \leq t \leq 1$ is said to be the *line segment* joining vectors x_0 and y_0 . Of course, this name is associated with the geo-

metrical representation of this set in spaces of directed line segments.

Suppose H intersects some plane. Then according to the consequence arising from Theorem 45.1 the intersection is either a straight line or a single vector. If the intersection is a straight line, then it certainly coincides with the straight line H . But this means that if a straight line intersects a plane, then the straight line is either entirely in the plane or it and the plane have only one vector in common.

The concept of hyperplane makes sense in any vector space, but we shall use it only in spaces with scalar product.

Consider a hyperplane H . Let it be formed by translating an $(m-1)$ -dimensional subspace L by a vector x_0 . The orthogonal complement $L^\perp$ is in this case a one-dimensional subspace. Denote by n any one of its basis vectors. A vector z is in H if and only if the vector $z - x_0$ is in L . That condition holds in turn if and only if $z - x_0$ is orthogonal to n , i.e. if

$$(n, z - x_0) = 0. \quad (46.4)$$

Thus we have obtained an equation satisfied by all vectors of H . To give a hyperplane in the form of this equation, it suffices to indicate any vector n , orthogonal to the direction subspace, and a translation vector x_0 .

The explicit form of the equation substantially simplifies making various studies. Given vectors $n_1, n_2, \dots, n_k$ and $x_1, x_2, \dots, x_k$, investigate a plane R which is the intersection of the hyperplanes

$$\begin{aligned} (n_1, z - x_1) &= 0, \\ (n_2, z - x_2) &= 0, \\ &\dots \dots \dots \\ (n_k, z - x_k) &= 0. \end{aligned} \quad (46.5)$$

This problem may be regarded as a solution of the system of equations (46.5) for vectors z . Suppose that the intersection of the hyperplanes is not empty, i.e. that (46.5) has at least one solution z_0 . Then, as we know, the desired plane is defined by the following system as well:

$$\begin{aligned} (n_1, z - z_0) &= 0, \\ (n_2, z - z_0) &= 0, \\ &\dots \dots \dots \\ (n_k, z - z_0) &= 0 \end{aligned} \quad (46.6)$$

since any plane remains unaffected if the translation vector is replaced by any other vector of the plane.

A vector $y = z - z_0$ is an arbitrary vector of the intersection L of the direction subspaces of all k hyperplanes. It is obvious that vectors y of the subspace L satisfy the following system:

$$\begin{aligned}(n_1, y) &= 0, \\(n_2, y) &= 0, \\&\dots\dots\dots \\(n_k, y) &= 0.\end{aligned}\tag{46.7}$$

Giving the intersection L as system (46.7) allows an easy solution of the question of the dimension of L . As is seen from the system itself, the subspace L is the orthogonal complement of the span of a system of vectors $n_1, n_2, \dots, n_k$. Let r be the rank of that system. Then the dimension of L and hence of R is $m - r$, where m is the dimension of the space. In particular, if $n_1, n_2, \dots, n_k$ are linearly independent, then the dimension of plane (46.5) is $m - k$. It is of course assumed that the plane exists, i.e. that system (46.5) has at least one solution. To give the subspace L defined by system (46.7) it suffices to indicate its basis, i.e. any system of $m - r$ linearly independent vectors orthogonal to $n_1, n_2, \dots, n_k$.

Equation (46.4) of a hyperplane can be written in a somewhat different form. Let $(n, x_0) = b$. Then

$$(n, z) = b\tag{46.8}$$

will define the same hyperplane as equation (46.4). Note that the general equations (43.3) and (43.5) of a straight line and of a plane are essentially given in the same form. It is important to stress that *any* equation of the form (46.8) can be reduced to the form (46.4) by a suitably chosen vector x_0 . To do this it suffices, for example, to take x_0 in the following form:

$$x_0 = \alpha n.$$

Substituting this expression in (46.4) and comparing with (46.8) we conclude that we must have

$$\alpha = \frac{b}{(n, n)}.$$

We can now conclude that if a system of the form (46.5) defines some plane, then the same plane may also be defined by a system of the following form:

$$\begin{aligned}(n_1, z) &= b_1, \\(n_2, z) &= b_2, \\&\dots\dots\dots \\(n_k, z) &= b_k\end{aligned}\tag{46.9}$$

for appropriate numbers $b_1, b_2, \dots, b_k$. The converse is obviously true too. System (46.9) defines the same plane as (46.5) does for appropriate vectors $x_1, x_2, \dots, x_k$.

A straight line and a plane are hyperplanes in spaces V_2 and V_3 respectively. We have established earlier the relation of the distance from a point to these hyperplanes to the result of a substitution of the coordinates of the point in the general equations. There is a similar relation in the case of an arbitrary hyperplane.

Let H be a hyperplane given by equation (46.4). Denote as before by $\rho(v, H)$ the distance between a vector v and H . Taking into account equation (46.4) we get

$$(n, v - x_0) = (n, v - x_0) - (n, z - x_0) = (n, v - z)$$

for any vector z in H . According to the Cauchy-Buniakowski-Schwarz inequality

$$|(n, v - x_0)| \leq |n| |v - z|. \quad (46.10)$$

Therefore

$$|v - z| \geq \frac{|(n, v - x_0)|}{|n|}.$$

If we show that H contains one and only one vector, z^* , for which (46.10) becomes an equation, then this will mean, first, that

$$\rho(v, H) = \frac{|(n, v - x_0)|}{|n|} \quad (46.11)$$

and, second, that the value $\rho(v, H)$ is attained only on that single vector z^* .

Denote by L the direction subspace of H . It is clear that any vector orthogonal to L is collinear with n , and vice versa. Inequality (46.10) becomes an equation if and only if n and $v - z$ are collinear, i.e. $v - z = \alpha n$ for some number α . Let equality hold for two vectors, z_1 and z_2 , of H , i.e.

$$v - z_1 = \alpha_1 n,$$

$$v - z_2 = \alpha_2 n.$$

It follows that

$$z_1 - z_2 = (\alpha_2 - \alpha_1) n.$$

Consequently, $z_1 - z_2 \perp L$. But $z_1 - z_2 \in L$ as the difference of two vectors of a hyperplane. Therefore $z_1 - z_2 = 0$ or $z_1 = z_2$.

Denote by z_0 a vector in H orthogonal to L . As we know, that vector exists and is unique. We write a vector z as a sum

$$z = z_0 + y,$$

where $y \in L$. A vector v can be represented as

$$v = f + s,$$

where $f \in L$, i.e. $s \perp L$. Now

$$v - z = (s - z_0) + (f - y).$$

If we take $z = z_0 + f$, then

$$v - z = s - z_0.$$

The vector $h = s - z_0$ is orthogonal to L , and formula (46.11) is established.

We have simultaneously proved that any vector v of a space can be represented *uniquely* as a sum

$$v = z + h,$$

where z is a vector in H and h is orthogonal to L . By analogy with spaces of directed line segments a vector z in the decomposition is called the *projection* of v onto H and h is called the *perpendicular* from v to H . The process of obtaining a vector z from v is termed the *projection* of v onto H . If a hyperplane is given by (46.4), then n is the *normal vector* to the hyperplane. Given vectors x_0 and n there is only *one* hyperplane containing the vector x_0 and orthogonal to the vector n .

Exercises

1. Prove that any plane distinct from the entire space may be given as the intersection of hyperplanes (46.9).

2. Prove that a sum of hyperplanes is a hyperplane if and only if the hyperplanes to be added are parallel.

3. Prove that a product of a hyperplane by a nonzero number is a hyperplane.

4. Under what conditions are a straight line and a hyperplane parallel?

5. Derive the formula for the distance between a vector and a straight line given by equation (46.1).

47. The half-space

Associated with the concepts of a straight line and of the hyperplane of a body is the notion of the so-called convex sets. Since these are widely used in various fields of mathematics, we shall examine some of them.

A set of vectors of a real vector space is said to be *convex* if together with every pair of vectors it contains the entire line segment joining them.

Examples of convex sets are a single vector, a line segment, a straight line, a subspace, a plane, a hyperplane and many others.

Suppose a hyperplane is given in a real space by

$$(n, z) - b = 0.$$

A set of vectors z satisfying

$$(n, z) - b < 0 \quad (47.1)$$

or

$$(n, z) - b > 0 \quad (47.2)$$

is called an *open half-space*. Half-space (47.1) is *negative* and (47.2) is *positive*.

Theorem 47.1. *A half-space is a convex set.*

Proof. Take two vectors, x_0 and y_0 , and let

$$\Phi_1 = (n, x_0) - b, \quad \Phi_2 = (n, y_0) - b.$$

If z is any vector of the straight line through x_0 and y_0 , then

$$z = x_0 + t(y_0 - x_0).$$

When $0 \leq t \leq 1$ we obtain vectors of the line segment joining x_0 and y_0 . We have

$$(n, z) - b = \Phi_1(1 - t) + \Phi_2 t. \quad (47.3)$$

If Φ_1 and Φ_2 have the same sign, i.e. x_0 and y_0 are in the same half-space, then the right-hand side of (47.3) will also have the same sign for all values of t satisfying $0 \leq t \leq 1$.

Thus any hyperplane divides a vector space into three nonintersecting convex sets, the hyperplane itself and two open half-spaces.

Suppose that x_0 and y_0 are in different half-spaces, i.e. that Φ_1 and Φ_2 have different signs. Formal transformation of relation (47.3) results in the following equation:

$$(n, z) - b = (\Phi_2 - \Phi_1) \left(t - \frac{1}{1 - \Phi_2/\Phi_1} \right).$$

It follows that the straight line through x_0 and y_0 intersects the hyperplane. The intersection is defined by

$$t = \frac{1}{1 - \Phi_2/\Phi_1}$$

satisfying $0 \leq t \leq 1$. So

If two vectors are in different half-spaces, then the line segment joining those vectors intersects the hyperplane determining the half-spaces.

Considering this property it is easy to understand what the half-spaces of spaces of directed line segments are. In the plane the terminal points of vectors of a half-space are on the same side of a straight line and in space they are on the same side of a plane.

Along with open half-spaces *closed half-spaces* are not infrequently considered. They are defined as sets of vectors z satisfying

$$(n, z) - b \leq 0 \quad (47.4)$$

or

$$(n, z) - b \geq 0. \quad (47.5)$$

Half-space (47.4) is *nonpositive* and (47.5) is *nonnegative*. Of course, closed half-spaces are also convex sets.

Theorem 47.2. *An intersection of convex sets is a convex set.*

Proof. Obviously it suffices to consider the case of two sets U_1 and U_2 . Let $U = U_1 \cap U_2$ be their intersection. Take any two vectors x_0 and y_0 in U and denote by S the line segment joining them. The vectors x_0 and y_0 are in both U_1 and U_2 . Therefore in view of the convexity of U_1 and U_2 the line segment S is entirely in both U_1 and U_2 , i.e. it is in U .

This theorem plays an important part in the study of convex sets. In particular, it allows us to say that a nonempty set of vectors z , each satisfying the system of inequalities

$$\begin{aligned} (n_1, z) - f_1 &\geq 0, \\ (n_2, z) - f_2 &\geq 0, \\ &\dots\dots\dots \\ (n_k, z) - f_k &\geq 0, \end{aligned}$$

is a convex set. Such systems of inequalities are the basic element in the description of many production planning problems, management problems and the like.

Exercises

1. Prove that a set of vectors z satisfying $(z, z) \leq \alpha$ is convex.
2. Prove that if a vector z is in hyperplane (46.8), then the vector $z + n$ is in the positive half-space.
3. Prove that if hyperplanes are given by the normed equations (44.1) and (44.2), then the origin is always in the nonpositive half-space.

48. Systems of linear equations

We again turn to the study of systems of linear algebraic equations, this time, however, in connection with the problem of the intersection of hyperplanes.

Consider a real or complex space K of dimension m . Suppose a scalar product is introduced in it. Choose some orthonormal basis and let $n_1, n_2, \dots, n_k$ be normal vectors to hyperplanes $H_1, H_2, \dots, H_k$ of (46.9) given by their coordinates

$$\begin{aligned} n_1 &= (a_{11}, a_{12}, \dots, a_{1m}), \\ n_2 &= (a_{21}, a_{22}, \dots, a_{2m}), \\ &\dots\dots\dots \\ n_k &= (a_{k1}, a_{k2}, \dots, a_{km}) \end{aligned} \quad (48.1)$$

and is called the *augmented matrix of the system*. Note, in particular, that the rank of the matrix of the system coincides with that of the system of vectors (48.1).

Theorem 48.1. (Kronecker-Capelli). *For a system of linear algebraic equations to be compatible it is necessary and sufficient that the rank of the augmented matrix of the system should equal that of the matrix of the system.*

Proof. We use the notation of Section 22. Up to arrangement vectors $a_1, a_2, \dots, a_m, b$ are columns of the matrices under consideration. Since the rank of a matrix coincides with that of the system of its column vectors, to prove the theorem it suffices to show that the system is compatible if and only if the rank of the system $a_1, a_2, \dots, a_m$ coincides with that of the system $a_1, a_2, \dots, a_m, b$.

Suppose system (48.2) is compatible. This means that equation (22.1) holds for some collection of numbers $z_1, z_2, \dots, z_m$, i.e. that b is a linear combination of vectors $a_1, a_2, \dots, a_m$. But it follows that any basis of the system $a_1, a_2, \dots, a_m$ is also a basis of the system $a_1, a_2, \dots, a_m, b$, i.e. that the ranks of both systems coincide.

Now let the ranks of the systems coincide. Choose some basis from $a_1, a_2, \dots, a_m$. It will be a basis of the system $a_1, a_2, \dots, a_m, b$ as well. Consequently, b is linearly expressible in terms of some of the vectors of the system $a_1, a_2, \dots, a_m$. Since it can be represented as a linear combination of all vectors $a_1, a_2, \dots, a_m$ too, this means that system (48.2) is compatible. Thus the theorem is proved.

The system of linear algebraic equations (48.2) is said to be *nonhomogeneous* if the right-hand sides are not all zero. Otherwise the system is said to be *homogeneous*. Any homogeneous system is always compatible since one of its solutions is $z_1 = z_2 = \dots = z_m = 0$. A system obtained from (48.2) by replacing all the right-hand sides with zeros is called the *reduced homogeneous system*. If (48.2) is compatible, then each of its solutions is a *particular* or *partial solution*. The totality of particular solutions is the *general solution of the system*.

Using the previously obtained facts about planes and systems (46.6), (46.7) and (46.9) describing the intersection of hyperplanes we can make a number of conclusions concerning the general solution of a system of linear algebraic equations. Namely.

*The general solution of a reduced homogeneous system forms in a space P_m a subspace of dimension $m - r$, where r is the rank of the matrix of the system. Any basis of that subspace is called a *fundamental system of solutions*.*

The general solution of a nonhomogeneous system is a plane in a space P_m obtained by translating the general solution of the reduced homogeneous system to the amount of any particular solution of the nonhomogeneous system.

The difference between any two particular solutions of a nonhomogeneous system is a particular solution of the reduced homogeneous system.

Among the particular solutions of a nonhomogeneous system there is only one orthogonal to all the solutions of the reduced homogeneous system. That solution is called normal.

For a compatible system to have a unique solution it is necessary and sufficient that the rank of the matrix of the system should equal the number of unknowns.

For a homogeneous system to have a nonzero solution it is necessary and sufficient that the rank of the matrix of the system should at least equal the number of unknowns.

This study of systems of linear algebraic equations used the concept of determinant only indirectly, mainly through the notion of the rank of a matrix. The determinant plays a much greater role in the theory of systems of equations, however.

Let the matrix of a system be a square one. For the rank of the matrix of the system to be less than the number of unknowns it is necessary and sufficient that the determinant of the system should be zero. Therefore

A homogeneous system has a nonzero solution if and only if the determinant of the system is zero.

Suppose now that the determinant of a system is nonzero. This means that the rank of the matrix of the system equals m . The rank of the augmented matrix cannot be less than m . But it cannot be greater than m either since there is no minors of order $m + 1$. Hence the ranks of both matrices are equal, i.e. the system is necessarily compatible in this case. Moreover, it has a unique solution. Thus

If the determinant of a system is nonzero, then the system always has a unique solution.

In terms of the intersection of hyperplanes this fact may be restated as follows:

If the normal vectors of hyperplanes form a basis of a space, then the intersection of the hyperplanes is not empty and contains only one vector.

Denote by d the determinant of a system and by d_j a determinant differing from d only in the j th column being replaced in it by a column of the right-hand sides $b_1, b_2, \dots, b_m$. Then the unique solution of the system can be obtained by the formulas

$$z_j = \frac{d_j}{d}, \quad j = 1, 2, \dots, n. \quad (48.3)$$

Indeed, let A_{ij} denote the algebraic adjunct of an element a_{ij} of the determinant of the system. Expanding d_j by the j th column we get

$$d_j = \sum_{i=1}^n b_i A_{ij}.$$

The Limit in Vector Space

49. Metric spaces

One of the basic concepts of mathematical analysis is the notion of limit. It is based on the fact that for the points of the number line the concept of "closeness" or more precisely of the distance between points is defined.

Comparison for "closeness" can be introduced in sets of a quite different nature. We have already defined in Section 29 the distance between vectors of vector spaces with scalar product. It was found that it possesses the same properties (29.5) as those the distance between points of the number line has. The distance between vectors was defined in terms of a scalar product which in turn was introduced axiomatically.

It is natural to attempt to introduce *axiomatically* the distance itself by requiring that properties (29.5) should necessarily hold.

Notice that many fundamental facts of the theory of limit in mathematical analysis are not associated with the fact that algebraic operations are defined for numbers. We therefore begin by extending the concept of distance to arbitrary sets of elements that are not necessarily vectors of a vector space.

A set is said to be a *metric space* if every pair of its elements is assigned a nonnegative real number called a *distance*, with the following axioms holding:

- (1) $\rho(x, y) = \rho(y, x)$,
- (2) $\rho(x, y) > 0$ if $x \neq y$; $\rho(x, y) = 0$ if $x = y$,
- (3) $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$

for any elements x, y and z . These axioms are called axioms of a *metric*, the first being the *axiom of symmetry* and the third the *triangle axiom*.

Formally any set of elements in which the equality relation is defined can be converted into a metric space by setting

$$\rho(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases} \quad (49.1)$$

It is easy to verify that all the axioms of a metric hold.

The vector x_0 of a metric space X is said to be the *limit* of a sequence $\{x_n\}$ of elements $x_1, x_2, \dots, x_n, \dots$ of X if the sequence of distances $\rho(x_0, x_1), \rho(x_0, x_2), \dots, \rho(x_0, x_n), \dots$ converges to zero. In this case we write

$$x_n \rightarrow x_0$$

or

$$\lim_{n \rightarrow \infty} x_n = x_0$$

and call the sequence $\{x_n\}$ *convergent in X* or simply *convergent*.

Notice that the same sequence of elements of the same set X may be convergent or nonconvergent depending on what metric is introduced in X . Suppose, for example, that in a metric space X some convergent sequence $\{x_n\}$ is chosen consisting of mutually distinct elements. We change the metric in X by introducing it according to (49.1). Now $\{x_n\}$ is no longer convergent. Indeed, suppose $x_n \rightarrow x'_0$, i.e. $\rho(x_n, x'_0) \rightarrow 0$. With the new metric this is possible if and only if all elements of $\{x_n\}$, except for a finite number of them, coincide with x'_0 . The contradiction obtained confirms the above assertion.

The following two properties are shared by any convergent sequence:

If $\{x_n\}$ converges, then so does and has the same limit any of its subsequences. A sequence may have no more than one limit.

The first property is obvious. Suppose $\{x_n\}$ has two limits, x_0 and y_0 . Then for any arbitrarily small number $\varepsilon > 0$ we can find a number N such that

$$\rho(x_0, x_n) < \frac{\varepsilon}{2}, \quad \rho(y_0, x_n) < \frac{\varepsilon}{2}$$

for every $n > N$. But from this, using the triangle axiom, we find

$$\rho(x_0, y_0) \leq \rho(x_0, x_n) + \rho(x_n, y_0) < \varepsilon.$$

By virtue of the arbitrariness of ε this means that $\rho(x_0, y_0) = 0$, i.e. $x_0 = y_0$.

A *sphere* $S(a, r)$ in a metric space X is the set of all elements $x \in X$ satisfying the condition

$$\rho(a, x) < r. \quad (49.2)$$

An element a is called the *centre* of the sphere and r is its *radius*. Any sphere with centre in a is a *neighbourhood* of a . A set of elements is said to be *bounded* if it is entirely in some sphere.

It is easy to see that x_0 is the limit of $\{x_n\}$ if and only if any neighbourhood of x_0 contains all the elements of $\{x_n\}$ beginning with some index.

In a metric space it is possible to introduce many of the other major concepts dealt with in number sets. Thus if a set $M \subset X$ is given, then an element $x \in X$ is said to be a *cluster point* (or *limit point*

or *accumulation point*) of that set if any neighbourhood of x contains at least one element of M distinct from x . The set obtained by joining to M all its cluster points is called the *closure* of M and designated $\bar{M}$. The set M is said to be *closed* if $M = \bar{M}$.

Consider the cluster points of sphere (49.2). We show that they all satisfy the condition

$$\rho(a, x) \leq r. \quad (49.3)$$

Indeed, suppose there is at least one cluster point x' for sphere (49.2) such that $\rho(a, x') > r$. By the definition of a cluster point any neighbourhood of x' must contain at least one element of (49.2) distinct from x' . But a neighbourhood with radius $0.5(\rho(a, x') - r)$ clearly contains no such element. Accordingly:

A set $\bar{S}(a, r)$ of all elements x satisfying (49.3) is a closed sphere.

Exercises

1. Prove that if $x_n \rightarrow x$, then $\rho(x_n, z) \rightarrow \rho(x, z)$ for any element z .

2. Will the set of all real numbers be a metric space if for numbers x and y we set

$$\rho(x, y) = \arctan |x - y|?$$

3. Can a set consisting of a finite number of elements have cluster points?

50. Complete spaces

A sequence $\{x_n\}$ of elements of a metric space is said to be a *fundamental* or *Cauchy sequence* if given any number $\varepsilon > 0$ we can find N such that $\rho(x_n, x_m) < \varepsilon$ for $n, m > N$.

Any fundamental sequence is bounded. Indeed, given ε choose N according to the definition and take a number $n_0 > N$. All elements of the sequence, beginning with x_{n_0} , are clearly in a sphere with centre x_{n_0} and of radius ε . The entire collection of the elements, however, is in a sphere with centre x_{n_0} and of radius equal to the maximum of the numbers

$$\varepsilon, \rho(x_1, x_{n_0}), \dots, \rho(x_{n_0-1}, x_{n_0}).$$

If a sequence is convergent, then it is fundamental. Let a sequence $\{x_n\}$ converge to x_0 . Then given any $\varepsilon > 0$ there is N such that

$$\rho(x_n, x_0) < \frac{\varepsilon}{2}$$

for $n > N$. By the triangle axiom

$$\rho(x_n, x_m) \leq \rho(x_n, x_0) + \rho(x_0, x_m) < \varepsilon$$

for $n, m > N$, which precisely means that $\{x_n\}$ is fundamental.

For the set of all reals the converse is also true. That is, any fundamental sequence is convergent. In general, however, this is not true, which is exemplified by a metric space with at least one cluster point eliminated.

A metric space is said to be *complete* if any fundamental sequence in it is convergent.

In complete metric spaces a theorem holds similar to the theorem on embedded line segments for real numbers. Given some sequence of spheres, we shall say that they are *embedded* into one another if every subsequent sphere is contained inside the preceding one.

Theorem 50.1. *In a complete metric space X let $\{\bar{S}(a_n, \varepsilon_n)\}$ be a sequence of closed spheres embedded into one another. If the sequence of their radii tends to zero, then there is a unique element in X which is in all those spheres.*

Proof. Consider a sequence $\{a_n\}$. Since $\bar{S}(a_{n+p}, \varepsilon_{n+p}) \subset \bar{S}(a_n, \varepsilon_n)$ for any $p \geq 0$, we have $a_{n+p} \in \bar{S}(a_n, \varepsilon_n)$. Consequently,

$$\rho(a_{n+p}, a_n) \leq \varepsilon_n$$

which implies that $\{a_n\}$ is fundamental.

The space X is complete and therefore $\{a_n\}$ tends to some limit a in X . Take any sphere $\bar{S}(a_h, \varepsilon_h)$. This sphere contains all terms of $\{a_n\}$, beginning with a_h . By virtue of the closure of the spheres the limit of $\{a_n\}$ is also in $\bar{S}(a_h, \varepsilon_h)$. Thus a is in all the spheres.

Assume further that there is another element, b , that is also in all the spheres. By the triangle axiom

$$\rho(a, b) \leq \rho(a, a_n) + \rho(a_n, b) \leq 2\varepsilon_n.$$

Since ε_n may be taken as small as we like, this means that $\rho(a, b) = 0$, i.e. that $a = b$.

The most important examples of complete spaces are the sets of real and complex numbers. We assume that the distance between numbers coincides with the absolute value of their difference. The completeness of the set of reals is proved in the course of mathematical analysis. We show the completeness of the set of complex numbers.

Assume that complex numbers are given in algebraic form. The distance between numbers

$$z = a + ib, \quad v = c + id$$

is introduced by the rule

$$\rho(z, v) = |z - v|, \quad (50.1)$$

where

$$|z - v|^2 = (a - c)^2 + (b - d)^2. \quad (50.2)$$

It is obvious that the axioms of a metric hold.

Consider a sequence $\{z_k = a_k + ib_k\}$ of complex numbers. Let it be fundamental. Given $\varepsilon > 0$ there is N such that for all $n, m > N$

$$|z_n - z_m| < \varepsilon.$$

From (50.2) it follows that

$$|a_n - a_m| < \varepsilon, \quad |b_n - b_m| < \varepsilon, \quad (50.3)$$

i.e. that $\{a_k\}$ and $\{b_k\}$ are also fundamental. By virtue of the completeness of the set of reals there are a and b such that

$$a_k \rightarrow a, \quad b_k \rightarrow b.$$

Proceeding in (50.3) to the limit we get

$$|a_n - a| \leq \varepsilon, \quad |b_n - b| \leq \varepsilon.$$

On setting

$$z = a + ib,$$

we find that

$$\rho(z_n, z) \leq \sqrt{2}\varepsilon$$

for all $n > N$. But this means that the fundamental sequence $\{z_k\}$ is convergent.

As a consequence, $\{z_k = a_k + ib_k\}$ converges to a number $z = a + ib$ if and only if $\{a_k\}$ and $\{b_k\}$ converge to a and b .

The complete space of complex numbers has much in common with the space of real numbers. In particular, any bounded sequence of complex numbers has a convergent subsequence. Indeed, this is true for any bounded sequence of reals. It is also obvious that the boundedness of $\{z_k = a_k + ib_k\}$ implies the boundedness of $\{a_k\}$ and $\{b_k\}$. Since $\{a_k\}$ is bounded, it has a convergent subsequence $\{a_{k_n}\}$. Consider a sequence $\{b_{k_n}\}$. It is bounded and therefore it too has a subsequence $\{b_{k_{n_n}}\}$. It is clear that $\{a_{k_{n_n}}\}$ will be convergent. Consequently, so is $\{z_{k_{n_n}}\}$.

In the complex space, just as in the real one, the concept of infinitely large sequence can be introduced. Namely, a sequence $\{z_k\}$ is said to be *infinitely large* if given an arbitrarily large number A we can find N such that $|z_k| > A$ for every $k > N$. It is obvious that we can always choose an infinitely large subsequence of any unbounded sequence.

Exercises

1. Will the set of all real numbers be a complete space if for numbers x and y we set

$$\rho(x, y) = \arctan |x - y|?$$

2. Prove that any closed set of a complete space is a complete space.

3. Will any closed set of an arbitrary metric space be necessarily a complete space?

4. Construct a metric such that the set of all complex numbers is not a complete space.

51. Auxiliary inequalities

We establish some inequalities to be used in our immediate studies. Take a positive number α and consider the exponential function $y = \alpha^x$ (Fig. 51.1). Let x_1 and x_2 be two distinct reals. Draw a straight line through points with coordinates (x_1, α^{x_1}) and (x_2, α^{x_2}) . Taking into account the properties of the exponential function we conclude that if the independent variable on the closed interval $[x_1, x_2]$ is changed, none of its points will lie higher than the points of the constructed straight line.

Now let $x_1 = 0$ and $x_2 = 1$. Then the equation of the straight line under consideration will be $y = \alpha x + (1 - x)$. Consequently,

$$\alpha^x \leq \alpha x + (1 - x) \quad (51.1)$$

for $0 \leq x \leq 1$.

We shall say that positive numbers p and q are *conjugate* if they satisfy the relation

$$\frac{1}{p} + \frac{1}{q} = 1. \quad (51.2)$$

It is clear that $p, q > 1$.

For any positive numbers a and b , the number $a^p b^{-q}$ will also be positive and it can be taken as α of (51.1). If we assume $x = p^{-1}$, then $1 - x = q^{-1}$. Now from (51.1) we get

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \quad (51.3)$$

for any positive a and b and conjugate p and q . It is obvious that in fact this inequality holds for all nonnegative a and b .

Consider two arbitrary vectors x and y in the space R_n or C_n . Let these vectors be given by their coordinates

$$x = (x_1, x_2, \dots, x_n),$$

$$y = (y_1, y_2, \dots, y_n).$$

We establish the so-called *Hölder's inequality*

$$\sum_{k=1}^n |x_k y_k| \leq \left(\sum_{k=1}^n |x_k|^p \right)^{1/p} \left(\sum_{k=1}^n |y_k|^q \right)^{1/q}. \quad (51.4)$$

Notice that if at least one of the vectors x and y is zero, then Hölder's inequality is obviously true. It may therefore be assumed that $x \neq 0$ and $y \neq 0$. Let the inequality hold for some nonzero vec-

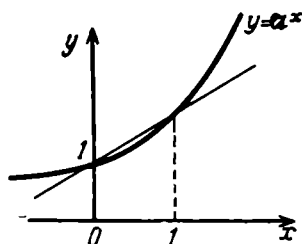


Fig. 51.1

tors x and y . Then it does for vectors λx and μy , with any λ and μ . Therefore it suffices to prove it for the case where

$$\sum_{k=1}^n |x_k|^p = \sum_{k=1}^n |y_k|^q = 1. \quad (51.5)$$

Now putting $a = |x_k|$ and $b = |y_k|$ in (51.3) and summing over k from 1 to n we get in view of (51.2) and (51.5)

$$\sum_{k=1}^n |x_k y_k| \leq 1.$$

But this is Hölder's inequality for (51.5).

We now proceed to prove *Minkowski's inequality* which, for any vectors x and y from $\mathbf{R}_n$ or $\mathbf{C}_n$, says that

$$\left(\sum_{k=1}^n |x_k + y_k|^p \right)^{1/p} \leq \left(\sum_{k=1}^n |x_k|^p \right)^{1/p} + \left(\sum_{k=1}^n |y_k|^p \right)^{1/p} \quad (51.6)$$

for every $p \geq 1$.

Minkowski's inequality is obvious for $p = 1$, since the absolute value of the sum of two numbers is at most the sum of their absolute values. Moreover, it clearly holds if at least one of the vectors x and y is zero. Therefore we may restrict ourselves to the case $p > 1$ and $x \neq 0$. We write the identity

$$(|a| + |b|)^p = (|a| + |b|)^{p-1} |a| + (|a| + |b|)^{p-1} |b|.$$

Setting $a = x_k$ and $b = y_k$ and summing over k from 1 to n we get

$$\begin{aligned} \sum_{k=1}^n (|x_k| + |y_k|)^p \\ = \sum_{k=1}^n (|x_k| + |y_k|)^{p-1} |x_k| + \sum_{k=1}^n (|x_k| + |y_k|)^{p-1} |y_k|. \end{aligned}$$

Applying to each of the two sums at the right Hölder's inequality and considering that $(p-1)q = p$ we get

$$\begin{aligned} \sum_{k=1}^n (|x_k| + |y_k|)^p \\ \leq \left(\sum_{k=1}^n (|x_k| + |y_k|)^p \right)^{1/q} \left(\sum_{k=1}^n |x_k|^p \right)^{1/p} + \left(\sum_{k=1}^n |y_k|^p \right)^{1/p}. \end{aligned}$$

On dividing both sides of the inequality by

$$\left(\sum_{k=1}^n (|x_k| + |y_k|)^p \right)^{1/q},$$

we find that

$$\left(\sum_{k=1}^n (|x_k| + |y_k|)^p\right)^{1/p} \leq \left(\sum_{k=1}^n |x_k|^p\right)^{1/p} + \left(\sum_{k=1}^n |y_k|^p\right)^{1/p}$$

from which we at once obtain inequality (51.6).

Exercises

1. Derive the Cauchy-Buniakowski-Schwarz inequality from Hölder's.
2. Study Hölder's inequality for $p \rightarrow \infty$.
3. Study Minkowski's inequality for $p \rightarrow \infty$.

52. Normed spaces

We have arrived at the concept of metric space by concentrating on a single property of a set, that of distance. Similarly, by concentrating on operations in a set we have arrived at the concept of vector space. We now discuss vector spaces with a metric.

It is obvious that if the concept of distance is in no way connected with operations on elements, it is impossible to construct an interesting theory whose facts would join together algebraic and metric notions. We shall therefore impose additional conditions on the metric introduced in a vector space.

In fact we have already encountered metric vector spaces. These are, for example, a Euclidean and a unitary space with metric (29.4). The need for such a metric, however, arises not always by far. Introducing a scalar product actually means introducing not only distance between elements but also angles between them. In a vector space it is most often required to introduce only an acceptable definition of distance. The most important vector spaces of this kind are the so-called normed spaces.

A real or complex vector space X is said to be a *normed space* if each vector $x \in X$ is assigned a real number $\|x\|$ called the *norm* of a vector x , the following axioms holding:

- (1) $\|x\| > 0$ if $x \neq 0$, $\|0\| = 0$,
- (2) $\|\lambda x\| = |\lambda| \|x\|$,
- (3) $\|x + y\| \leq \|x\| + \|y\|$

for any vectors x and y and any number λ . The second axiom is called the *axiom of the absolute homogeneity of a norm* and the third is the *triangle inequality axiom*.

From the axiom of the absolute homogeneity of a norm it follows that for any nonzero vector x we can find a number λ such that the

norm of a vector λx equals unity. To do this it suffices to take $\lambda = \|x\|^{-1}$. A vector whose norm equals unity will be called a *normed vector*.

The triangle inequality for a norm yields one useful relation. We have $\|x\| \leq \|y\| + \|x - y\|$ for any x and y . Hence $\|x\| - \|y\| \leq \|x - y\|$. Interchanging x and y we find $\|y\| - \|x\| \leq \|x - y\|$. Therefore

$$\|x\| - \|y\| \leq \|x - y\|. \quad (52.1)$$

A normed space is easy to convert into a metric one by setting

$$\rho(x, y) = \|x - y\|. \quad (52.2)$$

Indeed, $\rho(x, y) = 0$ means $\|x - y\| = 0$, which according to axiom (1) means that $x = y$. The symmetry of the distance introduced is obvious. Finally, the triangle inequality for a distance is a simple consequence of the triangle inequality for a norm. That is, $\rho(x, y) = \|x - y\| = \|(x - z) + (z - y)\|$

$$\leq \|x - z\| + \|z - y\| = \rho(x, z) + \rho(z, y).$$

Notice that

$$\|x\| = \rho(x, 0). \quad (52.3)$$

Metric (52.2) defined in a vector space possesses also the following properties:

$$\rho(x + z, y + z) = \rho(x, y)$$

for any $x, y, z \in X$, i.e. distance remains unaffected by a translation of vectors and

$$\rho(\lambda x, \lambda y) = |\lambda| \rho(x, y)$$

for any vectors $x, y \in X$ and any number λ , i.e. distance is an absolutely homogeneous function.

If in a metric vector space X some metric satisfies these two additional requirements, then X may be regarded as a normed space if a norm is defined by equation (52.3) for any $x \in X$.

Taking into account the relations of Section 29 it is easy to establish that a vector space with scalar product is a normed space. It should be assumed that the norm of a vector is its length.

It is possible to give other examples of introducing a norm. In a vector space let vectors be given by their coordinates in some basis. If $x = (x_1, x_2, \dots, x_n)$, then we set

$$\|x\|_p = \left(\sum_{k=1}^n |x_k|^p \right)^{1/p}, \quad (52.4)$$

where $p \geq 1$. That the first two axioms hold for a norm is obvious and the third axiom is seen to hold from Minkowski's inequality

(51.6). Most commonly used are the following norms:

$$\|x\|_1 = \sum_{k=1}^n |x_k|, \quad \|x\|_2 = \left(\sum_{k=1}^n |x_k|^2 \right)^{1/2}, \quad \|x\|_\infty = \max_{1 \leq k \leq n} |x_k|. \quad (52.5)$$

The second of the norms is often called a *Euclidean norm* and designated $\|x\|_E$.

In what follows we shall consider only normed spaces with metric (52.2). The convergence of a sequence of vectors in such a metric is called the *convergence in the norm*, the boundedness of a set of vectors is called the *boundedness in the norm*, etc.

Exercises

1. Prove that there is a sequence of vectors whose norms form an infinitely large sequence.

2. Prove that for any numbers λ_i and vectors e_i

$$\left\| \sum_{i=1}^n \lambda_i e_i \right\| \leq \sum_{i=1}^n |\lambda_i| \|e_i\|.$$

3. Prove that if $x_n \rightarrow x$, $y_n \rightarrow y$ and $\lambda_n \rightarrow \lambda$, then

$$\|x_n\| \rightarrow \|x\|, \quad x_n + y_n \rightarrow x + y, \quad \lambda_n x_n \rightarrow \lambda x.$$

53. Convergence in the norm and coordinate convergence

In a real or complex finite dimensional vector space it is possible to introduce another concept of convergence besides the convergence in the norm. Consider a space X and let $e_1, e_2, \dots, e_n$ be its basis. For any sequence $\{x_m\}$ of vectors of X there are expansions

$$x_m = \sum_{k=1}^n \xi_k^{(m)} e_k. \quad (53.1)$$

If for the vector

$$x_0 = \sum_{k=1}^n \xi_k^{(0)} e_k \quad (53.2)$$

there are limiting relations

$$\lim_{m \rightarrow \infty} \xi_k^{(m)} = \xi_k^{(0)} \quad (53.3)$$

for every $k = 1, 2, \dots, n$, then we shall say that there is a *coordinate convergence* of $\{x_m\}$ to a vector x_0 .

Coordinate convergence is quite natural. If two vectors are "close", then it may be assumed that so should the corresponding coordi-

nates in the expansion with respect to the same basis. Finite dimensional normed spaces are noteworthy for the fact that the notions of convergence in the norm and of coordinate convergence are *equivalent* in them.

It is easy to show that coordinate convergence implies convergence in the norm. Indeed, let (53.3) hold. Using the axioms of the absolute homogeneity of a norm and of the triangle inequality we conclude from (53.1) and (53.2) that

$$\|x_m - x_0\| = \left\| \sum_{k=1}^n (\xi_k^{(m)} - \xi_k^{(0)}) e_k \right\| \leq \sum_{k=1}^n |\xi_k^{(m)} - \xi_k^{(0)}| \|e_k\| \rightarrow 0.$$

The proof of the converse is essentially more involved.

Lemma 53.1. *If in a finite dimensional normed space a sequence of vectors is norm bounded, then the numerical sequences of all the coordinates in the expansion of vectors with respect to any basis are bounded too.*

Proof. Let each vector of a sequence $\{x_m\}$ be represented in the form (53.1). We introduce the notation

$$\sigma_m = \sum_{k=1}^n |\xi_k^{(m)}|$$

and prove that $\{\sigma_m\}$ is bounded.

Suppose this is not the case. Then we may choose an infinitely large subsequence $\{\sigma_{m_p}\}$. Set

$$y_p = \frac{1}{\sigma_{m_p}} x_{m_p}.$$

If

$$y_p = \sum_{k=1}^n \eta_k^{(p)} e_k,$$

then of course

$$\eta_k^{(p)} = \frac{\xi_k^{(m_p)}}{\sigma_{m_p}}$$

for every $k = 1, 2, \dots, n$ and every m_p and we get

$$\sum_{k=1}^n |\eta_k^{(p)}| = 1. \quad (53.4)$$

Sequences $\{\eta_k^{(p)}\}$ are bounded since by (53.4) $|\eta_k^{(p)}| \leq 1$. Hence it is possible to choose a subsequence of vectors $\{y_{p_1}\}$ such that $\{\eta_k^{(p_1)}\}$ is convergent, i.e.

$$\lim_{p_1 \rightarrow \infty} \eta_k^{(p_1)} = \eta_k$$

for some number η_1 . It is possible to choose in turn a subsequence $\{y_{p_2}\}$ of $\{y_{p_1}\}$ such that

$$\lim_{p_1 \rightarrow \infty} \eta_1^{(p_2)} = \eta_2$$

for some number η_2 . As before

$$\lim_{p_1 \rightarrow \infty} \eta_1^{(p_2)} = \eta_1.$$

Continuing the process we choose a subsequence $\{y_{p_n}\}$ of $\{y_p\}$ such that there are limits

$$\lim_{p_n \rightarrow \infty} \eta_k^{(p_n)} = \eta_k \quad (53.5)$$

for every $k = 1, 2, \dots, n$. By (53.4)

$$\sum_{k=1}^n |\eta_k| = 1. \quad (53.6)$$

Since coordinate convergence implies convergence in the norm, limiting relations (53.5) mean that

$$\lim_{p_n \rightarrow \infty} \|y_{p_n} - y\| = 0, \quad (53.7)$$

where

$$y = \sum_{k=1}^n \eta_k e_k.$$

By virtue of (53.6) y must not equal zero. On the other hand,

$$\|y_{p_n}\| = \frac{\|x_{m_{p_n}}\|}{\sigma_{m_{p_n}}} \rightarrow 0,$$

since the sequence $\{x_{m_{p_n}}\}$ is norm bounded and $\{\sigma_{m_{p_n}}\}$ is infinitely large. Hence it follows from (53.7) that $\|y\| = 0$, i.e. that y is a zero vector. This contradiction proves the lemma.

Theorem 53.1. *In a finite dimensional normed space convergence in the norm implies coordinate convergence.*

Proof. Let $\{x_m\}$ be a sequence of vectors converging in the norm to a vector x_0 . It is obvious that it suffices to consider the case where $x_0 = 0$ and $\{x_m\}$ contains no zero vectors. We represent vectors x_m as (53.1). The sequence of vectors

$$y_m = \frac{1}{\|x_m\|} x_m$$

will be norm bounded and by Lemma 53.1 the sequences of numbers

$$\eta_k^{(m)} = \frac{\xi_k^{(m)}}{\|x_m\|}$$

must be bounded for every $k = 1, 2, \dots, n$. Since $\|x_m\| \rightarrow 0$, this is possible if and only if $\xi_k^{(m)} \rightarrow 0$ for every k . But this means that there is coordinate convergence of the sequence $\{x_m\}$ to the vector x_0 .

Coordinate convergence can be efficiently used in *theoretical* studies, but in *practical* applications it is more convenient to use convergence in the norm. This is mainly because in the study of vector spaces of high dimension it is hard to deal with a large number of coordinate sequences. Besides we do not always know at least one basis. But even if we do, its use most often results in unjustly lengthy calculations.

Exercises

1. Is the requirement that a space should be finite dimensional essential in proving the equivalence of the two kinds of convergence?

2. Prove that if some set of vectors of a finite dimensional space is bounded in one norm, then it will be bounded in any other norm.

3. Prove that if in a finite dimensional space $x_n \rightarrow x$ in one norm, then $x_n \rightarrow x$ in any other norm.

54. Completeness of normed spaces

Finite dimensional normed spaces are spaces where many statements similar to those associated with the concept of limit in number sets are true. Consider some of them.

Lemma 54.1. *It is possible to choose a subsequence of any bounded sequence of vectors of a finite dimensional normed space, convergent in that space.*

Proof. Let $\{x_m\}$ be an arbitrary norm bounded sequence. Represent vectors x_m as (53.1). By Lemma 53.1 sequences $\{\xi_k^{(m)}\}$ are bounded for every $k = 1, 2, \dots, n$. Choose, in the same way as in the proof of Lemma 53.1, a subsequence $\{x_{m_n}\}$ of $\{x_m\}$ such that there are limiting relations $\xi_k^{(m_n)} \rightarrow \xi_k^{(0)}$ for every k . It follows that $\{x_{m_n}\}$ converges in the norm to vector (53.2).

This lemma is similar to the well-known Bolzano-Weierstrass theorem of mathematical analysis. It is of great importance for studies of any finite dimensional normed spaces. As an illustration, we prove some statements.

Theorem 54.1. *Any finite dimensional normed space is complete.*

Proof. Let $\{x_m\}$ be a fundamental sequence. It is bounded. Choose a convergent subsequence $\{x_{m_n}\}$ and denote by x_0 its limit. We have

$$\|x_m - x_0\| \leq \|x_m - x_{m_n}\| + \|x_{m_n} - x_0\|.$$

Take an arbitrary number $\varepsilon > 0$. Since $\{x_m\}$ is fundamental, there is N_1 such that $\|x_m - x_{m_n}\| < \varepsilon/2$ for $m, m_n > N_1$. Since the sequence $\{x_{m_n}\}$ converges to x_0 , there is N_2 such that $\|x_{m_n} - x_0\| < \varepsilon/2$ for $m_n > N_2$. If N is the maximum number of N_1 and N_2 , for $m > N$

$$\|x_m - x_0\| < \varepsilon.$$

The number ε is arbitrary. Hence the fundamental sequence $\{x_m\}$ converges in the norm to the vector x_0 .

Lemma 54.2. Any finite dimensional subspace X_0 of a normed space X is a closed set.

Proof. Consider in a normed space X a finite dimensional subspace X_0 . Let a vector $x \in X$ be cluster point for X_0 . This means that there is a sequence $\{x_m\}$ of vectors of X_0 , distinct from x , such that $\|x_m - x\| \rightarrow 0$. The sequence $\{x_m\}$ is bounded and therefore it is possible to choose a subsequence $\{x_{m_p}\}$ of $\{x_m\}$, converging, in view of the completeness of X_0 , to some vector $x_0 \in X_0$. Now we have

$$\|x - x_0\| \leq \|x - x_{m_p}\| + \|x_{m_p} - x_0\| \rightarrow 0,$$

i.e. $x = x_0$.

Lemma 54.3. Let X be a normed space and let X_0 be its finite dimensional subspace distinct from it. There is a normed vector $x \notin X_0$ such that $\|x - x_0\| \geq 1$ for any vector $x_0 \in X_0$.

Proof. The subspace X_0 does not coincide with X and therefore there is a vector $x' \notin X_0$. Since X_0 is closed, we have

$$\inf_{x_0 \in X_0} \|x' - x_0\| = d > 0. \quad (54.1)$$

By the definition of the greatest lower bound there is a vector $x_0^{(h)}$ in X_0 for which

$$d \leq \|x' - x_0^{(h)}\| \leq \frac{d}{1-2^{-h}}.$$

The sequence $\{x_0^{(h)}\}$ is bounded. Choose a subsequence $\{x_0^{(h_p)}\}$ of it, converging, in view of the completeness of X_0 , to some vector $x'_0 \in X_0$. For that vector obviously

$$\|x' - x'_0\| = d. \quad (54.2)$$

Set

$$x = \frac{1}{d} (x' - x'_0).$$

It is clear that $\|x\| = 1$. Moreover, if $x_0 \in X_0$, then by (54.1)

$$\|x - x_0\| = \left\| \frac{1}{d} x' - \frac{1}{d} x'_0 - x_0 \right\| = \frac{1}{d} \|x' - (x'_0 + dx_0)\| \geq \frac{1}{d} d = 1,$$

since the vector $x'_0 + dx_0$ is in X_0 .

Incidentally, we have proved that the lower bound (54.1) is reached on at least one vector $x'_0 \in X_0$. In $\|x - x_0\| \geq 1$ equality clearly holds for $x_0 = 0$.

To conclude, Lemma 54.1, which plays so great a part in finite dimensional spaces, holds in *no* infinite dimensional space. Namely, we have

Lemma 54.4. *If it is possible to choose a convergent subsequence of any bounded sequence of vectors of a normed space X , then X is finite dimensional.*

Proof. Suppose the contrary. Let X be an infinite dimensional space. Choose an arbitrary normed vector x_1 and denote by L_1 its span. By Lemma 54.3 there is a normed vector x_2 such that $\|x_2 - x_1\| \geq 1$. Denote by L_2 the span of x_1 and x_2 . Continuing the reasoning we find a sequence $\{x_n\}$ of normed vectors satisfying $\|x_n - x_k\| \geq 1$ for every $k < n$. Hence it is impossible to choose any convergent subsequence of $\{x_n\}$. This contradicts the hypothesis of the lemma and therefore the assumption that X is infinite dimensional was false.

Exercises

1. Prove that a plane in a normed finite dimensional space¹ is a closed set.
2. Prove that a set of vectors x of a finite dimensional space satisfying $\|x\| \leq \alpha$ is a closed set.
3. Prove that in a closed bounded set of vectors of a finite dimensional space there are vectors on which both the lower and the upper bound of values of any norm are attained.
4. Prove that given any two norms, $\|x\|_I$ and $\|x\|_{II}$, in a finite dimensional space there are positive α and β such that

$$\alpha \|x\|_I \leq \|x\|_{II} \leq \beta \|x\|_I$$

for every vector x . The numbers α and β are independent of x .

55. The limit and computational processes

In complete metric spaces the concept of limit is widely used in constructing and justifying various computational processes. We consider as an example one method of solving systems of linear algebraic equations.

Given a system of two equations in two unknowns, assume that it is compatible and has a unique solution. For simplicity of presenta-

tion suppose that all coefficients are real. Each of the equations

$$\begin{aligned}a_{11}x + a_{12}y &= f_1, \\ a_{21}x + a_{22}y &= f_2\end{aligned}$$

of the system defines in the plane a straight line. The point M of the intersection of the straight lines gives a solution of the system (Fig. 55.1).

Take a point M_0 lying on none of the straight lines in the plane. Drop from it a perpendicular to either straight line. The foot M_1 of the perpendicular is closer to M than M_0 is, since a projection is always smaller than the inclined line. Drop then a perpendicular from M_1 to the other straight line. The foot M_2 of that perpendicular is still closer to the solution. Successively projecting now onto one straight line now onto the other, we obtain a sequence $\{M_k\}$ of points of the plane converging to M . It is important to note that the sequence constructed converges for any choice of the initial point M_0 .

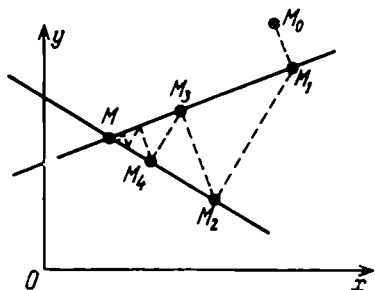


Fig. 55.1

This example suggests how to construct a computational process to solve a system of linear algebraic equations of the general form (48.2). We replace the problem by an equivalent problem of finding vectors of the intersection of a system of hyperplanes (46.9). Suppose the hyperplanes contain at least one common vector and assume for simplicity that the vector space is real.

Choose a vector v_0 and project it onto the first hyperplane. Project the resulting vector v_1 onto the second hyperplane and so on. This process determines some sequence $\{v_p\}$. Let us investigate that sequence.

Basic to the computational process is projecting some vector v_p onto the hyperplane given by equation (46.8). It is clear that v_{p+1} satisfies that equation and is related to v_p by the equation

$$v_{p+1} = v_p + tn$$

for some number t . Substituting v_{p+1} in (46.8) determines t . From this we get

$$v_{p+1} = v_p + \left(\frac{b_p - (n, v_p)}{(n, n)} \right) n.$$

This formula says that all vectors of the sequence $\{v_p\}$ are in the plane obtained by translating the span $L(n_1, n_2, \dots, n_k)$ to the amount of a length of the vector v_0 . But all vectors which are in the

intersection of hyperplanes (46.9) are in the plane obtained by translating the orthogonal complement $L^\perp(n_1, n_2, \dots, n_k)$. There is a unique vector z_0 that is in both planes.

If we prove that some subsequence of $\{v_p\}$ converges to some vector that is in hyperplanes (46.9), then by the closure of a plane it is to z_0 that it converges. Moreover, so does the entire sequence $\{v_p\}$.

For any r vectors $z_0 - v_{r+1}$ and $v_{r+1} - v_r$ are orthogonal and therefore by the Pythagorean theorem

$$\rho^2(z_0, v_r) = \rho^2(z_0, v_{r+1}) + \rho^2(v_r, v_{r+1}).$$

Summing the equations obtained over r from 0 to $p-1$ we find

$$\rho^2(z_0, v_0) = \rho^2(z_0, v_p) + \sum_{r=0}^{p-1} \rho^2(v_r, v_{r+1}).$$

Consequently

$$\sum_{r=0}^{p-1} \rho^2(v_r, v_{r+1}) \leq \rho^2(z_0, v_0)$$

from which we conclude that

$$\rho(v_p, v_{p+1}) \rightarrow 0. \quad (55.1)$$

Denote by H_r the hyperplane in the r th row of (46.9). It is clear that the distance from v_p to H_r is not greater than the distance between v_p and any vector of H_r . By the construction of $\{v_p\}$, among any k consecutive vectors of $\{v_p\}$ there is necessarily a vector that is in any of the hyperplanes. Using the triangle inequality and the limiting relation (55.1) we get

$$\begin{aligned} \rho(v_p, H_r) &\leq \rho(v_p, v_{p+1}) + \rho(v_{p+1}, v_{p+2}) + \\ &\quad \dots + \rho(v_{p+k-1}, v_{p+k}) \rightarrow 0 \end{aligned} \quad (55.2)$$

for every $r = 1, 2, \dots, k$.

The sequence $\{v_p\}$ is obviously bounded. Choose some convergent subsequence of it. Let that subsequence converge to a vector z'_0 . Proceeding in (55.2) to the limit we find that

$$\rho(z'_0, H_r) = 0$$

for every $r = 1, 2, \dots, k$. But, as already noted earlier, the vector z'_0 must coincide with z_0 . Consequently, $\{v_p\}$ converges to z_0 .

Exercises

1. Were the concepts of completeness and closure actually used in the above investigation?

2. How are other solutions of the system to be found if there are any?

3. How will the process behave itself if the system is incompatible?

PART II

Linear Operators

CHAPTER 7

Matrices and Linear Operators

56. Operators

A major point in creating the foundations of mathematical analysis is the introduction of the concept of function. By definition, to specify a function it is necessary to indicate two sets, X and Y , of real numbers and to formulate a rule assigning to each number $x \in X$ a unique number $y \in Y$. That rule is a single-valued function of a real variable x given on the set X .

In realizing the general idea of functional dependence it is not at all necessary to require that X and Y should be sets of real numbers. Understanding by X and Y various sets of elements we arrive at the following definition generalizing the concept of function.

A rule assigning to each element x of some nonempty set X a unique element y of a nonempty set Y is called an *operator*. A result y of applying an operator A to an element x is designated

$$y = A(x), \quad y = Ax \quad (56.1)$$

and A is said to be an operator from X to Y or to map X into Y .

The set X is said to be the *domain* of A . An element y of (56.1) is the *image* of an element x and x is the *inverse image* of y . The collection T_A of all images is the *range* (or *image*) of A . If each element $y \in Y$ has only one inverse image, then operator (56.1) is said to be 1-1. An operator is also called a *mapping*, *transformation* or *operation*.

In what follows we shall mainly consider the so-called *linear operators*. Their distinctive features are as follows. First, the domain of a linear operator is always some vector space or linear subspace. Second, the properties of a linear operator are closely related to operations on vectors of a vector space. As a rule in our study of linear operators we shall assume that spaces are given over a field of real or complex numbers. Unless otherwise stated, operator will mean linear operator. In the general theory of operators linear operators play as important a part as the straight line and the plane do in mathematical analysis. That is why they require a detailed study.

Let X and Y be vector spaces over the same field P . Consider an operator A whose domain is X and whose range is some set

in Y . The operator A is said to be *linear* if

$$A(\alpha u + \beta v) = \alpha Au + \beta Av \quad (56.2)$$

for any vectors $u, v \in X$ and any numbers $\alpha, \beta \in P$.

We have already repeatedly encountered linear operators. According to (9.8) a linear operator is the magnitude of a directed line segment. Its domain is the set of all directed line segments of an axis and its range is the set of all real numbers. Also a linear operator is by (21.2) an isomorphism between two vector spaces. Fix in a vector space with a scalar product some subspace L . We obtain two linear operators if each vector of the space is assigned either its projection onto L or the perpendicular from that vector to L . This follows from (30.5) and (30.6).

An operator assigning to each vector x of X the zero vector of Y is obviously a linear operator. It is called the *zero operator* and designated 0 . So

$$0 = 0x.$$

Assigning to each vector $x \in X$ the same vector x yields a linear operator E from X to X . This is called the *identity* or *unit operator*. By definition

$$x = Ex.$$

Let A be some operator from X to Y . We construct a new operator B according to the prescription $Bx = -Ax$. The resulting operator B is also a linear operator from X to Y . It is said to be *opposite* to A .

Finally fix an arbitrary number α and assign to each vector $x \in X$ a vector $\alpha x \in X$. The resulting operator is certainly a linear operator. It is called a *scalar operator*. When $\alpha = 0$, we obtain a zero operator, and when $\alpha = 1$, we obtain an identity operator.

We shall soon describe a general method for constructing linear operators, and now we point out some of their characteristic features. By (56.2)

$$A\left(\sum_{i=1}^p \alpha_i x_i\right) = \sum_{i=1}^p \alpha_i Ax_i$$

for any vectors x_i and any numbers α_i . In particular, it follows that any linear operator A transforms a zero vector into a zero vector, i.e.

$$0 = A0.$$

The range T_A of a linear operator A is a *subspace* of Y . If $z = Au$ and $w = Av$, then a vector $\alpha z + \beta w$ is clearly the image of a vector $\alpha u + \beta v$ for any numbers α and β . Hence the vector $\alpha z + \beta w$ is in the range of A . The dimension of T_A is called the *rank* of A and denoted by r_A .

Along with T_A consider the set N_A of all vectors $x \in X$ satisfying

$$Ax = 0.$$

It is also a *subspace* and is called the *kernel* or *null space* of A . The dimension n_A of the kernel is called the *nullity* of A .

The rank and nullity are not independent characteristics of a linear operator A . Let X have a dimension m . Decompose it as a direct sum

$$X = N_A + M_A, \quad (56.3)$$

where N_A is the kernel of A and M_A is any complementary subspace. Take a vector $x \in X$ and represent it as a sum

$$x = x_N + x_M,$$

where $x_N \in N_A$ and $x_M \in M_A$. If $y = Ax$, then since A is linear and $Ax_N = 0$

$$y = Ax_M.$$

Hence any vector from T_A has at least one inverse image in M_A .

In fact, that inverse image is *unique* in M_A . Suppose for some vector $y \in T_A$ we have two inverse images, $x'_M, x''_M \in M_A$. Since M_A is a subspace, $x'_M - x''_M \in M_A$. Since x'_M and x''_M are the inverse images of the same vector y , $x'_M - x''_M \in N_A$. Subspaces M_A and N_A have only a zero vector in common. Therefore $x'_M - x''_M = 0$, i.e. $x'_M = x''_M$.

Thus the operator A establishes a *1-1 correspondence* between the vectors of T_A and M_A . By virtue of the linearity of A this correspondence is an isomorphism. Hence the dimensions of T_A and M_A coincide and are equal to r_A . Now it follows from (56.3) that

$$r_A + n_A = m. \quad (56.4)$$

Note that the linear operator A establishes an isomorphism between T_A and *any* M_A of X that in the direct sum with the kernel of A constitutes the entire space X . It may therefore be assumed that every linear operator A generates an entire family of other linear operators. First, it is the zero operator defined on the kernel N_A , i.e. the one from N_A to 0. Second, it is a set of linear operators from the subspaces M_A complementary to the kernel to the subspace T_A . It is a very important fact that each of the new operators coincides on its domain with A . If $N_A = 0$, then $M_A = X$ and the entire second set of operators coincides with A . If $N_A = X$, however, then A is a zero operator. We shall continue this discussion later on.

Exercises

Prove that the following operators are linear.

1. A basis is given in a vector space X . An operator A assigns to each vector $x \in X$ its coordinate with a fixed index.
2. A vector x_0 is fixed in a space X with a scalar product. An operator A assigns to each vector $x \in X$ a scalar product (x, x_0) .
3. A vector x_0 is fixed in a space V_2 . An operator A assigns to each vector $x \in V_2$ a vector product $[x, x_0]$.
4. A space X is formed by polynomials with real coefficients. An operator A assigns to each polynomial its k th derivative. It is called an *operator of k -fold differentiation*.
5. In a space of polynomials dependent on a variable t , an operator A assigns to each polynomial $P(t)$ a polynomial $t \cdot P(t)$.
6. A space X is decomposed as a direct sum of subspaces S and T . Represent each vector $x \in X$ as a sum $x = u + v$, where $u \in S$ and $v \in T$. An operator A assigns to a vector x a vector u . It is called an *operator of projection onto S parallel to T* .

57. The vector space of operators

Fix two vector spaces, X and Y , over the same field P and consider the set ω_{XY} of all linear operators from X to Y . In ω_{XY} we can introduce the operations of operator addition and of multiplication of an operator by a number from P turning thereby ω_{XY} into a vector space.

Two operators, A and B , from X to Y are said to be *equal* if

$$Ax = Bx$$

for each vector $x \in X$. It is easy to verify that the equality relation of the operators is an equivalence relation. The equality of operators is designated

$$A = B.$$

An operator C is said to be the *sum* of operators A and B from X to Y if

$$Cx = Ax + Bx$$

for each $x \in X$. A sum of operators is designated

$$C = A + B.$$

By definition it is possible to add any operators from X to Y . If A and B are linear operators of ω_{XY} , then so is their sum. For any vectors $u, v \in X$ and any numbers $\alpha, \beta \in P$ we have

$$\begin{aligned} C(\alpha u + \beta v) &= A(\alpha u + \beta v) + B(\alpha u + \beta v) = \alpha Au \\ &\quad + \beta Av + \alpha Bu + \beta Bv = \alpha(Au + Bu) + \beta(Av + Bv) \\ &= \alpha Cu + \beta Cv. \end{aligned}$$

The addition of operators is an *algebraic operation*. It is also associative. Indeed, let A , B and C be three arbitrary linear operators of ω_{XY} . Then for any vector $x \in X$

$$\begin{aligned} ((A + B) + C)x &= (A + B)x + Cx = Ax + Bx + Cx \\ &= Ax + (Bx + Cx) = Ax + (B + C)x = (A + (B + C))x. \end{aligned}$$

But this means that

$$(A + B) + C = A + (B + C).$$

The addition of operators is a *commutative operation*. If A and B are any operators of ω_{XY} and x is any vector of X , then

$$(A + B)x = Ax + Bx = Bx + Ax = (B + A)x,$$

i.e.

$$A + B = B + A.$$

Now it is easy to show that the set ω_{XY} , with the operator addition introduced in it, is an Abelian group. It has at least one zero element, for example a zero operator. Each element of ω_{XY} has at least one opposite element, for example an opposite operator. Everything else follows from Theorem 7.1.

It follows from the same theorem that the addition of operators has an inverse. We shall call it *subtraction* and use the generally accepted notation and properties.

An operator C is said to be the *product of an operator A from X to Y by a number λ from a field P* if

$$Cx = \lambda \cdot Ax$$

for each $x \in X$. This product is designated

$$C = \lambda A.$$

A product of a linear operator of ω_{XY} by a number is again a linear operator of ω_{XY} . Indeed, for any vectors $u, v \in X$ and any numbers $\alpha, \beta \in P$ we have

$$\begin{aligned} C(\alpha u + \beta v) &= \lambda A(\alpha u + \beta v) = \lambda(\alpha Au + \beta Av) = \alpha(\lambda Au) \\ &\quad + \beta(\lambda Av) = \alpha Cu + \beta Cv. \end{aligned}$$

It is not hard to show that the addition of operators and the multiplication of an operator by a number satisfy all properties that define a vector space. Hence the set ω_{XY} of all linear operators from a vector space X to a vector space Y forms a new vector space. It follows that from the point of view of the operations of multiplication of an operator by a number and of addition and subtraction of operators all the rules of equivalent transformations of operator algebraic expressions are valid. In what follows these rules will no longer be stated explicitly.

Note that nowhere did we use the relation between vector spaces X and Y . They may be both distinct and coincident. The set ω_{XX} of linear operators from a space X to the same space X will be one of the main objects of our studies. We shall call them *linear operators in X* .

Exercises

1. Prove that multiplying an operator by a nonzero number leaves its rank and nullity unaffected.

2. Prove that the rank of a sum of operators is at most a sum of the ranks of the summands.

3. Prove that a set of linear operators of ω_{XY} whose ranges are in the same subspace is a linear subspace.

4. Prove that a system of two nonzero operators of $\omega_{\Lambda Y}$ whose ranges are distinct is linearly independent.

5. Prove that a space of linear operators in V_1 is one-dimensional.

58. The ring of operators

Consider three vector spaces X , Y and Z over the same field P . Let A be an operator from X to Y and let B be an operator from Y to Z .

An operator C from X to Z is said to be a *product of B by A* if

$$Cx = B(Ax)$$

for each vector $x \in X$. A product of B and A is designated

$$C = BA.$$

A product of linear operators is again a linear operator. For any vectors $u, v \in X$ and any numbers $\alpha, \beta \in P$ we have

$$\begin{aligned} C(\alpha u + \beta v) &= B(A(\alpha u + \beta v)) = B(\alpha Au + \beta Av) \\ &= \alpha B(Au) + \beta B(Av) = \alpha Cu + \beta Cv. \end{aligned}$$

The multiplication of operators is not an algebraic operation if only because product is not defined for any pair of operators. Nevertheless if realizable operator multiplication possesses quite definite properties. Namely:

- (1) $(AB)C = A(BC),$
- (2) $\lambda(BA) = (\lambda B)A = B(\lambda A),$
- (3) $(A + B)C = AC + BC,$
- (4) $A(B + C) = AB + AC$

(58.1)

for any operators A , B and C and any number λ from P if of course the corresponding expressions are defined.

The proofs of all these properties are similar, and therefore we shall restrict ourselves to the study of the first property. Let X , Y , Z and U be fixed vector spaces; and let A , B and C be any linear operators, where A is an operator from X to Y , B is operator from Y to Z , and C is an operator from Z to U . Observe first of all that in equation (1) both operators, $(AB)C$ and $A(BC)$, are defined. For any vector $x \in X$ we have

$$((AB)C)x = AB(Cx) = A(B(Cx)),$$

$$(A(BC))x = A(BCx) = A(B(Cx)),$$

which shows the validity of equation (1).

Again consider the set ω_{XX} of linear operators in X . For any two operators of ω_{XX} both a sum and a product are defined. According to properties (3) and (4) both operations are connected by the distributive law. The set ω_{XX} of linear operators is therefore a *ring*. It will be shown in what follows that a ring of operators is *noncommutative*. It may of course happen that for some particular pair of operators A and B the relation $AB = BA$ does hold. Such operators will be called *commutative*. In particular, an identity operator is commutative with any operator.

In a ring of linear operators, as in any other ring, a product of any operator by a zero operator is again a zero operator. The distributive law relates to multiplication not only a sum of operators but also their difference. A ring of linear operators is at the same time a vector space and therefore for a difference of operators we have

$$A - B = A + (-1)B.$$

Property (2) of (58.1) describes the relation of multiplication of operators in a ring to multiplication by a number. Remaining valid of course are all the relations following from the properties of vector spaces.

Exercises

1. In a space of polynomials in t , denote by D an operator of differentiation and by T an operator of multiplication by t . Prove that $DT \neq TD$. Find the operator $DT - TD$.

2. Fix some operator B of the space ω_{XX} . Prove that the set of operators A such that $BA = 0$ is a subspace in ω_{XX} .

3. Prove that the rank of a product of operators is not greater than the rank of each of the factors.

4. Prove that the nullity of a product of operators is not less than the nullity of each of the factors.

5. Prove that in the ring ω_{XX} of linear operators there are zero divisors.

59. The group of nonsingular operators

Linear operators in a space X form an Abelian group relative to addition. But we can find among such operators sets that are groups relative to multiplication. These are connected with the so-called nonsingular operators.

An operator in a vector space is said to be *nonsingular* if its kernel consists only of a zero vector. An operator that is not nonsingular is termed *singular*.

Nonsingular operators are, for example, the identity operator and the scalar operator, provided it is not zero. Sometimes it is possible to associate with an operator A in X some nonsingular operator even in the case where A is singular. Indeed, let T_A be the range of A and let N_A be its kernel. If T_A and N_A have no nonzero vectors in common, then by (56.4) we have

$$X = N_A + T_A.$$

As already noted, an operator A generates many other operators from any subspace complementary to N_A to the subspace of values T_A . In the case considered A generates an operator from T_A to T_A . That operator is nonsingular since it sends to zero only the zero vector of T_A .

Nonsingular operators possess many remarkable properties. For such operators the nullity is zero and therefore it follows from (56.4) that the rank of a nonsingular operator equals the dimension of the space. If a nonsingular operator A is an operator in X , then its range T_A coincides with X . Thus each vector of X is the image of some vector of X . This property of a nonsingular operator is *equivalent* to its definition.

An important property of the nonsingular operator is the uniqueness of the inverse image for any vector of a space. Indeed, suppose that for some vector y there are two inverse images, u and v . This means that

$$Au = y, \quad Av = y.$$

But then

$$A(u - v) = 0.$$

By the definition of a nonsingular operator its kernel consists only of a zero vector. Therefore $u - v = 0$, i.e. $u = v$. This property is also *equivalent* to the definition of a nonsingular operator. Actually it has already been noted in Section 56.

A product of any finite number of nonsingular operators is also a nonsingular operator. Obviously it suffices to prove this assertion for two operators. Let A and B be any nonsingular operators in the same space X . Consider the equation

$$BAx = 0. \quad (59.1)$$

According to the definition of operator multiplication this equation means that

$$B(Ax) = 0.$$

The operator B is nonsingular and therefore it follows from the last equation that $Ax = 0$. But A is also nonsingular and therefore $x = 0$. So only a zero vector satisfies (59.1), i.e. the operator BA is nonsingular.

A sum of nonsingular operators is not necessarily a nonsingular operator. If A is a nonsingular operator, then so is the operator $(-1)A$. But the sum of these operators is a zero operator which is singular.

Consider a set of nonsingular operators in the same vector space. On that set the multiplication of operators is an *algebraic* and *associative operation*. Among nonsingular operators is also the identity operator E which plays the part of unity. Indeed it is easy to verify that for any operator A in X

$$AE = EA = A.$$

If we show that for any nonsingular operator A there is a nonsingular operator such that a product of it and A is an identity operator, then this will mean that the set of all nonsingular operators is a group relative to multiplication.

Let A be a nonsingular operator. As we know, for each vector $y \in X$ there is one and only one vector $x \in X$ connected with y by the relation

$$y = Ax. \quad (59.2)$$

Consequently, it is possible to assign to each vector $y \in X$ a unique vector $x \in X$ such that y is its image. The relation constructed is some operator. It is called the *inverse* of the operator A and designated A^{-1} . If (59.2) holds, then

$$x = A^{-1}y. \quad (59.3)$$

We prove that an inverse operator is linear and nonsingular.

A product is defined for any operators and not only for linear operators. Therefore it follows from the definition of an inverse operator that

$$A^{-1}A = AA^{-1} = E. \quad (59.4)$$

To prove these equations it suffices to apply the operator A^{-1} to both sides of (59.2) and the operator A to both sides of (59.3).

Take any vectors $u, v \in X$ and any numbers $\alpha, \beta \in P$ and consider the vector

$$z = A^{-1}(\alpha u - \beta v) - \alpha A^{-1}u - \beta A^{-1}v.$$

Now apply A to both sides of the equation. From the linearity of A and from (59.4) we conclude that $Az = 0$. Since A is nonsingular, this means that $z = 0$. Consequently,

$$A^{-1}(\alpha u + \beta v) = \alpha A^{-1}u + \beta A^{-1}v,$$

i.e. A^{-1} is linear.

It is easy to show that A^{-1} is nonsingular. For any vector y from the kernel of A^{-1} we have

$$A^{-1}y = 0.$$

Apply A to both sides of this equation. Since A is a linear operator, $A0 = 0$. Considering (59.4) we conclude that $y = 0$. So the kernel of A^{-1} consists only of a zero vector, i.e. A^{-1} is a nonsingular operator.

Thus the set of nonsingular operators is a *group relative to multiplication*. It will be shown somewhat later that this group is *non-commutative*.

Using nonsingular operators it is possible to construct commutative groups too. Let A be an arbitrary operator in a space X . For any positive integer p we define the p th power of A by the equation

$$A^p = \overbrace{A \cdot A \dots A}^p, \quad (59.5)$$

where the right-hand side contains p multipliers. By virtue of the associativity of the operation of multiplication the operator A^p is uniquely defined. Of course it is linear.

For any positive integers p and r it follows from (59.5) that

$$A^p A^r = A^{p+r}. \quad (59.6)$$

If it is assumed by definition that

$$A^0 = E$$

for any operator A , then formula (59.6) will hold for any nonnegative integers p and r .

Suppose A is a nonsingular operator. Then so is the operator A^r for any nonnegative r . Hence there is an inverse operator for A^r . By (7.2) and (59.5) we have

$$(A^r)^{-1} = (A^{-1})^r = \overbrace{A^{-1} A^{-1} \dots A^{-1}}^r. \quad (59.7)$$

Also it is assumed by definition that

$$A^{-r} = (A^r)^{-1}.$$

Taking into account formulas (59.5) and (59.7) and the fact that $AA^{-1} = A^{-1}A$, it is not hard to prove the relation

$$A^p A^{-r} = A^{-r} A^p$$

for any nonnegative integers p and r . This means that formula (59.6) holds for any integers p and r .

Now take a nonsingular operator A and make up a set ω_A of operators of the form A^p for all integers p . On that set the multiplication of operators is an *algebraic* and, as (59.6) implies, *commutative operation*. Every operator A^p has an inverse equal to A^{-p} . Contained in ω_A is also an identity operator E . Hence ω_A is a *commutative group relative to multiplication*. It is called a *cyclic group* generated by the operator A .

Exercises

1. Prove that if for two linear operators A and B of ω_{XX} we have $AB = E$, then both operators are nonsingular.

2. Prove that for operators A and B of ω_{XX} to be nonsingular it is necessary and sufficient that so should operators AB and BA .

3. Prove that if an operator A is nonsingular and a number $\alpha \neq 0$, then the operator αA is also nonsingular and $(\alpha A)^{-1} = (1/\alpha)A^{-1}$.

4. Prove that $T_A \subset N_A$ if and only if $A^3 = 0$.

5. Prove that for any operator A

$$N_A \subseteq N_{A^2} \subseteq N_{A^3} \subseteq \dots, \quad T_A \supseteq T_{A^2} \supseteq T_{A^3} \supseteq \dots$$

6. Prove that an operator P is a projection operator if and only if $P^2 = P$. What are subspaces N_P and T_P ?

7. Prove that if P is a projection operator, then $E - P$ is also a projection operator.

8. Prove that if an operator A satisfies $A^m = 0$ for some positive integer m , then the operator $\alpha E - A$ is nonsingular for any number $\alpha \neq 0$.

9. Prove that a linear operator A for which $E + \alpha_1 A + \alpha_2 A^2 + \dots + \alpha_n A^n = 0$ is nonsingular.

10. Prove that if A is a nonsingular operator, then either all operators in the cyclic group ω_A are distinct or some power of A coincides with the identity operator.

60. The matrix of an operator

We discuss one general method of constructing a linear operator from an m -dimensional space X to an n -dimensional space Y . Suppose the vectors of a basis $e_1, e_2, \dots, e_m$ of X are assigned some vectors $f_1, f_2, \dots, f_m$ of Y . Then *there is a unique linear operator* A from X to Y which sends every vector e_k to a corresponding vector f_k .

Suppose that the desired operator A exists. Take a vector $x \in X$ and represent it as

$$x = \xi_1 e_1 + \xi_2 e_2 + \dots + \xi_m e_m.$$

Then

$$Ax = A\left(\sum_{k=1}^m \xi_k e_k\right) = \sum_{k=1}^m \xi_k A e_k = \sum_{k=1}^m \xi_k f_k.$$

The right-hand side of the relations is uniquely defined by x and the images of the basis. Therefore the equation obtained proves the uniqueness of the operator A if it exists. On the other hand, we can define the operator A by this equation, i.e. we can put

$$Ax = \sum_{k=1}^m \xi_k f_k.$$

It is easy to verify that the operator obtained is a linear operator from X to Y sending every vector e_k to a corresponding vector f_k . The range T_A of A coincides with the span of the system of vectors $f_1, f_2, \dots, f_m$.

We can now draw an important conclusion: *a linear operator A from a space X to a space Y is completely defined by the collection of images*

$$Ae_1, Ae_2, \dots, Ae_m$$

of any fixed basis

$$e_1, e_2, \dots, e_m$$

of X .

Fix a basis $e_1, e_2, \dots, e_m$ in X and a basis $q_1, q_2, \dots, q_n$ in Y . A vector e_1 is sent by A to some vector Ae_1 of Y which, as any vector of that space, can be expanded with respect to the basis vectors

$$Ae_1 = a_{11}q_1 + a_{21}q_2 + \dots + a_{n1}q_n.$$

Similarly

$$Ae_2 = a_{12}q_1 + a_{22}q_2 + \dots + a_{n2}q_n,$$

$$\dots \dots \dots \dots \dots \dots \dots \dots \dots$$

$$Ae_m = a_{1m}q_1 + a_{2m}q_2 + \dots + a_{nm}q_n.$$

The coefficients a_{ij} of these relations define an $n \times m$ matrix A_{qe}

$$A_{qe} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix}$$

called the *matrix of the operator A relative to the chosen bases*.

The columns of the matrix of the operator are the coordinates of the vectors $Ae_1, Ae_2, \dots, Ae_m$ relative to the basis $q_1, q_2, \dots, q_n$. To determine an element a_{ij} of the matrix of A we should apply A to the vector e_j and take the i th coordinate of the image Ae_j . If by $\{x\}_i$ we denote for brevity the i th coordinate of a vector x , then $a_{ij} = \{Ae_j\}_i$. This method of determining the elements of the matrix of an operator will be made use of in what follows.

Consider a vector $x \in X$ and its image $y = Ax$. We show how the coordinates of the vector y can be expressed in terms of the coordinates of x and the elements of the matrix of the operator. Let

$$x = \sum_{j=1}^m \xi_j e_j, \quad y = \sum_{i=1}^n \eta_i q_i. \quad (60.1)$$

Obviously

$$Ax = A\left(\sum_{j=1}^m \xi_j e_j\right) = \sum_{j=1}^m \xi_j A e_j = \sum_{j=1}^m \xi_j \sum_{i=1}^n a_{ij} q_i = \sum_{i=1}^n \left(\sum_{j=1}^m \xi_j a_{ij}\right) q_i.$$

Comparing the right-hand side of these equations with expansion (60.1) for y we conclude that the equations

$$\sum_{j=1}^m a_{ij} \xi_j = \eta_i$$

must hold for $i = 1, 2, \dots, n$, i.e. that

$$\begin{aligned} a_{11}\xi_1 + a_{12}\xi_2 + \dots + a_{1m}\xi_m &= \eta_1, \\ a_{21}\xi_1 + a_{22}\xi_2 + \dots + a_{2m}\xi_m &= \eta_2, \\ &\vdots \\ a_{n1}\xi_1 + a_{n2}\xi_2 + \dots + a_{nm}\xi_m &= \eta_n. \end{aligned} \quad (60.2)$$

Thus, given fixed bases in X and Y , every linear operator generates relations (60.2) connecting the coordinates of image and inverse image. To determine the coordinates of the image from the coordinates of the inverse image it suffices to calculate the left-hand sides of (60.2). To determine the coordinates of the inverse image from the known coordinates of the vector y we have to solve the system of linear algebraic equations (60.2) for the unknowns $\xi_1, \xi_2, \dots, \xi_m$. The matrix of the system coincides with the matrix of the operator.

Relations (60.2) establish a deep connection between linear operators and systems of linear algebraic equations. In particular, it follows from (60.2) that the rank of an operator coincides with that of the matrix of the operator and that its kernel coincides with the number of fundamental solutions of the reduced homogeneous system. This fact trivially yields (56.4) and a number of other formulas.

We shall fairly often turn to the connection between linear algebraic equations and linear operators. But we shall first prove that there is a 1-1 correspondence between operators and matrices, which properly speaking determine systems of the form (60.2). We have already shown that every operator A determines some matrix A_{qe} given fixed bases. Now take an $n \times m$ matrix A_{qe} . With bases in X and Y fixed, relations (60.2) assign to each vector $x \in X$ some vector $y \in Y$. It is easy to verify that this correspondence is a linear operator. We construct the matrix of a given operator in the same bases. All the

coordinates of a vector e_j are zero, except the j th coordinate which is equal to unity. It follows from (60.2) that the coordinates of the vector Ae_j coincide with the elements of the j th column of A_{qe} and therefore $\{Ae_j\}_i = a_{ij}$. Hence the matrix of the constructed operator coincides with the original matrix A_{qe} .

So every $n \times m$ matrix is the matrix of some linear operator from an m -dimensional space X to an n -dimensional space Y , with bases in X and Y fixed. A 1-1 correspondence is thus established between linear operators and rectangular matrices given any fixed bases. Of course both the vector spaces and the matrices are considered over the same field P .

Consider some examples. Let 0 be a zero operator. We have

$$\{0e_j\}_i = \{0\}_i = 0.$$

Hence all elements of the matrix of a zero operator are zero. Such a matrix is called a *zero matrix* and designated 0 .

Now take an identity operator E . For this operator we find

$$\{Ee_j\}_i = \{e_j\}_i = \begin{cases} 1 & \text{if } i=j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Therefore the matrix of an identity operator has the following form. It is a square matrix whose principal diagonal has unities and whose other elements are zeros. The matrix of an identity operator is called an *identity* or *unit matrix* and denoted by E .

We shall fairly often deal with yet another type of matrix. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be arbitrary numbers from a field P . We construct a square matrix Λ , with the given numbers along the principal diagonal and all other elements zero, i.e.

$$\Lambda = \begin{pmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{pmatrix}.$$

Matrices of this form are called *diagonal*. If all diagonal elements are equal, then the matrix is said to be a *scalar matrix*. In particular, an identity matrix is a scalar matrix. Rectangular matrices constructed in a similar way will also be called diagonal. If we consider relations (60.2), we shall easily see what the action of a linear operator with a matrix Λ is. This operator "stretches" the i th coordinate of any vector by a factor of λ_i for every i .

Exercises

1. In a space of polynomials of degree not higher than n a basis $1, t, t^2, \dots, t^n$ is fixed. Of what form is the matrix of the operator of differentiation in that basis?

2. In a space X an operator P of projection onto a subspace S parallel to a subspace T is given. Fix in X any basis made up as a union of the bases of S and T . Of what form are the matrices of P and $E - P$ in that basis?

3. Let A be a linear operator from X to Y . Denote by M_A a subspace in X complementary to the kernel N_A and by R_A a subspace in Y complementary to T_A . How will the matrix of A change if in choosing bases in X and Y we use bases of some or all of the indicated subspaces?

61. Operations on matrices

As we have shown, given fixed bases in spaces every linear operator is uniquely defined by its matrix. Therefore the operations on operators discussed earlier lead to quite definite operations on matrices. In the questions of interest to us now the choice of basis plays no part and therefore operators and their matrices will be denoted by the same letters with no indices relating to bases.

Let two equal operators from an m -dimensional space X to an n -dimensional space Y be given. Since equal operators exhibit sameness in all situations, they will have the same matrix. This justifies the following definition.

Matrices A and B of the same size $n \times m$ with elements a_{ij} and b_{ij} are said to be *equal* if

$$a_{ij} = b_{ij}$$

for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$. The equality of matrices is designated

$$A = B.$$

Suppose now that A and B are two operators from X to Y . Consider an operator $C = A + B$. Denote the elements of the matrices of the operators respectively by c_{ij} , a_{ij} and b_{ij} . According to the foregoing $c_{ij} = \{Ce_j\}_i$. Considering the definition of a sum of operators and the properties of the coordinates of vectors relative to the operations on them we get

$$\begin{aligned} c_{ij} = \{Ce_j\}_i &= \{(A + B)e_j\}_i = \{Ae_j + Be_j\}_i \\ &= \{Ae_j\}_i + \{Be_j\}_i = a_{ij} + b_{ij}. \end{aligned}$$

Therefore:

A sum of two matrices A and B of the same size $n \times m$ with elements a_{ij} and b_{ij} is a matrix C of the same size with elements c_{ij} if

$$c_{ij} = a_{ij} + b_{ij}$$

for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$. A sum of matrices is designated

$$C = A + B.$$

A difference of two matrices A and B of the same size $n \times m$ with elements a_{ij} and b_{ij} is a matrix C of the same size with elements c_{ij} if

$$c_{ij} = a_{ij} - b_{ij}$$

for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$. A difference of matrices is designated

$$C = A - B.$$

Consider an operator A from X to Y and an operator $C = \lambda A$ for some number λ . If a_{ij} and c_{ij} are elements of the matrices of the operators, then

$$c_{ij} = \{Ce_j\}_i = \{\lambda Ae_j\}_i = \lambda \{Ae_j\}_i = \lambda a_{ij},$$

and we arrive at the following definition:

A product of an $n \times m$ matrix A with elements a_{ij} by a number λ is an $n \times m$ matrix C with elements c_{ij} if

$$c_{ij} = \lambda a_{ij}$$

for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$. A product of a matrix by a number is designated

$$C = \lambda A.$$

Let an m -dimensional space X and an n -dimensional space Y be given over the same field P . As proved earlier, given fixed bases in X and Y there is a 1-1 correspondence between the set ω_{XY} of all operators from X to Y and the set of all $n \times m$ matrices with elements from P . Since operations on matrices were introduced in accordance with operations on operators, the set of $n \times m$ matrices, just as the set ω_{XY} , is a vector space.

It is easy to show one of the bases of the space of matrices. It is, for example, a system of matrices $A^{(kp)}$ for $k = 1, 2, \dots, n$ and $p = 1, 2, \dots, m$, where the elements $a_{ij}^{(kp)}$ of a matrix $A^{(kp)}$ are defined by the following equations:

$$a_{ij}^{(kp)} = \begin{cases} 1 & \text{if } i=k, j=p, \\ 0 & \text{otherwise.} \end{cases}$$

In the space ω_{XY} a basis is a system of operators with matrices $A^{(kp)}$. From this we conclude that the vector space of operators from X to Y is a finite dimensional space and that its dimension is equal to mn .

Let X, Y and Z be vector spaces, let A be an operator from X to Y and let B be an operator from Y to Z . Also let m, n and p be the

dimensions of X , Y and Z respectively. Assume that bases $e_1, \dots, e_m$, $q_1, \dots, q_n$ and $r_1, \dots, r_p$ are fixed in X , Y and Z . The operator A has an $n \times m$ matrix with elements a_{ij} , with

$$Ae_j = \sum_{s=1}^n a_{sj}q_s.$$

The operator B has a $p \times n$ matrix with elements b_{ih} , with

$$Bq_s = \sum_{h=1}^p b_{hs}r_h.$$

Investigating the matrix of the operator $C = BA$ we conclude that it must be $p \times m$ and that its elements are as follows:

$$\begin{aligned} c_{ij} &= \{Ce_j\}_i = \{BAe_j\}_i = \left\{B\left(\sum_{s=1}^n a_{sj}q_s\right)\right\}_i \\ &= \left\{\sum_{s=1}^n a_{sj}Bq_s\right\}_i = \left\{\sum_{s=1}^n a_{sj} \sum_{h=1}^p b_{hs}r_h\right\}_i = \left\{\sum_{h=1}^p \left(\sum_{s=1}^n b_{hs}a_{sj}\right)r_h\right\}_i = \sum_{s=1}^n b_{is}a_{sj}. \end{aligned}$$

This formula suggests the following definition:

A product of a $p \times n$ matrix B with elements b_{ih} and an $n \times m$ matrix A with elements a_{ij} is a $p \times m$ matrix C with elements c_{ij} if

$$c_{ij} = \sum_{s=1}^n b_{is}a_{sj} \quad (61.1)$$

for $i = 1, 2, \dots, p$ and $j = 1, 2, \dots, m$. A product of matrices is designated

$$C = BA.$$

Thus a product is defined only for the matrices in which the number of columns of the left factor is equal to the number of rows of the right factor. The element of the matrix of the product at the intersection of the i th row and the j th column is equal to the sum of the products of all the elements of the i th row of the left factor by the corresponding elements of the j th column of the right factor.

We recall once again that there is a 1-1 correspondence between linear operators and matrices. Operations on matrices were introduced according to operations on operators. Therefore the operation of matrix multiplication is connected by relations (58.1) with the operations of matrix addition and of multiplication of a matrix by a number.

We have already noted that a ring of operators and the group of all nonsingular operators in a vector space are noncommutative. To prove this it obviously suffices to find two square matrices A and B

such that $AB \neq BA$. Take, for example,

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

It is obvious that

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

and the noncommutativity of multiplication is proved.

The operation of matrix multiplication provides a convenient way of writing relations of the type (60.2). Denote by x_e an $m \times 1$ matrix made up of the coordinates of a vector x and y_q an $n \times 1$ matrix made up of the coordinates of a vector y . Then relations (60.2) are equivalent to a single matrix equation

$$A_{qe} x_e = y_q \quad (61.2)$$

which is called a *coordinate equation* corresponding to the *operator equation*

$$Ax = y$$

and relates in matrix form the coordinates of inverse image and image by means of the matrix of the operator.

It is important to note that the coordinate and the operator equation look quite alike from the notational point of view if of course the indices are dropped and the *symbol* Ax is understood as a product of A by x . Since the notation and the properties of operations on matrices and operators coincide, any transformation of an operator equation leads to the same transformation of the coordinate equation. Therefore formally it makes no difference whether we deal with matrix equations or with operator equations.

In what follows we shall actually draw no distinction between operator and coordinate equations. Moreover, *all new notions and facts that hold for operators will as a rule be extended to matrices, unless otherwise noted.*

Exercises

1. Prove that operations on matrices are related to the operation of transposition by

$$\begin{aligned} (\alpha A)' &= \alpha A', & (A + B)' &= A' + B', \\ (AB)' &= B'A', & (A)' &= A. \end{aligned}$$

2. Prove that every linear operator of rank r can be represented as a sum of r linear operators and cannot be represented as a sum of a smaller number of operators of rank 1.

3. Prove that an $n \times m$ matrix has rank 1 if and only if it can be represented as a product of two nonzero, $n \times 1$ and $1 \times m$, matrices.

4. Suppose that for fixed matrices A and B we have

$$AC = BC$$

for any matrix C . Prove that $A = B$.

5. Find the general form of a square matrix commutative with a given diagonal matrix.

6. Prove that for a matrix to be a scalar matrix it is necessary and sufficient that it should be commutative with all square matrices.

7. A sum of the diagonal elements of a matrix A is called the *trace* of A and designated $\text{tr } A$. Prove that

$$\begin{aligned}\text{tr } A &= \text{tr } A', & \text{tr } (\alpha A) &= \alpha \cdot \text{tr } A, \\ \text{tr } (A + B) &= \text{tr } A + \text{tr } B, & \text{tr } (BA) &= \text{tr } (AB).\end{aligned}$$

8. Prove that a real matrix A is zero if and only if $\text{tr } (AA') = 0$.

62. Matrices and determinants

Matrices play a very important part in the study of linear operators, with determinants not infrequently used as an auxiliary tool. We discuss now some questions connected with matrices and determinants.

Let A be a nonsingular operator in a space X . Its rank coincides with the dimension of X . According to formulas (60.2) this means that the rank of the system of columns of the matrix of the operator coincides with the number of them. This is possible if and only if the determinant of the matrix of an operator is zero. So

An operator in a vector space is nonsingular if and only if the determinant of its matrix is nonzero.

This property of a nonsingular operator justifies the following definitions:

A square matrix is said to be *nonsingular* if its determinant is nonzero and *singular* otherwise.

Of course, relying on the corresponding properties of nonsingular operators we can say that a product of nonsingular matrices is again a nonsingular matrix, that all nonsingular matrices form a group relative to multiplication, that every nonsingular matrix generates a cyclic group and so on. Their connection with nonsingular operators allows us to say that every nonsingular matrix A has a unique matrix A^{-1} such that

$$A^{-1}A = AA^{-1} = E. \quad (62.1)$$

The matrix A^{-1} is called the *inverse* of the matrix A .

Using the concept of determinant it is possible to express the explicit form of the elements of A^{-1} in terms of the minors of the matrix A . Formulas (40.5) to (40.9) provide a basis for this. Taking into account formula (61.1) for an element of a product of two ma-

trices we conclude that equations (62.1) are satisfied by the matrix

$$A^{-1} = \begin{pmatrix} \frac{A_{11}}{d} & \frac{A_{21}}{d} & \dots & \frac{A_{m1}}{d} \\ \frac{A_{12}}{d} & \frac{A_{22}}{d} & \dots & \frac{A_{m2}}{d} \\ \dots & \dots & \dots & \dots \\ \frac{A_{1m}}{d} & \frac{A_{2m}}{d} & \dots & \frac{A_{mm}}{d} \end{pmatrix}.$$

Here d is the determinant of a matrix A ; A_{ij} is the cofactor of a_{ij} , an element of A . By virtue of the uniqueness of the inverse matrix it is the only form it can have.

We introduce abbreviated notation for the minors of an arbitrary matrix A . A minor on the rows $i_1, i_2, \dots, i_p$ and columns $j_1, j_2, \dots, j_p$ is designated

$$A \begin{pmatrix} i_1, i_2, \dots, i_p \\ j_1, j_2, \dots, j_p \end{pmatrix}.$$

In addition, it is assumed that if any indices in the upper (lower) row in the notation of a minor coincide, then this means that so do the corresponding rows (columns) of the minor itself.

Theorem 62.1 (Cauchy-Binet formula). *Let a square $n \times n$ matrix C be equal to the product of two rectangular matrices A and B , of size $n \times m$ and $m \times n$ respectively, with $m \geq n$. Then*

$$C \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{pmatrix} = \sum_{1 \leq k_1 < k_2 < \dots < k_n \leq m} A \begin{pmatrix} 1 & 2 & \dots & n \\ k_1 & k_2 & \dots & k_n \end{pmatrix} B \begin{pmatrix} k_1 & k_2 & \dots & k_n \\ 1 & 2 & \dots & n \end{pmatrix}. \quad (62.2)$$

Proof. Denote by a_{ij} , b_{ij} and c_{ij} elements of A , B and C . According to the definition of a matrix product we have

$$c_{ij} = \sum_{s=1}^m a_{is} b_{sj}.$$

Substituting for the elements of C their expressions and using the linear property of the determinant for column vectors we find

$$\det \begin{pmatrix} c_{11} & \dots & c_{1n} \\ c_{n1} & \dots & c_{nn} \end{pmatrix} = \det \begin{pmatrix} \sum_{s_1=1}^m a_{1s_1} b_{s_1 1} & \sum_{s_2=1}^m a_{1s_2} b_{s_2 2} & \dots & \sum_{s_n=1}^m a_{1s_n} b_{s_n n} \\ \dots & \dots & \dots & \dots \\ \sum_{s_1=1}^m a_{ns_1} b_{s_1 1} & \sum_{s_2=1}^m a_{ns_2} b_{s_2 2} & \dots & \sum_{s_n=1}^m a_{ns_n} b_{s_n n} \end{pmatrix}$$

$$\begin{aligned}
&= \sum_{s_1=1}^m \det \begin{pmatrix} a_{1s_1}b_{s_11} & \sum_{s_2=1}^m a_{1s_2}b_{s_22} & \dots & \sum_{s_n=1}^m a_{1s_n}b_{s_nn} \\ \dots & \dots & \dots & \dots \\ a_{ns_1}b_{s_11} & \sum_{s_2=1}^m a_{ns_2}b_{s_22} & \dots & \sum_{s_n=1}^m a_{ns_n}b_{s_nn} \end{pmatrix} \\
&= \sum_{s_1=1}^m \sum_{s_2=1}^m \det \begin{pmatrix} a_{1s_1}b_{s_11} & a_{1s_2}b_{s_22} & \dots & \sum_{s_n=1}^m a_{1s_n}b_{s_nn} \\ \dots & \dots & \dots & \dots \\ a_{ns_1}b_{s_11} & a_{ns_2}b_{s_22} & \dots & \sum_{s_n=1}^m a_{ns_n}b_{s_nn} \end{pmatrix} \\
&\dots = \sum_{s_1=1}^m \sum_{s_2=1}^m \dots \sum_{s_n=1}^m \det \begin{pmatrix} a_{1s_1}b_{s_11} & a_{1s_2}b_{s_22} & \dots & a_{1s_n}b_{s_nn} \\ \dots & \dots & \dots & \dots \\ a_{ns_1}b_{s_11} & a_{ns_2}b_{s_22} & \dots & a_{ns_n}b_{s_nn} \end{pmatrix} \\
&= \sum_{s_1=1}^m \sum_{s_2=1}^m \dots \sum_{s_n=1}^m A \begin{pmatrix} 1 & 2 & \dots & n \\ s_1 & s_2 & \dots & s_n \end{pmatrix} b_{s_11} b_{s_22} \dots b_{s_nn}. \quad (62.3)
\end{aligned}$$

Each of the indices $s_1, s_2, \dots, s_n$ is independent of the others and may take on any values from 1 to m and so the expression obtained is a sum of m^n terms. In that sum those terms are zero at least two of whose indices are the same, since so are the corresponding minors of the matrix A . All the other terms can be divided into groups of $n!$ terms each, regarding as a group all terms the values of whose indices form the same collection of numbers.

Denote by $k_1, k_2, \dots, k_n$ the values of the indices $s_1, s_2, \dots, s_n$ arranged in increasing order. Let

$$\varepsilon(s_1, s_2, \dots, s_n) = (-1)^N,$$

where N is the number of transpositions required for a permutation $s_1, s_2, \dots, s_n$ to be transformed into $k_1, k_2, \dots, k_n$. Then within one group of values of the indices $s_1, s_2, \dots, s_n$ the sum of the corresponding terms in (62.3) will be equal to

$$\begin{aligned}
&\sum \varepsilon(s_1, s_2, \dots, s_n) A \begin{pmatrix} 1 & 2 & \dots & n \\ k_1 & k_2 & \dots & k_n \end{pmatrix} b_{s_11} b_{s_22} \dots b_{s_nn} \\
&= A \begin{pmatrix} 1 & 2 & \dots & n \\ k_1 & k_2 & \dots & k_n \end{pmatrix} \sum \varepsilon(s_1, s_2, \dots, s_n) b_{s_11} b_{s_22} \dots b_{s_nn} \\
&= A \begin{pmatrix} 1 & 2 & \dots & n \\ k_1 & k_2 & \dots & k_n \end{pmatrix} B \begin{pmatrix} k_1 & k_2 & \dots & k_n \\ 1 & 2 & \dots & n \end{pmatrix}.
\end{aligned}$$

It is from this relation that (62.2) follows.

with respect to the vectors $e_1, e_2, \dots, e_m$. The coefficients p_{ij} define a matrix

$$P = \begin{pmatrix} p_{11} & p_{12} & \dots & p_{1m} \\ p_{21} & p_{22} & \dots & p_{2m} \\ \dots & \dots & \dots & \dots \\ p_{m1} & p_{m2} & \dots & p_{mm} \end{pmatrix},$$

called the *coordinate transformation matrix* for a change from a basis $e_1, e_2, \dots, e_m$ to a basis $f_1, f_2, \dots, f_m$.

Take a vector $x \in X$ and expand it with respect to the vectors of both bases. Let

$$x = \sum_{i=1}^m \xi_i e_i = \sum_{i=1}^m \eta_i f_i.$$

By (63.1) we have

$$\sum_{i=1}^m \xi_i e_i = \sum_{i=1}^m \eta_i f_i = \sum_{i=1}^m \eta_i \sum_{j=1}^m p_{ji} e_j = \sum_{j=1}^m \left(\sum_{i=1}^m \eta_i p_{ji} \right) e_j = \sum_{i=1}^m \left(\sum_{j=1}^m \eta_j p_{ij} \right) e_i.$$

Comparing the coefficients of e_i on the left and the right of these relations we find

$$\xi_i = \sum_{j=1}^m p_{ij} \eta_j \quad (63.2)$$

for $i = 1, 2, \dots, m$. These formulas are called *coordinate transformation formulas*. As before denote by x_e and x_f $m \times 1$ matrices made up of the coordinates of the vector x in the corresponding bases. Formulas (63.2) show that

$$x_e = P x_f. \quad (63.3)$$

A coordinate transformation matrix must be *nonsingular*, since otherwise there will be linear dependence among its columns and hence among the vectors $f_1, f_2, \dots, f_m$. Of course, any nonsingular matrix is the matrix of some coordinate transformation defined by equation (63.3). Multiplying (63.3) on the left by the matrix P^{-1} we get

$$x_f = P^{-1} x_e.$$

Now let $e_1, \dots, e_m, f_1, \dots, f_m$ and $r_1, \dots, r_m$ be three bases in a vector space X . A change from the first basis to the third can be effected in two ways: either directly from the first to the third basis or from the first to the second and then from the second to the third. It is not hard to establish the connection between the corresponding coordinate transformation matrices. By (63.3)

$$x_e = P x_f, \quad x_f = R x_r, \quad x_e = S x_r.$$

From the first two relations it follows that

$$x_e = Px_f = P(Rx_r) = (PR)x_r,$$

which implies that

$$S = PR.$$

Thus, with coordinate transformations successively carried out, the matrix of the resulting transformation is equal to the product of the matrices of the constituent transformations.

Again consider a linear operator A from X to Y . Choose two bases, $e_1, \dots, e_m$ and $f_1, \dots, f_m$, in X and two bases, $q_1, \dots, q_n$ and $t_1, \dots, t_n$, in Y . Corresponding to the same operator A in the first pair of bases is the coordinate equation

$$y_q = A_{qe}x_e \quad (63.4)$$

and in the second

$$y_t = A_{tf}x_f. \quad (63.5)$$

Accordingly we have two matrices A_{qe} and A_{tf} for the same operator A .

Denote by P a coordinate transformation matrix for a change from the basis $e_1, \dots, e_m$ to the basis $f_1, \dots, f_m$ and by Q a coordinate transformation matrix for a change from $q_1, \dots, q_n$ to $t_1, \dots, t_n$. We have

$$x_e = Px_f, \quad y_q = Qy_t. \quad (63.6)$$

Substituting these expressions for x_e and y_q in (63.4) we find

$$Qy_t = A_{qe}Px_f,$$

which yields

$$y_t = (Q^{-1}A_{qe}P)x_f.$$

Comparing this with (63.5) we conclude that

$$A_{tf} = Q^{-1}A_{qe}P. \quad (63.7)$$

This is the desired relation connecting the matrices of the same operator in different bases.

Exercises

1. Prove that the rank of the matrix of an operator is not affected by a change to other bases.

2. Prove that the determinant of the matrix of an operator in a vector space is independent of the choice of basis.

3. What correspondence can be established between nonsingular operators in a space X and transformations of coordinates in the same space?

4. Let us say that two bases of the same real space are of the same sign if the determinant of their coordinate transformation matrix is positive. Prove that all bases can be divided into two classes of bases of the same sign.

5. Let us say that one class of bases of the same sign is left-handed and the other is right-handed. Compare these classes with those described in Section 34.

64. Equivalent and similar matrices

Corresponding to every linear operator A from a space X to a space Y is a set of its matrices defined by the possibility of choosing different bases in X and Y . The structure of that set is essentially different according as X coincides with Y or does not.

Two rectangular matrices A and B of the same size are said to be *equivalent* if there are two nonsingular square matrices R and S such that

$$B = RAS.$$

It follows from (63.7) that two matrices corresponding to the same linear operator when different bases are chosen in X and Y are always equivalent. It is not hard to see that the converse is also true. That is, two equivalent matrices always correspond to the same linear operator in suitably chosen bases. Thus, corresponding to every linear operator mapping X into Y is a class of equivalent matrices.

Theorem 64.1. *For two rectangular matrices of the same size to be equivalent it is necessary and sufficient that they should have the same rank.*

Proof. Multiplying any matrix by nonsingular matrices leaves its rank unaffected and therefore equivalent matrices have the same rank. Now let two matrices of the same size have the same rank. We prove that they are equivalent. We prove even more, that is that every matrix of rank r is equivalent to a matrix

$$I_r = \begin{pmatrix} 1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix}.$$

$\underbrace{\hspace{10em}}_r$

Let a rectangular $n \times m$ matrix be given. It defines some linear operator A mapping a space X with a basis $e_1, e_2, \dots, e_m$ into a space Y with a basis $q_1, q_2, \dots, q_n$. Denote by r the number of linearly independent vectors among the images of the vectors of the basis $Ae_1, Ae_2, \dots, Ae_m$. We may assume without loss of generality that it is the vectors $Ae_1, Ae_2, \dots, Ae_r$ that are linearly independent, since this can be achieved by a proper numbering of the basis vectors. The remaining vectors $Ae_{r+1}, \dots, Ae_m$ can be linearly expressed in

terms of them,

$$Ae_k = \sum_{j=1}^r c_{kj} Ae_j \quad (64.1)$$

for $k = r + 1, \dots, m$. We define a new basis $f_1, f_2, \dots, f_m$ in X as follows

$$f_k = \begin{cases} e_k, & k = 1, 2, \dots, r, \\ e_k - \sum_{j=1}^r c_{kj} e_j, & k = r + 1, \dots, m. \end{cases} \quad (64.2)$$

Then by (64.1)

$$Af_k = 0 \quad (64.3)$$

for $k = r + 1, \dots, m$. Set, further,

$$Af_j = t_j \quad (64.4)$$

for $j = 1, 2, \dots, r$. Vectors $t_1, t_2, \dots, t_r$ are by assumption linearly independent. Supplement them with some vectors $t_{r+1}, \dots, t_n$ to a basis in Y and consider the matrix of the operator A in the new bases $f_1, \dots, f_m$ and $t_1, \dots, t_n$. The coefficients of the k th column of the matrix coincide with the coordinates of the vector Af_k in the basis $t_1, \dots, t_n$. According to (64.3) and (64.4) the matrix of A will coincide with I_r .

The original matrix and I_r correspond to the same operator and therefore they are equivalent. Hence all matrices of the same rank are equivalent to I_r and therefore to one another.

While proving the theorem we answered a very important question: *How are bases in spaces X and Y to be chosen for the matrix of the linear operator to have the simplest form?* Besides we have shown an explicit form of that simplest matrix.

So simple and effective an answer has turned out to be possible because bases in X and Y could be chosen *independently* of each other. Now let A be an operator in a space X . Of course, we could again consider images and inverse images in different bases, but it is not natural now since both images and inverse images are in the same space. Using different bases would greatly hamper the study of the action of the operator on the vectors of the space X . If there is one basis, then the matrices P and Q in (63.6) coincide. Hence, corresponding to every linear operator in a vector space is a class of matrices connected by the relations

$$B = P^{-1}AP \quad (64.5)$$

for different nonsingular matrices P . Such matrices are called *similar* and a matrix P is called a *similarity transformation matrix*.

The question as to when two matrices can be similar is rather complicated and we shall get the answer much later. As complicated is the question of what the form of the simplest of all similar matrices is like. Devoted to these studies are the next two chapters.

Exercises

1. Prove that the equivalence and similarity criteria of matrices are equivalence relations.
2. Prove that similar matrices have the same trace and the same determinant.
3. Prove that under the same similarity transformation a cyclic group of nonsingular matrices goes over into a cyclic group.
4. Prove that under the same similarity transformation a linear subspace of matrices goes over into a linear subspace.
5. On a set of square matrices of the same size consider an operator of similarity transformation of those matrices using a fixed similarity transformation matrix. Prove that that operator is linear.
6. Prove that the set of all similarity transformation operators over the same set of square matrices of the same size forms a group relative to multiplication.

The Characteristic Polynomial

65. Eigenvalues and eigenvectors

Let A be a linear operator in a space X . This means that each vector $x \in X$ is assigned some vector $y = Ax$ of the same space X . It may turn out that for some nonzero vector x its image and inverse image are collinear. As we shall see in what follows such a situation substantially simplifies the study of the operator.

A number λ is said to be an *eigenvalue* and a nonzero vector x is said to be an *eigenvector* of a linear operator A if they are connected by the relation $Ax = \lambda x$.

Notice that if x is an eigenvector corresponding to an eigenvalue λ , then any collinear vector αx for $\alpha \neq 0$ will be also an eigenvector. If there are two eigenvectors, x and y , corresponding to an eigenvalue λ , then any nonzero vector of the form $\alpha x + \beta y$ will be an eigenvector. By definition a zero vector is not an eigenvector. Therefore the set X_λ of all eigenvectors which are linear combinations of any number of given eigenvectors corresponding to the same eigenvalue λ is not a subspace. If, however, we extend X_λ by joining a zero vector, then X_λ will become a subspace. We call it a *proper subspace* of A corresponding to the eigenvalue λ .

It is not hard to understand that eigenvectors of the operators 0 , E and αE are all the nonzero vectors of X . These operators have each only one eigenvalue, 0 , 1 and α , respectively, and hence at least one proper subspace coinciding with the entire space X . The projection operator P has two collections of eigenvectors, all vectors in the range of P and all vectors in the range of $E - P$. To the first collection there corresponds the eigenvalue $\lambda = 1$ and to the second there corresponds $-\lambda = 0$. Indeed, since $P^2 = P$, we have

$$P(Px) = P^2x = Px = 1 \cdot Px,$$

$$P((E - P)x) = (P - P^2)x = (P - P)x = 0 = 0 \cdot (E - P)x.$$

Consequently, a projection operator has at least two proper subspaces.

Theorem 65.1. *A system of eigenvectors $x_1, x_2, \dots, x_m$ of an operator A corresponding to mutually distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ is linearly independent.*

Proof. Eigenvectors are nonzero by definition and therefore the theorem is clearly true for $m = 1$. Let it be true for any system of $m - 1$ eigenvectors but false for vectors $x_1, x_2, \dots, x_m$. Then the system of those vectors will be linearly dependent, i.e. for some numbers $\alpha_1, \alpha_2, \dots, \alpha_m$ not all zero

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_m x_m = 0. \quad (65.1)$$

Suppose that $\alpha_1 \neq 0$. Applying A to (65.1) we get

$$\alpha_1 \lambda_1 x_1 + \alpha_2 \lambda_2 x_2 + \dots + \alpha_m \lambda_m x_m = 0. \quad (65.2)$$

On multiplying (65.1) by λ_m and subtracting it from (65.2) we find

$$\begin{aligned} \alpha_1 (\lambda_1 - \lambda_m) x_1 + \alpha_2 (\lambda_2 - \lambda_m) x_2 + \\ \dots + \alpha_{m-1} (\lambda_{m-1} - \lambda_m) x_{m-1} = 0. \end{aligned}$$

By induction hypothesis it follows that all coefficients of $x_1, x_2, \dots, x_{m-1}$ are zero. In particular, $\alpha_1 (\lambda_1 - \lambda_m) = 0$, which contradicts the hypothesis that $\lambda_1 \neq \lambda_m$ and the assumption that $\alpha_1 \neq 0$. Hence the system of vectors $x_1, x_2, \dots, x_m$ is linearly independent.

Corollary. *No linear operator in an m -dimensional space can have more than m mutually distinct eigenvalues.*

Of particular interest is the case when in an m -dimensional space the operator A has m mutually distinct eigenvalues. Then by Theorem 65.1 we can choose a basis of the space consisting entirely of eigenvectors of A .

A linear operator A in an m -dimensional space X is said to be an operator of a simple structure if it has m linearly independent eigenvectors.

The fact that of all linear operators we single out operators of a simple structure is very simply explained. These and only these operators have *diagonal matrices* in some basis. Indeed, let $x_1, x_2, \dots, x_m$ be linearly independent eigenvectors of an operator A . Take them as basis vectors of a space X and construct the matrix of A in that basis. We have

$$Ax_1 = \lambda_1 x_1,$$

$$Ax_2 = \lambda_2 x_2,$$

$$\dots$$

$$Ax_m = \lambda_m x_m.$$

We recall that column elements of the matrix of an operator coincide with the coordinates of the images of basis vectors. Therefore the matrix A_λ of the operator A has the following form in a basis con-

sisting of eigenvectors:

$$A_\lambda = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_m \end{pmatrix}.$$

If now A has in some basis $x_1, x_2, \dots, x_m$ a diagonal matrix with some, *not necessarily different*, numbers $\lambda_1, \lambda_2, \dots, \lambda_m$ on the principal diagonal, then $x_1, x_2, \dots, x_m$ are eigenvectors of A corresponding to eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$.

Thus operators of a simple structure, and operators of a simple structure alone, have diagonal matrices in some basis. That basis can be made up *only* of eigenvectors of the operator A . The action of any operator of a simple structure always reduces to a "stretching" of the coordinates of a vector in the given basis. If all linear operators had a simple structure, then the question of choosing a basis in which the matrix of an operator has the simplest form would have been completely solved. However, operators of a simple structure do not exhaust all linear operators.

Exercises

1. Let an operator A have an eigenvector x corresponding to an eigenvalue λ . Prove that for the operator

$$\alpha_0 E + \alpha_1 A + \dots + \alpha_n A^n,$$

where $\alpha_0, \alpha_1, \dots, \alpha_n$ are some numbers, the vector x is also an eigenvector but that it corresponds to an eigenvalue $\alpha_0 + \alpha_1 \lambda + \dots + \alpha_n \lambda^n$.

2. Prove that operators A and $A - \alpha E$ have the same eigenvectors for any operator A and any number α .

3. Prove that an operator A is nonsingular if and only if it has no zero eigenvalues.

4. Prove that operators A and A^{-1} have the same eigenvectors for any nonsingular operator A . What is the connection between the eigenvalues of the operators?

5. Prove that if an operator A is of a simple structure, then the operator

$$\alpha_0 E + \alpha_1 A + \dots + \alpha_n A^n$$

is also of a simple structure.

6. Prove that an operator of differentiation in a space of polynomials is not an operator of a simple structure. Find the eigenvectors and eigenvalues of that operator.

7. Consider a similarity transformation operator with a diagonal matrix. Prove that that operator is of a simple structure. Find all its eigenvectors and eigenvalues.

66. The characteristic polynomial

Not any linear operator has at least one eigenvector. Suppose, for example, that we have an operator in a space V_2 which turns every directed line segment about the origin

counterclockwise through an angle of 90° . It is obvious that in that case image and inverse image will never be collinear and the operator will have no eigenvector. To study the question of the existence of eigenvectors we first introduce an equation satisfied by all eigenvalues of a linear operator.

Let A be a linear operator in an m -dimensional space X over a field P . If A has an eigenvalue λ corresponding to an eigenvector x , then by definition $Ax = \lambda x$ or equivalently

$$(\lambda E - A)x = 0. \quad (66.1)$$

Since x is nonzero, from (66.1) it follows that the operator $\lambda E - A$ is singular. Thus the eigenvalues of A are only those numbers λ from P for which $\lambda E - A$ is singular.

Fix in X some basis $e_1, e_2, \dots, e_m$ and denote by A_e the matrix of A in that basis. The operator $\lambda E - A$ is singular if and only if so is its matrix $\lambda E - A_e$, i.e. if

$$\det(\lambda E - A_e) = 0. \quad (66.2)$$

Determining eigenvalues was independent of the choice of basis in X . Therefore the numbers λ from P satisfying (66.2) must not depend on the choice of basis either. It is in fact the left-hand side of (66.2) that is *independent of the choice of basis* for any λ , although formally this dependence is noted. Suppose in some other basis $f_1, f_2, \dots, f_m$ the operator A has a matrix A_f . According to (64.5) A_e and A_f are related by

$$A_f = Q^{-1}A_eQ$$

for some nonsingular matrix Q . Now for any λ from P we find

$$\begin{aligned} \det(\lambda E - A_f) &= \det(\lambda Q^{-1}EQ - Q^{-1}A_eQ) = \det(Q^{-1}(\lambda E - A_e)Q) \\ &= \det Q^{-1} \det(\lambda E - A_e) \det Q = (\det Q)^{-1} \det(\lambda E - A_e) \det Q \\ &= \det(\lambda E - A_e). \end{aligned}$$

Taking into account the expression for the determinant of the matrix in terms of its elements it is easy to see that the left-hand side of (66.2) can be represented as follows:

$$\det(\lambda E - A_e) = a_0 + a_1\lambda + \dots + a_{m-1}\lambda^{m-1} + a_m\lambda^m. \quad (66.3)$$

The coefficients $a_0, \dots, a_m$ are calculated in some way from the elements of A_e and are independent of λ . The maximum power of λ enters only into the product of the diagonal elements of $\lambda E - A_e$ and therefore

$$a_m = 1.$$

We show explicit expressions of two more coefficients. Namely

$$a_0 = (-1)^m \det A_e, \quad a_{m-1} = -\operatorname{tr} A_e.$$

In general, it may be assumed that expanding the determinant $\det(\lambda E - A_e)$ in powers of λ in different ways we should obtain expressions of the type of the right-hand side of (66.3) but with different coefficients a_i . It will be shown later on, however, that this assumption is not valid. The coefficients on the right of (66.3) are independent of the way they are calculated. Considering the independence of the determinant $\det(\lambda E - A_e)$ from the basis we conclude that all coefficients $a_0, \dots, a_{m-1}$ are in fact characteristics of the operator A . The function

$$f(\lambda) = a_0 + a_1\lambda + \dots + a_{m-1}\lambda^{m-1} + \lambda^m \quad (66.4)$$

is called the *characteristic polynomial of the operator A*.

Associated with every linear operator is a characteristic polynomial. The converse is also true. Every polynomial of the form (66.4) is a characteristic polynomial of some linear operator. This may be, for example, an operator whose matrix A_e in some basis has the following form:

$$A_e = \begin{pmatrix} -a_{m-1} & -a_{m-2} & \dots & -a_1 & -a_0 \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix}. \quad (66.5)$$

This is easy to see from a direct check, using the Laplace theorem for calculating $\det(\lambda E - A_e)$. A matrix of the form (66.5) is called a *Frobenius matrix*.

For a number λ from P to be an eigenvalue of an operator A it is necessary and sufficient that it should satisfy the equation

$$a_0 + a_1\lambda + \dots + a_{m-1}\lambda^{m-1} + \lambda^m = 0,$$

i.e. that it should be a root of the characteristic polynomial. Not in every field P any polynomial with coefficients from P has at least one root from P . As an illustration, $\lambda^2 + 1$ has no roots either in the field of rationals or in the field of reals.

A field P is said to be *algebraically closed* if any polynomial with coefficients from P has at least one root from P .

Thus if a linear operator acts in a space over an algebraically closed field, it must have at least one eigenvector. It is possible to construct various algebraically closed fields, but it is only one of them, the field of complex numbers, that is of the greatest practical value. To prove the algebraic closure of this field is the aim of our immediate studies.

Exercises

1. Find the characteristic polynomial for the zero and the identity operator.
2. Find the characteristic polynomial for the operator of differentiation.
3. Is the coincidence of characteristic polynomials an indication of the equality of the operators?
4. Prove that operators with matrices A and A' have the same characteristic polynomials.
5. Suppose that in some basis an operator has matrix (66.5). Find the coordinates of the eigenvectors in the same basis.
6. Prove that an operator with matrix (66.5) has a simple structure if and only if the characteristic polynomial has m mutually distinct roots.

67. The polynomial ring

In some exercises and examples we have already noted the algebraic properties of polynomials. In connection with the investigation of the characteristic polynomial we shall continue those studies.

Let P be an arbitrary field. Consider a set of polynomials, i.e. functions of the form

$$f(z) = a_0 + a_1z + \dots + a_nz^n \quad (67.1)$$

dependent on an independent variable z assuming values from P and having coefficients $a_0, \dots, a_n$ from P . A polynomial $f(z)$ is said to be a polynomial of degree n if $a_n \neq 0$ and all coefficients with larger indices are zero. The only polynomial without a definite degree is the one all of whose coefficients are zero. We shall call it a *zero polynomial* and designate 0.

Two polynomials are said to be *equal* if so are all their coefficients of equal powers of the independent variable.

Now let $f(z)$ and $g(z)$ be polynomials of degree n and s respectively. Also let

$$\begin{aligned} f(z) &= a_0 + a_1z + \dots + a_{n-1}z^{n-1} + a_nz^n, \\ g(z) &= b_0 + b_1z + \dots + b_{s-1}z^{s-1} + b_s z^s \end{aligned} \quad (67.2)$$

and suppose for definiteness that $n \geq s$. A *sum* $f(z) + g(z)$ of $f(z)$ and $g(z)$ is a polynomial

$$f(z) + g(z) = c_0 + c_1z + \dots + c_{n-1}z^{n-1} + c_nz^n,$$

where $c_i = a_i + b_i$ for $i \leq s$ and $c_i = a_i$ for $i > s$. The degree of the sum of the polynomials is n if $n > s$, but for $n = s$ it is less than n if $b_n = -a_n$.

A *product* $f(z) \cdot g(z)$ of $f(z)$ and $g(z)$ is a polynomial

$$f(z) \cdot g(z) = d_0 + d_1z + \dots + d_{n+s-1}z^{n+s-1} + d_{n+s}z^{n+s},$$

where

$$d_i = \sum_{k+l=i} a_k b_l$$

for $i = 0, 1, \dots, n + s$. A coefficient d_i is a sum of the products of the coefficients of $f(z)$ and $g(z)$ the sum of whose indices is i . For example,

$$d_0 = a_0 b_0, \quad d_{n+s} = a_n b_s.$$

From the last equation it follows that $d_{n+s} \neq 0$ and therefore the degree of the product of nonzero polynomials is equal to the sum of the degrees of the factors. Hence a product of nonzero polynomials is a nonzero polynomial.

A special case of a product of polynomials is a *product* $\alpha f(z)$ of a polynomial $f(z)$ by a number α , since a nonzero number can be regarded as a polynomial of degree 0.

A set of polynomials with the operations introduced above is a *commutative ring*. We shall not concern ourselves with checking that all the axioms hold.

Theorem 67.1. *For any polynomial $f(z)$ and nonzero polynomial $g(z)$ we can find unique polynomials $q(z)$ and $r(z)$ such that*

$$f(z) = g(z) q(z) + r(z), \quad (67.3)$$

with the degree of $r(z)$ lower than that of $g(z)$ or $r(z) = 0$.

Proof. Let $f(z)$ and $g(z)$ be polynomials of degrees n and s . If $n < s$ or $f(z) = 0$, then it is possible to set $q(z) = 0$ and $r(z) = f(z)$ in (67.3). Suppose therefore that $n \geq s$.

We represent $f(z)$ and $g(z)$ according to (67.2) and set

$$f(z) - \frac{a_n}{b_s} z^{n-s} g(z) = f_1(z). \quad (67.4)$$

Let the degree of $f_1(z)$ be n_1 and let its leading coefficient be $a_{n_1}^{(1)}$. It is clear that $n_1 < n$. If $n_1 \geq s$, then we set

$$f_1(z) - \frac{a_{n_1}^{(1)}}{b_s} z^{n_1-s} g(z) = f_2(z). \quad (67.5)$$

Denote by n_2 the degree of $f_2(z)$ and by $a_{n_2}^{(2)}$ its leading coefficient. If $n_2 \geq s$, then again we set

$$f_2(z) - \frac{a_{n_2}^{(2)}}{b_s} z^{n_2-s} g(z) = f_3(z) \quad (67.6)$$

and so on.

The degrees of $f_1(z), f_2(z), \dots$ are decreasing. Therefore in a finite number of steps we arrive at the following equation:

$$f_{k-1}(z) - \frac{a_{n_{k-1}}^{(k-1)}}{b_s} z^{n_{k-1}-s} g(z) = f_k(z), \quad (67.7)$$

where $f_k(z)$ is either zero or its degree n_k is less than s . After that the process is stopped.

Now adding all equations of the type (67.4) to (67.7) we get

$$f(z) - \left(\frac{a_n}{b_s} z^{n-s} + \frac{a_{n_1}^{(1)}}{b_s} z^{n_1-s} + \dots + \frac{a_{n_{k-1}}^{(k-1)}}{b_s} z^{n_{k-1}-s} \right) g(z) = f_k(z).$$

This means that the polynomials

$$q(z) = \frac{a_n}{b_s} z^{n-s} + \frac{a_{n_1}^{(1)}}{b_s} z^{n_1-s} + \dots + \frac{a_{n_{k-1}}^{(k-1)}}{b_s} z^{n_{k-1}-s}, \quad r(z) = f_k(z)$$

satisfy equation (67.3), with either $r(z) = 0$ or the degree of $r(z)$ less than that of $g(z)$.

We now prove that the polynomials $q(z)$ and $r(z)$ satisfying the hypothesis of the theorem are unique. Let there be other polynomials, $q'(z)$ and $r'(z)$, such that

$$f(z) = g(z) q'(z) + r'(z),$$

with either $r'(z) = 0$ or the degree of $r'(z)$ less than that of $g(z)$. Then

$$g(z) (q(z) - q'(z)) = r'(z) - r(z). \quad (67.8)$$

The polynomial on the right of this equation is either zero or its degree is less than that of $g(z)$. But the polynomial on the left of this equation has a degree not less than that of $g(z)$ for $q(z) - q'(z) \neq 0$. Therefore (67.8) is possible only if

$$q(z) = q'(z), \quad r(z) = r'(z).$$

This completes the proof of the theorem.

A polynomial $q(z)$ is called the *quotient* of $f(z)$ by $g(z)$ and $r(z)$ is the *remainder*. If the remainder is zero, then $f(z)$ is said to be *divisible* by $g(z)$ and $g(z)$ is said to be a *divisor* of $f(z)$.

Consider division of a nonzero polynomial $f(z)$ by a first-degree polynomial $(z - a)$. We have

$$f(z) = (z - a) q(z) + r(z). \quad (67.9)$$

Since the degree of $r(z)$ must be less than that of $(z - a)$, $r(z)$ is a polynomial of degree zero, i.e. a constant. That constant is easy to determine. On substituting on the right and left of (67.9) $z = a$ we find that $r(z) = f(a)$. So

$$f(z) = (z - a) q(z) + f(a). \quad (67.10)$$

For $f(z)$ to be divisible by $(z - a)$ it is necessary and sufficient that $f(a) = 0$. Numbers a such that $f(a) = 0$ are usually called *roots* of $f(z)$. Thus finding all linear divisors of a polynomial is equivalent to finding all its roots.

Formula (67.10) allows the following conclusion to be drawn. For any a from P a polynomial $f(z)$ of degree n can be uniquely represented as an expansion in powers of $(z - a)$:

$$f(z) = A_0 + A_1(z-a) + \dots + A_{n-1}(z-a)^{n-1} + A_n(z-a)^n, \quad (67.11)$$

where $A_0, \dots, A_n$ are numbers from P .

The existence of at least one expansion (67.11) can be established in a fairly simple way. On dividing $f(z)$ by $(z - a)$ we obtain a quotient $q_1(z)$ and a remainder A_0 related by

$$(z) = (z - a) q_1(z) + A_0. \quad (67.12)$$

If $q_1(z)$ is of degree zero, then expansion (67.11) is obtained. If, however, the degree of $q_1(z)$ is other than zero, then on dividing $q_1(z)$ by $(z - a)$ we have

$$q_1(z) = (z - a) q_2(z) + A_1. \quad (67.13)$$

Combining (67.12) and (67.13) we find

$$f(z) = (z-a)^2 q_2(z) + A_1(z-a) + A_0.$$

We again divide $q_2(z)$ by $(z - a)$, if necessary, and so on. Since the degrees of the quotients $q_1(z), q_2(z), \dots$ are successively decreasing, the process stops in n steps to yield expansion (67.11).

Suppose now that an expansion of the same form has been obtained in some other way and has coefficients $A'_0, \dots, A'_n$. On letting

$$q'_i(z) = A'_i + A'_{i+1}(z-a) + \dots + A'_n(z-a)^{n-i}$$

for $i = 0, 1, \dots, n$ we conclude that

$$q'_i(z) = (z-a) q'_{i+1}(z) + A'_i, \quad (67.14)$$

with $q'_0(z) = f(z)$ of course. Comparing (67.12) with (67.14), when $i = 0$, and considering the uniqueness of quotient and remainder we conclude that $A_0 = A'_0$ and $q_1(z) = q'_1(z)$. Similarly we can prove the equality of the other coefficients.

Exercises

1. Prove that there are no zero divisors in a polynomial ring.

2. Suppose that for some polynomials $f(z)$ $\varphi(z) = g(z) \varphi(z)$. Prove that if $\varphi(z) \neq 0$, then $f(z) = g(z)$.

3. Prove that nonzero polynomials $f(z)$ and $g(z)$ are divisible by each other if and only if $g(z) = \alpha f(z)$ for a nonzero number α .

4. Let each of the polynomials $f_1(z), \dots, f_h(z)$ be divisible by $\varphi(z)$. Prove that so is $f_1(z)g_1(z) + \dots + f_h(z)g_h(z)$, where $g_1(z), \dots, g_h(z)$ are arbitrary polynomials.

5. Prove that in expansions (67.1) and (67.11) for the same polynomial $f(z)$ the coefficients a_n and A_n coincide.

68. The fundamental theorem of algebra

We proceed to prove one of the most important statements, the theorem on the algebraic closure of the field of complex numbers. This theorem has application in various fields of mathematics. In particular, it underlies all further theory of linear operators. Following the established tradition we shall call it the *fundamental theorem of algebra*.

So we must prove that any polynomial of degree $n \geq 1$ with complex coefficients has at least one root, in general a complex root. We first consider polynomials of a special form. Namely,

$$f(z) = a - z^n. \quad (68.1)$$

Let us represent complex numbers z in the so-called *trigonometric form*

$$z = r (\cos \varphi + i \sin \varphi).$$

Here r is a nonnegative number called the *absolute value* or *modulus* of a number z , and φ is a real number called the *argument* of z . It is clear that for every number z its absolute value is uniquely defined. For nonzero numbers z their argument is defined up to a multiple of 2π ; for $z = 0$ the argument is not defined. Composing the product of two complex numbers

$$z = r (\cos \varphi + i \sin \varphi), \quad v = \rho (\cos \psi + i \sin \psi),$$

we find

$$zv = r\rho (\cos \varphi + i \sin \varphi) (\cos \psi + i \sin \psi) = r\rho (\cos (\varphi + \psi) + i \sin (\varphi + \psi)).$$

From this we deduce that

$$z^n = r^n (\cos n\varphi + i \sin n\varphi).$$

This equation is called *de Moivre's formula*. It provides an easy way for finding the roots of equation (68.1). Indeed, let a complex number a be represented in trigonometric form

$$a = \alpha (\cos \theta + i \sin \theta).$$

The equation

$$a - z^n = 0$$

in z is equivalent to

$$\alpha (\cos \theta + i \sin \theta) = r^n (\cos n\varphi + i \sin n\varphi)$$

in r and φ . But the last equation clearly has the following solutions:

$$r = + \sqrt[n]{\alpha}, \quad \varphi = \frac{\theta + 2k\pi}{n}$$

for $k = 0, 1, 2, \dots, n - 1$. Hence the complex numbers

$$a_k = + \sqrt[n]{\alpha} \left(\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right) \quad (68.2)$$

are the roots of (68.1). We shall call them the n th roots of a and designate them as

$$a_k = \sqrt[n]{\alpha}.$$

Now let $f(z)$ be a polynomial with complex coefficients. We consider it to be a complex function of the complex independent variable z . For such functions, as for real-valued functions of a real independent variable, it is possible to introduce the concepts of continuity, of derivative and so on. Not all of these notions will be equally necessary to us, but they are all based on the use of the completeness of the space of complex numbers.

A one-valued complex function $f(z)$ of a complex independent variable z is said to be *continuous at a point* z_0 if for any arbitrarily small number $\epsilon > 0$ we can find $\delta > 0$ such that for any complex number z satisfying

$$|z - z_0| < \delta$$

we have

$$|f(z) - f(z_0)| < \epsilon.$$

A function $f(z)$ continuous at each point of its domain is called *everywhere continuous* or simply *continuous*.

Lemma 68.1. *A polynomial $f(z)$ with complex coefficients is a continuous function of a complex independent variable z .*

Proof. Let

$$f(z) = a_0 + a_1 z + \dots + a_n z^n \quad (68.3)$$

and let z_0 be an arbitrary fixed complex number. Denote $h = z - z_0$. We show that for any arbitrarily small number $\epsilon > 0$ we can find $\delta > 0$ such that $|f(z) - f(z_0)| < \epsilon$ for $|h| < \delta$.

On expanding the polynomial $f(z)$ in powers of $(z - z_0)$ we get

$$f(z) = A_0 + A_1(z - z_0) + \dots + A_n(z - z_0)^n.$$

Since $A_0 = f(z_0)$ and $(z - z_0)$ is denoted by h , we have

$$f(z_0 + h) - f(z_0) = A_1 h + \dots + A_n h^n. \quad (68.4)$$

Hence

$$|f(z_0 + h) - f(z_0)| \leq |A_1| |h| + \dots + |A_n| |h|^n = A(|h|). \quad (68.5)$$

The real-valued function $A(|h|)$ is a polynomial with real coefficients $|A_i|$ in a real variable $|h|$. As is known from mathematical analysis, $A(|h|)$ is a continuous function everywhere and, in

particular, for $|h| = 0$. Since $A(0) = 0$, given $\varepsilon > 0$ we can find $\delta > 0$ such that for

$$|h| < \delta \quad (68.6)$$

we have

$$A(|h|) < \varepsilon.$$

Taking into account inequality (68.5) we conclude that if (68.6) holds so does the inequality

$$|f(z_0 + h) - f(z_0)| < \varepsilon.$$

Corollary. *The absolute value of a polynomial is a continuous function.*

This statement is immediate from the following relation:

$$\|f(z) - f(z_0)\| \leq |f(z) - f(z_0)|.$$

Corollary. *If a sequence of complex numbers $\{z_k\}$ converges to z_0 , then for any polynomial $f(z)$*

$$\lim_{k \rightarrow \infty} f(z_k) = f(z_0).$$

Lemma 68.2. *If a polynomial $f(z)$ of degree $n \geq 1$ does not vanish for $z = z_0$, then we can always find a complex number h such that*

$$|f(z_0 + h)| < |f(z_0)|.$$

Proof. Again consider expansion (68.4). Let A_k be the first nonzero coefficient among $A_1, A_2, \dots, A_n$. Take

$$h = t \sqrt[k]{-\frac{f(z_0)}{A_k}}, \quad (68.7)$$

where we take as the k th root any of its values and

$$0 \leq t \leq 1. \quad (68.8)$$

Let

$$B_p = A_p \left(\sqrt[k]{-\frac{f(z_0)}{A_k}} \right)^p.$$

Now from (68.4), taking into account (68.7) and (68.8), we find

$$\begin{aligned} |f(z_0 + h)| &= |f(z_0) - t^k f(z_0) + t^{k+1} B_{k+1} + \dots + t^n B_n| \\ &\leq |(1 - t^k) f(z_0)| + t^{k+1} |B_{k+1}| + \dots + t^n |B_n| \\ &= (1 - t^k) |f(z_0)| + t^{k+1} |B_{k+1}| + \dots + t^n |B_n| \\ &= |f(z_0)| + t^k (-|f(z_0)| + t |B_{k+1}| + \dots + t^{n-k} |B_n|) \\ &= |f(z_0)| + t^k B(t). \end{aligned}$$

Finally we have

$$||f(z_0 + h)| \leq |f(z_0)| + t^k B(t).$$

The function $B(t)$ is a polynomial with real coefficients and real independent variable t . It is a continuous function. But $B(0) = -|f(z_0)| < 0$ and therefore by virtue of the continuity of $B(t)$ there is t_0 within $0 < t_0 \leq 1$ such that $B(t_0)$ is also negative. For a complex number h defined by a number t_0 according to (68.7) we get

$$|f(z_0 + h)| \leq |f(z_0)| + t_0^h B(t_0) < |f(z_0)|.$$

Lemma 68.3. *For any polynomial $f(z)$ of degree $n \geq 1$ and any infinitely large sequence $\{z_k\}$ of complex numbers there is a limiting relation*

$$\lim_{k \rightarrow \infty} |f(z_k)| = +\infty. \quad (68.9)$$

Proof. Consider polynomial (68.3). For any $z \neq 0$ we find

$$|f(z)| \geq |a_n| |z|^n \left(1 - \left|\frac{a_0}{a_n}\right| |z|^{-n} - \dots - \left|\frac{a_{n-1}}{a_n}\right| |z|^{-1}\right). \quad (68.10)$$

Since $\{z_k\}$ is infinitely large,

$$\lim_{k \rightarrow \infty} |z_k| = +\infty.$$

The right-hand side of relation (68.10) is a real-valued function, and it is easy to see that

$$\lim_{|z_k| \rightarrow \infty} \left(1 - \left|\frac{a_0}{a_n}\right| |z_k|^{-n} - \dots - \left|\frac{a_{n-1}}{a_n}\right| |z_k|^{-1}\right) = 1.$$

But for the other factor of (68.10) we have

$$\lim_{|z_k| \rightarrow \infty} |a_n| |z_k|^n = +\infty.$$

Consequently, (68.9) is true.

Theorem 68.1 (fundamental theorem of algebra). *Any polynomial $f(z)$ of degree $n \geq 1$ with complex coefficients has at least one root, in general a complex root.*

Proof. Consider the set of all possible absolute values of $f(z)$. Since $|f(z)| \geq 0$, that set is bounded below. It is known from mathematical analysis that any bounded below nonempty set of real numbers has a greatest lower bound. Let it be l for the set of values $|f(z)|$. This means that for every natural k we can find a complex number z_k such that

$$0 \leq |f(z_k)| - l \leq 2^{-k}.$$

It follows that

$$\lim_{k \rightarrow \infty} |f(z_k)| = l. \quad (68.11)$$

Assuming $\{z_k\}$ to be unbounded it would be possible to choose an infinitely large subsequence of it, and according to Lemma 68.3 relation (68.11) could not hold. Therefore $\{z_k\}$ is bounded. Choose a subsequence $\{z_{k_v}\}$ and let

$$\lim_{k_v \rightarrow \infty} z_{k_v} = z_0.$$

According to a corollary of Lemma 68.1 the absolute value of a polynomial is a continuous function. Hence

$$|f(z_0)| = \lim_{k_v \rightarrow \infty} |f(z_{k_v})| = l.$$

If $l \neq 0$, then from Lemma 68.2 it follows that there is a number z_0 such that $|f(z_0)| < l$. This contradicts the fact that l is the greatest lower bound of the absolute values of the polynomial and therefore $l = 0$.

So we have shown that there is a complex number z_0 such that $|f(z_0)| = 0$ or equivalently

$$f(z_0) = 0.$$

This means that z_0 is a root of $f(z)$.

Exercises

1. Prove that the set of all n th roots of the complex number 1 forms a commutative group relative to multiplication.

2. Prove that for a sequence of complex numbers $\{z_k\}$ to be bounded it is necessary and sufficient that for at least one polynomial $f(z)$ of degree $n \geq 1$ the sequence $\{f(z_k)\}$ should be bounded.

3. Prove that for any polynomial $f(z)$ of degree $n \geq 1$ and for any complex number z_0 there is a complex number h such that $|f(z_0 + h)| > |f(z_0)|$.

4. Prove that all roots of polynomial (68.3) are in the ring

$$\left(1 + \max_{k \geq 0} \left| \frac{a_k}{a_0} \right| \right)^{-1} \leq |z| \leq \left(1 + \max_{k \leq n} \left| \frac{a_k}{a_n} \right| \right).$$

5. Try to "prove" the algebraic closure of a field of real numbers according to the same scheme as that for complex numbers. In what place has the "proof" no analogy?

69. Consequences of the fundamental theorem

There arise a variety of consequences from the fundamental theorem. Let us consider the most important of them.

A polynomial $f(z)$ of degree $n \geq 1$ with complex coefficients has at least one root z_1 . Therefore $f(z)$ has a factorization

$$f(z) = (z - z_1) \varphi(z),$$

where $\varphi(z)$ is a polynomial of degree $n - 1$. The coefficients of $\varphi(z)$ are again complex numbers. Consequently, $\varphi(z)$ has a root z_2 (if $n \geq 2$) and

$$\varphi(z) = (z - z_2) \psi(z)$$

from which it follows that

$$f(z) = (z - z_1)(z - z_2)\psi(z).$$

Continuing the process we obtain a representation of the polynomial as a product of linear multipliers:

$$f(z) = b(z - z_1)(z - z_2) \dots (z - z_n),$$

where b is some number. Removing the parentheses at the right and comparing the coefficients of the powers with the coefficients a_i of $f(z)$ we conclude that $b = a_n$.

There may be equal numbers among $z_1, z_2, \dots, z_n$. Suppose for simplicity that $z_1, \dots, z_r$ are mutually distinct and that each of the numbers $z_{r+1}, \dots, z_n$ is equal to one of the first numbers. Then $f(z)$ can be written as follows:

$$f(z) = a_n (z - z_1)^{k_1} (z - z_2)^{k_2} \dots (z - z_r)^{k_r}, \quad (69.1)$$

where $z_i \neq z_j$ for $i \neq j$ and

$$k_1 + k_2 + \dots + k_r = n.$$

Representation (69.1) is called a *canonical factorization of a polynomial* $f(z)$.

The canonical factorization is unique for a polynomial $f(z)$ up to an arrangement of factors. Indeed, suppose that along with factorization (69.1) there is another canonical factorization

$$f(z) = a_n (z - v_1)^{l_1} (z - v_2)^{l_2} \dots (z - v_m)^{l_m}.$$

Then

$$(z - z_1)^{k_1} (z - z_2)^{k_2} \dots (z - z_r)^{k_r} = (z - v_1)^{l_1} (z - v_2)^{l_2} \dots (z - v_m)^{l_m}. \quad (69.2)$$

Notice that the collection of numbers $z_1, \dots, z_r$ must coincide with the collection of numbers $v_1, \dots, v_m$. If, for example, z_1 is equal to none of the numbers $v_1, \dots, v_m$, then substituting $z = z_1$ in (69.2) we obtain zero at the left and a nonzero number at the right. So if there are two canonical factorizations of $f(z)$, then (69.2) may be only as follows:

$$(z - z_1)^{k_1} (z - z_2)^{k_2} \dots (z - z_r)^{k_r} = (z - z_1)^{l_1} (z - z_2)^{l_2} \dots (z - z_r)^{l_r}.$$

Suppose, for example, that $k_1 \neq l_1$ and let $k_1 > l_1$ for definiteness. By dividing the right- and left-hand sides of the last equation by the same divisor $(z - z_1)^{l_1}$ we get

$$(z - z_1)^{k_1 - l_1} (z - z_2)^{k_2} \dots (z - z_r)^{k_r} = (z - z_2)^{l_2} \dots (z - z_r)^{l_r}.$$

Substituting $z = z_1$ we again see that there is zero at the left and a nonzero number at the right. The uniqueness of the canonical factorization is thus proved.

If $k_i = 1$ in the canonical factorization (69.1), then a root z_i is said to be *simple*; if $k_i > 1$, then a root z_i is said to be *multiple*. A number k_i is the *multiplicity* of a root z_i . We can now draw a very important conclusion:

Any polynomial of degree $n \geq 1$ with complex coefficients has n roots, each root counted according to its multiplicity.

A polynomial of degree zero has no roots. The only polynomial that has arbitrarily many mutually distinct roots is the zero polynomial. These facts may be used to draw the following conclusion:

If two polynomials $f(z)$ and $g(z)$ whose degrees do not exceed n have equal values for more than n distinct values of the independent variable, then all the corresponding coefficients of those polynomials are equal.

Indeed, according to the assumption the polynomial $f(z) - g(z)$ has more than n roots. But its degree does not exceed n and therefore $f(z) - g(z) = 0$.

So a polynomial $f(z)$ whose degree is not greater than n is completely defined by its values for any $n + 1$ distinct values of the independent variable. This makes it possible to reconstruct the polynomial from its values. It is not hard to show an explicit form of this "reconstructing" polynomial. If for the values of the independent variable equal to $\alpha_1, \dots, \alpha_{n+1}$ a polynomial $f(z)$ assumes the values $f(\alpha_1), \dots, f(\alpha_{n+1})$, then

$$f(z) = \sum_{i=1}^{n+1} f(\alpha_i) \frac{(z - \alpha_1) \dots (z - \alpha_{i-1})(z - \alpha_{i+1}) \dots (z - \alpha_{n+1})}{(\alpha_i - \alpha_1) \dots (\alpha_i - \alpha_{i-1})(\alpha_i - \alpha_{i+1}) \dots (\alpha_i - \alpha_{n+1})}.$$

It is clear that the degree of the polynomial at the right does not exceed n , and at the points $z = \alpha_i$ it assumes the values $f(\alpha_i)$. The polynomial thus constructed is called *Lagrange's interpolation polynomial*.

Consider now a polynomial $f(z)$ of degree n and let $z_1, z_2, \dots, z_n$ be its roots repeated according to multiplicity. Then

$$f(z) = a_n (z - z_1)(z - z_2) \dots (z - z_n).$$

Multiplying the parentheses at the right, collecting similar terms and comparing the resulting coefficients with those in (68.3) we can derive the following equations:

$$a_{n-1}/a_n = -(z_1 + z_2 + \dots + z_n),$$

$$a_{n-2}/a_n = +(z_1 z_2 + z_1 z_3 + \dots + z_1 z_n + \dots + z_{n-1} z_n),$$

$$a_{n-3}/a_n = -(z_1 z_2 z_3 + z_1 z_2 z_4 + \dots + z_{n-2} z_{n-1} z_n),$$

$$a_1/a_n = (-1)^{n-1} (z_1 z_2 \dots z_{n-1} + \dots + z_2 z_3 \dots z_n),$$

$$a_0/a_n = (-1)^n z_1 z_2 \dots z_n.$$

These are called *Vieta's formulas* and express coefficients of the polynomial in terms of its roots.

On the right of the k th equation is the sum of all possible products, k roots each, taken with a plus or a minus sign according as k is even or odd.

For further discussion we shall need some consequences of the fundamental theorem of algebra, relating to polynomials with real coefficients. Let a polynomial

$$f(z) = a_0 + a_1 z + \dots + a_n z^n$$

with real coefficients have a complex (but not real) root v , i.e.

$$a_0 + a_1 v + \dots + a_n v^n = 0.$$

The last equation is not violated if all numbers are replaced in it by complex conjugate ones. However, the coefficients $a_0, \dots, a_n$ and the number 0, being reals, will remain unaffected by the replacement. Therefore

$$a_0 + a_1 \bar{v} + \dots + a_n \bar{v}^n = 0,$$

i.e. $f(\bar{v}) = 0$.

Thus if a complex (but not a real) number v is a root of a polynomial $f(z)$ with real coefficients, then so is the complex conjugate number $\bar{v}$.

It follows that $f(z)$ will be divisible by a quadratic trinomial

$$\varphi(z) = (z - v)(z - \bar{v}) = z^2 - (v + \bar{v})z + v\bar{v}$$

with real coefficients. Using this fact we prove that v and $\bar{v}$ have the same multiplicity.

Let them have multiplicities k and l respectively and let $k > l$, for example. Then $f(z)$ is divisible by the l th degree of the polynomial $\varphi(z)$, i.e.

$$f(z) = \varphi_l^l(z) \cdot q(z).$$

The polynomial $q(z)$, as a quotient of two polynomials with real coefficients, has also real coefficients. By assumption it must have a number v as its $(k - l)$ -fold root and must have no root equal to $\bar{v}$. According to what was proved above it is impossible and therefore $k = l$. Thus all complex roots of any polynomial with real coefficients are mutually complex conjugate. From the uniqueness of the canonical factorization we can draw the following conclusion:

Any polynomial with real coefficients can be represented, up to an arrangement of the factors, uniquely as a product of its leading coefficient and polynomials with real coefficients. Those polynomials have leading coefficients equal to unity and are linear, if they correspond to real roots, and quadratic, if they correspond to a pair of complex conjugate roots.

Finally, we proceed to the most important conclusion, that for the sake of which, properly speaking, the fundamental theorem of algebra was proved. Let A be a linear operator in a complex space. The eigenvalues of that operator, and they alone, are the roots of the characteristic polynomial. By the fundamental theorem A has at least one eigenvalue λ . Hence

Any linear operator in a complex vector space has at least one eigenvector.

Notice that if A is an operator in a real or rational space, this conclusion is *no longer valid*.

In reference to eigenvalues we shall apply the same terminology as in reference to roots of a polynomial. In particular, an eigenvalue will be said to be simple, if it is a simple root of a characteristic polynomial, and multiple otherwise. The multiplicity of an eigenvalue λ will be the multiplicity of λ as a root of the characteristic polynomial.

Exercises

1. Prove that if a complex number $a \neq 0$, then for any natural n there are only n distinct complex numbers whose n th power is equal to a .

2. What is the relation between the roots of $f(z)$ and $f(z - a)$, where a is a complex number?

3. Let a polynomial $f(z)$ of degree not greater than n with complex coefficients assume equal values for $n + 1$ distinct values of the independent variable. Prove that $f(z)$ is a polynomial of degree zero.

4. Prove that any polynomial of an odd degree with real coefficients has at least one real root.

5. Prove that a polynomial $f(z)$ has at least a root in each of the regions

$$|z| \leq \sqrt[n]{\left| \frac{a_0}{a_n} \right|}, \quad |z| \geq \sqrt[n]{\left| \frac{a_0}{a_n} \right|}.$$

6. Prove that an operator A has a simple structure if and only if there are as many linearly independent eigenvectors corresponding to each of its eigenvalues as is the multiplicity of λ .

The Structure of a Linear Operator

70. Invariant subspaces

All the studies to come next will be carried on under the hypothesis that the linear operator is given in a complex space X . As already noted earlier, this assumption ensures that every linear operator has at least one eigenvector.

A subspace L of a vector space X is said to be *invariant* under an operator A if for each vector x of L its image Ax is also in L .

Any linear operator has at least two *trivial* invariant subspaces, the zero subspace and the entire space X . Of vital importance are only nontrivial invariant subspaces. Among them are, for example, proper subspaces. Since in a complex vector space any operator clearly has at least one eigenvector, any operator in such a space must have at least one nontrivial invariant subspace.

It is easy to verify that for every operator A its domain T_A and kernel N_A are invariant subspaces. They are trivial if and only if A is nonsingular or zero.

If L is an invariant subspace, then there may be many ways to construct a complementary subspace M such that $X = L + M$. Among those complementary subspaces *there may be no* invariant subspace, however. But if there is at least one invariant complementary subspace, then we may speak of decomposing the space into a direct sum of invariant subspaces.

Knowledge of some invariant subspace and certainly of a decomposition of a space as a direct sum of invariant subspaces makes it possible to construct a basis in which the matrix of the operator has a simpler form. Let an operator A have in an m -dimensional space X an invariant subspace L of dimension n . Choose a basis $e_1, e_2, \dots, e_m$ in X so that its first n vectors are in L . Then the images $Ae_1, \dots, Ae_n$ of the vectors $e_1, \dots, e_n$ are in L and it is possible to expand them with respect to the vectors $e_1, \dots, e_n$ as the basis vectors of L . Consequently,

$$Ae_1 = a_{11}e_1 + a_{21}e_2 + \dots + a_{n1}e_n,$$

$$\dots \dots \dots$$

$$Ae_n = a_{1n}e_1 + a_{2n}e_2 + \dots + a_{nn}e_n.$$

Recall that the column elements of the matrix of an operator coincide with the coordinates of the images of basis vectors. Therefore

the matrix A_e of A in the basis $e_1, e_2, \dots, e_m$ will be of the form

$$A_e = \left(\begin{array}{ccc|ccc} a_{11} & \dots & a_{1n} & a_{1,n+1} & \dots & a_{1m} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} & a_{n,n+1} & \dots & a_{nm} \\ \hline 0 & \dots & 0 & a_{n+1,n+1} & \dots & a_{n+1,m} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & a_{m,n+1} & \dots & a_{mm} \end{array} \right).$$

As a rule matrices of such a type would be written in the so-called *block form*. Namely,

$$A_e = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}. \quad (70.1)$$

Here A_{11} is a square $n \times n$ matrix, A_{22} is a square $m - n$ matrix, 0 is a zero $(m - n) \times n$ matrix and A_{12} is an $n \times (m - n)$ matrix.

Suppose now that X is decomposed as a direct sum of invariant subspaces L and M . Choose a basis $e_1, e_2, \dots, e_m$ in X so that its first n vectors are in L and the remaining $m - n$ vectors are in M . In this case the images $Ae_1, \dots, Ae_n$ can be expanded only with respect to the vectors $e_1, \dots, e_n$ and the images $Ae_{n+1}, \dots, Ae_m$ only with respect to the vectors $e_{n+1}, \dots, e_m$. The matrix A_{12} in (70.1) will obviously be zero. Therefore the matrix A_e of A in the basis under consideration will have a still simpler form. Namely,

$$A_e = \begin{pmatrix} A_{11} & 0 \\ 0 & A_{22} \end{pmatrix}.$$

Suppose the action of the operator A is investigated only on the vectors of the invariant subspace L . If $x \in L$, then $Ax \in L$. Hence it may be assumed that A generates on L some other operator, $A|L$, defined by the equation

$$(A|L)x = Ax$$

for every $x \in L$. The operator $A|L$ is called the *induced operator generated by the operator A* . In relation to $A|L$ the operator A is called the *generator*. By virtue of the linearity of A the induced operator is also linear. It coincides with the generator A on L and is not defined outside L . Thus these operators mainly differ in their domains.

However artificial its introduction may seem, the induced operator provides a very convenient auxiliary tool in carrying out diverse studies. For example, the induced operator, as any other linear operator, has at least one eigenvector. But since its domain coincides with that of the generator, this means that

Any linear operator has at least one eigenvector in every invariant subspace.

If a space is decomposed as a direct sum of r invariant subspaces, then the linear operator has at least r linearly independent eigenvectors.

It is clear that any eigenvalue and any eigenvector of an induced operator are respectively the eigenvalue and the eigenvector of the generator. Less obvious is

Theorem 70.1. *The characteristic polynomial of an induced operator generated on a nontrivial subspace is a divisor of the characteristic polynomial of the generator.*

Proof. Let an induced operator $A|L$ be defined on an invariant subspace L . Again choose a basis $e_1, \dots, e_m$ of a space X so that vectors $e_1, \dots, e_n$ constitute a basis in L . If the matrix of the generator is A_e of (70.1), then the matrix of $A|L$ is A_{11} of (70.1). The characteristic polynomial is equal to $\det(\lambda E - A_e)$ for A and to $\det(\lambda E - A_{11})$ for $A|L$. Applying the Laplace theorem to expand the determinant $\det(\lambda E - A_e)$ by the first n columns we find

$$\begin{aligned} \det(\lambda E - A_e) &= \det \begin{pmatrix} \lambda E - A_{11} & -A_{12} \\ 0 & \lambda E - A_{22} \end{pmatrix} \\ &= \det(\lambda E - A_{11}) \det(\lambda E - A_{22}). \end{aligned}$$

This equation establishes the validity of the theorem.

Determining all eigenvalues of the operator A reduces to finding all roots of the characteristic polynomial. If A has a nontrivial invariant subspace, then by Theorem 70.1 this problem can be reduced to finding all roots of two polynomials of lower degree. If the induced operator itself has a nontrivial invariant subspace, then the process of factoring the characteristic polynomial can be continued.

Exercises

1. Prove that the sum and the intersection of invariant subspaces are invariant subspaces.

2. Prove that if A is a nonsingular operator, then any induced operator is also nonsingular.

3. Prove that if A is an operator of a simple structure, then any induced operator has also a simple structure.

4. In what case is an invariant subspace of an operator of a simple structure a direct sum of proper subspaces?

5. Prove that if at least one invariant subspace of an operator A has no complementary invariant subspace, then A cannot be an operator of a simple structure.

6. Prove that if A is an operator of a simple structure, then its range and kernel have no nonzero vectors in common.

7. Prove that if a subspace is invariant under an operator A , then it is also invariant under the operator $a_0E + a_1A + \dots + a_pA^p$.

71. The operator polynomial

One of the most important ways of constructing invariant subspaces of a linear operator is by using polynomials with complex coefficients.

Let A be some linear operator in a complex space X . Take a polynomial

$$\varphi(z) = a_0 + a_1 z + \dots + a_p z^p$$

with complex coefficients and consider a linear operator

$$\varphi(A) = a_0 E + a_1 A + \dots + a_p A^p.$$

It is an operator in X and is called an *operator polynomial* or a *polynomial in the operator A* .

Fix an operator A and construct the set of all operator polynomials in A . Since the set of all polynomials is a commutative ring, so is the set of all operator polynomials. In particular, it follows that

$$\varphi(A) A = A \varphi(A)$$

for any polynomial $\varphi(z)$. *The commutativity of the ring of operator polynomials plays an exceptionally important part in all further studies.*

It is easy to show that the range T_φ of any operator polynomial $\varphi(A)$ is an invariant subspace for the operator A . Indeed, let $x \in T_\varphi$. This means that $x = \varphi(A)y$ for some $y \in X$. By the permutability of A and $\varphi(A)$ we have

$$Ax = A\varphi(A)y = \varphi(A)(Ay).$$

Hence the vector Ax is the result of applying $\varphi(A)$ to the vector $Ay \in X$, i.e. $Ax \in T_\varphi$.

The kernel N_φ of the operator polynomial $\varphi(A)$ is also an invariant subspace for the operator A . If $x \in N_\varphi$, then $\varphi(A)x = 0$, but then

$$\varphi(A)(Ax) = A(\varphi(A)x) = A(0) = 0.$$

It has already been noted earlier that there is at least one eigenvector of the operator in any invariant subspace. Now it is possible to make a more precise statement. Namely,

If an eigenvalue of an operator A is (is not) a root of a polynomial $\varphi(z)$, then all eigenvectors of A corresponding to that eigenvalue are in the kernel (the range) of the operator $\varphi(A)$.

Indeed, let x be an eigenvector of an operator A corresponding to an eigenvalue λ . In the exercises to Section 65 it was stressed that x is also an eigenvector for $\varphi(A)$ but corresponds to an eigenvalue $\varphi(\lambda)$. Hence $\varphi(A)x = \varphi(\lambda)x$. If λ is a root of $\varphi(z)$, then $\varphi(\lambda) = 0$ and x is in the kernel of $\varphi(A)$. But if $\varphi(\lambda) \neq 0$, then $\varphi(\lambda)x$ is a nonzero vector and x is in the range of $\varphi(A)$.

We cannot prove that any invariant subspace of A is either the range or the kernel of some operator polynomial. This assertion is false in general, which is exemplified by the identity operator. We have $\varphi(E) = \varphi(1)E$ for any polynomial $\varphi(z)$, and therefore the operator $\varphi(E)$ is either zero or nonsingular. Hence range and kernel are always trivial subspaces for $\varphi(E)$. But an invariant subspace of the operator E is any subspace. Nevertheless each invariant subspace of A has a definite relation to operator polynomials in A . We have

Theorem 71.1. *Let L be an invariant subspace of an operator A . If all eigenvalues of an operator induced on L are roots of a polynomial $\varphi(z)$, then L is in the kernels of operators $\varphi^k(A)$ for every sufficiently large positive integral power k .*

Proof. Denote by $T'_1, T'_2, \dots$ the ranges of the operators induced on L using operator polynomials $\varphi^k(A)$, with $k = 1, 2, \dots$. The operator $\varphi(A)$ is singular on L , since its kernel contains at least all eigenvectors of A lying in L . Therefore $T'_1 \subset L$, $\dim T'_1 < \dim L$. The subspace T'_1 is invariant under A . If T'_1 is nonzero, then by Theorem 70.1 the characteristic polynomial of the operator induced on T'_1 using A is a divisor of the characteristic polynomial of the operator induced on L using A . Hence all eigenvalues of the operator induced on T'_1 are also roots of $\varphi(z)$. But it again follows that $T'_2 \subset T'_1$, $\dim T'_2 < \dim T'_1$ and so on. The dimensions of $T'_1, T'_2, \dots$ cannot decrease without limit. Beginning with some k therefore these subspaces will remain zero, which means that the theorem is true.

The above studies result in establishing an important fact concerning the existence of nontrivial invariant subspaces.

Theorem 71.2. *Any linear operator A in an m -dimensional complex space X has at least one invariant subspace of dimension $m - 1$.*

Proof. An operator A has at least one eigenvector x . Let it correspond to an eigenvalue λ . By what has been proved the range T_λ of the operator $A - \lambda E$ is an invariant subspace of A . But since $A - \lambda E$ is singular, the subspace T_λ has a dimension not greater than $m - 1$.

Consider now any subspace L of dimension $m - 1$ entirely containing T_λ . Any vector of X is transformed by $A - \lambda E$ into some vector of T_λ . Any vector of L therefore again goes over into a vector of L . Thus L is a subspace invariant under $A - \lambda E$ and of course invariant under A . Thus the theorem is proved.

Exercises

1. Let A be an operator of differentiation in a finite dimensional real space of polynomials. What is the operator $\varphi(A)$ for a polynomial $\varphi(z)$ with real coefficients?

2. Let $\varphi(z)$ be the characteristic polynomial of an induced operator generated by an operator A on an invariant subspace N . Prove that N is in the kernel of an operator $\varphi^k(A)$ for some positive integer k .

3. Prove that if all eigenvalues of an operator A are roots of a polynomial $\varphi(z)$, then $\varphi^k(A) = 0$ for some positive integer k .

4. Prove that the ring of operator polynomials generated by any operator has zero divisors.

5. Prove that if A is an operator of a simple structure, then $\varphi(A)$ is also an operator of a simple structure. Is the converse true?

72. The triangular form

Now we can solve the problem of reducing the matrix of an operator to one of the simplest forms, the so-called *triangular form*.

Theorem 72.1. *For any linear operator A in an m -dimensional space X there are invariant subspaces L_p of dimension p , $p = 0, 1, \dots, m-1, m$, such that*

$$L_0 \subset L_1 \subset \dots \subset L_{m-1} \subset L_m.$$

Proof. The existence of L_0 and L_m is obvious. By what was proved earlier an operator A has an invariant subspace L_{m-1} of dimension $m-1$.

Consider on L_{m-1} an induced operator. As any other operator on L_{m-1} it has an invariant subspace L_{m-2} of dimension $m-2$. But a subspace invariant under an induced operator is invariant under the generator A too. Thus the existence of L_{m-2} is proved. If we consider an induced operator on L_{m-2} , we can similarly establish the existence of L_{m-3} and so on.

The theorem is interesting mainly because of its matrix interpretation. Construct a basis $e_1, e_2, \dots, e_m$ of X using invariant subspaces L_p . As a vector e_1 take any nonzero vector in L_1 , as a vector e_2 take any nonzero vector in L_2 that is not in L_1 , and in general take as a vector e_p any nonzero vector in L_p that is not in L_{p-1} . Consider the matrix A_e of A in that basis. Since e_j is in L_j and L_j is invariant under A , the vector Ae_j must be a linear combination of only vectors $e_1, e_2, \dots, e_j$. In the expansion

$$Ae_j = a_{1j}e_1 + a_{2j}e_2 + \dots + a_{mj}e_m,$$

then the coefficient of e_i must be zero for every $i > j$. Hence the matrix of A is of the form

$$A_e = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1, m-1} & a_{1m} \\ 0 & a_{22} & \dots & a_{2, m-1} & a_{2m} \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & a_{m-1, m-1} & a_{m-1, m} \\ 0 & 0 & \dots & 0 & a_{mm} \end{pmatrix},$$

where $a_{ij} = 0$ for $i > j$.

A matrix all of whose elements under (above) the principal diagonal are zero is called a *right (left) triangular matrix*. In matrix terms

the result obtained implies that any square matrix is similar to a right triangular matrix.

The triangular form of matrix is widely used in proving diverse facts concerning linear operators. This is mainly due to the following property:

If an operator A has in some basis a triangular matrix A_e , then the diagonal elements of A_e coincide with the eigenvalues of A even taking into account their multiplicities.

Indeed, using the Laplace theorem we find that the characteristic polynomial of A_e is equal to

$$\det(\lambda E - A_e) = \prod_{i=1}^m (\lambda - a_{ii}),$$

which implies the validity of the above assertion.

Much of the further theory of linear operators is devoted to improving the result just obtained, that on reducing the matrix of an operator to triangular form. The simplest form possible the matrix of an operator may have is the diagonal form. As we know, only the matrices of operators of a simple structure can be reduced to this form. However, the triangular form is not the simplest for operators of not a simple structure either.

Exercises

1. Prove that any square matrix is similar to a left triangular matrix.

2. Prove that a set of left (or right) triangular matrices forms a ring.

3. Prove that a set of nonsingular left (or right) triangular matrices forms a group.

4. Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be the eigenvalues of an operator A written out in succession according to multiplicity. Prove that, taking into account their multiplicities, the eigenvalues of an operator $\varphi(A)$ for any polynomial $\varphi(x)$ are $\varphi(\lambda_1), \varphi(\lambda_2), \dots, \varphi(\lambda_m)$.

5. Prove that if all diagonal elements of a triangular $m \times m$ matrix A are zero, then $A^m = 0$.

6. Let a triangular matrix be similar to a diagonal matrix. Prove that the similarity transformation matrix may be chosen to be left (or right) triangular.

73. A direct sum of operators

A linear operator all of whose eigenvalues are equal is in a sense an exception. Nevertheless we shall show that it is such operators that any linear operator can be made up of.

Let a space X be represented as a direct sum of subspaces L and M . We define some operator B on L and some operator C on M . For any vector $x \in X$ there is a unique decomposition

$$x = x_L + x_M, \quad (73.1)$$

where $x_L \in L$ and $x_M \in M$.

An operator A defined by

$$Ax = Bx_L + Cx_M$$

is called a *direct sum* of B and C . If one of the subspaces L and M is trivial, then the direct sum is also called trivial.

It is easy to verify that A is a linear operator in X . We show that it can be represented only uniquely as a direct sum of operators defined on L and M . Indeed, for any vector $x \in L$ we have $Ax = Bx$. Similarly $Ax = Cx$ for any $x \in M$. This means that B coincides with the induced operator $A|L$ and C coincides with $A|M$.

Consider now an operator A in a space X . If X is decomposed in some way into a direct sum of subspaces L and M invariant under A , then the operator A itself can be decomposed as a direct sum. Indeed, construct $A|L$ and $A|M$. On decomposing again a vector $x \in X$ as a sum (73.1) we get

$$Ax = (A|L)x_L + (A|M)x_M.$$

In this case, by Theorem 70.1, the characteristic polynomial of A is equal to the product of the characteristic polynomials of $A|L$ and $A|M$.

The operator A can be decomposed as a direct sum using *any* operator polynomial $\varphi(A)$. Denote by N_k the kernel of an operator $\varphi^k(A)$. This is a subspace invariant under A and it is obvious that $N_1 \subset N_2 \subset \dots$. We first prove that if $N_k = N_{k+1}$ for some k , then $N_k = N_p$ for every $p > k$. Indeed, take any vector $x \in N_p$. Then $\varphi^p(A)x = 0$. On writing this as $\varphi^{k+1}(A)(\varphi^{p-k-1}(A)x) = 0$ we conclude that the vector $\varphi^{p-k-1}(A)x \in N_{k+1}$. By virtue of $N_k = N_{k+1}$ the same vector is in N_k . Consequently,

$$\varphi^k(A)(\varphi^{p-k-1}(A)x) = \varphi^{p-1}(A)x = 0,$$

i.e. the vector $x \in N_{p-1}$. The validity of the above assertion can now be established by induction on p .

The space X in which A is an operator is finite dimensional. Therefore the dimensions of subspaces N_k cannot increase without limit. Let q be the smallest positive integer for which $N_q = N_{q+1}$. Denote by T_q the range of an operator $\varphi^q(A)$ and consider any vector x common for subspaces T_q and N_q . We have $\varphi^q(A)x = 0$ and $x = \varphi^q(A)y$ for some vector $y \in X$. It follows that $\varphi^{2q}(A)y = 0$, i.e. $y \in N_{2q}$. But by what has been proved $N_q = N_{2q}$. Therefore $y \in N_q$, i.e. $x = \varphi^q(A)y = 0$.

Thus T_q and N_q have only a zero vector in common. In view of formula (56.3) this means that $X = T_q \dot{+} N_q$. Since T_q and N_q are invariant subspaces, the possibility of decomposing the operator is established.

As already noted earlier, all eigenvectors of A must be in T_q and N_q , with N_q containing the eigenvectors that correspond to the eigenvalues coinciding with some roots of a polynomial $\varphi(z)$ and T_q containing those for which the corresponding eigenvalues coincide with none of the roots of $\varphi(z)$. Since to every eigenvalue there corresponds at least one eigenvector, it follows that:

Each (none) of the roots of the characteristic polynomial of the operator induced on N_q (T_q) is (is not) a root of $\varphi(z)$.

A final characteristic of decompositions of an operator as a direct sum using operator polynomials is furnished by

Theorem 73.1. *Let the characteristic polynomial $f(z)$ of an operator A be decomposed as a product of polynomials $\varphi(z)$ and $\psi(z)$ having no roots in common. Then A can be decomposed uniquely as a direct sum of operators B and C with the characteristic polynomials $\varphi(z)$ and $\psi(z)$.*

Proof. Consider a decomposition of A as a direct sum obtained using a polynomial $\varphi(z)$. Since the product of the characteristic polynomials of the operators defining the direct sum coincides with the characteristic polynomial $f(z)$, the existence of at least one decomposition follows from the above studies.

Suppose now that there is another decomposition of the space X as a direct sum of invariant subspaces N and T . The induced operator on N has a characteristic polynomial $\varphi(z)$ and the operator on T has a polynomial $\psi(z)$. By Theorem 71.1, $N \subset N_k$ for every sufficiently large k , and therefore $N \subset N_q$. The operator $\varphi(A)$ is non-singular on T , and hence the set of the images of vectors from T relative to $\varphi(A)$ coincides with T . But this means that $T \subset T_k$ for every k . The subspaces N and T , as well as N_q and T_q in the direct sum, form the space X . Therefore inclusions $N \subset N_q$ and $T \subset T_q$ are possible only when $N = N_q$ and $T = T_q$. Thus the theorem is proved.

Let A be an operator in an m -dimensional space X . We represent the characteristic polynomial $f(z)$ of A as the canonical factorization

$$f(z) = (z - \lambda_1)^{k_1} (z - \lambda_2)^{k_2} \dots (z - \lambda_r)^{k_r}, \quad (73.2)$$

where $\lambda_1, \lambda_2, \dots, \lambda_r$ are mutually distinct eigenvalues and $k_1 + k_2 + \dots + k_r = m$. Consider the polynomials

$$(z - \lambda_1)^{k_1}, (z - \lambda_2)^{k_2}, \dots, (z - \lambda_r)^{k_r}.$$

They are divisors of the characteristic polynomial $f(z)$ and no pair of them have roots in common. By Theorem 73.1, there are invariant subspaces $R_1, R_2, \dots, R_r$ such that

$$X = R_1 \dot{+} R_2 \dot{+} \dots \dot{+} R_r.$$

The dimension of a subspace R_i is equal to k_i and the induced operator on R_i has a characteristic polynomial $(z - \lambda_i)^{k_i}$.

A subspace R_i is called a *root subspace* of A corresponding to an eigenvalue λ_i . Vectors of a root subspace are called *root vectors*. It

follows from what has been said that any operator can be decomposed as a direct sum of operators induced on root subspaces.

A root subspace R_i coincides with the kernel of the operator $((A - \lambda_i E)^{k_i})^q$ for some positive integer q . We show that in this case it is always possible to put $q = 1$. Consider operators $(A - \lambda_i E)^p$ for $p = 1, 2, \dots$. Let p_i be the smallest number for which the kernel of $(A - \lambda_i E)^{p_i}$ coincides with that of $(A - \lambda_i E)^{p_i+1}$. Then R_i will coincide with the kernel of $(A - \lambda_i E)^{p_i}$. Since the dimensions of the kernels of $(A - \lambda_i E)^p$ for $p = 1, 2, \dots$ are monotonically increasing and the dimension of R_i is equal to k_i , we have $p_i \leq k_i$.

Thus R_i corresponding to an eigenvalue λ_i of multiplicity k_i clearly coincides with the kernel of $(A - \lambda_i E)^{k_i}$.

Theorem 73.2 (Cayley-Hamilton). *If $f(z)$ is the characteristic polynomial of an operator A , then $f(A)$ is a zero operator.*

Proof. Let us represent the characteristic polynomial as the canonical factorization (73.2). Since the operator polynomial $f(A)$ contains the factor $(A - \lambda_i E)^{k_i}$ and any polynomials in the same operator are commutative, $f(A)x_i = 0$ for any vector x_i in R_i . Now take a vector x and represent it as $x = x_1 + x_2 + \dots + x_r$, where $x_i \in R_i$. It is now clear that $f(A)x = 0$, i.e. that $f(A)$ is a zero operator.

Of great interest is again the matrix interpretation of the results obtained. Compose a basis of the space as a successive combination of any bases of root subspaces $R_1, R_2, \dots, R_r$. Root subspaces are invariant and their direct sum coincides with X . Therefore the matrix A_e of A in the basis has the so-called *quasi-diagonal form*

$$\begin{pmatrix} A_{11} & & 0 \\ & A_{22} & \\ & & \ddots \\ 0 & & & A_{rr} \end{pmatrix}. \quad (73.3)$$

Each A_{ii} is a $k_i \times k_i$ matrix that is the matrix of the operator induced on a subspace R_i .

Exercises

1. Can an operator of differentiation in a finite dimensional space of polynomials be decomposed as a nontrivial direct sum?

2. Prove that a system of root vectors corresponding pairwise to distinct eigenvalues is linearly independent.

3. Prove that if an operator A is nonsingular, then $A^{-1} = \varphi(A)$ for some polynomial $\varphi(z)$.

4. An operator A is said to be *nilpotent* if $A^p = 0$ for some positive integer p . Prove that an operator is nilpotent if and only if all its eigenvalues are zero.

5. Let $\varphi(z)$ be a polynomial of the lowest degree for which $\varphi(A) = 0$. Prove that $\varphi(z)$ is a divisor of the characteristic polynomial of A .

74. The Jordan canonical form

A further simplification of the matrix of an operator as compared with the quasi-diagonal form (73.3) can be effected only by special construction of bases for each of the root subspaces. Root bases can of course be chosen so that each matrix A_{ii} in (73.3) is triangular. This form of the matrix of an operator is not the simplest either, however.

Let us study in more detail the structure of root subspaces. If $x \in R_i$, then $(A - \lambda_i E)^{k_i} x = 0$. But for every particular vector x the equation $(A - \lambda_i E)^m x = 0$ may well hold also for $m < k_i$. In particular, if x is an eigenvector corresponding to a multiple eigenvalue λ_i , then $(A - \lambda_i E)x = 0$, although $k_i \geq 2$.

The *height* of a root vector x is the smallest nonnegative integer m such that $(A - \lambda_i E)^m x = 0$.

All root vectors corresponding to an eigenvalue λ_i are of height not greater than the multiplicity of λ_i . Recall, however, that in general the heights of root vectors and the multiplicities of eigenvalues are two distinct notions. Thus, for example, for an operator of a simple structure there are no root vectors of height greater than unity at all, regardless of the multiplicities of the eigenvalues.

Let R_i be a root subspace corresponding to an eigenvalue λ_i of multiplicity k_i . Denote by t the maximum height of root vectors in R_i . It is clear that $t \leq k_i$. If a vector x is of height k , then a vector $(A - \lambda_i E)x$ will be of height $k - 1$. There are therefore root vectors of all heights from 0 to t in R_i .

For any $k \leq t$, denote by H_k the collection of all vectors whose heights are at most k . It is easy to show that H_k is a subspace in R_i . If $x, y \in H_k$, then $(A - \lambda_i E)^k x = (A - \lambda_i E)^k y = 0$. But then for any α and β we have $(A - \lambda_i E)^k (\alpha x + \beta y) = 0$, i.e. $\alpha x + \beta y \in H_k$. It is, further, obvious that

$$0 = H_0 \subset H_1 \subset \dots \subset H_{t-1} \subset H_t = R_i.$$

We denote the dimensions of these subspaces by m_k , $0 = m_0 < m_1 < \dots < m_{t-1} < m_t = k_i$.

Let $f_1, \dots, f_{p_1}$ be arbitrary linearly independent vectors from H_t such that the direct sum of their span and H_{t-1} is H_t . It is clear that they are root vectors of height t , that $p_1 = m_t - m_{t-1}$ and that no nonzero linear combination of the vectors $f_1, \dots, f_{p_1}$ is in H_{t-1} . Consider the collection of vectors

$$\begin{array}{ccccccc} f_1, & \dots, & & & f_{p_1}, \\ (A - \lambda_i E)f_1, & \dots, & & & (A - \lambda_i E)f_{p_1}, \\ (A - \lambda_i E)^2 f_1, & \dots, & & & (A - \lambda_i E)^2 f_{p_1}, \\ \dots & \dots & \dots & \dots & \dots \\ (A - \lambda_i E)^{t-1} f_1, & \dots, & & & (A - \lambda_i E)^{t-1} f_{p_1}. \end{array} \quad (74.1)$$

— p_1 subspaces are of dimension $t - 1$ and so on. The last columns define one-dimensional cyclic subspaces. *The entire root subspace R_t is a direct sum of the p_t cyclic subspaces.*

We write the matrix of the operator induced in a cyclic subspace. Suppose, for example, that vectors $e_1^{(1)}, e_1^{(2)}, \dots, e_1^{(t-1)}, e_1^{(t)}$ are taken as a basis. Since

$(A - \lambda_t E) e_1^{(1)} = 0, (A - \lambda_t E) e_1^{(2)} = e_1^{(1)}, \dots, (A - \lambda_t E) e_1^{(t)} = e_1^{(t-1)},$
we have

$$Ae_1^{(1)} = \lambda_t e_1^{(1)}, Ae_1^{(2)} = \lambda_t e_1^{(2)} + e_1^{(1)}, \dots, Ae_1^{(t)} = \lambda_t e_1^{(t)} + e_1^{(t-1)}.$$

Hence the matrix of the induced operator has the following form:

$$\begin{pmatrix} \lambda_t & 1 & 0 & \dots & 0 & 0 \\ 0 & \lambda_t & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \lambda_t & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda_t \end{pmatrix}.$$

Matrices of this form are called *Jordan canonical boxes*.

We shall now construct a basis of a space as a successive combination of the bases of root subspaces $R_1, R_2, \dots, R_r$. As a basis of each root subspace R_t we take vectors of the type (74.4) ordered in succession from bottom to top and from left to right. A space basis constructed in this way is called a *root basis*.

In a root basis the matrix J of an operator A assumes the so-called *Jordan canonical form*. It is a quasi-diagonal matrix made up of Jordan boxes. First come Jordan boxes corresponding to an eigenvalue λ_1 , in nonincreasing order of their sizes. Then, in the same order, come Jordan boxes corresponding to λ_2 and so on. Thus

$$J = \begin{pmatrix} \boxed{\begin{matrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ \dots & \dots & \dots \\ 0 & 0 & \dots \lambda_1 \end{matrix}} & & & 0 \\ & \ddots & & \\ & & \boxed{\lambda_1} & \\ & & \ddots & \\ & 0 & & \boxed{\begin{matrix} \lambda_r & 1 & 0 \\ 0 & \lambda_r & 0 \\ \dots & \dots & \dots \\ 0 & 0 & \dots \lambda_r \end{matrix}} & \\ & & & \ddots & \boxed{\lambda_r} \end{pmatrix} \quad (74.5)$$

In general some of the Jordan boxes of smaller sizes may be lacking, of course.

Specifying an operator in a vector space defines a class of similar matrices. The result obtained implies that any square matrix can be reduced by similarity transformation to a Jordan canonical form. It is clear that two square matrices of the same size are similar if and only if they have identical Jordan forms. Given a fixed basis therefore

Two square matrices of the same size define the same operator in a complex space if and only if they have identical Jordan forms.

Exercises

1. Let x be a root vector of height v corresponding to an eigenvalue λ_l of an operator A . Prove that if λ_l is a root of multiplicity p of a polynomial $\varphi(z)$, then a vector $v = \varphi(A)x$ is a root vector of height $r = \max \{0, v - p\}$ corresponding to the same eigenvalue λ_l . What can be said about the vector v if λ_l is not a root of $\varphi(z)$?

2. Let x be a nonzero vector and let $\varphi(z)$ be a polynomial of the lowest degree such that $\varphi(A)x = 0$. Prove that $\varphi(z)$ is a divisor of the characteristic polynomial of A .

3. Prove that any square matrix can be reduced to a unique Jordan canonical form up to a permutation of Jordan boxes.

4. Prove that if a matrix is similar to the matrix J of (74.5), then it is similar to J' as well.

5. Prove that square matrices A and A' are the matrices of the same operator.

6. Let J be a Jordan canonical matrix. What is the form of matrices J^p for positive integers p ?

75. The adjoint operator

Now we proceed to study linear operators in a unitary space. Of course, all the results obtained earlier for operators in a complex space hold in this case too. We shall study therefore only the additional properties of operators connected with the concept of orthogonality. In some cases we shall also consider operators from one unitary space into another. The principal part in our studies will be played by the so-called adjoint operator.

Let X and Y be two unitary spaces. An operator A^* from Y to X is said to be *adjoint* to an operator A from X to Y if for any vectors $x \in X$ and $y \in Y$

$$(Ax, y) = (x, A^*y). \quad (75.1)$$

Theorem 75.1. *For any linear operator A there is an adjoint operator A^* which is unique.*

Proof. Choose in X some orthonormal basis $e_1, e_2, \dots, e_m$. Recall that for any vector $x \in X$ there is an expansion

$$x = \sum_{k=1}^m (x, e_k) e_k. \quad (75.2)$$

If A^* exists, then, by this formula, for any vector $y \in Y$

$$A^*y = \sum_{k=1}^m (A^*y, e_k) e_k$$

or considering (75.1)

$$A^*y = \sum_{k=1}^m \overline{(e_k, A^*y)} e_k = \sum_{k=1}^m \overline{(Ae_k, y)} e_k = \sum_{k=1}^m (y, Ae_k) e_k. \quad (75.3)$$

And this means that if A^* exists, then it is unique.

Now take (75.3) to be the definition of A^* . It is easy to verify that the operator A^* thus constructed is linear. It satisfies equation (75.1) too. Indeed, considering that the system $e_1, e_2, \dots, e_m$ is orthonormal and taking into account (75.2) and (75.3) we get for any vectors $x \in X$ and $y \in Y$

$$\begin{aligned} (Ax, y) &= (A \sum_{k=1}^m (x, e_k) e_k, y) = \sum_{k=1}^m (x, e_k) (Ae_k, y), \\ (x, A^*y) &= (\sum_{k=1}^m (x, e_k) e_k, \sum_{k=1}^m (y, Ae_k) e_k) \\ &= \sum_{k=1}^m (x, e_k) \overline{(y, Ae_k)} = \sum_{k=1}^m (x, e_k) (Ae_k, y). \end{aligned}$$

Thus the theorem is proved.

The adjoint operator A^* is connected with A by definite relations. Note some of them:

$$\begin{aligned} (A^*)^* &= A, \\ (A + B)^* &= A^* + B^*, \\ (\alpha A)^* &= \bar{\alpha} A^*, \\ (AB)^* &= B^* A^*, \\ (A^*)^{-1} &= (A^{-1})^*. \end{aligned} \quad (75.4)$$

Here the bar over α means complex conjugation. All the relations can be proved according to the same scheme. We shall prove in detail therefore only the first and the last property.

Consider an operator A and the adjoint operator A^* . The adjoint operator of A^* will in turn be an operator $(A^*)^*$. Now for any $x \in X$ and $y \in Y$ we have

$$(y, (A^*)^*x) = (A^*y, x) = \overline{(x, A^*y)} = \overline{(Ax, y)} = (y, Ax).$$

The left-hand side is equal to the right-hand side for any vector y . Hence $(A^*)^*x = Ax$. But since this equation holds for any x , this means that $(A^*)^* = A$.

Suppose now that A is an operator in X and is nonsingular. We first prove that A^* is also nonsingular. Let $A^*y = 0$. According to (75.3) it follows that

$$\sum_{k=1}^m (y, Ae_k) e_k = 0.$$

Since the system of vectors $e_1, \dots, e_m$ is a basis,

$$(y, Ae_k) = 0 \quad (75.5)$$

for every $k = 1, 2, \dots, m$. Since A is nonsingular, it converts any basis again into a basis. But then the system of vectors $Ae_1, \dots, Ae_m$ is also a basis and it follows from (75.5) that $y = 0$. Thus the kernel of A^* contains only a zero vector, i.e. A^* is nonsingular.

Take vectors $x, y \in X$. There are unique vectors u and v such that

$$Au = x, \quad A^*v = y.$$

We then find

$$(x, (A^{-1})^*y) = (A^{-1}x, y) = (u, A^*v) = (Au, v) = (x, (A^*)^{-1}y).$$

The left-hand side equals the right-hand side for any x . Hence $(A^{-1})^*y = (A^*)^{-1}y$. Since y is arbitrary, this means that $(A^{-1})^* = (A^*)^{-1}$.

Many compatible properties of operators A and A^* can be established from investigating the matrices of these operators. Choose an orthonormal basis $e_1, e_2, \dots, e_m$ in X and an orthonormal basis $q_1, q_2, \dots, q_n$ in Y . If X and Y coincide, it will be assumed that so do their bases. Suppose that a matrix A_{qe} with elements a_{ij} corresponds to A . Then

$$Ae_j = \sum_{i=1}^n a_{ij} q_i.$$

From this and (75.2) we conclude that

$$a_{ij} = (Ae_j, q_i). \quad (75.6)$$

Also suppose that corresponding to A^* in the same bases is a matrix A_{eq}^* with elements a_{ij}^* . By (75.6)

$$a_{ij}^* = (A^*q_j, e_i).$$

Comparing the elements a_{ij} and a_{ij}^* and considering (75.1) we find

$$a_{ij}^* = (A^*q_j, e_i) = \overline{(e_i, A^*q_j)} = \overline{(Ae_i, q_j)} = \overline{a_{ji}}.$$

This formula justifies the following definition:

An $m \times n$ matrix A^* with elements a_{ij}^* is said to be the *adjoint* of an $n \times m$ matrix A with elements a_{ij} if $a_{ij}^* = \overline{a_{ji}}$ for all i and j .

Thus, corresponding to adjoint operators in any orthonormal bases are adjoint matrices. Adjoint matrices clearly satisfy all relations (75.4). An adjoint matrix A^* is related to a matrix A by the operations of transposition and complex conjugation. That is,

$$A^* = (\bar{A}') = (\bar{A})^*. \quad (75.7)$$

Here the bar means that all matrix elements are replaced by their complex conjugates.

The rank of an operator coincides with that of its matrix. Therefore it follows from (75.7) that operators A and A^* have the same rank.

Denote by $N \subset X$, $N^* \subset Y$ and $T \subset Y$, $T^* \subset X$ respectively the kernels and ranges of operators A and A^* . If $x \in N$, then $Ax = 0$ and $(x, A^*y) = 0$. This means that the range of A^* is a subspace orthogonal to the kernel of A . Of course, the range of A is also orthogonal to the kernel of A^* . From the equality of the dimensions of the subspaces T and T^* and from relations of the type (56.4) we conclude that

$$X = N \oplus T^*, \quad Y = N^* \oplus T. \quad (75.8)$$

A basis $y_1, y_2, \dots, y_m$ of a unitary space X is said to be *dual* to a basis $x_1, x_2, \dots, x_m$ of the same space if

$$(x_i, y_j) = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases}$$

A dual basis is not infrequently used to study compatible properties of operators A and A^* in the same space. We first prove that any basis has a dual which is unique. Let $x_1, x_2, \dots, x_m$ be a basis. For any j , a vector y_j must be orthogonal to vectors $x_1, \dots, x_{j-1}$ and $x_{j+1}, \dots, x_m$ and hence to the span L_j constructed on those vectors. It follows that y_j lies in a one-dimensional subspace $L_j^\perp$. The normalization condition $(x_j, y_j) = 1$ defines it uniquely.

It is clear that a basis will be dual to itself if and only if it is orthonormal. The duality relation of bases is symmetrical and therefore it makes sense speaking of a pair of mutually dual bases. Mutually dual bases are called *biorthonormal*.

Theorem 75.2. *If in some basis an operator A has a matrix J , then in the basis dual to the given one the adjoint operator A^* has a matrix J^* .*

Proof. Let A and A^* have in an orthonormal basis $e_1, e_2, \dots, e_m$ corresponding matrices A_e and A_e^* and let A have a matrix J in a basis $x_1, x_2, \dots, x_m$. Denote by P a coordinate transformation matrix for a change from $e_1, e_2, \dots, e_m$ to $x_1, x_2, \dots, x_m$. Then by (64.5) we have

$$J = P^{-1}A_eP.$$

Applying matrix conjugation to the left- and right-hand sides of this we find

$$J^* = P^* A_e^* (P^{-1})^*$$

or equivalently

$$J^* = ((P^{-1})^*)^{-1} A_e^* ((P^{-1})^*).$$

This relation shows that the adjoint operator A^* has a matrix J^* in a basis $y_1, y_2, \dots, y_m$ for which the coordinate transformation matrix for a change from the basis $e_1, e_2, \dots, e_m$ is $(P^{-1})^*$. According to (63.3) the coordinates of the vectors $x_1, x_2, \dots, x_m$ in $e_1, e_2, \dots, e_m$ are column elements of the matrix P , and the coordinates of the vectors $y_1, y_2, \dots, y_m$ in $e_1, e_2, \dots, e_m$ are column elements of the matrix $(P^{-1})^*$. Calculating pairwise scalar products of vectors of the basis $x_1, x_2, \dots, x_m$ and vectors of $y_1, y_2, \dots, y_m$ is equivalent to calculating the elements of the matrix $P' (\overline{P^{-1}})^*$. But

$$P' (\overline{P^{-1}})^* = P' (\overline{\overline{P^{-1}}})' = P' (P^{-1})' = (P^{-1}P)' = E.$$

Hence the basis $y_1, y_2, \dots, y_m$ is dual to $x_1, x_2, \dots, x_m$.

Theorem 75.2 allows many consequences to be deduced. If, for example, J is a Jordan canonical matrix, then there are eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ along its diagonal. But the eigenvalues of the matrix J^* are $\bar{\lambda}_1, \bar{\lambda}_2, \dots, \bar{\lambda}_m$. Therefore the eigenvalues of A^* are all the complex conjugates of the eigenvalues of A . If A is an operator of a simple structure, then Theorem 75.2 makes it possible to say that the adjoint operator A^* has also a simple structure. Basis systems of the eigenvectors of A and A^* can be chosen so that they are biorthonormal and so on.

Exercises

1. Suppose the coordinates of the vectors of some basis of a Euclidean space in an orthonormal basis $e_1, e_2, \dots, e_m$ form the columns of a matrix A . Prove that the coordinates of the vectors of the dual basis in the same basis $e_1, e_2, \dots, e_m$ form the rows of a matrix A^{-1} .

2. How are the characteristic polynomials of operators A and A^* related?

3. Prove that if some subspace is invariant under an operator A , then its orthogonal complements is invariant under A^* .

4. Prove that any eigenvector of an operator A corresponding to an eigenvalue λ is orthogonal to any eigenvector of an operator A^* corresponding to an eigenvalue $\mu \neq \bar{\lambda}$.

5. Prove that any root vector of an operator A corresponding to an eigenvalue λ is orthogonal to any root vector of an operator A^* corresponding to an eigenvalue $\mu \neq \bar{\lambda}$.

Proof. Let A be a normal operator. Choose by Theorem 76.1 an orthonormal basis such that the matrix of the operator is triangular. In the same basis the operator A^* has corresponding to it an adjoint triangular matrix. Under the hypothesis A is normal, and therefore the matrices of A and A^* , in the chosen basis, must be commutative. According to the above remark these matrices are diagonal. So we have constructed an orthonormal basis in which the matrix of the operator has a diagonal form. This means that that basis is made up entirely of eigenvectors of the operator.

Suppose now that A has a basis system of orthonormal eigenvectors. Then in the basis made up of those vectors the matrix of A will be diagonal. But corresponding in the same basis to the operator A^* is an adjoint matrix that is obviously also diagonal. Diagonal matrices are always commutative, and therefore so are A and A^* .

In proving the theorem we have shown that if an operator A is normal, then in a basis made up of orthonormal eigenvectors not only the matrix of A but also the matrix of A^* is diagonal. This leads to

Corollary. *If A is a normal operator, then any orthonormal system of eigenvectors of A is an orthonormal system of eigenvectors of A^* , and vice versa.*

Corollary. *If A is a normal operator, then the eigenvalues of A and A^* corresponding to the eigenvector they have in common are complex conjugate.*

Indeed, if $Ax = \lambda x$ and $A^*x = \mu x$, then by (75.1) for any normed eigenvector x we have

$$\lambda = (\lambda x, x) = (Ax, x) = (x, A^*x) = (x, \mu x) = \bar{\mu}.$$

Of course, this fact holds for any operator A sharing eigenvectors with A^* . The normality of A ensures that there are common vectors.

The significance of normal operators in the general theory is accounted for by two circumstances. One is that they constitute one of the simplest classes of operators in a unitary space. The other is that investigation of an arbitrary operator not infrequently reduces to a study of normal operators.

Exercises

1. Let A be a linear operator and let α and β be complex numbers equal in absolute value. Prove that $\alpha A + \beta A^*$ is a normal operator.
2. Let A be a normal operator. Prove that for any polynomial $\varphi(z)$ the operator $\varphi(A)$ is normal.
3. Prove that for a normal operator any induced operator is normal.
4. Prove that an operator A is normal if and only if for any invariant subspace L its orthogonal complement $L^\perp$ is also invariant.
5. Let A be an operator of a simple structure in a complex space. Prove that A can always be made normal by an appropriate assignment of a scalar product in its space.

77. Unitary and Hermitian operators

Among the normal operators the most widely used are operators of two types, unitary and Hermitian operators.

A linear operator U is said to be *unitary* if its adjoint operator U^* coincides with its inverse U^{-1} , i.e.

$$UU^* = U^*U = E.$$

Theorem 77.1. *A normal operator U is unitary if and only if all its eigenvalues are equal to unity in absolute value.*

Proof. Let U be a unitary operator. Take any of its eigenvalues λ and the corresponding normed eigenvector x . We have

$$1 = (x, x) = (x, U^*Ux) = (Ux, Ux) = (\lambda x, \lambda x) = \lambda \cdot \bar{\lambda} (x, x) = |\lambda|^2.$$

Suppose now that all eigenvalues of the normal operator U are equal to unity in absolute value. Let $x_1, \dots, x_m$ denote orthonormal eigenvectors of U and $\lambda_1, \dots, \lambda_m$ its eigenvalues. Under the hypothesis, $|\lambda_i| = 1$ for every i . Recall that for the adjoint operator U^* , $x_1, \dots, x_m$ remain eigenvectors but correspond to the eigenvalues $\bar{\lambda}_1, \dots, \bar{\lambda}_m$. Take a vector x and expand it with respect to the eigenvectors of U

$$x = \alpha_1 x_1 + \dots + \alpha_m x_m.$$

Now

$$\begin{aligned} U^*Ux &= U^*(Ux) = U^*(\alpha_1 \lambda_1 x_1 + \dots + \alpha_m \lambda_m x_m) \\ &= \alpha_1 \lambda_1 \bar{\lambda}_1 x_1 + \dots + \alpha_m \lambda_m \bar{\lambda}_m x_m = \alpha_1 x_1 + \dots + \alpha_m x_m = x. \end{aligned}$$

Since x is an arbitrary vector, this means that $U^*U = E$. Similarly for $UU^* = E$.

Theorem 77.2. *An operator U is unitary if and only if for any two vectors their scalar product equals that of their images.*

Proof. Let U be a unitary operator. Then for any two vectors x and y we have

$$(x, y) = (x, U^*Uy) = (Ux, Uy). \quad (77.1)$$

Suppose now that given some operator U equations (77.1) hold for any vectors x and y . It follows that

$$(x, (U^*U - E)y) = 0.$$

Since x and y are arbitrary, this means that $U^*U = E$. The operator U is nonsingular, for otherwise the equation $U^*U = E$ would be impossible. Hence the operator U^{-1} exists. Multiplying $U^*U = E$ by U on the left and by U^{-1} on the right we obtain another equation, $UU^* = E$. So U is a unitary operator.

Corollary. *An operator U is unitary if and only if either $UU^* = E$ or $U^*U = E$.*

Corollary. *Any unitary operator carries any orthonormal system of vectors again into an orthonormal system.*

Corollary. *If a linear operator U carries any orthonormal basis again into an orthonormal basis, then U is a unitary operator.*

Indeed, let $x_1, \dots, x_m$ be an orthonormal basis and let $Ux_i = y_i$ and $y_1, \dots, y_m$ be also an orthonormal basis. Take two vectors, x and y . If

$$x = \sum_{i=1}^m \alpha_i x_i, \quad y = \sum_{i=1}^m \beta_i x_i,$$

then

$$(x, y) = \sum_{i=1}^m \alpha_i \bar{\beta}_i.$$

By the linearity of U

$$Ux = \sum_{i=1}^m \alpha_i y_i, \quad Uy = \sum_{i=1}^m \beta_i y_i.$$

Therefore again

$$(Ux, Uy) = \sum_{i=1}^m \alpha_i \bar{\beta}_i.$$

So equations (77.1) hold for any vectors x and y .

Notice that we could define a unitary operator as an *isometric operator*, i.e. an operator preserving the lengths of all vectors. This follows from Theorem 77.2 and from the easily verifiable relation

$$(x, y) = \frac{|x+y|^2 - |x-y|^2 + i|x+iy|^2 - i|x-iy|^2}{4}.$$

A linear operator H is said to be *Hermitian* or *self-adjoint* if it coincides with its adjoint, i.e.

$$H = H^*.$$

Theorem 77.3. *A normal operator H is Hermitian if and only if all its eigenvalues are real numbers.*

Proof. Let H be a Hermitian operator. Take any of its eigenvalues λ and the corresponding normed eigenvector x . We have

$$\lambda = (\lambda x, x) = (Hx, x) = (x, H^*x) = (x, Hx) = (x, \lambda x) = \bar{\lambda},$$

i.e. λ is a real number. Suppose now that the normal operator H has real eigenvalues. Then in a basis made up of orthonormal eigenvectors of H the matrices of H and H^* will coincide. Hence so do the operators themselves, i.e. H is a Hermitian operator.

A Hermitian operator H is said to be *nonnegative* (*positive definite*) if for any (nonzero) vector x

$$(Hx, x) \geq 0 \quad (> 0).$$

Theorem 77.4. *A Hermitian operator H is nonnegative (positive definite) if and only if all its eigenvalues are nonnegative (positive).*

Proof. Choose an orthonormal basis made up of eigenvectors $x_1, \dots, x_m$ of a Hermitian operator H . Then it follows from the expansion

$$x = \xi_1 x_1 + \dots + \xi_m x_m$$

for a vector x that

$$(Hx, x) = \lambda_1 |\xi_1|^2 + \dots + \lambda_m |\xi_m|^2.$$

Hence, if all eigenvalues of a Hermitian operator are nonnegative (positive), then the operator itself is also nonnegative (positive definite). Putting $x = x_i$ we get

$$(Hx_i, x_i) = \lambda_i$$

for every i . Therefore all eigenvalues of a nonnegative (positive definite) operator are nonnegative (positive).

It follows from the foregoing that a positive definite operator is a nonsingular nonnegative operator. Among all the Hermitian operators nonnegative and positive definite operators play an especially important role. We note some of their properties.

If H and S are positive definite operators, then the operator $\alpha H + \beta S$ is positive definite for any nonnegative numbers α and β not both zero.

Indeed, the operator $\alpha H + \beta S$ is Hermitian for any real numbers α and β . If, however, those numbers are nonnegative and are not both zero, then

$$((\alpha H + \beta S)x, x) = \alpha (Hx, x) + \beta (Sx, x) > 0$$

for $x \neq 0$.

If an operator H is positive definite, then H^{-1} is also a positive definite operator.

Indeed, since $H = H^*$, we have $H^{-1} = (H^*)^{-1} = (H^{-1})^*$, i.e. the operator H^{-1} is Hermitian. The eigenvalues of H^{-1} are inverses of the eigenvalues of H . Therefore they are positive and H^{-1} is positive definite.

*If H is positive definite and A is nonsingular operator, then A^*HA and AHA^* are positive definite operators.*

It is easy to verify that they are Hermitian. By the nonsingularity of A we have $Ax \neq 0$ and $A^*x \neq 0$ for any $x \neq 0$. Therefore

$$(A^*HAx, x) = (HAx, Ax) > 0, \quad (AHA^*x, x) = (HA^*x, A^*x) > 0$$

for $x \neq 0$. In particular, it follows that for any nonsingular opera-

tor A the operators A^*A and AA^* are positive definite. But if A is a singular operator, then A^*A and AA^* are nonnegative.

For any nonnegative operator H there is a nonnegative operator S such that $S^2 = H$.

Indeed, let $\lambda_1, \dots, \lambda_m$ be the eigenvalues of H and $x_1, \dots, x_m$ the corresponding orthonormal eigenvectors. Then $Hx_i = \lambda_i x_i$ for every i . Let S be defined by the equations $Sx_i = \sqrt{\lambda_i} x_i$. The operator S is nonnegative, since it has a basis system of orthonormal eigenvectors $x_1, \dots, x_m$ corresponding to nonnegative eigenvalues $\sqrt{\lambda_1}, \dots, \sqrt{\lambda_m}$. Besides, $S^2 x_i = Hx_i = \lambda_i x_i$. Thus, S^2 and H coincide on the vectors of the basis $x_1, \dots, x_m$ and therefore they do on all vectors, i.e. $S^2 = H$.

A nonnegative operator S is said to be the *principal square root* of a nonnegative operator H if $S^2 = H$.

It is important to stress that all eigenvectors of S and H coincide.

Indeed, suppose $\lambda_1, \dots, \lambda_r$ and $\sqrt{\lambda_1}, \dots, \sqrt{\lambda_r}$ are the various eigenvalues of H and S respectively. Denote by $X_i (Y_i)$, $i = 1, 2, \dots, r$, the proper subspace of the operator $H (S)$ containing all eigenvectors corresponding to an eigenvalue $\lambda_i (\sqrt{\lambda_i})$. The direct sums of the proper subspaces $X_1, \dots, X_r$ and $Y_1, \dots, Y_r$ coincide with the entire space. Therefore

$$\dim X_1 + \dots + \dim X_r = \dim Y_1 + \dots + \dim Y_r. \quad (77.2)$$

It is clear that $Y_i \subset X_i$ for every i , i.e. $\dim Y_i \leq \dim X_i$. Hence (77.2) can hold only if for every i we have $\dim Y_i = \dim X_i$, i.e. $Y_i = X_i$.

So the eigenvalues and eigenvectors of S are uniquely defined by H . Since S is a Hermitian operator, this means that the principal root of H can be only *unique*.

Exercises

1. Prove that the set of all unitary operators in a given unitary space forms a group relative to multiplication.

2. Prove that the set of all Hermitian operators in a given unitary space forms a group relative to addition.

3. Let an operator A be Hermitian and B positive definite. Prove that the eigenvalues of the operators BA and $B^{-1}A$ are real.

4. Prove that if A and B are positive definite operators, then all eigenvalues of the operator BA are positive.

5. Prove that if A and B are commutative positive definite operators, then the operator BA is also positive definite.

6. Prove that if A is a positive definite operator in a unitary space, then the function $(x, y)_A = (Ax, y)$ satisfies all the scalar product axioms.

78. Operators A^*A and AA^*

If A is an operator from a unitary space X to a unitary space Y , then an operator A^*A is defined in X and an operator AA^* in Y . These operators will play an important role in our further studies. Therefore we shall now proceed to investigate them.

From the first and fourth properties of (75.4) it follows that A^*A and AA^* are Hermitian operators. Moreover, they are nonnegative, since for any vectors $x \in X$ and $y \in Y$ we have

$$\begin{aligned}(A^*Ax, x) &= (Ax, Ax) \geq 0, \\ (AA^*y, y) &= (A^*y, A^*y) \geq 0.\end{aligned}$$

Therefore there is a nonnegative operator G in X and a nonnegative operator F in Y such that

$$A^*A = G^2, \quad AA^* = F^2.$$

The operators G and F satisfying these relations are unique.

Whatever the operator A , the operator A^*A has an *orthonormal system* of eigenvectors $x_1, x_2, \dots, x_m$. The operator A always carries that system into some *orthogonal system*. Indeed, let

$$A^*Ax_k = \rho_k^2 x_k, \quad \rho_k \geq 0 \quad (78.1)$$

for all $k = 1, 2, \dots, m$. Then

$$(Ax_k, Ax_l) = (A^*Ax_k, x_l) = \rho_k^2 (x_k, x_l) = 0$$

for $k \neq l$. In addition, for every k

$$|Ax_k| = \rho_k,$$

and therefore the vector Ax_k is nonzero if and only if the eigenvalue ρ_k^2 of A^*A is nonzero.

The *nonzero vector* Ax_k is an eigenvector of AA^* and corresponds to the eigenvalue ρ_k^2 . Indeed, by (78.1)

$$AA^*(Ax_k) = A(A^*Ax_k) = A(\rho_k^2 x_k) = \rho_k^2 Ax_k.$$

Thus all nonzero eigenvalues of A^*A are eigenvalues of AA^* . The converse is also true of course. Therefore the *nonzero eigenvalues* of A^*A and AA^* always *coincide*.

Eigenvalues of A^*A and AA^* will be denoted by $\rho_1^2, \rho_2^2, \dots$. It may be assumed without loss of generality that

$$\rho_1^2 \geq \rho_2^2 \geq \dots \geq \rho_i^2 > 0$$

and that the other eigenvalues ρ_k^2 are zero. It is obvious that the eigenvalues of A^*A and AA^* differ only in the multiplicity of the ze-

ro eigenvalue. The multiplicity of A^*A is $(m - t)$ and that of AA^* is $(n - t)$.

The principal square roots of the common eigenvalues of A^*A and AA^* are called *singular* (or *principal*) *values* of A .

Using eigenvectors of A^*A and AA^* it is possible to construct such orthonormal bases in spaces X and Y with the aid of which it is easy to describe and investigate operators A and A^* . Take as a basis in X an orthonormal system $x_1, \dots, x_m$ of eigenvectors of A^*A . It follows from (75.8) that vectors $x_1, \dots, x_t$ form a basis in T^* and that vectors $x_{t+1}, \dots, x_m$ form a basis in N . The orthonormal basis $y_1, \dots, y_n$ in Y is constructed as follows. As $y_1, \dots, y_t$ we take the vectors obtained after the normalization of $Ax_1, \dots, Ax_t$. These vectors form a basis in T . As $y_{t+1}, \dots, y_n$ we take any orthonormal basis in N^* . It is clear that $y_1, \dots, y_n$ are eigenvectors for AA^* and form a basis in Y . Considering that $|Ax_k| = \rho_k$ we now find

$$Ax_k = \begin{cases} \rho_k y_k, & k \leq t, \\ 0, & k > t. \end{cases} \quad (78.2)$$

Multiplying these equations by A^* and taking into account (78.1) we get

$$A^*y_k = \begin{cases} \rho_k x_k, & k \leq t, \\ 0, & k > t. \end{cases} \quad (78.3)$$

The orthonormal bases in X and Y connected with A and A^* by relations (78.2) and (78.3) are called *singular bases*.

If X and Y are distinct spaces, then the matrix of A can be written in singular bases. Denote it by Λ . By (78.2) it is as follows:

$$\Lambda = \begin{Bmatrix} \rho_1 & & & & 0 \\ & \rho_2 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & \rho_t \\ 0 & & & & & 0 \\ & & & & & & \ddots \end{Bmatrix}. \quad (78.4)$$

If X and Y coincide, then singular bases are not used as a rule to write the matrix of an operator. Relations (78.2) and (78.3) hold again, however.

Exercises

1. Prove that it is the kernels of operators $A, A^*A (A^*, AA^*)$ and the ranges of $A, AA^* (A^*, A^*A)$ that coincide.
2. Prove that if $\dim X > \dim Y$ ($\dim X < \dim Y$), then $A^*A (AA^*)$ is a singular operator.

3. Prove that singular values remain unaffected by multiplication of an operator A by any unitary operators.

4. Let A be an operator in a space X and let all its singular values be mutually distinct. Prove that singular bases are uniquely defined up to multiplication of each of the vectors by a number equal to unity in absolute value.

5. Prove that the singular values of a normal operator coincide with the moduli of its eigenvalues.

6. Prove that the singular values of an operator A^{-1} are inverses of the singular values of an operator A and that the singular bases of both operators coincide.

7. Let A be an operator in an m -dimensional unitary space X . Denote by $\lambda_1, \dots, \lambda_m$ its eigenvalues and by $\rho_1, \dots, \rho_m$ its singular values. Prove that

$$\sum_{k=1}^m |\lambda_k|^2 \leq \sum_{k=1}^m \rho_k^2, \quad \prod_{k=1}^m |\lambda_k| = \prod_{k=1}^m \rho_k.$$

8. Prove that if $|\lambda_k| = \rho_k$ for all $k = 1, 2, \dots, m$, then the operator is normal.

79. Decomposition of an arbitrary operator

One of the circumstances determining the significance of the unitary and the Hermitian operator is the possibility of using them to represent an arbitrary linear operator.

Let A be a linear operator in a unitary space X . We show that it can always be represented as

$$A = H_1 + iH_2, \quad (79.1)$$

where H_1 and H_2 are Hermitian operators. Indeed, if this decomposition exists,

$$A^* = H_1 - iH_2.$$

But then

$$H_1 = \frac{1}{2}(A + A^*), \quad H_2 = \frac{1}{2i}(A - A^*).$$

It is these formulas that define decomposition (79.1). Since

$$H_1 H_2 - H_2 H_1 = \frac{1}{2i}(A^* A - A A^*),$$

the normality of A implies that H_1 and H_2 are commutative, and vice versa.

Let $x_1, \dots, x_m$ be an orthonormal system of eigenvectors of the operator $A^* A$. According to (78.2) there is an orthonormal system $y_1, \dots, y_m$ of eigenvectors of $A A^*$ such that

$$A x_k = \rho_k y_k \quad (79.2)$$

for all k . Now let linear operators F and U be defined in a space X by the following equations on basis systems of vectors:

$$U x_k = y_k, \quad F y_k = \rho_k y_k. \quad (79.3)$$

Relations (79.2) and (79.3) imply that the following decomposition is obtained:

$$A = FU. \quad (79.4)$$

Here F is a nonnegative Hermitian operator, since it has a basis orthonormal system of eigenvectors $y_1, y_2, \dots, y_m$ and nonnegative eigenvalues $\rho_1, \rho_2, \dots, \rho_m$. The operator U is unitary, since it carries the orthonormal system of vectors $x_1, x_2, \dots, x_m$ into the orthonormal system $y_1, y_2, \dots, y_m$. Note that (79.4) yields

$$AA^* = F^2, \quad (79.5)$$

i.e. F is the principal square root of AA^* .

Decomposition (79.4) is called a *polar factorization* of an operator A . By virtue of the uniqueness of the principal root the operator F in a polar factorization will always be unique. The operator U will be unique only when the operator A is nonsingular. In that case $U = F^{-1}A$.

Again there is a direct connection between the normality of an operator A and the commutativity of the components of its polar factorization. Indeed, let $UF = FU$ for some operator A . Then

$$A^*A = U^*F^*FU = F^*U^*UF = F^2,$$

which together with (79.5) means that the operator A is normal.

Suppose now that A is a normal operator, i.e. that $A^*A = AA^*$. By (79.4) $A = FU$. Hence $A^* = U^*F^*$. The normality condition of the operator leads to $U^*F^2U = F^2$ or

$$F^2U = UF^2.$$

Taking into account the second of the relations (79.3) we get

$$F^2(Uy_k) = \rho_k^2(Uy_k)$$

for all $k = 1, 2, \dots, m$, i.e. Uy_k are eigenvectors for the operator F^2 . As noted earlier, F^2 and F have the same eigenvectors. Therefore

$$(FU)y_k = F(Uy_k) = \rho_k(Uy_k)$$

for all $k = 1, 2, \dots, m$. On the other hand, by the second of the relations (79.3)

$$(UF)y_k = U(Fy_k) = U(\rho_k y_k) = \rho_k(Uy_k).$$

These equations show that the operators FU and UF coincide on the basis system of vectors $y_1, y_2, \dots, y_m$. Hence $UF = FU$.

Exercises

1. Prove that if an operator is normal, then the eigenvalues of the operator $H_1(H_2)$ of (79.1) are the real (imaginary) parts of the eigenvalues of the operator A .

2. Prove that if A is a normal operator, then the eigenvalues of the operator F (the independent variables of the eigenvalues of the operator U) of (79.4) are the absolute values of the eigenvalues (the independent variables of the nonzero eigenvalues) of A .

3. Prove that if the operator A is normal, then both operators in decomposition (79.1) have the same eigenvectors as A . What can be said about the eigenvectors of the components of decomposition (79.4)?

80. Operators in the real space

Additional difficulties arise in investigating linear operators in a real space. They are mainly due to the fact that not every linear operator in a real space has at least one eigenvector.

Of course, if the characteristic polynomial of an operator in a real space has only real roots, then there is close similarity in theory. In fact only terminology changes. That is, the words "complex, unitary, Hermitian" are replaced respectively by "real, orthogonal, symmetric". If, however, the characteristic polynomial has, in addition, complex roots, then the study of such an operator becomes a more complicated matter.

Let the real space R be given. Consider the set of all possible pairs $(x; y)$ of vectors x and y from R . We define operations on those pairs. It is assumed that

$$(x; y) + (u; v) = (x + u; y + v)$$

for any two pairs and that for any complex number $\xi + i\eta$ and any pair $(x; y)$

$$(\xi + i\eta)(x; y) = (\xi x - \eta y; \eta x + \xi y).$$

It is easy to verify that the set of all pairs of vectors from R with the operations thus introduced is a complex space C .

The constructed space C has the same dimension as the space R . Indeed, let $e_1, e_2, \dots, e_m$ be a basis in R . For any pair of vectors u and v from R we have

$$\left. \begin{aligned} u &= \alpha_1 e_1 + \dots + \alpha_m e_m, \\ v &= \beta_1 e_1 + \dots + \beta_m e_m, \end{aligned} \right\} \quad (80.1)$$

where α_i and β_i are real numbers. But it follows that

$$(u; v) = \sum_{k=1}^m (\alpha_k + i\beta_k)(e_k; 0). \quad (80.2)$$

The system $(e_1; 0), \dots, (e_m; 0)$ is linearly independent. Therefore the dimension of C is equal to m .

For any basis $e_1, \dots, e_m$ in R and any real numbers $\alpha_1, \dots, \alpha_m$

$$\sum_{k=1}^m (\alpha_k + i0)(e_k; 0) = \left(\sum_{k=1}^m \alpha_k e_k; 0 \right).$$

Hence there is a 1-1 correspondence between all vectors u from $\mathbf{R}$ and all pairs of the form $(u; 0)$ from $\mathbf{C}$. Moreover, this correspondence is an isomorphism if restricted to the operations with real numbers.

If all pairs of the form $(u; 0)$ are identified with vectors u from $\mathbf{R}$, then it follows from (80.1) and (80.2) that the space $\mathbf{C}$ may be considered as a set of elements

$$w = u + iv,$$

where $u, v \in \mathbf{R}$. It should be remembered, of course, that in fact the elements u and v are pairs $(u; 0)$ and $(v; 0)$ and that multiplication by a number i and addition are carried out according to the definitions introduced above. When $v = 0$ we obtain elements of $\mathbf{R}$. It is natural to consider $\mathbf{R}$ to be some set of $\mathbf{C}$. Elements of the form $u + i0$ will be called *real* and elements $u + iv$ and $u - iv$ *complex conjugate*.

The space $\mathbf{C}$ is called the *complexification of the real space* $\mathbf{R}$.

In solving various problems in a Euclidean space we can proceed in a similar way to obtain a unitary space. Consider the complexification $\mathbf{C}$ of a Euclidean space $\mathbf{R}$. For any two vectors

$$z = x + iy, \quad w = u + iv$$

from $\mathbf{C}$ it is assumed by definition that

$$(z, w) = ((x, u) + (y, v)) + i((y, u) - (x, v)).$$

It is not hard to establish that the space $\mathbf{C}$ with such a scalar product is unitary. The scalar product for any two vectors from $\mathbf{R}$ is preserved.

Let A be an operator in $\mathbf{R}$. Construct a new operator $\hat{A}$ in $\mathbf{C}$ equal to A on $\mathbf{R}$. To do this we set

$$\hat{A}(u + iv) = Au + iAv.$$

It is clear that $\hat{A}$ is a linear operator and that $\hat{A}u = Au$ for every vector $u \in \mathbf{R}$.

The operator $\hat{A}$ is called the *complexification of the operator* A .

Now instead of studying the operator A in the real space $\mathbf{R}$ it is possible to consider the operator $\hat{A}$ in the complex space $\mathbf{C}$ and investigate it in $\mathbf{R}$ as a set of $\mathbf{C}$. This device is most often used when some fact in the complex space has no analogue in the real space.

Suppose that a real basis is given in $\mathbf{C}$. Then in that basis the matrix of the complexification $\hat{A}$ is real and coincides with the matrix of the operator A in the same basis. It follows that the characteristic polynomial of $\hat{A}$ coincides with that of A and hence has real coefficients. It is obvious that

If the characteristic polynomial of an operator A in the real space $\mathbf{R}$ has a real root, then that root is an eigenvalue of A and has at least one real eigenvector corresponding to it.

Consider now some complex root λ of the characteristic polynomial of A . It is an eigenvalue of $\hat{A}$ and has some eigenvector w corresponding to it. Since the characteristic polynomial of $\hat{A}$ has real coefficients, $\hat{A}$ will also have the complex conjugate eigenvalue $\bar{\lambda}$. The operator $\hat{A}$ carries complex conjugate vectors into complex conjugate vectors. Therefore it follows from $\hat{A}w = \lambda w$ that $\hat{A}\bar{w} = \bar{\lambda}\bar{w}$. Hence the complex conjugate eigenvalues of $\hat{A}$ have the corresponding complex conjugate vectors.

If $\lambda \neq \bar{\lambda}$, then the vectors w and $\bar{w}$ are linearly independent as eigenvectors corresponding to distinct eigenvalues.

Consider vectors x and y defined as follows in terms of w and $\bar{w}$:

$$x = \frac{1}{2}(w + \bar{w}), \quad y = \frac{1}{2i}(w - \bar{w}). \quad (80.3)$$

It is easy to verify that they are real. Moreover, it is not hard to see that if $\lambda = \mu + iv$, then

$$Ax = \mu x - vy, \quad Ay = vx + \mu y.$$

Therefore the span in $\mathbf{R}$ constructed on vectors (80.3) is an invariant subspace of A . The matrix of the induced operator on that subspace in basis (80.3) is as follows:

$$\begin{pmatrix} \mu & v \\ -v & \mu \end{pmatrix}.$$

Hence the characteristic polynomial of the induced operator is $(z - \mu)^2 + v^2$ or equivalently $z^2 - (\lambda + \bar{\lambda})z + \lambda\bar{\lambda}$. Note that in the invariant subspace constructed A has no eigenvector for $v \neq 0$. Thus we have arrived at an important conclusion. Namely:

If the characteristic polynomial of an operator A in the real space $\mathbf{R}$ has a complex (not real!) root, then that root has in $\mathbf{R}$ a corresponding two-dimensional invariant subspace of A containing no eigenvectors.

This conclusion is as important for the study of operators in a real space as is the fact of the existence of at least one eigenvector for the study of operators in a complex space. Choosing in a suitable way bases in the space $\mathbf{R}$ we can reduce the matrix of an operator to a form resembling in a sense either the diagonal form or the triangular form or the Jordan canonical form. This method of investigating the operator is employed comparatively rarely, since real canonical forms lack many merits of complex canonical forms. It is much easier and more fruitful to investigate the complexification of an operator.

Exercises

1. Prove that the range (the kernel) of an operator $\hat{A}$ is a complexification of the range (the kernel) of an operator A .
2. Let a complexification $\hat{A}$ have a simple structure. Prove that a basis can be chosen in R such that the matrix of an operator A has a quasi-diagonal form with 1×1 and 2×2 matrices along the diagonal.
3. Prove that in the real space R of dimension m any operator has an invariant subspace of dimension $m - 1$ or $m - 2$.
4. What is the counterpart of Theorem 72.1 in a real space?
5. Prove that any linear operator in a real space of odd dimension has at least one eigenvector.

81. Matrices of a special form

We have discussed some operators of a special form. It is natural to suggest that the matrices of those operators should also have some specificity.

A square complex matrix U is said to be *unitary* if its adjoint U^* coincides with its inverse U^{-1} , i.e.

$$UU^* = U^*U = E.$$

We recall that in an orthonormal basis the adjoint operator has a corresponding adjoint matrix. Hence the matrix of a unitary operator in an orthonormal basis is unitary.

Suppose that in a unitary space any two orthonormal bases are given. We construct a coordinate transformation matrix for a change from one of the bases to the other. According to (63.3) the matrix columns are made up of the coordinates of the vectors of the second basis relative to the first. But of the same form is also the matrix of a linear operator transforming the vectors of the first basis into those of the second. According to the second corollary of Theorem 77.2 that operator is unitary. Therefore

A coordinate transformation matrix for a change from an orthonormal basis to an orthonormal basis is unitary.

We shall say that two matrices are *unitarily similar* if they are similar and the similarity transformation matrix is unitary. It follows from the properties of the unitary operator that any unitary matrix is unitarily similar to a diagonal matrix with diagonal elements equal to unity in absolute value.

It is easy to write the relations defining the elements of a unitary matrix. Let U be an $m \times m$ matrix. We denote by u_{ij} its elements. Then it follows from $UU^* = E$ that

$$\sum_{k=1}^m u_{ik} \bar{u}_{jk} = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases}$$

Similarly from $U^*U = E$ we get

$$\sum_{k=1}^m u_{ki} \bar{u}_{kj} = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases}$$

Thus the systems of row vectors and column vectors of any unitary matrix are orthonormal systems.

A real unitary matrix U is called *orthogonal*. It is defined by the following relations:

$$UU' = U'U = E.$$

All properties of orthogonal matrices follow from those of unitary matrices.

A square complex matrix H is said to be *Hermitian* or *self-adjoint* if it coincides with its adjoint, i.e.

$$H = H^*.$$

Thus the matrix of a Hermitian operator in an orthonormal basis is Hermitian.

It follows from the properties of the Hermitian operator that any Hermitian matrix is unitarily similar to a real diagonal matrix. If h_{ij} are elements of the Hermitian matrix H , then

$$h_{ij} = \bar{h}_{ji}$$

for all i and j . It follows in particular that the diagonal elements of any Hermitian matrix are real.

A real Hermitian matrix H is called *symmetric*. It is defined by the following relation:

$$H = H'.$$

Note that any symmetric matrix is orthogonally similar to a real diagonal matrix.

A square matrix is said to be *normal* if it is commutative with its adjoint.

According to this definition the matrix of a normal operator in an orthonormal basis is normal. Taking into account the properties of a normal operator it is easy to see that any complex normal matrix is unitarily similar to a diagonal matrix.

Matrices of a special form play an important role in constructing various computational algorithms. Nevertheless we shall not be concerned with their detailed study. All the properties of these matrices are virtually a reflection of similar properties of corresponding operators.

Exercises

1. Prove that any complex matrix is unitarily similar to a triangular matrix.

2. Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be the eigenvalues of a matrix A , each eigenvalue repeated according to multiplicity. Prove that

$$\sum_{i=1}^m |\lambda_i|^2 \leq \text{tr}(A^*A). \quad (81.1)$$

3. Prove that equality holds in (81.1) if and only if the matrix A is normal.

4. Using the Binet-Cauchy formula prove that for any matrix A the principal minors of the matrix A^*A are nonnegative.

5. Prove that the sum of the squares of the absolute values of all minors of a unitary matrix in any fixed rows and columns is equal to unity.

6. Prove that any rectangular matrix A can be represented as $A = Q\Lambda S$, where Q and S are unitary matrices and Λ is a diagonal matrix with nonnegative elements.

Metric Properties of an Operator

82. The continuity and boundedness of an operator

We have introduced the concept of linear operator as some generalization of the notion of function. Assuming that in spaces a metric is defined, it is possible to draw an analogy with the boundedness of a function, the continuity of a function, etc. When studying these questions we shall always assume that the operator acts from an m -dimensional normed space X to an n -dimensional normed space Y . If X does not coincide with Y , then the norms in both spaces can be introduced independently of each other.

An operator A from X to Y is said to be *continuous at a point* $x_0 \in X$ if the condition $x_k \rightarrow x_0$ implies $Ax_k \rightarrow Ax_0$ for any sequence $\{x_k\}$ in X . If the operator is continuous at each point of X , then it is said to be *everywhere continuous* or simply *continuous*.

Theorem 82.1. *A linear operator in arbitrary finite dimensional normed spaces is continuous.*

Proof. We take a vector $x_0 \in X$ and choose any basis $e_1, e_2, \dots, e_m$ in X . We have

$$x_0 = \xi_1^{(0)} e_1 + \dots + \xi_m^{(0)} e_m.$$

Suppose $x_k \rightarrow x_0$ and

$$x_k = \xi_1^{(k)} e_1 + \dots + \xi_m^{(k)} e_m.$$

By Theorem 53.1 convergence in the norm implies coordinate convergence. Therefore $\xi_s^{(k)} \rightarrow \xi_s^{(0)}$ for every s . But

$$Ax_0 = \xi_1^{(0)} Ae_1 + \dots + \xi_m^{(0)} Ae_m$$

and in addition

$$Ax_k = \xi_1^{(k)} Ae_1 + \dots + \xi_m^{(k)} Ae_m.$$

Now the convergence of $\xi_s^{(k)} \rightarrow \xi_s^{(0)}$ for every s will imply the convergence of $Ax_k \rightarrow Ax_0$ in the norm of Y .

An operator A is said to be *bounded* if there is a constant M such that $\|Ax\| \leq M \|x\|$ for any vector $x \in X$.

Theorem 82.2. *A linear operator in arbitrary finite dimensional normed spaces is bounded.*

Proof. Suppose an operator A is not bounded in some case. Then there is a sequence $\{x_k\}$ of nonzero vectors such that

$$\|Ax_k\| \geq k \|x_k\|.$$

Consider a sequence of vectors

$$y_k = \frac{1}{k \|x_k\|} x_k.$$

It converges to zero, since

$$\|y_k\| = \frac{1}{k \|x_k\|} \|x_k\| = \frac{1}{k} \rightarrow 0.$$

On the other hand,

$$\|Ay_k\| = \frac{1}{k \|x_k\|} \|Ax_k\| \geq 1.$$

This means that $\{Ay_k\}$ does not converge to zero, i.e. that A is not continuous at zero. This contradiction with Theorem 82.1 completes the proof.

It is natural to pose the question concerning the *smallest* of the constants M satisfying $\|Ax\| \leq M \|x\|$ for all vectors x . Since the set of those constants is bounded below by zero, the smallest constant clearly exists. It is called the *norm of the operator A* and designated $\|A\|$. By definition the norm of an operator has the following two properties:

(1) for any vector x in X

$$\|Ax\| \leq \|A\| \cdot \|x\|, \quad (82.1)$$

(2) for every number $\epsilon > 0$ there is a vector $x_\epsilon \in X$ such that

$$\|Ax_\epsilon\| \geq (\|A\| - \epsilon) \|x_\epsilon\|. \quad (82.2)$$

We prove that

$$\|A\| = \sup_{\|x\| \leq 1} \|Ax\| \quad (82.3)$$

or equivalently that

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} \quad (82.4)$$

if of course $\dim X > 0$.

We take a vector x satisfying $\|x\| \leq 1$. Then it follows from (82.1) that

$$\|Ax\| \leq \|A\| \|x\| \leq \|A\|.$$

Consequently

$$\sup_{\|x\| \leq 1} \|Ax\| \leq \|A\|. \quad (82.5)$$

We further take any vector x_ε according to (82.2) and construct a vector

$$y_\varepsilon = \frac{1}{\|x_\varepsilon\|} x_\varepsilon.$$

Then

$$\|Ay_\varepsilon\| = \frac{1}{\|x_\varepsilon\|} \|Ax_\varepsilon\| \geq \frac{1}{\|x_\varepsilon\|} (\|A\| - \varepsilon) \|x_\varepsilon\| = \|A\| - \varepsilon.$$

Since $\|y_\varepsilon\| = 1$, we have

$$\sup_{\|x\| \leq 1} \|Ax\| \geq \|Ay_\varepsilon\| \geq \|A\| - \varepsilon.$$

By virtue of the arbitrariness of ε we get

$$\sup_{\|x\| \leq 1} \|Ax\| \geq \|A\|. \quad (82.6)$$

Now from (82.5) and (82.6) we obtain relation (82.3) which was to be established.

We shall soon show that *the norm of an operator plays an exceptionally important role in introducing a metric in the space of linear operators. It is the explicit form (82.3) that will be essential.*

Exercises

1. Prove that on a bounded closed set of vectors the supremum and infimum of the norms of the values of a linear operator are attained.

2. Prove that a linear operator carries any bounded closed set again into a bounded closed set.

3. Is the assertion of the preceding exercise true if the boundedness requirement of a set is dropped?

4. Prove that in (82.3) the supremum is attained on a set of vectors satisfying $\|x\| = 1$ provided $\dim X > 0$.

5. Let A be an operator in a space X . Prove that A is nonsingular if and only if there is a number $m > 0$ such that $\|Ax\| \geq m\|x\|$ for any $x \in X$.

83. The norm of an operator

A set ω_{XY} of linear operators from X to Y is a finite dimensional vector space. If that space is real or complex, then it can be converted into a complete metric space by introducing a norm in it in some way.

To introduce a norm in a space of linear operators the same methods can be used as those employed in any other vector space. Of most interest in this case, however, are only the norms in ω_{XY} that are sufficiently closely related to those in X and Y . One of the most important classes of such norms is the class of the so-called compatible norms.

If for each operator of ω_{XY}

$$\|Ax\| \leq \|A\| \cdot \|x\|$$

for all $x \in X$, then the operator norm is said to be *compatible* with the vector norms in X and Y .

The advantage of compatible norm is easy to see from the following example. Suppose that λ is an eigenvalue of an operator A in X and that x is the corresponding eigenvector. Then $Ax = \lambda x$ and therefore

$$|\lambda| \cdot \|x\| = \|\lambda x\| = \|Ax\| \leq \|A\| \cdot \|x\|.$$

Hence $|\lambda| \leq \|A\|$. So we have obtained a very important conclusion:

The moduli of the eigenvalues of a linear operator do not exceed any of its compatible norms.

This example shows that to obtain the best estimates it is desirable that the smallest of the compatible norms should be used. It is clear that all compatible norms are bounded below by expression (82.3). If we show that this expression satisfies the norm axioms, then it will precisely be the smallest of the compatible norms. This justifies both the name of expression (82.3) and the notation used.

It is obvious that for any operator A the expression $\|A\|$ is nonnegative. If $\|A\| = 0$, i.e. if

$$\sup_{\|x\| \leq 1} \|Ax\| = 0,$$

then $\|Ax\| = 0$ for every vector x whose norm does not exceed unity. But then, by the linearity of the operator, $Ax = 0$ for every x . Hence $A = 0$. For any operator A and any λ we have

$$\|\lambda A\| = \sup_{\|x\| \leq 1} \|\lambda Ax\| = |\lambda| \sup_{\|x\| \leq 1} \|Ax\| = |\lambda| \cdot \|A\|.$$

And finally for any two operators A and B in ω_{XY}

$$\begin{aligned} \|A+B\| &= \sup_{\|x\| \leq 1} \|Ax+Bx\| \leq \sup_{\|x\| \leq 1} (\|Ax\| + \|Bx\|) \\ &\leq \sup_{\|x\| \leq 1} \|Ax\| + \sup_{\|x\| \leq 1} \|Bx\| = \|A\| + \|B\|. \end{aligned}$$

All these relations precisely mean that (82.3) is a norm in a space of linear operators. Norm (82.3) is called an operator norm *subordinate* to the vector norms in X and Y .

A subordinate norm has a very important property relative to the operation of operator multiplication, too. Let A be an operator from X to Y and B an operator from Y to Z . As is known, this defines an operator BA . Considering the compatibility of subordinate norms

we find

$$\begin{aligned}\|BA\| &= \sup_{\|x\| \leq 1} \|(BA)x\| = \sup_{\|x\| \leq 1} \|B(Ax)\| \\ &\leq \sup_{\|x\| \leq 1} (\|B\| \cdot \|Ax\|) = \|B\| \sup_{\|x\| \leq 1} \|Ax\| = \|B\| \cdot \|A\|.\end{aligned}$$

Thus any subordinate norm of an operator has the following four basic properties. For any operators A and B and any number λ

$$\begin{aligned}(1) \quad &\|A\| > 0 \text{ if } A \neq 0; \|0\| = 0, \\ (2) \quad &\|\lambda A\| = |\lambda| \|A\|, \\ (3) \quad &\|A + B\| \leq \|A\| + \|B\|, \\ (4) \quad &\|BA\| \leq \|B\| \cdot \|A\|.\end{aligned}\tag{83.1}$$

To note a further property, for the identity operator E

$$(5) \quad \|E\| = 1.$$

This follows from (82.3), since $Ex = x$ for any vector x .

In the general case a subordinate norm of an operator depends both on the norm in X and on the norm in Y . If both spaces are unitary, then we may take as a norm in them the length of the vectors. The corresponding subordinate norm of the operator is called the *spectral norm* and designated $\|\cdot\|_2$. So for any operator A from X to Y

$$\|A\|_2^2 = \sup_{(x, x) \leq 1} (Ax, Ax).\tag{83.2}$$

We investigate some properties of spectral norm.

The spectral norm remains unaffected by the multiplication of an operator by any unitary operators.

Let V and U be arbitrary unitary operators in X and Y respectively. Consider the operator $B = UAV$. We have

$$\begin{aligned}\|B\|_2^2 &= \sup_{(x, x) \leq 1} (Bx, Bx) = \sup_{(x, x) \leq 1} (UAVx, UAVx) \\ &= \sup_{(x, x) \leq 1} (AVx, U^*UAVx) = \sup_{(x, x) \leq 1} (AVx, AVx) \\ &= \sup_{(Vx, Vx) \leq 1} (AVx, AVx) = \sup_{(v, v) \leq 1} (Av, Av) = \|A\|_2^2.\end{aligned}$$

Assigning a spectral norm in the form (83.2) establishes its relation to singular values of the operator A . Let $x_1, x_2, \dots, x_m$ be an orthonormal system of eigenvectors of the operator A^*A and let $\rho_1^2, \rho_2^2, \dots, \rho_m^2$ be its eigenvalues. It may be assumed without loss of generality that

$$\rho_1 \geq \rho_2 \geq \dots \geq \rho_m = 0.\tag{83.3}$$

We represent a vector $x \in X$ as

$$x = \alpha_1 x_1 + \dots + \alpha_m x_m,\tag{83.4}$$

then

$$(x, x) = \sum_{i=1}^m |\alpha_i|^2.$$

As noted in Section 78, the system $x_1, x_2, \dots, x_m$ is carried by an operator A into an orthogonal system, with

$$(Ax_i, Ax_i) = \rho_i^2$$

for every i . Hence

$$(Ax, Ax) = \sum_{i=1}^m |\alpha_i|^2 \rho_i^2,$$

which yields

$$\|A\|_2^2 = \sup_{\sum_{i=1}^m |\alpha_i|^2 \leq 1} \sum_{i=1}^m |\alpha_i|^2 \rho_i^2. \quad (83.5)$$

It is clear that under (83.3)

$$\|A\|_2^2 \leq \rho_1^2.$$

But for the vector x_1 the right-hand side of (83.5) takes on the value ρ_1^2 . Therefore

$$\|A\|_2^2 = \rho_1^2.$$

Thus

The spectral norm of an operator A is equal to its maximum singular value.

We recall that for a normal operator A its singular values coincide with the moduli of its eigenvalues. Hence the spectral norm of a unitary operator is equal to unity and the spectral norm of a non-negative operator is equal to its largest eigenvalue.

Exercises

1. Prove that for any eigenvalue λ of an operator A

$$|\lambda| \leq \inf_k \|A^k\|^{1/k}.$$
2. Let $\varphi(z)$ be any polynomial with nonnegative coefficients. Prove that

$$\|\varphi(A)\| \leq \varphi(\|A\|).$$
3. Prove that $\|A\| \geq \|A^{-1}\|^{-1}$ for any nonsingular operator A . When does equality hold in the spectral norm case?

84. Matrix norms of an operator

The spectral norm is virtually the only subordinate norm of an operator the calculation of which is not explicitly connected with bases. If, however, in spaces, in which operators are given, some bases are fixed, the possibility of introducing operator norms is greatly extended.

So we again consider linear operators from a space X to a space Y . Suppose we fix a basis $e_1, e_2, \dots, e_m$ in X and a basis $g_1, g_2, \dots, g_n$ in Y . Expanding a vector $x \in X$ with respect to a basis we get

$$x = x_1 e_1 + \dots + x_m e_m. \quad (84.1)$$

Now it is possible to introduce a norm in X by formula (52.3), for example, or in some other way in terms of the coefficients of expansion (84.1). Similarly it is possible to introduce a norm in Y .

The most common are norms of the form (52.4). Therefore we shall study operator norms subordinate to, and compatible with, those norms. Moreover, it will be assumed that norms of the same type are introduced in both X and Y . It is obvious that the corresponding norms of an operator A must be somehow related to the elements a_{ij} of the matrix of the operator in the chosen basis.

We first establish expressions for operator norms subordinate to the 1-norms and ∞ -norms of (52.4). We have

$$\begin{aligned} \|A\|_1 &= \sup_{\|x\|_1 \leq 1} \|Ax\|_1 = \sup_{\|x\|_1 \leq 1} \left(\sum_{i=1}^n \left| \sum_{j=1}^m a_{ij} x_j \right| \right) \\ &\leq \sup_{\|x\|_1 \leq 1} \left(\sum_{i=1}^n \sum_{j=1}^m |a_{ij}| |x_j| \right) \leq \sup_{\|x\|_1 \leq 1} \left(\sum_{j=1}^m |x_j| \sum_{i=1}^n |a_{ij}| \right) \\ &\leq \left(\max_{1 \leq j \leq m} \sum_{i=1}^n |a_{ij}| \right) \left(\sup_{\|x\|_1 \leq 1} \|x\|_1 \right) = \max_{1 \leq j \leq m} \sum_{i=1}^n |a_{ij}|. \end{aligned}$$

We now show that for some vector x satisfying the condition $\|x\|_1 \leq 1$, $\|Ax\|_1$ coincides with the right-hand side of the relation obtained.

Let the largest value at the right be reached when $j = l$. Then all the inequalities become equations, for $x = e_l$, for example. So

$$\|A\|_1 = \max_{1 \leq j \leq m} \sum_{i=1}^n |a_{ij}|.$$

Similarly for the other norm:

$$\|A\|_\infty = \sup_{\|x\|_\infty \leq 1} \|Ax\|_\infty = \sup_{\|x\|_\infty \leq 1} \left(\max_{1 \leq i \leq n} \left| \sum_{j=1}^m a_{ij} x_j \right| \right)$$

$$\leq \sup_{\|x\|_\infty \leq 1} \left(\max_{1 \leq i \leq n} \sum_{j=1}^m |a_{ij}| |x_j| \right) \leq \sup_{\|x\|_\infty \leq 1} \left(\left(\max_{1 \leq i \leq n} \sum_{j=1}^m |a_{ij}| \right) \times \left(\max_{1 \leq j \leq m} |x_j| \right) \right) = \left(\max_{1 \leq i \leq n} \sum_{j=1}^m |a_{ij}| \right) \left(\sup_{\|x\|_\infty \leq 1} \|x\|_\infty \right) = \max_{1 \leq i \leq n} \sum_{j=1}^m |a_{ij}|.$$

Suppose the largest value at the right is reached when $i = l$. We take a vector x with coordinates $x_j = |a_{lj}| / a_{lj}$, if $a_{lj} \neq 0$, and with $x_j = 1$, if $a_{lj} = 0$. It is not hard to verify that for that vector all the inequalities become equations. Hence

$$\|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^m |a_{ij}|.$$

To find an operator norm subordinate to the 2-norms of (52.4) we proceed as follows. We introduce in X and Y a scalar product in a way similar to (32.1). Then the 2-norms of (52.4) will coincide with the length of the vector. Therefore the subordinate norm is nothing but the spectral norm of an operator corresponding to the given scalar product. The bases for the chosen scalar products become orthonormal and therefore in these bases an adjoint operator will have a corresponding adjoint matrix. If we let A_{qe} denote the matrix of an operator A , then it follows from the foregoing that

The operator norm subordinate to 2-norms is equal to the maximum singular value of A_{qe} .

The norms we have considered are some functions of the matrix of an operator. Not only subordinate but also compatible norms can we construct in this way. One of the most important compatible norms is the so-called *Euclidean norm*. It will be designated $\|\cdot\|_E$. If in the chosen bases an operator A has a matrix A_{qe} with elements a_{ij} , then by definition

$$\|A\|_E = \left(\sum_{i=1}^n \sum_{j=1}^m |a_{ij}|^2 \right)^{1/2}.$$

The right-hand side of this is the norm in an $n \times m$ -dimensional space of linear operators. That the first three properties of (83.1) hold is therefore beyond doubt. Of great importance is the fact that for a Euclidean norm the fourth property of (83.1) is also valid. To prove this we use a Cauchy-Buniakowski-Schwarz inequality of the type (27.5).

Let us consider vector spaces X , Y and Z of dimensions m , n and p respectively. Let A be an operator from X to Y and B an operator from Y to Z . By a_{ij} and b_{ij} we denote the elements of the matrices

of the operators in the chosen bases. We have

$$\begin{aligned}\|BA\|_{\mathbf{E}} &= \left(\sum_{i=1}^p \sum_{j=1}^m \left| \sum_{k=1}^n b_{ik} a_{kj} \right|^2 \right)^{1/2} \leq \left(\sum_{i=1}^p \sum_{j=1}^m \left(\sum_{k=1}^n |b_{ik}| |a_{kj}| \right)^2 \right)^{1/2} \\ &\leq \left(\sum_{i=1}^p \sum_{j=1}^m \left(\sum_{k=1}^n |b_{ik}|^2 \right) \left(\sum_{l=1}^n |a_{lj}|^2 \right) \right)^{1/2} \\ &= \left(\left(\sum_{i=1}^p \sum_{k=1}^n |b_{ik}|^2 \right) \left(\sum_{l=1}^n \sum_{j=1}^m |a_{lj}|^2 \right) \right)^{1/2} = \|B\|_{\mathbf{E}} \cdot \|A\|_{\mathbf{E}}.\end{aligned}$$

In the general case a Euclidean norm is not subordinate. Its compatibility with 2-norms can be proved in the same way as the property just considered.

A direct check makes it possible to establish important formulas for the Euclidean norm. Namely,

$$\|A\|_{\mathbf{E}} = \text{tr}(A_{qe}^* A_{qe}) = \text{tr}(A_{qe} A_{qe}^*). \quad (84.2)$$

We can now draw the following conclusions.

An adjoint matrix in orthonormal bases has a corresponding adjoint operator. We transform the chosen bases into orthonormal bases if we introduce in X and Y scalar products in a way similar to (32.1). Since the trace of a matrix is equal to the sum of its eigenvalues, it follows from (84.2) that

The square of a Euclidean norm of an operator is equal to the sum of the squares of its singular values.

If scalar products are introduced in X and Y , it is possible to speak of unitary operators. It is for these unitary operators that it is easy to show that

A Euclidean norm is not affected by the multiplication of an operator by any unitary operators.

Indeed, as noted in the exercises to Section 78, singular values remain unaffected by the multiplication by unitary operators, and the Euclidean norm can be expressed only in terms of singular values.

In most applications connected with norms, not so much an explicit assignment of an operator norm is important as the fact that properties (83.1) hold. An operator norm can therefore be defined axiomatically in terms of its matrix. Choose in the spaces, in which the operators are given, some bases, then each operator will have a corresponding matrix. We assign to each matrix a number designated as $\|\cdot\|$ and suppose that conditions (83.1) hold as axioms. A number $\|\cdot\|$ will be called a *matrix norm*. If now each operator is assigned the norm of its matrix, it is clear that this introduces a norm in the space of the operators. Conditions (83.1) obviously hold for the operators, too. The converse is also true. Given fixed bases, any operator norm generates a matrix norm. These matrix norms will be designated by similar symbols $\|\cdot\|_2$, $\|\cdot\|_{\infty}$, etc. It is obvious that

we may also require axiomatically that the norm should be compatible.

The above examples show that it is practically feasible to assign an operator norm axiomatically in terms of a matrix norm. In what follows, speaking of matrix and operator norms we shall always assume that they are compatible and that conditions (83.1) hold.

Exercises

1. Prove that, given any norm, for a unit matrix

$$\|E\| \geq 1. \quad (84.3)$$
2. Let $\lambda_1, \dots, \lambda_m$ be the eigenvalues of a matrix A . Prove that

$$\inf_B \|B^{-1}AB\|_E = \sum_{k=1}^m |\lambda_k|^2.$$

Compare this equation with (81.1).

85. Operator equations

One of the most important problems of algebra is that of solving linear algebraic equations. We have often met with this problem in the present course. We now consider it from the viewpoint of the theory of linear operators.

Given system (60.2) with elements from a field P of real or complex numbers, take any m -dimensional space X and n -dimensional space Y over the same field P and fix some bases in them. Then relations (60.2) will be equivalent to a single matrix equation of the type (61.2) which in turn is equivalent to an operator equation

$$Ax = y. \quad (85.1)$$

Here A is an operator from X to Y with the same matrix in the chosen bases as that of system (60.2). Vectors $x \in X$ and $y \in Y$ have in the chosen bases the coordinates $(\xi_1, \dots, \xi_m)$ and $(\eta_1, \dots, \eta_n)$ respectively.

Thus instead of a system of linear algebraic equations we may consider equations (85.1). The problem is to determine all vectors $x \in X$ satisfying (85.1) for a given operator A and a given vector $y \in Y$. An equation of the form (85.1) is called an *operator equation*, a vector y is a *right-hand side* and a vector x is a *solution*. Of course, all the properties of a system of equations are automatically carried over to operator equations and vice versa.

The Kronecker-Capelli theorem formulates a necessary and sufficient condition for a system to be solved in terms of the rank of a matrix. This is not very convenient since one is not allowed to notice the deep connection existing between systems and equations of other types.

Let X and Y be unitary spaces. Then an operator A^* is defined. Equation (85.1) is called the *basic nonhomogeneous equation*, and the equation

$$A^*u = v$$

is the *adjoint nonhomogeneous equation*. If the right-hand sides are zero, then the corresponding equations are called *homogeneous*. The following statement is true:

Either the basic nonhomogeneous equation has a solution for any right-hand side or the adjoint homogeneous equation has at least one nonzero solution.

Indeed, let r denote the rank of an operator A . The operator A^* will have the same rank. Two cases are possible: either $r = n$ or $r < n$. In the former case the range of A is of dimension n and hence it coincides with Y . Therefore the basic nonhomogeneous equation must have a solution for any right-hand side. In the same case the nullity of the adjoint operator is equal to zero and therefore the kernel has no nonzero solutions, i.e. the adjoint homogeneous equation has no nonzero solutions. If $r < n$, then the range of A does not coincide with Y and the basic nonhomogeneous equation cannot have a solution for any right-hand side. The kernel of the adjoint operator consists not only of a zero vector and therefore the adjoint homogeneous equation must have nonzero solutions.

The above statement is of particular importance when X and Y coincide. Now the existence of a solution of the basic nonhomogeneous equation for any right-hand side implies the nonsingularity of the operator A . In this case therefore we have the so-called

Fredholm Alternative. *Either the basic nonhomogeneous equation always has a unique solution for any right-hand side or the adjoint homogeneous equation has at least one nonzero solution.*

Fredholm Theorem. *For a basic nonhomogeneous equation to be solvable, it is necessary and sufficient that its right-hand side should be orthogonal to all solutions of the adjoint homogeneous equation.*

Proof. Let N^* denote the kernel of an operator A^* and T the range of A . If the basic nonhomogeneous equation is solvable, then its right-hand side $y \in T$. In view of (75.8) it follows that $y \perp N^*$, i.e. that $(y, u) = 0$ for every vector u satisfying $A^*u = 0$. Now let $(y, u) = 0$ for the same vectors u . Then $y \perp N^*$ and by (75.8) $y \in T$. But this means that there is a vector $x \in X$ such that $Ax = y$, i.e. that the basic nonhomogeneous equation is solvable.

Exercises

1. Prove that the equation $A^*Ax = A^*y$ is solvable.
2. Prove that the equation $(A^*A)^px = (A^*A)^qy$ is solvable for any positive integers p and q .
3. Give a geometrical interpretation of the Fredholm alternative and theorem.

86. Pseudosolutions and the pseudoinverse operator

Prescribing arbitrarily an operator A and a right-hand side y may result in equation (85.1) having no solution. Obviously, this is only due to *what exactly we mean by a solution of an equation*.

Take a vector $x \in X$ and consider a vector $r = Ax - y$ called the *discrepancy* of the vector x . For x to be a solution of (85.1) it is necessary and sufficient that its discrepancy should be zero. In turn, for the discrepancy to be zero it is necessary and sufficient that its length should be zero. Thus all solutions of (85.1), and they alone, satisfy the equation

$$|Ax - y|^2 = 0.$$

Since the zero value of the length of the discrepancy is the smallest, the finding of solutions of equation (85.1) may be regarded as the problem of finding such vectors x for which the following expression attains the smallest value:

$$\Phi_0(x) = |Ax - y|^2. \quad (86.1)$$

The right-hand side of the expression is called the *functional of discrepancy*. Finding the vectors minimizing the functional of discrepancy makes sense also when no solution of (85.1) exists. This justifies the following definition:

A *pseudosolution* (or *generalized solution*) of equation (85.1) is any vector $x \in X$ for which the functional of discrepancy attains its smallest value. The shortest pseudosolution is called a *normal pseudosolution*.

We show that a normal pseudosolution always exists and is unique. Fix in X and Y singular bases $x_1, \dots, x_m$ and $y_1, \dots, y_n$. Let

$$x = \sum_{k=1}^m \alpha_k x_k, \quad y = \sum_{p=1}^n \beta_p y_p. \quad (86.2)$$

Considering relations (78.2) we find that

$$Ax - y = \sum_{k=1}^m \rho_k \alpha_k y_k - \sum_{p=1}^n \beta_p y_p.$$

It is assumed as before that the singular values $\rho_1, \dots, \rho_l$ are non-zero and that the rest are zero. Since singular bases are orthonormal, we have

$$\Phi_0(x) = \sum_{k=1}^l |\rho_k \alpha_k - \beta_k|^2 + \sum_{p=l+1}^n |\beta_p|^2.$$

It is obvious that the smallest value of the functional of discrepancy is attained on those vectors x whose last $m - t$ coordinates α_k are arbitrary and whose first t coordinates are defined by

$$\alpha_k = \beta_k / \rho_k. \quad (86.3)$$

The normal pseudosolution will be as follows:

$$x_0 = \sum_{k=1}^t \frac{\beta_k}{\rho_k} x_k. \quad (86.4)$$

We recall that vectors $x_{t+1}, \dots, x_m$ form the basis of the kernel N of the operator A . Therefore the set of all pseudosolutions is a plane in X whose direction subspace coincides with N and whose translation vector coincides with any pseudosolution. *A normal pseudosolution is the only vector of that plane that is orthogonal to N .*

Using relations (78.2) and (78.3) it is easy to show that pseudosolutions, and they alone, satisfy

$$A^*Ax = A^*y. \quad (86.5)$$

Indeed, write vectors x and y as expansions (86.2). We have

$$A^*Ax = \sum_{k=1}^t \rho_k^2 \alpha_k x_k, \quad A^*y = \sum_{p=1}^t \rho_p \beta_p x_p.$$

It follows that solutions of equation (86.5) are only those vectors x whose first t coordinates α_k are calculated according to (86.3) and whose last $m - t$ coordinates are arbitrary.

Thus, *if the solvability of (85.1) is not guaranteed, then we can always replace the solution of the equation by the solution of (86.5). In addition, a minimization of the functional of discrepancy for (85.1) is ensured.*

The inverse operator plays an important part in carrying out various studies. However, it was defined only for the nonsingular operator and we have no analogue as yet for the singular operator and for the operator from one space to another. This analogue can be constructed on the basis of pseudosolutions.

Suppose that A is an operator from X to Y . Then each vector $y \in Y$ can be assigned a unique vector $x_0 \in X$ which is the normal pseudosolution of (85.1). This correspondence defines some operator A^+ from Y to X called the *pseudoinverse* (or *generalized inverse*) of A . So by definition

$$x_0 = A^+y \quad (86.6)$$

for any $y \in Y$. It is clear that if the operator A is nonsingular, then the *pseudoinverse of A coincides with its inverse*. We investigate the properties of the pseudoinverse.

Suppose along with (86.6) we have $u_0 = A^+v$ for some vector $v \in Y$. Consider the vector $\alpha y + \beta v$ for any α and β . If we take it as a right-hand side of (85.1), then the vector $\alpha x_0 + \beta u_0$ will clearly satisfy a corresponding equation of the type (86.5) and therefore it will be a pseudosolution. Since x_0 and u_0 are orthogonal to the kernel of A , so is the vector $\alpha x_0 + \beta u_0$. Hence it is the normal pseudosolution. The *linearity* of the pseudoinverse operator is thus established.

The properties of the pseudoinverse operator are easy to establish if we consider its action on the vectors of singular bases. By (86.4) we have

$$A^+y_k = \begin{cases} \rho_k^{-1}x_k, & k \leq t, \\ 0, & k > t. \end{cases} \quad (86.7)$$

It follows that

The domain, kernel and range of the pseudoinverse operator and those of the adjoint operator coincide.

Using (78.2), (78.3) and (86.7) it is possible to obtain various relations connecting the operators A , A^* and A^+ . We note some of them:

- (1) $(A^*)^+ = (A^+)^*$,
- (2) $(A^+)^+ = A$,
- (3) $(AA^+)^* = AA^+$, $(AA^+)^2 = AA^+$,
- (4) $(A^+A)^* = A^+A$, $(A^+A)^3 = A^+A$,
- (5) $AA^+A = A$.

These relations can be proved according to the same scheme. As an example, we therefore consider in more detail only the first and the third.

Comparing (78.2) and (86.7) we take as an operator A the adjoint operator A^* . Since (78.3) holds for this operator, we have

$$(A^*)^+x_k = \begin{cases} \rho_k^{-1}y_k, & k \leq t, \\ 0, & k > t. \end{cases}$$

Now, proceeding from (86.7), we apply a relation similar to (78.3) to the operator $(A^+)^*$. Then

$$(A^+)^*x_k = \begin{cases} \rho_k^{-1}y_k, & k \leq t, \\ 0, & k > t. \end{cases}$$

Thus the operators $(A^*)^+$ and $(A^+)^*$ coincide on the basis $x_1, \dots, x_m$ and therefore they are equal.

Taking into account (78.2) and (86.7) we conclude that for the operator AA^+

$$AA^+y_k = \begin{cases} 1 \cdot y_k, & k \leq t, \\ 0, & k > t. \end{cases} \quad (86.8)$$

This means that AA^+ has an orthonormal system of eigenvectors $y_1, \dots, y_n$ and real eigenvalues 1 and 0, i.e. that it is Hermitian. This proves the first equation in the relations of group (3). The second is obvious from (86.8).

Exercises

1. What is the pseudoinverse of a zero operator?
2. Let X and Y be distinct spaces. Write the matrix of the pseudoinverse of an operator in singular bases and compare it with (78.4).
3. Let U and V be unitary operators in X and Y respectively. Prove that

$$(VAU)^* = U^*A^*V^*.$$

4. Prove that there are operators K in X and L in Y such that

$$A^+ = KA^* = A^*L.$$

Describe the action of the operators K and L .

5. Prove that the pseudoinverse of an operator is uniquely defined by the conditions

$$AA^+A = A,$$

$$A^+ = KA^* = A^*L.$$

6. Prove that all pseudosolutions, and they alone, are solutions of the equation

$$Ax = AA^+y.$$

7. Give a geometrical interpretation of pseudosolutions.

87. Perturbation and nonsingularity of an operator

We have repeatedly emphasized that small changes in the basis, coordinate vectors, matrix elements and the like may result in changes of many properties connected with the concept of linear dependence. This notion plays a decisive role in the entire theory of linear operators, so it is very important to study the influence of small changes in operators themselves on their properties.

As an auxiliary tool in solving diverse questions one has not infrequently to use an operator *almost equal to an identity operator*. By this we shall mean an operator in a space X of the form $E + A$, where $\|A\| < 1$ for some norm.

If λ is any eigenvalue of an operator A , then $1 + \lambda$ is an eigenvalue of the operator $E + A$. Since $|\lambda| \leq \|A\|$, by virtue of $\|A\| < 1$ all eigenvalues of A are less than unity in absolute value. Hence all eigenvalues of $E + A$ are nonzero and the operator is nonsingular.

Thus if $\|A\| < 1$, there is an operator $(E + A)^{-1}$. If, however, the operator $E + A$ is singular, then $\|A\| \geq 1$ for any norm.

For any number α less than unity in absolute value we have the limiting relation

$$(1 + \alpha)^{-1} = \lim_{p \rightarrow \infty} \alpha_p,$$

where

$$\alpha_p = \sum_{h=0}^p (-\alpha)^h.$$

We show that a similar relation holds for the operator $(E + A)^{-1}$ too, if $\|A\| < 1$. Consider a sequence $\{A_p\}$ of operators

$$A_p = \sum_{h=0}^p (-A)^h.$$

It is easy to verify that

$$(E + A) A_p = E - (-A)^{p+1},$$

so

$$\|(E + A) A_p - E\| = \|A^{p+1}\|. \quad (87.1)$$

Formally this equation is true for $p = -1$ as well, if it is assumed that $A_{-1} = 0$. Also we have

$$\begin{aligned} \|(E + A) A_p - E\| &= \|(A_p - (E + A)^{-1}) + A(A_p - (E + A)^{-1})\| \\ &\geq \| \|A_p - (E + A)^{-1}\| - \|A\| \cdot \|A_p - (E + A)^{-1}\| \| \\ &= (1 - \|A\|) \|A_p - (E + A)^{-1}\|. \end{aligned}$$

Now, considering (87.1), we obtain for $p = -1$ an estimate of the norm of the operator $(E + A)^{-1}$, i.e.

$$\|(E + A)^{-1}\| \leq \frac{\|E\|}{1 - \|A\|}.$$

For any subordinate norm $\|E\| = 1$ and hence in this case

$$\|(E + A)^{-1}\| \leq \frac{1}{1 - \|A\|}. \quad (87.2)$$

For $p \geq 0$ we obtain an estimate of the deviation of the operator A_p from the operator $(E + A)^{-1}$. Namely,

$$\|A_p - (E + A)^{-1}\| \leq \frac{\|A\|^{p+1}}{1 - \|A\|}. \quad (87.3)$$

By virtue of the condition $\|A\| < 1$ this means that $\{A_p\}$ converges to $(E + A)^{-1}$. If A_p is assumed to be an approximation to $(E + A)^{-1}$, then formula (87.3) gives an estimate of the accuracy of the approximation.

Let A be any nonsingular operator. Consider an operator $A + e_A$, where e_A is an arbitrary operator. We shall call e_A the *perturbation of the operator A* , and $A + e_A$ is a perturbed operator. We show under what conditions on the value of the perturbation norm the perturbed operator is nonsingular. We shall be concerned only with *small* values of the perturbation norm.

The operator A is nonsingular and therefore there is an operator A^{-1} . Hence

$$A + e_A = A(E + A^{-1}e_A).$$

It follows that $A + e_A$ is nonsingular if and only if so is the operator $E + A^{-1}e_A$. This condition clearly holds if

$$\|A^{-1}e_A\| < 1$$

for some norm. Of course it holds if $\|A^{-1}\| \|e_A\| < 1$.

Thus a *perturbed operator is nonsingular for all perturbations satisfying*

$$\|e_A\| < \|A^{-1}\|^{-1}. \quad (87.4)$$

When an operator A is perturbed to an amount of e_A , the inverse operator A^{-1} acquires a perturbation equal to $(A + e_A)^{-1} - A^{-1}$. We denote by

$$\delta A = \frac{\|e_A\|}{\|A\|}, \quad \delta A^{-1} = \frac{\|(A + e_A)^{-1} - A^{-1}\|}{\|A^{-1}\|} \quad (87.5)$$

the values of *relative perturbations* of A and A^{-1} . When condition (87.4) holds, the operator $E + A^{-1}e_A$ is nonsingular and therefore

$$\begin{aligned} A + e_A)^{-1} - A^{-1} &= ((A + e_A)^{-1}A - E)A^{-1} \\ &= ((A^{-1}(A + e_A))^{-1} - E)A^{-1} = ((E + A^{-1}e_A)^{-1} - E)A^{-1}. \end{aligned}$$

From formula (87.3), for $p = 0$ we find that

$$\|(A + e_A)^{-1} - A^{-1}\| \leq \frac{\|A^{-1}e_A\| \cdot \|A^{-1}\|}{1 - \|A^{-1}e_A\|} \leq \frac{\|A^{-1}\|^2 \|e_A\|}{1 - \|A^{-1}\| \|e_A\|}.$$

Now, using symbols (87.5), we obtain the following estimate:

$$\delta A^{-1} \leq \frac{\nu_A \delta A}{1 - \nu_A \delta A}, \quad (87.6)$$

where

$$\nu_A = \|A^{-1}\| \cdot \|A\|. \quad (87.7)$$

The number ν_A is called the *condition number* of the operator A . Although it depends on the choice of norm, it can never be very small. From

$$E = A^{-1}A$$

and (84.3) we conclude that

$$1 \leq \|E\| \leq \|A^{-1}\| \cdot \|A\| = v_A.$$

Formula (87.6) shows that a small relative perturbation of an operator A results in a small relative perturbation of A^{-1} only when the condition number of A is not too large as compared with unity. This number will occur in other problems too.

Suppose that given a nonsingular operator A we are to solve the operator equation

$$Ax = y. \quad (87.8)$$

Consider the perturbed equation

$$(A + e_A) \tilde{x} = y + e_y. \quad (87.9)$$

If condition (87.4) holds, then the perturbed equation (87.9) and the original equation (87.8) will have unique solutions $\tilde{x}$ and x . We evaluate their difference.

Along with (87.5) and (87.7) we introduce the corresponding symbols for *relative perturbations* in x and y , i.e.

$$\delta x = \frac{\|\tilde{x} - x\|}{\|x\|}, \quad \delta y = \frac{\|e_y\|}{\|y\|}.$$

We have

$$x = A^{-1}y, \quad \tilde{x} = (A + e_A)^{-1}(y + e_y).$$

From this we find

$$\tilde{x} - x = ((E + A^{-1}e_A)^{-1} - E)A^{-1}y + (E + A^{-1}e_A)^{-1}A^{-1}e_y$$

and further

$$\begin{aligned} \|\tilde{x} - x\| &\leq \|(E + A^{-1}e_A)^{-1} - E\| \cdot \|x\| \\ &\quad + \|(E + A^{-1}e_A)^{-1}\| \cdot \|A^{-1}\| \cdot \|e_y\|. \end{aligned}$$

It is assumed that subordinate norm is used. Taking into account estimates (87.2) and (87.3), as well as the inequality $\|y\| \leq \|A\| \times \|x\|$, we get

$$\begin{aligned} \|\tilde{x} - x\| &\leq \frac{\|A^{-1}\| \cdot \|e_A\| \cdot \|x\|}{1 - \|A^{-1}\| \cdot \|e_A\|} + \frac{\|A^{-1}\| \cdot \|e_y\|}{1 - \|A^{-1}\| \cdot \|e_A\|} \\ &\leq \frac{\|A^{-1}\| \cdot \|A\| \cdot \frac{\|e_A\|}{\|A\|}}{1 - \|A^{-1}\| \cdot \|A\| \cdot \frac{\|e_A\|}{\|A\|}} \|x\| + \frac{\|A^{-1}\| \cdot \|A\| \cdot \frac{\|e_y\|}{\|y\|}}{1 - \|A^{-1}\| \cdot \|A\| \cdot \frac{\|e_A\|}{\|A\|}} \|x\|. \end{aligned}$$

In symbols

$$\delta x \leq \frac{v_A}{1 - v_A \delta A} (\delta A + \delta y). \quad (87.10)$$

This formula again gives the value of the condition number and again it is important from the viewpoint of stability that it *should not be too large*.

Exercises

1. Prove that a condition number expressed in terms of spectral norm is equal to the ratio of the maximum singular value to the minimum singular value.

2. There are operators with the smallest condition number. What are these operators if spectral norm is used?

3. Prove that multiplication of an operator by unitary operators leaves its condition number expressed in terms of spectral or Euclidean norm unchanged.

4. Prove that for any nonsingular operators A and B

$$\frac{\|B^{-1} - A^{-1}\|}{\|B^{-1}\|} \leq v_A \frac{\|A - B\|}{\|A\|}.$$

5. What causes the large instability of the system of n vectors described in Section 22? Evaluate the condition number of the operator whose matrix columns coincide with the coordinates of n vectors (22.7).

88. Stable solution of equations

Formula (87.10) shows that for an operator almost equal to a singular operator *large perturbations are possible in a solution even for small perturbations in the operator and the right-hand side*. It may seem that this is due only to the fact that it is not always that a solution itself exists. However, the situation with finding pseudosolutions is similar.

Indeed, let an operator be in a two-dimensional space. Suppose that in some orthonormal basis a system of linear algebraic equations of the following form corresponds to (85.1):

$$1 \cdot x_1 + 0 \cdot x_2 = 1,$$

$$0 \cdot x_1 + 0 \cdot x_2 = 1.$$

It is easy to find that the normal pseudosolution u_0 has the following coordinates:

$$u_0 = (1, 0).$$

It may well be that the perturbed equation should lead in the same basis to a system

$$1 \cdot x_1 + 0 \cdot x_2 = 1,$$

$$0 \cdot x_1 + \epsilon \cdot x_2 = 1,$$

where the number ϵ , although small, will nevertheless be other than zero. Now the normal pseudosolution $u_0^{(\epsilon)}$ of the perturbed equation

has the coordinates

$$u_0^{(e)} = (1, e^{-t}).$$

For small ε the vectors u_0 and $u_0^{(e)}$ not only differ very much but are even *almost orthogonal*.

If our equation has more than one pseudosolution, then in the general case small perturbations in the operator and the right-hand side will always result in large perturbations in the normal pseudosolution. Nevertheless we show that *despite the instability of many concepts connected with operator equations a normal pseudosolution can be stably determined*.

Let A be an operator from X to Y and suppose that equation (85.1) is to be solved. Similarly to the functional of discrepancy we consider the so-called *regularizing functional*

$$\Phi_\alpha(x) = \alpha |x|^2 + |Ax - y|^2, \quad (88.1)$$

where the number $\alpha \geq 0$. It is clear that for $\alpha = 0$ the functional coincides with the functional of discrepancy and attains its minimum on the pseudosolutions of (85.1). We find on what vectors the regularizing functional attains its minimum for $\alpha > 0$. Using expansions (86.2) we find

$$\Phi_\alpha(x) = \sum_{h=1}^t (\alpha |\alpha_h|^2 + |\rho_h \alpha_h - \beta_h|^2) + \alpha \sum_{h=t+1}^m |\alpha_h|^2 + \sum_{p=t+1}^n |\beta_p|^2.$$

It follows that for the minimum to be attained it is necessary to take the zero values of the last coordinates $\alpha_{t+1}, \dots, \alpha_m$ and to minimize for each $k \leq t$ the expression

$$\alpha |\alpha_k|^2 + |\rho_k \alpha_k - \beta_k|^2.$$

This yields for $k \leq t$

$$\alpha_k = \frac{\rho_k \beta_k}{\alpha + \rho_k^2}.$$

Thus a minimum value of the regularizing functional (88.1) is attained for every $\alpha > 0$ on a *unique vector*

$$x_\alpha = \sum_{h=1}^t \frac{\rho_h \beta_h}{\alpha + \rho_h^2} x_h. \quad (88.2)$$

A comparison of formulas (86.4) and (88.2) makes it possible to establish some relations connecting x_α and x_0 . We have for $\rho, \alpha > 0$

$$\begin{aligned} 0 \leq \frac{|\beta|^2}{\rho^2} - \frac{\rho^2 |\beta|^2}{(\alpha + \rho^2)^2} &= \frac{|\beta|^2 \alpha^2 + 2 |\beta|^2 \alpha \rho^2}{\rho^2 (\alpha + \rho^2)^2} \\ &= \frac{2\alpha |\beta|^2 \left(\frac{\alpha}{2} + \rho^2 \right)}{\rho^2 (\alpha + \rho^2)^2} \leq \frac{2\alpha |\beta|^2}{\rho^4}, \end{aligned}$$

hence

$$|x_\alpha|^2 \leq |x_0|^2 \leq |x_\alpha|^2 + 2\alpha\eta^2, \quad (88.3)$$

where

$$\eta^2 = \sum_{k=1}^t \frac{|\beta_k|^2}{\rho_k^4}.$$

We then find

$$x_0 - x_\alpha = \alpha \sum_{k=1}^t \frac{\beta_k}{\rho_k(\alpha + \rho_k^2)} x_k,$$

from which we conclude that

$$|x_0 - x_\alpha| \leq \alpha\gamma, \quad (88.4)$$

where

$$\gamma^2 = \sum_{k=1}^t \frac{|\beta_k|^2}{\rho_k^6}.$$

Consequently

$$\lim_{\alpha \rightarrow +0} x_\alpha = x_0.$$

Thus, for small values of α the vector x_α may serve as an *approximation* to the normal pseudosolution x_0 .

We expand the vectors x_α and x_0 with respect to singular bases in a way similar to (86.2). A direct check easily shows that x_α satisfies

$$(A^*A + \alpha E)x_\alpha = A^*y. \quad (88.5)$$

For $\alpha > 0$ the operator $A^*A + \alpha E$ is positive definite and therefore there is an operator $(A^*A + \alpha E)^{-1}$, i.e.

$$x_\alpha = (A^*A + \alpha E)^{-1} A^*y. \quad (88.6)$$

On x_α the minimum value of functional (88.1) is attained and therefore $\Phi_\alpha(x_\alpha) \leq \Phi_\alpha(x_0)$. Taking into account (88.3) and (88.4) yields

$$|Ax_\alpha - y|^2 \leq |Ax_0 - y|^2 + \alpha(|x_0|^2 - |x_\alpha|^2) \leq |Ax_0 - y|^2 + 2\alpha^2\eta^2. \quad (88.7)$$

In addition $\Phi_\alpha(x_\alpha) \leq \Phi_\alpha(0)$, from which it follows that

$$|x_\alpha| \leq \frac{|y|}{\alpha^{1/2}}.$$

Together with (88.6) this means that, given any operator A and any vector y , for $\alpha > 0$

$$|(A^*A + \alpha E)^{-1} A^*y| \leq \frac{|y|}{\alpha^{1/2}}. \quad (88.8)$$

In practice, when solving (85.1), the operator A and the right-hand side y are usually given inexactly and one has instead to consider the perturbed operator $\tilde{A}$ and the right-hand side $\tilde{y}$. If in X and Y one uses the length of vectors as a norm, then the spectral norm of the operators is subordinate to it. We shall therefore assume that

$$\|A - \tilde{A}\|_2 \leq \bar{\varepsilon}_A, \quad \|y - \tilde{y}\| \leq \bar{\varepsilon}_y. \quad (88.9)$$

Determining an approximate solution $\tilde{x}_\alpha$ using the perturbed $\tilde{A}$ and $\tilde{y}$ leads to the following equation:

$$(\tilde{A}^* \tilde{A} + \alpha E) \tilde{x}_\alpha = \tilde{A}^* \tilde{y}. \quad (88.10)$$

From (88.5) and (88.10) we find

$$\begin{aligned} (\tilde{A}^* \tilde{A} + \alpha E)(\tilde{x}_\alpha - x_\alpha) &= A^* (Ax_\alpha - y) - \tilde{A}^* (\tilde{A}x_\alpha - \tilde{y}) \\ &= (A - \tilde{A})^* (Ax_\alpha - y) - \tilde{A}^* ((\tilde{A} - A)x_\alpha - (\tilde{y} - y)). \end{aligned}$$

This means that the difference $\tilde{x}_\alpha - x_\alpha$ is a solution of the equation with operator $(\tilde{A}^* \tilde{A} + \alpha E)$ and the right-hand side of the form $z = u + \tilde{A}^* v$, where

$$\begin{aligned} u &= (A - \tilde{A})^* (Ax_\alpha - y), \\ v &= -((\tilde{A} - A)x_\alpha - (\tilde{y} - y)). \end{aligned}$$

Therefore

$$\tilde{x}_\alpha - x_\alpha = (\tilde{A}^* \tilde{A} + \alpha E)^{-1} u + (\tilde{A}^* \tilde{A} + \alpha E)^{-1} \tilde{A}^* v.$$

Now we evaluate the norms of both summands in this equation.

The eigenvalues of the operator $(\tilde{A}^* \tilde{A} + \alpha E)$ are at least α . Hence the eigenvalues of $(\tilde{A}^* \tilde{A} + \alpha E)^{-1}$ are at most α^{-1} . For a positive definite operator its spectral norm coincides with its maximum eigenvalue, i.e.

$$\|(\tilde{A}^* \tilde{A} + \alpha E)^{-1}\|_2 \leq \alpha^{-1}.$$

Considering (88.7) and (88.9) we have

$$\begin{aligned} \|(\tilde{A}^* \tilde{A} + \alpha E)^{-1} u\| &\leq \|(\tilde{A}^* \tilde{A} + \alpha E)^{-1}\|_2 \|u\| \\ &\leq \frac{\bar{\varepsilon}_A}{\alpha} \|Ax_\alpha - y\| \leq \frac{\bar{\varepsilon}_A}{\alpha} (\|Ax_0 - y\|^2 + 2\alpha^2 \eta^2)^{1/2}. \end{aligned}$$

To evaluate the second summand we use formulas (88.3), (88.8) and (88.9). We find

$$\|(\tilde{A}^* \tilde{A} + \alpha E)^{-1} \tilde{A}^* v\| \leq \frac{\|v\|}{\alpha^{1/2}} \leq \frac{1}{\alpha^{1/2}} (\bar{\varepsilon}_A \|x_0\| + \bar{\varepsilon}_y).$$

So

$$\|\tilde{x}_\alpha - x_\alpha\| \leq \frac{\bar{\varepsilon}_A}{\alpha} (\|Ax_0 - y\|^2 + 2\alpha^2\eta^2)^{1/2} + \frac{1}{\alpha^{1/2}} (\bar{\varepsilon}_A \|x_0\| + \bar{\varepsilon}_y).$$

The total error of the computed pseudosolution $\tilde{x}_\alpha$ is

$$\|x_0 - \tilde{x}_\alpha\| \leq \|x_0 - x_\alpha\| + \|\tilde{x}_\alpha - x_\alpha\| \leq \alpha\gamma + \frac{\bar{\varepsilon}_A}{\alpha} (\|Ax_0 - y\|^2 + 2\alpha^2\eta^2)^{1/2} + \frac{1}{\alpha^{1/2}} (\bar{\varepsilon}_A \|x_0\| + \bar{\varepsilon}_y). \quad (88.11)$$

The right-hand side of this is independent of the perturbed $\tilde{A}$ and $\tilde{y}$. There is therefore an α such that the right-hand side attains its minimum. That value of α will ensure almost the best approximation $\tilde{x}_\alpha$ to the exact normal pseudosolution x_0 .

Suppose that $\bar{\varepsilon}_A$ and $\bar{\varepsilon}_y$ are values of an order of ε and that ε is sufficiently small. If the original equation (85.1) has a solution, then $Ax_0 - y = 0$. In this case the right-hand side of (88.11) is, according to the nature of its dependence on α and ε , a function of the form

$$\alpha + \varepsilon + \frac{\varepsilon}{\alpha^{1/2}}.$$

For $\alpha = \varepsilon^{2/3}$ it takes on a value of an order of $\varepsilon^{2/3}$. If, however, the original equation has no solution, then $Ax_0 - y \neq 0$. Now the right-hand side of (88.11) is a function of the form

$$\alpha + \frac{\varepsilon}{\alpha} + \frac{\varepsilon}{\alpha^{1/2}}.$$

For $\alpha = \varepsilon^{1/2}$ it takes on a value of an order of $\varepsilon^{1/2}$.

Thus, if the input data in (85.1) are prescribed up to an order of ε , then the normal pseudosolution can be determined up to an order of $\varepsilon^{2/3}$, if the original equation is solvable, and up to $\varepsilon^{1/2}$ otherwise.

The parameter α ensuring the required approximation $\tilde{x}_\alpha$ cannot be found only from the perturbed $\tilde{A}$ and $\tilde{y}$. This is mainly due to the fact that conditions (88.9) do not guarantee the continuity of the normal pseudosolution in a given range of the operator and the right-hand side. To determine the parameter α use is usually made of additional information about the solution. In some problems no guaranteed closeness to the normal pseudosolution is required and it is considered sufficient to determine stably a minimum of the functional of discrepancy. In such problems it is a somewhat simpler matter to determine α . Despite the importance of these questions we shall not dwell on them, since they are beyond the scope of this book.

Exercises

1. Prove that η in estimate (88.3) is the norm of the normal solution of

$$A^*A (A^*A)^{1/2}x = A^*y.$$

2. Prove that γ in estimate (88.4) is the norm of the normal solution of

$$(A^*A)^2x = A^*y.$$

3. Prove that the difference $x_\alpha - x_\beta$ satisfies

$$(A^*A + \alpha E)(A^*A - \beta E)(x_\alpha - x_\beta) = (\beta - \alpha) A^*y.$$

4. Compare (88.11) and (87.10). What can be said about estimate (88.11) in the case of a nonsingular operator A ?

5. To what accuracy can a normal pseudosolution be computed if $A = 0$?

89. Perturbation and eigenvalues

In the general case the perturbation of an operator leads to changes in all of its eigenvalues and eigenvectors. Since the study of this relation is very complicated, we restrict ourselves to some illustrations. It is *more convenient* to describe this problem in terms of the matrices of operators rather than operators themselves.

Let B be matrix of a simple structure and H a matrix such that

$$H^{-1}BH = \Lambda, \quad (89.1)$$

where Λ is a diagonal matrix of eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$. Consider a perturbed matrix $B + \varepsilon_B$ and some of its eigenvalues λ . The matrix $B + \varepsilon_B - \lambda E$ is singular and therefore so is the matrix

$$H^{-1}(B + \varepsilon_B - \lambda E)H = (\Lambda - \lambda E) + H^{-1}\varepsilon_B H.$$

Two cases are possible:

(1) $\lambda = \lambda_i$ for some i ,

(2) $\lambda \neq \lambda_i$ for every i .

In the second case the matrix $\Lambda - \lambda E$ is nonsingular, so

$$(\Lambda - \lambda E) + H^{-1}\varepsilon_B H = (\Lambda - \lambda E)(E + (\Lambda - \lambda E)^{-1} H^{-1}\varepsilon_B H).$$

The matrix that is the second factor is singular. This means that any norm of the matrix $(\Lambda - \lambda E)^{-1} H^{-1}\varepsilon_B H$ must at least be equal to unity. In particular

$$\|(\Lambda - \lambda E)^{-1} H^{-1}\varepsilon_B H\|_2 \geq 1.$$

Hence

$$\max_{1 \leq i \leq n} |(\lambda_i - \lambda)^{-1}| \|H^{-1}\|_2 \|\varepsilon_B\|_2 \|H\|_2 \geq 1$$

or

$$\min_{1 \leq i \leq n} |\lambda_i - \lambda| \leq \|H^{-1}\|_2 \|e_B\|_2 \|H\|_2.$$

In the first case this inequality also holds and therefore always

$$|\lambda_i - \lambda| \leq v_H \|e_B\|_2 \quad (89.2)$$

at least for one value of i . Here

$$v_H = \|H^{-1}\|_2 \|H\|_2$$

is the condition number of the matrix H expressed in terms of spectral norm.

The relation obtained means that whatever the perturbation e_B of the matrix B is, for any eigenvalue λ of the perturbed matrix $B + e_B$ there is an eigenvalue λ_i of B such that we have inequality (89.2). Notice that we *nowhere required that e_B should be small*. Relation (89.2) may be interpreted somewhat differently. Namely:

The eigenvalues of a perturbed matrix are in the region which is the union of all disks with centres at λ_i and of radius $v_H \|e_B\|_2$.

The columns of the matrix H are eigenvectors of the matrix B . It follows from (89.2) therefore that as a general measure of sensitivity of eigenvalues to the perturbation of a matrix we could apparently take the condition number of the matrix H of eigenvectors (rather than of the matrix B itself). The matrix H satisfying (89.1) is not unique, since the eigenvectors are defined up to arbitrary factors. It will be assumed that H is always chosen so that its value v_H is a minimum one. We recall that in any case $v_H \geq 1$.

If B is a normal matrix and, in particular, Hermitian or unitary, then we may take H to be a unitary matrix. Then $v_H = 1$ and consequently

$$|\lambda_i - \lambda| \leq \|e_B\|_2. \quad (89.3)$$

We consider in somewhat greater detail the case of a Hermitian matrix B with Hermitian perturbation e_B . Now we can show that:

Every disk with centre at λ_i and of radius $\|e_B\|_2$ contains at least one eigenvalue of a perturbed matrix.

Indeed, let us agree to consider a matrix $B + e_B$ as the "original" matrix and the matrix $B = (B + e_B) - e_B$ as a "perturbed" matrix with perturbation equal to $-e_B$. Repeating the above calculations word for word we obtain a formula similar to (89.3) but with the eigenvalues of B and $B + e_B$ reversed. This means that for any eigenvalue λ_i of the "perturbed" matrix B there must be at least one eigenvalue λ of the "original" matrix $B + e_B$ for which (89.3) holds.

If the eigenvalues of B are simple, then for a sufficiently small perturbation e_B all the disks become *separated* and then each disk will contain *one and only one* eigenvalue of the perturbed matrix.

Formula (89.3) shows that the eigenvalues of normal matrices possess a considerable stability to perturbations. In the general problem of determining eigenvalues, however, this phenomenon is an exception rather than a rule.

Consider as an example the case, an "extreme" case in a sense, when the matrix B consists of a single Jordan canonical box. We may agree to assume that all eigenvectors of such a matrix are collinear, that the matrix consisting of eigenvectors is singular and that consequently its condition number equals "infinity". So, let B be an $m \times m$ matrix of the form

$$B = \begin{pmatrix} \lambda_0 & 1 & & 0 \\ & \lambda_0 & 1 & \\ & & \lambda_0 & 1 \\ & & & \ddots & \ddots \\ 0 & & & & \lambda_0 & 1 \\ & & & & & \lambda_0 \end{pmatrix}.$$

It is obvious that its characteristic polynomial is $(\lambda - \lambda_0)^m$.

Now take such a matrix of perturbation ε_B in which only one element, that in position $(m, 1)$, is nonzero and equal to ε . The characteristic polynomial of the perturbed matrix is $(\lambda - \lambda_0)^m - \varepsilon$. The eigenvalues of the perturbed matrix are therefore a distance of $|\varepsilon|^{1/m}$ away from those of the original matrix. If, for example, $m = 20$, $\varepsilon = 10^{-10}$ and λ_0 is of an order of unity, then any practical stability is out of question.

It is important to understand that the instability of eigenvalues is not necessarily due to the presence of multiple eigenvalues, nor is it of course to the presence of Jordan boxes. Let us consider a 20×20 matrix B :

$$B = \begin{pmatrix} 20 & & 20 & & 0 \\ & 19 & & 20 & \\ & & 18 & & 20 \\ & & & \ddots & \\ 0 & & & & 2 & 20 \\ & & & & & 1 \end{pmatrix}.$$

It is a triangular matrix and therefore its eigenvalues are the diagonal elements. On the face of it they are sufficiently well separated and there seem to be no grounds to expect instability. But let us add perturbation ε to the zero element in position $(20, 1)$. The free term of the characteristic polynomial will change by an amount of $20^{19}\varepsilon$. Since a product of eigenvalues is equal to a free term, the eigenvalues themselves must change very greatly.

Still more complicated questions arise in the study of the stability of eigenvectors. It is clear that if an eigenvalue λ of a matrix B is perturbation unstable, then the corresponding eigenvector x clearly cannot be stable, since B , λ and x are connected by the linear relations $Bx = \lambda x$.

It is important to note, however, that even if the eigenvalues remain unaffected by perturbation, not only may the eigenvectors be unstable, but their number may also change. For example, the first of the matrices

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & \varepsilon \\ 0 & 0 & 1 \end{pmatrix}$$

has three linearly independent eigenvectors, and the second has two, although their eigenvalues are equal. Theoretically this phenomenon is due only to the presence of multiple eigenvalues in the original matrix. But *under conditions of approximate assignment of a matrix it is hard, if not impossible, to decide which eigenvalues are to be considered multiple and which simple.*

Questions concerning the stability of eigenvalues, eigenvectors and root vectors are among the most complicated in the sections of algebra [connected] with computations.

Exercises

1. Let B be a matrix of a simple structure but with multiple eigenvalues. Prove that for any arbitrarily small $\varepsilon > 0$ there is a perturbation e_B satisfying $\|e_B\| < \varepsilon$ such that the matrix $B + e_B$ is no longer of a simple structure.

2. Let a matrix B have mutually distinct eigenvalues and let $d > 0$ be the smallest distance between them. Prove that there is a perturbation e_B satisfying $\|e_B\|_2 > d$ such that the matrix $B + e_B$ is not of a simple structure.

3. Now let B be a Hermitian matrix. Prove that if a Hermitian perturbation e_B satisfies the condition $\|e_B\|_2 < d/2$, then the matrix $B + e_B$ has mutually distinct eigenvalues.

4. Finally, let B be a non-Hermitian matrix with mutually distinct eigenvalues. Prove that there is a number r satisfying $0 < r \leq d$ such that the matrix $B + e_B$ is of a simple structure provided $\|e_B\|_2 \leq r$.

5. Try to establish a more exact relation between the numbers r and d .

PART III

Bilinear Forms

CHAPTER 11

Bilinear and Quadratic Forms

90. General properties of bilinear and quadratic forms

Consider numerical functions $\varphi(x, y)$ of two independent vector variables x and y of some vector space K_n over a number field P , taking on values from P . A function $\varphi(x, y)$ is said to be a *bilinear form* if for any vectors $x, y, z \in K_n$ and any number $\alpha \in P$

$$\begin{aligned}\varphi(x + z, y) &= \varphi(x, y) + \varphi(z, y), & \varphi(\alpha x, y) &= \alpha \varphi(x, y), \\ \varphi(x, y + z) &= \varphi(x, y) + \varphi(x, z), & \varphi(x, \alpha y) &= \alpha \varphi(x, y).\end{aligned}\quad (90.1)$$

The first two of the relations (90.1) imply the linearity of $\varphi(x, y)$ in the first independent variable, the last two imply the linearity in the second independent variable.

It is easy to verify that a sum of two bilinear forms, as well as a product of a bilinear form by a number, is again a bilinear form. Therefore the set of all bilinear forms over the same space K_n assuming values from the same number field P is a vector space. The "zero" of the given space is a bilinear form $0(x, y)$ such that $0(x, y) = 0$ for all x and y . The form $0(x, y)$ is called a *zero bilinear form*.

We have already encountered a function of this form. Comparing (27.1) and (90.1) it is easy to notice that a scalar product in a Euclidean space is a bilinear form. Recalling the important role played by the scalar product in the study of Euclidean spaces and of linear operators in them it may be suggested that a study of bilinear forms may turn out to be useful.

A special place among the bilinear forms is occupied by symmetric and skew-symmetric bilinear forms. A bilinear form $\varphi(x, y)$ is said to be *symmetric* if for any vectors $x, y \in K_n$

$$\varphi(x, y) = \varphi(y, x).$$

If, however, for any $x, y \in K_n$

$$\varphi(x, y) = -\varphi(y, x),$$

then the bilinear form is said to be *skew-symmetric*.

Any skew-symmetric bilinear form $\varphi(x, y)$ assumes a zero value

when its independent variables coincide. Indeed, since $\varphi(x, x) = -\varphi(x, x)$, we have $\varphi(x, x) = 0$. Somewhat unexpected is another fact connected with the values of a symmetric bilinear form when its independent variables coincide. Namely, any symmetric bilinear form $\varphi(x, y)$ is uniquely defined by its values when its independent variables coincide. Indeed, let x and y be any vectors from K_n . Taking into account the symmetry of $\varphi(x, y)$, we have

$$\varphi(x + y, x + y) = \varphi(x, x) + \varphi(y, y) + 2\varphi(x, y), \quad (90.2)$$

whence

$$\varphi(x, y) = \frac{1}{2} \{ \varphi(x + y, x + y) - \varphi(x, x) - \varphi(y, y) \}. \quad (90.3)$$

This formula proves the validity of the above assertion, since the right-hand side of the relation is a symmetric bilinear form.

A bilinear form is uniquely decomposable into a sum of a symmetric and a skew-symmetric bilinear form. In explicit form

$$\varphi(x, y) = \frac{1}{2} \{ \varphi(x, y) + \varphi(y, x) \} + \frac{1}{2} \{ \varphi(x, y) - \varphi(y, x) \}. \quad (90.4)$$

It is easy to verify that the first two terms at the right yield a symmetric bilinear form and the last two yield a skew-symmetric form. Assuming the existence of some other decomposition we shall have to conclude, on substituting equal independent variables, that the symmetric part of the decomposition is uniquely defined and that hence so is the decomposition as a whole.

If the bilinear form is not symmetric, then instead of (90.2) we shall have

$$\varphi(x + y, x + y) = \varphi(x, x) + \varphi(y, y) + \varphi(x, y) + \varphi(y, x).$$

Consequently

$$\begin{aligned} & \frac{1}{2} \{ \varphi(x, y) + \varphi(y, x) \} \\ &= \frac{1}{2} \{ \varphi(x + y, x + y) - \varphi(x, x) - \varphi(y, y) \}. \end{aligned} \quad (90.5)$$

Comparing this relation with (90.3) we conclude that for a nonsymmetric bilinear form its symmetric part is uniquely defined by the values of the form when its independent variables coincide.

Along with bilinear forms we shall consider the so-called quadratic forms. Let $\varphi(x, y)$ be a bilinear form in a space K_n . A *quadratic form* is a numerical function $\varphi(x, x)$ of a single independent vector variable $x \in K_n$ obtained from $\varphi(x, y)$ by replacing the vector y with x .

In general it is impossible to reconstruct uniquely from a quadratic form the bilinear form that has generated it. But, as formula (90.3) implies, there is one and only one symmetric bilinear form from which the original quadratic form can be obtained. That bilinear form is

called *polar* relative to a given quadratic form. The set of all bilinear forms generating the same quadratic form can be obtained by adding the polar bilinear form and an arbitrary skew-symmetric form. In using bilinear forms for the study of the properties of quadratic forms it suffices therefore to consider only symmetric bilinear forms.

The impossibility of reconstructing a bilinear form from a quadratic form is explained by the fact that the quadratic form gives no information about the skew-symmetric part of any bilinear form.

Lemma 90.1. *Skew-symmetric bilinear forms, and these forms alone, assume zero values for all coinciding independent variables.*

Proof. We have already noted that if $\varphi(x, y)$ is skew-symmetric, then $\varphi(x, x) = 0$ for every x . If, however, $\varphi(x, x) = 0$ for every x , then from (90.5) it follows that $\varphi(x, y) + \varphi(y, x) = 0$ for all vectors x and y , i.e. the bilinear form $\varphi(x, y)$ is skew-symmetric.

A comparison of the properties of a scalar product and relations (90.1) shows that in a unitary space strictly speaking a scalar product is not a bilinear form. In a complex space, closely related to a scalar product are Hermitian bilinear forms. A numerical function $\varphi(x, y)$ is said to be a *Hermitian bilinear form* if for any vectors $x, y, z \in K_n$ and any number α from the complex field P

$$\varphi(x + z, y) = \varphi(x, y) + \varphi(z, y), \quad \varphi(\alpha x, y) = \alpha \varphi(x, y),$$

$$\varphi(x, y + z) = \varphi(x, y) + \varphi(x, z), \quad \varphi(x, \alpha y) = \bar{\alpha} \varphi(x, y).$$

Here the bar stands for complex conjugation.

Again a sum of two Hermitian bilinear forms, as well as a product of a Hermitian bilinear form by a number, is a Hermitian bilinear form. The set of all Hermitian bilinear forms over the complex space assuming complex values is therefore a complex vector space.

A Hermitian bilinear form is said to be *Hermitian-symmetric* if for any vectors $x, y \in K_n$

$$\varphi(x, y) = \overline{\varphi(y, x)}.$$

If for any $x, y \in K_n$

$$\varphi(x, y) = -\overline{\varphi(y, x)},$$

then the form is called *skew-Hermitian*. On coinciding vectors the skew-Hermitian form assumes pure imaginary values and the Hermitian-symmetric form assumes real values. Now any Hermitian bilinear form is uniquely defined by its values when its independent variables coincide. But instead of (90.3) the following relation is true:

$$\begin{aligned} \varphi(x, y) = \frac{1}{4} \{ & \varphi(x+y, x+y) - \varphi(x-y, x-y) \\ & + i\varphi(x+iy, x+iy) - i\varphi(x-iy, x-iy) \}. \end{aligned} \quad (90.6)$$

From this it follows in particular that

Of the Hermitian bilinear forms only the zero form assumes zero values when all its independent variables coincide.

In this case, too, a Hermitian bilinear form can be uniquely represented as a sum of a Hermitian-symmetric and a skew-Hermitian form, with

$$\varphi(x, y) = \frac{1}{2} \{ \varphi(x, y) + \overline{\varphi(y, x)} \} + \frac{1}{2} \{ \varphi(x, y) - \overline{\varphi(y, x)} \}. \quad (90.7)$$

The proofs of the facts for Hermitian forms are much the same as the corresponding proofs for bilinear forms.

A quadratic *Hermitian form* is a numerical function $\varphi(x, x)$ of a single independent vector variable $x \in K_n$ obtained from a Hermitian bilinear function $\varphi(x, y)$ by replacing the vector y with x . Unlike quadratic forms, a Hermitian quadratic form allows a unique reconstruction of the Hermitian bilinear form that generates it. The reconstruction is carried out according to formula (90.6), and the corresponding bilinear form is also called *polar* relative to the original quadratic form.

The possibility of reconstructing uniquely a Hermitian bilinear form from the Hermitian quadratic form generated by it is due to a close relation of Hermitian-symmetric to skew-Hermitian bilinear forms.

Lemma 90.2. *If $\varphi(x, y)$ is a Hermitian-symmetric (skew-Hermitian) bilinear form, then $\psi(x, y) = i\varphi(x, y)$ is a skew-Hermitian (Hermitian-symmetric) bilinear form.*

Proof. Suppose, for example, $\varphi(x, y)$ is Hermitian-symmetric. Then for all vectors x and y we have

$$\psi(x, y) = i\varphi(x, y) = \varphi(ix, y) = \overline{\varphi(y, ix)} = -i\overline{\varphi(y, x)} = -\overline{\psi(y, x)},$$

i.e. $\psi(x, y)$ is skew-Hermitian. The case of a skew-Hermitian form $\varphi(x, y)$ can be considered in a similar way.

In what follows we shall more often be concerned with Hermitian quadratic forms generated by Hermitian-symmetric bilinear forms.

Lemma 90.3. *Of the Hermitian bilinear forms only symmetric forms generate real Hermitian quadratic forms.*

Proof. As already noted earlier, Hermitian-symmetric forms assume real values when their independent variables coincide. Suppose now that a Hermitian quadratic form $\varphi(x, x)$ assume only real values. According to (90.6), for a polar bilinear form $\varphi(x, y)$ we have

$$\begin{aligned} \varphi(y, x) &= \frac{1}{4} \{ \varphi(y+x, y+x) - \varphi(y-x, y-x) \\ &\quad + i\varphi(y+ix, y+ix) - i\varphi(y-ix, y-ix) \} \\ &= \frac{1}{4} \{ \varphi(x+y, x+y) - \varphi(x-y, x-y) + i\varphi(x-iy, x-iy) \\ &\quad - i\varphi(x+iy, x+iy) \} = \frac{1}{4} \{ \overline{\varphi(x+y, x+y)} - \overline{\varphi(x-y, x-y)} \\ &\quad + i\overline{\varphi(x+iy, x+iy)} - i\overline{\varphi(x-iy, x-iy)} \} = \overline{\varphi(x, y)}. \end{aligned}$$

Corollary. *Of the Hermitian bilinear forms only skew-symmetric forms generate pure imaginary Hermitian quadratic forms.*

Corollary. *No Hermitian nonsymmetric bilinear form can generate a real Hermitian quadratic form.*

As follows from the properties of linearity of bilinear and Hermitian bilinear forms in each independent variable, $\varphi(0, 0) = 0$ for any quadratic form $\varphi(x, x)$. In the general case, however, there may also be nonzero vectors x such that $\varphi(x, x) = 0$. These vectors will be called *isotropic*. The concept of isotropy is connected only with quadratic form. Therefore vectors isotropic for one quadratic form may be nonisotropic for another and vice versa. In particular, Lemma 90.1 implies that for a quadratic form generated by a skew-symmetric bilinear form all vectors in K_n , except the zero vector, are isotropic.

Of the ordinary and Hermitian real forms the most widely used are the forms that assume values of the same sign for all independent vector variables. A real quadratic form $\varphi(x, x)$ is said to be *positive definite* if $\varphi(x, x) > 0$ for every $x \neq 0$. The form is said to be *non-negative* if $\varphi(x, x) \geq 0$ for every $x \neq 0$. Similar definitions can be obtained for *nonpositive* and *negative definite quadratic forms*.

It is only positive definite and negative definite quadratic forms as a rule that are called *forms of constant signs*. But sometimes this term is also applied to nonnegative and nonpositive quadratic forms. To avoid confusion, whenever necessary positive definite and negative definite quadratic forms will be called *forms strictly of constant signs*.

If a quadratic form is a form of constant signs, then the ordinary or Hermitian bilinear form generating it will also be said to be positive definite, nonnegative and so on.

If a real quadratic form $\varphi(x, x)$ is strictly of constant signs, then it has no isotropic vectors. In the case of real bilinear and Hermitian-symmetric bilinear forms $\varphi(x, y)$ the corresponding quadratic forms will be real, and the converse is true for them. That is, we have

Theorem 90.1. *Let a quadratic form $\varphi(x, x)$ be generated by a real bilinear or Hermitian-symmetric bilinear form $\varphi(x, y)$. If $\varphi(x, x)$ has no isotropic vectors, then it is strictly of constant signs.*

Proof. As already noted, the quadratic form $\varphi(x, x)$ is real. In both cases it assumes on collinear vector values of the same sign. Suppose $\varphi(x, x)$ is not strictly of constant signs. Then we can find linearly independent vectors u and v such that $\varphi(u, u) > 0$ and $\varphi(v, v) < 0$. For any real number α

$$\begin{aligned} \varphi(u + \alpha v, u + \alpha v) &= \varphi(u, u) + \alpha(\varphi(u, v) + \varphi(v, u)) \\ &\quad + \alpha^2 \varphi(v, v). \end{aligned} \quad (90.8)$$

The right-hand side of this is a second-degree polynomial in α . It has real coefficients, which follows from the reality of $\varphi(x, x)$

and Lemma 90.3. Since $\varphi(u, u)$ and $\varphi(v, v)$ have opposite signs, polynomial (90.8) will have two real roots. Let α_0 be one of them. This means that $\varphi(u + \alpha_0 v, u + \alpha_0 v) = 0$. However, the vector $u + \alpha_0 v$ is nonzero by virtue of linear independence of u and v , so the vanishing on it of the quadratic form is impossible under the hypothesis of the theorem. This contradiction completes the proof.

It is no chance that we restricted our discussion in Theorem 90.1 to quadratic forms generated only by the real bilinear and the Hermitian-symmetric bilinear form. No other bilinear form can lead to a real quadratic form. Actually it only remains to consider the bilinear form in the complex space. But it is impossible for such a bilinear form to generate a real quadratic form not identically zero. If for some vector u the quadratic form takes on a nonzero real value $\varphi(u, u)$, then $\varphi(\alpha u, \alpha u) = \alpha^2 \varphi(u, u)$ will be a complex number for any complex α with a nonzero real and a pure imaginary part. So

For real quadratic forms to have no isotropic vectors it is necessary and sufficient that they should be strictly of constant signs.

A complex bilinear form always generates a quadratic form with isotropic vectors provided it is defined on a vector space of dimension greater than unity. Indeed, assuming that this is not the case we can always find linearly independent vectors u and v such that $\varphi(u, u) \neq 0$ and $\varphi(v, v) \neq 0$. But according to (90.8) the vector $u + \alpha v$ will be isotropic under a suitable choice of complex number α . A Hermitian bilinear complex form can generate a quadratic form having no isotropic vectors. It follows from our studies that

For a quadratic form generated by a Hermitian bilinear form to have no isotropic vectors it is sufficient that the real (or imaginary) part of the quadratic form should be strictly of constant signs.

Exercises

1. Prove that given any bilinear form $\varphi(x, y)$, equations $\varphi(0, y) = \varphi(x, 0) = 0$ for any $x, y \in K_n$.
2. Find the dimension and a basis of a vector space of bilinear forms.
3. Prove that sets of symmetric and skew-symmetric bilinear forms constitute subspaces in the vector space of all bilinear forms.
4. Prove that the space of all bilinear forms is a direct sum of subspaces of symmetric and skew-symmetric bilinear forms.
5. Prove that the set of all quadratic forms constitutes a vector space. Find its dimension and a basis.
6. Are the following sets of quadratic forms linear subspaces:
the quadratic forms of constant signs,
the quadratic forms assuming real values,
the quadratic forms having no isotropic vectors,
the quadratic forms for which all vectors of a given set are isotropic?
7. Prove that, given any quadratic form in a normed space, there is a number α such that for every x

$$|\varphi(x, x)| \leq \alpha \|x\|^2.$$

8. Let $\varphi(x, x)$ be a quadratic form strictly of constant signs and $\psi(x, x)$ an arbitrary quadratic form. Prove that there is a number β such that for every x

$$|\psi(x, x)| \leq \beta \varphi(x, x).$$

9. Prove that a quadratic form is not strictly of constant signs if and only if the set of isotropic vectors and the zero vector form a linear subspace.

10. Consider Exercises 1 to 9 for Hermitian bilinear and quadratic forms. Do all the assertions remain valid?

11. Suppose that in a complex space K_n some subspace L consists only of isotropic vectors of a Hermitian bilinear form $\varphi(x, y)$ and a zero vector. Prove that $\varphi(u, v) = 0$ for any vectors $u, v \in L$.

91. The matrices of bilinear and quadratic forms

We investigate a bilinear form $\varphi(x, y)$ in a space K_n . Choose in K_n two fixed bases, $e_1, e_2, \dots, e_n$ and $q_1, q_2, \dots, q_n$, and let

$$x = \sum_{i=1}^n \xi_i e_i, \quad y = \sum_{j=1}^n \eta_j q_j.$$

Then by property (90.1) we have

$$\varphi(x, y) = \varphi\left(\sum_{i=1}^n \xi_i e_i, \sum_{j=1}^n \eta_j q_j\right) = \sum_{i=1}^n \sum_{j=1}^n \varphi(e_i, q_j) \xi_i \eta_j. \quad (91.1)$$

Denote as before by x_e and y_q $n \times 1$ matrices made up of the coordinates of vectors x and y in the corresponding bases and by G_{eq} an $n \times n$ matrix with elements $g_{ij}^{eq} = \varphi(e_i, q_j)$. Relation (91.1) implies that

$$\varphi(x, y) = x'_e G_{eq} y_q. \quad (91.2)$$

Thus, given fixed bases in K_n , the bilinear form can be represented in matrix form (91.2).

The matrix G_{eq} is called the *matrix of the bilinear form* and is uniquely defined given fixed bases. Assuming that for $\varphi(x, y)$ there is another similar representation with some matrix F_{eq} besides (91.2) and taking $x = e_i$ and $y = q_j$, we at once get $f_{ij}^{eq} = \varphi(e_i, q_j)$, i.e. $F_{eq} = G_{eq}$.

Note that the right-hand side of (91.2) defines some bilinear form whatever the matrix G_{eq} . The validity of (90.1) is immediate from the corresponding properties of matrix operations. Thus, given fixed bases in K_n there is a 1-1 correspondence between bilinear forms and quadratic matrices.

Changing the bases in K_n affects the matrix of the bilinear form, of course. Let P be the coordinate transformation matrix for a change from $e_1, e_2, \dots, e_n$ to $f_1, f_2, \dots, f_n$ and Q the coordinate transformation matrix for a change from $q_1, q_2, \dots, q_n$ to $t_1, t_2, \dots, t_n$.

By (63.3)

$$x_e = Px_f, \quad y_q = Qy_t \quad (91.3)$$

and therefore it follows from (91.2) that

$$\varphi(x, y) = x'_e G_{eq} y_q = x'_f P' G_{eq} Q y_t.$$

But on the other hand

$$\varphi(x, y) = x'_f G_{ft} y_t.$$

Consequently

$$G_{ft} = P' G_{eq} Q. \quad (91.4)$$

Since P and Q are nonsingular, in accordance with the terminology introduced in Section 64 we shall call matrices G_{ft} and G_{eq} equivalent. As shown earlier, equivalent matrices of the same size, and only such matrices, have the same rank. This means that the rank of the matrix of a bilinear form is independent of the choice of bases and is a characteristic of the form itself. We shall call it the *rank* of the bilinear form. A bilinear form is said to be *nonsingular* if so is its matrix. A characteristic of a bilinear form is also the difference between the dimension of the space K_n and the rank of the form. We shall call it the *nullity* of the bilinear form.

It follows from the results of Section 64 that all matrices of the same rank are equivalent to a diagonal matrix with elements 0 and 1. In terms of bilinear forms this means that for an arbitrary form of rank r we can always find bases $f_1, f_2, \dots, f_n$ and $t_1, t_2, \dots, t_n$ such that the form is of the simplest type. That is, if

$$x = \sum_{i=1}^n \tau_i f_i, \quad y = \sum_{j=1}^n \nu_j t_j,$$

then

$$\varphi(x, y) = \sum_{i=1}^r \tau_i \nu_i.$$

A separate choice of bases for each variable of the bilinear form is made fairly rarely. It is more usual to choose a common basis. Let $e_1, e_2, \dots, e_n$ be some basis of K_n and

$$x = \sum_{i=1}^n \xi_i e_i, \quad y = \sum_{j=1}^n \gamma_j e_j.$$

In this case, as in (91.1), we obtain the following representation of a bilinear form:

$$\varphi(x, y) = \sum_{i=1}^n \sum_{j=1}^n \varphi(e_i, e_j) \xi_i \gamma_j$$

or in matrix notation

$$\varphi(x, y) = x'_e G_e y_e. \quad (91.5)$$

Here G_e is a matrix with elements $g_{ij}^{(e)} = \varphi(e_i, e_j)$. It is G_e that will throughout be called the matrix of a bilinear form. If again P is a coordinate transformation matrix for a change from $e_1, e_2, \dots, e_n$ to $f_1, f_2, \dots, f_n$, then according to (91.4) matrices G_e and G_f of the same bilinear form $\varphi(x, y)$ will be related by

$$G_f = P' G_e P. \quad (91.6)$$

The matrices G_e and G_f related by (91.6), with P nonsingular, are called *congruent*. Congruent matrices are always equivalent. In general the converse is not true, of course.

What was said about bilinear forms carries over with slight changes to Hermitian bilinear forms. Every Hermitian form can be represented uniquely in matrix notation

$$\varphi(x, y) = x'_e G_{eq} \bar{y}_q,$$

with $e_1, e_2, \dots, e_n$ and $q_1, q_2, \dots, q_n$ fixed. Under a change to $f_1, f_2, \dots, f_n$ and $t_1, t_2, \dots, t_n$, instead of (91.4) we have

$$G_{ft} = P' G_{eq} \bar{Q}.$$

If the independent variables of the Hermitian bilinear form are given in a single basis, then the matrix notation of the form is similar to (91.6). That is,

$$\varphi(x, y) = x'_e G_e \bar{y}_e. \quad (91.7)$$

Under a change to a new basis the matrices of the form are related by

$$G_f = P' G_e \bar{P}$$

and we shall say that they are *Hermitian-congruent*.

Now we can establish a relation between the type of a bilinear form and the type of its matrix. If the form is symmetric, then for any basis $e_1, e_2, \dots, e_n$

$$g_{ij}^{(e)} = \varphi(e_i, e_j) = \varphi(e_j, e_i) = g_{ji}^{(e)},$$

i.e. $G_e = G'_e$ and the matrix G_e of the form $\varphi(x, y)$ is *symmetric*. If, however, the form is skew-symmetric, then

$$g_{ij}^{(e)} = \varphi(e_i, e_j) = -\varphi(e_j, e_i) = -g_{ji}^{(e)},$$

i.e. $G_e = -G'_e$. In this case the matrix G_e is also called *skew-symmetric*.

The converse is also true. If in some basis the matrix of a form is symmetric (skew-symmetric), then so is the bilinear form generating

it. Let $G_e = G'_e$, then

$$\varphi(y, x) = y'_e G_e x_e = (y'_e G_e x_e)' = x'_e G'_e y_e = x'_e G_e y_e = \varphi(x, y).$$

If, however, $G'_e = -G_e$, then

$$\varphi(y, x) = y'_e G_e x_e = (y'_e G_e x_e)' = x'_e G'_e y_e = -x'_e G_e y_e = -\varphi(x, y).$$

Similar assertions hold also for the relation of the Hermitian bilinear form to its matrix. If the form is Hermitian-symmetric, then

$$g_{ij}^{(e)} = \varphi(e_i, e_j) = \overline{\varphi(e_j, e_i)} = \overline{g_{ji}^{(e)}},$$

i.e. $G_e = G_e^*$ and the matrix G_e of the form $\varphi(x, y)$ is *Hermitian*. If the form is skew-Hermitian, then

$$g_{ij}^{(e)} = \varphi(e_i, e_j) = -\overline{\varphi(e_j, e_i)} = -\overline{g_{ji}^{(e)}},$$

i.e. $G_e = -G_e^*$. In this case G_e is said to be *skew-Hermitian*.

The converse statements are also true. Let $G_e = G_e^*$. Then for the generating Hermitian bilinear form we have

$$\varphi(y, x) = y'_e G_e \bar{x}_e = (y'_e G_e \bar{x}_e)' = \bar{x}'_e G'_e y_e = \overline{x'_e G_e^* y_e} = \overline{x'_e G_e y_e} = \overline{\varphi(x, y)}.$$

For the case $G_e = -G_e^*$ we find

$$\varphi(y, x) = y'_e G_e \bar{x}_e = (y'_e G_e \bar{x}_e)' = \bar{x}'_e G'_e y_e = \overline{x'_e G_e^* y_e} = -\overline{x'_e G_e y_e} = -\overline{\varphi(x, y)}.$$

The matrix of a zero bilinear form consists only of zero elements, i.e. is a zero matrix. It is the only matrix that is simultaneously symmetric and skew-symmetric, as is the zero form.

We have already noted that there is a very close connection between symmetric bilinear and quadratic forms. It is especially obvious on the matrix level. For a bilinear form $\varphi(x, y)$ the matrix relation (91.5) holds. For the corresponding quadratic form we have

$$\varphi(x, x) = x'_e G_e x_e. \quad (91.8)$$

For a fixed basis $e_1, e_2, \dots, e_n$, given any matrix G_e , (91.8) defines some quadratic form. The matrix G_e in (91.8) is now called not the matrix of a bilinear form but the *matrix of a quadratic form*.

While for bilinear forms there is a 1-1 correspondence between the forms and their matrices given a fixed basis in K_n , there is no longer such a correspondence now. Every quadratic form can be given by the entire set of its matrices. This set contains only one symmetric matrix and the difference between any two matrices of a given set is a skew-symmetric matrix.

Thus any ordinary quadratic form can always be given by a symmetric matrix. Changing to a different basis affects the matrices of the quadratic form according to (91.6). We therefore conclude again that

problems of investigating symmetric bilinear and quadratic forms are closely related. For Hermitian quadratic forms this is no longer the case, since there is a 1-1 correspondence between them and Hermitian bilinear forms and there is a 1-1 correspondence between their matrices.

As with bilinear forms, the *rank* of a quadratic form is the rank of its matrix in any basis. If the matrix of a quadratic form is nonsingular, then the quadratic form is also called *nonsingular*.

Essentially the study of bilinear forms is the study of their matrices in different bases or equivalently the study of a class of congruent matrices. All our immediate studies therefore will be concerned with the investigation of classes of congruent and Hermitian-congruent matrices.

A number of properties for such classes follow at once from the preceding results. Thus a matrix congruent with a symmetric (skew-symmetric) matrix will necessarily be symmetric (skew-symmetric). In particular, symmetric is a matrix congruent with a diagonal matrix. From this we conclude that a nonzero symmetric matrix is never congruent with a skew-symmetric matrix although it may be equivalent to that matrix, and that a nonzero skew-symmetric matrix can never be congruent with a diagonal matrix. A matrix Hermitian-congruent with a Hermitian (skew-Hermitian) matrix is necessarily Hermitian (skew-Hermitian). Of the diagonal matrices it is only the matrix with real (pure imaginary) elements that can be Hermitian (skew-Hermitian).

In accordance with decompositions (90.4) and (90.7) of bilinear and Hermitian bilinear forms we obtain decompositions of an arbitrary matrix as a sum of a symmetric and a skew-symmetric matrix as well as that of a Hermitian and a skew-Hermitian matrix. These decompositions can be written out in explicit form:

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A'),$$

$$A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*).$$

If A is the matrix of a bilinear form, then the first terms of the right-hand sides are the matrices of the symmetric parts of the bilinear form and the second terms are the matrices of the skew-symmetric parts of the same form.

We shall often carry over to matrices without comment the terminology introduced for bilinear and quadratic forms. For example, we shall call a matrix *positive definite*, meaning by this that it is a matrix of a positive definite form and so on.

One of the major problems connected with the bilinear form is that of determining the simplest form its matrix can be reduced to by changing the basis and finding the appropriate basis. This problem

will be called the problem of *transforming* a bilinear form or the problem of *reducing* it to the simplest form.

In matrix interpretation the transformation problem can be stated as follows:

Given a matrix A find a nonsingular matrix P such that the matrix

$$C = P'AP \quad (91.9)$$

congruent with A has the simplest form.

Essentially this results from factoring the matrix, since it follows from (91.9) that

$$A = (P^{-1})'CP^{-1}.$$

For Hermitian bilinear forms, of course, instead of (91.9) we shall consider the transformation

$$C = P' A \bar{P}. \quad (91.10)$$

Computationally it is important that the matrix P in (91.9) and (91.10) should not be very complicated. That is because in finding new coordinates of vectors in terms of the old coordinates according to (63.3) one has to solve a system of linear algebraic equations with a matrix P and it is necessary that that solution should be carried out sufficiently fast. In some cases it is more convenient to seek the matrix P^{-1} instead of P .

Some other forms of notation for bilinear and quadratic forms may be used besides those considered above. Sometimes we shall give them in explicit form:

$$\Phi = \sum_{i=1}^n \sum_{j=1}^n a_{ji} x_i y_j, \quad F = \sum_{i=1}^n \sum_{j=1}^n a_{ji} x_i x_j. \quad (91.11)$$

This notation can be simplified. For example, let the space be real. Then so are both the bilinear form and the matrix A of coefficients a_{ji} . We introduce a space R_n whose elements are the column vectors

$$x = (x_1, x_2, \dots, x_n)', \quad y = (y_1, y_2, \dots, y_n)'$$

and suppose that the scalar product is introduced as a sum of pairwise products of coordinates. Now we can write:

$$\Phi = (Ax, y), \quad F = (Ax, x). \quad (91.12)$$

For Hermitian bilinear forms written as

$$\Phi = \sum_{i=1}^n \sum_{j=1}^n a_{ji} x_i \bar{y}_j, \quad F = \sum_{i=1}^n \sum_{j=1}^n a_{ji} x_i \bar{x}_j$$

we again have (91.12) if of course the scalar product is introduced as a sum of the products of the coordinates of the first vector by the complex conjugate coordinates of the second vector.

Exercises

1. Prove that the determinant of a Hermitian matrix is a real number.

2. What kind of number is the determinant of a skew-Hermitian matrix?

3. Prove that the rank of a skew-symmetric matrix is an even number.

4. Bilinear forms $\varphi(x, y)$ and $\overline{\varphi(y, x)}$ are in general different. What can be said about their matrices?

5. Prove that the rank of a sum of bilinear forms does not exceed the sum of the ranks of the summands.

6. Prove that every bilinear form of rank r can be represented as a sum of r bilinear forms of rank 1.

7. Prove that every bilinear form $\varphi(x, y)$ of rank 1 can be represented as

$$\varphi(x, y) = \varphi(x, a) \cdot \varphi(b, y)$$

for some vectors a and b . Is this representation unique?

92. Reduction to canonical form

Before proceeding to the study of various areas of application of bilinear and quadratic forms we consider a general method of congruence and Hermitian congruence transformation of matrices to a simple form.

Given a square $n \times n$ matrix A , find a nonsingular matrix P such that a matrix $C = P'AP$ has a sufficiently simple form. Under a Hermitian congruence transformation, it is the matrix $C = P'AP$ that must have a simple form. We shall now describe a general transformation method suitable for all matrices A . Differences between the congruence and Hermitian congruence transformations are insignificant. To be definite, we shall therefore assume that it is the congruence transformation of the matrix that holds.

The method consists in constructing a sequence of matrices $A_0 = A, A_1, A_2, \dots, A_s$, where each subsequent matrix is congruent with the preceding matrix, i.e.

$$A_{k+1} = P'_{k+1} A_k P_{k+1}$$

for some matrix P_{k+1} . Since the congruence relation is transitive, the last matrix, A_s , will be congruent with the original matrix A . The principle of constructing a sequence of matrices A_k relies on obtaining in the matrix A_{k+1} more zero elements for every k than there are in A_k . Moreover, each time we compute the matrix P_{k+1} from A_k we shall require that not only should there appear new zero elements in A_{k+1} but also that there should remain all zero elements obtained at all the preceding steps.

The transformation of a matrix A_k into A_{k+1} will be called a basic step of the method. Every basic step may consist of several auxiliary steps. They will all be reduced to elementary operations: interchanging of matrix columns (rows), addition to one column (row) of an-

other column (row) multiplied by a number, multiplication of a column (row) by a number. We describe the auxiliary steps in terms of transformations of a matrix A into a matrix $C = P'AP$ congruent with it, dropping for simplicity the index k .

A. In a matrix A the element $a_{11} \neq 0$. There is a nonsingular matrix P such that for the elements of the first column of a matrix $C = P'AP$

$$C_{j1} = \begin{cases} a_{11}, & j = 1, \\ 0, & j \neq 1. \end{cases} \quad (92.1)$$

The matrix P differs from a unit matrix in having a different first row, with

$$p_{1j} = \begin{cases} 1, & j = 1, \\ -\frac{a_{j1}}{a_{11}}, & j \neq 1. \end{cases} \quad (92.2)$$

Multiplying A on the left by a matrix P' does not affect the first row of A and makes zero all off-diagonal elements of the first column of a matrix $P'A$. Multiplying $P'A$ on the right by P does not affect the first column of $P'A$.

Note one important fact. All matrix minors in the upper left-hand corner of a matrix will be called *principal minors*. Since the matrix P is right triangular and each of its diagonal elements is equal to unity, of all minors in the first r columns only the principal minor is non-zero, it is equal to unity. All principal minors will therefore coincide in the matrices A and C . Indeed, using the Binet-Cauchy formula we find

$$\begin{aligned} C \begin{pmatrix} 1 & 2 & \dots & r \\ 1 & 2 & \dots & r \end{pmatrix} &= \sum_{1 \leq k_1 < k_2 < \dots < k_r \leq n} P' \begin{pmatrix} 1 & 2 & \dots & r \\ k_1 & k_2 & \dots & k_r \end{pmatrix} \\ &\times AP \begin{pmatrix} k_1 & k_2 & \dots & k_r \\ 1 & 2 & \dots & r \end{pmatrix} = AP \begin{pmatrix} 1 & 2 & \dots & r \\ 1 & 2 & \dots & r \end{pmatrix} = \sum_{1 \leq k_1 < k_2 < \dots < k_r \leq n} \\ &\times A \begin{pmatrix} 1 & 2 & \dots & r \\ k_1 & k_2 & \dots & k_r \end{pmatrix} \cdot P \begin{pmatrix} k_1 & k_2 & \dots & k_r \\ 1 & 2 & \dots & r \end{pmatrix} = A \begin{pmatrix} 1 & 2 & \dots & r \\ 1 & 2 & \dots & r \end{pmatrix}. \end{aligned}$$

We shall use this remark later on.

B. In a matrix A the element a_{11} is 0, but some element a_{jj} is other than 0, where $j > 1$. There is a nonsingular matrix P such that for a matrix $C = P'AP$ the element $c_{11} = a_{jj}$ is other than 0. The matrix P differs from a unit matrix only in the four elements at the intersections of rows and columns with indices 1, j . In those positions P is of the form $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Multiplying A on the right by P interchanges in A

the columns with indices 1, j . Multiplying the matrix AP on the left by P' interchanges in AP the rows with indices 1, j .

C. In a matrix A all diagonal elements are zero, but there are indices j, l , where $j < l$, such that $a_{lj} + a_{jl} \neq 0$. There is a nonsingular matrix P such that for a matrix $C = P'AP$ the element $c_{jj} = a_{lj} + a_{jl}$ is other than 0. The matrix P differs from a unit matrix in one element $p_{lj} = 1$. Multiplying A on the right by P adds to the j th column of A its l th column. Multiplying the matrix AP on the left by P' adds to the j th row of AP its l th row.

D. A matrix A is nonzero skew-symmetric, the element a_{12} is 0, but some element a_{jl} is other than 0, where $j < l$. There is a nonsingular matrix P such that in a skew-symmetric matrix $C = P'AP$ the element $c_{12} = a_{jl}$ is other than 0. The matrix P is represented as a product $P = P_1 \cdot P_2$. The matrices P_1 and P_2 differ from unit matrices in the four elements at the intersections of rows and columns with indices 1, j and 2, l respectively. In those positions P_1 and P_2 are of the form $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. As already stated, multiplying on the right by these matrices interchanges the columns and multiplying on the left interchanges the rows.

E. The matrix of the third-order principal minor of a matrix A is of the form

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & a_{23} \\ 0 & a_{32} & 0 \end{pmatrix}, \quad (92.3)$$

where the elements a_{11} , a_{23} and a_{32} are nonzero. There is a nonsingular matrix P such that the first three principal minors in $C = P'AP$ are nonzero. The matrix P differs from a unit matrix in one element p_{31} , which may be any number save 0, $-a_{12}a_{32}^{-1}$ and $-a_{11}a_{13}^{-1}$. Multiplying A on the right by P adds to the first column of A its third column multiplied by p_{31} . Multiplying AP on the left by P' adds to the first row of AP its third row multiplied by p_{31} .

F. A matrix A is skew-symmetric, the element a_{12} is other than 0. There is a nonsingular matrix P such that for the elements of the first two columns of a matrix $C = P'AP$

$$c_{j1} = \begin{cases} -a_{12}, & j = 2, \\ 0, & j \neq 2, \end{cases} \quad c_{j2} = \begin{cases} a_{12}, & j = 1, \\ 0, & j \neq 1. \end{cases}$$

Since under a congruence transformation a skew-symmetric matrix goes over into a skew-symmetric matrix, similar relations will hold also for the elements of the first two rows of C . The matrix P can be represented as a product $P = P_1 \cdot P_2$. The matrix P_1 differs from a

unit matrix in having a different second row, with

$$p_{2j}^{(1)} = \begin{cases} 0, & j = 1, \\ 1, & j = 2, \\ \frac{a_{j1}}{a_{12}}, & j > 2. \end{cases}$$

The matrix P_2 differs from a unit matrix only in having a different first row, with

$$p_{1j}^{(2)} = \begin{cases} 1, & j = 1, \\ 0, & j = 2, \\ -\frac{a_{j2}}{a_{12}}, & j > 2. \end{cases}$$

Multiplying A on the left by P'_1 does not affect the first two rows and the second column of A and makes zero all the elements of the first column of $P'_1 A$ save the first two. Multiplying $P'_1 A$ on the left by P'_2 does not affect the first two rows and the first column of $P'_1 A$ and makes zero all the elements of the second column of $P'_1 A$ save the first two. Multiplying $P'_1 A$ on the right by P does not affect the first two columns of $P'_1 A$.

G. Suppose a matrix A has for some partition into blocks the structure

$$A = \left(\begin{array}{c|c} A_{11} & A_{12} \\ \hline 0 & A_{22} \end{array} \right), \quad (92.4)$$

where A_{11} and A_{22} are square blocks. If P_{22} is a nonsingular matrix whose size is that of A_{22} , then the matrix

$$C = \left(\begin{array}{c|c} A_{11} & A_{12}P_{22} \\ \hline 0 & P'_{22}A_{22}P_{22} \end{array} \right)$$

is congruent with A . Moreover, $C = P'AP$, where

$$P = \left(\begin{array}{c|c} E & 0 \\ \hline 0 & P_{22} \end{array} \right).$$

A direct check of all the assertions made in the descriptions of the auxiliary steps presents no particular difficulty, and it is left for the reader as an exercise to show their validity.

The method as a whole is carried out as follows. At the first basic step the matrix A is reduced to the form (92.4), where A_{11} is a nonsingular 1×1 or 2×2 matrix. If a matrix A_k , $k \geq 1$, is of the form (92.4), then at the next basic step the matrix in the lower right-

hand corner is also reduced to the form (92.4) and a general congruence transformation is carried out according to step G. The matrix A_{k+1} can again be represented in the form (92.4) but the block in the upper left-hand corner will be not only nonsingular for it but will also have a greater size than for the matrix A_k . The process is repeated until at some step in the matrix A , there appears in (92.4) a zero block in the lower right-hand corner or the size of the block in the upper left-hand corner is $n \times n$. The resulting transformation matrix is a left-to-right product of the transformation matrices of all the steps.

The form of A , depends on whether the matrix A is skew-symmetric or not. So does the composition of the basic steps and the auxiliary steps.

Whatever the structure of a basic step, its aim is to obtain the next portion of zeros in the matrix to be transformed. If the original matrix is not skew-symmetric, then zeros are always obtained using an auxiliary step A, and steps B to C are necessary only for it to be prepared. But if the original matrix is skew-symmetric, then zeros are obtained using step F, and it is step D that is preparatory. We describe the basic step of the method also in terms of the transformation of a matrix A and we begin with a nonskew-symmetric matrix A .

At the first basic step the matrix to be transformed is nonskew-symmetric. If the element $a_{11} \neq 0$ and all off-diagonal elements of the first column are zero, then nothing changes and we assume that the basic step has been carried out. We take a unit matrix as a transformation matrix P . In general, however, we carry out the first of the auxiliary steps A to C that can be made. If this happens to be step B or C, then after it we must carry out step A or both steps, B and A. We take as a transformation matrix P a left-to-right product of all transformation matrices of the actually made auxiliary steps. As a result of the first basic step, in the transformed matrix A_1 all the off-diagonal elements of the first column will be zero, i.e. A_1 will have a block structure of the form (92.4).

The difference of all the other steps from the first is due to the fact that the matrix to be transformed may turn out to be skew-symmetric. If it is not, then the basic step to be made next does not differ in anything from the first. If however, the matrix to be transformed is skew-symmetric, then under any congruence transformation it remains skew-symmetric and it is impossible to obtain a nonzero element in the upper left-hand corner using this matrix alone. A way out is based on transforming an extended lower diagonal block.

Until a skew-symmetric matrix is found, the block in the upper left-hand corner of (92.4) for matrices A_k will be a right triangular matrix with nonzero diagonal elements. If the elements in positions (1, 2) and (2, 1) of the skew-symmetric matrix are nonzero in the lower right-hand corner, then for the matrix A_k which is the next to

be transformed we change representation (92.4) by decreasing by unity the size of the block in the upper left-hand corner. Now the 3×3 matrix in the upper left-hand corner of the new lower diagonal block will have the form (92.3) and we can carry out an auxiliary step E. After that it is possible to carry out three times in succession step A. Indeed, as we have noted, making step A does not affect the principal minors of the matrix. Hence in the given case, after carrying out step A, in the lower right-hand corner of the new matrix the first two principal minors will be nonzero. It is clearly possible therefore to make another step A. A similar reasoning shows that step A can be carried out a third time. Having made a step "back" we were enabled to move three steps "forward". If necessary, step D is carried out before step E.

Thus, if A is not a skew-symmetric matrix, then the above method allows us to construct a nonsingular matrix P such that the congruent matrix $P'AP$ will have the following structure:

$$P'AP = \left(\begin{array}{c|c} M & N \\ \hline 0 & 0 \end{array} \right). \quad (92.5)$$

Here M is a right triangular matrix with nonzero diagonal elements and the size of M is equal to the rank of A .

If A is a skew-symmetric matrix, then all the basic steps of the method, including the first, are carried out according to the same scheme. Suppose we have already obtained a matrix A_k of the form (92.4) and there is a nonsingular block-diagonal matrix with skew-symmetric 2×2 blocks in the upper left-hand corner. Since under a congruence transformation a skew-symmetric matrix goes over into a skew-symmetric matrix, the block $A_{1,2}$ in (92.4) is zero. We first have to obtain nonzero elements in positions (1, 2) and (2, 1) of the skew-symmetric matrix in the lower right-hand corner. It is possible that to do this should require an auxiliary step D. We further carry out step F, which adds to the diagonal another nonsingular skew-symmetric 2×2 block, and proceed to the next basic step. Now too the process is continued until at some step in the matrix A_s there appears in representation (92.4) a zero block in the lower right-hand corner or the size of the block in the upper left-hand corner is $n \times n$.

So if A is a skew-symmetric matrix, then in this case the method allows us to construct a nonsingular matrix P such that $P'AP$ has the following structure:

$$P'AP = \left(\begin{array}{c|c} M & 0 \\ \hline 0 & 0 \end{array} \right). \quad (92.6)$$

Here M is a block-diagonal matrix with nonsingular skew-symmetric 2×2 blocks. The size of M equals the rank of A .

For a Hermitian congruence transformation the general scheme of the method remains the same. The process, however, turns out to be even simpler than for the ordinary congruence transformation if the auxiliary step C is replaced by the following.

C'. In a matrix A all diagonal elements are zero but there are indices j, l , where $j < l$, such that among the elements a_{lj} and a_{jl} there is at least one nonzero element. There is a nonsingular matrix P such that for a matrix $C = P'AP$ one of the diagonal elements c_{jj} and c_{ll} is nonzero. That is, $c_{jj} = a_{jl} + a_{lj}$ and $c_{ll} = i(a_{jl} - a_{lj})$. The matrix P differs from a unit matrix in two elements, $p_{lj} = 1$ and $p_{jl} = i$. Multiplying A on the right by $\bar{P}$ adds to the j th column of A its l th column and to the l th column of A its j th column multiplied by $-i$. Multiplying $A\bar{P}$ on the left by P' adds to the j th row of $A\bar{P}$ its l th row and to the l th row of $A\bar{P}$ its j th row multiplied by i .

Now there is no need for steps D to F of the general method, as we shall never go beyond step C'. Moreover, formulas (92.2) remain unchanged.

Thus, if A is a nonzero matrix, then the method allows us to construct a nonsingular matrix P such that a matrix $P'AP$ Hermitian-congruent with A will have the following structure:

$$P'AP = \left(\begin{array}{c|c} M & N \\ \hline 0 & 0 \end{array} \right) \quad (92.7)$$

Here M is a right triangular matrix with nonzero diagonal elements. The size of M equals the rank of A .

The forms of matrices (92.5) to (92.7) are called *canonical forms* for the operations of congruence transformation. A *canonical basis* is also any basis in which the original matrix has such a form. Matrices of the forms (92.5) and (92.7) are by themselves called *right trapezoidal matrices*. Similarly defined are *left trapezoidal matrices*.

Note a number of interesting conclusions arising from the canonical forms of matrices. As we have already said, a congruence transformation preserves the symmetry and skew-symmetry of matrices. If one of these properties is a feature of the original matrix, it must be inherited by the canonical form. In addition to what has been said it can be concluded therefore that

A symmetric matrix is congruent with a diagonal matrix.

A Hermitian matrix is Hermitian-congruent with a real diagonal matrix.

A skew-Hermitian matrix is Hermitian-congruent with a pure imaginary diagonal matrix.

In all these cases reduction to canonical form is effected particularly simply, since there cannot arise a need to carry out even one of the auxiliary steps D to F.

For matrices of the canonical forms (92.5) and (92.6) we may perform yet another congruence transformation with a diagonal matrix and have the nonzero elements determining the nonsingularity of the block M equal either $+1$ or -1 . Such a canonical form of a matrix and the corresponding basis are called *normal*. It is clear that multiplying on the right (left) by a diagonal matrix results in multiplying the columns (rows) by the diagonal elements of the transformation matrix. We again describe the transformation in terms of the auxiliary step with a matrix A .

H. A real nonskew-symmetric matrix A of rank r has the canonical form (92.5). There is a real diagonal matrix P such that the nonzero diagonal elements c_{jj} of a matrix $C = P'AP$ are equal to $\text{sgn } a_{jj}$, with

$$p_{jj} = \begin{cases} (a_{jj} \text{sgn } a_{jj})^{-1/2}, & j \leq r, \\ 1, & j > r. \end{cases}$$

A real (complex) skew-symmetric matrix A of rank r has the canonical form (92.6). There is a real (complex) diagonal matrix P such that the nonzero upper off-diagonal elements of the matrix $C = P'AP$ equal $+1$ and the nonzero lower off-diagonal elements equal -1 , with

$$p_{jj} = \begin{cases} 1, & j \text{ is odd,} \\ a_{j-1,j}^{-1}, & j \text{ is even.} \end{cases}$$

A complex nonskew-symmetric matrix A of rank r has the canonical form (92.5). There is a complex diagonal matrix P such that the nonzero diagonal elements c_{jj} of $C = P'AP$ equal 1 , with

$$p_{jj} = \begin{cases} a_{jj}^{-1/2}, & j \leq r, \\ 1, & j > r. \end{cases}$$

A Hermitian congruence transformation with a diagonal matrix is rarely employed, since it can change only the absolute values of the elements determining the nonsingularity of the block M in (92.7) but cannot make the complex diagonal elements real.

Exercises

1. Prove that if reduction to canonical form using a matrix P is carried out according to the above method, then $\det P = \pm 1$.
2. What does the matrix equation

$$\begin{pmatrix} 1 & 1 \\ 1 & -i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -i \end{pmatrix} - \begin{pmatrix} 1-i & 0 \\ 2 & 1 \end{pmatrix} \quad (92.8)$$

mean in terms of the canonical form?

3. To what form can a nonskew-symmetric matrix be reduced using congruence transformation if the auxiliary step E is excluded?

4. What form would a transformation matrix P have if each basic step of the above method consisted only of the auxiliary step A?

5. Prove that any right triangular matrix is congruent with a left triangular matrix. What is the simplest form of the transformation matrix?

6. Prove that any nonsingular matrix of odd size is congruent with a nonsingular right triangular matrix.

7. Let G be the matrix of a positive definite bilinear form. Prove that given its elements g_{ij} , for all i and j

$$g_{ii} > 0, \quad (g_{ij} + g_{ji})^2 < 4g_{ii}g_{jj}.$$

8. Let G be the matrix of a negative definite bilinear form. Prove that given its elements g_{ij} , for all i and j

$$g_{ii} < 0, \quad (g_{ij} + g_{ji})^2 < 4g_{ii}g_{jj}.$$

9. Prove that the matrices of all symmetric positive (negative) definite bilinear forms are congruent.

10. Prove that for a matrix G to be the matrix of an alternating bilinear form it is sufficient that there should be diagonal elements with opposite signs in it.

93. Congruence and matrix decompositions

The general method of congruence transformation of a matrix to canonical form does not always make it possible to predict the form the coordinate transformation matrix for a change to canonical basis will have. Under some additional constraints on the original matrix, however, this question can be given a quite definite answer.

Suppose that in a matrix A all principal minors, except perhaps the highest-order minor, i.e. the determinant of A , are nonzero. We show that it is always possible to represent such a matrix as a product

$$A = LDU, \quad (93.1)$$

where L is a left triangular matrix with unit diagonal elements, D is a diagonal matrix and U is a right triangular matrix with unit diagonal elements, i.e.

$$A = \begin{pmatrix} 1 & & & \\ l_{21} & 1 & & 0 \\ & \ddots & \ddots & \\ l_{n1} & l_{n2} & \dots & 1 \end{pmatrix} \begin{pmatrix} d_{11} & & & \\ & d_{22} & & 0 \\ & & \ddots & \\ 0 & & & d_{nn} \end{pmatrix} \begin{pmatrix} 1 & u_{12} & \dots & u_{1n} \\ & 1 & \dots & u_{2n} \\ & & \ddots & \\ 0 & & & 1 \end{pmatrix}.$$

Equating the elements of A and the products of LDU we get

$$a_{ij} = \sum_{p=1}^{\min(i, j)} l_{ip} d_{pp} u_{pj}. \quad (93.2)$$

Now we find from (93.2) successively all the unknown elements of the matrices of decomposition (93.1). Namely,

$$\begin{aligned}
 d_{11} &= a_{11}, \\
 u_{1j} &= \frac{a_{1j}}{d_{11}}, \quad l_{j1} = \frac{a_{j1}}{d_{11}}, \quad j > 1, \\
 d_{ii} &= a_{ii} - \sum_{p=1}^{i-1} l_{ip} d_{pp} u_{pi}, \quad i > 1, \\
 u_{ij} &= \frac{a_{ij} - \sum_{p=1}^{i-1} l_{ip} d_{pp} u_{pj}}{d_{ii}}, \\
 l_{ji} &= \frac{a_{ji} - \sum_{p=1}^{j-1} l_{jp} d_{pp} u_{pi}}{d_{ii}}, \quad i > 1, j > i.
 \end{aligned} \tag{93.3}$$

We apply to (93.1) the Binet-Cauchy formula. Recall that among the minors of the left triangular matrix L in the first r rows only the principal minor is nonzero, it is equal to unity. A similar assertion holds for the matrix U , if the rows are replaced by columns, of course. Therefore

$$\begin{aligned}
 A \begin{pmatrix} 1 & 2 & \dots & r \\ 1 & 2 & \dots & r \end{pmatrix} &= \sum_{1 \leq k_1 < k_2 < \dots < k_r \leq n} L \begin{pmatrix} 1 & 2 & \dots & r \\ k_1 & k_2 & \dots & k_r \end{pmatrix} \\
 &\times DU \begin{pmatrix} k_1 & k_2 & \dots & k_r \\ 1 & 2 & \dots & r \end{pmatrix} = DU \begin{pmatrix} 1 & 2 & \dots & r \\ 1 & 2 & \dots & r \end{pmatrix} = d_{11} d_{22} \dots d_{rr}.
 \end{aligned}$$

Hence

$$d_{11} = a_{11}, \quad d_{ii} = \frac{A \begin{pmatrix} 1 & 2 & \dots & i \\ 1 & 2 & \dots & i \end{pmatrix}}{A \begin{pmatrix} 1 & 2 & \dots & i-1 \\ 1 & 2 & \dots & i-1 \end{pmatrix}}. \tag{93.4}$$

Under the assumption the principal minors of A are nonzero. Therefore so are all the diagonal elements d_{ii} in (93.4) except perhaps the last one.

We shall fairly often deal with decompositions (93.1) for symmetric and Hermitian matrices. If again all the principal minors of A , except perhaps the last one, are nonzero, then a symmetric matrix can always be represented as a product

$$A = S' D S \tag{93.5}$$

and a Hermitian matrix can be represented as a product

$$A = S' D \bar{S}. \tag{93.6}$$

Here S is a right triangular matrix with unit diagonal elements and D is a diagonal matrix, i.e.

$$S = \begin{pmatrix} 1 & s_{12} & \dots & s_{1n} \\ & 1 & \dots & s_{2n} \\ & & \ddots & \\ 0 & & & 1 \end{pmatrix}, \quad D = \begin{pmatrix} d_{11} & & & 0 \\ & d_{22} & & \\ & & \ddots & \\ 0 & & & d_{nn} \end{pmatrix}.$$

Completely in accordance with (93.3) we now have

$$\begin{aligned} d_{11} &= a_{11}, \quad s_{1j} = \frac{a_{1j}}{d_{11}}, \quad j > 1, \\ d_{ii} &= a_{ii} - \sum_{p=1}^{i-1} d_{ip} s_{pi}^2, \quad i > 1, \\ s_{ij} &= \frac{a_{ij} - \sum_{p=1}^{i-1} d_{ip} s_{pi} s_{pj}}{d_{ii}}, \quad j > i, \end{aligned} \quad (93.7)$$

for decomposition (93.5) and

$$\begin{aligned} d_{11} &= a_{11}, \quad s_{1j} = \frac{\bar{a}_{1j}}{d_{11}}, \quad j > 1, \\ d_{ii} &= a_{ii} - \sum_{p=1}^{i-1} d_{ip} |s_{pi}|^2, \quad i > 1, \\ s_{ij} &= \frac{a_{ij} - \sum_{p=1}^{i-1} d_{ip} s_{pi} \bar{s}_{pj}}{d_{ii}}, \quad j > i, \end{aligned}$$

for decomposition (93.6). Formulas (93.4) remain valid.

Decompositions (93.1), (93.5) and (93.6) are extensively used to solve diverse problems of linear algebra. As to congruence transformations of the matrix these decompositions lead to the following relations:

$$\begin{aligned} (L^{-1})' AL^{-1} &= DUL^{-1}, & (L^{-1})' A\overline{L^{-1}} &= DUL\overline{L^{-1}}, \\ S^{-1} AS^{-1} &= D, & S^{-1} A\overline{S^{-1}} &= D. \end{aligned}$$

The matrices DUL^{-1} and $DUL\overline{L^{-1}}$ are right triangular and the matrices D are diagonal; the zero element on their principal diagonals may be only the last. Therefore we have again obtained the already familiar canonical forms of matrices under a congruence transformation. Now we can say, however, that coordinate transformation matrices for a change to canonical basis are right triangular, since so are the matrices L^{-1} and S^{-1} . The above decompositions themselves give coordinate transformation matrices L' and S for a change

from the canonical basis to the original basis, which are also right triangular.

In the case of a symmetric matrix the described process of decomposition is closely related to the so-called *Jacobi algorithm* for transforming a quadratic form to canonical form. The only difference is that in the Jacobi algorithm we find the matrix S^{-1} instead of S . Notice that S is much easier to find than S^{-1} .

Congruence transformations with a right triangular matrix are among the simplest, but still sufficiently general transformations to be applied to a wide class of matrices. Of certain interest therefore is a description of the class of matrices that can be reduced to canonical form using transformations with a right triangular matrix.

Lemma 93.1. *If a rectangular matrix A is representable in the block form*

$$A = \begin{pmatrix} B & Q \\ \vdots & \vdots \\ R & T \end{pmatrix}, \quad (93.8)$$

where B is a square nonsingular $r \times r$ matrix, then the rank of A is r if and only if

$$T = RB^{-1}Q. \quad (93.9)$$

Proof. We multiply A on the left by a nonsingular block matrix

$$V = \begin{pmatrix} E & 0 \\ -RB^{-1} & E \end{pmatrix},$$

where the corresponding blocks have the same size as in (93.8). Then

$$VA = \begin{pmatrix} B & Q \\ \vdots & \vdots \\ 0 & T - RB^{-1}Q \end{pmatrix}.$$

The matrices A and VA have the same rank but it is equal to r if and only if $T - RB^{-1}Q = 0$.

Now we can describe the desired class of matrices. It turns out to be closely related to matrices of the form (93.8) and (93.9).

Theorem 93.1. *For a nonskew-symmetric matrix A to be reducible to canonical form using a congruence transformation with a right triangular matrix it is necessary and sufficient that the number of the first nonzero principal minors of A should equal its rank.*

Proof. Necessity. Let a nonskew-symmetric matrix A be reducible to canonical form (92.5) using a right triangular matrix P . It is clear that the number of the first nonzero principal minors in A cannot be greater than the size of the block M . Applying the Binet-Cauchy formula and considering that there is no nonzero minor in

the first columns of P , except the principal minor, we get

$$A \begin{pmatrix} 1 & 2 & \dots & s \\ 1 & 2 & \dots & s \end{pmatrix} \left\{ P \begin{pmatrix} 1 & 2 & \dots & s \\ 1 & 2 & \dots & s \end{pmatrix} \right\}^2 = M \begin{pmatrix} 1 & 2 & \dots & s \\ 1 & 2 & \dots & s \end{pmatrix}$$

for every s not greater than the size of M . Since the principal minors of M and P are nonzero, the number of the first nonzero principal minors of A equals its rank.

Sufficiency. Suppose the number of the first nonzero principal minors of A and its rank equal r . We represent A in the block form (93.8), where the size of the block B is equal to $r \times r$. Since all principal minors of the matrix B are nonzero, it follows from the foregoing that it can be represented as $B = LDU$ similarly to (93.1). We construct a block matrix

$$P = \begin{pmatrix} L^{-1'} & -B^{-1'}R' \\ 0 & E \end{pmatrix}.$$

A direct check shows that

$$P'AP = \begin{pmatrix} DUL^{-1'} & L^{-1}(-BB^{-1'}R' + Q) \\ 0 & 0 \end{pmatrix}.$$

The matrix $DUL^{-1'}$ is nonsingular right triangular and P is nonsingular right triangular, and hence A is reducible in the required way to canonical form.

For a congruence transformation of a skew-symmetric matrix and a Hermitian congruence transformation of an arbitrary matrix the corresponding statements can be proved in a similar way and we shall restrict ourselves to their formulation.

Theorem 93.2. *For a skew-symmetric matrix A of rank r to be reducible to canonical form using a congruence transformation with a right triangular matrix it is necessary and sufficient that the number of the first nonzero principal minors of even order of A should equal $r/2$.*

Theorem 93.3. *For a matrix A to be reducible to canonical form using a Hermitian congruence transformation with a right triangular matrix it is necessary and sufficient that the number of the first nonzero principal minors of A should equal its rank.*

Congruence and Hermitian congruence transformations of a matrix are not in general similarity transformations. However, if for some class of matrix P one of the following groups of relations holds

$$PP' = P'P = E, \quad PP^* = P^*P = E, \quad (93.10)$$

then the congruence transformation becomes a similarity transformation and in order to make studies we may use the earlier obtained results relating to the similarity of matrices. As we already know, the first group of relations in (93.10) is satisfied by real orthogonal

matrices, and the second group of relations is satisfied by complex unitary matrices. Recalling the results of Sections 76 to 81 for orthogonal and unitary similarities we therefore conclude that the following statements are true:

Any real symmetric or skew-symmetric matrix can be reduced to canonical form using a congruence transformation with an orthogonal matrix.

Any complex matrix can be reduced to canonical form using a Hermitian congruence transformation with a unitary matrix.

These statements are mainly of theoretical interest, since in practice orthogonal and unitary transformation matrices are difficult to find, especially for $n \geq 5$.

Exercises

1. Prove that if decompositions (93.1), (93.5) and (93.6) exist, then they are unique.

2. Prove that if all the minors of a matrix A in the lower right-hand corner (except perhaps the minor of the highest order) are nonzero, then there is a unique decomposition $A = LDC$, where L is a right triangular matrix, U is a left triangular matrix, each with unit diagonal elements, and D is a diagonal matrix.

3. Prove that for the elements d_{ii} of the matrix D of Exercise 2

$$d_{nn} = a_{nn}, \quad d_{ii} = \frac{A \begin{pmatrix} i, i+1, \dots, n \\ i, i+1, \dots, n \end{pmatrix}}{A \begin{pmatrix} i-1, i+2, \dots, n \\ i-1, i+2, \dots, n \end{pmatrix}}, \quad i < n.$$

4. Into what triangular factors can a matrix be factored if its minors in the lower left-hand (upper right-hand) corner are nonzero?

5. Suppose for the elements a_{ij} of a matrix A

$$a_{ij} = 0, \quad k < j - l, \quad j - i < l, \quad (93.11)$$

given some numbers $l < k$. Such a matrix is called a *band matrix*. Prove that if for a band matrix A decomposition (93.1) holds, then

$$l_{ij} = 0, \quad j - i < l, \quad u_{ij} = 0, \quad j - i > k.$$

6. A matrix A is said to be *tridiagonal* if it satisfies conditions (93.11) for $k = 1$ and $l = -1$. What form have formulas (93.3) and (93.7) for a tridiagonal matrix?

7. A matrix A is said to be *right (left) almost triangular* if it satisfies conditions (93.11) for $k = n$ and $l = -1$ ($k = 1$ and $l = -n$). What form have formulas (93.3) for almost triangular matrices?

8. What number of arithmetical operations is required for the various forms of matrices to obtain decompositions of the type (93.1)?

9. How are decompositions (93.1), (93.5) and (93.6) to be applied to solve systems of linear algebraic equations?

94. Symmetric bilinear forms

Discussing bilinear and quadratic forms we have often paid particular attention both to symmetric bilinear forms and to bilinear forms generating real quadratic forms. Only two

kinds of bilinear forms simultaneously satisfy both conditions, these are the real symmetric and the Hermitian-symmetric bilinear form. The matrices of these forms are in any basis a real symmetric or a Hermitian matrix respectively. A congruence transformation reduces both forms of matrices to a diagonal real normal form.

As we have seen, various congruence transformations can reduce the same matrix to canonical form. In general therefore the canonical form is not uniquely defined. The question naturally arises: What do the different canonical forms to which the same matrix is reducible have in common? We know that the rank of a matrix does not depend on transformation. Whatever the method of reducing to canonical form is therefore, the number of the last zero rows will be the same. Much more can be said concerning the real symmetric and the Hermitian matrix. The canonical form of these matrices can be characterized by the number of positive and negative terms it contains. There is an important

Theorem 94.1 (the law of inertia for quadratic forms). *The number of positive terms and that of negative terms in the canonical form of a real symmetric matrix under the ordinary congruence transformation and of a Hermitian matrix under a Hermitian congruence transformation do not depend on the method of reduction.*

Proof. Let some matrix A satisfy the hypotheses of the theorem. Consider a quadratic form F with a matrix A of rank r in variables $x_1, x_2, \dots, x_n$ and suppose that two methods have been used to reduce it to the normal form

$$F = y_1^2 - y_2^2 + \dots + y_k^2 - y_{k+1}^2 - y_{k+2}^2 - \dots - y_r^2 \\ = z_1^2 + z_2^2 + \dots + z_l^2 - z_{l+1}^2 - z_{l+2}^2 - \dots - z_r^2. \quad (94.1)$$

Since the change from the variables $x_1, x_2, \dots, x_n$ to $y_1, y_2, \dots, y_n$ was effected using a nonsingular linear transformation, the new variables will be linearly expressible in terms of the old variables, with the determinant of the inverse transformation matrix nonzero. So

$$y_l = \sum_{s=1}^n b_{ls} x_s, \quad \det \begin{pmatrix} b_{11} & \dots & b_{1n} \\ \dots & \dots & \dots \\ b_{n1} & \dots & b_{nn} \end{pmatrix} \neq 0. \quad (94.2)$$

Similarly

$$z_l = \sum_{t=1}^n c_{lt} x_t, \quad \det \begin{pmatrix} c_{11} & \dots & c_{1n} \\ \dots & \dots & \dots \\ c_{n1} & \dots & c_{nn} \end{pmatrix} \neq 0. \quad (94.3)$$

Suppose that $k < l$ and write the system of equations

$$y_1 = y_2 = \dots = y_k = z_{l+1} = \dots = z_n = 0. \quad (94.4)$$

If the left-hand sides of these equations are replaced by their expressions in (94.2) and (94.3), a system $n - l + k$ of homogeneous linear equations in n unknowns $x_1, x_2, \dots, x_n$ is obtained. The number of equations in that system is smaller than that of unknowns, so the system has a nonzero real solution $\alpha_1, \alpha_2, \dots, \alpha_n$.

Now replace in (94.1) all the variables by their expressions in (94.2) and (94.3) and then substitute the numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ for $x_1, x_2, \dots, x_n$. If for brevity we denote by $y_i(\alpha)$ and $z_j(\alpha)$ the values of the variables y_i and z_j after such a substitution, then taking into account (94.4) relation (94.1) becomes

$$-y_{l+1}^2(\alpha) - \dots - y_r^2(\alpha) - z_1^2(\alpha) - \dots - z_l^2(\alpha).$$

It follows that

$$z_1(\alpha) = \dots = z_l(\alpha) = 0. \quad (94.5)$$

On the other hand, according to the choice of the numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ we have

$$z_{l+1}(\alpha) = \dots = z_r(\alpha) = \dots = z_n(\alpha) = 0. \quad (94.6)$$

Thus the system of n homogeneous linear equations

$$z_i = 0, \quad i = 1, 2, \dots, n,$$

in n unknowns $x_1, x_2, \dots, x_n$ has, by (94.5) and (94.6), a nonzero solution $\alpha_1, \alpha_2, \dots, \alpha_n$, i.e. the determinant of the system must be zero. This contradicts (94.3). We arrive at a similar contradiction assuming $l < k$. Hence $l = k$ and the theorem is proved.

Any real ordinary (Hermitian) quadratic form in a real (complex) vector space has a unique real symmetric (complex Hermitian) matrix in any basis. These matrices satisfy the hypotheses of Theorem 94.1. Whatever the basis, the number of positive and negative terms in the canonical form of a matrix is invariant for a quadratic form and is called its positive and negative *index of inertia* respectively. The difference between its positive and negative indices is called the *signature* of the quadratic form. We can now formulate some useful corollaries of Theorem 94.1.

Corollary. *A quadratic form is positive (negative) definite if and only if the positive (negative) index of inertia is equal to n .*

Corollary. *A quadratic form is of constant signs if and only if one of the indices of inertia is zero.*

The law of inertia permits some classification of real quadratic forms to be given. We shall say that two quadratic forms are *affinely equivalent* if for each of them we can choose a basis such that the matrices of those quadratic forms are the same. In this case we shall also say that a nonsingular transformation converts one quadratic form into the other. It is easy to verify that the affine equivalence of quadratic forms is an equivalence relation and that two quadratic

forms are equivalent if and only if their matrices are congruent in the same basis. It follows from the law of inertia therefore that all real quadratic forms in a vector space K_n can be grouped into non-overlapping classes, each consisting only of affinely equivalent quadratic forms. A class is characterized by rank and signature. This grouping is called an *affine classification* of real quadratic forms.

Given any rank r of quadratic forms a given classification always has two "extreme" classes, the classes with signatures $+r$ and $-r$. The first class comprises all nonnegative quadratic forms of rank r , the second comprises all nonpositive quadratic forms of rank r . Both classes taken together contain all rank- r quadratic forms of constant signs and only these forms.

In general the constancy of signs of a quadratic form is easy to establish by reducing the form to canonical form in one of the ways described above. Of considerable interest in some cases, however, are also direct criteria of the constancy of signs. Taking into account the great significance of these very quadratic forms we shall carry out additional studies for them, confining our discussion mainly to quadratic forms in a real space. We shall again assume that the matrix of a quadratic form is real symmetric. For the case of a complex space the results of the studies will be the same and the proofs differ in minor details.

Theorem 94.2 (Sylvester's criterion). *For a quadratic form to be positive definite it is necessary and sufficient that all principal minors of the matrix of that form should be positive.*

Proof. Necessity. Let a quadratic form with a matrix A be positive definite. Then there is a nonsingular transformation with a matrix P reducing the form to a sum of squares. According to (91.9) this means that $E = P'AP$ or $A = (P^{-1})'P^{-1}$. Using the Binet-Cauchy formula we find

$$\begin{aligned} A \begin{pmatrix} 1 & 2 & \dots & s \\ 1 & 2 & \dots & s \end{pmatrix} \\ = \sum_{1 \leq k_1 < k_2 < \dots < k_s \leq n} (P^{-1})' \begin{pmatrix} 1 & 2 & \dots & s \\ k_1 & k_2 & \dots & k_s \end{pmatrix} \cdot P^{-1} \begin{pmatrix} k_1 & k_2 & \dots & k_s \\ 1 & 2 & \dots & s \end{pmatrix} \\ = \sum_{1 \leq k_1 < k_2 < \dots < k_s \leq n} \left(P^{-1} \begin{pmatrix} k_1 & k_2 & \dots & k_s \\ 1 & 2 & \dots & s \end{pmatrix} \right)^2. \end{aligned}$$

Since P is nonsingular, the first s columns contain at least one nonzero minor. Hence for every s the right-hand side of the equation obtained is positive.

Sufficiency. Suppose now that all principal minors of the matrix A of some quadratic form are positive. We reduce that form to canoni-

cal form using the transformation defined by formulas (93.7). Under the hypotheses of the theorem and according to formulas (93.4) all the coefficients of the canonical form will be positive, i.e. the quadratic form is positive definite.

Corollary. *For a quadratic form to be negative definite it is necessary and sufficient that all principal minors of odd order should be negative and all principal minors of even order should be positive.*

The proof follows from Sylvester's criterion and from the fact that if A is the matrix of a negative definite quadratic form, then $-A$ is the matrix of a positive definite quadratic form.

Theorem 94.3 (Jacobi's criterion). *For a quadratic form to be positive definite it is necessary and sufficient that all coefficients of the characteristic polynomial of the matrix of the form should be nonzero and have alternating signs.*

Proof. Necessity. As already noted, a transformation of variables with an orthogonal matrix can reduce a given quadratic form to canonical form, where coefficients are the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of the matrix of the form. Under the hypotheses of the theorem the eigenvalues must be positive. The characteristic polynomial $f(\lambda)$ equals

$$f(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n) = \lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0$$

and all of its coefficients are nonzero and have alternating signs, which is immediate from Vieta's formulas for the coefficients a_i .

Sufficiency. Let the coefficients of the characteristic polynomial be nonzero and have alternating signs. The roots of this polynomial will be real as the eigenvalues of a symmetric matrix and it remains to show that they are positive. Suppose this statement has been proved for all polynomials of degree $n-1$. Since all coefficients $f'(\lambda)$ are nonzero and have alternating signs, under the assumption $f'(\lambda)$ has the $(n-1)$ th positive root. It is known from mathematical analysis that if a polynomial has only real roots, then they are separated by the roots of a derivative. Therefore $f(\lambda)$ has at least $n-1$ positive roots. The last root will also be positive since the product of roots is positive.

Criteria for nonnegative and nonpositive quadratic forms are much more complicated, and this is mainly due to the fact that in these cases the matrices of the forms are singular. One of the main ways of investigating the constancy of signs of a quadratic form involves reducing its matrix to a symmetric form (93.8) and (93.9) and studying that form. Matrices of constant signs being closely related, we restrict our consideration to nonnegative matrices only.

We shall say that a matrix H is a *permutation matrix* if in each of its rows and in each of its columns there is only one nonzero element and all the nonzero elements equal unity. It is clear that multiplying an arbitrary matrix A on the right by a permutation matrix H

interchanges the columns in A and multiplying on the left interchanges the rows.

Lemma 94.1. *For an arbitrary nonsingular matrix A there is a permutation matrix H such that in the matrix AH all principal minors are nonzero.*

Proof. The matrix A is nonsingular. Hence there is at least one nonzero element in its first row. Interchanging an appropriate column with the first column makes the first-order principal minor nonzero. Suppose that by interchanging the columns we have made all principal minors up to the k th order nonzero. If interchanging the last $n - k$ columns cannot yield a nonzero principal minor of order $k + 1$, this means that in the first $k + 1$ rows of A there is no nonzero minor of order $k + 1$, i.e. that A must be singular. This contradiction proves the lemma.

Theorem 94.4. *For a quadratic form of rank r with a matrix A to be nonnegative it is necessary and sufficient that there should be a permutation matrix H such that in the matrix $H'AH$ the first r principal minors are positive.*

Proof. **Necessity.** Let a quadratic form of rank r with a matrix A be nonnegative. Then there is a nonsingular matrix P such that $A = (P^{-1})'E_rP^{-1}$, where E_r is a diagonal matrix whose first r elements equal unity and whose other elements are zero. According to Lemma 94.1 there is a permutation matrix H such that all principal minors of the matrix $P^{-1}H$ are nonzero.

Using the Binet-Cauchy formula we find for $1 \leq s \leq r$

$$\begin{aligned} H'AH & \begin{pmatrix} 1 & 2 & \dots & s \\ 1 & 2 & \dots & s \end{pmatrix} \\ &= \sum_{1 \leq k_1 < k_2 < \dots < k_s \leq n} (P^{-1}H)' \begin{pmatrix} 1 & 2 & \dots & s \\ k_1 & k_2 & \dots & k_s \end{pmatrix} \cdot (E_r P^{-1}H) \begin{pmatrix} k_1 & k_2 & \dots & k_s \\ 1 & 2 & \dots & s \end{pmatrix} \\ &= \sum_{1 \leq k_1 < k_2 < \dots < k_s \leq r} \left\{ (P^{-1}H) \begin{pmatrix} k_1 & k_2 & \dots & k_s \\ 1 & 2 & \dots & s \end{pmatrix} \right\}^2 \\ &\geq \left\{ (P^{-1}H) \begin{pmatrix} 1 & 2 & \dots & s \\ 1 & 2 & \dots & s \end{pmatrix} \right\}^2 > 0. \end{aligned}$$

Sufficiency. Suppose that for a quadratic form of rank r with a matrix A there is a permutation matrix H such that in the matrix $H'AH$ the first r principal minors are positive. By Theorem 93.1 $H'AH$ can be reduced to canonical form using a transformation with a triangular matrix. According to (93.4) the nonzero coefficients of the canonical form of $H'AH$ and hence of A are positive, i.e. the quadratic form is nonnegative.

As to quadratic forms that are not of constant signs, there are no theorems closely similar to Theorems 94.1 and 94.4 for them. There is only

Theorem 94.5. *If a quadratic form has a symmetric matrix A of the form (93.8) and (93.9), then its indices of inertia coincide with those of the "truncated" quadratic form defined by the matrix B of (93.8).*

Proof. By Theorem 93.1 the matrix A can be reduced to canonical form using a transformation with a right triangular matrix, with relations (93.4) holding for the nonzero coefficients of the canonical form. But the matrix B of the "truncated" quadratic form also satisfies the hypotheses of Theorem 93.1 and for the coefficients of its canonical form we again have relations (93.4). Therefore the indices of inertia of the quadratic forms defined by A and B coincide.

The particular interest we have shown for quadratic forms of constant signs is accounted for by their vast area of application. One of the major applications is the introduction of a metric in a vector space. Any bilinear form polar to some positive definite quadratic form may be regarded as a scalar product and hence we can turn a vector space into a Euclidean or a unitary space, using it. The validity of the axioms for these spaces is obvious. Of no less importance for introducing a metric, especially a metric on subspaces, are also non-negative forms. As an example of using the constancy of signs we prove the validity of

Theorem 94.6. *For a nonsingular Hermitian bilinear form to be reducible to diagonal form it is sufficient that its symmetric (or skew-symmetric) part should be strictly of constant signs.*

Proof. Consider the case of a positive definite symmetric part. Let A be the matrix of the bilinear form. Then a matrix $(1/2)(A + A^*)$ will be the matrix of the symmetric part. Since the symmetric part is positive definite, the matrix $(1/2)(A + A^*)$ is Hermitian-congruent with a unit matrix. Hence there is a nonsingular matrix S such that

$$\frac{1}{2} S' (A + A^*) \bar{S} = E. \quad (94.7)$$

We show that the matrix $S' A \bar{S}$ is normal. From (94.7) we have

$$\bar{S} S' = 2 (A + A^*)^{-1}.$$

Therefore

$$\begin{aligned} (S' A \bar{S}) (S' A \bar{S})^* - (S' A \bar{S})^* (S' A \bar{S}) &= S' (A \bar{S} S' A^* - A^* \bar{S} S' A) \\ &= 2 S' (A (A + A^*)^{-1} A^* - A^* (A + A^*)^{-1} A) \bar{S} \\ &= 2 S' ((A^{*-1} (A + A^*) A^{-1})^{-1} - (A^{-1} (A + A^*) A^{*-1})^{-1}) \bar{S} \\ &= 2 S' ((A^{*-1} + A^{-1})^{-1} - (A^{*-1} + A^{-1})^{-1}) \bar{S} = 0. \end{aligned}$$

By virtue of normality $S'AS$ is reducible to diagonal form using a Hermitian congruence transformation with a unitary matrix.

Thus the matrix A of a Hermitian bilinear form is Hermitian-congruent with a diagonal matrix, which was to be shown. The proofs of the other cases are similar.

Exercises

1. Prove that if all principal minors of a real symmetric or a complex Hermitian matrix are nonzero, then the number of its positive and negative eigenvalues coincides respectively with that of positive and negative terms of sequence (93.4).

2. Prove that if a matrix is positive definite, then any diagonal minor is positive.

3. Prove that a symmetric matrix of rank r always has at least one diagonal minor of order r not equal to zero.

4. Prove that the maximum element of a positive definite matrix is on the principal diagonal.

5. Prove that A is a positive definite matrix if for every i

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|.$$

6. Prove that for any symmetric matrix A of rank r there is a permutation matrix H such that among the first r principal minors of the matrix $H'AH$ there are no two adjacent zero minors and the minor of order r is nonzero.

7. Prove that the matrix $H'AH$ of Exercise 6 can be represented as $H'AH = S'DS$, where S is a right triangular matrix with unit diagonal elements and D is a block-diagonal matrix with 1×1 and 2×2 blocks.

8. Prove that any nonnegative matrix of rank r can be represented as a sum of r nonnegative matrices of rank 1.

9. Let A and B be positive definite matrices with elements a_{ij} and b_{ij} . Prove that a matrix C with elements $c_{ij} = a_{ij}b_{ij}$ is also positive definite.

95. Second-degree hypersurfaces

Closely related to the study of real quadratic forms is another study, that concerned with second-degree hypersurfaces. Wishing to stress the geometrical character of many of the properties of hypersurfaces, in what follows we shall nearly always call vectors points of a space R_n .

A *second-degree hypersurface* f in R_n is a set of points whose coordinates $x_1, x_2, \dots, x_n$ satisfy the equation

$$\sum_{i=1}^n \sum_{j=1}^n a_{ji} x_i x_j - 2 \sum_{k=1}^n b_k x_k + c = 0, \quad (95.1)$$

where a_{ji} , b_k and c are real numbers.

We simplify the notation. As in the case of quadratic forms, we shall assume that a matrix A with coefficients a_{ji} is symmetric. By b we denote a vector with coordinates $b_1, b_2, \dots, b_n$. We introduce

in R_n a scalar product as a sum of pairwise products of coordinates. We can now regard a second-degree hypersurface f in R_n as a set of points x of a Euclidean space R_n satisfying the equation

$$(Ax, x) - 2(b, x) + c = 0 \quad (95.2)$$

or, A being symmetric, the equation

$$(x, Ax) - 2(b, x) + c = 0.$$

To begin with, we consider relative positions of second-degree hypersurfaces and straight lines. Take a straight line in R_n . Let it pass through a point x_0 and have a direction vector l . Points x of that straight line are defined by

$$x = x_0 + lt \quad (95.3)$$

for all possible real numbers t . Substituting the expression for x in (95.2) we get

$$t^2 (Al, l) - 2t ((b, l) - (Al, x_0)) + (Ax_0, x_0) - 2(b, x_0) + c = 0. \quad (95.4)$$

Thus the points of intersection of the straight line (95.3) with hypersurface (95.2) are given by the roots of the quadratic equation (95.4).

We shall say that the straight line (95.3) with direction vector l has a *nonasymptotic (asymptotic) direction* relative to hypersurface (95.2) if $(Al, l) \neq 0$ ($(Al, l) = 0$).

Consider any straight line with a nonasymptotic direction l intersecting a hypersurface. The points of intersection determine on every such line a segment, which by analogy with elementary geometry will be called a *chord*. We denote by L the set of the midpoints of all chords. If the ends of a chord collapse at a point, then that point will be regarded also as the midpoint of the chord. We show that L is in some hyperplane.

The ends of any chord are given by the values of a parameter t coinciding with the roots of (95.4). Therefore the midpoint of a chord is given by the value of t equal to a half-sum of the roots. According to Vieta's formulas this yields

$$t = \frac{(b, l) - (Al, x_0)}{(Al, l)}. \quad (95.5)$$

If z_0 is the midpoint of a chord, then

$$z_0 = x_0 + \frac{(b, l) - (Al, x_0)}{(Al, l)} l.$$

Now we have

$$\begin{aligned} (Al, z_0) &= \left(Al, x_0 + \frac{(b, l) - (Al, x_0)}{(Al, l)} l \right) \\ &= (Al, x_0) + \frac{(b, l) - (Al, x_0)}{(Al, l)} (Al, l) = (b, l). \end{aligned}$$

So the midpoints of all chords satisfy the equation

$$(Al, x) = (b, l). \quad (95.6)$$

Since its right-hand side is independent of x_0 , according to (46.8) this equation defines a hyperplane whose normal vector is equal to Al .

Hyperplane (95.6) is called a *diametrical hyperplane* conjugate to a direction l relative to hypersurface (95.2).

The explicit form of the equation of a diametrical hyperplane allows us to establish a number of important properties of second-degree hypersurfaces. Let A be a nonsingular matrix. Then for any linearly independent vectors $l_1, l_2, \dots, l_n$ so are vectors $Al_1, Al_2, \dots, Al_n$. Suppose further that all directions $l_1, l_2, \dots, l_n$ are nonasymptotic. This will clearly hold, for example, when a quadratic form (Ax, x) is positive definite. Hence it is possible to construct a system of n diametrical hyperplanes conjugate to $l_1, l_2, \dots, l_n$. The hyperplanes will have in common a unique point x^* . From (95.6) we now get

$$(Ax^* - b, l_i) = 0$$

for $i = 1, 2, \dots, n$. By virtue of the linear independence of the vectors l_i this means that $Ax^* - b = 0$, i.e. that the point x^* is nothing but a solution of the system of linear algebraic equations

$$Ax = b. \quad (95.7)$$

The solution of the system with a nonsingular matrix is unique, so the constructed point x^* is in fact independent of the choice of vectors $l_1, l_2, \dots, l_n$.

Simple calculations show that for any point x^*

$$(Ax, x) - 2(b, x) + c = (A(x - x^*), x - x^*) + 2(Ax^* - b, x - x^*) + (Ax^*, x^*) - 2(b, x^*) + c. \quad (95.8)$$

If, however, x^* is a solution of system (95.7), then with respect to such a point hypersurface (95.2) has an important symmetry property. That is, for any x the left-hand side of (95.2) assumes the same values at the points

$$x = x^* + (x - x^*), \quad x' = x^* - (x - x^*). \quad (95.9)$$

It follows in particular that both points x and x' are or are not simultaneously on hypersurface (95.2). The equation

$$x^* = \frac{1}{2}(x + x')$$

allows us to call the point x^* the *centre of symmetry* of the hypersurface. If on (95.2) there is at least one point of R_n , then the centre of symmetry is said to be *real*. Otherwise it is called *imaginary*.

Now let x^* be the centre of symmetry, i.e. for any x the left-hand side of (95.2) assumes the same values at the points x and x' . Hence

$$(Ax, x) - 2(b, x) + c = (Ax', x') - 2(b, x') + c.$$

According to (95.8) and (95.9) this is possible only if for any x

$$(Ax^* - b, x - x^*) \equiv 0.$$

But the last identity holds if and only if $Ax^* - b = 0$, i.e. if x^* is a solution of system (95.7). Notice that here we have never assumed either the nonsingularity of the matrix A or the presence of any other of its properties besides symmetry. Therefore:

For a system $Ax = b$ to have a solution it is necessary and sufficient that hypersurface (95.2) should have a centre of symmetry. The set of all solutions coincides with that of all centres of symmetry.

Thus there emerges a far-reaching connection between systems of linear algebraic equations and second-degree hypersurfaces. It is widely used in constructing a host of computational algorithms. Construction of a system of diametrical hyperplanes is central to a large group of methods among the so-called methods of conjugate directions. These will be discussed in the last chapter.

In general investigation of second-degree hypersurfaces may be based on reducing them to canonical form in much the same way as quadratic forms are. But in addition to linear nonsingular transformations of variables, translations are required.

Consider any transformation of variables $x = Py$ reducing a quadratic form (Ax, x) to normal form. In variables $y_1, y_2, \dots, y_n$ the equation of a hypersurface will have the form

$$y_1^2 + \dots + y_k^2 - y_{k+1}^2 - \dots - y_r^2 - 2d_1 y_1 - \dots - 2d_r y_r - 2d_{r+1} y_{r+1} - \dots - 2d_n y_n + c = 0.$$

Now we translate the variables by the formulas

$$z_i = \begin{cases} y_i - d_i, & 1 \leq i \leq k, \\ y_i + d_i, & k+1 \leq i \leq r, \\ y_i, & r+1 \leq i \leq n. \end{cases}$$

In these variables the equation is as follows:

$$z_1^2 + \dots + z_k^2 - z_{k+1}^2 - \dots - z_r^2 - 2d_{r+1} z_{r+1} - \dots - 2d_n z_n + p = 0.$$

Let one of the numbers $d_{r+1}, \dots, d_n$, for example d_n , be other than 0. Set

$$v_i = \begin{cases} z_i, & i < n, \\ d_{r+1} z_{r+1} + \dots + d_n z_n, & i = n, \end{cases}$$

and then make another translation

$$u_i = \begin{cases} v_i, & i < n, \\ v_i - \frac{p}{2}, & i = n. \end{cases}$$

Now the equation of a hypersurface is as follows:

$$\pm u_1^2 \pm u_2^2 \pm \dots \pm u_r^2 - 2u_n = 0, \quad 1 \leq r \leq n-1. \quad (95.10)$$

If there is no nonzero number among $d_{r+1}, \dots, d_n, p$, then the equation of a hypersurface assumes the form

$$\pm u_1^2 \pm u_2^2 \pm \dots \pm u_r^2 = 0, \quad 1 \leq r \leq n. \quad (95.11)$$

And finally if $d_{r+1}, \dots, d_n$ are zero but $p \neq 0$, then putting $u_i = z_i/|p|^{1/2}$ for every i we obtain one more form of the equation of a hypersurface. Namely,

$$\pm u_1^2 \pm u_2^2 \pm \dots \pm u_r^2 \pm 1 = 0, \quad 1 \leq r \leq n. \quad (95.12)$$

Because of the law of inertia for quadratic forms, surfaces given by different equations of the form (95.10) to (95.12) cannot be converted into one another using linear transformation of variables and translation. As different one should regard equations that cannot be converted into one another by multiplying by (-1) and changing the indices of the coordinates. As in the case of quadratic forms, we have again obtained a subdivision of all second-degree hypersurfaces into nonoverlapping classes.

Not infrequently reduction of second-degree hypersurfaces to canonical form uses only operations of translation and linear transformations of variables with orthogonal matrices. This is mainly due to both types of the transformations leaving unchanged the distances between points. In this case the canonical forms will be somewhat different, although on the whole they are obtained in the same way as those considered above. For example, in the case of R_2 a second-degree hypersurface can be reduced only to one of the following forms:

- I. $\lambda_1 x^2 + \lambda_2 y^2 + a_0 = 0,$
- II. $\lambda_2 y^2 + b_0 x = 0,$
- III. $\lambda_1 x^2 + a_0 = 0,$

(95.13)

in the case of R_3 it can be reduced to one of the following forms:

- I. $\lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2 + a_0 = 0,$
- II. $\lambda_1 x^2 + \lambda_2 y^2 + b_0 z = 0,$
- III. $\lambda_1 x^2 + \lambda_2 y^2 + a_0 = 0,$
- IV. $\lambda_1 y^2 + b_0 x = 0,$
- V. $\lambda_1 x^2 + a_0 = 0.$

(95.14)

In all equations (95.13) and (95.14) the coefficients of the variables written out are nonzero. The free term may be zero. According to the well-established terminology hypersurfaces in R_2 will be called *second-degree curves* and those in R_3 *second-degree surfaces*. Considering the interests of many branches of mathematics we shall study in more detail second-degree curves and surfaces using their canonical forms (95.13) and (95.14).

Exercises

1. Let A be a positive definite matrix. Prove that on the solution of the system $Ax = b$ the expression $(Ax, x) - 2(b, x)$ reaches its minimum.

2. Let A be a positive definite matrix. Prove that on the straight line (95.3) the expression $(Ax, x) - 2(b, x)$ reaches its minimum for the value of t in (95.5).

3. Prove that for any direction to be nonasymptotic for hypersurface (95.2) it is necessary and sufficient that the quadratic form (Ax, x) should be either positive definite or negative definite.

4. What symmetry property has a diametrical hyperplane conjugate to a direction l if l is an eigenvector of a matrix A corresponding to a nonzero eigenvalue?

5. Prove that the system $Ax = b$ has no solution if and only if hypersurface (95.2) can be reduced to canonical form (95.10).

96. Second-degree curves

We shall study second-degree curves using equations (95.13). Let the equation of a straight line be of the form

$$\lambda_1 x^2 + \lambda_2 y^2 + a_0 = 0. \quad (96.1)$$

I.1. The number a_0 is not zero; the numbers λ_1 and λ_2 are opposite in sign to a_0 . We write (96.1) as

$$\frac{x^2}{\left(\sqrt{-\frac{a_0}{\lambda_1}}\right)^2} + \frac{y^2}{\left(\sqrt{-\frac{a_0}{\lambda_2}}\right)^2} = 1$$

and set

$$a = \sqrt{-\frac{a_0}{\lambda_1}}, \quad b = \sqrt{-\frac{a_0}{\lambda_2}}. \quad (96.2)$$

Under the hypothesis a and b are real numbers, so (96.1) is equivalent to

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \quad (96.3)$$

The curve described by this equation is called an *ellipse* (Fig. 96.1) and the equation is called the *canonical equation* of an ellipse. We show some properties of the ellipse. An ellipse is a bounded curve. It follows from equation (96.3) that for all points of an ellipse we

have $|x| \leq a$ and $|y| \leq b$. An ellipse has two axes of symmetry, the x axis and the y axis, and a centre of symmetry, the origin. This follows from the fact that apart from a point with coordinates (x, y) an ellipse contains points with coordinates $(x, -y)$, $(-x, y)$ and $(-x, -y)$. The axes of symmetry are called the *principal axes* of the ellipse and the centre of symmetry is the *centre* of the ellipse. If $a > b$, then the x axis is called the *major axis* of the ellipse and the y axis is the *minor axis*. The points of intersection of the principal axes of the ellipse with the ellipse itself are called the *vertices* of the ellipse. When $a = b$, the ellipse is a circle of radius a with centre at the origin. Suppose for definiteness that $a > b$ and let

$$c^2 = a^2 - b^2. \quad (96.4)$$

The points F_1 and F_2 with coordinates $(-c, 0)$ and $(+c, 0)$ are called the *foci* of the ellipse.

Theorem 96.1. *The sum of the distances from any point of an ellipse to its foci is a constant value equal to $2a$.*

Proof. For any point $M(x, y)$ of an ellipse

$$y^2 = b^2 - \frac{b^2 x^2}{a^2}.$$

For the same point

$$\begin{aligned} \rho(M, F_2) &= ((x-c)^2 + y^2)^{1/2} = \left(x^2 - 2xc + c^2 + b^2 - \frac{b^2 x^2}{a^2} \right)^{1/2} \\ &= \left(x^2 \left(1 - \frac{b^2}{a^2} \right) - 2xc + a^2 \right)^{1/2} = \left(\frac{x^2 c^2}{a^2} - 2xc + a^2 \right)^{1/2} \\ &= \left(\left(-\frac{c}{a}x + a \right)^2 \right)^{1/2} = -\frac{c}{a}x + a. \end{aligned}$$

The last equation holds, since $-cx/a + a > 0$ in view of the fact that $|x| \leq a$ and $c/a < 1$. Further

$$\begin{aligned} \rho(M, F_1) &= ((x+c)^2 + y^2)^{1/2} = \left(x^2 + 2xc + c^2 + b^2 - \frac{b^2 x^2}{a^2} \right)^{1/2} \\ &= \left(x^2 \left(1 - \frac{b^2}{a^2} \right) + 2xc + a^2 \right)^{1/2} = \left(\frac{x^2 c^2}{a^2} + 2xc + a^2 \right)^{1/2} \\ &= \left(\left(\frac{c}{a}x + a \right)^2 \right)^{1/2} = \frac{c}{a}x + a. \end{aligned}$$

Finally we have

$$\rho(M, F_1) + \rho(M, F_2) = -\frac{c}{a}x + a + \frac{c}{a}x + a = 2a.$$

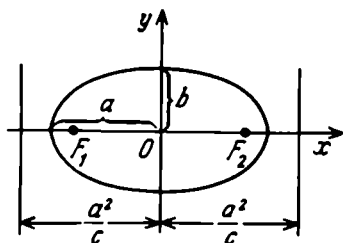


Fig. 96.1

I.2. The number a_0 is not zero; the numbers λ_1 , λ_2 and a_0 have the same sign. Let

$$a = \sqrt{\frac{a_0}{\lambda_1}}, \quad b = \sqrt{\frac{a_0}{\lambda_2}}. \quad (96.5)$$

Then (96.1) is equivalent to

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = -1. \quad (96.6)$$

It is clear that there is no point in the plane satisfying (96.6). It is usual to say that (96.6) is the equation of an *imaginary ellipse*.

I.3. The number a_0 is zero; the numbers λ_1 and λ_2 have the same sign. Let

$$a = \sqrt{\frac{1}{|\lambda_1|}}, \quad b = \sqrt{\frac{1}{|\lambda_2|}}.$$

Then (96.1) is equivalent to

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0. \quad (96.7)$$

It is clear that only the origin satisfies (96.7). It is usual to say that (96.7) is the equation of a *degenerate ellipse*.

I.4. The number a_0 is not zero; the numbers λ_1 and λ_2 are opposite in sign. Introducing new coefficients similar to (96.2) and (96.5) we reduce (96.1) to an equivalent equation (up to a designation of the variables)

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1. \quad (96.8)$$

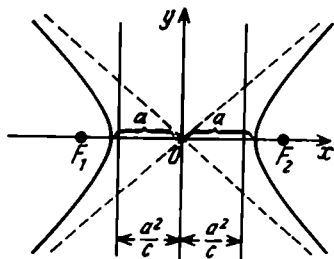


Fig. 96.2

The curve described by this equation is called a *hyperbola* (Fig. 96.2) and the equation is called the *canonical equation* of a hyperbola. Unlike the ellipse, the hyperbola is an unbounded curve. As in the ellipse, the axes of symmetry of the hyperbola are the coordinate axes and its centre of symmetry is the origin. The axes of symmetry are the *principal axes* of the hyperbola and the centre of symmetry is the *centre* of the hyperbola. One of the principal axes (the x axis) intersects with the hyperbola at two points, which are called the *vertices* of the hyperbola. It is called the *transverse* (or *real*) *axis* of the hyperbola. The other axis (the y axis) has no points in common with the hyperbola and is therefore called the *imaginary* (or *conjugate*) *axis* of the hyperbola. Let

$$c^2 = a^2 + b^2.$$

The points F_1 and F_2 with coordinates $(-c, 0)$ and $(+c, 0)$ are called the *foci* of the hyperbola.

Theorem 96.2. *The absolute value of the difference of the distances from any point of a hyperbola to its foci is a constant value equal to $2a$.*

Proof. For any point $M(x, y)$ of a hyperbola

$$y^2 = -b^2 + \frac{b^2 x^2}{a^2}.$$

For the same point

$$\begin{aligned} \rho(M, F_2) &= ((x-c)^2 + y^2)^{1/2} = \left(x^2 - 2xc + c^2 - b^2 + \frac{b^2 x^2}{a^2}\right)^{1/2} \\ &= \left(x^2 \left(1 - \frac{b^2}{a^2}\right) - 2xc + a^2\right)^{1/2} = \left(\frac{x^2 c^2}{a^2} - 2xc + a^2\right)^{1/2} \\ &= \left(\left(\frac{c}{a}x - a\right)^2\right)^{1/2} = \left|\frac{c}{a}x - a\right|. \end{aligned}$$

Further

$$\begin{aligned} \rho(M, F_1) &= ((x+c)^2 + y^2)^{1/2} = \left(x^2 + 2xc + c^2 - b^2 + \frac{b^2 x^2}{a^2}\right)^{1/2} \\ &= \left(x^2 \left(1 + \frac{b^2}{a^2}\right) + 2xc + a^2\right)^{1/2} = \left(\frac{x^2 c^2}{a^2} + 2xc + a^2\right)^{1/2} \\ &= \left(\left(\frac{c}{a}x + a\right)^2\right)^{1/2} = \left|\frac{c}{a}x + a\right|. \end{aligned}$$

For all points of a hyperbola we have $|x| \geq a$ and $c/a > 1$. Therefore

$$\begin{aligned} \rho(M, F_2) &= \begin{cases} \frac{c}{a}x - a & \text{for } x > 0, \\ -\frac{c}{a}x + a & \text{for } x < 0, \end{cases} \\ \rho(M, F_1) &= \begin{cases} \frac{c}{a}x + a & \text{for } x > 0, \\ -\frac{c}{a}x - a & \text{for } x < 0. \end{cases} \end{aligned}$$

Finally

$$|\rho(M, F_1) - \rho(M, F_2)| = 2a.$$

Consider the part of the hyperbola that is in the first quarter. For that part $x \geq a$ and $y \geq 0$, and equation (96.8) is equivalent to

$$y = \frac{b}{a} \sqrt{x^2 - a^2},$$

assuming of course $b > 0$ and $a > 0$. It is easy to see that this function can be represented in the following form:

$$y = \frac{b}{a}x - \frac{ba}{x + \sqrt{x^2 - a^2}}. \quad (96.9)$$

Along with function (96.9) consider the equation of a straight line

$$y' = \frac{b}{a} x. \quad (96.10)$$

Let $M(x, y)$ and $M'(x, y')$ denote a point of hyperbola (96.9) and that of the straight line (96.10) having the same abscissa x . With x increasing without limit, the difference

$$y' - y = \frac{ba}{x + \sqrt{x^2 - a^2}},$$

remaining positive, is monotonically decreasing and vanishing. Hence M and M' converge but M always remains below M' .

A similar property holds for the other parts of the hyperbola too, the role of the straight line (96.10) being played by one of the straight lines

$$y = \frac{b}{a} x, \quad y = -\frac{b}{a} x. \quad (96.11)$$

These are called the *asymptotes* of the hyperbola.

Notice that we said that (96.8) was the equation of a hyperbola. However, another equation also called the *equation* of a hyperbola is known from the school course.

According to (96.11) we make a change of coordinates

$$x' = \frac{x}{a} - \frac{y}{b},$$

$$y' = \frac{x}{a} + \frac{y}{b}.$$

From (96.8) we have

$$\left(\frac{x}{a} - \frac{y}{b}\right) \left(\frac{x}{a} + \frac{y}{b}\right) = 1.$$

Hence in the new coordinate system (not rectangular, in general) the equation of a hyperbola has the form

$$x'y' = 1 \quad (96.12)$$

or

$$y' = \frac{1}{x'}.$$

This is just the familiar school-book equation. Equation (96.12) is called the equation of a hyperbola in its asymptotes.

1.5. The number a_0 is zero; the numbers λ_1 and λ_2 are opposite in sign. After a standard change of coefficients we get

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \quad (96.13)$$

equivalent to (96.1). From (96.13) we get

$$\left(\frac{x}{a} - \frac{y}{b}\right) \left(\frac{x}{a} + \frac{y}{b}\right) = 0$$

or

$$y = \frac{b}{a}x, \quad y = -\frac{b}{a}x. \quad (96.14)$$

Thus (96.13) is the equation of a curve splitting into *two intersecting straight lines* (96.14).

Consider now the second equation of (95.13). It has the form

$$\lambda_2 y^2 + b_0 x = 0. \quad (96.15)$$

II.6. Both numbers λ_2 and b_0 are non-zero. Let

$$2p = -\frac{b_0}{\lambda_2} \neq 0.$$

Now (96.15) is equivalent to the following equation:

$$y^2 = 2px. \quad (96.16)$$

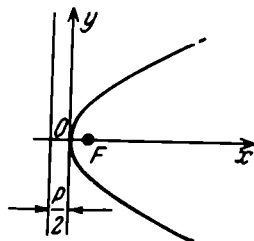


Fig. 96.3

The curve described by this equation is called a *parabola* (Fig. 96.3) and the equation is called the *canonical equation* of a parabola. It may be assumed without loss of generality that $p > 0$, since for $p < 0$ we obtain a curve symmetric with respect to the y axis. Like the hyperbola, the parabola is an unbounded curve. It has only one axis of symmetry, the x axis, and no centre of symmetry. The point of intersection of the axis of the parabola with the parabola itself is called the *vertex* of the parabola. The point F with coordinates $(p/2, 0)$ is called the *focus* of the parabola. A straight line L given by

$$x = -\frac{p}{2} \quad (96.17)$$

is called the *directrix* of the parabola.

Theorem 96.3. *The distance from any point of a parabola to its directrix is equal to that from the same point to the focus of the parabola.*

Proof. We have for any point $M(x, y)$ of a parabola

$$\rho(L, M) = x + \frac{p}{2},$$

and further

$$\begin{aligned} \rho(F, M) &= \left(\left(x - \frac{p}{2}\right)^2 + y^2 \right)^{1/2} = \left(\left(x - \frac{p}{2}\right)^2 + 2px \right)^{1/2} \\ &= \left(x^2 - px + \frac{p^2}{4} + 2px \right)^{1/2} = \left(x^2 + px + \frac{p^2}{4} \right)^{1/2} \\ &= \left(\left(x + \frac{p}{2}\right)^2 \right)^{1/2} = x + \frac{p}{2}, \end{aligned}$$

since $x \geq 0$ and $p > 0$.

Finally, consider the third equation of (95.13). It has a quite simple form:

$$\lambda_1 x^2 + a_0 = 0. \quad (96.18)$$

III.7. *The number a_0 is not zero; the number λ_1 is opposite in sign to a_0 .* Let

$$a^2 = -\frac{a_0}{\lambda_1}.$$

Then the equation of a curve (96.18) is equivalent to

$$x^2 - a^2 = 0 \quad (96.19)$$

or

$$x = a, \quad x = -a. \quad (96.20)$$

Hence the equation of a curve (96.19) is the equation of a curve splitting into *two parallel lines* (96.20).

III.8. *The number a_0 is not zero; the number λ_1 coincides in sign with a_0 .* Let

$$a^2 = \frac{a_0}{\lambda_1}.$$

Then (96.18) is equivalent to

$$x^2 + a^2 = 0. \quad (96.21)$$

It is clear that there is no point in the plane whose coordinates satisfy that equation. It is usual to say that (96.21) is the equation of *two imaginary straight lines*.

III.9. *The number a_0 is zero.* In this case (96.18) is equivalent to

$$x^2 = 0. \quad (96.22)$$

By analogy with (96.19) it is usual to say that (96.22) defines *two coincident straight lines*, each defined by

$$x = 0.$$

Notice that for all points of an ellipse or a hyperbola we have the following equations:

$$\begin{aligned} \rho(M, F_2) &= \frac{c}{a} \left| x - \frac{a^2}{c} \right|, \\ \rho(M, F_1) &= \frac{c}{a} \left| x + \frac{a^2}{c} \right|. \end{aligned} \quad (96.23)$$

Straight lines α_i ($i = 1, 2$) defined by

$$x - \frac{a^2}{c} = 0, \quad x + \frac{a^2}{c} = 0 \quad (96.24)$$

are called the *directrices* of the ellipse and of the hyperbola. We shall label the directrix and the focus with the same index if they

are in the same half-plane given by the y axis. We can now show that:

The ratio of distances $\rho(M, F_1)$ and $\rho(M, \alpha_1)$ is constant for all points M of an ellipse, a hyperbola and a parabola.

For the parabola this statement follows from Theorem 96.3. For the ellipse and the hyperbola it follows from (96.23) and (96.24). The ratio

$$e = \frac{\rho(M, F_1)}{\rho(M, \alpha_1)}$$

is called the *eccentricity*. We have:

$$e = \frac{c}{a} = \left(1 - \frac{b^2}{a^2}\right)^{1/2} < 1$$

for the ellipse,

$$e = \frac{c}{a} = \left(1 + \frac{b^2}{a^2}\right)^{1/2} > 1$$

for the hyperbola,

$$e = 1$$

for the parabola.

Exercises

1. What is a diametrical hyperplane conjugate to a given direction for second-degree curves?

2. Write the equation of a tangent for the ellipse, the hyperbola and the parabola.

3. Prove that a light ray issuing from one focus of an ellipse passes, after a mirror reflection from a tangent, through the other focus.

4. Prove that a light ray issuing from the focus of a parabola passes, after a mirror reflection from a tangent, parallel to the axis of the parabola.

5. Prove that a light ray issuing from one focus of a hyperbola, after a mirror reflection from a tangent, appears to issue from the other focus.

97. Second-degree surfaces

We now proceed to study second-degree surfaces given as equations (95.14). We first consider the equation

$$\lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2 + a_0 = 0. \quad (97.1)$$

I.1. *The number a_0 is not zero; the numbers λ_1 , λ_2 and λ_3 are all opposite in sign to a_0 . A standard change of coefficients yields*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1. \quad (97.2)$$

The surface described by this equation is called an *ellipsoid* (Fig. 97.1) and equation (97.2) is called the *canonical equation* of an ellipsoid. It follows from (97.2) that the coordinate planes are *planes*

of symmetry and that the origin is a *centre of symmetry*. The numbers a , b and c are the *semiaxes* of the ellipsoid. An ellipsoid is a bounded surface contained in a parallelepiped $|x| \leq a$, $|y| \leq b$, $|z| \leq c$. The curve of intersection of the ellipsoid with any plane is an ellipse. Indeed, such a curve of intersection is a second-degree curve. By virtue of the boundedness of the ellipsoid that curve is bounded but the only bounded second-degree curve is the ellipse.

I.2. The number a_0 is not zero; the numbers λ_1 , λ_2 , λ_3 and a_0 are all of the same sign. A standard change of coefficients yields

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1. \quad (97.3)$$

There is no point in space whose coordinates satisfy this equation. Equation (97.3) is said to be the equation of an *imaginary ellipsoid*.

I.3. The number a_0 is zero; the numbers λ_1 , λ_2 and λ_3 are all of the same sign. We have

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0. \quad (97.4)$$

This equation holds only for the origin. Equation (97.4) is said to be the equation of a *degenerate ellipsoid*.

I.4. The number a_0 is not zero; the numbers λ_1 and λ_2 have the same sign opposite to that of λ_3 and a_0 . A standard change of coefficients yields

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1. \quad (97.5)$$

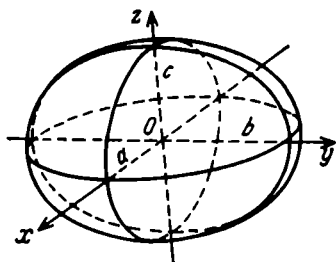


Fig. 97.1

The surface described by this equation is called a *hyperboloid of one sheet* (Fig. 97.2) and the equation is the *canonical equation* of a hyperboloid of one sheet. It follows from (97.5) that the coordinate planes are *planes of symmetry* and the origin is a *centre of symmetry*. Consider a curve L_n of intersection of the one-sheeted hyperboloid

with planes $z = h$. The equation of a projection of such a curve onto the x, y plane is obtained from (97.5) if we assume $z = h$ in it. It is easy to see that that curve is an ellipse

$$\frac{x^2}{a^{*2}} + \frac{y^2}{b^{*2}} = 1,$$

where

$$a^* = a \sqrt{1 + h^2/c^2} \quad \text{and} \quad b^* = b \sqrt{1 + h^2/c^2},$$

its size increasing without limit for $h \rightarrow +\infty$. Sections of the one-sheeted hyperboloid by the y, z and x, z planes are hyperbolas.

Thus the hyperboloid of one sheet is a surface consisting of one sheet and resembling a tube extending without limit in the positive and negative directions of the z axis.

I.5. The number a_0 is not zero; the numbers λ_1 , λ_2 and a_0 are opposite in sign to λ_3 . Similarly to (97.5) we have

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1. \quad (97.6)$$

The surface described by this equation is called a *hyperboloid of two sheets* (Fig. 97.3) and the equation is the *canonical equation* of a hyperboloid of two sheets. The coordinate planes are *planes of symmetry* and the origin is a *centre of symmetry*. Curves L_h of intersection of the two-sheeted hyperboloid with planes $z = h$ are ellipses the equations of whose projections onto the x, y plane have the form

$$\frac{x^2}{a^{*2}} + \frac{y^2}{b^{*2}} = 1,$$

where $a^* = a\sqrt{-1 + h^2/c^2}$ and $b^* = b\sqrt{-1 + h^2/c^2}$. It follows that the cutting plane $z = h$ begins to intersect the two-sheeted hyperboloid only when $|h| \geq c$. In the region between the planes $z = -c$ and $z = +c$ there are no points of the surface under consideration. By virtue of symmetry with respect to the x, y plane the surface consists of two sheets lying outside that region. The sections of the two-sheeted hyperboloid by the y, z and x, z planes are hyperbolas.

I.6. The number a_0 is zero; the numbers λ_1 and λ_2 are opposite in sign to λ_3 . We have

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0. \quad (97.7)$$

The surface defined by this equation is called an *elliptic cone* (Fig. 97.4) and the equation is the *canonical equation* of an elliptic cone. The coordinate planes are *planes of symmetry* and the origin is a *centre of symmetry*. Curves L_h of intersection of the elliptic cone with planes $z = h$ are ellipses. If a point $M(x_0, y_0, z_0)$ is on the surface of the cone, then (97.7) is satisfied by the coordinates of a point $M_t(tx_0, ty_0, tz_0)$ for any number t . Hence the entire straight line passing through M_0 and the origin is *entirely* on the given surface.

We now proceed to consider the second equation of (95.14). We have

$$\lambda_1 x^2 + \lambda_2 y^2 + b_0 z = 0.$$

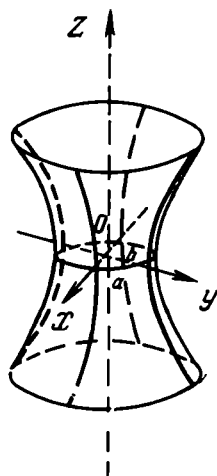


Fig. 97.2

II.7. The numbers λ_1 and λ_2 are of the same sign. It may be assumed without loss of generality that b_0 is opposite in sign, since if b_0 coincides in sign with λ_1 and λ_2 , we obtain a surface symmetric with

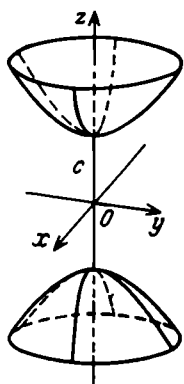


Fig. 97.3

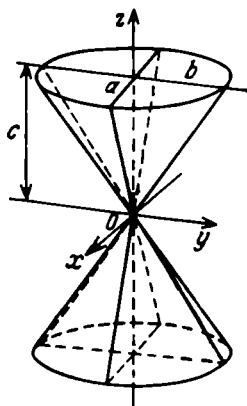


Fig. 97.4

respect to the x, y plane. A standard change of coefficients yields

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}. \quad (97.8)$$

The surface described by this equation is called an *elliptic paraboloid* (Fig. 97.5) and the equation is the *canonical equation* of an elliptic paraboloid. For this surface the x, z and y, z planes are *planes of symmetry* and there is no centre of symmetry. The elliptic paraboloid lies in the half-space $z \geq 0$. Curves L_h of intersection of the elliptic paraboloid with planes $z = h$, $h > 0$, are ellipses whose projections onto the x, y plane are defined by

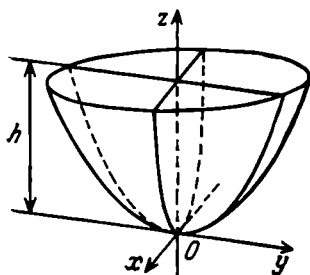


Fig. 97.5

$$\frac{x^2}{a^{*2}} + \frac{y^2}{b^{*2}} = 1,$$

where $a^* = a\sqrt{h}$ and $b^* = b\sqrt{h}$. It follows that as h increases the ellipses increase without limit, i.e. an elliptic paraboloid is an infinite cup.

Sections of an elliptic paraboloid by the planes $y = h$ and $x = h$ are parabolas. For example, the plane $x = h$ intersects the surface in the parabola

$$z - \frac{h^2}{a^2} = \frac{y^2}{b^2}$$

lying in the plane $x = h$

II.8. The numbers λ_1 and λ_2 are of opposite signs. A typical surface for this case is defined by the equation

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

The surface described by this equation is called a *hyperbolic paraboloid* (Fig. 97.6) and the equation is the *canonical equation* of a hyperbolic paraboloid. The x, z and y, z planes are *planes of symmetry* and there is no centre of symmetry. Curves of intersection of the hyperbolic paraboloid with planes $z = h$ are hyperbolas

$$\frac{x^2}{a^{*2}} - \frac{y^2}{b^{*2}} = 1,$$

where $a^* = a\sqrt{h}$ and $b^* = b\sqrt{h}$, for $h > 0$, and hyperbolas

$$-\frac{x^2}{a^{*2}} + \frac{y^2}{b^{*2}} = 1,$$

where $a^* = a\sqrt{-h}$ and $b^* = b\sqrt{-h}$, for $h < 0$. The plane $z = 0$

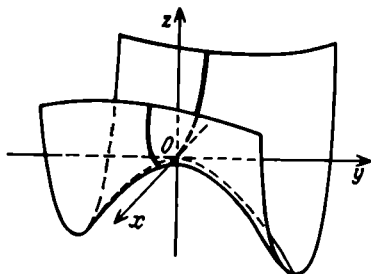


Fig. 97.6

intersects the hyperbolic paraboloid in two straight lines

$$y = \pm \frac{b}{a} x.$$

All surfaces defined by Equations III to V of (95.14) are independent of z . Therefore projections onto the x, y plane of curves of intersection of those surfaces with planes $z = h$ are also independent of h . Such surfaces are called *cylinders* with added adjectives: *elliptic*, *hyperbolic*, etc. depending on the form of projection of the surface onto the x, y plane.

Theorem 97.1. *There are two distinct straight lines through each point of a one-sheeted hyperboloid and a hyperbolic paraboloid, lying entirely on those surfaces.*

Proof. Consider a hyperboloid of one sheet given by its canonical equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1. \quad (97.10)$$

For any α and β not both zero a pair of planes

$$\alpha \left(\frac{x}{a} + \frac{z}{c} \right) = \beta \left(1 - \frac{y}{b} \right), \quad \beta \left(\frac{x}{a} - \frac{z}{c} \right) = \alpha \left(1 + \frac{y}{b} \right) \quad (97.11)$$

determines some straight line Γ . It is easy to verify that the given straight line Γ lies entirely on surface (97.10). Moreover, there is one straight line of family Γ through each point of that surface.

Indeed, look at (97.11) as a system of two equations

$$\alpha \left(\frac{x}{a} + \frac{z}{c} \right) - \beta \left(1 - \frac{y}{b} \right) = 0,$$

$$\alpha \left(1 + \frac{y}{b} \right) - \beta \left(\frac{x}{a} - \frac{z}{c} \right) = 0$$

in α and β . The determinant of the system is zero if and only if a point $M(x, y, z)$ is on hyperbola (97.10). The rank of the matrix of the system is obviously equal to unity. Hence α and β are determined up to proportionality. But this precisely means that there is a unique straight line Γ through each point of a hyperboloid.

Similarly we can see that through each point of a hyperboloid there is a unique straight line Γ^* determined by the planes

$$v \left(\frac{x}{a} + \frac{z}{c} \right) = \lambda \left(1 + \frac{y}{b} \right), \quad \lambda \left(\frac{x}{a} - \frac{z}{c} \right) = v \left(1 - \frac{y}{b} \right).$$

The straight lines Γ and Γ^* are distinct. The same reasoning shows that the hyperbolic paraboloid

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

is covered by two distinct families of straight lines Π and Π^* determined by the planes

$$\alpha z = \beta \left(\frac{x}{a} + \frac{y}{b} \right), \quad \beta = \alpha \left(\frac{x}{a} - \frac{y}{b} \right)$$

and

$$vz = \lambda \left(\frac{x}{a} - \frac{y}{b} \right), \quad \lambda = v \left(\frac{x}{a} + \frac{y}{b} \right).$$

Exercises

1. What is a diametrical hyperplane conjugate to a given direction for second-degree surfaces?
2. Write the equations of a tangential plane for the various second-degree surfaces.
3. Investigate the optical properties of second-degree surfaces.

Bilinear Metric Spaces

98. The Gram matrix and determinant

Let $\varphi(x, y)$ be some bilinear form introduced in a vector space K_n over a number field P . The space K_n is said to be *bilinear metric* if each pair of vectors x and y from K_n is assigned a number (x, y) from P called a scalar product, with

$$(x, y) = \varphi(x, y).$$

If a bilinear form in a complex space K_n is Hermitian, then K_n is said to be *Hermitian bilinear metric*. In these cases we shall also say that a bilinear metric is introduced in the vector space.

Some similarity can be seen between bilinear metric spaces and Euclidean and unitary spaces considered earlier. It should be stressed from the outset, however, that there are significant differences. Comparing the definitions of a scalar product in Euclidean and unitary spaces with that of a bilinear form it is not hard to observe that in bilinear metric spaces a scalar product may in general not be symmetric and positive definite.

The study of Euclidean and unitary spaces reduced to the investigation of additional properties of both the spaces and operators in them arising from bilinear forms defining scalar products. The same problem faces the study of bilinear metric spaces. The necessity of introducing a weaker definition of a scalar product results from the fact that it is by far not always that bilinear functions studied simultaneously with vectors of a space and operators possess the symmetry property, to say nothing of that of positive definiteness.

Many definitions and facts will be the same both for ordinary bilinear spaces and for Hermitian bilinear metric spaces. Where no confusion arises, the word "Hermitian" will therefore be dropped and the appropriate calculations will be performed only for bilinear spaces, tacitly assuming that for Hermitian spaces they are performed in a similar way.

The main method of investigating any vector space is to expand a vector with respect to a given system of vectors and to study the expansion depending on various factors. In a general vector space there is no tool using which we could find an expansion but in a bilinear metric space the scalar product turns out to be such a tool.

system (98.2) must have a nonzero solution. Hence the determinant of the matrix of that system, i.e. the Gramian of the system $x_1, x_2, \dots, x_m$, is zero.

Theorem 98.1. *If a quadratic form (x, x) has no isotropic vectors, then the Gramian is not zero if and only if its system of vectors is linearly independent.*

Proof. Necessity. Let the Gramian of a system of vectors $x_1, x_2, \dots, x_m$ be nonzero. Assuming that that system is linearly dependent, the Gramian must be zero by Lemma 98.2, which is impossible under the hypothesis.

Sufficiency. Suppose the system of vectors is linearly independent. If the Gramian is zero, then according to the corollary of Lemma 98.1 there must be an isotropic vector. Since this is impossible under the hypothesis, the Gramian is not zero.

Corollary. *If a quadratic form (x, x) is strictly of constant signs, then the Gramian is zero if and only if the system of vectors is linearly dependent.*

Corollary. *If a bilinear form (x, y) is symmetric and a quadratic form (x, x) is strictly of constant signs, then for any two vectors x and y we have the Cauchy-Buniakowski-Schwarz inequality*

$$|(x, y)|^2 \leq (x, x)(y, y), \quad (98.5)$$

equality holding if and only if x and y are linearly dependent.

Under the hypotheses of this statement the Gramian will be positive for linearly independent vectors x and y , according to Sylvester's criterion or its corollary, and zero for linearly dependent vectors, according to Lemma 98.2. In both cases inequality (98.5) holds. If, however, equality holds in (98.5), then x and y will be linearly dependent according to the preceding corollary, since their Gramian is zero.

Consider the following simple but sufficiently important properties of the Gramian. They not only lead to numerous consequences but also not infrequently allow a clear-cut geometrical interpretation of them to be given.

Property 1. *The Gramian remains unaffected by an interchange of any two vectors in a system $x_1, x_2, \dots, x_m$.*

Indeed, if any two vectors x_i and x_j are interchanged in $x_1, x_2, \dots, x_m$, then so are the i th and j th columns and the i th and j th rows in the Gramian. The Gramian changes sign twice, i.e. as a result it will remain unchanged.

Property 2. *The Gramian remains unaffected by addition to any vectors of a system $x_1, x_2, \dots, x_m$ of any linear combination of the remaining vectors.*

Obviously it suffices to consider the case where the vector x_1 is changed, since the other cases reduce to that case in view of Property 1. Let a vector $\alpha_2 x_2 + \dots + \alpha_m x_m$ be added to x_1 . Suppose the

bilinear form (x, y) is ordinary. It is easy to verify that the new Gramian is obtained from the old Gramian by adding to the first row the second row multiplied by α_2 , etc., up to the last row multiplied by α_m and adding to the first column the second column multiplied by α_2 , etc., up to the last column multiplied by α_m . As is known, this leaves the Gramian unaffected. If the bilinear form (x, y) is Hermitian, then the columns are multiplied by $\bar{\alpha}_2, \dots, \bar{\alpha}_m$.

Property 3. *If any vector of a system $x_1, x_2, \dots, x_m$ is multiplied by a number α , then the Gramian is multiplied by α^2 , if the bilinear form (x, y) is ordinary, and by $|\alpha|^2$, if (x, y) is Hermitian.*

Again it is sufficient to consider the case where x_1 is changed. But multiplying x_1 by α leads to the multiplication of the first row and the first column of the Gramian by α in the case of the ordinary bilinear form (x, y) . If, however, (x, y) is Hermitian, then the first row of the Gramian is multiplied by α and the first column is multiplied by $\bar{\alpha}$. It is from this that the property follows.

Property 4. *If each of the vectors $x_1, x_2, \dots, x_m$ is left (right) orthogonal to all the preceding vectors, then for the Gramian we have*

$$G(x_1, x_2, \dots, x_m) = \prod_{i=1}^m (x_i, x_i). \quad (98.6)$$

Indeed, the left (right) orthogonality of each vector of the system $x_1, x_2, \dots, x_m$ to all the preceding vectors results in the Gram matrix being right (left) triangular. But the determinant of a triangular matrix is equal to the product of its diagonal elements, whence (98.6).

Especially interesting properties of the Gram matrix and the Gramian arise in the cases where the bilinear form (x, y) is real symmetric or Hermitian-symmetric, and positive definite. Of course these cases imply simply that a bilinear metric space K_n is in fact Euclidean or, respectively, unitary.

In a Euclidean and a unitary space the Gram matrix for any basis system is the matrix of a positive definite quadratic form (x, x) . According to Sylvester's criterion all principal minors of the Gram matrix are positive. Since any linearly independent system of vectors can be supplemented to a basis, it follows that we have

Lemma 98.3. *In a Euclidean and a unitary space the Gramian for any linearly independent system of vectors is positive.*

In a Euclidean space the Gramian has a very simple geometrical interpretation. This is stated by

Theorem 98.2. *In a Euclidean space the Gramian $G(x_1, x_2, \dots, x_m)$ of a system of vectors $x_1, x_2, \dots, x_m$ equals the square of the volume $V^2(x_1, x_2, \dots, x_m)$ of that system.*

Proof. Consider a real-valued function $G^{1/2}(x_1, x_2, \dots, x_m)$ of m independent vector variables $x_1, x_2, \dots, x_m$. It satisfies Properties A and B of (36.3) according to Properties 2 and 3 of the Gramian.

In a Euclidean space each vector of any orthonormal system of vectors is orthogonal to all the preceding vectors of the system. According to (98.6) therefore $G^{1/2}(x_1, x_2, \dots, x_m)$ satisfies Property C of (36.3) as well. But it now follows from Theorem 36.1 that $G^{1/2}(x_1, x_2, \dots, x_m)$ coincides with the volume of the system of vectors.

Corollary. For any system of vectors $x_1, x_2, \dots, x_m$ of a Euclidean space

$$0 \leq G(x_1, x_2, \dots, x_m) \leq \prod_{i=1}^m (x_i, x_i),$$

the equation at the left holding if and only if the system of vectors is linearly dependent and the equation at the right holding if and only if either the system of vectors is orthogonal or there is a zero vector in it.

The validity of the statement follows from the first corollary of Theorem 98.1 and from the property of the volume of a system of vectors described by Hadamard's inequality (36.1).

Corollary. For any system of vectors $x_1, x_2, \dots, x_m$ of a Euclidean space

$G(x_1, \dots, x_l, x_{l+1}, \dots, x_m) \leq G(x_1, \dots, x_l) \cdot G(x_{l+1}, \dots, x_m)$, equality holding if and only if either the sets of vectors $x_1, \dots, x_l$ and $x_{l+1}, \dots, x_m$ are orthogonal or one of them is a linearly dependent system.

The proof is based on a simple analysis of formula (35.4). Recall only the following. If $L_1 \subseteq L_2$, where L_1 and L_2 are any subspaces, then $|\text{ort}_{L_2} x| \leq |\text{ort}_{L_1} x|$ for any vector x , equality holding only if $x \perp L_2$.

Exercises

1. Are the problems of finding expansions (98.1) and of solving system (98.2) equivalent?
2. What yields the solution of system (98.2) if the vector x is not in the span of vectors $x_1, \dots, x_m$?
3. What does the Gram matrix (98.3) look like if:
the vectors $x_1, \dots, x_m$ are mutually orthogonal,
each of the vectors $x_1, \dots, x_m$ is left (right) orthogonal to all the preceding (subsequent) vectors,
each of the vectors $x_1, \dots, x_m$ is left (right) orthogonal to all the subsequent (preceding) vectors,
each of the vectors $x_{l+1}, \dots, x_m$ is left (right) orthogonal to each of the vectors $x_1, \dots, x_l$?
4. How does the Gram matrix change under elementary transformations of a system of vectors?
5. Prove that if in an ordinary bilinear metric space $(x, y) = 0$ always implies $(y, x) = 0$, then the scalar product is given by either a symmetric or a skew-symmetric bilinear form.
6. Is the statement of Exercise 5 true for a Hermitian bilinear metric space?

7. Let G be the Gram matrix for some basis in a nonsingular Hermitian bilinear metric space K_n . Prove that for an operator U with matrix $G^{-1}G'$ in the same basis for each vector $x \in K_n$

$$(Ux, Ux) = (x, x).$$

8. Prove that for any linear operator A in a Euclidean or a unitary space K_n the ratio

$$k(A) = \frac{G(Ax_1, \dots, Ax_m)}{G(x_1, \dots, x_m)}$$

is independent of the vectors $x_1, \dots, x_m$ and equals the product of the squares of the moduli of the eigenvalues of A .

9. Prove that for any linearly independent system of vectors $x_1, \dots, x_m$ of a Euclidean or a unitary space and any vector z

$$\frac{G(x_1, \dots, x_m, z)}{G(x_1, \dots, x_m)} \leq \frac{G(x_1, \dots, x_{m-1}, z)}{G(x_1, \dots, x_{m-1})}.$$

99. Nonsingular subspaces

Any linear subspace L of K_n can be regarded as a bilinear metric space relative to the same scalar product that is introduced in K_n . In general the nonsingularity of K_n does not imply that of L and vice versa.

Theorem 99.1. *For all subspaces of K_n to be nonsingular it is necessary and sufficient that the quadratic form (x, x) should have no isotropic vectors.*

Proof. Necessity. Let all subspaces in K_n be nonsingular. Then so are all one-dimensional subspaces. But the Gram matrices for nonzero vectors x coincide with the scalar product (x, x) which must be nonzero by virtue of the nonsingularity of one-dimensional subspaces.

Sufficiency. Suppose $(x, x) \neq 0$ for every $x \neq 0$. Consider any subspace L and a basis $x_1, x_2, \dots, x_m$ in it. By Theorem 98.1 the Gramian for this system is nonzero, i.e. L is nonsingular.

Corollary. *For all subspaces in a bilinear metric space K_n to be nonsingular it is necessary and sufficient that so should all of its one-dimensional subspaces.*

Corollary. *In any ordinary complex bilinear metric space there are singular one-dimensional subspaces.*

To prove this statement it suffices to recall that in an ordinary complex bilinear metric space any quadratic form has isotropic vectors.

When a quadratic form has isotropic vectors, there are both singular and nonsingular subspaces in the bilinear metric space. If a bilinear form (x, y) has a rank r , then it is clear that there can be no nonsingular subspaces of dimension greater than r . But nonsingular subspaces of dimension r exist. Such, for example, is the subspace spanned by those vectors of the canonical basis for which the Gram matrix coincides with the matrix M of (92.5).

We shall say that a set of vectors F of a bilinear metric space K_n is *right*, *left orthogonal* or simply *orthogonal* to a set of vectors G of K_n if for every pair of vectors x and y , where $x \in F$ and $y \in G$, we have a similar orthogonality relation. It is clear that the set of all the vectors of K_n right (left) orthogonal to each vector of F is a subspace. It is called the *right (left) orthogonal complement* of F and designated $F^\perp$ (${}^\perp F$).

In a Euclidean and a unitary space, subspaces ${}^\perp K_n$ and $K_n^\perp$ coincide and consist only of a zero vector. In bilinear metric spaces they may be distinct and do not necessarily consist only of a zero vector. The subspaces ${}^\perp K_n$ and $K_n^\perp$ are called respectively the *left* and *right null* subspaces in K_n .

Observe that for any set of vectors F there are always inclusions $K_n^\perp \subseteq F^\perp$ and ${}^\perp K_n \subseteq {}^\perp F$ and that for any vectors for ${}^\perp K_n$ or $K_n^\perp$ the Gram matrices turn out to be zero.

Theorem 99.2. *The dimensions of the left and right null subspaces are the same and equal the nullity of a bilinear form (x, y) .*

Proof. Choose in K_n some basis $x_1, x_2, \dots, x_m$. Take a vector x of $K_n^\perp$ and represent it as an expansion with respect to the basis according to (98.1). The condition that x should be in $K_n^\perp$ is equivalent to the conditions of the right orthogonality of x to each vector of the basis. But these conditions lead to a solution of a homogeneous system of the type (98.2) for determining the coefficients of the expansion. It is known (see Section 48) that the set of solutions of that system is a subspace whose dimension is equal to the nullity of the Gram matrix or equivalently to the nullity of the bilinear form (x, y) . The proof for the left null subspace is similar.

Corollary. *For K_n to be nonsingular it is necessary and sufficient that the right and left null subspaces should consist only of a zero vector.*

In Euclidean and unitary spaces any subspace is orthogonal to its orthogonal complement and determines decomposition of the entire space not only into a direct, but even into an orthogonal, sum of those subspaces. Similar facts do not always hold in bilinear metric spaces.

Theorem 99.3. *Let L be a subspace in K_n . For a decomposition*

$$K_n = L \dot{+} L^\perp = L^\perp \dot{+} L \quad (99.1)$$

to exist it is necessary and sufficient that L should be nonsingular.

Proof. **Necessity.** Suppose that decompositions (99.1) hold. We shall look at L as a bilinear metric space with the same scalar product as in K_n . The intersection $L \cap L^\perp$ is the right null subspace in L . Since sums (99.1) are direct, that subspace contains only a zero vector. According to the corollary of Theorem 99.2 this means that L is nonsingular.

Sufficiency. If L is nonsingular, then $L \cap L^\perp$ will contain only a zero vector and it is necessary to show that any vector $x \in K_n$ can

be represented as $x = u \div v$, where $u \in L$ and $v \in L^\perp$. Take some basis $x_1, x_2, \dots, x_m$ in L . For the desired decomposition $x = u \div v$ to exist it is necessary and sufficient that there should be a vector u in L such that $x - u$ is right orthogonal to the vectors $x_1, x_2, \dots, x_m$. Again we obtain a system of linear algebraic equations with a Gram matrix to determine the coefficients of the expansion of u with respect to $x_1, x_2, \dots, x_m$. That matrix is nonsingular and the system has a solution, i.e. the vector u exists. Of course all that has been said concerning $L^\perp$ carries over completely to ${}^\perp L$.

Corollary. *If a nonsingular subspace L is of dimension m , then the dimension of $L^\perp$ and ${}^\perp L$ is $n - m$.*

To prove this it suffices to use equation (19.1) and recall that the dimension of $L \cap L^\perp$ and $L \cap {}^\perp L$ is zero.

Corollary. *If a nonsingular subspace L has a maximum dimension, then $L^\perp = K_n^\perp$ and ${}^\perp L = {}^\perp K_n$.*

Indeed, let the rank of the bilinear form (x, y) be r . As we have already noted, the subspace L will be of dimension r and $L^\perp$ and ${}^\perp L$ will be of dimension $n - r$. But $K_n^\perp$ and ${}^\perp K_n$ are also of dimension $n - r$ and, moreover, $K_n^\perp \subseteq L^\perp$ and ${}^\perp K_n \subseteq {}^\perp L$. Therefore $K_n^\perp = L^\perp$ and ${}^\perp K_n = {}^\perp L$.

As to decompositions of the type (99.1) into orthogonal sums, Theorem 99.3 yields

Corollary. *Let L be a nonsingular subspace of maximum dimension. Decompositions (99.1) will be orthogonal if and only if the left and right null subspaces coincide.*

Indeed, if decompositions (99.1) are orthogonal, then $L^\perp$ is not only right orthogonal but also left orthogonal to L , i.e. $L^\perp \subseteq {}^\perp L$. Similarly we have ${}^\perp L \subseteq L^\perp$. Hence $L^\perp = {}^\perp L$. Since L is of maximum dimension, this means that $K_n^\perp = {}^\perp K_n$. If, however, the null subspaces coincide, then it follows that $L^\perp = {}^\perp L$, i.e. that $L^\perp$ and ${}^\perp L$ are both right and left orthogonal to L , and that decompositions (98.5) are orthogonal.

We can now give the answer to the question: What is the connection between expansion (98.1) and the solution of system (98.2)? Let vectors $x_1, \dots, x_m$ form a basis of a nonsingular subspace L . By Theorem 99.3 there are direct decompositions (99.1). Therefore each vector x of a bilinear metric space K_n can be uniquely represented as $x = u \div v$, where $u \in L$ and $v \in {}^\perp L$. Recall that a vector u is the projection of a vector x onto L parallel to the subspace ${}^\perp L$. If we solve the system of linear algebraic equations (98.2) and compose a vector

$$u = \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_m x_m, \quad (99.2)$$

then it is that vector that will be the projection of x onto L parallel to ${}^\perp L$. Indeed, u is in L and by (98.2) the difference $x - u$ is left orthogonal to the vectors $x_1, \dots, x_m$. Hence $x - u$ is in ${}^\perp L$. It is

clear that in order to obtain the projection of x onto L parallel to $L^\perp$ it is necessary to solve the following system:

$$\begin{aligned} \alpha_1(x_1, x_1) + \alpha_2(x_1, x_2) + \dots + \alpha_m(x_1, x_m) &= (x_1, x), \\ \alpha_1(x_2, x_1) + \alpha_2(x_2, x_2) + \dots + \alpha_m(x_2, x_m) &= (x_2, x), \\ &\vdots \\ \alpha_1(x_m, x_1) + \alpha_2(x_m, x_2) + \dots + \alpha_m(x_m, x_m) &= (x_m, x) \end{aligned} \quad (99.3)$$

and then calculate the desired projection according to (99.2). In the case of a Hermitian bilinear metric space the coefficients α_j in (99.2) are replaced by $\bar{\alpha}_j$.

Exercises

1. Describe all nonsingular subspaces of maximum dimension.

2. Prove that for any set L there are inclusions

$$L \subseteq {}^\perp(L^\perp), \quad L \subseteq ({}^\perp L)^\perp.$$

In what cases does equality hold in these formulas?

3. Prove that if L is a nonsingular subspace of maximum dimension in a space K_n , then

$${}^\perp(L^\perp) - ({}^\perp L)^\perp = K_n.$$

4. Prove that if a scalar product is given by a symmetric or a skew-symmetric bilinear form, then for any set F we have $F^\perp = -F$.

5. What is the connection between expansion (98.1) and the solution of system (98.2) if the Gram matrix of a system $x_1, x_2, \dots, x_m$ is singular?

6. Can there be a basis made up of isotropic vectors in a nonsingular space?

7. What can be said about the scalar product if the projections onto a fixed subspace L parallel to $-L$ and $L^\perp$ coincide for all vectors?

8. What can be said about the scalar product if the projections of a fixed vector onto the entire subspace L parallel to $-L$ and $L^\perp$ coincide?

9. Let L be a nonsingular subspace of a Hermitian bilinear metric space K_n of rank $r < n$. Prove that the following statements are equivalent:

the subspace ${}^\perp L$ is of dimension $n < r$,

the subspace $L^\perp$ is of dimension $n - r$,

the subspace L is of dimension r ,

the subspaces $L^\perp$ and $K_n^\perp$ coincide,

the subspaces ${}^\perp L$ and ${}^\perp K_n$ coincide,

the subspace ${}^\perp L$ consists of isotropic vectors and a zero vector,

the subspace $L^\perp$ consists of isotropic vectors and a zero vector,

the scalar product on ${}^\perp L$ is zero,

the scalar product on $L^\perp$ is zero.

10. What form will the Gram matrices have for the bases made up of bases of a nonsingular subspace L and the subspace $-L$ ($L^\perp$)?

100. Orthogonality in bases

In bilinear metric spaces the bases are non-equivalent. There are such among them for which systems (98.2) can be solved and studied particularly simply. This happens, for

example, when much of a Gram matrix consists of zero elements. Depending on the form the Gram matrices have we shall consider different classes of bases in bilinear metric spaces.

The simplest matrices are diagonal matrices. Diagonal Gram matrices occur if and only if the bases consist of mutually orthogonal vectors. Such bases will be called *orthogonal*. A system of vectors formed by an orthogonal basis in their span will be called an *orthogonal system*.

Orthogonal bases can be defined in different ways. Definition in terms of mutual orthogonality is not always convenient to check, especially when the vectors of a basis are constructed successively starting from the first. It is sometimes useful therefore to employ the following definition:

A basis $e_1, e_2, \dots, e_n$ is said to be orthogonal if each of its vectors is orthogonal to all the preceding vectors.

A Gram matrix for vectors that satisfy this definition is diagonal, so both definitions are equivalent. In the general case a basis may contain both nonisotropic and isotropic vectors. Vectors of an orthogonal basis can always be interchanged so that nonisotropic vectors are the first and isotropic vectors are the last. The diagonal form of the Gram matrix is preserved of course.

Not all bilinear metric or Hermitian bilinear metric spaces have orthogonal bases. If there is at least one orthogonal basis, then this means that the matrix of the bilinear form (x, y) in the given basis has a diagonal form. Hence the matrix of (x, y) must be congruent with a diagonal matrix in any other basis. The converse is also true of course. Therefore

For an orthogonal basis to exist in a bilinear metric or Hermitian bilinear metric space it is necessary and sufficient that the matrix of the bilinear form (x, y) should be congruent with a diagonal matrix. The set of all orthogonal bases coincides up to an interchange of vectors with the set of canonical bases of (x, y) .

Relying on our earlier studies of bilinear forms we can now say that of the ordinary bilinear metric spaces only those spaces have orthogonal bases in which the basic bilinear form (x, y) is symmetric. As to Hermitian bilinear metric spaces, it is spaces with a Hermitian and skew-Hermitian basic bilinear form (x, y) , as well as those with a bilinear form (x, y) having the real or imaginary part of the quadratic form (x, x) of constant signs, that have orthogonal bases.

Note from the outset one fundamental difference between ordinary and Hermitian bilinear metric spaces with orthogonal bases. In an ordinary bilinear metric space K_n the existence of an orthogonal basis implies the symmetry of the scalar product (x, y) and this in turn ensures the existence of an orthogonal basis in any subspace of K_n . In a Hermitian bilinear metric space, in general the existence

of an orthogonal basis in the space itself does not automatically imply the existence of an orthogonal basis in any of its subspaces. But if a scalar product is given by a Hermitian-symmetric or skew-Hermitian bilinear form, then the consequence is again valid.

Consider any orthogonal basis $e_1, e_2, \dots, e_n$ of a bilinear metric space K_n . There are as many isotropic and as many nonisotropic vectors as are respectively the deficiency and rank of K_n . Taking into account the law of inertia for quadratic forms we conclude that if the bilinear form (x, y) is real symmetric or Hermitian-symmetric, then there will be the same number of vectors with positive and negative values of (e_i, e_i) in any orthogonal basis. Each number is invariant for all orthogonal bases in K_n . Accordingly we shall speak of a positive and a negative index, as well as signature, of spaces with a symmetric form (x, y) . For bilinear metric spaces with a nonsymmetric form (x, y) , we shall speak only of the rank and deficiency of the spaces.

If K_n is a nonsingular space, then no orthogonal basis $e_1, e_2, \dots, e_n$ has isotropic vectors. In this case, for any vector $x \in K_n$

$$x = \sum_{j=1}^n \frac{(x, e_j)}{(e_j, e_j)} e_j. \quad (100.1)$$

Indeed, performing a right scalar multiplication of

$$x = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_n e_n \quad (100.2)$$

successively by the vectors $e_1, e_2, \dots, e_n$ we get

$$\alpha_j = \frac{(x, e_j)}{(e_j, e_j)}$$

for every j . The vectors of an orthogonal basis in a nonsingular space can be normed to yield an *orthonormal basis*. For an orthonormal basis $e_1, e_2, \dots, e_n$ there are relations $|(e_j, e_j)| = 1$ for every j .

In singular spaces, there must be isotropic vectors among the vectors of any basis. Therefore representation (100.1) no longer holds for expansion (100.2) of vectors of a space. Orthogonal bases prove to be sufficiently useful in these spaces too, however. As an illustration of their use, consider

Theorem 100.1. *If there is an orthogonal basis in a space with a scalar product, then the right and left null subspaces coincide.*

Proof. Let there be an orthogonal basis $e_1, e_2, \dots, e_n$ in a space K_n of rank r . It is assumed that the vectors $e_1, \dots, e_r$ are nonisotropic and $e_{r+1}, \dots, e_n$ are isotropic. We take a vector $x \in K_n$ and expand it according to (100.2). Using representation (100.2) and taking into account the orthogonality of the basis and the isotropy of the vectors $e_{r+1}, \dots, e_n$ it is easy to establish that $(x, e_j) = (e_j, x) = 0$ for $r < j \leq n$. Hence $e_{r+1}, \dots, e_n$ are simultaneously in the right and

in the left null subspace. But $e_{r+1}, \dots, e_n$ are linearly independent as vectors of a basis and equal in number the dimension of the null subspaces, so both null subspaces coincide.

Corollary. *If in a bilinear metric space the scalar product is given by a symmetric or Hermitian-symmetric bilinear form, then its right and left null subspaces coincide.*

Corollary. *In any orthogonal basis isotropic vectors, and only isotropic vectors, form the basis of the null subspace in common.*

Corollary. *If in a space with a scalar product there is an orthogonal basis, then that space can be decomposed as an orthogonal sum of any nonsingular subspace of maximum dimension and a null subspace.*

The last corollary actually means that the study of any singular spaces with orthogonal bases reduces to studying separately nonsingular subspaces with orthogonal bases and subspaces on which the scalar product is zero.

To know an orthogonal basis in a space is not only to be able to find an orthogonal basis in the nonsingular subspace of maximum dimension but also to obtain an explicit expansion of the orthogonal projection of any vector onto that subspace with respect to its orthogonal basis. Indeed, let $e_1, e_2, \dots, e_n$ be an orthogonal basis in K_n , let $e_1, \dots, e_r$ be nonisotropic vectors and let $e_{r+1}, \dots, e_n$ be isotropic vectors. Denote by L the subspace spanned by the vectors $e_1, \dots, e_r$. It is clear that it is nonsingular and of maximum dimension, that $L^\perp = {}^\perp L$ and in addition

$$K_n = L \oplus L^\perp.$$

Any vector x in K_n can be represented in a unique way as a sum $x = u + v$, where $u \in L$ and $v \in L^\perp$. Here u is called the left orthogonal projection of x onto a subspace L and v is the left perpendicular to that subspace. We write for x expansion (100.2) with respect to $e_1, e_2, \dots, e_n$. Formula (100.1) no longer holds. Observe, however, that the first r terms in (100.2) form a vector u and the last $n - r$ terms form a vector v . Performing a right scalar multiplication of equation (100.2) successively by $e_1, \dots, e_r$ we get

$$u = \sum_{j=1}^r \frac{(x, e_j)}{(e_j, e_j)} e_j.$$

The projection v of a vector x onto the null subspace is defined very simply

$$v = x - \sum_{j=1}^r \frac{(x, e_j)}{(e_j, e_j)} e_j.$$

The only thing that cannot be done now is to find the expansion of v with respect to the vectors $e_{r+1}, \dots, e_n$ using a scalar product, although the expansion itself does exist.

As we have already said, not all bilinear metric and Hermitian bilinear metric spaces have orthogonal bases. This circumstance leads us to seek other classes of bases, more convenient from the point of view of the scalar product given in space. A solution is suggested by the canonical form of the matrix of a bilinear form.

A basis $e_1, e_2, \dots, e_n$ is said to be *pseudoorthogonal* if each of its vectors is left orthogonal to all the preceding vectors and each of its isotropic vectors is left orthogonal to all vectors of the basis. The system of vectors forming a pseudoorthogonal basis in their span will be called a *pseudoorthogonal system*.

Observe that in this definition the left orthogonality of vectors to all the preceding vectors can be replaced by the right orthogonality of vectors to all the subsequent vectors. This gives the same conditions.

The Gram matrix for the vectors of a pseudoorthogonal basis is *right trapezoidal*. If the vectors of the basis are interchanged so that the nonisotropic vectors are the first vectors and the isotropic vectors are the last vectors, then the Gram matrix not only remains right trapezoidal but will also have the canonical form (92.5). Our earlier studies on reducing the matrix of a bilinear form to canonical form give a complete answer to the question as to when there is a pseudoorthogonal basis. Namely,

There is a pseudoorthogonal basis in any Hermitian bilinear metric space, as well as in any ordinary bilinear metric space, except for spaces with a skew-symmetric bilinear form (x, y) . The set of all pseudoorthogonal bases coincides up to an interchange of vectors with the set of canonical bases of (x, y) .

Every orthogonal basis is pseudoorthogonal. An ordinary bilinear metric space cannot contain simultaneously an orthogonal basis and a pseudoorthogonal basis that is not orthogonal. That is because the existence of at least one orthogonal basis implies the symmetry of all Gram matrices. A right trapezoidal matrix may be symmetric only if it is diagonal. A Hermitian bilinear metric space can contain simultaneously both an orthogonal basis and a pseudoorthogonal basis that is not orthogonal. This means that a right trapezoidal complex matrix may be Hermitian congruent with a diagonal matrix, which is also supported by example (92.8).

If K_n is a nonsingular space, then none of the pseudoorthogonal bases has isotropic vectors, since a right trapezoidal matrix can be nonsingular if and only if it is a right triangular matrix with nonzero diagonal elements. In a nonsingular space, for the coefficients α_j of expansion (100.2) of a vector x with respect to the vectors of a pseudoorthogonal basis $e_1, e_2, \dots, e_n$ we obtain a system of linear algebraic equations with a left triangular matrix. Indeed, multiplying equation (100.2) successively on the right by $e_1, e_2, \dots, e_n$

a pseudoorthogonal basis and subspaces on which the scalar product is zero.

Any vector of K_n can be uniquely represented as a sum $x = u + v$, where $u \in L$, $v \in L^\perp$. If for a vector x we write expansion (100.2), then we again obtain a system of the type (100.3), but now with a left trapezoidal matrix instead of a nonsingular left triangular matrix, to determine the coefficients α_j . Nevertheless we can determine from that system the first coefficients $\alpha_1, \dots, \alpha_r$ and we have

$$u = \alpha_1 e_1 + \alpha_2 e_2 + \dots + \alpha_r e_r,$$

i.e. the projection of x onto L is determined completely relying on the knowledge only of a pseudoorthogonal basis in L . Again $v = x - u$ and again we cannot find an expansion of the vector v with respect to the vectors $e_{r+1}, \dots, e_n$ using a scalar product.

A pseudoorthogonal basis is a sufficiently general type of basis, since almost all spaces have such a basis. As we already know, there is no basis of this type only in ordinary bilinear metric spaces with a skew-symmetric form (x, y) . For these spaces the most convenient type of basis is obvious, it is certainly the canonical basis of Gram matrices. In general it is possible to introduce a type of basis covering all the above types of bases and existing in any space with a scalar product. Few new factors follow, however, and we shall not discuss it now.

• Besides a single basis with certain orthogonality relations between its vectors we shall sometimes deal with pairs of similar bases.

A basis $f_1, f_2, \dots, f_n$ is said to be *left (right) dual* to a basis $e_1, e_2, \dots, e_n$ if $(f_i, e_j) = 0$ ($(e_j, f_i) = 0$) for $i \neq j$, with (f_i, e_i) ((e_i, f_i)) equalling 1 or 0 for every i .

A basis $f_1, f_2, \dots, f_n$ is said to be *left (right) pseudodual* to a basis $e_1, e_2, \dots, e_n$ if $(f_i, e_j) = 0$ ($(e_j, f_i) = 0$) for $j < i$, with $(f_i, e_i) = 1$ ($(e_i, f_i) = 1$), and for every j , with $(f_i, e_j) = 0$ ($(e_j, f_i) = 0$).

It is easy to see that the matrix of a bilinear form (x, y) is diagonal in a pair of dual bases and right (left) trapezoidal in a pair of pseudodual bases. The questions of the existence and construction of dual and pseudodual bases are closely related to equivalence transformations (91.4) of the matrix of a bilinear form (x, y) as well as to the factorization of that matrix. We shall turn to a detailed study of such bases only as need arises. For the present we shall restrict ourselves to their brief discussion.

Theorem 100.2. *In any nonsingular space each basis has a right and a left dual basis, which are unique.*

Proof. Consider a basis $e_1, e_2, \dots, e_n$ in a nonsingular space K_n and let G_e be the matrix of a bilinear form (x, y) in that basis. According to (91.4) finding a left (right) dual basis to $e_1, e_2, \dots, e_n$ is equivalent to determining a matrix P (Q) for which $P'G_e$ (G_eQ) is

a unit matrix. Then $P(Q)$ is the coordinate transformation matrix for a change from the basis $e_1, e_2, \dots, e_n$ to the dual basis. Since the space is nonsingular, so is the matrix G_e and there is a unique solution: $P = G_e^{-1}(Q = G_e^{-1})$.

Corollary. *In any nonsingular space each basis has a left and a right pseudodual basis.*

Indeed, every left (right) dual basis is simultaneously a left (right) pseudodual basis.

Taking into account the form of the matrix of a bilinear form (x, y) it is easy to establish that if in a nonsingular space we change from a left (right) dual basis to another basis, one with a left triangular coordinate transformation matrix with unit diagonal elements, then the new basis is left (right) pseudodual.

Exercises

1. Let a scalar product be symmetric. Is the number of vectors with positive, negative and zero values of (e_i, e_i) invariant for nonorthogonal bases $e_1, e_2, \dots, e_n$?

2. How can any real or complex vector space be converted into a bilinear metric space with a symmetric scalar product with a given rank and signature?

3. An orthogonal basis has no isotropic vectors in a nonsingular space. Can there be a basis consisting of isotropic vectors in such a space?

4. Prove that as functions of vectors of a bilinear metric space orthogonal projection and perpendicular are linear operators.

5. What form has the Gram matrix for a pseudoorthogonal basis if the right and left null spaces coincide?

6. Prove that in any ordinary or Hermitian bilinear metric space there is a basis in which the Gram matrix is a right block-triangular matrix with 1×1 and 2×2 blocks along the diagonal.

7. How can the coefficients of an expansion of a vector with respect to a basis for which some dual or pseudodual basis is known be determined?

8. Prove that in a nonsingular space the coordinate transformation matrix for a change from one basis, pseudodual to a given basis, to any other pseudodual basis of the same name is left triangular.

101. Operators and bilinear forms

If we have a linear operator in an ordinary or Hermitian bilinear metric space, then of course all the results obtained earlier for operators in a real or a complex space hold. We shall therefore study only additional properties of operators due to the presence in space of a scalar product.

One of the major objects is the adjoint operator. In a Euclidean and a unitary space the adjoint operator was introduced using a scalar product, but in investigating its properties wide use was made of the existence in space of an orthonormal basis. Now we cannot take this way, since there may be no orthogonal basis in the general bilinear metric space. We shall make our studies in the Hermitian bilinear metric space. Changes for the ordinary bilinear metric space are very simple.

An operator A^* ($*A$) in a Hermitian bilinear metric space K_n is said to be a *right (left) adjoint operator* to an operator A in K_n if for any vectors $x, y \in K_n$

$$(Ax, y) = (x, A^*y) \quad ((x, Ay) = (*Ax, y)). \quad (101.1)$$

Take a basis $e_1, e_2, \dots, e_n$ in K_n and let G_e be the Gram matrix for that basis. Denote by A_e the matrix of A in the basis $e_1, e_2, \dots, e_n$ and by A_e^* and $*A_e$ the matrices of A^* and $*A$ if they exist.

Theorem 101.1. *For any linear operator A in a nonsingular Hermitian bilinear metric space there are unique adjoint operators A^* and $*A$, with*

$$A_e^* = \bar{G}_e^{-1} \bar{A}_e' \bar{G}_e', \quad *A_e = G_e^{-1} A_e' G_e'. \quad (101.2)$$

Proof. If A^* exists, then according to (101.1), in matrix notation of the type (61.2) and (91.7), we have

$$(Ax, y) = (Ax)_e' G_e \bar{y}_e = x_e' (A_e' G_e) \bar{y}_e,$$

$$(x, A^*y) = x_e' G_e (\overline{A^*y})_e = x_e' (G_e \bar{A}_e^*) \bar{y}_e.$$

The right-hand sides of these relations must coincide for all vectors x_e and y_e , so $A_e' G_e = G_e \bar{A}_e^*$, whence follows the first of the equations (101.2). Similarly,

$$(x, Ay) = x_e' G_e (\bar{A}y)_e = x_e' (G_e \bar{A}_e) \bar{y}_e,$$

$$(*Ax, y) = (*Ax)_e' G_e \bar{y}_e = x_e' (*A_e' G_e) \bar{y}_e,$$

so $G_e \bar{A}_e = *A_e' G_e$ and we obtain the second of the equations (101.2).

Equations (101.2) imply that if adjoint operators exist, then they are unique. Take now those equations as a form of assigning the right and left adjoint operators. It is easy to verify directly that the operators thus constructed are linear and satisfy relations (101.1).

Corollary. *If a Hermitian bilinear form (x, y) is symmetric or skew-symmetric, then the right and left adjoint operators coincide.*

Indeed, in these cases $G_e = \pm \bar{G}_e'$ for any basis $e_1, e_2, \dots, e_n$. According to (101.2) we now conclude that $A_e^* = *A_e$.

It follows from the corollary that the right and left adjoint operators coincide in a unitary space. But this fact can be established in another way. If in a unitary space an orthonormal basis $e_1, e_2, \dots, e_n$ is taken, then $G_e = E$ and we obtain the well-known equations $A_e^* = *A_e = \bar{A}_e'$.

The adjoint operators are connected with A by certain relations. Note some of them, for example, for the right adjoint operator:

$$\begin{aligned} (A + B)^* &= A^* + B^*, \\ (\alpha A)^* &= \bar{\alpha} A^*, \\ (AB)^* &= B^* A^*, \\ (A^*)^{-1} &= (A^{-1})^*. \end{aligned} \quad (101.3)$$

For the left adjoint operator the relations are similar. All relations can be proved according to the same scheme using representations (101.2) for the matrices of adjoint operators. Therefore we shall prove only the validity of the last property. We have

$$(A_e^*)^{-1} = \bar{G}_e^{-1} (\bar{A}_e')^{-1} (\bar{G}_e^{-1})^{-1} = \bar{G}_e^{-1} (\bar{A}_e^{-1})' \bar{G}_e = (A_e^{-1})^*.$$

Comparing formulas (75.4) and (101.3) we can see the absence in (101.3) of the analogue of the first of the relations (75.4). It now looks like this:

$$(*A)^* = *(A^*) = A. \quad (101.4)$$

To prove its validity we again turn to representations (101.2) and get

$$(*A_e)^* = \bar{G}_e^{-1} (*A_e')' \bar{G}_e = \bar{G}_e^{-1} (\bar{G}_e^{-1'} \bar{A}_e' G_e')' \bar{G}_e = \bar{G}_e^{-1} \bar{G}_e A_e G_e^{-1'} \bar{G}_e = A_e,$$

$$*(A_e^*) = G_e^{-1'} (\bar{A}_e^*)' G_e' = G_e^{-1'} (\bar{G}_e^{-1} \bar{A}_e' G_e')' G_e' = G_e^{-1'} G_e' A_e G_e^{-1'} G_e' = A_e,$$

i.e. relations (101.4) do hold.

Theorem 101.2. *If in a nonsingular Hermitian bilinear metric space an operator A has in some basis a matrix J , then in a right (left) dual basis the operator A^* ($*A$) has a matrix J^* .*

Proof. Let A have a matrix J in a basis $e_1, e_2, \dots, e_n$. Consider a right dual basis $f_1, f_2, \dots, f_n$. Denote by $\bar{G}_e, G_f$ and $G_{ef} = E$ the matrices of a bilinear form (x, y) in the corresponding bases. If P is a coordinate transformation matrix for a change from the first basis to the second, then we have

$$G_e = G_{ef} \bar{P}^{-1} = \bar{P}^{-1}, \quad G_f = P' G_{ef} = P'$$

and then, taking into account (63.7) and (101.2), we get

$$A_f^* = \bar{G}_f^{-1} \bar{A}_f' \bar{G}_f = \bar{G}_f^{-1} (\bar{P}^{-1} J P)' \bar{G}_f = \bar{G}_f^{-1} \bar{G}_f J' \bar{G}_f^{-1} \bar{G}_f = J^*.$$

If, however, A has a matrix J in a basis $f_1, f_2, \dots, f_n$, then for that basis the basis $e_1, e_2, \dots, e_n$ is the left dual basis and we now find

$$*A_e = G_e^{-1'} \bar{A}_e' G_e' = G_e^{-1'} (\bar{P} A_f P^{-1})' G_e' = G_e^{-1'} G_e' J' G_e^{-1'} G_e' = J^*.$$

This theorem is as significant in the study of adjoint operators in Hermitian bilinear metric spaces as Theorem 75.2 is in unitary spaces. It follows from it in particular that the right and left adjoint operators A^* and $*A$ have the same eigenvalues, complex conjugate to those of A , that the right and left adjoint operators A^* and $*A$ have a simple structure, if A has a simple structure, and so on.

Besides a scalar product (x, y) other Hermitian bilinear forms can be given in a Hermitian bilinear metric space. Consider, for example, functions of the form (Ax, y) and (x, Ay) , where A is an arbitrary linear operator. It is not hard to see that they are Hermitian bilinear

forms. Different operators give different forms in any nonsingular space K_n . Indeed, if A and B are different operators, then $Ax \neq Bx$ at least for one vector x . Suppose $(Ax, y) = (Bx, y)$ for each $y \in K_n$. It follows that $((A - B)x, y) = 0$ for each $y \in K_n$, i.e. that $(A - B)x \in {}^\perp K_n$. But in a nonsingular space the subspace ${}^\perp K_n$ consists only of a zero vector, so $Ax = Bx$.

Theorem 101.3. *In a nonsingular Hermitian bilinear metric space K_n any Hermitian bilinear form $\varphi(x, y)$ can be uniquely represented as*

$$\varphi(x, y) = (Ax, y) = (x, By),$$

where A and B are some linear operators in K_n .

Proof. Choose in K_n some basis $e_1, e_2, \dots, e_n$ and let G_e be the Gram matrix in that basis and Φ_e the matrix of the form $\varphi(x, y)$. We have

$$\begin{aligned}\varphi(x, y) &= x'_e \Phi_e \bar{y}_e = x'_e \Phi_e G_e^{-1} G_e \bar{y}_e \\ &= (G_e^{-1'} \Phi'_e x_e)' G_e \bar{y}_e = x'_e G_e G_e^{-1} \Phi_e \bar{y}_e = x'_e G_e (\bar{G}_e^{-1} \bar{\Phi}_e \bar{y}_e).\end{aligned}$$

Now the matrices A_e and B_e of the desired operators A and B are defined by

$$A_e = G_e^{-1'} \Phi'_e, \quad B_e = \bar{G}_e^{-1} \bar{\Phi}_e. \quad (101.5)$$

The uniqueness of A and B was proved earlier.

An adjoint operator can be defined using a scalar product. Therefore, if different scalar products are introduced in a vector space, then the same linear operator will have different adjoint operators. Suppose in a vector space together with a scalar product given by a bilinear form (x, y) we introduce scalar products given by forms (Mx, y) and (x, My) . Label with a subscript M on the left (right) adjoint operators relating to the scalar product (Mx, y) ((x, My)).

Theorem 101.4. *For any operator A and a nonsingular operator M*

$$\begin{aligned}{}_M A^* &= (MAM^{-1})^*, & {}^* M A &= M^{-1} ({}^* A) M, \\ A^* {}_M &= M^{-1} A^* M, & {}^* A {}_M &= {}^* (MAM^{-1}).\end{aligned} \quad (101.6)$$

Proof. Choose some basis $e_1, e_2, \dots, e_n$ and let G_e and M_e be the matrices of a bilinear form (x, y) and of the operator M in that basis respectively. According to (101.5) the matrix of a bilinear form (Mx, y) is equal to $M'_e G_e$. According to (101.2) we now find

$$\begin{aligned}{}_M A^* &= (\overline{M'_e G_e})^{-1} \bar{A}'_e (\overline{M'_e G_e}) = \bar{G}_e^{-1} (\bar{M}_e^{-1'} \bar{A}'_e \bar{M}'_e) \bar{G}_e \\ &= \bar{G}_e^{-1} (\overline{M_e A_e M_e^{-1}})' \bar{G}_e = (MAM^{-1})^*,\end{aligned}$$

$$\begin{aligned}{}^* M A_e &= (M'_e G_e)^{-1'} \bar{A}'_e (M'_e G_e)' = M_e^{-1} G_e^{-1'} \bar{A}'_e G'_e M \\ &= M_e^{-1} (G_e^{-1'} \bar{A}'_e G'_e) M_e = M_e^{-1} ({}^* A_e) M_e.\end{aligned}$$

These matrix equations prove the validity of the first group of the operator equations (101.6). The second group follows trivially from the first, if we take into account the equation $(x, My) = (*Mx, y)$ and relations (101.2).

There are different types of operators in the Hermitian bilinear metric space. An operator A is said to be *Hermitian* or *self-adjoint*, if for any $x, y \in K_n$

$$(Ax, y) = (x, Ay),$$

and *skew-Hermitian* or *skew-adjoint*, if

$$(Ax, y) = -(x, Ay).$$

Hence respectively

$$A = A^* = *A, \quad A = -A^* = -*A.$$

An operator A is said to be *isometric* if for any $x, y \in K_n$

$$(Ax, Ay) = (x, y).$$

This leads to the equations

$$*AA = A^*A.$$

In an ordinary bilinear metric space the analogues of the Hermitian and the skew-Hermitian operator are called a *symmetric* and a *skew-symmetric* operator respectively. In what follows we shall often deal with operators defined by the equation

$$A^* = \alpha E + \beta A \quad (101.7)$$

for some numbers α and β .

By far not all properties of operators of a special form carry over from the unitary to the Hermitian bilinear metric space, although they do have something in common. We shall not discuss all these questions.

Exercises

1. How are the characteristic polynomials of operators A , A^* and $*A$ related?
2. Let a subspace L be invariant under an operator A . Prove that the subspace $L^\perp$ (${}^\perp L$) is invariant under A^* ($*A$).
3. Prove that any eigenvector of an operator A corresponding to an eigenvalue λ is left (right) orthogonal to any eigenvector of A^* ($*A$) corresponding to an eigenvalue $\mu \neq \bar{\lambda}$.
4. Prove that any root vector of an operator A corresponding to an eigenvalue λ is left (right) orthogonal to any root vector of A^* ($*A$) corresponding to an eigenvalue $\mu \neq \bar{\lambda}$.
5. Prove that the eigenvalues of a Hermitian (skew-Hermitian) operator corresponding to nonisotropic eigenvectors are real (pure imaginary).
6. Prove that the moduli of the eigenvalues of an isometric operator corresponding to nonisotropic eigenvectors equal unity.

7. Suppose in a nonsingular space the scalar product is Hermitian-symmetric. Prove that if A is a Hermitian (skew-Hermitian) operator, then the bilinear form (Ax, y) is Hermitian-symmetric (skew-symmetric).

8. Suppose in a nonsingular space the scalar product is Hermitian-symmetric. Prove that if the bilinear form (Ax, y) is Hermitian-symmetric (skew-symmetric), then the operator A is Hermitian (skew-Hermitian).

9. How do the statements of Exercises 7 and 8 change if the scalar product is skew-Hermitian?

10. Prove that if an operator A satisfying condition (101.7) has at least two distinct eigenvalues, then $|\beta| = 1$.

102. Bilinear metric isomorphism

It was shown in the study of Euclidean and unitary spaces that to within isomorphism there is only one space of each dimension n . For bilinear metric spaces the situation is more complicated.

We introduce the concept of isomorphism. We shall say that ordinary or Hermitian bilinear metric spaces over the same number field are *isomorphic* if they are isomorphic as vector spaces and the scalar products of pairs of corresponding vectors are equal to each other.

It follows from this definition that in isomorphic spaces the Gram matrices of the systems of corresponding vectors coincide. The converse is also true. If in bilinear metric spaces over the same number field there are such bases in which Gram matrices coincide, then those spaces are isomorphic. Indeed, by establishing a correspondence between bases with equal Gram matrices we ensure that scalar products coincide for any pair of vectors in the bases and hence for any pair of vectors.

Theorem 102.1. *Ordinary (Hermitian) bilinear metric spaces over the same number field are isomorphic if and only if the Gram matrices of arbitrary bases of those spaces are congruent (Hermitian-congruent).*

Proof. *Necessity.* The Gram matrices of all bases of the same space are congruent and they coincide on the corresponding bases of different spaces. By virtue of the transitivity of the congruence relation, the Gram matrices of arbitrary bases of isomorphic spaces are congruent.

Sufficiency. If the Gram matrices of arbitrary bases of bilinear metric spaces are congruent, then there are bases in different spaces on which the Gram matrices coincide. But then the spaces are isomorphic.

The theorem says that the problem of classifying bilinear metric spaces is equivalent to that of classifying bilinear forms to within congruence. Consider some classes of bilinear metric spaces.

A real bilinear metric space K_n is said to be *pseudo-Euclidean* if the scalar product is given by a nonsingular symmetric bilinear form.

For an arbitrary basis of a pseudo-Euclidean space the Gram matrix is real symmetric and, as we know, congruent with a diagonal

matrix with elements ± 1 . This means that in every pseudo-Euclidean space there is a basis in which a scalar product (x, y) of vectors x and y with coordinates $\xi_1, \dots, \xi_n$ and $\eta_1, \dots, \eta_n$ is given by the formula

$$(x, y) = \xi_1 \eta_1 + \dots + \xi_s \eta_s - \xi_{s+1} \eta_{s+1} - \dots - \xi_n \eta_n.$$

To within isomorphism pseudo-Euclidean spaces are defined by their two characteristics: dimension and signature, a positive and a negative index and so on. Of particular interest to physics among the pseudo-Euclidean spaces is four-dimensional space with a positive index equal to unity. This is the so-called *Minkowski space-time* or *Minkowski universe*. It is the space-time of special relativity.

A real bilinear metric space K_n is said to be *symplectic* if the scalar product is given by a nonsingular skew-symmetric bilinear form.

The Gram matrix for any symplectic space is skew-symmetric and therefore congruent with a block-diagonal matrix with blocks of

the form $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Consequently the dimension of a symplectic

space is always even and, to within isomorphism, there is only one symplectic space of a given even dimension. There is a basis in such a space in which the scalar product of vectors x and y with coordinates $\xi_1, \dots, \xi_n$ and $\eta_1, \dots, \eta_n$ has the form

$$(x, y) = \xi_1 \eta_2 - \xi_2 \eta_1 + \dots + \xi_{n-1} \eta_n - \xi_n \eta_{n-1}.$$

A complex bilinear metric space K_n is said to be *complex Euclidean* if the scalar product is given by a nonsingular symmetric bilinear form.

For any basis its Gram matrix is complex symmetric and congruent with a unit matrix. To within isomorphism there is only one complex Euclidean space of each dimension. In any complex Euclidean space there is a basis in which the scalar product of vectors x and y is as follows:

$$(x, y) = \xi_1 \eta_1 + \xi_2 \eta_2 + \dots + \xi_n \eta_n.$$

A complex Hermitian bilinear metric space is said to be *pseudounitary* if the scalar product is given by a nonsingular Hermitian-symmetric bilinear form.

The Gram matrix for any pseudounitary space is Hermitian. It is Hermitian-congruent with a real diagonal matrix with elements ± 1 . There is always a basis therefore in which the scalar product of vectors x and y has the form

$$(x, y) = \xi_1 \bar{\eta}_1 + \dots + \xi_s \bar{\eta}_s - \xi_{s+1} \bar{\eta}_{s+1} - \dots - \xi_n \bar{\eta}_n.$$

where $\xi_1, \dots, \xi_n$ and $\eta_1, \dots, \eta_n$ are the coordinates of x and y . Again to within isomorphism a pseudounitary space is uniquely defined by its two characteristics: dimension and signature, a positive and a negative index and so on.

Exercises

1. Prove that in isomorphic spaces to orthogonal (pseudoorthogonal, dual, pseudodual) bases there correspond orthogonal (pseudoorthogonal, dual, pseudodual) bases.

2. Prove that in isomorphic spaces to nonsingular subspaces there correspond nonsingular subspaces.

3. Prove that in isomorphic spaces perpendicular and projection go over into perpendicular and projection respectively.

4. Prove that in isomorphic spaces the Gramians of the corresponding systems of vectors are equal.

One of the major concepts associated with any bilinear metric space is that of orthogonality. We have often seen what important role orthogonal systems of vectors and especially orthogonal bases play in the study of Euclidean and unitary spaces. Of no less significance is the role played by bases with orthogonal vectors in other spaces. Up to now, however, most of our reasoning has been connected with proving the existence of such systems and not with the processes of constructing them. The only exception in a sense is the general method of transforming the matrices of bilinear forms to canonical form and the related construction of canonical bases. In view of the importance of orthogonal, pseudoorthogonal and other similar systems for designing diverse computational algorithms we consider now a general process of constructing such systems in bilinear metric spaces.

(1) for any $k \geq 1$ the spans L_k of vectors $e_1, \dots, e_k$ and $f_1, \dots, f_k$ coincide,

(2) the basis $f_1, \dots, f_n$ is pseudoorthogonal.

Let $(e_1, e_1) \neq 0$ and put $f_1 = e_1$. Suppose a system of pseudoorthogonal vectors $f_1, \dots, f_k$ has already been constructed, with the span of those vectors and that of $e_1, \dots, e_k$ coinciding and $(f_i, f_i) \neq 0$ for $1 \leq i \leq k$. We shall seek a vector f_{k+1} in the form

$$f_{k+1} = e_{k+1} + \sum_{l=1}^k \alpha_{l,k+1} f_l, \quad (103.1)$$

where $\alpha_{1, k+1}, \dots, \alpha_{k, k+1}$ are unknown coefficients. The conditions of the left orthogonality of f_{k+1} to $f_1, \dots, f_k$ give the following system of linear algebraic equations to determine $\alpha_{1, k+1}, \dots$

[illegible]

The matrix of the system is left triangular. Under the assumption its diagonal elements are nonzero, so system (103.2) has a unique solution. It is clear that the vector f_{k+1} thus constructed and the vectors $f_1, \dots, f_k$ form together a pseudoorthogonal system and that their span coincides with that of $e_1, \dots, e_{k+1}$. The system of vectors $f_1, \dots, f_{k+1}$ is linearly independent, for so is the system $f_1, \dots, f_k, e_{k+1}$.

We continue the process further. If it turns out that for every i (f_i, f_i) are nonzero, then the resulting system of vectors $f_1, \dots, f_n$ will be the desired pseudoorthogonal basis. Of course, we can now norm the vectors $f_1, \dots, f_n$ and obtain a pseudoorthonormal basis.

A useful consequence follows from (103.1). We rewrite it as

$$e_{k+1} = \left(- \sum_{i=1}^k \alpha_{i,k+1} f_i \right) + f_{k+1}.$$

The vector in the parentheses lies in L_k and the vector f_{k+1} is in ${}^\perp L_k$ by construction, so the solution of system (103.2) gives in fact a decomposition of each vector e_{k+1} into the projection and left perpendicular relative to L_k .

The process is greatly simplified if the scalar product is given by a Hermitian symmetric bilinear form. In this case the conditions $(f_i, f_j) = 0$ for $j < i$ imply that the conditions $(f_i, f_j) = 0$ hold for $j \neq i$. System (103.2) therefore becomes a system with a diagonal matrix and we have

$$\alpha_{i,k+1} = - \frac{(e_{k+1}, f_i)}{(f_i, f_i)}$$

for every i . The constructed basis $f_1, f_2, \dots, f_n$ will be not only pseudoorthogonal but it will also be orthogonal.

The only thing that may prevent the construction of a pseudoorthogonal basis $f_1, \dots, f_n$ from $e_1, \dots, e_n$ is that one of the scalar products (f_i, f_i) , $i < n$, may vanish. Such a situation is called degenerate. No degenerate situation will obviously set in if the quadratic form (x, x) has no isotropic vectors, if, for example, it is strictly of constant signs. Indeed, equation $(f_i, f_i) = 0$ is then possible only for $f_i = 0$. But $f_i \neq 0$ for every i , since vectors $f_1, \dots, f_i$ are linearly independent. Hence the process can now be realized under any choice of basis $e_1, \dots, e_n$.

In many problems it is not necessary to preserve the relation of the new basis $f_1, \dots, f_n$ to the original basis $e_1, \dots, e_n$, since it is required to construct only some pseudoorthogonal basis in space. It is then necessary whenever $(f_i, f_i) = 0$ occurs to replace e_i by another vector and compute again a vector f_i , repeating the procedure until the condition $(f_i, f_i) \neq 0$ holds. The vectors $f_1, \dots, f_{i-1}$ remain unchanged.

Proof. Necessity. Let the process be implementable, i.e. let relation (103.6) hold. The matrix G_f is nonsingular, since it is the Gram matrix of a nonsingular bilinear form (x, y) for a basis. Therefore all its diagonal elements are nonzero. Applying the Binet-Cauchy formula we get

$$G_e \begin{pmatrix} 1 & 2 & \dots & r \\ 1 & 2 & \dots & r \end{pmatrix} = A' G_f \bar{A} \begin{pmatrix} 1 & 2 & \dots & r \\ 1 & 2 & \dots & r \end{pmatrix} = G_f \begin{pmatrix} 1 & 2 & \dots & r \\ 1 & 2 & \dots & r \end{pmatrix} \neq 0$$

for every r .

Sufficiency. Let the principal minors of the Gram matrix G_e be nonzero. Hence according to (93.1) there is a decomposition $G_e = L_e D_e U_e$, where L_e is a left triangular matrix with unit diagonal elements, D_e is a diagonal matrix with nonzero elements and U_e is a right triangular matrix with unit diagonal elements. It is easy to see that the matrix

$$G_f = \bar{U}_e^{-1'} G_e U_e^{-1} = \bar{U}_e^{-1'} L_e D_e$$

is a left triangular matrix whose diagonal elements coincide with those of D_e . If we now take as a matrix A a right triangular matrix $\bar{U}_e$ with unit diagonal elements, then relations (103.5) will hold for the basis $f_1, f_2, \dots, f_n$. It is this basis that will be constructed according to process (103.1) and (103.2), which can easily be seen from a direct check.

If the scalar product is given by a symmetric Hermitian bilinear form, then the matrix G_e will be Hermitian, as well as the matrix G_f . But it follows that G_f is diagonal. This fact has already been noted. Comparing (93.5) and (103.6) we conclude that in this case the orthogonalization process virtually coincides completely with the process of obtaining decomposition (93.5).

If K_n is a unitary space, then the orthogonalization process determines not only the factorization of the Gram matrix into triangular factors but also the decomposition of the matrix of coordinates as a product of the unitary and the right triangular factor. Indeed, choose an orthonormal basis $q_1, q_2, \dots, q_n$ and denote by D_q a diagonal matrix made up of the lengths of the columns of the matrix F_q of (103.7). We now have $E_q = (F_q D_q^{-1}) (D_q A)$. The matrix $D_q A$ is right triangular. But G_q and $G_f (D_q^{-1})^2$ are unit matrices. In this case according to (103.8) $(F_q D_q^{-1})' (\overline{F_q D_q^{-1}}) = E$, i.e. $F_q D_q^{-1}$ is a unitary matrix.

The fact that the basis $t_1, t_2, \dots, t_n$ is left pseudodual to the basis $e_1, e_2, \dots, e_n$ implies that for the bilinear form (x, y) determining the scalar product in K_n conditions $(t_i, e_j) = 0$ for $j < i$ and $(t_i, e_i) = 1$ for every i hold. In other words, this implies that for a pair of bases $e_1, e_2, \dots, e_n$ and $t_1, t_2, \dots, t_n$ the matrix G_{t_e} of the bilinear form (x, y) is a right triangular matrix with unit

diagonal elements. From this we conclude that the coordinate transformation matrix Q^{-1} for a change from the original basis $q_1, q_2, \dots, q_n$ to $t_1, t_2, \dots, t_n$ is right triangular. Now the diagonal elements will not equal unity, however, since the vectors $t_1, t_2, \dots, t_n$ have undergone normalization. We have

$$G_{te} = Q^{-1'} G_{qe}$$

and further

$$G_{qe} = Q' G_{te}.$$

The process of constructing a basis pseudodual to a given basis also proves to be closely related to factorization (93.1) of a matrix into triangular factors.

Theorem 103.2. *For process (103.3) and (103.4) of constructing a basis $t_1, t_2, \dots, t_n$, left pseudodual to $e_1, e_2, \dots, e_n$, starting from $q_1, q_2, \dots, q_n$ to be implementable it is necessary and sufficient that the matrix G_{qe} of a bilinear form (x, y) should have nonzero principal minors in the bases $q_1, q_2, \dots, q_n$ and $e_1, e_2, \dots, e_n$.*

The proof of this theorem is omitted since it is almost a word-for-word repetition of the proof of the preceding theorem.

To conclude, the orthogonalization processes considered carry over completely to ordinary bilinear metric spaces. There are changes in some details associated with complex conjugation. Besides, it is more complicated to eliminate degenerate situations.

Exercises

1. What is the geometrical interpretation of the orthogonalization process?

2. Prove that if the orthogonalization process is applied to a linearly dependent system $e_1, e_2, \dots, e_n$, then $f_k = 0$ for some $k \leq n$.

3. Let a quadratic form (x, x) have no isotropic vectors. How is a basis of the given system of vectors to be constructed using the orthogonalization process?

4. Prove that if the orthogonalization process is carried out in a Euclidean or a unitary space, then $|f_k| \leq |e_k|$ for every k , equality holding if and only if a vector e_k is orthogonal to vectors $e_1, \dots, e_{k-1}$.

5. Let the coordinates of vectors $e_1, \dots, e_n$ form a triangular matrix in some orthonormal basis of a Euclidean or a unitary space. How is the matrix of coordinates affected by the orthogonalization process?

6. Is it possible to construct a dual basis using the orthogonalization process?

7. How is the orthogonalization process to be applied to obtain a right pseudodual basis?

8. Prove that the decomposition of a complex nonsingular matrix into a product of a unitary matrix and a right triangular matrix is unique if it is required that the diagonal elements of the triangular matrix should be positive.

9. How is the orthogonalization process to be applied in order to solve a system of linear algebraic equations?

10. Let a space K_n be singular. How is a pseudoorthogonal basis of a non-singular subspace of maximum dimension to be constructed using the orthogonalization process?

11. Does the construction of pseudoorthogonal systems of vectors become simpler in a singular space if the quadratic form (x, x) is of constant sign?

104. Orthogonalization of a power sequence

In orthogonalization processes the coordinate transformation matrix for a change from the old basis to a new basis is always triangular. However, if the original basis is chosen in a special way, it is possible to obtain significantly simpler representations for the coordinate transformation matrix and hence simpler orthogonalization processes.

Let A be some operator in a nonsingular Hermitian bilinear metric space K_n . Take a nonzero vector x and consider a sequence of vectors

$$x, Ax, A^2x, \dots, A^{k-1}x. \quad (104.1)$$

We shall call such sequences *power sequences* generated by the vector x .

In any power sequence some number of the first vectors is linearly independent. Suppose k is the largest of such numbers. This means that there are numbers $\alpha_0, \alpha_1, \dots, \alpha_k$, with $\alpha_k \neq 0$, such that

$$\alpha_0 x + \alpha_1 Ax + \dots + \alpha_k A^k x = 0. \quad (104.2)$$

Denote by $\varphi(\lambda) = \alpha_k \lambda^k + \dots + \alpha_1 \lambda + \alpha_0$ a polynomial of degree k . Clearly (104.2) is equivalent to

$$\varphi(A)x = 0. \quad (104.3)$$

There are many polynomials for which relations of the type (104.3) hold. In particular, such a polynomial is the characteristic polynomial of A . But there is clearly a polynomial of the lowest degree among them. It is called the *minimum polynomial* annihilating the vector x . It is clear that its degree equals the maximum number of the first vectors of the power sequence (104.1) that form the linearly independent system or equivalently is a unity less than the minimum number of the first vectors that form the linearly dependent system.

The degree of the minimum polynomial turns out to be closely related to the expansion of the vector x with respect to the root basis of the operator A by the heights of the root vectors and the number of mutually distinct eigenvalues. That is, we have

Lemma 104.1. *The degree of a minimum polynomial annihilating a vector x equals the sum of the maximum heights of the root vectors of*

an operator A present in the expansion of x with respect to a root basis and corresponding to mutually distinct eigenvalues.

Proof. We represent a vector x as a sum

$$x = u_1 + u_2 + \dots + u_s, \quad (104.4)$$

where $u_1, \dots, u_s$ lie in different cyclic subspaces of A . Since different cyclic subspaces have no common vectors except the zero vector, for equation (104.3) to hold it is necessary and sufficient that the equations $\varphi(A) u_i = 0$ should hold for every i . If u_i is a root vector of height m_i and corresponds to an eigenvalue λ_i , then the equation $\varphi(A) u_i = 0$ will hold if and only if the polynomial $\varphi(\lambda)$ is divisible by $(\lambda - \lambda_i)^{m_i}$, where $r \geq m_i$. In this case $\varphi(A) u_j = 0$ not only for $j = i$ but also for all those j for which the vectors u_j correspond to the eigenvalues coinciding with λ_i and have heights not greater than r . Let $\lambda_{i_1}, \dots, \lambda_{i_p}$ be mutually distinct eigenvalues corresponding to the vectors $u_1, \dots, u_s$ of (104.4) and let $m_{i_1}, \dots, m_{i_p}$ be the maximum heights of the root vectors $u_1, \dots, u_s$ corresponding to the eigenvalues coinciding with $\lambda_{i_1}, \dots, \lambda_{i_p}$. Then

$$\varphi(\lambda) = (\lambda - \lambda_{i_1})^{m_{i_1}} \dots (\lambda - \lambda_{i_p})^{m_{i_p}}$$

will be the minimum polynomial annihilating the vector x . Thus the lemma is proved.

Suppose the vectors $e_i = A^{i-1}x$ for $1 \leq i \leq k$ are linearly independent. Apply to this system the process described earlier for obtaining a pseudoorthogonal system of vectors f_i , assuming of course that the process is feasible. If the operator A is in no way related to the scalar product introduced in K_n , then there is little hope for any simplification of the process. The situation sharply changes, however, if A satisfies relation (101.7), if for example it is a self-adjoint operator in a unitary space.

Theorem 104.1. *If an operator A satisfies relation (101.7), vectors $e_i = A^{i-1}x$ are linearly independent for $1 \leq i \leq k$ and vectors $f_1, \dots, f_k$ are obtained from $e_1, \dots, e_k$ using the pseudoorthogonalization process, then the following relations hold*

$$\begin{aligned} f_1 &= x, \\ f_2 &= Af_1 - \alpha_1 f_1, \\ f_{i+1} &= Af_i - \alpha_i f_i - \beta_{i-1} f_{i-1}, \quad i > 1, \end{aligned} \quad (104.5)$$

where

$$\begin{aligned} \alpha_i &= \frac{(Af_i, f_i)}{(f_i, f_i)}, \quad \beta_{i-1} = \frac{(Af_i, f_{i-1})}{(f_{i-1}, f_{i-1})}, \quad i > 1, \\ \alpha_i &= \frac{(f_{i-1}, f_{i-1})(Af_i, f_i) - (Af_i, f_{i-1})(f_{i-1}, f_i)}{(f_{i-1}, f_{i-1})(f_i, f_i)}, \quad i > 1. \end{aligned} \quad (104.6)$$

Suppose that there are n linearly independent vectors in the power sequence. Applying the orthogonalization process it is possible to

Exercises

1. Prove that the minimum polynomial annihilating a vector x is a divisor of the characteristic polynomial.

2. Prove that the minimum polynomial annihilating a vector x is unique up to a scalar multiplier.

3. Prove that if A is a Hermitian operator and K_n is a unitary space then formulas (104.6) become as follows:

$$\alpha_l = \frac{(Af_l, f_l)}{(f_l, f_l)},$$

$$\beta_{l-1} = \frac{(Af_l, f_{l-1})}{(f_{l-1}, f_{l-1})} = \frac{(f_l, Af_{l-1})}{(f_{l-1}, f_{l-1})} = \frac{(f_l, f_l)}{(f_{l-1}, f_{l-1})} > 0.$$

4. Prove that under the conditions of Exercise 3 there is a diagonal matrix D such that for the matrix A_f of (104.8) the matrix $D^{-1}A_fD$ is real symmetric tridiagonal.

5. Prove that if condition (101.7) holds then the matrices of bilinear forms (Ax, y) and (x, Ay) in the basis $f_1, \dots, f_n$ are right almost triangular.

6. Prove that if conditions of Exercise 3 hold then the matrices of bilinear forms (Ax, y) and (x, Ay) in the basis $f_1, \dots, f_n$ are Hermitian tridiagonal.

7. Prove that if K_n is a singular space then using processes (104.5) and (104.6) it is possible to construct a pseudoorthogonal basis of a nonsingular subspace of maximum dimension.

105. Methods of conjugate directions

Constructing orthogonal, pseudoorthogonal, etc. systems of vectors, especially by making use of power sequences, provides great possibilities for developing various numerical methods of solving equations of the form

$$Ax = b, \quad (105.1)$$

where A is an operator in a vector space K_n , b is a given vector and x is the desired vector.

We have repeatedly touched on various aspects of this problem. We now describe a large group of numerical methods of solving equation (105.1) known under the general name of methods of conjugate directions. They are all based on orthogonalization processes of power sequences. Suppose for simplicity of presentation that the operator A is nonsingular and that hence equation (105.1) always has a unique solution. We assume that the space K_n is complex and that the scalar product in it is given using a symmetric positive definite Hermitian bilinear form, i.e. that K_n is a unitary space.

We take any nonsingular operators C and B and let $s_1, \dots, s_n$ be some CAB -pseudoorthogonal system of vectors, i.e. let

$$(CABs_i, s_i) \neq 0, \quad (CABs_i, s_k) = 0, \quad k < i,$$

for every i . Denote by x_0 the initial vector and let

$$\begin{aligned}x &= x_0 + B \sum_{j=1}^n a_j s_j, \\x_i &= x_0 + B \sum_{j=1}^i a_j s_j, \\r_i &= Ax_i - b.\end{aligned}\tag{105.2}$$

Then from the relations

$$x_i = x_{i-1} + a_i B s_i \tag{105.3}$$

it follows that

$$r_i = r_{i-1} + a_i A B s_i. \tag{105.4}$$

It is easy to show that for the chosen CAB -pseudoorthogonal system $s_1, \dots, s_n$

$$(Cr_i, s_k) = 0, \quad 1 \leq k \leq i. \tag{105.5}$$

Indeed

$$r_i = Ax_i - b = A(x_i - x) = - \sum_{j=i+1}^n a_j A B s_j$$

and further

$$(Cr_i, s_k) = - \sum_{j=i+1}^n a_j (CAB s_j, s_k) = 0$$

for every $k \leq i$.

We assume that the system of vectors $s_1, \dots, s_n$ is constructed parallel to the system $r_0, \dots, r_{n-1}$ using the process of its CAB -pseudoorthogonalization. Set $s_1 = r_0$ and for every i we have

$$s_{i+1} = r_i + \sum_{k=1}^i \beta_{k, i+1} s_k. \tag{105.6}$$

As always the conditions of the left CAB -pseudoorthogonality of a vector s_{i+1} to vectors $s_1, \dots, s_i$ yield a left triangular system to determine the coefficients $\beta_{k, i+1}$. In this case r_k is a linear combination of vectors $s_1, \dots, s_{k+1}$. Hence the scalar product (Cr_i, r_k) is a linear combination of numbers $(Cr_i, s_1), \dots, (Cr_i, s_{k+1})$ and it is zero by (105.5) for $k < i$, i.e.

$$(Cr_i, r_k) = 0, \quad k < i. \tag{105.7}$$

This means that if $(Cr_i, r_i) \neq 0$ for every i then the sequence of vectors r_i is C -pseudoorthogonal. In an n -dimensional vector space a C -pseudoorthogonal system cannot contain more than n nonzero vectors. At some step of the computational process therefore one of the discrepancies vanishes and we obtain an exact solution of equation (105.1).

To implement the process, it is necessary for us to determine the coefficients a_i in (105.2) and the coefficients $\beta_{k, i+1}$ in (105.6). The coefficients a_i can always be found in a very simple way. According to (105.4), (105.5) and (105.7) we have

$$a_i = -\frac{(Cr_{i-1}, r_{i-1})}{(CABs_i, r_{i-1})} = -\frac{(Cr_{i-1}, s_i)}{(CABs_i, s_i)}. \quad (105.8)$$

In general computation of the coefficients $\beta_{k, i+1}$ is much more complicated. But if the operators A , B and C are connected by the relation

$$(CABC^{-1})^* = \alpha E + \beta AB \quad (105.9)$$

for some numbers α and β , then among all the coefficients $\beta_{k, i+1}$ only $\beta_{i, i+1}$ can be nonzero. Let

$$s_{i+1} = r_i + b_i s_i. \quad (105.10)$$

The coefficient b_i is uniquely determined from the condition of the left CAB -orthogonality of the vector s_{i+1} to s_i , which, considering (105.9) and (105.10), yields

$$b_i = -\frac{(CABr_i, s_i)}{(CABs_i, s_i)} = -\bar{\beta} \frac{(Cr_i, ABs_i)}{(CABs_i, s_i)}. \quad (105.11)$$

Suppose that computing a sequence of vectors s_i by formulas (105.10) and (105.11) we have shown that the sequence $s_1, \dots, s_i$ forms a CAB -pseudoorthogonal system. This is obviously true for $i = 2$. Taking into account (105.9) we get for $k < i$ from (105.4) to (105.7)

$$\begin{aligned} (CABs_{i+1}, s_k) &= (CABr_i, s_k) + b_i (CABs_i, s_k) = ((CABC^{-1})Cr_i, s_k) \\ &= (Cr_i, (CABC^{-1})^* s_k) = (Cr_i, (\alpha E + \beta AB)s_k) = \bar{\alpha} (Cr_i, s_k) + \bar{\beta} (Cr_i, ABs_k) \\ &= \bar{\beta} \left(Cr_i, \frac{1}{a_k} (r_k - r_{k-1}) \right) = \frac{\bar{\beta}}{a_k} \{ (Cr_i, r_k) - (Cr_i, r_{k-1}) \} = 0. \end{aligned}$$

Thus, (105.9) holding, the solution of the operator equation (105.1) can be effected according to the following prescription:

$$\begin{aligned} s_1 &= r_0, \\ r_i &= r_{i-1} + a_i ABs_i, \\ s_{i+1} &= r_i + b_i s_i, \\ x_i &= x_{i-1} + a_i Bs_i. \end{aligned} \quad (105.12)$$

Here x_0 is an arbitrary initial vector, the coefficients a_i and b_i are computed according to (105.8) and (105.11). Letting $u_i = Bs_i$,

the process will be as follows:

$$\begin{aligned}u_i &= Br_0, \\r_i &= r_{i-1} + a_i Au_i, \\u_{i+1} &= Br_i + b_i u_i, \\x_i &= x_{i-1} + a_i u_i,\end{aligned}$$

with

$$\begin{aligned}a_i &= -\frac{(Cr_{i-1}, r_{i-1})}{(CAu_i, r_{i-1})} = -\frac{(B^{-1*}Cr_{i-1}, u_i)}{(B^{-1*}CAu_i, u_i)}, \\b_i &= -\frac{(B^{-1*}CABr_i, u_i)}{(B^{-1*}CAu_i, u_i)} = -\bar{\beta} \frac{(Cr_i, Au_i)}{(B^{-1*}CAu_i, u_i)}.\end{aligned}$$

It is these processes that are called *methods of conjugate directions*.

From formulas (105.4) and (105.10) we conclude that vectors r_i and s_{i+1} are linear combinations of vectors of the same power sequence

$$r_0, AB r_0, \dots, (AB)^i r_0. \quad (105.13)$$

Moreover, they are obtained from it using C - and CAB -pseudoorthogonalizations respectively. This result has consequences of exceptional importance.

If not all the components are present in the expansion of the vector r_0 with respect to the Jordan canonical basis of the operator AB , then the vanishing of the discrepancy occurs earlier than at the n th step. The process terminates especially rapidly if AB has a simple structure and a large number of coinciding eigenvalues. That is, if in the expansion of r_0 with respect to the eigenvectors of the matrix of AB the nonzero components correspond to m mutually distinct eigenvalues, then $r_m = 0$.

By Theorem 104.1 trinomial relations of the type (104.5) must hold for vectors s_i and r_i . They can be obtained directly from (105.4) and (105.10). Namely,

$$\begin{aligned}s_{i+1} &= a_i AB s_i + (1 + b_i) s_i - b_{i-1} s_{i-1}, \quad i > 1, \\r_{i+1} &= a_{i+1} AB r_i + \left(1 + \frac{b_i a_{i+1}}{a_i}\right) r_i - \frac{b_i a_{i+1}}{a_i} r_{i-1}, \quad i \geq 1.\end{aligned} \quad (105.14)$$

From these we can obtain other relations, for example this:

$$x_{i+1} = x_{i-1} + \omega_{i+1} (\alpha_i Br_i + x_i - x_{i-1}),$$

where ω_{i+1} and α_i are suitably chosen numbers.

In view of what has been said concerning sequence (105.13) note the following peculiarity of condition (105.9). On the face of it this condition differs from those of the type (101.7). But if we take into account (101.6), then it is easy to show that (105.9) is in fact also a condition of the type (101.7), in relation to two scalar products

$(CABx, y)$ and (Cx, y) for that matter. Indeed, observe that the adjoint operator in (105.9) is connected with the basic scalar product of a unitary space, whereas the orthogonality of vectors s_i and r_i is ensured in relation to the scalar products $(CABx, y)$ and (Cx, y) respectively

$$\begin{aligned} CAB(AB)^* &= (CAB \cdot AB \cdot (CAB)^{-1})^* = (CABC^{-1})^* = \alpha E + \beta AB, \\ c(AB)^* &= (CABC^{-1})^* = \alpha E + \beta AB. \end{aligned}$$

The implementation of methods of conjugate directions can be prevented only by the vanishing of one of the scalar products, $(CABs_i, s_i)$ or (Cr_{i-1}, r_{i-1}) , before the discrepancy vanishes. If $(CABs_i, s_i) = 0$, then the coefficients a_i and b_i cannot be computed. If, however, $(Cr_{i-1}, r_{i-1}) = 0$, then this leads to a zero coefficient a_i , to the coincidence of the nonzero discrepancies r_{i-1} and r_i and hence to the equation $(CABs_{i+1}, s_{i+1}) = 0$ holding. Such a situation can be avoided by choosing a new initial vector x_0 . If the operators CAB and C are positive definite, then the above degenerations are impossible and the computational process runs without complications. If CAB is positive definite, then the methods of conjugate directions acquire further interesting properties.

The closeness of a vector z to the solution of (105.1) can be judged by the smallness of the square of some norm of the difference $e = x - z$. To that end it is convenient to use the so-called *generalized error functionals* of the form (Re, e) , where R is any positive definite operator, for example $B^{-1*}CA$. That operator is positive definite, since it is connected with the operator CAB by the relation $B^{-1*}CA = B^{-1*}(CAB)B^{-1}$. We have

Theorem 105.1. *If an operator CAB is positive definite, then among the totality of vectors of the form $z = x_0 + Bs$, where s is in the span of vectors $s_1, \dots, s_i$, a vector x_i gives a minimum of the generalized error functional*

$$\varphi(z) = (B^{-1*}CAe, e).$$

Proof. Since CAB is positive definite, the system of vectors s_i is CAB -orthogonal. We represent a vector z as a decomposition similar to (105.2) for a vector x :

$$z = x_0 + B \sum_{j=1}^i h_j s_j.$$

We have

$$\begin{aligned} \varphi(z) &= (B^{-1*}CA(x-z), x-z) \\ &= (B^{-1*}CAB \left(\sum_{j=1}^n a_j s_j - \sum_{j=1}^i h_j s_j \right), B \left(\sum_{j=1}^n a_j s_j - \sum_{j=1}^i h_j s_j \right)) \end{aligned}$$

$$\begin{aligned}
&= (CAB \left(\sum_{j=1}^n a_j s_j - \sum_{j=1}^i h_j s_j \right), \sum_{j=1}^n a_j s_j - \sum_{j=1}^i h_j s_j) \\
&= \sum_{j=1}^i |a_j - h_j|^2 (CAB s_j, s_j) + \sum_{j=i+1}^n |a_j|^2 (CAB s_j, s_j).
\end{aligned}$$

From this we conclude that a minimum of the error functional is attained for $h_j = a_j$, $j \leq i$, i.e. for $z = x_i$.

The error functional cannot be determined in practical calculations, since it depends on the solution x , which is unknown. However, it differs only in the presence of a constant term from another functional:

$$\psi(z) = (B^{-1*}CAz, z) - 2\operatorname{Re}(B^{-1*}Cb, z),$$

which can be calculated. Indeed,

$$\begin{aligned}
\varphi(z) &= (B^{-1*}CA(x - z), x - z) = (B^{-1*}CAx, x) \\
&\quad - (B^{-1*}CAx, z) - (B^{-1*}CAz, x) + (B^{-1*}CAz, z) \\
&= (B^{-1*}CAz, z) - (B^{-1*}Cb, z) - \overline{(B^{-1*}Cb, z)} \\
&\quad + (B^{-1*}Cb, x) = \psi(z) + (B^{-1*}Cb, x).
\end{aligned}$$

Finally we note some classes of operators for which condition (105.9) holds.

1. All operators A , B and C are Hermitian, with $B = C$. Condition (105.9) holds for $\alpha = 0$ and $\beta = 1$:

$$(CABC^{-1})^* = (BA)^* = A^*B^* = 0 \cdot E + 1 \cdot AB.$$

2. Operators CAB and C are Hermitian too. Condition (105.9) again holds for $\alpha = 0$ and $\beta = 1$:

$$(CABC^{-1})^* = C^{-1*}(CAB)^* = C^{-1}CAB = 0 \cdot E + 1 \cdot AB.$$

3. An operator C is commutative with AB , and AB is normal and its spectrum lies on a straight line. The last condition implies that $AB = \gamma E + \delta H$ for some Hermitian operator H . We now find

$$\begin{aligned}
(CABC^{-1})^* &= (CC^{-1}AB)^* = (AB)^* = (\gamma E + \delta H)^* \\
&= \bar{\gamma}E - \bar{\delta}H = \frac{\bar{\gamma}\delta - \bar{\delta}\gamma}{\delta} E + \frac{\bar{\delta}}{\delta} (\gamma E + \delta H) = \frac{2\operatorname{Im}(\bar{\gamma}\delta)}{\delta} E + \frac{\bar{\delta}}{\delta} AB.
\end{aligned}$$

4. Represent an operator A as $A = M + N$, where $M = M^*$ and $N = -N^*$. If M is a nonsingular operator, then put $B = C = M^{-1}$. Condition (105.9) holds for $\alpha = 2$ and $\beta = -1$:

$$\begin{aligned}
(CABC^{-1})^* &= (M^{-1}(M+N))^* = (M - N)M^{-1} \\
&= 2E - (M + N)M^{-1} = 2 \cdot E - 1 \cdot AB.
\end{aligned}$$

5. If in the decomposition $A = M + N$ it is N that is a nonsingular operator, put $B = C = N^{-1}$. Condition (105.9) holds for $\alpha = 2$ and $\beta = -1$:

$$(CABC^{-1})^* = (N^{-1}(M + N))^* = -(M - N)N^{-1} \\ = 2E - (M + N)N^{-1} = 2 \cdot E - 1 \cdot AB.$$

Exercises

1. Prove that the matrix of a bilinear form $(CABx, y)$ is:
 right triangular in a basis $s_1, \dots, s_n$,
 right almost triangular in a basis $r_0, \dots, r_{n-1}$,
 left triangular in bases $s_1, \dots, s_n$ and $r_0, \dots, r_{n-1}$ if C is a Hermitian operator.

2. How does the form of the matrix of the bilinear form $(CABx, y)$ of Exercise 1 change if CAB is a Hermitian operator?

3. Prove that the matrix of a bilinear form (Cx, y) is:
 right triangular in a basis $r_0, \dots, r_{n-1}$,
 right triangular in bases $r_0, \dots, r_{n-1}$ and $s_1, \dots, s_n$,
 right triangular in bases $ABs_1, \dots, ABs_n$ and $s_1, \dots, s_n$,
 right almost triangular in bases $ABr_0, \dots, ABr_{n-1}$ and $r_0, \dots, r_{n-1}$.

4. How does the form of the matrix of the bilinear form (Cx, y) of Exercise 3 change if C is a Hermitian operator?

5. Prove that if condition (105.9) is replaced by the following:

$$(CABC^{-1})^* = \alpha_0 E + \alpha_1 AB + \dots + \alpha_p (AB)^p, \quad p \geq 1,$$

then relation (105.10) will be like this:

$$s_{l+1} = r_l + b_l s_l + b_{l-1} s_{l-1} + \dots + b_{l-p+1} s_{l-p+1}.$$

6. Prove that

$$a_l = -\frac{(Cr_{l-1}, r_{l-1})}{(CABs_l, s_l)}.$$

7. Prove that if CAB and C are Hermitian operators then

$$b_l = \frac{(Cr_l, r_l)}{(Cr_{l-1}, r_{l-1})}.$$

8. Prove that if CAB and C are Hermitian and positive definite operators then $a_l < 0$ and $b_l > 0$ for every l .

9. Prove that the matrix of an operator AB in a basis made up of vectors $s_1, \dots, s_n$ or $r_0, \dots, r_{n-1}$ has a tridiagonal form.

10. How do the methods of conjugate directions run if A is a singular operator?

106. Main variants

We discuss the best known variants of the methods of conjugate directions. Theoretically they all fit into scheme (105.12) described above. Practical calculations, however, are sometimes carried out using somewhat different algorithms.

Method of conjugate gradients. In this method A is a Hermitian positive definite operator, $B = C = E$ and (105.9) holds for $\alpha = 0$

and $\beta = 1$. The positive definiteness of the operators $CAB = A$ and $C = E$ guarantees that the computational process runs without degeneration. At each step of the method the error functional with matrix A is minimized. The computational scheme of the method has the form

$$\begin{aligned} s_1 &= r_0, \\ r_t &= r_{t-1} + a_t A s_t, \\ s_{t+1} &= r_t + b_t s_t, \\ x_t &= x_{t-1} + a_t s_t, \end{aligned}$$

where

$$a_t = -\frac{(r_{t-1}, r_{t-1})}{(A s_t, r_{t-1})} = -\frac{(r_{t-1}, s_t)}{(A s_t, s_t)} < 0, \quad b_t = -\frac{(r_t, A s_t)}{(A s_t, s_t)} = \frac{(r_t, r_t)}{(r_{t-1}, r_{t-1})} > 0.$$

In the method of conjugate gradients vectors r_t form an orthogonal system, and vectors s_t form an A -orthogonal system.

Method of AA^* -minimum iterations. In this method A is an arbitrary nonsingular operator, $B = A^*$, $C = E$ and (105.9) holds for $\alpha = 0$ and $\beta = 1$. The positive definiteness of $CAB = AA^*$ and $C = E$ guarantees that the computational process runs without degeneration. At each step of the method the error functional with matrix E , i.e. the square of the Euclidean norm of the error, is minimized. The computational scheme here has the form

$$\begin{aligned} u_1 &= A^* r_0, \\ r_t &= r_{t-1} + a_t A u_t, \\ u_{t+1} &= A^* r_t + b_t u_t, \\ x_t &= x_{t-1} + a_t u_t, \end{aligned}$$

where

$$a_t = -\frac{(r_{t-1}, r_{t-1})}{(A u_t, r_{t-1})} = -\frac{(r_{t-1}, u_t)}{(u_t, u_t)} < 0, \quad b_t = -\frac{(r_t, A u_t)}{(u_t, u_t)} = \frac{(r_t, r_t)}{(r_{t-1}, r_{t-1})} > 0.$$

In the method of AA^* -minimum iterations vectors r_t and u_t form orthogonal systems.

Method of A^*A -minimum iterations. In this method A is an arbitrary nonsingular operator, $B = A^*$, $C = AA^*$ and (105.9) holds for $\alpha = 0$ and $\beta = 1$. The positive definiteness of $CAB = (AA^*)^2$ and $C = AA^*$ guarantees that the computational process runs without degeneration. At each step of the method the error functional with matrix A^*A , i.e. the square of the Euclidean norm of the discrepancy vector, is minimized. The computational scheme of the method has the form

$$\begin{aligned} u_1 &= A^* r_0, \\ r_t &= r_{t-1} + a_t A u_t, \\ u_{t+1} &= A^* r_t + b_t u_t, \\ x_t &= x_{t-1} + a_t u_t, \end{aligned}$$

where

$$a_i = -\frac{(A^*r_{i-1}, A^*r_{i-1})}{(Au_i, Au_i)} < 0, \quad b_i = \frac{(A^*r_i, A^*r_i)}{(A^*r_{i-1}, A^*r_{i-1})} > 0.$$

In the method of A^*A -minimum iterations vectors A^*r_i and Au_i form orthogonal systems.

Method of complete Hermitian decomposition. In this method A is an arbitrary nonsingular operator. We represent it as a sum $A = M + N$, where $M = M^*$ and $N = -N^*$. If either M or N is nonsingular, we set $B = C = M^{-1}$ or $B = C = N^{-1}$ respectively. Condition (105.9) holds for $\alpha = 2$ and $\beta = -1$. If M (or iN) is an operator of constant signs, the process runs without degeneration. For example, let $M > 0$ and $B = C = M^{-1}$. The operator C will be positive definite and therefore $(Cz, z) = (M^{-1}z, z) > 0$ for any nonzero vector z . Consider now the operator $CAB = M^{-1} + M^{-1}NM^{-1}$. For any $z \neq 0$

$$(CABz, z) = (M^{-1}z, z) + (M^{-1}NM^{-1}z, z) \neq 0,$$

since the first scalar product at the right is real and, in view of the positive definiteness of the operator, positive and the second scalar product is pure imaginary in view of the fact that the operator $M^{-1}NM^{-1}$ is skew-Hermitian. For the case $B = C = M^{-1}$ the computational scheme of the method has the form

$$\begin{aligned} Mu_1 &= r_0, \\ r_i &= r_{i-1} + a_i Au_i, \\ Mv_i &= r_i, \\ u_{i+1} &= v_i + b_i u_i, \\ x_i &= x_{i-1} + a_i u_i, \end{aligned}$$

where

$$a_i = -\frac{(r_{i-1}, u_i)}{(Au_i, u_i)}, \quad b_i = \frac{(v_i, Au_i)}{(Au_i, u_i)}.$$

If $B = C = N^{-1}$, then the computational scheme and the formulas for the coefficients a_i and b_i remain the same, except that M is replaced by N of course.

Method of incomplete Hermitian decomposition. In this method A is a Hermitian positive definite operator. We represent it as a sum $A = M + N$, where $M = M^*$ and $N = N^*$. If M is nonsingular, then we set $B = C = M^{-1}$. Condition (105.9) holds for $\alpha = 0$ and $\beta = 1$. If M is positive definite, the process runs without degeneration. At each step of the method the error functional with matrix A is minimized. The computational scheme remains the same as in the case of the complete decomposition method.

Speeding up the computational process. As we have already noted, the methods of conjugate directions allow a solution to be found especially rapidly if the operator AB has few mutually distinct eigenvalues. Use is made of this fact to construct various devices for speeding up the solution of equation (105.1), which are based on the following idea.

Suppose the operator A can be represented as a sum $A = M + N$, where the operator M determines the "principal" part of A and allows a simple solution of equations of the type (105.1) with M at the left. Now instead of (105.1) we shall solve the equation

$$(E + NM^{-1})y = b, \quad (106.1)$$

where $Mx = y$. If in some reasonable sense the operator M is close to A , then most eigenvalues of N and hence of the operator NM^{-1} are close to zero or zero. Applying the methods of conjugate directions to equation (106.1) in this case results in finding rapidly a solution.

Observe that it is this idea that underlies the method of incomplete Hermitian decomposition, which in many cases proves to be more efficient than the classical variant of the method of conjugate gradients. It all depends on how lucky one is in decomposing the operator A .

We shall not discuss in detail the computational schemes of speeding up processes, since they rely too heavily on the use of particular features of the operator A .

Exercises

1. Under what conditions is it appropriate to apply one variant or another of the method of conjugate directions?

2. How many iterations are required to implement the different variants of the methods of conjugate directions for an operator A of the form $E + R$, where the operator R is of rank r ?

3. Assuming an operator A to be a matrix evaluate the number of arithmetical operations required for solving a system of linear algebraic equations by the methods of conjugate directions.

4. The matrix of an operator A is Hermitian and differs from a tridiagonal matrix in a small number of its elements. Which of the variants of the methods of conjugate directions is it appropriate to apply in this case?

5. Let $P_0(t)$, $P_1(t)$, $\dots$ be some sequence of polynomials. Choose a vector x_0 and construct a sequence of vectors $x_0, x_1, \dots$ by the rule

$$x_{k+1} = x_k - BP_k(AB)(Ax_k - b), \quad k \geq 0. \quad (106.2)$$

How do the expansions of the discrepancies $r_0, r_1, \dots$ with respect to the Jordan canonical basis of the operator AB change as k increases depending on the choice of a sequence of polynomials?

6. How is sequence (106.2) to be used for the purpose of constructing an initial vector for the methods of conjugate directions that ensures that a solution is obtained in a smaller number of iterations?

7. Which of the systems of vectors in each of the particular variants of the methods of conjugate directions are up to normalization A -pseudodual?

107. Operator equations and pseudoduality

The methods of conjugate directions are not the only methods of solving the operator equation

$$Ax = b \quad (107.1)$$

that rely on the use of bilinear forms. Vast possibilities for creating methods arise from constructing systems of vectors dual or pseudodual to some bilinear form related to the operator A of equation (107.1).

We again assume that A is a nonsingular operator in a unitary space K_n . Consider a bilinear form (Ax, y) and suppose that systems of vectors $u_1, u_2, \dots, u_n$ and $v_1, v_2, \dots, v_n$, A -pseudodual up to normalization, have been obtained for it in some way, i.e. that

$$(Au_i, v_i) \neq 0, \quad (Au_i, v_k) = 0, \quad k < i \quad (107.2)$$

for every $i \leq n$. We show that knowing A -pseudodual systems of vectors makes it possible to construct the process of finding a solution of equation (107.1).

We choose vector x_0 . Since A -pseudodual systems are linearly independent, there is a decomposition

$$x = x_0 + \sum_{j=1}^n a_j u_j. \quad (107.3)$$

If

$$x_i = x_0 + \sum_{j=1}^i a_j u_j,$$

then similarly to (105.3) and (105.4) we have

$$x_i = x_{i-1} + a_i u_i, \quad r_i = r_{i-1} + a_i A u_i. \quad (107.4)$$

Further

$$r_i = Ax_i - b = A(x_i - x) = - \sum_{j=i+1}^n a_j A u_j,$$

and according to the second conditions in (107.2) we find that

$$(r_i, v_k) = - \sum_{j=i+1}^n a_j (A u_j, v_k) = 0$$

for every $k \leq i$. So

$$(r_i, v_k) = 0 \quad (107.5)$$

for $k \leq i$. This makes it possible to determine coefficients a_i from (107.4). That is,

$$a_i = - \frac{(r_{i-1}, v_i)}{(A u_i, v_i)}. \quad (107.6)$$

According to the first conditions in (107.2) the denominator on the right of (107.6) is nonzero.

It follows from (107.5) that the vector r_n is left orthogonal, and by the symmetry of a scalar product simply orthogonal, to the linearly independent vectors $v_1, v_2, \dots, v_n$, i.e. that $r_n = 0$ and the vector x_n is a solution of equation (107.1).

The above methods of solving equation (107.1) are known as methods of dual directions. The number of different methods is infinite in the full sense of the word since there are an infinite number of different A -pseudodual pairs of systems of vectors. The methods of conjugate directions considered earlier obviously belong to this group.

In general there is no analogue of Theorem 105.1 for methods of dual directions, even for a positive definite operator A . The behaviour of errors $e_k = x - x_k$ in these methods is described only by the weak, and yet useful,

Theorem 107.1. *Let P_k be an operator of projection onto the subspace spanned by vectors $u_1, \dots, u_k$ parallel to the subspace spanned by vectors $u_{k+1}, \dots, u_n$. Then*

$$e_k = (E - P_k) e_0. \quad (107.7)$$

Proof. By formula (107.3) we have the following expansion for the error e_0 of an initial vector x_0 :

$$e_0 = x - x_0 = \sum_{j=1}^n a_j \mu_j.$$

But by the definition of the projection operator

$$P_k e_0 = \sum_{j=1}^k a_j \mu_j.$$

The right-hand side of this equation is nothing but $x_k - x_0$. Therefore

$$P_k e_0 = x_k - x_0 = (x - x_0) - (x - x_k) = e_0 - e_k,$$

which proves the theorem.

Interesting results associated with A -pseudodual systems can be obtained by considering the matrix interpretation of the above methods.

We assume that the space K_n is not only unitary but also arithmetical, which is admissible by virtue of the isomorphism of finite dimensional vector spaces. The entire reasoning above remains valid, only the terminology is changed: equation (107.1) becomes a system of linear algebraic equations, the operators are replaced by matrices, and by vectors column vectors are meant. Denote by U (V) a matrix whose columns are vectors $u_1, \dots, u_n$ ($v_1, \dots, v_n$). Then the fact

that those vectors satisfy relations (107.2) implies that the matrix

$$C = V^*AU$$

is nonsingular left triangular. From this we obtain the following factorization of the matrix A :

$$A = V^{-1}CU^{-1}. \quad (107.8)$$

So a knowledge of A -pseudodual, up to normalization, systems of vectors allows us to solve the system of linear algebraic equations (107.1) with error estimates (107.7) and to obtain factorization (107.8) of the matrix A into factors among which there is one triangular factor. We show that the converse is also true. That is, any method of solving systems of linear algebraic equations, based on factoring a matrix into factors among which there is at least one that is triangular, determines some A -pseudodual, up to normalization, systems of vectors. Hence implementing such methods according to schemes (107.3) to (107.6), it is possible to use estimates (107.7).

Consider a matrix P obtained from a unit matrix by reversing its columns or equivalently its rows. It is easy to verify that multiplying an arbitrary matrix C on the right by P reverses the order of the columns of C and multiplying a matrix CP on the left by P reverses the rows of CP . The elements f_{ij} of the matrix $F = PCP$ are therefore connected with the elements c_{ij} of C by the relation

$$f_{ij} = c_{n-i+1, n-j+1}.$$

A number of useful consequences follow. Let us label the diagonals of a matrix parallel to the principal matrix upwards in succession by the numbers $-(n-1), -(n-2), \dots, 0, \dots, (n-2), (n-1)$. The diagonal with the zero index is the principal diagonal. In such an indexing the elements of the k th diagonal are defined by the relation $j-i=k$. If the matrix C satisfies the conditions

$$c_{ij} = 0, \quad k < j-i, \quad j-i < l$$

for some numbers $l \leq k$, then for the matrix $F = PCP$

$$f_{ij} = 0, \quad -l < j-i, \quad j-i < -k.$$

Hence under the transformation $F = PCP$ a diagonal matrix remains diagonal, a right (left) triangular matrix becomes left (right) triangular, a right (left) bidiagonal matrix becomes left (right) bidiagonal, and so on.

Suppose now that use is being made of some method of solving a system of linear algebraic equations (107.1) based on a preliminary factorization of a matrix A :

$$A = QCR, \quad (107.9)$$

where C is a triangular matrix. It may be assumed without essential loss of generality that C is left triangular, since otherwise instead of (107.9) we should consider the factorization

$$A = (QP) (PCP) (PR),$$

where according to the foregoing the matrix PCP must be left triangular. The desired matrices U and V defining A -pseudodual, up to normalization, systems of vectors $u_1, \dots, u_n$ and $v_1, \dots, v_n$ can be given by the equations

$$U = R^{-1}, \quad V = Q^{-1*}.$$

Observe that in factorization (107.9) generated by some numerical method the matrices Q and R are as a rule sufficiently simple. Most often they are unitary or triangular matrices as well as matrices differing from triangular matrices in having permuted rows and columns. Matrices R^{-1} and Q^{-1*} can therefore be found without particular difficulties. At all events the total computational costs of finding them are much lower than those of obtaining factorization (107.9). This is characteristic of such widely known methods as the Gauss methods, the square root method, the Jordan method, the orthogonalization method, the method of reflections, the method of revolutions; methods based on reducing a system to a bidiagonal form and on obtaining normed decompositions; methods of conjugate directions, and so on.

Thus most of the existing numerical methods of solving operator equations (107.1) in a finite dimensional space are in fact methods of constructing A -pseudodual systems of vector. Despite the diversity of their specific modes all these methods can be analyzed from a general standpoint based on Theorem 107.1.

Exercises

1. Assume an operator to be a matrix and vectors of a space to be column vectors. Prove that in the methods of dual directions successive errors are related by

$$e_k = (E - S_k) e_{k-1},$$

where

$$S_k = \frac{u_k v_k^* A}{v_k^* A u_k}. \quad (107.10)$$

2. Prove that operators S_k satisfy the equations

$$S_k^2 = S_k, \quad S_i S_k = 0,$$

$$S_i (E - S_k) (E - S_{k-1}) \dots (E - S_1) = 0, \quad i < k.$$

3. Prove that the operators S_k of (107.10) and the operator P_k of (107.7) are related by

$$P_k = (E - S_k) (E - S_{k-1}) \dots (E - S_1).$$

As already said, the matrix of A in the basis made up of vectors $f_1, f_2, \dots, f_n$ has a tridiagonal form. If in finding a basis $f_1, f_2, \dots, f_n$ there were early terminations of the process of orthogonalization of the power sequence, then the matrix splits into a number of tridiagonal matrices of smaller size.

The positiveness of β_i allows the form of tridiagonal matrix (104.8) to be simplified. Take a nonsingular diagonal matrix D with elements α_i and perform a similarity transformation with matrix (104.8). In the matrix $D^{-1}AD$ the diagonal elements will remain the same as in the matrix A , the elements $\beta_i \alpha_i^{-1} \alpha_{i+1}$ will take the place of the off-diagonal elements β_i , and the elements symmetric with respect to them will equal $\alpha_i \alpha_{i+1}^{-1}$. If we take

$$\alpha_1 = 1, \quad \alpha_{i+1} = \frac{\alpha_i}{\beta_i^{1/2}}, \quad i \geq 1, \quad (108.2)$$

then the matrix D is real and the matrix $D^{-1}AD$ is real symmetric tridiagonal. Since β_i are positive, by (108.2) the diagonal elements of D and hence the off-diagonal elements of $D^{-1}AD$ are also positive.

Thus the problem of determining the eigenvalues and eigenvectors of any Hermitian operator can always be reduced to that of determining the eigenvalues and eigenvectors of a real symmetric tridiagonal matrix A with nonzero off-diagonal elements:

$$A = \begin{pmatrix} \alpha_1 & \gamma_1 & & & \\ \gamma_1 & \alpha_2 & \gamma_2 & & 0 \\ & \gamma_2 & \alpha_3 & \gamma_3 & \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ & & \gamma_{n-2} & \alpha_{n-1} & \gamma_{n-1} \\ 0 & & & \gamma_{n-1} & \alpha_n \end{pmatrix}.$$

Such matrices are called *Jacobian matrices*.

One of the most efficient numerical methods of finding the eigenvalues of a Jacobian matrix is based on the law of inertia for quadratic forms. But before proceeding to describe it we consider some properties of Jacobian matrices.

Theorem 108.1. *All eigenvalues of a Jacobian matrix are simple.*

Proof. Suppose the eigenvalue λ is multiple. Then the rank of the matrix $A - \lambda E$ must not be greater than $n - 2$. But it is clearly not less than $n - 1$, since the off-diagonal elements are nonzero and hence so is the minor in the first $n - 1$ columns and the last $n - 1$ rows. The contradiction obtained proves the theorem.

Corollary. *If a symmetric tridiagonal matrix has an eigenvalue of multiplicity p , then there are at least $p - 1$ elements equal to zero among its upper off-diagonal elements.*

If A is a Jacobian matrix, then the matrix $A - \lambda E$ remains Jacobian for all real λ . Denote by $\sigma_1(\lambda), \dots, \sigma_n(\lambda)$ the principal

minors of $A - \lambda E$. It is clear that for all r , $\sigma_r(\lambda)$ is a polynomial of degree r coinciding up to a sign with the characteristic polynomial of the matrix of the minor of order r for A . Since A is a symmetric Jacobian matrix, all the roots of polynomials $\sigma_r(\lambda)$ are real and simple. Expanding a minor $\sigma_r(\lambda)$ by the last row or the last column we obtain the following recurrence relations:

$$\begin{aligned}\sigma_0(\lambda) &= 1, \quad \sigma_1(\lambda) = \alpha_1 - \lambda, \\ \sigma_r(\lambda) &= (\alpha_r - \lambda) \sigma_{r-1}(\lambda) - \gamma_{r-1}^2 \sigma_{r-2}(\lambda), \\ 2 &\leq r \leq n.\end{aligned}\tag{108.4}$$

Theorem 108.2. *For all $r > 1$ no polynomials $\sigma_r(\lambda)$ and $\sigma_{r-1}(\lambda)$ have common roots.*

Proof. Suppose for some r the number λ is a common root of polynomials $\sigma_r(\lambda)$ and $\sigma_{r-1}(\lambda)$. Then it follows from (108.4) that λ is a root of $\sigma_{r-2}(\lambda)$, since $\gamma_{r-1} \neq 0$. Proceeding with this reasoning we arrive at the conclusion that λ is a root of $\sigma_0(\lambda)$. But $\sigma_0(\lambda)$ has no roots, and therefore no adjacent polynomials have common roots.

Corollary. *If λ is a root of a polynomial $\sigma_r(\lambda)$, then $\sigma_{r-1}(\lambda)$ and $\sigma_{r+1}(\lambda)$ are nonzero and have opposite signs.*

Let λ be a root of none of the polynomials $\sigma_1(\lambda), \dots, \sigma_n(\lambda)$. Calculate for that λ the values of all polynomials from formulas (108.4) and consider the alternation of signs in the sequence

$$\sigma_0(\lambda), \sigma_1(\lambda), \sigma_2(\lambda), \dots, \sigma_n(\lambda).\tag{108.5}$$

Taking into account the law of inertia and the connection between the principal minors of a matrix, the coefficients of the canonical form and the eigenvalues we can say that the number $n_-(\lambda)$ of sign alternations in sequence (108.5) equals the number of the eigenvalues of the matrix A strictly less than λ .

The presence of zero terms in (108.5) does not lead to any difficulties. Indeed, choosing ε sufficiently small, we can make all polynomials $\sigma_r(\lambda - \varepsilon)$ nonzero, while maintaining the signs of the nonzero minors of (108.5). According to the corollary of Theorem 108.2 the signs of the minors that were zero do not affect the general number of sign alternations in (108.5). Therefore all zero terms of sequence (108.5), except $\sigma_n(\lambda)$, may be assigned arbitrary signs. If, however, $\sigma_n(\lambda) = 0$, then λ is an eigenvalue of A . The number of sign alternations $n_-(\lambda)$ in sequence (108.5) without $\sigma_n(\lambda)$ is again equal to the number of the eigenvalues of A strictly less than λ .

Suppose the eigenvalues λ of A are indexed in algebraically decreasing order, i.e.

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n.\tag{108.6}$$

We show how to determine the k th eigenvalue λ_k . Let numbers a_0 and b_0 be known such that

$$b_0 > a_0, \quad n_-(a_0) < k, \quad n_-(b_0) \geq k.$$

Then λ_k is clearly in the half-interval $[a_0, b_0)$. Observe that we may take any number less than $-\|A\|$ as a_0 and any number greater than $\|A\|$ as b_0 . Now set

$$c_0 = \frac{1}{2}(a_0 + b_0)$$

and determine $n_-(c_0)$. If $n_-(c_0) < k$, then λ_k is in the half-interval $[c_0, b_0)$. If $n_-(c_0) \geq k$, then λ_k is in $[a_0, c_0)$. Therefore we can always find a half-interval, half the previous one, containing λ_k . Proceeding with this process we obtain a system of embedded half-intervals $[a_s, b_s)$ containing λ_k , with

$$(b_s - a_s) = 2^{-s}(b_0 - a_0).$$

This allows the eigenvalue λ_k to be localized to any required precision.

The described method of determining the eigenvalues of a tri-diagonal matrix is called the *method of bisections* and sequence (108.5) is called the *Sturm sequence*. Once any eigenvalue λ has been determined, the eigenvectors relating to it are determined as solutions of a homogeneous system of linear algebraic equations with a matrix $A - \lambda E$.

Another method of finding the eigenvalues of a Hermitian operator A is based on some extreme properties of the related quadratic form (Ax, x) .

It is again assumed that the eigenvalues of A are arranged according to (108.6). Suppose that for the eigenvalues $\lambda_{i_1}, \lambda_{i_2}, \dots, \lambda_{i_r}$ orthogonal eigenvectors $x_{i_1}, x_{i_2}, \dots, x_{i_r}$ are known. Denote by L_r the span of those eigenvectors and let λ_{r_1} be the largest of the eigenvalues of A .

Theorem 108.3.

$$\lambda_{r_1} = \max_{\substack{x \neq 0 \\ x \perp L_r}} \frac{(Ax, x)}{(x, x)}, \quad (108.7)$$

and any vector for which a maximum is attained will be an eigenvector corresponding to λ_{r_1} .

Proof. The quotient on the right of (108.7) will remain unchanged if the vector x is multiplied by any nonzero number. Instead of (108.7) we may therefore investigate the following equation:

$$\lambda_{r_1} = \max_{\substack{(x, x) = 1 \\ x \perp L_r}} (Ax, x). \quad (108.8)$$

Choose in K_n an orthogonal basis made up of eigenvectors of the operator A , including the vectors $x_{i_1}, \dots, x_{i_r}$. Then in this basis

equation (108.8) assumes the following $\bar{\lambda}_r$ form:

$$\lambda_{r_1} = \max_{\substack{|\alpha_1|^2 + \dots + |\alpha_n|^2 = 1 \\ \alpha_{i_1} = \dots = \alpha_{i_r} = 0}} \sum_{j=1}^n \lambda_j |\alpha_j|^2, \quad (108.9)$$

where $\alpha_1, \dots, \alpha_n$ are the coordinates of a vector x in the expansion with respect to the chosen basis.

The unitary space K_n is complete, the set of vectors satisfying the condition $(x, x) = 1$ is bounded, and therefore there are vectors on which the maximum in (108.8) is attained. Let $\hat{\alpha}_1, \dots, \hat{\alpha}_n$ be the coordinates of one of such vectors. We show that if $\lambda_j \neq \lambda_{r_1}$, then $\hat{\alpha}_j = 0$. Indeed, if $\lambda_j > \lambda_{r_1}$, the equation $\hat{\alpha}_j = 0$ follows from the condition $x \perp L_r$, since eigenvalues greater than λ_{r_1} may exist only among $\lambda_{i_1}, \dots, \lambda_{i_r}$. If $\lambda_j < \lambda_{r_1}$, then the coordinate $\hat{\alpha}_j$ cannot be nonzero, since otherwise we could obtain a larger value of the sum in (108.9) by taking $|\hat{\alpha}_{r_1}|^2 + |\hat{\alpha}_j|^2$ and 0 instead of the numbers $|\hat{\alpha}_{r_1}|^2$ and $|\hat{\alpha}_j|^2$.

Thus the largest value of the sum in (108.9) may be attained only for those $\hat{\alpha}_1, \dots, \hat{\alpha}_n$ for which $\hat{\alpha}_j \neq 0$ only when $\lambda_j = \lambda_{r_1}$. Hence the maximum value of the quotient in (108.7) is attained on eigenvectors corresponding to λ_{r_1} . Of course equation (108.7) follows too.

This theorem shows a way of constructing a numerical method of finding the eigenvalues and the eigenvectors of the operator A , which is based on seeking the maxima of the function

$$\theta(x) = \frac{(Ax, x)}{(x, x)}$$

called the *Rayleigh quotient*. We shall restrict ourselves to a brief discussion of the method.

Take a vector x and consider the behaviour of the Rayleigh quotient in the neighbourhood of that vector. Simple transformations show that for any vector l and a small ε the following representation holds:

$$\theta(x + \varepsilon l) = \theta(x) + \frac{2\varepsilon}{(x, x)} \operatorname{Re} (Ax - \theta(x)x, l) + O(\varepsilon^2). \quad (108.10)$$

If $\operatorname{Re} (Ax - \theta(x)x, l) \neq 0$, then for sufficiently small real ε whose sign coincides with that of the real part of the scalar product we get $\theta(x + \varepsilon l) > \theta(x)$, the inequality being opposite for small ε of the opposite sign.

Suppose the vector x is not an eigenvector. Then $Ax - \theta(x)x \neq 0$ and we may set

$$l = Ax - \theta(x)x. \quad (108.11)$$

The scalar product under the sign of the real part in (108.10) is positive and we find from (108.11) for l that

$$\theta(x + \varepsilon l) = \theta(x) + \frac{2\varepsilon}{(x, x)}(l, l) + O(\varepsilon^2).$$

We can now choose for a small positive ε a vector $y = x + \varepsilon l$ such that $\theta(y) > \theta(x)$.

Proceeding in a similar way we obtain, under some additional conditions not to be discussed here, a maximum of the Rayleigh quotient and hence a maximum eigenvalue and the corresponding eigenvector. If at some step it is found that the vector l of (108.11) is zero, then this means that x is an eigenvector and there corresponds an eigenvalue to it equaling $\theta(x)$. In this case, it is not possible at all to guarantee the increase of the Rayleigh quotient relying only on formula (108.10). For this reason the eigenvectors of the operator A are called *stationary points of the Rayleigh quotient* in the terminology of mathematical analysis.

If one or several eigenvectors are known, then in order to determine another eigenvector we shall again seek the maximum of the Rayleigh quotient, yet not in the entire space but only in the orthogonal complement of the subspace spanned by the previously found eigenvectors.

Exercises

1. Prove that each root of the polynomial $\sigma_{r-1}(\lambda)$ of (108.4) lies between two adjacent roots of the polynomial $\sigma_r(\lambda)$.

2. Suppose a tridiagonal symmetric matrix has several zero off-diagonal elements. Does this mean that the matrix has multiple eigenvalues?

3. Prove that, with the coefficient α_n of matrix (108.3) increasing (decreasing) without limit, all the eigenvalues except the maximum (minimum) one remain bounded.

4. Prove that, with the absolute value of the coefficient β_{n-1} of (108.3) increasing without limit, all the eigenvalues except the maximum and the minimum one remain bounded.

5. A Jacobian matrix of size $2n+1$ with elements $\alpha_i = n-i+1$ and $\beta_i = 1$ has no multiple eigenvalues. Prove that its maximum eigenvalue differs from the one closest to it by an amount of an order of $(n!)^{-2}$.

6. Prove that

$$\lambda_1 = \max_{x \neq 0} \frac{(Ax, x)}{(x, x)}, \quad \lambda_n = \min_{x \neq 0} \frac{(Ax, x)}{(x, x)}.$$

7. What does the quantity

$$\rho = \min_{\substack{x \neq 0 \\ x \perp L_r}} \frac{(Ax, x)}{(x, x)}$$

mean in the notation of Theorem 108.3?

Conclusion

This textbook has provided ample enough material for the reader to comprehend both the theoretical basis of linear algebra and the numerical methods it employs. However, because of the peculiarities of individual syllabuses and the limited lecture time allowed for this course some sections of this text may escape the reader's attention. We shall therefore characterize briefly all the material presented.

Linear algebra as a science studies sets of special structure and functions of them. In general similar problems face other areas of mathematics, for example mathematical analysis. A characteristic feature explicitly of linear algebra is that the sets are always finite dimensional vector spaces and the functions are linear operators.

The general properties of vector spaces are discussed in Sections 10 and 13 to 21, and the general properties of linear operators are treated in Sections 56 to 61 and 63 to 74. The information presented in those sections can be obtained in various ways, including direct methods employing no concepts or tools but the most elementary. Yet one of the additional concepts does deserve comment. It is the determinant.

As a numerical function given on systems of vectors, the determinant is a relatively simple object. Nevertheless it possesses many important properties. They have made it a widely used tool significantly facilitating various studies. Besides, the determinant is very often employed in constructing numerical methods. All this has led us to give the concept of determinant sufficiently much attention by considering its geometrical and algebraic properties in Sections 34 to 42 and 62. As a tool of study the determinant is used in this book to prove diverse assertions.

Another numerical function of two independent vector variables, the scalar product, defines two major classes of vector spaces called Euclidean and unitary. The basic new concept in these spaces is that of orthogonality. Sections 27 to 33 discuss the properties of vector spaces due to the scalar product, and Sections 75 to 81 describe the properties of linear operators due to the scalar product.

Systems of linear algebraic equations play an exceptionally important role throughout mathematics, not only in linear algebra. Devoted to the study of the various aspects of them are Sections 22, 45, 46 and 48.

As a rule only the material listed constitutes the basis of a course in linear algebra, with a course in analytic geometry added as a separate unit. In the present text the necessary facts from analytic

geometry are given not in isolation but intermittently with the corresponding facts from linear algebra. Such a presentation of the material has allowed us to attain a number of advantages: to reduce many proofs of the same type in both courses, to emphasize the geometrical interpretation of abstract algebraic notions, such as vector space, plane in a vector space, determinant, systems of linear algebraic equations and so on.

Linear algebra is to a great extent enriched in new facts if the concept of distance between vectors and that of limit of a sequence of vectors are introduced into vector spaces. The necessity of introducing them is also dictated by the requirements of numerical methods. The metric properties of vector spaces are studied in Sections 49 to 54, and those of linear operators are considered in Sections 82 to 84. Of course all these facts are usually given in functional analysis, but as a rule many results important for finite dimensional vector spaces remain unaccentuated.

Numerical solution of problems in linear algebra is nearly always accompanied by the appearance of round-off errors. Therefore the computer to be must realize what changes in the properties of different objects of linear algebra result from small changes in vectors and operators. The effects of small perturbations are treated in Sections 33, 87 and 89.

Properties of many objects of linear algebra may be reversed even by arbitrarily small perturbations. Thus a linearly dependent system of vectors may become linearly independent or increase its rank, an operator with Jordan structure may become one of a simple structure, a compatible system of linear algebraic equations may become incompatible and so on. All these facts give rise to exceptionally great difficulties in practical solutions of problems.

It is imperative that the reader should once again peruse Section 22 to study the example given there and to ponder on the questions posed at the end of the section.

Despite the instability of many notions of linear algebra its problems can be solved in a stable way. To demonstrate this the book includes a description of a stable method of solving systems of linear algebraic equations. Its theoretical justification and general scheme are given in Sections 85, 86 and 88.

The last part of the book is devoted to the description and investigation of various questions relating to bilinear and quadratic forms. These numerical functions play a very important role in linear algebra and are closely related to the construction of numerical methods. Sections 90 to 94 deal with the general properties of bilinear forms and with the connection of their transformations with matrix decompositions, Sections 98 to 101 discuss the extended concept of orthogonality and Sections 103 to 108 consider uses of bilinear forms in computational processes.

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